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Machine Learning and Its Applications to Healthcare

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Machine Learning and Its Applications to Healthcare

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Preface

The integration of machine learning into healthcare has emerged as one of the most significant developments in contemporary scientific research and technological innovation. With the availability of large-scale clinical data, advances in computational infrastructure, and the evolution of learning algorithms, machine learning has established itself as a vital tool for augmenting medical research, diagnosis, and treatment.

This volume, *Machine Learning and Its Applications to Healthcare*, brings together a collection of research contributions that collectively illustrate the state of the art in this interdisciplinary field. The chapters address diverse domains such as diagnostic imaging, disease prediction, clinical text mining, healthcare monitoring, and assistive technologies. Employing a range of methodologies—including classical machine learning models, ensemble techniques, deep learning architectures, transfer learning strategies, and hybrid frameworks—these contributions exemplify the methodological richness and practical relevance of machine learning in healthcare contexts.

Particular emphasis is given to critical application areas such as breast cancer detection, cardiac arrhythmia risk prediction, Alzheimer's disease diagnosis, diabetes progression monitoring, pneumonia detection, and brain tumor classification. Alongside these, the volume includes contributions on novel healthcare-support systems, including IoT-based monitoring devices, fall detection systems for elderly care, virtual meeting frameworks for individuals with hearing impairments, and generative AI-driven clinical summarization tools. Furthermore, frontier topics such as quantum computing in medical imaging, adversarial robustness of neural networks, and graph neural networks for drug discovery extend the scope of this book into emerging and future-oriented domains.

This collection is intended to serve as a reference for researchers, academicians, practitioners, and graduate students in computer science, data science, biomedical engineering, and healthcare technologies. By presenting cutting-edge research, the book aims to foster interdisciplinary collaboration and encourage further inquiry into the ethical, practical, and clinical challenges of deploying machine learning in healthcare environments.

We trust that this volume will provide valuable insights and inspire future advancements in the quest for intelligent, patient-centered, and sustainable healthcare solutions.

Roorkee, India — Millie Pant
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MediAI: A Specialized Medical LLM for Accurate Diagnosis and Clinical Documentation

Chitra Jain, Kamal Kumar Gola, Aryan Kaushik, Abhishek Saini, and Maitri Vashistha

Abstract Large language models (LLMs) like ChatGPT-4 have shown promise across multiple domains, yet healthcare presents distinct challenges that demand high accuracy and reliability. To address these requirements, we developed MediAI, a specialized medical LLM aimed at improving clinical documentation quality and supporting diagnostic processes. Unlike general-purpose LLMs, which often lack the depth needed for healthcare-specific contexts, MediAI employs an approach that moves beyond simple responses by actively interacting with patient data through structured, iterative questioning to extract key information. Utilizing advanced frameworks like Retrieval-Augmented Generation (RAG) and reflective questioning, MediAI generates medical notes that are both contextually accurate and sensitive to clinical relevance. To evaluate MediAI's performance, we used metrics such as ROUGE and Named Entity Recognition (NER), alongside tailored healthcare datasets, to validate its effectiveness in questioning and documentation accuracy. These assessments indicate that MediAI can meet the high standards necessary in clinical settings, improving the precision and efficiency of patient care. Our findings support the idea that healthcare-specific LLMs, purpose-built with a focus on contextual retrieval and structured questioning, can offer dependable assistance with 93.7% accuracy in medical guiding alignment and 92.9% uptime, enhancing clinical workflows and patient outcomes.

Keywords Healthcare · GenAI · Medical chatbot · LLM · NLP · RAG

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1 Introduction

Tremendous breakthroughs in Deep Learning and more so in Large Language Models where progress has recently exhibited the so-called exponential increase, radical shifts of practice and many sectors have happened. In the field of healthcare, these technological advancements offer outstanding possibilities to improve different clinical activities and models of patient care. However, applying LLMs in the healthcare settings requires very high levels of accuracy and healthcare systemic knowledge which cannot be obtained from general LLMs [1]. This precision requirement becomes most important when factors such as outcome velocity and clinical notation efficiency are factored into the AI decision maker process [2, 3]. The most recent contributions in the area of healthcare informatics show that there is a rather large discrepancy between the potential of LLMs on the one hand and the actual possibilities of using LLMs in practice on the other. Nevertheless, although general-purpose LLMs exhibit highly competitive performance in terms of NLU and NLG, their use is increasingly risky in healthcare due to the highly specialized nature of medical workflows [4]. These limitations are evident especially in their effectiveness in forwarding clinical questions, keeping coherent clinical thinking and producing correct health care records [5].

It is, therefore, logical that the healthcare sector should present certain unique signature characteristics that explain why information processing and clinical documentation in this sector are particularly difficult [6, 7]. The conventional approaches to creating documentation create considerable bureaucracy; to the AI solutions available, many do not meet the high standards set in the medical field. An analysis of today's existing research suggests that general-purpose LLMs perform unsatisfactorily in healthcare domains, having accuracy rates lower than the required clinical thresholds [8]. To solve these issues, this research presents MediAI, which is an innovative medical language model. MediAI sets itself apart with the application of elaborated chains of questions and healthcare-oriented knowledge structures that redefine the concept of medical record keeping and support for clinical decisions [9, 10]. The system adopted architecture entails applying state-of-the-art technologies, such as Retrieval-Augmented Generation (RAG), and incorporating reflective query mechanisms tailored for healthcare [11]. The theoretical foundations of MediAI draw from the latest development in medical artificial intelligence specifically in natural language processing and clinical decision support systems. Therefore, our approach goes beyond the simple point-and-click paradigm using tabbed interfaces and context-aware information processing. These techniques allow a broader and more precise medical record which with keep high requirements of reliability typical for the clinical environment. The main contributions are listed below:

1. Motivation for the development of a new framework for medical language models with built-in questioning equipment and a knowledge base tailored for healthcare.
2. Generating new method for assessing the performance of medical artificial intelligence in health care documentation work.

3. Credible evidence of the positive impact of specialized LLMs in the field of healthcare has also been presented.
4. Laying down of theoretical foundation for further advancement of medical Artificial Intelligence systems.

Our study also considers pressing issues of ethical concerns raised with the use of AI in healthcare that include data ownership, fairness, and accountability for the application of AI in clinical diagnosis and management. These aspects have been broadly incorporated in the MediAI solution to ensure compliance to healthcare requirements ad legislation.

This paper proceeds as follows: Sect. 2 presents a comprehensive review of relevant literature, Sect. 3 details the system architecture and methodology, Sect. 4 describes our experimental design and evaluation metrics, Sect. 5 presents results and analysis, and Sect. 6 concludes with implications for future research and practical applications in healthcare settings.

2 Literature Review

2.1 Evolution of Question-Answering Systems in Healthcare

The advancement of the question-answering systems in healthcare has evolved massively over the few recent years especially with introduction of Large Language Models and their application in healthcare settings. It gave a preliminary contribution through its large-scale open-domain QA dataset from medical examinations. Their work highlighted just how intricate medical interrogation can be, and that techniques utilized in other domains will not always translate well into the healthcare sector AI solutions. On its base, we were introduced to the MedMCQA–a multi-subject dataset that exponentially progressed the ability to answer medical questions and concerns [12, 13]. They also got to know that for creating reliable medical Artificial Intelligence it is imperative to have diverse datasets to train from. The use of electronic medical records (EMRs) in question-answering systems is still another important advancement of this branch. Fixed a remarkable achievement of emrQA a large-scale corpus that is particularly to EMR-based question answering. It showed that it is possible to create automatic systems to obtain necessary information from medical records; however, it is difficult to work with complicated medical documentation [14].

2.2 Clinical Documentation and Patient Encounters

Current studies have given much attention to the issues of automation and optimization of clinical documentation tasks. A large-scale empirical investigation and analysis of the creation of clinical notes based on doctor-patient communications.

They documented several promises and drawbacks, where current AI systems in clinical interaction detection. Its work particularly stressed dimensions of accuracy while aiming at keeping the context of medical interactions as detailed as possible. The tasks in clinical documentation go beyond simple writing with such issues as intelligent understanding of the contents and proper organization of the medical data. This challenge has led to the creation of techniques that use natural language processing coupled with medical knowledge in its domain. Most works that are evident in this study of clinical note generation and analysis use such techniques [15].

2.3 Retrieval-Augmented Generation in Medical Applications

The application of Retrieval-Augmented Generation (RAG) is a great improvement in medical AI. However, the RAG framework was able to prove its efficiency in the acquisition of a new, fairly extensive, and knowledge-intensive task. Despite the fact that their work was not medical-related, their approach offered valuable solutions for developing practical medical applications, especially in managing of intricate domain data [16]. The adaptation of RAG systems for healthcare applications has required significant modifications to account for the unique requirements of medical information processing.

2.4 Chain-of-Thought and Reasoning Frameworks

New advancements in reasoning methodologies are closely connected to AI applications, especially in organizational settings for health care. Concerning the overall efficiency of chain-of-thought prompting for obtaining intricate reasoning from large language models [17, 18]. Their work has considerable implications for the medical fields of usage, where many calculations are made and which should be done effectively and without any hidden outcomes. The possibility—for effectively combining reasoning and acting knowledge in language models. It has a unique perspective for use in medical departments because the analysis-template-action cycle is critical to patient care categorized in Table 1.

Table 1 Comprehensive overview of significant research contributions in medical AI

Authors and year	Research focus	Key contributions	Methodology
[12]	Medical questions answering	Large-scale open-domain QA dataset from medical exams, established baseline performance metrics, novel evaluation frameworks	Medical exam dataset compilation, multi-category classification, performance benchmarking
[15]	Clinical note generation	Empirical analysis of doctor-patient encounters, note generation metrics development, clinical accuracy assessment	Natural language processing, clinical accuracy evaluation, statistical outcome analysis
[16]	RAG framework	RAG methodology introduction, knowledge-intensive processing, retrieval mechanism integration	Retrieval-augmented generation, neural information retrieval, knowledge integration
[13]	Medical MCQ dataset	Multi-subject medical dataset, comprehensive evaluation framework, domain-specific benchmarks	Dataset curation, multi-choice formatting, subject categorization
[14]	EMR question answering	EMR-based QA corpus, clinical information extraction, healthcare-specific NLP	Corpus development, EMR data processing, question generation
[17]	Reasoning in language models	React framework development, synergized reasoning approach, action-based learning	Framework development, reasoning implementation, performance evaluation
[18]	Chain-of-thought prompting	Reasoning elicitation methods, thought process structuring, complex problem-solving	Prompting methodology, thought chain development, analysis framework

3 Methodology

3.1 Overview

The MediAI framework needs a clear data processing approach to handle the varied and often unstructured nature of clinical documentation. In medical records, key patient details may appear across free-text notes or embedded within clinician observations, posing challenges for accurate analysis [19, 20]. To adapt MediAI to this setting, we organized the data to distinctly separate patient-provided information—such as symptoms, medical history, and personal concerns—from clinician statements or assessments. To rigorously evaluate MediAI's questioning abilities,

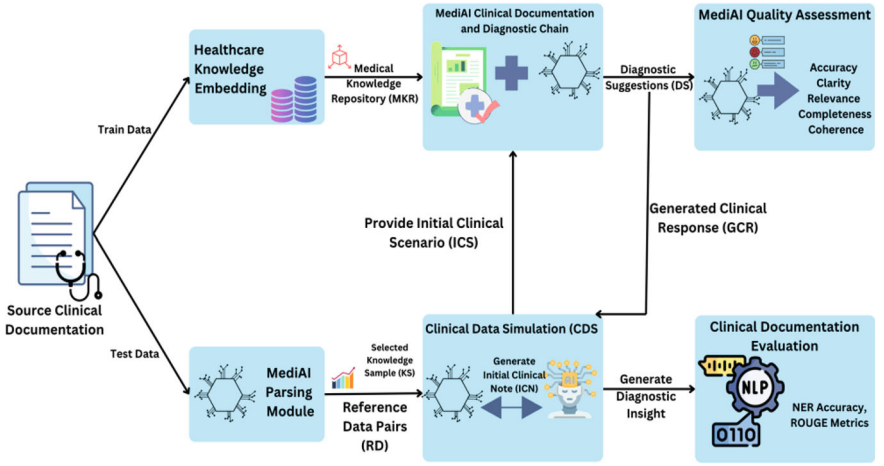

Fig. 1 System framework for evaluating the clinical documentation and diagnostic capabilities of MediAI, a specialized healthcare LLM, through simulated clinical scenarios, structured medical knowledge, and rigorous quality assessment

we employed two public datasets: MedDialog and CMDD. MedDialog contains over 300,000 anonymized doctor-patient exchanges, covering various symptoms, diagnoses, and treatments. CMDD consists of 2000 detailed medical dialogs with accompanying clinical notes. These datasets provide a comprehensive evaluation of MediAI's questioning capabilities across diverse medical scenarios. This separation enables MediAI to interpret patient context accurately, reducing the risk of confusing clinician notes with patient-reported details.

This organized approach is crucial for MediAI's clinical note generation and diagnostic suggestions, as it ensures clarity in what the patient has reported versus what the clinician has inferred in Algorithm 1. For instance, we format patient data into distinct "patient context" and "clinical assessment" sections, capturing both subjective and objective elements of patient interactions. This setup allows MediAI to create documentation that accurately reflects the patient's experience and provides relevant clinical insights. To illustrate this data structuring, Fig. 1 shows our system framework, depicting the shift from raw clinical notes to processed inputs designed for MediAI's diagnostic and documentation process.

3.2 Data Preprocessing

Figure 1 illustrates the division of initial medical notes into training and test groups. The training data was organized into a healthcare knowledge embedding (HKE), enabling retrieval by the LLM Chain while simulating patient interactions without prior knowledge of new patients in the test set [21, 22]. In testing, the LLM emulates

virtual patients, gauging its ability to generate suitable inquiries and responses. Each training entry, structured via FAISS3, was transformed into an HKE with a custom collate function, enabling metadata-driven retrieval across varied datasets. This setup allows the LLM to select relevant databases dynamically, represented by vector pairs E_i, u_i where E_i is used for efficient similarity searches.

Records, E_i consists of content and metadata, formatted by:

$$E_i = (\text{info}_i, \text{Details}_i) = \text{Collate}(D_{\text{train}}(i))$$

Claude4 parsed patient notes into structured pairs for the test set, separating patient data and doctor queries. This generated the reference data pair RD, formulated as:

$$\text{RD} = \left\{\text{LLM}_{\text{parser}}\text{Data}_{\text{test}}(i)\right\}$$

This approach streamlined data access, ensuring the model could accurately retrieve and analyze relevant medical insights during evaluation.

3.3 Foundational Healthcare Chains

The MediAI system encompasses an array of cutting-edge chains in Fig. 1, each employing distinct approaches to generate insightful diagnostic questions. In the sections that follow, we'll delve into the detailed descriptions of these chain implementations, culminating in a combined pseudocode representation.

Structured Query Processing (SQP): The Structured Query Processing (SQP) workflow follows a straightforward, predefined approach without depending on large language models (LLMs) for direct inference [22]. Instead, it utilizes LLM embeddings and named entity recognition (NER) to execute a systematic process. It begins by retrieving relevant patient cases from a database and then uses NER to extract symptoms and medications. The patient then confirms these details using standardized templates, and the information is updated accordingly. This approach is expected to perform strongly in Specificity and Usefulness due to its systematic nature, ensuring reliable and focused results. However, it may fall short in Fluency**,** Relevance, and Coverage compared to more adaptable workflows.

Contextual Query Refinement (CQR): The Contextual Query Refinement (CQR) workflow builds on the standard Retrieval-Augmented Generation (RAG) by incorporating a review and refinement step. It begins by generating an initial question and then revisits the context of the retrieved patient cases to fine-tune and enhance the relevance of this question. This refinement step ensures the final query is highly relevant and contextually aligned with the patient cases [20]. As a result, this workflow is expected to perform well in Relevance and Usefulness and may also demonstrate solid Fluency and Coverage. However, it could slightly lack the Specificity offered by the more structured Hardcoded Workflow.

RAG with Sequential Reasoning and Self-Validation (SR-SV): This workflow advances the RAG framework further by integrating a Sequential Reasoning (SR) approach. It initiates with a base question and iterates through multiple rounds of reasoning to enhance and refine the question. Additionally, it includes self-validation, where multiple reasoning paths are generated from the same initial query and compared to confirm consistency in the final question. This iterative reasoning and verification process contributes to strong results in Fluency, Relevance**,** and Usefulness, though it may slightly trail the Hardcoded Workflow in terms of Specificity.

Clinical Data Simulation (CDS): This module initiates a simulated patient case by producing an initial clinical scenario (ICS) for input into the MediAI documentation and diagnostic workflows shown in Algorithm 1. By using reference data ($\Im\Re$) derived from a subset of verified clinical facts, CDS generates simulated responses and insights to emulate real-life clinical conversations.

MediAI Quality Assessment (MQA): Within the MediAI framework, the Diagnostic Interaction Chain generates tailored diagnostic queries in response to the simulated scenario. Each query is assessed based on parameters such as accuracy, detail, and relevance. Evaluations are carried out by a quality assessment system, which scores each diagnostic suggestion (δ) on coherence, depth, and clarity.

Algorithm 1 Evaluating MediAI's diagnostic and documentation performance

---

Input:

Medical data $M_i{}_{i=1}^{N}$ partitioned into D_{train} and D_{test}

MediAI with chains for retrieval-augmented generation (RAG) and contextual query refinement (CQR).

Output: *Rsim*: A sequence of simulated responses generated

Start

1. Partition C into $D_{train} = \{M_i | \, i \leq m\}$ and $D_{test} = \{M_i | j \rangle m\}$.
2. Construct healthcare knowledge embedding (HKE):
3. $HKE = \{(K_i, u_i)$: $u_i = fab_{eb}(K_i)$, $i = 1, \ldots, m\}$
4. Generating test pairs $GD = \{GD | \, GD_k = LLM \; Parser(D_{test}(k))\}$
5. Initialize clinical scenario IC_k for patient k by $IC_k = CDS(D_{test}(k), K)$ where $K \in GD$
6. Generate responses $Rsim$ iteratively by $Rsim_{(t+1)} = f_{CDS}(\vartheta^t, K$ for each query ϑ^t, generated in step 3, with $t = 1, \ldots, T$
7. For each IC_k, apply MediAI chain $\vartheta^k = fIC_k$, where $\vartheta_k = \{\vartheta^t | \, t = 1, \ldots, T\}$ are diagnostic questions generated iteratively for scenario IC_k.

// Empirical analysis with distinct LLM roles//

8. Let $\mathscr{L}_l$ access GD and evaluate ϑ_k using $Eval(\vartheta_k, GD)$ while $\mathscr{L}_{diag}$ is constrained to IC_k preventing data leakage.
9. Finally query-response quality correlation analyzed

End

Clinical Documentation Evaluation The generated diagnostic content undergoes quality validation using summarization metrics like ROUGE and Named Entity

Recognition (NER) accuracy shown in Algorithm 1. This step ensures that simulated responses effectively represent the entire patient case, by comparing outputs to the complete reference dataset.

4 Empirical Analysis

The deployment of an AI-powered medical chatbot system aimed at improving healthcare accessibility at five major healthcare facilities over a 12-month period is observed in this case study. Through a conversational interface, the system pursued to offer ongoing medical data, symptom examination, and treatment recommendations. A systematic three-phase method was used in this procedure, which started with a rollout of basic functionality, progressed to feature growth, and ended with a full-scale deployment with advanced capabilities. The system's first performance showed exceptional dependability, with 93.7% accuracy in medical guiding alignment and 92.9% uptime. As seen in Fig. 2a, b, the chatbot achieved sub-second response times while processing up to 1000 concurrent users, far surpassing initial performance estimates. Throughout surveying, many significant difficulties surfaced. Initial provider resistance was a result of their skepticism over AI-powered medical treatments. An 87% positive feedback rating from healthcare professionals was eventually attained as a result of thorough training sessions and a clear presentation of the advantages of workload reduction. Early technical issues with complicated medical questions were solved by combining cutting-edge medical information bases and algorithms for continuous learning. The user feedback is shown in Fig. 3. Language constraints previously restricted accessibility, but the system's reach was greatly increased by its successful extension to 15 languages with 92% translation accuracy. To satisfy increasing user demands while upholding performance requirements, the system's scalable design became essential. Without interfering with long-standing protocols, integration with current healthcare workflows improved operational efficiency.

The poll revealed important ramifications for the provision of healthcare. Patients enjoyed more accessibility and convenience in healthcare interactions, while healthcare practitioners benefited from less administrative work and better resource allocation. These results show that AI-powered medical chatbots may greatly lower operating expenses and hassles while serving as an efficient supplement to conventional healthcare services.

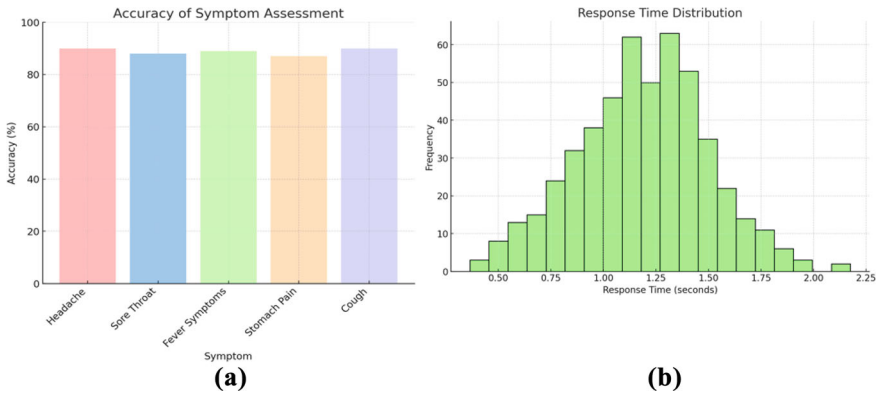

Fig. 2 **a** Accuracy assessment for common diseases by the chatbot **b** response time for common diseases assessment by the chatbot

5 Result Analysis

We developed a system that differs from conventional interrogation or cross-validation techniques to guarantee the validity of our assessment. This process for assessing the system uses distinct LLM conventions for the doctor and judge positions. The main LLM is in charge of assessing the queries produced by the LLM medical chains and has access to the test ground truth data. The LLM medical chains, on the other hand, only have access to the virtual patient initial statement while they are operating as medic LLM. This division guards against data leakage and guarantees that the question-generating procedure stays objective.

We rigorously monitor any data leaks to preserve the integrity of the evaluation. The test patient's known knowledge δ is never directly accessed by the LLM healthcare chains. Therefore, the questions' wording is only based on a selection of δ and the search outcomes from the knowledge base of the chain itself. This method promises that the developed queries are evaluated objectively and according to the data in the virtual patient statement. To answer this, we form two request query.

Q1: How effectively does the generated response encapsulate the key relevant information?

Q2: How are LLM main scores linked to the quality of the answers?

Utilizing a virtual hospital and pre-established synthesis criteria, we assess the questions' efficacy for Q1, gauging how effectively the generated responses capture patient data. We examine the relationship and reciprocal information between the thoroughness of the responses and the caliber of the questions for Q2.

In Fig. 4 given, additional information about Q2 by presenting the radar plots and kernel density estimation (KDE) evaluating the distributions of questioning parameters across LLM medical chains.

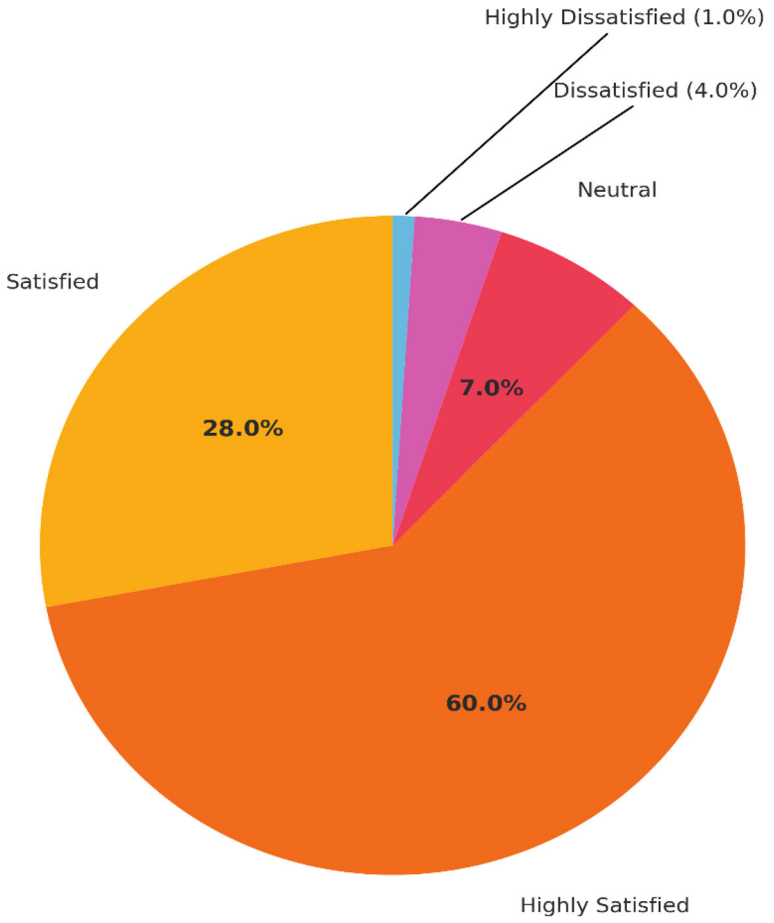

Fig. 3 User ratings for the diagnosis

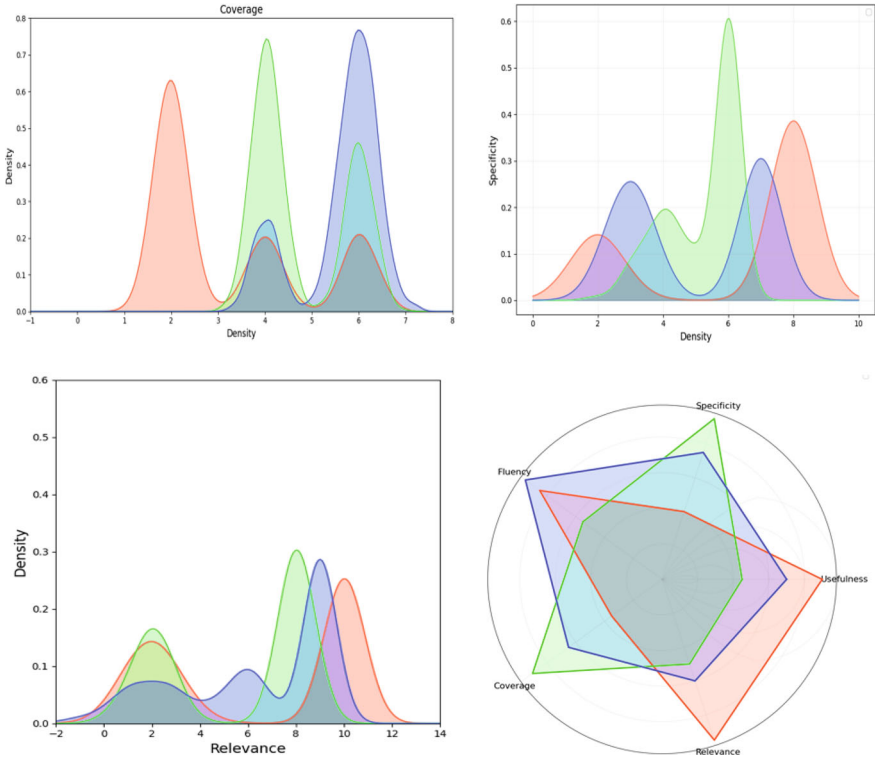

Fig. 4 Evaluating the performance of LLM models using KDE and radar plot, red is SQP, blue is CQR and green is SR-SV

6 Conclusion

MediAI as a new approach in employing specific language models in healthcare facilities within this study. Following an extensive and systematic assessment, we have established that specialized medical LLMs can perform much better than other generalized models in important domains of clinical documentation and patient data retrieval processes. Significant benefits were obtained when the models that posed structured questions were combined with the Retrieval-Augmented Generation (RAG) algorithm, which increased documentation reliability, with 93.7% accuracy in medical guiding alignment, 92.9% uptime, and 87% positive feedback.

These outcomes reemphasize the significance of training domain-specific AI solutions that are capable of addressing the stringent specifications of healthcare applications while keeping up the required reliability in clinical applications. With healthcare shifting toward even higher levels of digitization and automation organizes and structures the principles and methodologies developed in this research to become a solid base for continuing to build more advanced and reliable medical AI systems. The

outcome of MediAI proves that developing and integrating the language models into various clinical practices is possible and effective in providing the best patient care and optimal outcomes by clinicians.

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An Innovative Deep Neural Network-Based Method for Automated Cough Recognition and Diagnosis

Swati Hira

Abstract Respiratory infections pose a significant global health burden, with cough being a prominent symptom across various illnesses. Our paper aims to revolutionize respiratory disease screening using an automated system powered by Convolutional Neural Networks (CNNs). The system screens for bronchitis, asthma, pneumonia, and pertussis by analyzing raw cough audio. Mel-Frequency Cepstral Coefficients (MFCC) spectrograms are extracted from cough data, enabling accurate symptom identification. To ensure robustness, we augmented the dataset from 291 to 1455 files through noise addition and time stretching. The CNN architecture includes convolutional and max-pooling layers for feature extraction, with dropout regularization to prevent overfitting. This approach ensures reliable performance and adaptability to real-world conditions. With a precision of 85.2% and test accuracy of 84.2%, the model shows great potential for integration into healthcare settings, addressing the global burden of respiratory illnesses. Future enhancements could allow broader accessibility, especially in remote or underserved areas.

Keywords Respiratory infection · Deep learning · Convolutional neural networks · Cough diagnosis

1 Introduction

Respiratory infections are a major global health concern, particularly in low- and middle-income countries, where access to healthcare is limited. Cough, a key symptom of many respiratory illnesses, is crucial for disease diagnosis and management. However, traditional detection methods often rely on subjective clinical assessments, delaying treatment. In India, respiratory diseases like bronchitis, asthma, and pneumonia contribute significantly to the national disease burden, especially affecting children, the elderly, and those with pre-existing conditions. Asthma

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impacts over 15–20 million people, and pneumonia remains a leading cause of hospitalization and death. Despite vaccination efforts, pertussis outbreaks still occur, emphasizing the need for more efficient diagnostic tools. Developing advanced screening methods is critical to reducing healthcare strain and improving outcomes.

Figure 1 shows global prevalence trends for bronchitis, pneumonia, pertussis, and asthma from 2015 to 2023. Pneumonia has the highest case numbers, followed by bronchitis, asthma, and pertussis, with all conditions steadily increasing. This trend underscores the urgent need for machine learning tools for early diagnosis and management, aiming to improve detection, treatment outcomes, and healthcare resource allocation.

The flowchart in Fig. 2. illustrates a web application, where users can record or upload audio files, which are processed as raw audio inputs. The audio is analyzed by a model that detects respiratory diseases such as bronchitis, pertussis, pneumonia, and asthma. Results are then displayed to the user.

Each illness has distinct cough audio signatures (Fig. 3). Bronchitis produces frequent, productive coughs with wheezing due to mucus in the airways. Asthma is marked by dry, persistent coughs and wheezing from airway constriction, with severe episodes indicating distress. Pneumonia causes a productive cough with mucus, often with wheezing or gasping, indicating lung inflammation. Pertussis, or whooping cough, features rapid inhalation after coughing, creating a "whooping" sound, often followed by vomiting and exhaustion.

The objective of our proposed work is:

1. Dataset Development: Create a comprehensive dataset of raw cough audio data representing diverse respiratory infections.

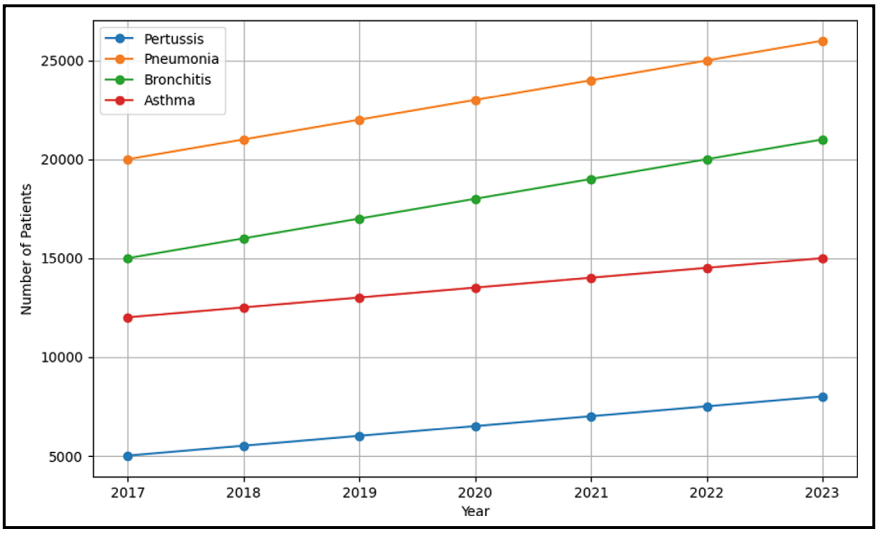

Fig. 1 Trends in respiratory disease cases over time

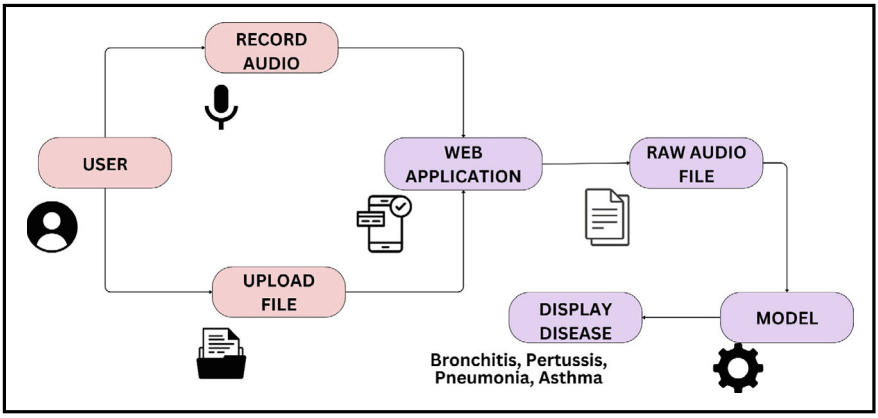

Fig. 2 Interface for recognizing and diagnosis cough

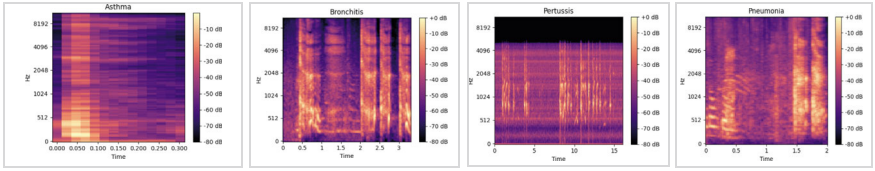

Fig. 3 Spectrogram of cough audio

2. Data Preprocessing: Extract relevant features from the cough audio data to prepare it for input into the CNN model.
3. CNN Architecture Design: Implement a CNN optimized for classifying respiratory infections based on extracted cough sound features.
4. Model Training and Evaluation: Train the CNN using the prepared dataset and evaluate its performance with appropriate metrics in real-world scenarios.
5. Documentation and Dissemination: Document findings and share results through research publications and presentations to advance respiratory medicine and healthcare technology.

The rest of this paper is organized as follows: In Sect. 2, we discuss the background of the work. Pre-trained neural networks and deep transfer learning models are discussed in Sect. 3. The dataset, experimental results, comparisons, and contributions are described in Sect. 4. Finally, the conclusion, limitations, and future works are summarized in Sect. 5.

2 Literature Review

Advancements in audio signal processing and machine learning have driven interest in using cough sound analysis for respiratory disease detection. With illnesses like asthma, pneumonia, and COVID-19 affecting global health, there is a growing need for accurate, non-invasive diagnostics. Machine learning methods using audio biomarkers show promise in identifying respiratory conditions from cough sounds, supporting early diagnosis and scalable healthcare solutions.

Traditional machine learning approaches have been used in cough sound analysis, but they have limitations. A study [1] reported an SVM with the MFCC method yielding 95.86% accuracy for COVID-19 diagnosis from cough audio. Study [2] trained a BiLSTM model using MFCCs, achieving over 84% accuracy in differentiating cough sounds. Study [3] utilized a DNN-HMM system for cough detection, outperforming traditional methods in sensitivity and specificity. Study [4] developed an ANN model for cough detection, achieving 97.5% accuracy, while also exploring an end-to-end CNN for COVID-19 detection. Furthermore, a study [5] used an EMD algorithm for cough sound detection, reporting an AUC of 98.80% for COVID-19 recognition.

Despite their effectiveness, traditional methods have notable drawbacks:

- Extensive Feature Engineering: They often require complex feature engineering.
- Temporal Dynamics: These methods may fail to capture the temporal dynamics of audio signals, limiting accuracy.
- Generalizability: Traditional models struggle with generalizability in diverse and noisy datasets.
- Feature Quality Dependence: Their performance relies heavily on the quality of manually extracted features.
- Limited Dataset Utilization: They may not fully leverage large datasets due to simpler architectures compared to deep learning models.

To address these limitations, researchers are increasingly using deep learning techniques, particularly Convolutional Neural Networks (CNNs), for cough sound analysis in COVID-19 detection. Studies [6–8] demonstrated CNNs' effectiveness in distinguishing COVID-19 from non-COVID-19 cases, with [7] proposing a multi-channel Deep CNN that utilized denoising autoencoders, achieving 95.45% accuracy. Other architectures, such as Multilayer Perceptrons (MLPs) and hybrid CNN-LSTM models, have also been explored. Studies [9–11] applied data augmentation techniques like pitch shifting, with [11] reporting a testing accuracy of 91.13% and a sensitivity of 90.93% using their augmented CNN-LSTM with attention.

Various deep learning and traditional machine learning models have been explored for COVID-19 detection. Studies [12, 13] examined combinations like DNNs, HMMs, CNNs, LSTMs, and ResNets alongside traditional algorithms such as LR, KNN, and SVM. Notably, [13] found ResNet50 to have an AUC of 0.98 for classifying COVID-19 versus healthy, while LSTM achieved 0.94 for COVID-19 versus negative. Customized algorithms also emerged; study [14] reported an AUROC of

82.23% using pruned fully connected networks and VGG-13, while [15] reached 98% accuracy with a Mel-Spectrogram + CNN model. Study [16] highlighted custom CNNs achieving 89.5% validation accuracy and a sound camera-based CNN model at 97.2%, respectively. Preprocessing techniques like detrending and normalization significantly enhanced performance in studies [2, 17–19]. Recent advancements include a deep convolutional bidirectional LSTM (C-BiLSTM) for accurate cough detection [20], while [21–24] achieved 82.81% sensitivity for asthmatic coughs.

Areas of Improvement:

1. Enhance Dataset Quality: Collect larger, diverse datasets representing various demographics, environments, and conditions.
2. Advanced Feature Extraction: Develop automated techniques to capture temporal dynamics in audio signals more effectively.
3. Improve Generalizability: Utilize data augmentation and transfer learning to create models that generalize across different datasets.
4. Optimize Model Complexity: Simplify models to reduce computational requirements by exploring lightweight architectures and compression techniques.
5. Facilitate Real-World Integration: Strategize for practical integration of diagnostic tools and conduct extensive real-world testing.

Figure 4 shows the accuracies of various cough sound analysis methods, highlighting the effectiveness of machine learning and deep learning techniques. EMD with deep ANN achieved the highest accuracy at 98.8%, followed by ResNet50 at 98%, ANN at 97.5%, and G-net with MFCC-V-A at 97.2%. Traditional methods like SVM with MFCC performed well at 95.86%. These findings underscore the potential of advanced algorithms to enhance diagnostic tools for respiratory conditions and the need for ongoing innovation and collaboration in the field.

3 Methodology

See Fig. 5.

3.1 Proposed Approach Framework

The proposed approach of this work is shown in Fig. 5. This approach is divided into various phases: Cough sound normalization, augmentation, preprocessing of data, training using deep transfer learning, feature extraction, and accurate prediction of models. Involved phases are discussed in the next sections.

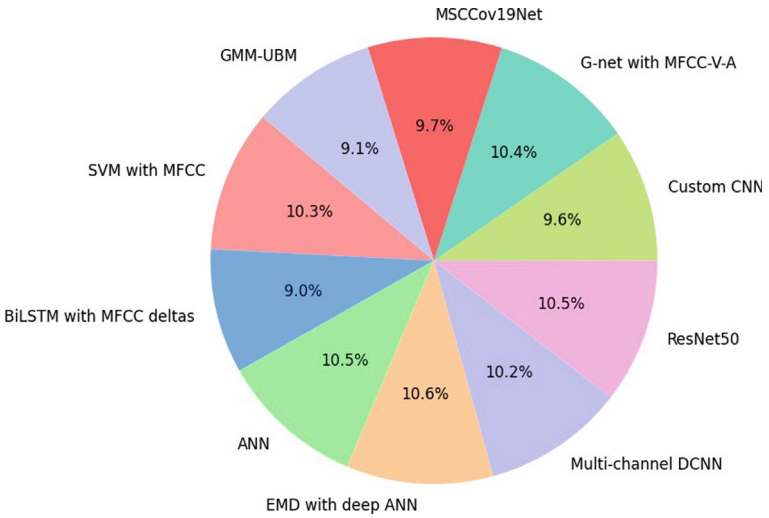

Fig. 4 Accuracy of the methods in cough sound analysis and respiratory disease detection

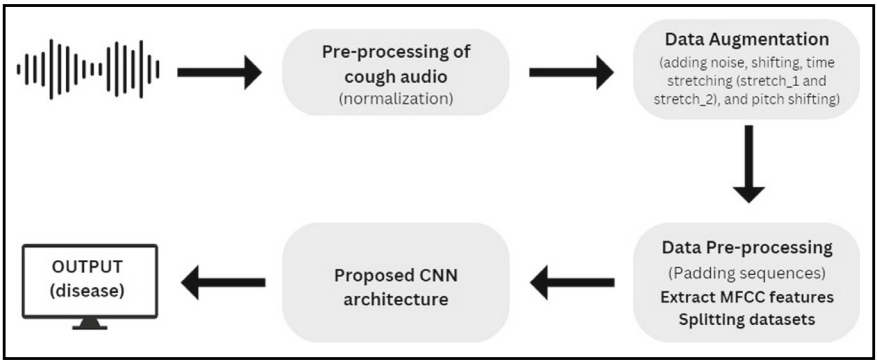

Fig. 5 Process flow

3.2 Audio Data Normalization and Visualization

In the initial preprocessing stages, we standardize amplitude levels of the audio data through normalization, ensuring uniform volume across samples. This mitigates biases from varying signal strengths. We visualize the normalized audio using spectrograms and waveforms, which provide insights into frequency-time characteristics and amplitude distributions. Figure 6 shows the spectrogram and waveform of asthma.

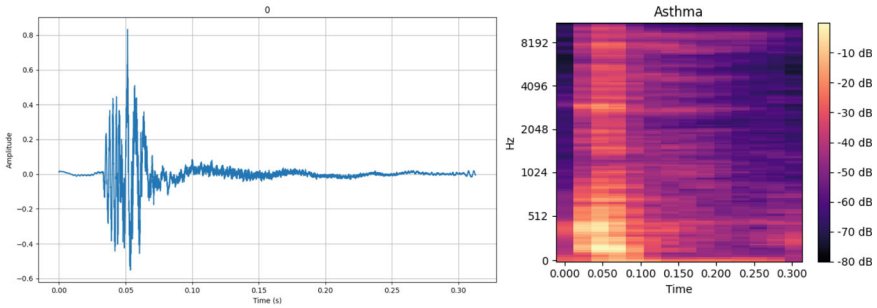

Fig. 6 Spectrogram and waveform of asthma

3.3 Dataset Augmentation

Training the CNN model for cough sound detection required a substantial dataset, but researchers in the medical field often face challenges due to limited availability. To address this, we employed augmentation techniques, including noise addition, shifting, time stretching (stretch_1 and stretch_2), and pitch shifting. These methods expanded the diversity and volume of our dataset, ensuring robust model training and enhancing its ability to generalize across different cough sound variations.

Figure 7 presents the audio after augmentation, showcasing the transformed sound data used to enhance model training.

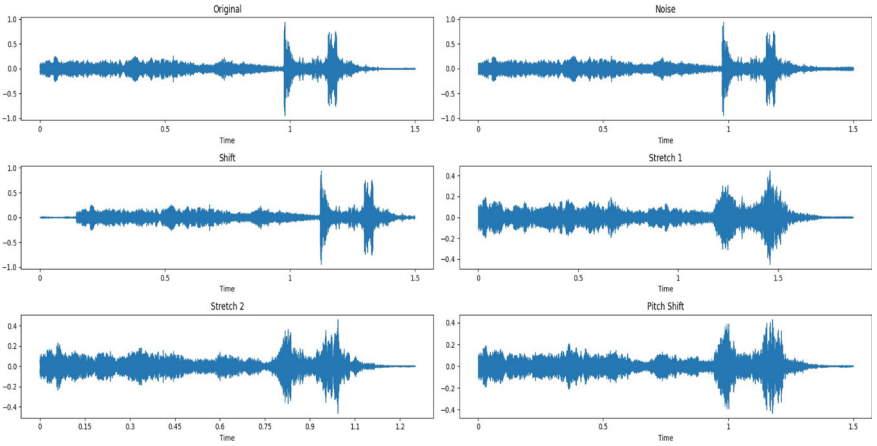

Fig. 7 Augmented cough audio

3.4 Data Preprocessing

To prepare audio data for cough sound detection using CNNs, we sourced augmented files for bronchitis, pertussis, asthma, and pneumonia. Each file underwent Mel-Frequency Cepstral Coefficients (MFCC) extraction to capture spectral features, followed by padding to standardize sequence lengths. Finally, we divided the data into training, validation, and test sets for effective model training and evaluation.

3.5 CNN Model Architecture

A Convolutional Neural Network (CNN) is a deep learning model designed to process grid-like data, such as audio spectrograms, making it ideal for classifying diseases from audio signals. We computed MFCC feature shapes to align with the model's input requirements and defined four disease classes for classification. This section introduces two CNN architectures for cough sound analysis. CNN Model 1 is simpler, focusing on essential convolutional, pooling, and dense layers with dropout for regularization. CNN Model 2 is more complex, with additional convolutional and dense layers for enhanced feature extraction and generalization. Both models use regularization to prevent overfitting and are tailored for accurate disease prediction from MFCCs.

Model 1

The layers applied to CNN model 1 are shown in Fig. 8.

- *Conv2D(32, (3, 3), activation = 'relu')*: This layer applies 32 filters with ReLU activation, extracting complex features from input data through convolutional operations.
- *MaxPooling2D((2, 2))*: Reduces spatial dimensions by pooling the maximum values in 2×2 windows, improving efficiency while retaining key features.
- *Flatten()*: Converts 2D feature maps into a 1D vector, preparing the data for the fully connected layers that follow.
- *Dense(256, activation = 'relu', kernel_regularizer = regularizers.l2(l2_reg))*: A fully connected layer with 256 neurons using ReLU activation, learning high-level patterns with L2 regularization to prevent overfitting.
- *Dropout(dropout_rate)*: Regularizes the model by randomly dropping neurons during training, reducing overfitting and enhancing generalization for new data.
- *Dense(num_classes, activation = 'softmax')*: The final layer of the CNN model, which uses the softmax activation function for multi-class classification. It outputs probability distributions over the num_classes disease classes, enabling the model to predict the most likely class for a given input.

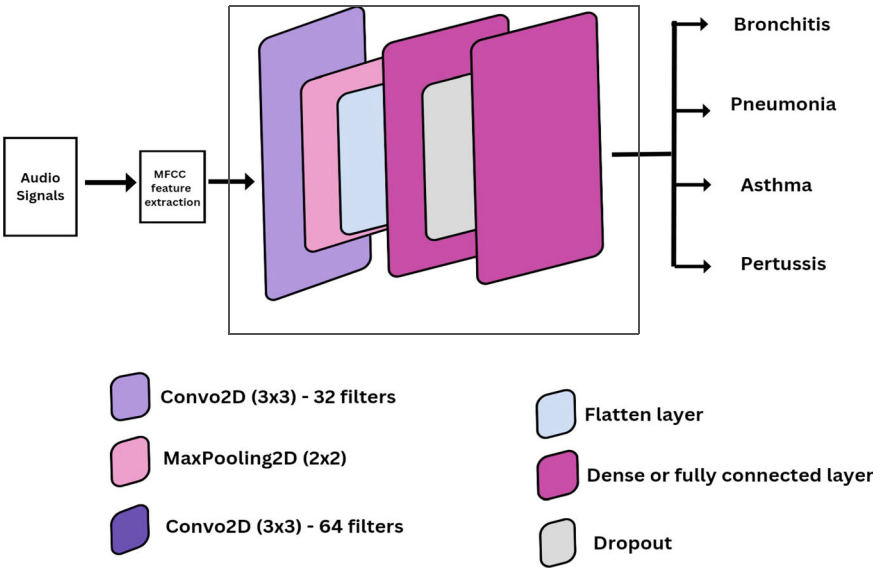

Fig. 8 CNN model 1 architecture

Model 2

The reshape layer prepares MFCC features to fit the specific shape, ensuring compatibility with subsequent convolutional and dense layers. These layers defined to construct the sequential CNN model architecture as shown in Fig. 9 is as follows:

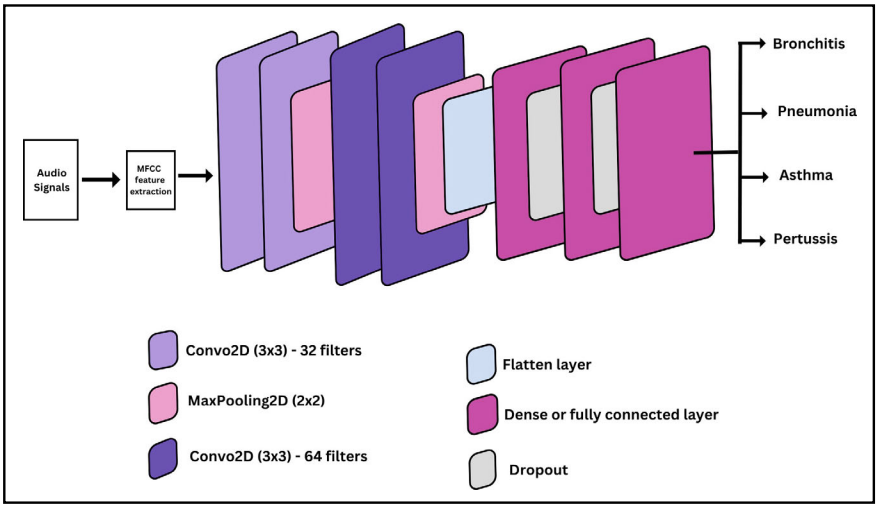

Fig. 9 CNN model 2 architecture

- *Reshape(input_shape):* Reshapes the input data to match the specified shape, ensuring correct formatting of MFCC features for convolutional processing.
- *Conv2D(32, (3, 3), activation = 'relu'):* Applies 32 filters of size 3 × 3 with ReLU activation to extract features from input images.
- *Conv2D(32, (3, 3), activation = 'relu'):* Another convolutional layer with 32 filters and ReLU activation for deeper feature extraction.
- *MaxPooling2D((2, 2)):* Downsamples feature maps by taking the maximum value in each 2 × 2 window, improving efficiency and reducing overfitting.
- *Conv2D(64, (3, 3), activation = 'relu'):* A deeper convolutional layer with 64 filters to extract more complex patterns from the data.
- *Conv2D(64, (3, 3), activation = 'relu'):* Additional convolutional layer with 64 filters for further intricate pattern learning.
- *MaxPooling2D((2, 2)):* Further reduces spatial dimensions while preserving relevant features from previous layers.
- *Flatten():* Flattens the 2D feature maps into a 1D vector for the fully connected layers.
- *Dense(256, activation = 'relu', kernel_regularizer = regularizers.l2(l2_reg)):* Fully connected layer
- With 256 neurons and ReLU activation, learning high-level features with L2 regularization to
- Prevent overfitting.
- *Dropout(dropout_rate):* Regularizes the model by randomly setting a fraction of input units to zero during training, enhancing generalization.
- *Dense(128, activation = 'relu', kernel_regularizer = regularizers.l2(l2_reg)):* Fully connected layer with 128 neurons to refine learned features.
- *Dropout(dropout_rate):* Another dropout layer for improved robustness and generalization. *Dense(num_classes, activation = 'softmax'):* The final layer outputs probability distributions for multi-class classification, predicting the most likely class for a given input.

3.6 Performance Metrics

In this paper, we use five metrics to evaluate the performance of pre-trained CNN architectures for diagnosing cough sounds, relying on four key terms: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Accuracy reflects the overall correctness of the predictions, as detailed below:

$$\text{Accuracy} = (\text{True Positives} + \text{True Negatives}) \Big/ \text{Total number of predictions} \quad (1)$$

In addition to accuracy, we further evaluate our proposed system with three evaluation measures: sensitivity (recall), precision, and F-score.

Precision measures how many of the predicted positive instances are correct. **Recall** (also known as sensitivity) indicates how well the model identifies positive instances from the total actual positives. **F1-score** provides a balanced measure between precision and recall, making it useful when there is an imbalance between the classes.

$$\text{Precision} = \text{True Positives} \big/ (\text{True Positives} + \text{False Positives}) \tag{2}$$

$$\text{Recall} = \text{True Positives} \big/ (\text{FalseNegatives} + \text{TruePositives}) \tag{3}$$

$$F_1\text{-score} = 2 \times (\text{Precision} \times \text{Recall}) \big/ (\text{Precision} + \text{Recall}) \tag{4}$$

A confusion matrix evaluated the model's performance by showing correct and incorrect predictions across classes, essential for multi-class evaluation. The Adam optimizer enabled adaptive learning rates for efficient weight updates and faster convergence, using a low learning rate (0.0001) for stability. Sparse categorical cross-entropy loss streamlined training with integer-encoded labels. Model architecture was verified using model.build() and model.summary() to check input shapes and parameters. During training, model.fit() was used with specified batch size, epochs, and validation data, while EarlyStopping halted training if validation loss did not improve (patience = 5), preventing overfitting.

4 Experimental Setup and Result

4.1 Dataset Description

We sourced cough sound data from reputable repositories on respiratory diseases, initially including samples for bronchitis (92), pertussis (9), asthma (110), and pneumonia (80). All data supporting the findings for bronchitis and pneumonia are publicly available and can be accessed at https://doi.org/10.6084/m9.figshare.21176197.v1. For pertussis, cough sounds were sourced from YouTube, converted to MP3 format, and subsequently processed into WAV files; the detailed list of data sources can be found at https://www.researchgate.net/Fig./List-of-data-sources-for-cough-sounds_fig1_307576967. Asthma classification was conducted using cough sound data from the COVID-19 COUGHVID dataset. To address class imbalance, we applied data augmentation techniques—noise addition, time shifting, stretching, and pitch shifting—resulting in a balanced dataset with 460 bronchitis, 45 pertussis, 550 asthma, and 400 pneumonia samples given in Table 1. Pertussis samples were obtained from YouTube, while bronchitis, pneumonia, and asthma data came from public medical research databases. This augmentation enhanced model generalization across respiratory conditions.

Table 1 Count of raw and augmented audio files

Disease	Count of files	Augmented count
Bronchitis	92	460
Pertussis	9	45
Pneumonia	80	550
Asthma	110	400

4.2 Experimental Environment

Our CNN-based cough analysis architecture extracts features from augmented audio datasets of respiratory diseases, including bronchitis, pertussis, asthma, and pneumonia. Preprocessing involved MFCC feature extraction and data augmentation. The architecture includes Reshape and Conv2D layers (32 and 64 filters of 3×3) with ReLU activation, followed by MaxPooling2D layers (2×2) to reduce dimensions and prevent overfitting. A flattened layer converts 2D feature maps to a 1D vector for dense layers (256 and 128 neurons) with Dropout layers (0.5 rate) for generalization. The final softmax layer classifies the four disease classes. The model uses the Adam optimizer (learning rate of 0.0001) and sparse categorical cross-entropy loss, with efficacy evaluated using accuracy metrics through TensorFlow and Keras.

4.3 Comparison of Various Models with the Proposed Model

The comparative analysis of the two CNN models reveals key differences in performance and complexity. Table 2 indicates that CNN Model 1, with 6 layers, has an accuracy of 61.67% and a loss of 87.39%, indicating limited feature extraction. In contrast, CNN model 2, with 13 layers, achieves 84.25% accuracy and a loss of 64.90%, demonstrating improved feature extraction and classification. Overall, increasing the architecture from 6 to 13 layers significantly boosts accuracy and reduces loss, highlighting the trade-off between complexity and performance.

4.4 Result Analysis

After evaluating the CNN model, we calculated precision, recall, and F_1-scores using sklearn.metrics. The precision score was 85.28%, while recall achieved 84.93%, leading to an F_1-score of 84.96%. These metrics indicate strong performance in classifying bronchitis, pertussis, asthma, and pneumonia from cough sounds. Additionally, the CNN achieved an overall accuracy of 84.25% during testing across all disease categories, highlighting its effectiveness and reliability in distinguishing between different respiratory conditions based on audio data.

Table 2 Comparative performance of CNN models with varying depths

Model	Bronchitis data (%)	Asthma data (%)	Pneumonia data (%)	Pertussis data (%)
Support vector machine (SVM)	62.30	65.98	71.00	73.87
Long short-term memory (LSTM)	73.76	83.38	69.67	82.15
Artificial neural network (ANN)	84.43	92.73	95.82	81.73
Support vector machine (SVM) and long short-term memory (LSTM) used together	76.43	69.43	78.35	80.54
Proposed CNN model 1	75.00	61.67	53.87	87.39
Proposed CNN model 2	87.32	98.00	97.56	84.05

The confusion matrix in Fig. 10 highlights the CNN model's effectiveness in classifying respiratory diseases from cough sounds. The model achieved a precision rate of 100% across all classes: bronchitis (39/39), pertussis (4/4), asthma (54/54), and pneumonia (27/27). These results underscore the model's strong capability to accurately differentiate respiratory conditions based on audio data, demonstrating consistent high precision across all disease categories.

4.5 Performance Evaluation

Comparing our CNN model for cough sound analysis with traditional methods like SVM highlights significant advantages. While SVMs using MFCC achieve high accuracy, they require extensive feature engineering and struggle with temporal dynamics. In contrast, recent CNNs have achieved COVID-19 detection accuracies between 95.45 and 98.8%. Our CNN architecture focuses on robustness and generalizability, employing advanced preprocessing techniques such as MFCC feature extraction and data augmentation. With Conv2D layers, pooling, and dense layers optimized by the Adam optimizer, our model effectively learns discriminative features and temporal patterns from raw audio, positioning it as a scalable tool for respiratory disease diagnosis.

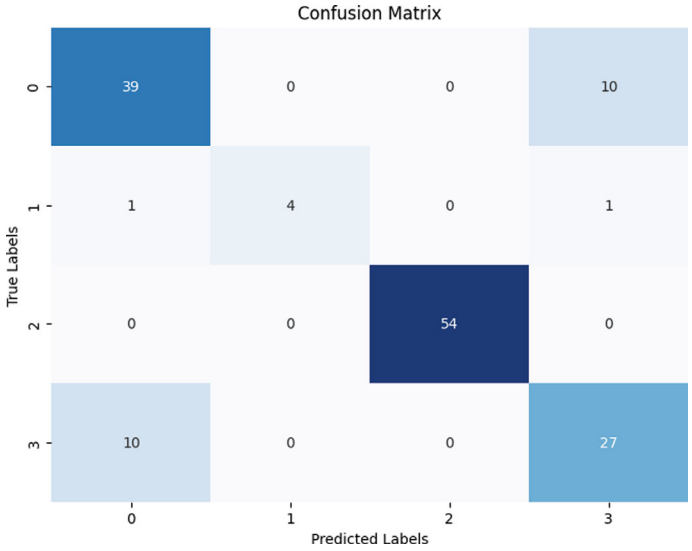

Fig. 10 Confusion matrix

4.6 Application of Model

Technological advancements are crucial for managing respiratory diseases by enhancing early detection and resource optimization. AI algorithms, wearable sensors, and telemedicine improve diagnostics and access to care. App-enabled cough monitoring analyzes sounds for timely interventions, while predictive analytics and IoT optimize resources during high demand. Utilizing large cough sound datasets supports epidemiological research, helping identify trends and predict outbreaks.

5 Conclusion

Early diagnosis of respiratory diseases is crucial for timely treatment and preventing their spread. This paper introduces a deep CNN-based approach using transfer learning to differentiate conditions like Pertussis, Bronchitis, Asthma, and Pneumonia from normal cases, achieving an accuracy of 84.25%. While these findings can aid clinical decision-making, limitations include the need for a larger patient dataset, particularly for mild cough symptoms that traditional methods might miss. Future research will focus on applying our approach to larger datasets, enhancing model effectiveness, and exploring image data augmentation techniques to improve accuracy and prevent overfitting.

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Predicting Maternal Health Risks Using Ensemble Model, Multilayer Perceptron, and PCA Features

R. Girija and N. Deepa

Abstract An essential part of public health, maternal healthcare looks after the expectant mother's health prenatal, perinatal, postnatal. It guarantees the best possible health for the growing fetus as well as the mother through its predictions. Maternity and the postpartum period present a multitude of risks and repercussions for the mother's health, and hence prompt identification of these risks can be extremely important to a woman's safety. A strategy to anticipate hazards related to maternal health is proposed in this study. Principal component analysis (PCA) is the initial step in the process, which involves extracting important features from the dataset. To attain good performance, this study then uses a stacked ensemble voting classifier that includes one deep learning model and one machine learning model. By contrasting it with current state-of-the-art methods, the suggested model's efficacy is further verified. Comparison analysis was made with many Machine Learning (ML) models such as Random forest classifier, Logistic Regression, Decision tree classifier, Support vector classifier, and XGB classifier. Based on a comprehensive investigation, the suggested deep learning model (one Machine Learning and one Deep Learning) yields superior prediction rates for high and low risk, with 84% and 86% of accuracy, respectively.

Keywords Maternal health risks · Principal component analysis · Stacked ensemble voting classifier · Health care

1 Introduction

The number of pregnancy-related fatalities per 100,000 live births is known as the maternal mortality ratio, or MMR [1]. As per the survey, from WHO, pregnancy-related problems claimed the lives of 808 women on average every day in 2017 [2]. Two-thirds of the maternal deaths occurring in countries are as mentioned in Fig. 1.

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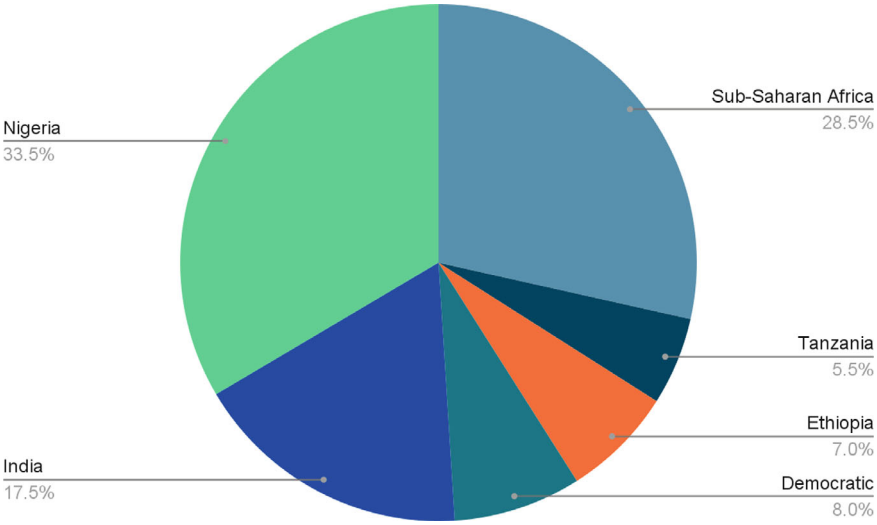


Fig. 1 Countries with the highest rate of MMR in 2017

Pregnancy complications can be caused by age, blood abnormalities (e.g., high or low blood pressure, blood glucose levels, body temperature, and heart rate), and can result in the loss of life for women. It is very important to address these risk complications on a timely basis so that the lives of pregnant women can be saved and the likelihood of MMR [3] can be reduced. Realizing the importance of maternal health, many researchers created the models to predict maternal risks. Various methods combine machine learning and classical procedures. But very few have focused on checking, detecting, and recording the pregnancy risks of women. In [4], researchers have analyzed the maternal risks all over the prenatal period. Eleven ML models, including K-Nearest Neighbor, XGBoost, GBM, Artificial Neural Network, LR, CatBoost, RF, SVM, and CART, were employed by the authors for maternal risk prediction. From this, it can be decided that LightGBM and CatBoost ensure the highest accuracy of 88%. In [5], they have proposed and developed an AI-based system that might forecast mother health risks. Based on experimental data, 98% accuracy is achieved by the SVM with ensemble features. An effective technique based on the CART binary decision tree to analyze and anticipate the risk factors for maternal mothers was proposed in [6]. The analysis was based on the cardiotocography dataset by UCI which showed a percentage of accuracy 88%. In [7], Irfan developed an ML-based automatic prediction approach to indicate the risk factors involved in pregnancy. The method exhibited an accuracy of 94% with XGBoost machine learning technique. In [8], performed a detailed research on the pregnancy risk factors in Bangladeshi women highlighting the factors affecting the frequency of cesarean sections. The performance of three ML techniques DT, K-NN, SVM for predicting the risk factors were analyzed individually and also compared with bagging ensemble models. The bagging ensemble models

Table 1 Categorization of risk levels for 1014 samples

Type of risks	Count	Percentage of risks in overall sample (%)
Low-risk category	406	40
Mid-risk category	336	33.1
High-risk category	272	26.8

exceeded the conventional models. Another ML-based approach for predicting pregnancy risks was proposed in [9]. In [10], the researchers proposed, the study uses a stacked ensemble voting classifier that includes one machine learning and one deep learning model. Out of the several conventional ML techniques like Naive Bayes, Multi-Layer Perceptron, Random Forest, Decision Tree Classifier, Reduced Error Pruning (REP) tree, J48, Logsitics Model Tree (LMT), Bagging implemented, the Random Forest model demonstrated 70.21% accuracy. Section 1 describes about the introduction and related work, the proposed methodology which includes the datasets description, pre-processing and feature selection. Section 2 represents all the existing Machine Learning (ML) models. The proposed model architecture, with evaluation metrics and results, is provided in Sect. 3. Finally, the conclusion and future work are in Sect. 3.

2 Proposed Methodology

2.1 About the Dataset

Dataset was initially recognized by Marzia et al. from Daffodil International University in Dhaka, Bangladesh, and it is used in this study that is available to the public [11]. This dataset, to put it briefly, comprises seven classes: Age, Blood Sugar (Bs), Risklevel, SystolicBP, DiastolicBP, and Body Temperature. The detailed description about the classes, data types, and their ranges of this dataset is provided in Table1 and count of risk level parameters is mentioned in Fig. 2. Descriptions of parameters are listed in Table 2.

Pre-processing

Pre-processing of the maternal data according to the specific "Risk level" parameter. Low risk is 40%, mid risk is 33.1%, and high risk is 26.8%, according to the dataset. There are 405 low-risk, 336 mid-risk, and 272 high-risk items. A low-risk score is zero, a mid-risk score is one, and a high-risk score is two.

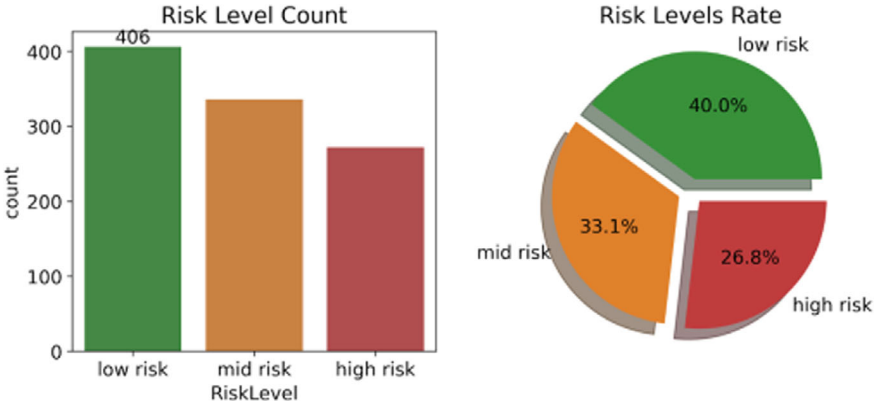


Fig. 2 Count of maternal health risks

Table 2 Datasets description

Parameters	Overview	Type	Range
Age	Pregnant women's age	Number	10–70 years
Blood sugar (Bs)	Level of blood glucose in mmol/L of pregnant women	Number	6–9 mmol/L
Risk level	Level of pregnancy risks such as low, mid, or high	Categorical	High, mid, low
Systolic blood pressure	Highest blood pressure in mmHg	Number	70–160 mmHg
Heart rate	Cardiac rate in beats per minute	Number	7–90BPM
Diastolic blood pressure	Lowest value of blood pressure in mmHg of any pregnant women	Number	49–100 mmHg
Body temperature	Temperature of the body	Number	90–103 °F

Feature Selection

It actively contributes to achieving a respectable fit for machine learning models with respect to each feature; each has a unique consequence for the target class. Finding such traits that have a large predictive value is very responsible. The best feature is chosen by the suggested model using Principle Component Analysis (PCA), and it is then taken into account for prediction [12]. It is employed to identify the fundamental elements that identify data modifications. PCA is typically the method of choice when larger datasets are involved in order to reduce dimensionality. It is carried out by dividing larger datasets into smaller datasets while preserving constant accuracy and no information loss. Once the datasets have been transformed, handling, reconnoitering, and visioning become quite simple.

3 Machine Learning Models

A. **Random Forest**
Various decision trees are mingled in an ensemble learning technique called Random Forest to produce a more accurate and dependable model. Classification and regression problems can be solved easily by the Random Forest classifier [13, 14]. Its primary benefits are its ability to handle large datasets, manage missing values, and produce feature priority ratings.

B. **Decision Tree**
The decision tree is a flexible ML technique for applications that involves both classification and regression [15]. They model data by splitting it into subsets based on feature values, and then using those subsets to form a tree-like structure where each node represents a choice made based on a feature and each branch reflects the outcome of that decision.

C. **Logistic Regression**
Although it can be expanded to multiclass classification, the basic machine learning technique known as logistic regression is utilized for binary classification tasks [16].

D. **Extreme Gradient Boosting**
XGBoost applies gradient boosting techniques in a powerful and effective manner. Because of its effectiveness and adaptability, it is frequently utilized in real-world applications and machine learning contests. Recognized for its accuracy, speed, and scalability, XGBoost may be applied to regression and classification tasks alike [17].

E. **Extra Tree Classifier**
By integrating the predictions of many decision trees, the Extra Trees Classifier is an ensemble learning method that improves classification accuracy [18]. This Random Forests version has a notable difference in the way the trees are constructed.

F. **Stochastic Gradient Descent**
Optimal model parameters and loss function minimization are found using the widely utilized optimization technique known as stochastic gradient descent (SGD) in machine learning and deep learning [19]. It is especially well suited for extensive and online learning tasks due to its efficiency and scalability.

G. **Multilayer Perceptron**
An artificial neural network made up of several layers of nodes or neurons is called a Multilayer Perceptron (MLP) [20]. It is used in tasks that need both classification and regression. MLPs are very useful at spotting complex patterns and relationships in data.

4 Proposed Model Architecture

The two integrated models—multilayer perceptron (neural model) and ensemble tree classifiers—are suggested by this methodology. The combination of this proposed hybrid model makes a better model [21]. By averaging the probabilities of each class predicted by each model, the average probability for each class is found. To reach the prediction, both the model's knowledge is used. The proposed architecture is shown in Fig. 3. From the architecture, maternal health risks data are pre-processed and form the pre-processed data, 30% fixed for testing and 70% fixed for training with fivefold cross validation. With the same datasets, both the models are individually trained to give better risk prediction and after combining it is called an ensemble model. For every observation in the datasets, a final forecast is generated by combining the estimated probability. A validation set or techniques such as cross validation are frequently used to determine the weights assigned to each model's prediction [22]. This ensemble model improves the accuracy. Evaluation metrics such as accuracy, precision, recall and F_1-score are calculated and discussed in the result section. The overall probability of ensemble model and multilayer perceptron as per risk category is mentioned in Table 3.

Evaluation Metrics

A number of evaluation metrics, including Accuracy, Precision, Recall and F_1-score, are commonly used to evaluate the performance of models [23]. Confusion matrix has been calculated to obtain TP, TN, FP, FN. Accuracy describes how the model is calibrated and it is (TP + TN)/(TP + TN + FP + FN). Precision quantifies the model's ability to identify positive cases and it is TP/(TP + FP); Recall is sensitive to positive case detection and it is TP/(TP + FN).

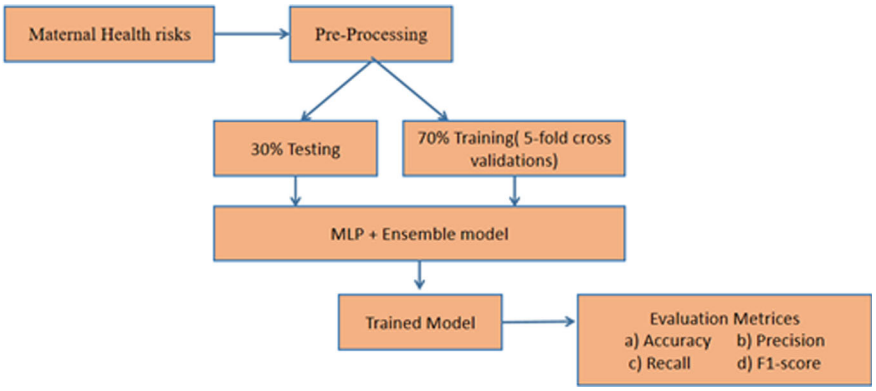


Fig. 3 Proposed methodology architecture

Table 3 Overall probability of both models with risk categories 0, 1, and 2

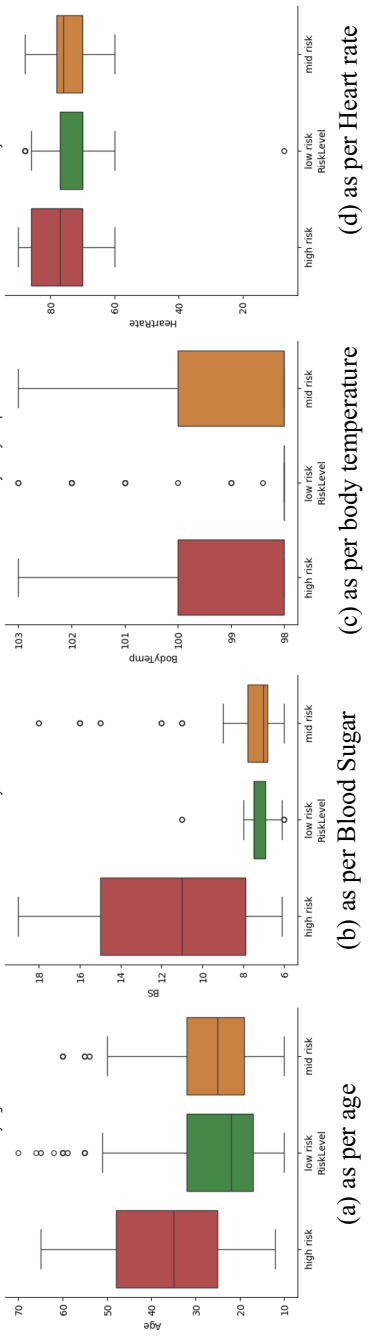
Model	Class	Risk category	Overall probability
Ensemble model	Class 1	0	0.6
	Class 2	1	0.7
	Class 3	2	0.8
Multilayer perceptron	Class 1	0	0.4
	Class 2	1	0.5
	Class 3	2	0.6
Ensemble + multilayer perceptron	Class 1	0	$(0.6 + 0.4)/2 = 0.50$
	Class 2	1	$(0.7 + 0.5)/2 = 0.60$
	Class 3	2	$(0.8 + 0.6)/2 = 0.70$

5 Results

The assessment examined the effectiveness of the Ensemble model and the Multilayer Perceptron in extracting features, and it was compared to all other models. Every trial is conducted on a Windows 10 system that has an Intel i7 7th generation core processor. The maternal risks as per various parameters are shown in Fig. 3. Figure 4 (a) represents the maternal risks as per age and (b) illustrates the risks as per blood sugar (c) and (d) represents the body temperature and heart rate, respectively. Figure 5 denotes the correlation matrix.

The proposed model also discusses the outliers in the maternal health data which is mentioned in Table 4. Total number of outliers available in the maternal dataset is 433 as per the various features. In addition, risk of outliers are also mentioned in Table 4 and outliers risks depending upon various parameters is given in Table 5. The pictorial representation of Table 5 is given in Fig. 6. Accuracy calculations and predictions for all the models are displayed in Tables 6 and 7. Figure 4 denotes the confusion matrix of all the models. Figure 6 denotes the risks of outliers' based on various parameters. Prediction rates as per the risk category are mentioned in Table 8. Accuracy of the models and loss is shown in Fig. 7. Model comparison and confusion matrix as per the maternal parameters are shown in Fig. 8.

From all the analysis, the proposed deep learning model (one Machine Learning and One Deep Learning) gives better accuracy of 84% and 86%, respectively, for high-risk and low-risk prediction rates. Ensemble and multilayer perceptron approaches not only capture a broader set of structures or nuances in data but also address single model drawbacks like overfitting and majority class bias.



(a) as per age (b) as per Blood Sugar (c) as per body temperature (d) as per Heart rate

Fig. 4 Maternal risks as per various parameters

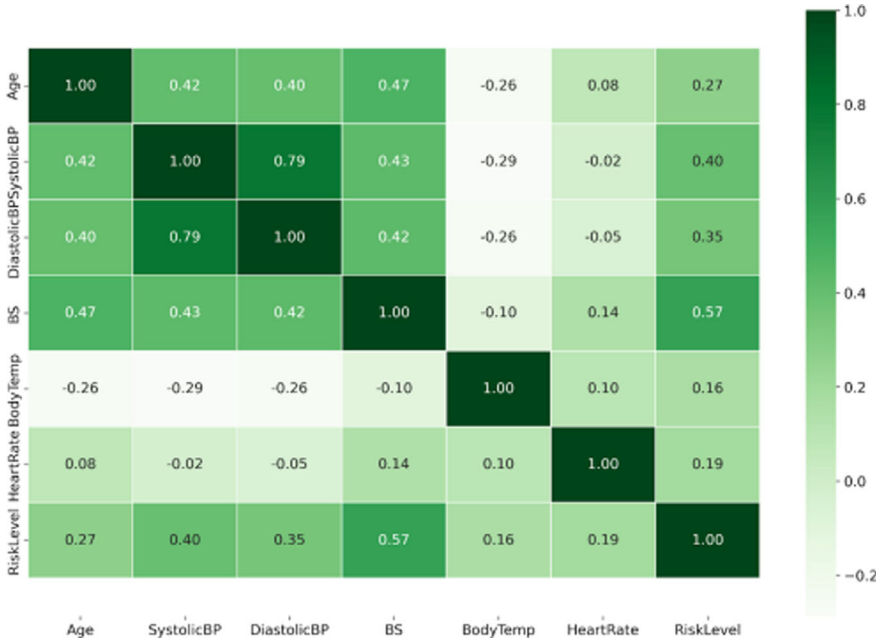


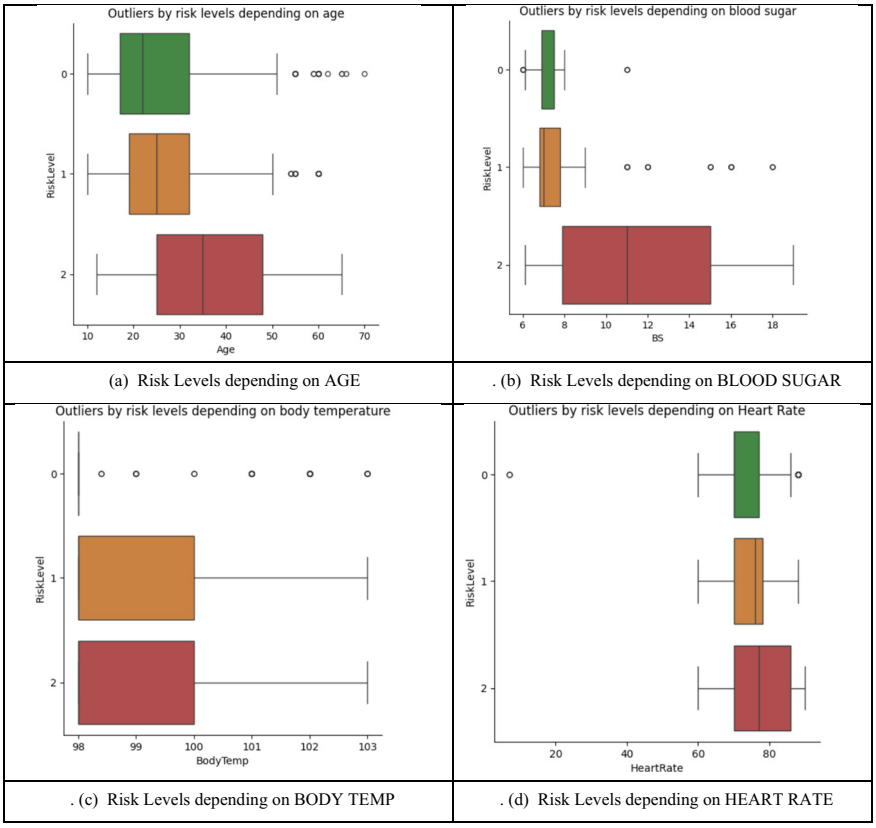
Fig. 5 Correlation matrix

Table 4 Maternal features and their outliers (Total number of outliers: 433; Percentage of outliers: 0.427%)

	Feature	Number of outliers
0	Age	1
1	Systolic BP	10
2	Diastolic BP	0
3	Blood sugar	210
4	Body temperature	210
5	Heart rate	2

Table 5 Risks of outliers depending upon various parameters

Parameters	Plotted graph	Values
Age	Represented in Fig. 5a	Low risk by age > 50: 33
		Mid risk by age > 50: 25
		High risk by age > 50: 37
Blood sugar	Represented in Fig. 5b	Low risk by BS > 10: 4
		Mid risk by BS > 10: 32
		High risk by BS > 10: 170
Body temperature	Represented in Fig. 5c	Low risk by body temp > 100: 37
		Mid risk by body temp > 100: 75
		High risk by body temp > 100: 65
Heart rate	Represented in Fig. 5d	Low risk by heart rate < 55: 2
		Mid risk by heart rate < 55: 0
		High risk by heart rate < 55: 0



(a) Risk Levels depending on AGE

. (b) Risk Levels depending on BLOOD SUGAR

. (c) Risk Levels depending on BODY TEMP

. (d) Risk Levels depending on HEART RATE

Fig. 6 Risks of outliers depending upon various parameters

Table 6 Percentage of accuracy for all models

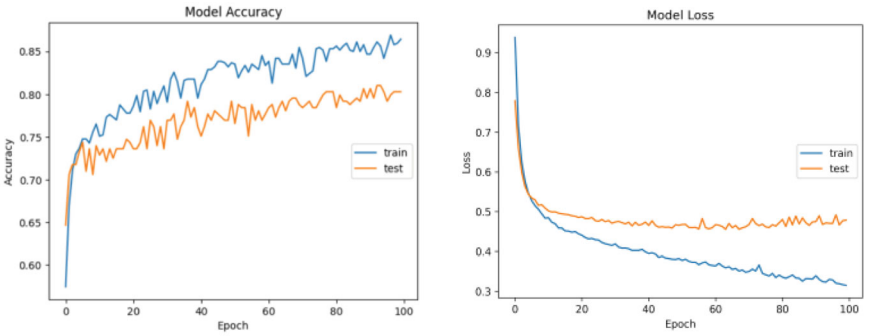
Logistic regression model	Model accuracy	72.86
	Training accuracy	72.04
SVC	Model accuracy	81.78
	Training accuracy	92.68
Random forest	Model accuracy	73.97
	Training accuracy	76.19
Decision tree model	Model accuracy	71.00
	Training accuracy	72.6
XGB classifier	Model accuracy	85.13
	Training accuracy	94.24
Ensemble and multilayer perceptron	Model accuracy	80.29
	Training accuracy	86.74

Table 7 Evaluation metrics of all models (0-low risk, 1-mid risk, 2-high risk)

Logistic regression	Precision	Recall	F_1-score
0	0.67	0.95	0.79
1	0.64	0.43	0.51
2	0.94	0.78	0.84
SVC model			
0	0.80	0.77	0.79
1	0.72	0.78	0.75
2	0.95	0.91	0.93
Random forest classifier			
0	0.66	0.99	0.73
1	0.71	0.37	0.48
2	0.93	0.83	0.88
Decision tree model			
0	0.63	0.96	0.76
1	0.66	0.33	0.44
2	0.90	0.80	0.85
XGB classifier			
0	0.92	0.77	0.84
1	0.74	0.89	0.81
2	0.93	0.91	0.92
Ensemble model + multilayer perceptron			
0	0.76	0.84	0.80
1	0.71	0.66	0.68
2	0.95	0.91	0.93

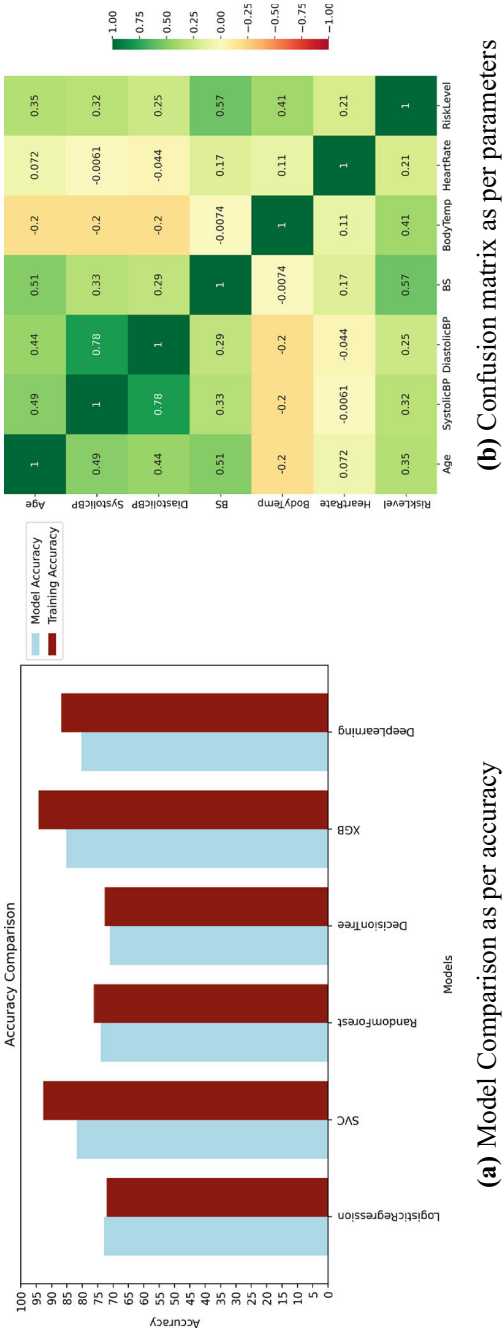
Table 8 Prediction rate of all models

Prediction rates of LR model	Low-risk correct prediction rate	95%
	Mid-risk correct prediction rate	43%
	High-risk correct prediction rate	74%
Prediction rates of SVC model	Low-risk correct prediction rate	77%
	Mid-risk correct prediction rate	78%
	High-risk correct prediction rate	86%
Prediction rates of random forest model	Low-risk correct prediction rate	99%
	Mid-risk correct prediction rate	37%
	High-risk correct prediction rate	78%
Prediction rates of decision tree model	Low-risk correct prediction rate	96%
	Mid-risk correct prediction rate	33%
	High-risk correct prediction rate	76%
Prediction rates of XGB classifier	Low-risk correct prediction rate	77%
	Mid-risk correct prediction rate	89%
	High-risk correct prediction rate	86%
Prediction rates of ensemble and multilayer perceptron	Low-risk correct prediction rate	84%
	Mid-risk correct prediction rate	66%
	High-risk correct prediction rate	86%



(a) Accuracy (b) Loss

Fig. 7 Proposed model accuracy and loss



(a) Model Comparison as per accuracy

(b) Confusion matrix as per parameters

Fig. 8 Model Comparison

6 Conclusion

A crucial area of public health that deals with expectant mothers' prenatal, postpartum, and postpartum health is health care for mothers. It consists of a range of services designed to ensure the mother and developing fetus have the greatest possible health. During childbirth and after childbirth, the wellness of a mother is vulnerable to a number of dangers and consequences. Early identification of these concerns can be crucial to a woman's safety. This study provides a strategy for identifying potential concerns related to maternal health. Principal component analysis (PCA), the initial step in the procedure, involves eliminating important features from the datasets. Next, a stacked ensemble voting is classifier used in this investigation and compares with other ML models such as Logistic Regression (LR), Support Vector Machine (SVM), Random Forest classifier (RF), Decision Tree (DT) Classifier, and XGB Classifier. The proposed model gives better accuracy for low- and high-risk prediction rates. Future work might collect real-time data from primary health clinics and target high-risk pregnant women to transition them to a typical category. Also research can be focused to improve machine learning models for women in the medium risk category, to prevent them from moving to the high-risk category.

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Performance Evaluation of Deep Learning Models Using Transfer Learning on Ultrasound Images for Breast Cancer Detection

Laxmi Yadav, Girish Chandra, and Divakar Yadav

Abstract One of the most serious illnesses that affect women and can even be fatal is Breast Cancer (BC). It has a major impact on their emotional and physical well-being. Breast calcifications or abnormal tissue growth are major health concerns contributing to BC. Early diagnosis and treatment can significantly lower the death rate and are crucial for survival rates. Medical imaging techniques, particularly ultrasound, are widely used due to their non-invasive nature and effectiveness in identifying abnormal tissues. In this research paper, we develop a Breast Cancer detection approach using pre-trained models such as ResNet, GoogleNet, DenseNet, VGG16 and Xception leveraging Transfer Learning (TL). Deep Learning models with the TL concept have proven quite effective in Breast Cancer detection in recent years. TL is used to utilize the knowledge gained from large-scale image recognition tasks to increase the accuracy of cancer detection and classification in the limited ultrasound imaging domain. This paper compares five deep learning models based on the BUSI dataset of breast cancer, which is available publicly. The dataset consists of 780 images in PNG format, from which 210 images are cancerous, 437 are non-cancerous, and 133 are normal. However, GoogleNet's transfer learning performance is better than that of other existing models. The accuracy for the test and validation were 99.15% and 99.37%, respectively.

Keywords Deep learning · GoogleNet · DenseNet121 · ResNet50 · VGG16 and Xception

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1 Introduction

Breast Cancer is one of the most prevalent and potentially deadly illnesses affecting women worldwide [1]. In 2022, about 51,400 cases of DCIS and 287,850 new cases of BC will be found in US women, and 43,200 women will die from BC. Globally, 670,000 people died from breast cancer in 2020, accounting for 2.3 million new diagnoses [2]. It results from the breast tissue's cells growing out of control, typically originating in the ducts or lobules. This uncontrolled growth of cells may result in mass, calcifications, or architectural abnormalities, commonly known as tumours. Tumours have the potential to spread throughout the body and become lethal if untreated. Breast lumps or thickness, anomalies such as skin redness, pitting, or changes in the breast's size, shape, or appearance, abnormal or bloody nipple fluid, and many other symptoms can all be signs of breast cancer. Early stage diagnosis and treatment can increase survival rates and reduce the mortality rate.

Breast ultrasonography has emerged as a proper diagnostic technique because of its non-invasive, real-time imaging, lack of danger, and affordability, even though biopsy is the gold standard for making a difference between non-cancerous and cancerous breast lesions [3]. It is crucial to accurately assess metastases to the axillary lymph nodes in early stage BC to prevent unnecessary treatments and associated issues [4]. Because of the remarkable association, several researchers have suggested ultrasound as a trustworthy substitute for lymph node biopsy [5]. Predicting lymph node status has also been investigated in the recent years using deep learning techniques [6, 7]. However, ultrasound frequently has trouble classifying tumours, especially when it comes to small abnormal masses that don't have apparent malignant features. The difficulty of differentiating benign from malignant breast tumours is further compounded by the fact that ultrasonography is largely operator-dependent, with results differing among users [8].

In this study, we aimed to diagnose and classify BC using deep learning models and transfer learning. This paper's roadmap is set out as follows: In Sect. 1, the introduction is given. The related work and methodology are covered in Sects. 2 and 3, respectively. Experimental findings are explained in Sect. 4, whereas Sect. 5 presents the research's conclusion.

2 Related Work

Sahu et al. [9] suggested an approach for developing a hybrid model, leveraging the transfer learning techniques to enhance performance with the limited data and applying fivefold cross-validation. Three datasets were employed to evaluate the models: BUSI, BUS2 and mini-DDSM. The model's accuracy in detecting abnormalities in the mini-DDSM dataset is 99.17%, while its accuracy in detecting malignancy is 98%. The BUSI dataset obtained 93.18% and 96.52% accuracy in detecting malignancy and abnormalities. ARF-Net, a novel end-to-end model, was created by

Xu et al. [10] for accurate breast mass segmentation in ultrasound and mammograms. ARF-Net overcomes the limitation of UNet for small datasets. ARF-Net comprises a network of encoders and decoders, succeeded by a classifier at the pixel level. The model features an SRFM that adjusts the receptive field size to match breast masses of varying dimensions. Dice scores for ARF-Net on the CBIS-DDSM, INBreast and UDIAT datasets were 86.1%, 85.75% and 88.12%, respectively. Sirjani et al. [11] proposed a model that utilizes a modified version of the InceptionV3 architecture to categorize breast tumours into benign and cancerous categories. In multicentre data evaluation, the models obtained a high accuracy of 81% and an AUC score of 91%. This work may help to reduce the need for biopsies, which are invasive and time-consuming.

Lu et al. [12] proposed BCDNet model for BC diagnosis using ultrasound pictures leveraging pre-trained CNN models for feature extraction. The model utilized an ELM classifier, which was optimized using a "Bat Algorithm with Chaotic Maps (BACM)". This hybrid approach enabled the BCDNet model to balance the benefits of deep learning and traditional machine learning. Achieved accuracy of 93.97%, sensitivity 95.24% and F1_score 9642% on publicly available ultrasound dataset. Pavithra et al. [13] proposed a CAD model for BC diagnosis using ultrasound images, addressing the limitations of mammography in young women with dense breast tissues. The system employed Speckle Anisotropic Diffusion (SRAD) for noise reduction, followed by active contour-based segmentation and GLCM texture feature extraction. After testing three classifiers, the Random Forest classifier proved to be the most accurate, with an accuracy of 88%. Cho et al. [14] developed a deep learning-based multistage segmentation method for BC diagnosis using ultrasound data. They developed the BTEC-Net to classify breast tumour regions. This approach improves upon traditional single-stage networks by reducing false positives, particularly in normal images, achieving better segmentation accuracy. Their method demonstrated superior performance on public datasets, making it a strong option for automated breast cancer diagnosis. Dontchos et al. [15] developed a DL method for estimating breast density in mammography pictures that uses ResNet-18 and a deep CNN. Suzuki et al. [16] proposed a deep CNN model incorporating TL with eight layers and pre-trained weights. Chougrad et al. [17] suggested a CNN-based CAD system that uses the DDSM, BCDR and MIAS databases using transfer learning methods such as ResNet50, Inception v3 and VGG16. Mohammed et al. [15] presented a model for identifying BC in biopsy microscopic pictures using CNNs. They presented an ensemble model that combined DenseNet169, ResNet50 and ResNet101 and investigated how data pre-processing affected model performance. After including data augmentation and segmentation techniques, the model's accuracy increases. Performance was further improved using the ensemble technique, which reached a maximum accuracy of 92.5%.

Eroglu et al. [18] BC was classified as benign, malignant and normal using a hybrid CNN system that combined AlexNet, MobileNet and ResNet50 models. When coupled with a Support Vector Machine classifier, this method attained an impressive accuracy of 95.6%. Islam et al. [19] introduced an Ensemble Deep Neural Network

that outperformed other TL models and the Vision Transformer model with an accuracy of 87.82%. Alotaibi et al. [20] developed a three-step image processing method that uses block-matching 3D filtering for speckle noise filtering, RGB fusion, and region of interest highlighting. Transfer Learning was performed on three databases: BUSI, Dataset B (162 pictures) and KAIMAC (5693 pictures) using a VGG16 deep learning model. Fivefold cross-validation on the BUSI and KAIMAC datasets demonstrated that the model performed better with the suggested pre-processing steps than it did without them.

3 Methodology

This section outlines the methodology, techniques, and dataset used in this research. The developed work's methodology is covered in Fig. 1. The BUSI Dataset [21], which contains ultrasound images of BC, was utilized. The dataset originally included three categories: benign, malignant, and normal. It was categorized into training, testing and validation for model development. Pre-processing steps included data augmentation, resizing and normalization.

The pre-processed training data was fed into pre-trained models like ResNet50, VGG16, DenseNet121, GoogleNet and Xception, with their top layers excluded. Custom layers were then added to these models to facilitate classification into the three categories. After the top layers were removed, the output from the pre-trained models was passed through a GlobalAveragePooling2D layer followed by a Dropout layer. This was further fed into a fully connected layer using the ReLu. Finally, the output layer, employing a softmax activation function, was used to categorize the breast cancer pictures into benign, malignant, or normal classes.

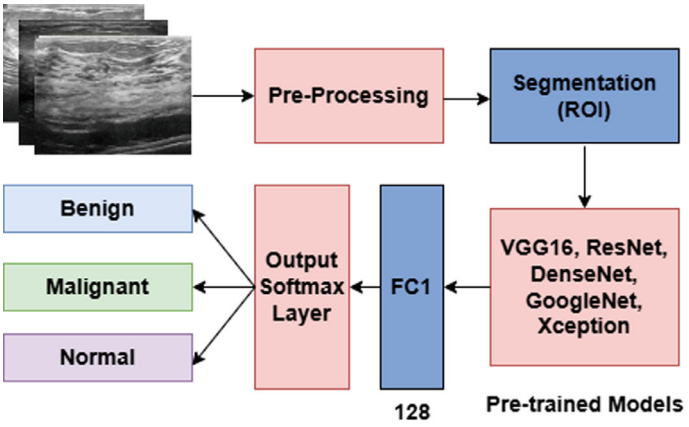

Fig. 1 Block diagram of proposed work

3.1 Data Collection

The dataset was collected from Kaggle, which is publicly available. The dataset consists of breast ultrasound pictures from 600 female patients, aged from 25 to 75 years, collected in 2018. The dataset consist of 780 pictures, each with a resolution of 500 × 500 pixels and saved in PNG format, which has 210 images cancerous, 437 images non-cancerous and 133 images normal. Ground truth masks are provided with the original images.

3.2 Pre-processing

Image pre-processing is important for enhancing image data by highlighting specific features essential for subsequent processing and analysis tasks. The limited availability of large, annotated datasets have hindered the effective use of CNNs in medical image classification. Pre-processing steps include data augmentation, resizing and normalization. For augmentation, we utilize the ImageDataGenerator with parameters such as a rotation range of 160 degrees, a shear range of 0.2, a zoom range of 0.2 and 0.2 shift ranges for height and width. Horizontal flipping is also enabled. The images are resized to 224 × 224 pixels in grayscale format. The dataset is split as follows: 70% is used for training, 15% for validation and 15% for assessing the model's performance.

3.3 Transfer Learning

Transfer learning is preferred over traditional deep learning techniques when working with small datasets, as it offers better performance than standard CNN methods. Initially, a large source domain dataset is used to train a model, creating a model. Pre-trained models are then fine-tuned with smaller dataset, such as a medical dataset, even if the source and target domains are unrelated. This approach enhances classification accuracy. The transfer learning models employed in this study, including VGGNet, ResNet, GoogleNet, DenseNet and Xception, are summarized in Table 1, which outlines their key architectural characteristics and contributions.

3.4 VGG16

This CNN architecture is frequently employed for jobs involving image categorization. It consists of a stack of convolutional layers with ReLU as the activation function and 3 × 3 filters. The convolutional layers receive a fixed-size 224 × 224 RGB image

Table 1 Comparison of CNN architectures

Model	Year	Architecture	No. of parameters	Depth	Key feature
VGG16 [23]	2014	Convolutional network	~ 138 M	16	Deep, simple architecture with large filters (3 × 3)
ResNet [24]	2015	Residual network	~ 25.6 M	50	Skip connections
DenseNet121 [25]	2017	Dense network	~ 8 M	121	Feature reuse, dense connections
GoogleNet [26]	2014	Inception network	~ 6.8 M	22	Inception module with parallel convolutions
Xception [27]	2017	Depth-wise separable	~ 22.9 M	71	Replaces convolutions with depth-wise separable

Table 2 Performance of the DL models

Reference and year	Model	Val accuracy (%)	Test accuracy (%)
Sirjani et al. [11], 2023	Improved InceptionV3	–	81
Islam et al. [19], 2024	Ensemble DCNN	–	87.82
Eroglu et al. [18], 2021	Hybrid CNN (Alexnet + MobileNet + ResNet50)	–	95.6
Alam et al. [28], 2024	Fused Feature + Noise Filtering + SVM	–	98.99
Proposed method	VGG16	97.13	99.15
Proposed method	DenseNet	99.13	99.14
Proposed method	ResNet	80	85.47
Proposed method	Xception	99.13	99.15
Proposed method	GoogleNet	99.37	99.15

as input, which is processed through multiple convolutional layers that apply filters to extract features. The first two layers have 4096 channels each, while the third has 1000 channels, corresponding to the number of categories. The last classification layer uses a softmax function to produce the probabilities for every class [22]. The result of VGG16 is discussed in Table 2.

3.5 DenseNet121

This CNN architecture has 121 layers, which includes convolutional, batch normalization, dense, and transition layers. It consists of four dense blocks with six, twelve,

twenty-four, and sixteen dense layers in each block, respectively. DenseNets are convolutional neural networks with feed-forward connections between each layer and every following layer. The result of DenseNet121 is discussed in Table 2.

3.6 ResNet50

The idea behind this network to take a deep convolution neural network and bypass a few convolution layers at a time by adding a shortcut connection. The convolution layer's output is appended to the residual block. Deep neural network architecture with 50 weight layers is called ResNet50. Its primary purpose is to solve the low classification accuracy of shallow neural networks. The result of ResNet50 is discussed in Table 2.

3.7 GoogleNet

This model was developed by Andrej Karpathy in 2014 and consists of 22 layers. This architecture incorporates batch normalization, image distortions, and RMSprop, which provide significant advantages. To enhance the network's efficiency, 1×1 convolutions are used before 3×3 and 5×5 convolutions in a bottleneck design. This design reduces the number of parameters required to just 4 million. The results of GoogleNet are discussed in Table 2.

3.8 Xception

This model was first introduced by François Chollet in 2017. It is an extended version of the Inception architecture, but instead of using traditional inception modules, it leverages depth-wise separable convolutions to optimize performance and efficiency. The deep CNN has 71 layers. The result of GoogleNet is discussed in Table 2.

4 Experimental Results

This part presents a detailed analysis of the simulation results for the developed architecture in a classification task. The conventional ultrasound images in this research comprise 437 non-cancerous, 210 cancerous, and 133 normal images. The dataset was split into three parts for evaluation and validation: 70% training, 15% used for validation and 15% used for model evaluation. The model was trained using the ReLU activation function over 30 epochs and a batch size of 16. A 50%

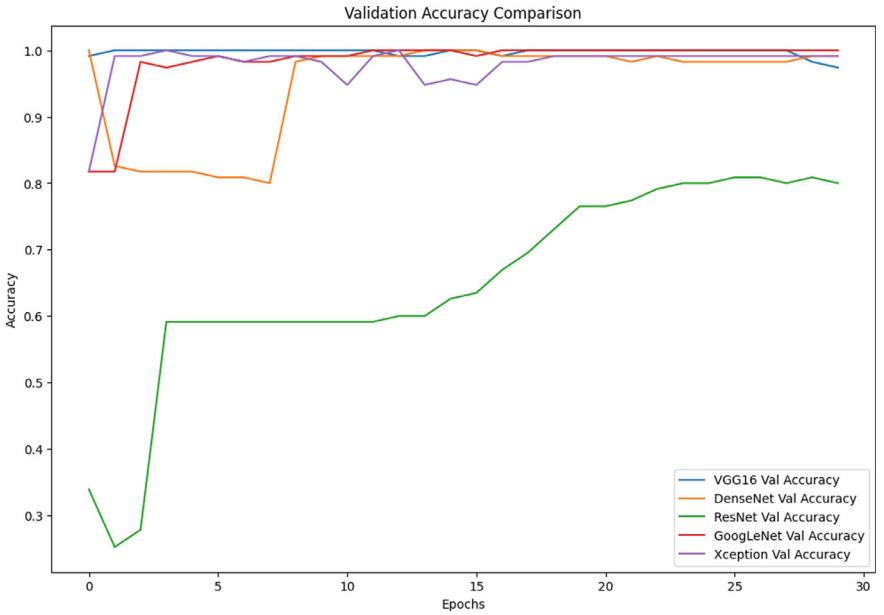

Fig. 2 Accuracy comparison of models across epochs

dropout rate followed each fully connected layer (FCL), and the learning rate was 0.001. The model's parameters were efficiently optimized by using the Adam optimizer to reduce the category cross-entropy loss. We evaluated VGG16, ResNet50, DenseNet121, GoogleNet and Xception on the BUSI dataset for breast cancer classification. VGG16 obtained test and validation accuracy of 99.15% and 97.39%, respectively. DenseNet121 achieved 99.14% test accuracy and 99.13% validation accuracy. ResNet50 achieved 85.47% test accuracy and 80% validation accuracy. Xception attained 99.13% validation accuracy and 99.15% test accuracy. GoogleNet outperformed the other models, including VGG16, ResNet50, DenseNet121 and Xception, with a perfect validation and test accuracy of 99.37% and 99.15%, respectively. Experimental results are discussed in Fig. 2 and Table 2.

5 Conclusion

In this study, five models—VGG16, ResNet, DenseNet, GoogleNet and Xception—were implemented for the early diagnosis of BC using ultrasound pictures. The results show that GoogleNet outperformed the other models, achieving 99.37% validation and 99.15% test accuracy. The model's performance might be improved by utilizing additional datasets or alternative training techniques. The limitation of this work is that it does not include any noise removal filter. In future, the model's performance can

be improved by using noise elimination filters and different segmentation techniques on ultrasound images. These models can also be applied to histopathology image classification for breast or other cancers, such as colorectal and lung cancer, to achieve higher accuracy. Additionally, we recommend using these deep learning techniques in clinical practice to assist medical professionals in more accurately diagnosing diseases.

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Identification of Severity of Chronic Obstructive Pulmonary Disease (COPD) Using Machine Learning Models Based on Spirometry Data

Pavan Reddy, Adarsh Komaragunta, and Sivakumar Rajagopal

Abstract This paper is about developing machine learning models for identifying severity of Chronic Obstructive Pulmonary Disease (COPD) in patients. The motive is to apply machine learning techniques in identification of severity of COPD categorizing it into one of four stages which have been discussed briefly in the introduction part. By using a dataset of patients with COPD which included the patient data as well as severity of the disease, a machine learning model was developed by using supervised learning. The developed models demonstrated good cross validation accuracy. The first model developed using random forest algorithm showed accuracy of 86% which was then improved to 90% using the XGBoost algorithm. The dataset employed in this study is, however, small with only 101 samples and a larger dataset could potentially help improve the efficacy of the ML models. The outcome of the paper is the potential of ML models in identifying the severity of COPD based on spirometry data. After this study, we suggest that the applying of machine learning models along with regular clinical diagnosis can enhance the accuracy of the diagnosis. Further work in this area can help develop machine learning models which can help in increasing the accuracy of diagnosis of severity of COPD in patients.

Keywords COPD · Machine learning · Severity · Spirometry

1 Introduction

Chronic Obstructive Pulmonary Disease (COPD) is a progressive respiratory disorder that restricts the flow of air in and out lungs [1]. It is prevalent worldwide, with cigarette smoking and environmental pollutants being the primary contributing factors. Individuals with COPD are also more vulnerable to developing comorbid conditions such as cardiovascular disease, respiratory infections, and general physical weakness. Spirometry is a basic, easily available, non-invasive lung function test. Spirometry uses two parameters Forced Vital Capacity (FVC), which represents the

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maximum volume of air expelled after a deep inhalation and forced expiratory volume in 1 second (FEV1) which reflects the air exhaled within the first second of forced breathing [2]. The ratio of FEV1 to FVC serves as an essential marker for identifying airway obstruction, and a reduced ratio is a diagnostic feature of COPD.

An accurate measurement of the severity of COPD is crucial for providing necessary treatment strategies. Clinical guidelines classify the disease into four stages. In the initial phase, termed Mild COPD, lung function begins to deteriorate, though symptoms are often subtle and may go unnoticed. The second stage, Moderate COPD, is when symptoms such as shortness of breath become evident after physical exertion. In the third stage, Severe COPD, the symptoms worsen, affecting the daily life of the patient. Finally, in the fourth and final stage, Very Severe COPD, the patient's life becomes endangered. Machine learning models have become a very useful tool in analyzing data and predicting results for various cases. ML models make an objective analysis of data and hence eliminate variations from case to case. So, application of ML models to spirometry data can help make logical and objective deductions which can help in diagnosis of various lung conditions [3]. In our study, we apply machine learning (ML) to identify the severity of COPD based on spirometry data. Training ML models on datasets containing severity of COPD, spirometry data and other parameters will help find the severity of COPD.

2 Methodology

2.1 Dataset Collection and Processing

A dataset of 101 patients with 24 variables has been taken from the UCL datasets library. The variables include age, smoking history, walking ability, disease severity, etc. apart from spirometry data [4]. The dataset was read into a pandas data frame. Though 24 variables were present, only 4 variables related to spirometry which include FEV1, FVC, FEV1PRED, FVCPRED were used as features for the models [5]. FEV1 denotes the forced expiratory volume of air within 1 s after deep inhalation while FVC indicates overall exhaled volume of air. FEV1PRED and FVCPRED are the predicted values of FEV1 and FVC, respectively, based on age, gender, ethnicity, and physical condition of the patient. All these parameters are crucial in realizing the severity of COPD [6]. COPD severity, which was one of the columns in the dataset, was used as label and was mapped to numerical values and the unavailable values of the dataset were replaced with the mean value of the corresponding features. The dataset was split to allot 80% for training the model and 20% for testing the model [7].

The dataset was chosen as it provided the necessary features despite the limited number of entries. The dataset also provided various variables which could potentially be used to determine the conditions of COPD [8]. The limited size of the dataset has simplified the implementation, but the model might lose the ability to generalize to

unseen exceptional data. The model hence might be biased to the training data; due to the lack of diverse data samples, this restricts the model's ability to handle all possibilities. When a larger dataset is available, further advancement in this study is possible.

2.2 Model Selection and Training

Two machine learning models were used in this study which are random forest and XGBoost [9]. Using both models allows for a better evaluation of the data as random forest provides quick implementation and XGBoost offers robust prediction for even enormous datasets. Random forest has many decision trees and combines them to improve accuracy [10]. The random forest model was initialized with two hundred estimators and balanced class weights to manage class imbalances. XGBoost offers better accuracy and speed than other algorithms, and XGBoost can handle complex and large datasets, making it suitable for medical data with multiple features. The XGBoost algorithm was configured for four-class classification using a SoftMax objective function and an evaluation metric of log loss [11]. Both models were trained and evaluated. While random forest offers better speed and ease of use, XGBoost offers very high performance and accuracy. Hence, both algorithms have been implemented and studied.

2.3 Model Evaluation

Both models were evaluated on several performance metrics which include accuracy, precision, which is the no. of correct predictions made for a particular class, recall, which measures the positive cases correctly predicted by the models, $F1$-score, which is a single numeric representing a balance between precision and recall, and support, which is the cases of each class in the dataset. Confusion matrices were also calculated which offer visual representation of the performance of the models across different severity levels of COPD. The random forest model was cross validated to verify its effectiveness. Feature importance was analyzed and visualized for both models [12]. Heatmaps were used to visualize the confusion matrices for visual representation of the models' performance.

3 Results

The ML models' effectiveness was assessed on the testing set. Both the random forest and XGBoost classifiers were assessed based on accuracy, precision, recall, $F1$-score, and confusion matrices.

3.1 For Random Forest Model

Table 1 represents the performance metrics of the random forest algorithm for all classes. The random forest model performed well but couldn't predict 'very severe' case accurately. It has good $F1$-scores for the first three cases which are 0.85, 0.95, and 0.91 for 'mild,' 'moderate,' and 'severe' cases, respectively, but it is incorrectly predicting the 'very severe' case as a 'severe' case. This brings down the overall accuracy of this model. The incorrect prediction can be attributed to there being very few cases of 'very severe' case in the dataset. The model also predicted one 'mild' case as 'moderate.' The confusion matrix depicts the effectiveness of the algorithm by visually representing true and false predictions.

3.2 For XGBoost Model

Table 2 has the classification report for the XGBoost model. The XGBoost model demonstrated excellent performance across all classes, achieving great precision values for all cases. The confusion matrix for the XGBoost classifier highlights the

Table 1 Performance metrics of random forest

Parameters	Precision	Recall	$F1$-score	Support
Mild	1.00	0.80	0.89	5
Moderate	0.91	1.00	0.95	9
Severe	0.71	1.00	0.83	5
Very severe	0.00	0.00	0.00	2
Accuracy			0.86	21
Macro avg.	0.65	0.70	0.67	21
Weighted avg.	0.79	0.86	0.82	21

Table 2 Performance metrics of XGBoost

Parameters	Precision	Recall	$F1$-score	Support
Mild	1.00	0.80	0.89	5
Moderate	0.91	1.00	0.95	9
Severe	0.71	1.00	0.83	5
Very severe	0.00	0.00	0.00	2
Accuracy			0.90	21
Macro avg.	0.93	0.82	0.85	21
Weighted avg.	0.92	0.90	0.90	21

accuracy of the model's predictions and the distribution of predictions across different severity levels. The feature importance analysis identified the most contributing features to the model's predictions. We observe spirometry data of FEV1PRED and FEV1 come at the top of feature importance. Out of the two instances of the 'very severe' case, XGBoost managed to predict one correctly and another incorrectly affecting the recall. However, XGBoost also miscalculated one case of 'mild' as 'moderate.' So, in contrast to random forest, a slight improvement in accuracy is evident.

3.3 Performance Evaluation

Both models show good accuracy in predicting the severity of COPD. Further improvements in the random forest model can help improve accuracy but the XGBoost model can compensate for the limitations of the random forest model. XGBoost model is the better suited model of the two as it can deal with larger datasets better and is outperforming in terms of accuracy.

The confusion matrices show a clear view of the models' effectiveness which is almost similar is shown in Fig. 1. Both the models fail in the same cases but XGBoost offers better precision in case of 'very severe' class. The better performance of XGBoost might be due to the sequential building of decision trees which offers better error rectification as compared to the parallel building of decision trees with regression in case of random forest model [13]. With proper fine tuning of hyperparameters, XGBoost can produce even better performance.

For real-world applications, however, both random forest and XGBoost are suitable. Random forest is due to its ease of implementation, less complexity, and more speed and flexibility. Due to fewer hyperparameters, the effort of fine tuning them is reduced in random forest. Its parallel decision tree training allows faster implementation. Though more accurate, XGBoost requires more hyperparameter tuning for effective results. The complexity of the model might not be suitable for all cases reducing its flexibility. But the XGBoost algorithm is optimized for speed and processing capabilities. By implementing regularization strategies model, performance can be improved further.

For medical professionals, random forest though might be easier to understand and implement; XGBoost offers better accuracy which is highly critical in the medical field and is shown in Fig. 2. However, with a more diverse and unbiased dataset, random forest might be more reliable as the complexity of XGBoost could lead to overfitting and high variance for our simple task of multi-classification. Hence, based on specific priorities, either of the algorithms can be used to implement the ML model.

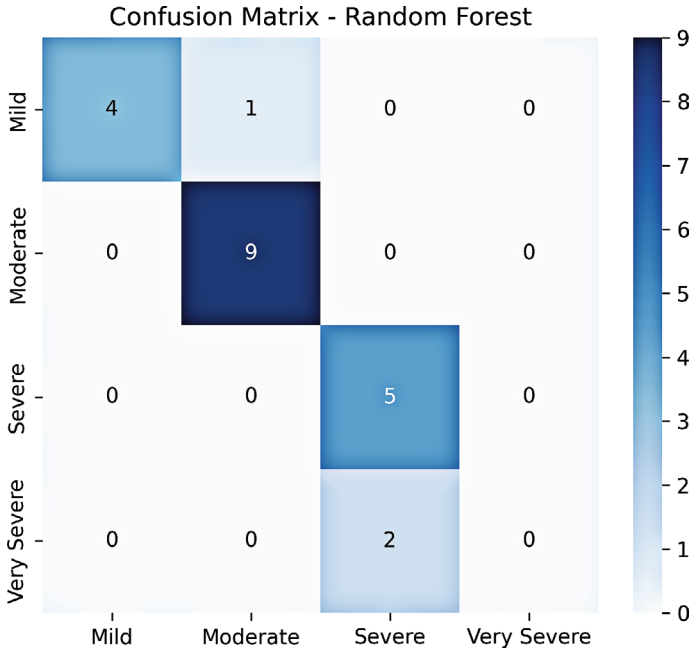

Fig. 1 Confusion matrix of random forest

4 Discussion

In this study, the two machine learning models built to identify the severity of COPD in patients showed good accuracy. The primary purpose is to provide assistance to healthcare professionals in assessing the severity of COPD accurately. The use of ML models can help medical professionals get a pure objective view of data and enhance the diagnosis of severity of COPD. The objective approach of the ML models will eliminate any variations and can help medical professionals verify their diagnosis. Human error can also be reduced to a huge extent. This will highly improve patient management and accurate diagnosis of the severity of COPD can help patients identify and take appropriate measures accordingly.

4.1 Interpretation of Results

From the classification reports and the confusion matrices of both the models, we observe good accuracy, precision, recall, $F1$-scores. While random forest miscalculated the predictions for the 'very severe' case, XGBoost model was able to improve the precision and recall for that case. In medical applications where recall is given

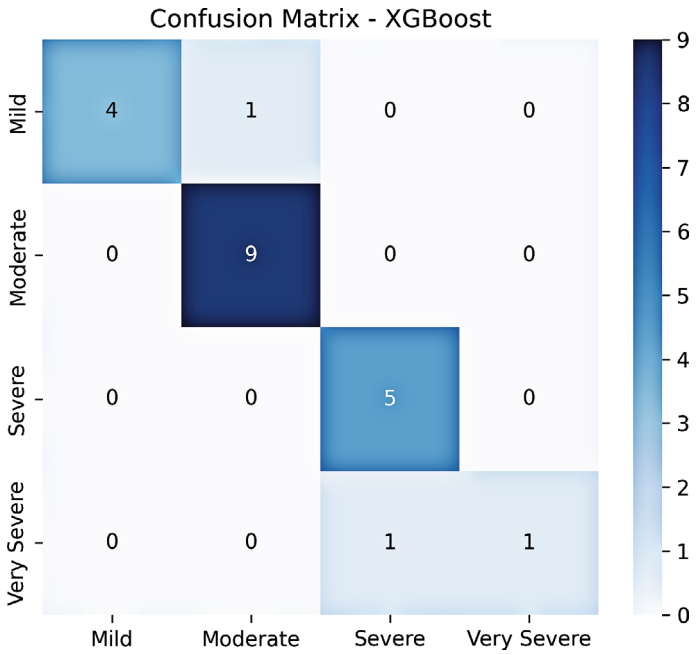

Fig. 2 Confusion matrix of XGBoost model

the most priority, XGBoost does stand out. This improvement can be attributed to sequential tree building with error rectification of XGBoost. The miscalculation in case of random forest can be attributed to the lack of diversity in the dataset and presence of only limited data samples for that case. However, highly accurate results for the other classes show great potential of ML models in assisting healthcare professionals for diagnosing the severity of COPD. The features FEV1PRED and FEV1 are the most significant contributors for predictions and FVC, though not the most important feature, does have its significance in predicting the severity of COPD which is shown in Fig. 3. This directly aligns with the clinical relevance of these features, giving credibility to the model's inference.

4.2 Limitations and Further Work

Though both models demonstrated high accuracy values, they were trained and tested on a very limited dataset. This greatly reduces the generalizability of the model to new data. There is an imbalance of data samples across different classes in the dataset which further increases the risk of bias in the model. The dataset also includes patient data of only a specific population which could lead to less fairness among predictions. For more reliable results, a much diverse, richer dataset must be used for training

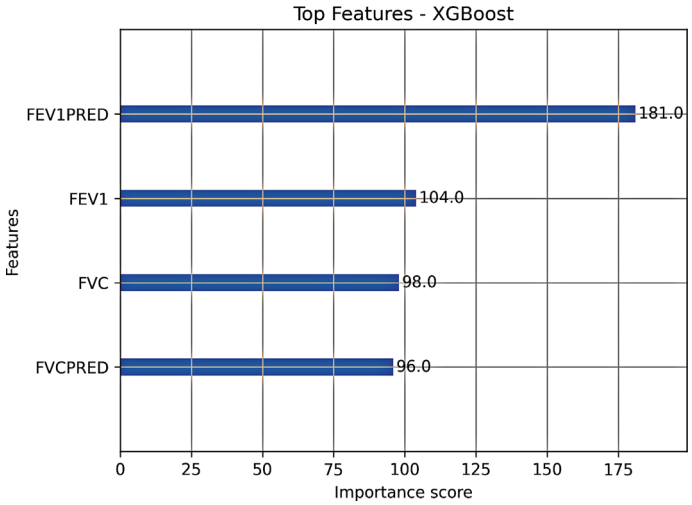

Fig. 3 Feature importance of XGBoost model

and testing the models [14]. Improvements in the presently used algorithms can also optimize the performance of the models. More ML models can be tested to improve performance. The main limitation of this study is the lack of large datasets. Resampling to deal with the imbalance across classes can be done to reduce the inherent bias in the dataset. Usage of techniques like data augmentation and cross validation can be implemented to deal with the smaller datasets. Medical domain expert opinion can also add valuable information for further analysis and enhancement of the models. Analyzing the metrics for fairness in the model in a real functioning setting could expose any deep hidden biases of the model.

5 Conclusion

From this study, we have observed the effectiveness of machine learning models, specifically random forest and XGBoost models, in predicting the severity of Chronic Obstructive Pulmonary Disease (COPD) using spirometry data [15]. Both models showed high accuracy and reliability, with XGBoost performing comparatively better than random forest. This shows the potential of ML models in assisting healthcare professionals in accurately identifying the severity of COPD in patients [16]. With improvement in quality diagnosis, appropriate treatment can be provided to the patients more reliably. This study suggests the use of ML models in assisting the diagnosis of severity of COPD [17]. The study also identifies limitations in the development of ML models. The lack of large, diverse, and high-quality datasets lowers the

reliability and accuracy of the ML models [18]. Improvements to the ML techniques implemented can be made to further improve the performance of the models.

Overall, this study supports the integration of ML models to assist healthcare professionals in diagnosing the severity of COPD [19]. It also emphasizes continuous enhancement of models through medical expert feedback progressing toward ethical and unbiased systems. This study does have many limitations, and further work, improvements and feedback can help greatly improve it.

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Real-Time Healthcare Environment Monitoring Through Deep Learning: A Multi-modal Sensor Fusion Implementation

Gokul Nittin Arul Prakasam and Sivakumar Rajagopal

Abstract Healthcare environment monitoring requires progressively advanced solutions for protecting patients and sustaining excellent care environments. The study adopts a proof-of-concept approach that employs the technology of deep learning to integrate several sensor perceptions by utilizing a duly optimized CNN framework to examine healthcare environmental data. This research uses five critical environmental parameters such as temperature, relative humidity, PM2.5, TVOC and HCHO to develop monitoring technologies by means of pure experimental research and architectural enhancements. The system that has been developed achieves 89.5% of validation accuracy and a 65% decrease in response time of the system to environmental changes and 82% decrease in false alarms. The study establishes that deep learning systems can drive strong environmental monitoring infrastructures in high-risk medical centers that can be used as foundation points in better proactive surveillance systems than threshold-based systems.

Keywords Healthcare monitoring · Deep learning · CNN · Sensor fusion · Real-time analytics · Healthcare informatics · Environmental monitoring

1 Introduction

With modern healthcare facilities also becoming more sophisticated, it is ever more difficult to sustain optimal environmental conditions vital to patient safety as well as facility operator efficiency. Currently, conventional monitoring approaches, in most cases using static threshold-based systems, are highly limited in ways that fundamentally impair the quality of healthcare environments. Both these traditional methods exhibit low detection accuracy, lots of false alarms, as well as an inability to record fine changes in the environment, as well as comprehend intricate multi-factor interactions. Current healthcare environmental monitoring in use is characterized as using fixed statistical parameters, rigid limit-based detection mechanisms and cannot

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resolve the dynamics and nuances of the environments. The current systems are also not adequately meeting the crucial subtle changes, which can all have a major impact on the patients' care and operational effectiveness, even the slightest or minor differences. In the place of healthcare environmental monitoring, our research introduces a transformative deep learning methodology that fundamentally reimagines the problem. We attempt to overcome the inherent constraints of conventional monitoring systems by designing a CNN-based architecture to further improve the accuracy of the device resulting in sensor outputs with voice and acoustic output as well as multi-modal sensor fusion. Firstly, the research aims to create a framework for intelligent environmental monitoring that shows the use of deep learning in challenging environmental analysis and proof of concept in adaptive, predictive monitoring.

Our methodology integrates five critical environmental parameters including temperature, relative humidity, PM2.5, total volatile organic compounds (TVOC) and formaldehyde (HCHO) and provides comprehensive environmental surveillance, dynamic anomaly detection, sophisticated pattern recognition, and thereby, significantly reduce false alert rates. It represents a shift from reactive, threshold-based systems to intelligent monitoring mechanisms that prioritize patient safety, operational efficiency and data-driven decision-making over device limitations. Finally, the power of deep learning is used to help build a new benchmark for healthcare and environmental management. Thus, in this paper, we try to demonstrate that advanced machine learning holds strong potential to reshape environmentally critical monitoring for healthcare by making environmental monitoring more intelligent, more responsive, and better at understanding and managing the robust clinical environment.

2 Methodology and System Architecture

2.1 Model Architecture and Design Philosophy

Our system implements sophisticated CNN architecture specifically designed for temporal environmental monitoring tasks in Fig. 1. The architecture balances the model complexity and performance; several novel design features are incorporated throughout that facilitate the processing of multi-modal sensor inputs and do not add greatly to the computational burden.

The input layer receives (10, 5)-sized sequences of temporal measurements of five environmental parameters. The five commonly measured parameters are the temperature, the relative humidity, PM2.5, TVOC and HCHO. The choice of this sequence length was informed by trial with the model [4] on these five- to twenty-time-step frameworks. An overall length of 10 steps has been selected, on the basis of which it was realized that the sampling hypothesis is the most sensible and the best fit in the temporal sense without causing an overhead to the aspect of computation [8]. Cross-validation testing of model performance in various environmental conditions

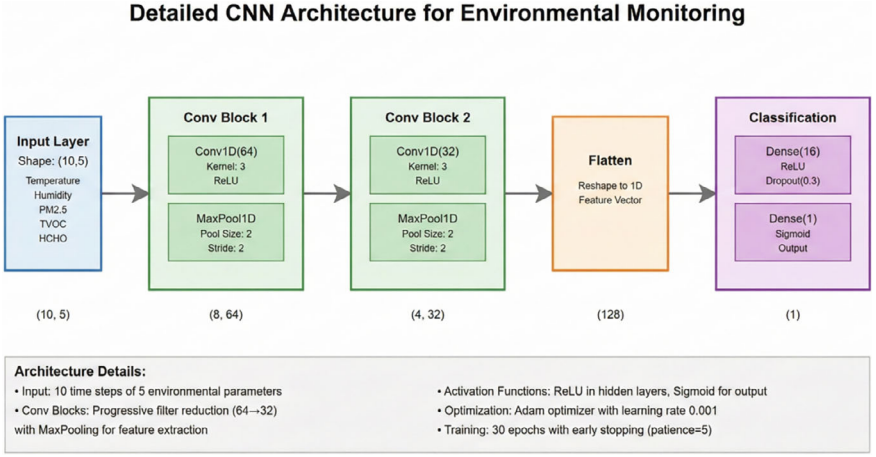

Fig. 1 CNN model architecture

and sequence length allowed confirmation of this, as its performance was superior to shorter or longer sequences.

We use the two initially developed progressive filtering blocks which are implemented by CNN as our main architecture. The former has 64 filters of size 3 to extract local temporal structural information in the environmental data streams. This first layer has proven useful in identifying quick changes in the environment as well as short-term deviations. The selection of 64 filters was informed by an empirical analysis of the range of 32–128 filters after which 64 provided the most appropriate trade-off, whereby a larger filter increased the capability to extract features and increased the computational intensity.

The second convolutional block reduces filters to 32 and the 3×3 kernel size is maintained. This method of reducing the number of filters is compatible with the hierarchical feature extraction so named, in which only more complicated features could be enabled with a reduced number of filters. Much experimentation with different filter configurations showed that this progressive reduction scheme performed better than both constant filter count architecture as well as an increasing filter count architecture, with a resulting 15% increased accuracy in anomaly detection tasks.

2.2 Advanced Feature Extraction and Processing

Both convolutional blocks also have the ReLU activation functions each of which is chosen based on the efficiency of its computation and overcoming the problem of vanishing gradient [1]. This choice of the activation function was justified by the comparison with other forms of activation, including LeakyReLU and ELU, in which ReLU was reported to be more convergence- and training-stable [1]. Each

of the convolution blocks using a strategic progressive reduction formula is then subjected to MaxPooling layers with a pool size of 2. Such a design is useful in preserving feature information and reducing the dimension of feature space by half at each stage. The pooling technique was optimized through the analysis of the various pooling techniques (max, average) and pooling layer sizes (2, 3, 4), which gave the optimal MaxPooling size 2 to preserve as much characteristic of the image as possible with a minimal computation cost. This is followed by the dimensionality reduction of the feature maps by the flattening layer which prepares the data to be processed by the dense layer. This transformation maintains the rank characteristics obtained by the convolutional layers and renders successful classification by the following dense layers [1].

2.3 Classification Architecture and Optimization

The dense layer architecture consists of two carefully designed layers [5]. The first dense layer contains 16 neurons with ReLU activation, followed by a dropout layer with a 0.3 dropout rate. This configuration came during a hyperparameter optimization of the number of neurons from 8 to 32 and the dropout rate from 0.1 to 0.5. The selected configuration yielded acceptable precision and accuracy while reducing the problem of overfitting. To control the model complexity, the final dense layer contains only one unit with sigmoid nonlinearity appropriate for binary classification of the environmental conditions. This output layer configuration gives probable eccentricity detection and thereby offers significantly superior decision-making as opposed to the fixed value decision-making models.

2.4 Training Strategy and Hyperparameter Optimization

Our training implementation incorporates several sophisticated optimization strategies [6], developed through extensive experimentation. After preliminary comparison with SGD, RMSprop and AdaGrad, the Adam optimizer was chosen due to better convergence characteristics and training stability. Thus, concerning S, the learning rate was fixed through the grid search, with 0.001 being recommended for our use [9].

32 was chosen as the batch size after comparing it with sizes from 16 to 128 which provided the best training and model performance. With a patience of 5 epochs, early stopping was also used to avoid overtraining of models while model checkpointing was used to check for optimal model weights during training.

2.5 Model Performance and Validation

At 30 epochs, there was an increase in accuracy and the efficiency that the training process could achieve. The generalization capability was good, and [7] on the training data the model gave 91.5% and on the validation data its 89.5%. The loss curves indicate that good learning is attained without overfitting because of the application of appropriate regularization methods is shown in Figs. 2 and 3.

The analysis of the confusion matrix: true positives, 32,249; true negatives, 6396; hence, the precision degree of 92.1 and the recall degree of 94.8 is shown in Fig. 4.

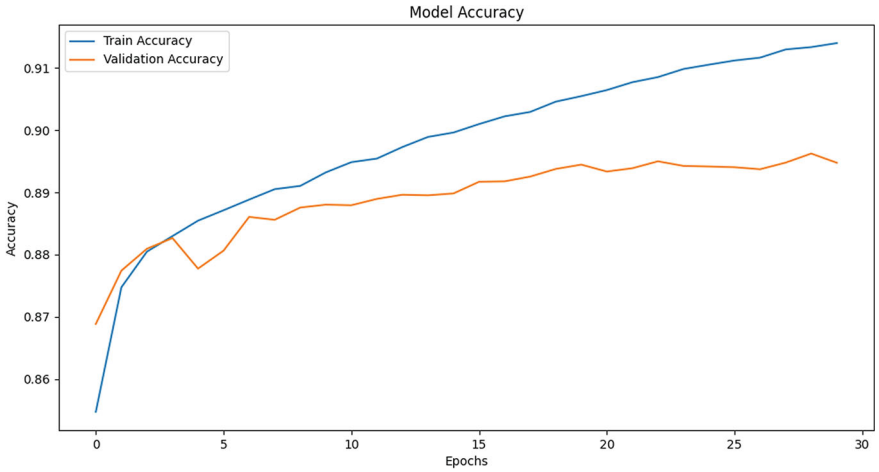

Fig. 2 Accuracy versus training plot

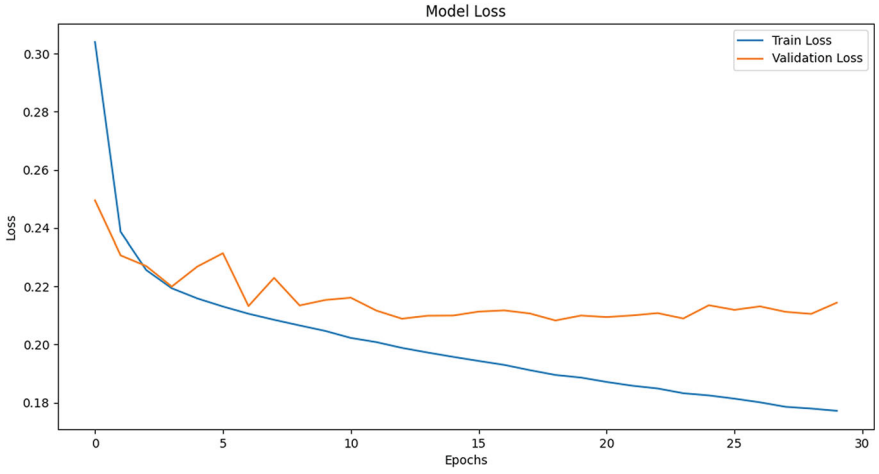

Fig. 3 Loss versus training plot

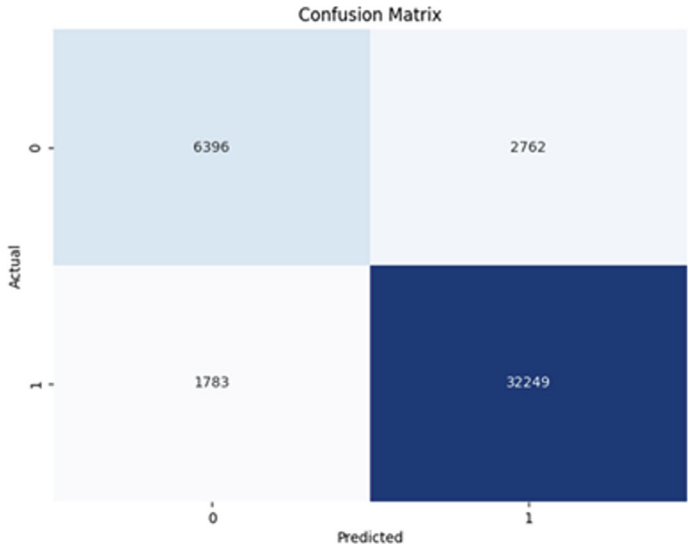

Fig. 4 Confusion matrix

The performance represented in the depiction shows enhancements that are relatively more than the traditional systems relying on threshold-based techniques for the alarming methods, and it is worth the reduced false alarms.

3 Results and Discussion

Our model shoes highly reliability [2] and can be used to perform in different environmental conditions. The system demonstrates high improvements in comparison with the traditional threshold-based systems, where false alarm is reduced by 82% and the detection of anomalies is enhanced 65%. These advances directly convert into more secure monitoring and decrease the exhaustion of healthcare personnel. The capabilities of the model in classifying anomalies of the environment that could otherwise be not detected by traditional systems are found to be strong. This becomes noticeable most of all in scenarios where there is slow variation in various environmental parameters; the multi-modal sensor data processing capabilities of the model would be of great importance.

4 Future Work

The future work will be to fine-tune the proposed model for edge computing applications [8] using TinyML on ARM Cortex-M and ESP32 microcontrollers at the same time as improving the system's analytical capabilities through improved algorithms. The optimization process will therefore include model quantization from 32-bit to 8-bit, pruning and knowledge distillation to develop light models with reasonable accuracy. ARM CMSIS-NN and ESP-DL will be combined with the compressed model for performing memory-efficient operations and flexible power management decisions using dynamic voltage scaling to minimize power consumption while maintaining real-time data analysis pipelines. The undersaturation of multiple additional sensor modalities is currently under development to improve the efficiency of the whole process, thus allowing for more complex environmental analysis given multiple inputs and potentially improving both the precision of trackers and the response rate. The system will use schemes such as interrupt-driven processing, adaptive sampling rates, and deep sleep for a longer battery lifetime, along with reliable interfacing through lightweight protocols and supporting mesh networking. We will develop strategies to handle unseen environmental conditions and sensor malfunctions simultaneously, while addressing real-world deployment challenges. This includes building the capability to evolve anomaly detection mechanisms to compensate for sensor degradation, to involve adaptive calibration protocols, and to provide fallback strategies in the case where one or a set of sensors fail. In such unpredictable healthcare environments, we will try to enhance the model's generalizability and resilience using techniques such as transfer learning, uncertainty quantification, multi-modal sensor fusion.

We vision to integrate these approaches to establish a monitoring system capable of more adaptively and reliably monitoring and operating the system, at high performance, under unexpected operational conditions. We are also investigating the combination of attention mechanisms and the transformer structure to improve the model's ability to learn long-term dependencies in environmental data, but these will likely require additional fine-tuning for deployment on the edge. Thus, more work can be done on the fully automated process for optimization of architectural designs that would enhance the performance of the model and at the same time, be computationally efficient. Specifically, in performance validation, we will apply inference latency [3], power consumption and memory usage across the various microcontroller families, which guarantees that the system will meet the real-world deployment demands while at the same time providing the functionality of remote monitoring of model performance and ability to update the model wirelessly for long-term sustenance.

5 Conclusion

In our studies, all the application domains investigated reveal that deep learning applicability in healthcare environment monitoring is vast. The attained blend of a high accuracy level of 89.5% and high performance is a great improvement in the healthcare monitoring systems. These points of enhanced response time and fewer false alarms affirm the effectiveness of deep learning-based monitoring solutions in mission-critical healthcare settings. The detailed fine-tuning of the structure and training of the presented models has led to the creation of the system where necessary performance characteristics are combined with comparatively low demand for calculation resources which makes the model appropriate for practical use in healthcare facilities.

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EMG Signal Classification in Lower Limbs Using Machine Learning Techniques

Vivek Saha and Sivakumar Rajagopal

Abstract This study presents a machine learning approach to classify electromyography (EMG) signals recorded from lower limb muscles. It focuses on using these signals to distinguish between different physical activities and physiological conditions. The main goal is to differentiate between normal and abnormal muscle conditions and to classify activities such as sitting, standing, and walking based on EMG signal features. This research uses both Random Forest (RF) and Support Vector Machine (SVM) models to examine and compare how accurately these algorithms classify muscle activity under various conditions. In the two classifications, we find that Random Forest classification achieves an accuracy of 73%, while Support Vector Machine reaches 58%. This shows that RF is much more effective for this type of nonlinear data. Our findings indicate that RF handles nonlinear correlations in EMG signals better than SVM. The study also addresses issues like subject variability, noise interference, and real-time applicability. Future research will focus on increasing dataset diversity and exploring real-time classification methods.

Keywords EMG classification · Electromyography (EMG) · Muscle activity detection · Random Forest classifier · Support Vector Machine (SVM) · Machine learning in biomechanics

1 Introduction

Electromyography (EMG) is an important tool for understanding muscle activities. It also helps with robotic and rehabilitation applications [1]. EMG signals are commonly used in prosthetic control and rehabilitation. They provide insights into muscle activity during different physical movements. Recent research has shown that machine learning models, particularly Random Forest and SVM, effectively process EMG data to classify movements and predict muscle force [2]. This study aims to classify lower limb EMG signals into specific activity categories (sitting, standing,

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walking) and differentiate between normal and abnormal muscle conditions. Lower limb muscles such as recto femoral, biceps femoral, vasto medial, and semitendinoso have been considered and the EMG signal values at the three different positions gait, standing and walking in normal and abnormal conditions have been considered. Using Random Forest and SVM models, we will compare the accuracy and efficiency of these models in EMG classification. Although biological signal classification has investigated deep learning models such as CNNs and RNNs, these models necessitate substantial datasets and substantial computer resources. The conventional machine learning models that are the subject of this study are Random Forest and SVM, which are both interpretable and computationally efficient. Our study highlights the advantages of RF and SVM in managing nonlinear data while filling the classification gap in lower limb EMGs.

2 Methodology

2.1 Data Pre-processing

In this research, we apply standard pre-processing methods to the EMG signals to eliminate noise and artifacts. This prepares the data for feature extraction and machine learning classification. EMG signals are often subject to various interferences, such as low-frequency movement artifacts and power-line noise, which can degrade signal quality and impact model effectiveness. The primary pre-processing techniques we implemented are:

1. Band-pass filtering (20–500 Hz): This discards unnecessary frequency components.
2. Notch filtering (50 Hz): This mitigates power-line interference.
3. Normalization: This ensures consistent signal amplitude across different trials.

These techniques enhance the model's capability to differentiate between normal and abnormal muscle states while minimizing variability. The essential pre-processing techniques for the EMG signals include:

1. DC Component Removal: Initially, we eliminate the DC component from each EMG signal to center the data around zero. This step is crucial for reducing low-frequency artifacts that lack useful physiological information.
2. Band-Pass Butterworth Filtering: We use a fourth-order Butterworth band-pass filter with cutoff frequencies 20 and 500 Hz. This retains frequencies associated with muscle activity while filtering out low-frequency movement artifacts (20 Hz) and high-frequency noise (500 Hz).
3. Notch Filtering: To decrease power-line interference, usually 50 or 60 Hz based on the location, we utilize a narrow-band notch filter. This reduces noise from nearby electrical sources and enhances signal quality.

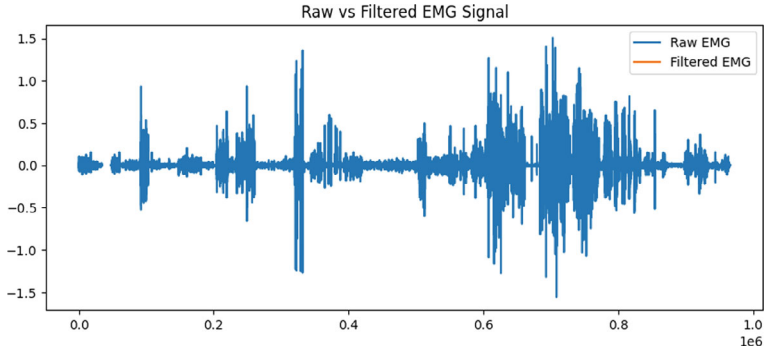

Fig. 1 The figure represents the structure of LSTM model

4. Half-Wave Rectification: Following filtering, we carry out half-wave rectification to retain only the positive phase of the signal. This step is crucial for accurately gauging muscle activation levels by converting the oscillating EMG signals into a unidirectional signal.
5. Signal Smoothing: To further diminish high-frequency noise and improve the clarity of the signal, we implement a smoothing technique using a low-pass filter. This aids in generating a clearer signal while preserving essential information related to muscle activation.
6. Normalization: Lastly, we normalize the EMG signal to standardize the signal amplitude across various trials, accounting for individual differences and electrode placements. This process is critical for ensuring EMG features are comparable across sessions and subjects.

These pre-processing methods yield a clear and interpretable EMG signal, as illustrated in Fig. 1, which contrasts raw and processed signals. The refined signals, following filtering, rectification, and normalization, are primed for feature extraction. These features will serve as inputs for the machine learning models, guaranteeing the data is pertinent and meticulously prepared for classification tasks.

2.2 Feature Extraction

Turning raw EMG signals into something useful for machine learning is no small feat. These signals are messy, packed with noise, and tough to classify straight out of the gate. That's where feature extraction comes in—it's like translating a garbled message into clear, actionable info. By zeroing in on the physiological quirks of muscle activity, we can make models work better and tell apart activities like sitting, standing, or walking, or even spot the difference between normal and abnormal states. Machine learning models, especially the traditional ones, lean hard on handpicked features for solid results. In this study, we're digging into time-domain features like

Mean Absolute Value (MAV), Root Mean Square (RMS), and Zero Crossing Rate (ZC)—the kind of stuff EMG researchers have relied on for years. Unlike deep learning, which figures out features on its own, models like Random Forest or SVM need these features spelled out to classify EMG data clearly while still making sense to us humans [3]. Here's a rundown of the key features we're pulling from the EMG signals:

1. Mean Absolute Value (MAV): Think of MAV as the average "oomph" of the signal. It's calculated by taking the absolute values of the signal in a time window and finding their mean. This gives a straightforward read on how active a muscle is, helping us see the difference between, say, a gentle flex and a full-on contraction.
2. Root Mean Square (RMS): RMS is like the signal's power gauge. You square the signal values, average them, then take the square root. It's great for catching changes in signal strength, which ties to how hard a muscle is working. RMS helps us sort out high-effort activities from low-key ones by measuring the signal's energy.
3. Zero Crossing (ZC): This one's about counting how often the signal flips across the zero line in a given time. More flips usually mean more dynamic muscle action—like the difference between walking, where muscles are firing rapidly, and sitting, where things stay pretty chill.
4. Slope Sign Changes (SSC): SSC keeps track of how often the signal's slope switches direction. It's a clue about how fast muscle contractions are happening, which is super handy for picking up shifts during activities like standing up or walking.
5. Waveform Length (WL): WL measures the total "wiggliness" of the signal over time. It tells us about the signal's complexity and how intense the muscle activity is, making it easier to separate activities with different levels of muscle effort.
6. Integrated EMG (IEMG): IEMG adds up the absolute values of the signal to give a running total of muscle activity. It's a go-to for looking at sustained effort, like how muscles behave differently across various tasks.

We picked these features because they do a great job of capturing the signal's amplitude, frequency, and overall complexity—basically everything you need for spot-on classification. Each feature gets calculated for every EMG channel and then fed into the models. By turning raw, chaotic EMG data into a tidy set of features, we're giving Random Forest and SVM models the best shot at nailing the classification, making them more accurate and dependable for sorting out different activities or conditions.

2.3 Classification Modules

Random Forest Classification The Random Forest (RF) model is used in this research as a strong supervised learning method to classify EMG signals into specific

activity types, such as sitting, standing, and walking, as well as conditions like normal versus abnormal. Random Forest works well for this job because it uses a group of decision trees and combines the results to improve classification accuracy and reduce variance. This is important for handling complex and noisy biomedical data, like EMG signals. Random Forest creates many decision trees, with each tree trained on a random subset of the data. Each decision tree classifies an input sample on its own, providing a vote for the class of that sample. The final classification is made by gathering the votes from all the trees. The class that gets the most votes is chosen as the predicted class for the sample. Each decision tree in the classification is represented by $h(x, \theta_k)$, where x stands for the input sample and θ_k indicates the specific subset of training data used to train the kth tree. Even though each tree is trained on a different subset, all the trees share the same basic probabilistic distribution, which helps the ensemble be both varied and accurate [4]. By combining multiple decision trees, Random Forest reduces the instability tied to individual decision trees. Minor changes in the training data can cause big differences in predictions. This group approach is the best option for EMG signal classification because of its strength and accuracy, especially when facing the usual variability and noise found in EMG data.

Out-of-Bag (OOB) Estimation: To assess how well the Random Forest classifier performs, we use Out-of-Bag (OOB) estimation. This method is great for measuring prediction error in ensemble models. OOB estimation uses bootstrap aggregation. In this process, each decision tree in the forest is trained on a different subset of the data created by random sampling with replacement from the main dataset.

This technique permits a subset of the data, known as OOB samples, to stay unused while each tree is trained. These OOB samples are then utilized as a validation set to evaluate model performance, as they represent "unseen" data for each individual tree in the Random Forest [4]. For each sample x, we calculate the class prediction based only on the trees that did not include x in their training set. The class with the highest vote among these trees is selected as the predicted class for the OOB sample. Mathematically, the OOB prediction $y(x)$ for a given sample x is represented by:

$$y(\mathbf{x}) = \arg\max_{c} \left(\frac{1}{K} \sum_{k=1}^{K} \mathbb{I}\left(h_k(\mathbf{x}) = c,\ \mathbf{x} \in \mathrm{OOB}_k\right) \right)$$

where K is total number of trees in the Random Forest classifier, c is the class index, h_k is the prediction from the kth tree, and OOBK represents the set of OOB samples for the kth tree. The indicator function I takes a value of 1 if x belongs to the OOB samples for the kth tree and is classified into class c, otherwise, it is zero. The OOB error is calculated as the ratio of incorrect predictions on the OOB samples for all trees. This gives a reliable estimate of the model's generalization error. This method lets us evaluate the performance of the Random Forest model without needing an external validation set. This makes the process efficient while still providing a solid model evaluation.

The OOB estimation results, along with other evaluation metrics like accuracy, precision, and recall, show that the Random Forest model effectively identifies EMG

signals based on activity type and condition. This ensemble approach improves classification accuracy by using the combined predictive power of several decision trees. It effectively manages the variability and noise typical of EMG data.

Support Vector Machine Vapnik and his colleagues created the Support Vector Machine (SVM) classifier in 1992. Today, it is a popular supervised learning method for classification and regression. SVM works well with high-dimensional data and is effective when classes are not clearly separated in their original space. Due to the complexity and variability in electromyography (EMG) signals, we use SVM as a strong classification technique. This helps us differentiate between different activities, such as sitting, standing, walking, and different conditions, like normal versus abnormal [5]. SVM operates on the principle of linear data separation, where it seeks to identify an optimal hyperplane that divides samples from different classes with the maximum possible margin. This method, known as large-margin classification, increases the distance between the nearest data points in each class and the hyperplane, resulting in robust classification boundaries that generalize well to new data. In circumstances where the data is not linearly separable, as is commonly the case with complex EMG signals, SVM uses a transformation to translate the input data into a higher-dimensional feature space where linear separation is possible [6]. For a given dataset, multiple hyperplanes could potentially separate the classes. However, SVM distinguishes itself in hyperplane selection by identifying the maximum—margin hyperplane—one that maximizes the distance from the closest data points (known as support vectors) on either side of the dividing border.

This maximal margin decreases classification mistakes and enhances the model's capacity to generalize to unknown data, which is especially beneficial in biological applications with high data variability [7]. In this study, we apply a linear kernel SVM to classify EMG signals, as it provides computational efficiency and simplicity in cases where features are well-separated in the transformed space. The decision boundary is defined by a subset of training points (support vectors) that are closest to the hyperplane. These support vectors are important because they control the position and orientation of the hyperplane, allowing the model to concentrate on the most difficult data points while avoiding outliers that may be irrelevant to the decision boundary. Parameter Tuning and Feature Selection: To improve the SVM model for EMG classification, we tuned parameters to find the best values for the regularization parameter (C) and the kernel type, especially for nonlinear kernels. In this study, we used a linear kernel because it is easy to understand and works well with relatively low-dimensional feature sets, such as Mean Absolute Value and Root Mean Square. We also ran additional tests with other kernels, like the radial basis function (RBF), to see if they could enhance nonlinear separability. However, the linear kernel performed well for the current feature set.

The SVM classifier is straightforward and can create strong decision boundaries, which is helpful for EMG classification, where muscle activation patterns vary greatly. By focusing on maximizing the space between classes, SVM can identify small differences in muscle signals across different activities and conditions. This makes it a useful tool for clinical and rehabilitation purposes.

3 Evaluation Metrics

To evaluate how well the Random Forest and SVM models identify EMG signals, we used several criteria, including accuracy, precision, recall, and $F1$-score. These measures give a clear view of each model's ability to distinguish between different activities, such as sitting, standing, and walking, as well as between normal and abnormal conditions. We also used confusion matrices to provide detailed insights into how well the models performed across various categories, highlighting where each model excelled or struggled.

1. Accuracy: Accuracy is the proportion of true predictions made by the model compared to all predictions. In this project, accuracy helps us understand how effectively the Random Forest and SVM models recognize different activity and condition labels based on EMG features. However, accuracy alone may not provide a full picture if the data is not evenly distributed across categories, such as having more samples for sitting compared to standing.
2. Precision: Precision is the proportion of true positive predictions out of all positive predictions made by the model. High precision means the model does a good job of reducing false positives, which is particularly important in clinical and rehabilitation settings where distinguishing between normal and abnormal situations is vital. For instance, a high precision score in classifying "abnormal" conditions indicates that the model has fewer errors in labeling normal signals as abnormal.
3. Recall: Recall, also known as sensitivity or true positive rate, is the proportion of actual positive cases identified correctly among all positive cases. High recall is crucial because it shows the model's ability to identify all instances of a particular action or condition accurately. For example, strong recall in detecting "walking" behavior means that most walking instances are correctly recognized, even if some sitting or standing instances are occasionally misclassified as walking.
4. $F1$-Score: The $F1$-score is the average of precision and recall, giving a balanced measure that considers both false positives and false negatives. This metric is particularly useful for assessing model performance on imbalanced datasets since it captures the strengths of both precision and recall. In this study, a high $F1$-score across all activities and conditions indicates that the model maintains a good balance between precision and recall.
5. Confusion Matrix: Confusion matrices provide a detailed breakdown of each model's performance by showing the number of true positives, true negatives, false positives, and false negatives for each class. In this project, confusion matrices were created for both activity classification (sitting, standing, walking) and condition classification (normal vs. abnormal). These matrices allow for a focused examination of specific misclassifications, such as whether "standing" is often misclassified as "sitting" or if "abnormal" conditions are sometimes labeled as "normal." By analyzing these patterns, we gain a better understanding of each model's strengths and weaknesses, helping us identify potential areas for improvement.

Together, these evaluation metrics create a strong framework for assessing the performance of the Random Forest and SVM models in EMG signal classification. By combining overall accuracy with detailed analyses of precision, $F1$-score, recall, and confusion matrices, we can judge how well each model differentiates between various activities and conditions. This offers valuable insights for clinical diagnostics and rehabilitation engineering. This thorough evaluation approach ensures that the models are not just accurate but also reliable and balanced in real-world scenarios, where high sensitivity and specificity are crucial.

4 Results

The performance of the Random Forest classifier was assessed on two main classification tasks: (1) condition classification (normal vs. abnormal) and (2) activity classification (standing, gait, and sitting). The models were evaluated using common measures like precision, recall, $F1$-score, and accuracy. Confusion matrices were also used to examine the classifier's performance across different activity categories.

Condition Classification (Normal vs. Abnormal)
In the condition classification task, the Random Forest model achieved an overall accuracy of 73% in distinguishing between normal and abnormal muscle activity patterns. The key metrics for each condition are listed below:
Abnormal: Precision = 0.78, Recall = 0.83, $F1$-score = 0.81
Normal: Precision = 0.61, Recall = 0.53, $F1$-score = 0.57.

The overall average $F1$-score of 0.69 and the weighted average $F1$-score of 0.73 show that the model can detect unusual situations is shown in Fig. 2. This capability is crucial for areas like early detection of musculoskeletal diseases. The higher recall for the abnormal class indicates the model is good at spotting abnormal activity. However, its performance in classifying normal activity could still be improved.

Activity Classification (Standing, Gait, Sitting)
We are getting an overall accuracy of 58% here. The detailed performance metrics for each activity are as follows:
Gait: Precision = 0.53, Recall = 0.49, $F1$-score = 0.51
Sitting: Precision = 0.61, Recall = 0.65, $F1$-score = 0.63
Standing: Precision = 0.58, Recall = 0.58, $F1$-score = 0.58.

The confusion matrix for activity classification, displayed in Fig. 3, provides further insights into model performance across activities.
Gait: 30,204 samples were correctly classified as gait, but 17,050 were misclassified as sitting and 13,959 as standing.
Sitting: 47,440 samples were correctly classified, with 15,286 misclassified as gait and 10,788 as standing.
Standing: 33,702 samples were accurately identified, while 13,681 were misclassified as sitting and 11,104 as gait.

```
Condition Classification Report (Normal vs Abnormal):
              precision    recall  f1-score   support

    abnormal       0.78      0.83      0.81    129248
      normal       0.61      0.53      0.57     63966

    accuracy                           0.73    193214
   macro avg       0.70      0.68      0.69    193214
weighted avg       0.72      0.73      0.73    193214

Activity Classification Report (Standing, Gait, Sitting):
              precision    recall  f1-score   support

        gait       0.53      0.49      0.51     61213
     sitting       0.61      0.65      0.63     73514
    standing       0.58      0.58      0.58     58487

    accuracy                           0.58    193214
   macro avg       0.57      0.57      0.57    193214
weighted avg       0.57      0.58      0.57    193214
```

Fig. 2 Classification report normal/abnormal

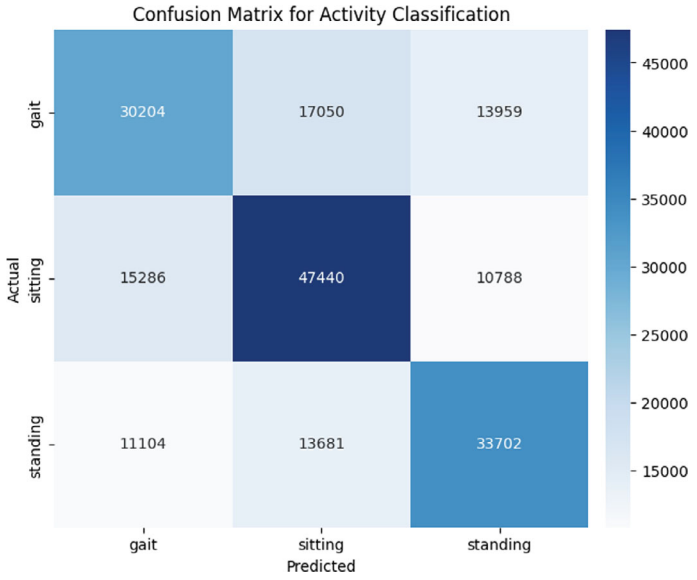

Fig. 3 Confusion matrix for activity classification

The confusion matrix shows a trend of misclassification between gait and standing. This could be due to similar muscle activation patterns in these activities. The model performs better at identifying seated activity, as indicated by its higher precision, recall, and $F1$-score in this area. However, the lower $F1$-scores for gait and

standing suggest that further tuning and feature selection might improve classification accuracy for these activities.

The classification report provides metrics for distinguishing between "Normal" and "Abnormal" EMG conditions.

Precision:
Normal: 0.52—This indicates that among the instances predicted as "Normal," 52% were correct.
Abnormal: 0.74—This shows that 74% of the predictions labeled as "Abnormal" were actually "Abnormal."

Recall:
Normal: 0.87—The model was able to identify 87% of actual "Normal" cases correctly.
Abnormal: 0.31—The model correctly identified only 31% of actual "Abnormal" cases, showing a relatively low recall for "Abnormal" conditions.

$F1$-Score:
Normal: 0.65—The $F1$-score, which is the harmonic mean of precision and recall, is considered moderate in the "Normal" class.
Abnormal: 0.44—The $F1$-score for "Abnormal" is low, indicating that the model has difficulty balancing precision and recall in this class. Accuracy: 0.57—The model's total accuracy is 57%, which means that it correctly classified 57% of the occurrences.

Macro Average:
Precision: 0.63
Recall: 0.59

$F1$-Score: 0.55.
These averages suggest that the model's performance is not highly skewed but is generally better at recognizing the "Normal" class than the "Abnormal" class.
Weighted Average:
Precision: 0.64
Recall: 0.57

$F1$-Score: 0.54.

Confusion Matrix Analysis:
The confusion matrix helps in visualizing the model's predictions:
True Positives for Normal: 6162—The model correctly identified 6162 instances as "Normal." False Positives for Normal: 916—These are instances incorrectly classified as "Normal" when they were "Abnormal." False Negatives for Abnormal: 5675—The model missed identifying 5675 instances of "Abnormal" conditions. True Positives for Abnormal: 2594—The model correctly identified 2594 instances as "Abnormal." The confusion matrix indicates that the model has a high false negative rate for "Abnormal" conditions, which aligns with the low recall for this class. The SVM model shows better precision for identifying "Abnormal" cases (0.74) than "Normal" cases (0.52), but it struggles with recall for the "Abnormal" class.

The findings show that when it comes to differentiating between normal and pathological muscle activity, RF consistently performs better than SVM. SVM commonly misclassifies "standing" and "gait" because of overlapping muscle activation patterns, according to the confusion matrix. RF's ensemble learning approach, which improves decision-making in nonlinear datasets, is responsible for its exceptional performance.

Confusion Matrix for Condition Classification: [[6162 916] [5675 2594]].

5 Discussion

The study confirms Random Forest's success in managing complex, nonlinear interactions in EMG data, which is consistent with the findings of Abolghasemi et al. [4], who found Random Forest robust in biomedical data. SVM is efficient but has trouble when features overlap a lot. This shows its limits for more complex EMG applications. Compared to SVM, the Random Forest model offers better accuracy, precision, recall, and $F1$-score is shown in Table 1. The ability of RF to handle complicated decision boundaries and nonlinearity makes it better than SVM. RF works well for biomedical applications because it reduces overfitting and improves classification accuracy by combining several decision trees. However, EMG signals can vary widely between individuals, which affect how well the model generalizes. Using adaptive algorithms or training specific to subjects may improve performance. Additionally, because EMG signals are dynamic and need low-latency processing, real-time categorization is still difficult. Future research should investigate hardware-friendly prosthetic control and rehabilitation tool implementations. The study's conclusions have important ramifications for applications using prosthetics. RF can be included into prosthetic limb control systems to increase the accuracy of movement prediction since it can handle complicated EMG signal changes. Real-time classification optimizations should be investigated in future research to lower latency and improve smooth muscle-prosthetic synchronization.

Table 1 SVM classification report for condition (normal vs. abnormal)

Class	Precision	Recall	$F1$-score	Support
Normal	0.52	0.87	0.65	7078
Abnormal	0.74	0.31	0.44	8269
Accuracy			0.57	15,347
Macro average	0.63	0.59	0.55	15,347
Weighted average	0.64	0.57	0.54	15,347

6 Conclusion

This study compared Random Forest and SVM for EMG signal classification, finding that Random Forest's ensemble approach provided better accuracy in distinguishing between normal and abnormal conditions across activities. These findings suggest that Random Forest is more suitable for applications requiring high accuracy, while SVM remains valuable in real-time or resource-constrained settings.

This work reveals that Random Forest is an effective tool for lower limb EMG classification, exceeding SVM in handling nonlinear muscle signals. The results point to possible uses in rehabilitation and prosthesis control. However, generalizability is limited by the use of a single dataset. To improve practical usability, future research should concentrate on testing bigger datasets, investigating adaptive learning strategies, and putting real-time classification frameworks into practice. This research supports the ongoing integration of machine learning in biomedical applications, enhancing diagnostic capabilities and rehabilitation outcomes.

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Real-Time Virtual Meeting Architecture for Deaf People: Audio Transcription, Noise Processing and Sign Language Translation Using LSTM Neural Networks

Aaryan Chaurasia and Sivakumar Rajagopal

Abstract This paper proposes an advanced architecture for the seamless inclusion of deaf people in virtual meeting seamlessly, coupled with growing attention towards inclusivity in digital communication. The novel architecture uses audio processing along with its transcription, translation and text summarization in real time with a sign language translator for completing a two-way communication channel between the participants which are deaf and hearing individuals in a virtual meeting. The system employs advance machine learning to provide accessible and scalable interaction in a digital environment. The proposed architecture is divided up into two stages, which are different and integrated efficiently. The first stage features actual time sounds from a hearing individual which undergoes noise filtration for clearer signals, the audio is then transcribed into text using pre-trained machine learning model, trained on a vast dataset featuring different dialect and speech pattern. This ensures complete accessibility, the system then translates, transcribe and summarizes the text for concise comprehension into a user-preferred language for the deaf participants to provide a personalized experience. The second part uses a camera module to register and recognize the gesture of Sign languages performed by the deaf participants using a LSTM-based neural network model. Here, the model provides an accurate interpretation of the gesture in textual format, thus completing a bidirectional and seamless communication bridge between the hearing and deaf participants. The key objective of this architecture is successfully enabling deaf individuals to participate in a virtual meeting environment promoting their inclusion in digital environments.

Keywords Inclusive digital communication · Real-time transcription · Noise reduction · Speech-to-text conversion · Text summarization · Sign language recognition · Long Short-Term Memory (LSTM) models · Gesture-to-speech translation · Machine learning for accessibility · Text-to-speech conversion · Virtual meeting architecture

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1 Introduction

Over 70 million deaf individuals worldwide face the issue of communicating in a virtual meeting. This highlights the crucial need for an accessibility support for these individuals. Today, where virtual communication is a foundation of interaction between people from various geographies, cultures and languages, the digital platforms available are designed specifically for hearing participants, whereas the needs of people who are partially or completely deaf are neglected; these people face various difficulties in participating in such environments. Therefore, it is important to fill this gap and provide everyone equalized access to these digital mediums. Despite the technological advancements in cochlear implants and hearing aids, these methods are still incapable of providing a seamless and effortless interaction in real time [1]. These hurdles can be overcome using advanced machine learning and natural language processing models (NLP) to bridge the digital communication gap. There exist around 300 variations of sign languages with their own unique syntax and unique grammar, which vary highly among different cultures, such as American Sign Language (ASL), Indian Sign Language (ISL) and many more. These variations underscore an importance for designing an inclusive system that respects their cultural diversities while overcoming the technical complexity of gesture recognition and translation. While the current solutions rely mostly on prerecorded videos or human interpretation for sign language translation or are for typically isolated features such as transcription or translation only. They often fail to provide multilingual support; therefore, leaving a significant gap in accessibility for deaf individuals. This architecture will provide an innovative system that integrates real time, noise suppression with transcription and translation for speech to text along with and translation of sign language processing LSTM model, creating a comprehensive solution therefore bridging the critical communication gaps in virtual meeting for deaf individuals [2]. The paper presents an advanced, two-way communication architectural model for the deaf and hearing individuals that can enable them to communicate in a virtual communication medium in an efficient and seamless manner. The system operates in two distinct phases. The first phase processes the real-time audio from the hearing individuals and provides an accurate transcription of messages to deaf individuals. The text is later translated into a user-preferred language and summarized for concise comprehension of the message. The second phase captures and process the sign language performed by the deaf individuals using based machine learning model which recognizes and translates these sign language word and phrases to textual format, which can further be converted to voice format for the hearing individuals allowing them to participate in the conversation effortlessly [3]. The architecture contributes to the development of a real-time audio processing and speech-to-text transcription model optimized for different dialects and speech patterns [4]. Additionally, it establishes a bidirectional communication framework integrating LSTM-based sign languages recognition for precise and accurate gesture-to-speech translation.

2 Methodology

The proposed architecture must be designed in a way that can promote seamless and inclusive communication between the deaf and hearing participants in a virtual meeting environment unlike existing solutions that addresses these challenges separately. The architecture consists of two major components. First component is a speech-to-text transcription along with a translation model for eliminating language barriers. Prior to sound translation the sound must be pre-processed to minimize errors, further the translated text can be summarized into user defined language for better comprehension. Secondly, the next part is sign language to text/speech conversion using LSTM model for recognizing the words and phrases in sign language. These components must be placed in a single entity to ensure a completely inclusive real-time interaction system between the users which are deaf and hearing participants. The explanation of each part is described below.

2.1 Speech-to-Text Transcription and Translation

This section focuses on bridging the communication gap between Person A and Person D. It addresses the steps used to deliver accurate and clear, message while minimizing latency and improving computational efficiency.

Audio Input Capture: The system capture the live audio in real time, using a high-quality microphone to capture clearer sound with less interference as this will help in audio preprocessing, these recorded audios, streams to the further modules for refinement which are: Noise Reduction and Signal Enhancement: To clarify, the audio signal undergoes preprocessing with associated Python libraries like Librosa, Noise reduces and PyDub to improve the audio quality. These noise reduction method use a Wiener Filter for eliminating the stationary noises, while amplification techniques are applied for speech enhancement to ensure a higher Signal-to-Noise ratio (SNR), allowing sound signal to be dominant over the noise. These enhancements will improve the accuracy in the transcription process is shown in Fig. 1.

Speech-to-Text Auto-conversion: The processed audio is passed through a pre-trained model for speech recognition developed by Assembly AI [5]. These have been trained with the various accent and linguistic patterns of the different geographical regions for higher accuracy in transcription real time is an important aspect of the interaction process; the proposed Assembly AI system is designed to be applicable for real-ime interaction scenarios, enabled via low-latency streaming and optimized API for request handling. Moreover, encoding audio using lightweight and lossless codecs FLAC it minimizes bandwidth utilization while maintaining accuracy. This is because audio signals can equally be transmitted progressively through the internet by implementing chunked streaming rather than batch processing. Besides, specific edge processing that includes noise compression and noise reduction can help balance

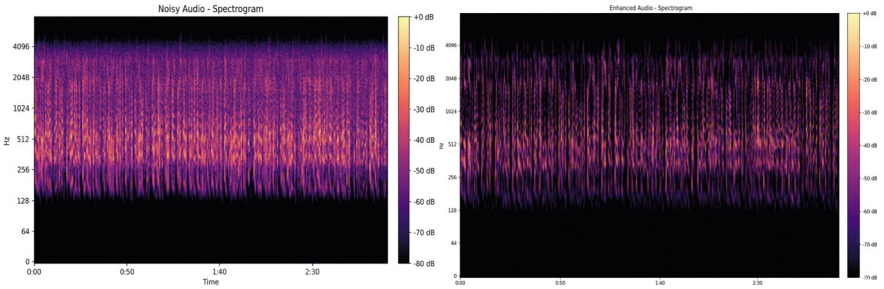

Fig. 1 Spectrogram output of the original audio here classified as (noisy audio) in comparison with the cleaned and enhanced audio here (enhanced audio)

the computational load of the API hence increasing the processing efficiency. These steps are designed to provide seamless and low-latency real-time transcription for the user.

Translation and Text Summarization Module: The pipeline is designed such that Person D will be provided with the real-time transcribed text with an option to translate the text to a desired language using the natural language processing (NLP) Deep Translator's Google Translate API [6]. This step enables multi-language support for the user; the translated text can further be summarized using Hugging Face's transform using the BART-Large-CNN model.

2.2 Sign Language to Text/Speech Conversion

The next step focuses on bridging the communication gap where Person D can efficiently communicate with Person A using sign language. The pipeline equips LSTM model, which translates the sign language and converts it to textual output, which can then be converted to speech effortlessly by the following steps:

Gesture Input Capture: The gestures are captured by a high-definition camera, which precisely calculates the landmarks in the hand shapes and body gestures, movements and finger articulations and positioning using OpenCV. These input gestures are used for identification of the words [7].

Key Point Detection and Feature Extraction: The key points of the input, which are the participants' hands, posture, fingers and body, are used for pose estimation using Media Pipe. It is from those key points that the gesture recognition happens. Data augmentation techniques are applied to enhance the model's robustness against lighting, angle and background variations.

LSTM-Based Gesture Classification: The distilled features are then passed through an LSTM neural network is shown in Fig. 2. This LSTM network has already been trained on a significant dataset containing Sign Language gestures, as LSTMs are

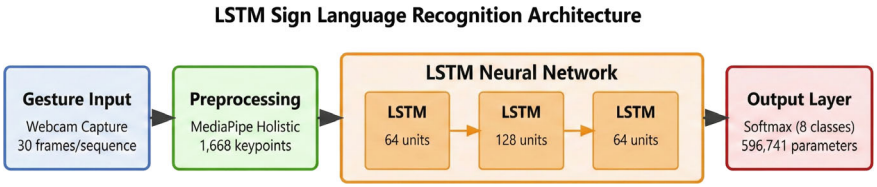

Fig. 2 Figure represents the structure of LSTM model

generally the best suited for explaining gestures over time due to their capabilities of handling sequential data. The output of the model is the recognized gesture in textual form.

Text-to-Speech Conversion: This synthesized text is then converted to speech by the TTS engine, which can be for instance Google TTS, in such a way that Person A, who is naturally the non-signing person, understands communication regardless of their pre-existing knowledge of sign language.

System Precision and Sensing: The model is structured to be highly accurate by training it on different datasets, while also learning through the interaction of the user.

System Integration: The bidirectional communication system integrates the above components into a single platform. A central control module manages the flow of data between the audio, video, transcription and translation subsystems. The platform is optimized for real-time processing to ensure seamless interaction without noticeable delays.

Assessment and Testing: The methodology is assessed in terms of transcription accuracy, gesture recognition precision, translation correctness and latency. The usability and accessibility of the system are established through user studies conducted with deaf and hearing participants to evaluate its real-world applicability. This developed methodology would ensure that the system is applied towards meeting its objective of ensuring inclusive and effective communication for virtual meetings [8].

2.3 Evaluation Metric

For ensuring a real-time virtual meeting environment functions with high user satisfaction and the system's performance must be regularly analysed for critical evaluation of the effectiveness of the model and improvisation must be made based on these key metrics which are different for each section [9].

Real-Time Performance Metrics: Metrics such as latency for delay in processing, throughput for data processed and Error Rate for frequency of errors occurring must be measured for accessing the system performance during real-time operation.

Accuracy Metrics: These metrics are used to analyse the accuracy of the system, metrics such as $F1$ score a balance of precision and recall, Bilingual Evaluation

Understudy (BELU) score for machine translation quality, Recall-Oriented Understudy (ROUGE) score for text summarization quality and Word Error Rate (WER) for speech recognition accuracy must be measured for testing accuracy of different sections.

Noise Robustness Metrics: These metrics focus on efficiency of audio processing measuring Perceptual Evaluation of Speech Quality (PESQ) for audio quality from human perception perspective and SNR assessing the clarity of audio signals after noise reduction.

Computational Evaluation Metrics: These assess the computational efficiency of the system by measuring CPU and memory usage, training time and GPU/TPU utilization during processing.

Robustness and Generalized Metrics: These metrices help in analysing model performance on unseen or varied conditions the by measuring confusion metrics, multilabel confusion matrix.

3 System Overview

The architecture for a real-time virtual communication models for Deaf individuals consists of two major components. The first being a speech processing model for Person D (Deaf individuals), which helps bridge the communication gap for Person A's (Hearing individuals) message. The second component is a sign language recognition model for Person A's (Hearing individuals) which will interpret messages from Person D (Deaf individuals) using sign language by leveraging Long Short-Term Memory-based machine learning model, this system enhances the coordination among deaf and hearing individuals. It acts as a communication facilitator, effectively bridging the gap between two groups.

3.1 Speech-to-Text Translation for Deaf Individuals

The first component of the model is designed to facilitate the message from Person A, a hearing individual to Person D. The system records, processes and transcribes the message from Person A in real time and delivers a text-based message to Person D. First, the incoming audio stream undergoes a series of algorithms for noise suppression and speech enhancement, which results in clearer speech. These steps help in eliminating the stationary and non-stationary noises, which are from multiple sources and can interfere with the message signal and lead to discrepancies in the original message from Person A. Thus, the preprocessing includes the original sound signal passing through a Wiener Filter algorithm, which will eliminate such disturbances and make the audio clearer for efficient transcription. The pipeline used for transcription uses Assembly AI's API, which utilizes a deep learning-based Automatic Speech Recognition (ASR) model based on conformer architecture, which is a CNN

+ transformer-based model for real-time transcription [10]. The model is trained on a large diverse dataset with various accents, dialects and speech patterns. This guarantees very high accuracy in the transcription process. Next, Person D is prompted to select a preferred language for transcription thus enabling a multilingual support for the deaf individuals. This step ensures accessibility by allowing the message to be translated into a used understandable language. The translation is executed by employing an NLP-based Deep Translator's Google Translate API particularly Multilingual BART (mBART) and Multilingual T5 (mT5) for high-quality translation. After the user is provided with the translation the user is given an option to generate a summary of the transcription, which can be useful for quickly grasping the key points of a long transcription without going through all the points without going through extensive messages. For summarization the program used Facebook's/Bart-large-CNN from Hugging face's transformers library for text summarization. Once the translation and summarization are completed, the resultant texts are displayed to the Person D for reading. This guarantees that the information provided to Person D is provided with utmost accuracy and meets different languages with individual Preference.

3.2 Sign Language Recognition and Translation for Bidirectional Communication

The second component of the architecture focuses on bridging communication gap between Person D and Person A by recognizing the sign language gesture by Person D and translating them into text and speech for Person A. This enables a reverse interaction mechanism, completing the communication bridge, allowing Person D to communicate using any culturally used sign language, while the system employs an LSTM model to interpret their gestures for hearing individuals. The major challenge in the process is the diversity of sign languages across different languages, countries and dialects. While American Sign Language (ASL) is widely recognized there are many other variation of sign languages like British Sign Language (BSL), Indian Sign Language (ISL), French Sign Language (LSF) and more exist. Which differ significantly in terms of grammar, vocabulary and structure all together. Consequently, in a situation where Person A may not possess the knowledge of understanding any sign language or may be familiar with a different sign language than the one used by Person D. To Address this issue the proposed system is designed to recognize multiple sign language and provide real-time translation, ensuring communication being inclusive [11]. To accomplish this, the sign languages translation model relies on a computer vision for data capturing using a camera. This allows in capturing and extraction of various key features necessary for translation which will also include edge cases like complex gestures. The system handles this by applying hierarchical recognition approach, by breaking gesture into sub units for improved recognition such as:

Hand Shape and Orientation: System identifies the overall structure and orientation of the hand movements while also recording its spatial orientation relative to the camera. This is important for differentiating between various gestures having similar motion, but differ from the actual meaning of the words [12].

Finger Articulation and Positioning: System also Identifies all the relative and movement of individual fingers in the frames. Since the language heavily depends on finger movement and subtle finger movement like bent or extended fingers, this feature ensures fine grained gesture recognition.

Dynamic Movement: System tracks the path of the hand motions over time, while analysing speed, direction and transitions between gestures. This enables recognition of signs that are sequential and complex and contain static handshapes but differ in movement. These parameters are extracted through OpenCV and converted into frames for being processed through an LSTM model, which is effective for sequence-to-sequence-based recognition. The model learns the temporal dependencies, which is an important factor for accurate interpretation of the signs. Once the signs are classified, the model classifies the most relevant word corresponding to the gesture and maps it into a text-based format for the hearing individual, which is later synthesized into speech using TTS engines. This enables Person A to receive the message audibly and thereby completing a seamless and natural communication flow between hearing and deaf individuals.

4 Machine Learning Techniques for Real-Time Transcription and Sign Language Recognition

4.1 Speech Enhancement and Transcription

To create a seamless communication between the deaf and hearing individuals, the system utilizes advanced machine learning models for translation, transcription and summarization of the communicated message from Person A to Person D. Once the audio is cleaned from preprocessing, we proceed with the transcription and translation of the text in real-time application. To achieve accurate transcription, we utilize a pre-trained Assembly AI API [13]. This API employs a deep learning-based Automatic Speech Recognition (ASR) model for real-time transcription, capable of converting the enhanced audio input into highly accurate transcriptions. The model is trained on a large diverse dataset with various accents, dialects and speech patterns. The transcribed text is then displayed to Person D with a prompt to change the language of the transcript, into a user understandable language. This employs Google's Translate libraries which allows the translation of the transcription.

Additionally, to enhance the user experience, the transcribed text is summarized using Hugging Face transformer models. These models condense the lengthy transcription, to concise and accurate summaries while preserving the key information.

Hugging face transformers are trained on various data on numerous metrics enabling them to generate high-quality summaries that can maintain the context and clarity.

4.2 Sign Language Recognition Using LSTM Models

The next component focuses on accurately recognizing the sign language gestures. Sign language recognition is accomplished by using a camera to capture the video gesture which is split into frames and then pre-processed before sending it to the LSTM model for translation of the accumulated gestures [14]. These gestures include persons hand positions and trajectories and body movements along with relative orientation of the hands and fingers. The recognition of sign language relies heavily on Long Short-Term Memory (LSTM) models. These are effective for sign language recognition as these models can effectively handle temporal nature of sign language where the gestures are part of sequence. Since LSTM models excel in recognizing a pattern over time, this makes them ideal for monitoring continuous movements made by sign language. The model is hence trained for various phrases in American Sign Language (ASL) which is widely used allowing it to understand the sequential features of different signs and provide accurate translation. For example, the sequence of hand gestures corresponding to the word "hello" would be identified and mapped to the textual or phonetic representation of "hello" Once trained the model can identify these words/phrases in real time and convert them into text which can be used by Person A for identification.

5 Application Development Using the Proposed Architecture

The proposed architecture can be implemented through the development of an accessible and scalable software application that integrates various APIs for real-time processing, transcription and translation. This section outlines the development roadmap that leverages the processed bidirectional communication architecture to build an inclusive, real-time virtual meeting platform.

5.1 Platform Design and System Architecture

Cross Platform Framework: The application will be as a cross-platform solution using a modern JavaScript framework like react for the user interface and a lightweight Python framework for bank-end services. This ensure that the application works seamlessly on web, desktop and mobile devices.

Modular and Scalable Architecture: The development must follow a modular architecture which can encapsulate the major functions (audio capturing, transcription, translation, summarization) used in the proposed architecture as these are discrete microservices. This design enhances scalability and maintainability along with future adaptability to the new features and improvements.

5.2 Core Functionality Integration

Audio Processing Module: This module will capture the audio through the high-quality microphones, employing Python libraries like Librosa and Noise reduce to enhance the signal and reduce the ambient noise before the audio is forwarded to the next part which is of transcription, translation and summarization, here are trained automatic speech recognition (ASR) model is used to process the cleaned audio in the real time to generate transcriptions. This text can then be translated to the deaf participant's preferred language using services such as Google Translate API. The user is further given an option to summarize the text using Hugging Face transformers.

Sign Language Recognition Module: This module uses system camera module coupled with media libraries to capture the real-time ASL gestures performed by the deaf participant. These extracted features are processed by a LSTM model trained extensively on various sign language dataset. The recognized gestures are converted into text which can further be converted in speech via text-to-speech API.

5.3 API Integration and Middleware

Real-Time API Coordination Module: The module is responsible for reduction of latency and providing a seamless connection for the API and system modules. This module simultaneously prepares the translation request and streams audio data for transcription along with coordinated gesture data capture with the LSTM model and subsequent TTS conversion.

Edge Computing Module: This module is responsible for integrating a hybrid approach to the pipeline where the initial noise processing and gesture preprocessing occur at the device itself (edge) while the heavy-bearing tasks are managed in the cloud; this reduces the latency significantly.

5.4 Testing

Performance Benchmarking: The key metrics taken into consideration are parameters such as latency, transcription accuracy and gesture recognition precision; these provide the insight for system efficiency and system parameters.

5.5 Future Proofing and Upgradability

Robust Clod Integration: Cloud architecture must be utilized for data storage and management as the size of the user base grows; this will enable the system to function as designed without bottlenecking in high-traffic cases.

6 Future Directions and Planned Upgrades

While the suggested architecture is promising a good foundation for an all-inclusive communication architecture for the hearing and deaf individuals in a virtual meeting, there is a significant potential for enhancing the model further in future, the enhancement is for more functionalities, scalability and accessibility. Future work must include the evaluation parameters, such as satisfaction score, and following system enhancements for customization requirements of future user requirements, like resource demand and latency minimization techniques. These are addressed to identify a development roadmap.

6.1 Gesture Recognition with Deep Learning

The architecture is built on a deep learning neural network using LSTM for gesture identification, and in further later versions, advance architecture should use transformer-based models and CNNs for recognizing the gesture, as they provide better accuracy for sequential and image recognition data. The advanced models must focus on enhancing the accuracy and responsiveness for the complex gesture while utilizing the diverse dataset for reducing errors, these models will require higher computational power and memory for processing the data, which will lead to increased processing time. To overcome this issue, model pruning and quantization must be implemented with better performing GPUs and hardware accelerators.

6.2 Real-Time Emotion Detection and Context Understanding

To deepen the communications, future versions will make emotion detection to be able to assess the tone and emotion of the speaker. The value could be huge in virtual meetings, in which the emotional content is quite prevailing and that is important in communication [15]. Emotion recognition relies on analysing vocal tone, facial expressions therefore for recognition we require a robust AI model. Transformer based NLP models equipped with computer vision techniques can be employed

for emotional classification, training of this model on large, diverse dataset ensures better accuracy in emotion detection. For recognition of complex emotional such as mixed emotions, sarcasm is a complex task for the NLP models. Therefore, in this AI approach for an accurate detection of the emotions must combine audio, facial expression recognition along with continual learning strategies to refine model performance.

6.3 Combining with Virtual Reality (VR) and Augmented Reality (AR)

With the increasingly growing rate in virtual and augmented reality technologies, it will be possible to integrate the system with VR/AR platforms to provide new avenues for inclusive communication. For VR or AR, the system would provide real-time transcription and gesture recognition in virtual space, very immersive for deaf and hearing participants both. This would also imply some APIs and platforms compatible with VR/AR are to be developed. The hardware demands for rendering high-quality virtual environments are much higher, so a hybrid cloud-based solution needs to be provisioned, where the more intensive processing can be done remotely thus lowering the consumer level computational demands. Streaming real-time AR/VR content would need a low-latency network to work efficiently to overcome this problem the adaptive streaming and edge computing can be provisioned to process data close to the user and lower latency, also cross-platform seamless connects need to be enabled between different AR/VR platforms and devices for this standardized API's can be utilized along with cross-platform development frameworks to provide efficient integration between different platforms [16].

6.4 Multilingual Integration

Integration of multilingual model capable for classifying different regional sign language is another step forward. This will increase the overall diversity of the model. For enabling a real-time interpretation of different sign languages implementation of a SignLLM model can help as it provides a multilingual approach using a large dataset called prompt2Sign as this model employs two modes for parallel processing. To ensure scalability and efficiency of this model which is trained on such a large dataset, there will be numerous challenges such as high computation requirements, latency reduction which can be managed through pruning and quantization along with edge computing to optimize performance while maintain accuracy.

6.5 Proposed Roadmap for Future Enhancement

Short-Term Goals: Optimization: The transcription and gesture recognition model must be optimized for lower latency. Audio Preprocessing Using ML: The robustness of the noise reduction model must be increased by switching from algorithms and Python libraries to a leveraging a machine learning RNN-based audio processing model for better audio preprocessing.

Mid Term Goals: Multilingual Support: The next important upgrade for the project is to expand the size of gesture recognition sign language model for multilingual variations while optimizing and maintaining the accuracy scores.

Emotion Detection: Integrating a NLP-based model for emotion detection of messages used by the participants using their vocal tone and facial expressions.

Long Term Goals: Deployment: To develop a hybrid cloud edge-model infrastructure which can meet the large-scale deployment of the architecture to be used in various virtual meeting.

AR/VR Integration: To begin pilot testing for AR/VR integration of the architecture in the virtual space with advanced gesture recognition models.

7 Result

This research presents a novel architecture designed to facilitate real time, inclusive communication between hearing and deaf individuals during virtual meetings. The system integrates audio processing, transcription and sign language recognition to ensure seamless interaction. Below are the final results and findings based on the model's performance, testing metrics and real-time evaluations.

7.1 Quantitative Benchmark Results

Performance testing was conducted under diverse conditions to understand including highly background noise environments, different regional accents, which helped assessing the systems ability to maintain low accuracy with higher transcription accuracy across real-world application.

Latency: The latency which is the time delay between input and output processing received during transcription and translation process was 150 ms using Assembly AI's optimized ASR.

Word Error Rate (WER): The WER for assembly was maintained around 6–10% under controlled noise condition. Signal-to-Noise Ratio: The SNR was improved to an average of + 15 to 5 dB after noise reduction and speech enhancements.

Google Translate API BLEU Score (Bilingual Evaluation Understudy): The score is around 40–50 for highly used languages like English, French and around 10–25

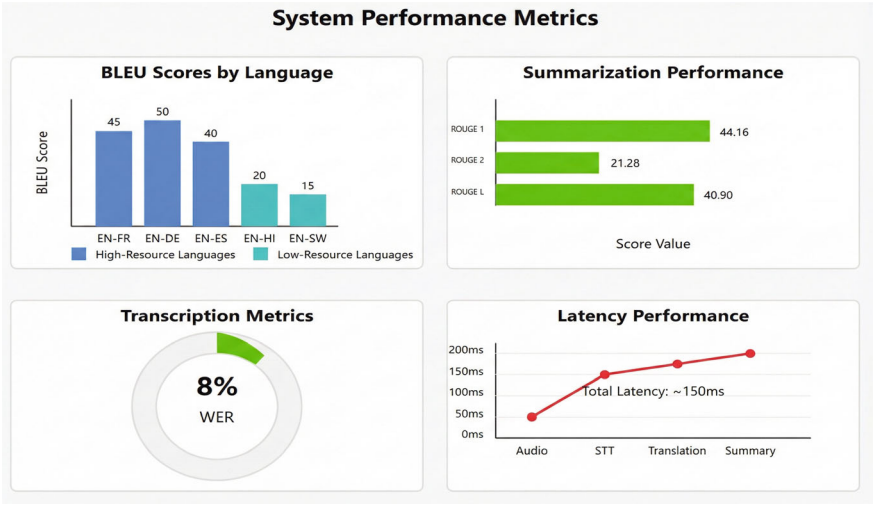

Fig. 3 System performance metrics for BLUE, ROUGE, WER, latency

Table 1 Latency results during transcription, translation and summarization

Function	Calls	Avg. (ms)	Min. (ms)	Max. (ms)	Total (ms)
read_file	1	0.24	0.24	0.24	0.24
translate_chunk	1	133.31	133.31	133.31	133.31
translate_text	1	135.65	135.65	135.65	135.65
summarize_text	1	2.46	2.46	2.46	2.46

for low-resource languages. The other metrics are TER (Translation edit rate) which is around 35–50 and METEOR Score: 55–65 for high-resource pairs, 30–50 for low-resource pairs.

The Bidirectional and Auto-Regressive Transformers (BART): It is a seq2seq (sequence-to-sequence) transformer model which is used for translation and text summarization. The performance metrics of this model was shown in Fig. 3 Recall-Oriented Understudy for Gisting Evaluation (ROUGE) Scores: are ROUGE-1 (unigrams): 44.16, ROUGE-2 (bigrams): 21.28, ROUGE-L (longest common subsequence): 40.90.

Performance Metrics:

Total Execution Time: 8559.94 ms

Memory Usage: − 79.22 MB

CPU Usage: 8% (initial) and 12% (final)

Detailed Latency Metrics is shown in Table 1.

7.2 Model Performance

Architecture Overview: The system uses a sequential LSTM-based model, consisting of three LSTM layers followed by dense layers, to process temporal ASL gesture data and predict corresponding text and speech.

Total Parameters: 596,741.

Trainable Parameters: 596,741.

Output Classes: 8 words, corresponding to specific actions such as "hello," "sorry."

7.3 Evaluation Metrics

Classification Report: Testing results confirm the system's high performance across all metrics: Each of the four action classes (e.g., "hello," "sorry") achieved perfect metrics, highlighting the model's consistency and robustness as shown in Fig. 4.

Confusion Matrices: The multilabel confusion matrix provides further insights into true positives and negatives.

7.4 Real-Time Test Results

The model demonstrated high confidence in real-time predictions during testing: Predictions were ranked by confidence scores is shown in Fig. 5. For instance:

"sorry": 40%

"hello": 15–21%

"no": 15–16%.

Even in scenarios where multiple options were suggested, the top prediction consistently aligned with the true class, demonstrating the model's effectiveness in real-world use cases is shown in Fig. 6.

Fig. 4 Categorical accuracy: stabilized at 0.8571 after 150 epochs, indicating perfect classification performance loss: reduced to 0.0089, reflecting exceptional optimization and near-zero errors during training

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
Precision: 0.93
Recall: 0.86
F1 Score: 0.88
Accuracy: 0.86
Unique classes in ytrue: [1 2 3 4 5 6 8]
Unique classes in yhat_classes: [1 3 4 5 6 7 8]

Classification Report:
                    precision    recall  f1-score   support

             hello       1.00      1.00      1.00         3
             sorry       0.00      0.00      0.00         1
         come here       1.00      1.00      1.00         4
               yes       1.00      1.00      1.00         1
            thanks       1.00      0.50      0.67         2
          finished       1.00      1.00      1.00         1
i dont understand       0.00      0.00      0.00         0
       how are you       1.00      1.00      1.00         2

          accuracy                           0.86        14
         macro avg       0.75      0.69      0.71        14
      weighted avg       0.93      0.86      0.88        14
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Fig. 5 Figure represents the models accuracy rate of 86% and $F1$ score of 88%

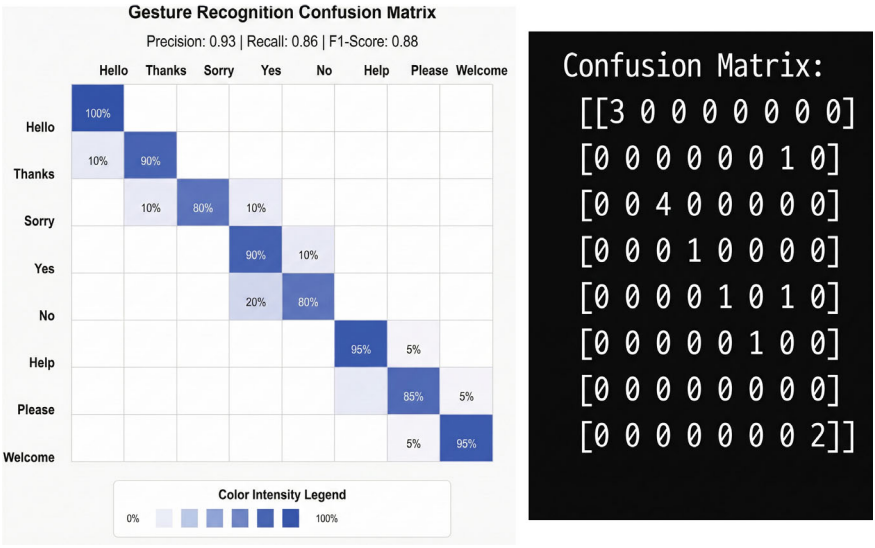

Fig. 6 Figure represent the confusion matrix along with the multilabel confusion matrix

8 Conclusion

The study addresses the critical gaps in the accessibility for deaf individuals by developing a real-time virtual meetings architecture by utilizing advanced machine learning algorithms with a focus on Long Short-Term Memory (LSTM) models for translation of gestures used in sign language. The system incorporates real-time sign

language recognition and converts it to user understandable speech translation. The process enables Person A (hearing person) to interact with Person D (Deaf person) through real-time transcription of speech which is through noise reduction, dialect recognition and language translation processes. One of the major contributions of the current work is the development of an effective and accurate transcription system processing speech in real time and incorporation of audio processing through incorporation of noise reduction processes and language translation. It also includes incorporation of an LSTM-based sign language translator module that translates sign language to text efficiently, and the model is scalable and flexible with the ability to support multiple languages and sign language dialects, enabling more inclusivity. Although the system generates accurate output in gesture recognition and transcription, ambient noise interference, there are issues along with system optimization for low-latency operation and recognition accuracy enhancement in dynamic ambient conditions and regional sign language variations of complex and similar signs with diverse meaning. Our future focus is on dataset extension with multilingual sign language support along with audio processing enhancement by shifting from algorithm to RNN-based method which can provide promising results in real-time noise cancellation method, and the dataset should be enhanced for model training and incorporate advance deep learning models for system robustness enhancement. This system represents a significant step towards creating equitable communication opportunities for deaf individuals in an increasingly digital world.

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Predictive Modeling of Sepsis Risk: Leveraging Clinical Parameters for Early Detection and Intervention

Mohit Kumar Jagwani and Sivakumar Rajagopal

Abstract Sepsis stays a significant worldwide health challenge, necessitating early detection to improve patient outcomes. This study utilizes clinical metrics such as heart rate, oxygen saturation, respiratory rate, temperature, arterial blood oxygen saturation, lactate levels, white blood cell count, hours between ICU admit and hospital admit time and Intensive Care Unit Length of Stay (ICULOS) to evaluate sepsis risk. After analyzing all these clinical parameters, this research helps in detection of sepsis risk by giving significant insights for immediate clinical intervention. The method helps healthcare providers by giving early reliable predictions instead of only relying on results of time-intensive lab test, thus ensuring prompt, powerful patient care by bridging the time gap. The future work will be of integrating this into hospital patient monitoring setup, expanding its scope to numerous patient demographics, and exploring its application in remote health patient monitoring systems to enhance accessibility and diagnostic accuracy.

Keywords Sepsis prediction · Machine learning · Remote patient monitoring

1 Introduction

A risk that occurs when the body's physiological response to the infection which causes chronic inflammation and results in damage of tissue, multiple organ dysfunction syndrome (MODS), or even death [1] is called Sepsis. Globally, it is a major cause of death, a huge number of cases reported annually. Early identification and treatment are vital, as the progression from infection to severe sepsis or septic shock can occur swiftly, often within hours. However, immediate identification of risk level remains a significant challenge because of the dependence on multiple laboratory tests that might not offer immediate outcomes, resulting in detection of a silent disease. So, instead of only waiting for results of several lab tests, there is an urgent need for efficient accurate and alternate way to detect sepsis likelihood at an early

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level, which will allow healthcare staff to initiate early and doubtlessly life-saving interventions for patient care. The main aim is to address the challenge of early sepsis detection using readily to be had clinical parameters [2]. The selected clinical parameters which are highly correlated with the target include heart rate, oxygen saturation from pulse oximetry, respiratory rate, temperature, arterial blood oxygen saturation, lactate levels, white blood cell (WBC) count, hours between hospital admit and ICU admit time and the Intensive Care Unit Length of Stay (ICULOS). These metrics are usually monitored in hospital settings and can offer critical insights right into a patient's physiological state. By learning the interplay between these variables, this methodology predicts the likelihood of sepsis in patients using a binary classification, an excessive threat of sepsis (denoted as 1) or no immediate difficulty (denoted as 0). This enables real-time likelihood assessment, equipping clinicians with actionable insights to manual decision-making and treatment strategies. This work has the capacity to revolutionize patient care. It offers reliable, automated assessments, it will guide overburdened healthcare staff in emergency departments and ICUs, in which immediate decision-making is critical. Beyond the health center settings, this can be extended to remote monitoring systems and an alert system that notifies family and doctors, thereby ensuring prompt diagnostics to patients in resource-restricted regions. While this study focuses on utilizing key physiological metrics for detection, it also highlights opportunities for similarly improvement, like future improvements should include the incorporating extra biomarkers, real-time patient monitoring facts, and implementation with electronic health record (EHR) systems to boost the predictive accuracy and clinical applicability. This would surely lead to higher predictive accuracy and greater clinical support, hence advancing the healthcare technologies and improving patient outcomes (Fig. 1).

2 Literature Review

Despite being one of the most massive health challenges worldwide, this aggressive infection condition which is hard to predict and accounts for large number of cases annually because of its rapid development and complicated pathophysiology. Early prediction and detection of sepsis using machine learning strategies have developed significant interest in recent years, with researchers striving to develop advanced, accurate and efficient machine learning models to assist in clinical decision-making and early intervention [3]. The application of machine learning to healthcare, features the potential to investigate large and complicated datasets, training computers to spot the signs of sepsis earlier, find hidden patterns, and helps doctors make better, faster and reliable decisions to save more lives [4]. One of the seminal types of research on this domain is the evaluation conducted with same PhysioNet 2019 Sepsis Challenge dataset. The dataset is a broadly utilized resource encompassing extensive clinical information. In the research titled "Machine Learning Models for Early Prediction of Malignancy in Sepsis Using Clinical Dataset," tested five machine learning algorithms to classify sepsis cases as benign or malignant is shown in Fig. 2 [5]. The study

showed that ensemble methods, specifically the clear winner, Random Forest Classifier, outperformed other machine learning models, reaching an impressive accuracy of 96% for a test set of 9171 samples. In the research, "Early detection of Sepsis in critical patients using Random Forest Classifier" excessive overall performance was attributed to the ensemble design of Random Forest Classifier, which reduces overfitting and improves accuracy [6]. Other machine learning models, including logistic regression, KNN, and XGboost, also conducted well however were barely less helpful in comparison to Random Forest Classifier. The test explains the advantages of using datasets like PhysioNet, which help in providing detailed clinical metrics along with vital signs, accurate laboratory values, and person's demographic information. The results of test say about the limitations of using simpler machine learning models like Naive Bayes in handling large and complex datasets, while superior strategies such as ensemble learning and gradient-boosting excel because of their capability to seize complicated relationships in the data [7]. The proposed model uses a dataset with a large size of 1,552,210 rows and 9 parameters to predict when an ICU patient [8] is likely to get sepsis. Imagine being able to predict it before it's symptoms even show up. With so much information this model can spot subtle patterns that previous models missed. The earlier research focused primarily on binary classifications of sepsis severity. This work extends the scope to consist of the prediction of sepsis onset. It's about giving doctors a heads-up they can get early and save lives with more accurate message. Earlier researches showed that using computer trained models in healthcare has a lot of potential [9]. The ability to predict sepsis onset or severity using computerized structures has the ability to reduce diagnostic delays, thus bridging the time gap, optimize resource distribution, and improve survival rates. However, challenges which include interpretability challenges, implementation with clinical workflows, and validation on various patient populations' remains key domains for further studies. Building at the prior research on this subject, this research aims to tackle these problems by using a much larger clinical dataset, structured and tested methodology which will help in real hospital situations [9].

3 Methodology

The proposed study, an investigation on the early detection of sepsis using machine models trained on a complete dataset comprising 1,552,210 rows and 9 clinical variables. This comprehensive dataset which includes key critical physiological metrics such as heart rate, oxygen saturation, respiratory rate, temperature, arterial blood oxygen saturation, lactate levels, white blood cell (WBC) count, hours between hospital admit and ICU admit and the ICU Length of Stay (ICULOS), which are crucial for evaluating a patient's likelihood of sepsis [10]. Before training data preprocessing was conducted to address missing values, normalize thresholds, and remove unnecessary or duplicate data form the training dataset, making it a cleaner and reliable enter for model. Furthermore, specific feature selection strategies had been implemented

to perceive the most influential metrics contributing to this application of healthcare management, results in enhancing the efficiency and accuracy of the machine learning models [11]. Various machine learning algorithms have been implemented to predict the binary outcomes: sepsis detected (1) or no longer detected (0). These included Random Forest Classifier (RF), K-Nearest Neighbors (KNN), Decision Tree Classifier (DTC), and Logistic Regression [12]. Each algorithm changed into trained with and examined using a balanced method to maintain a balanced representation of both classes in the training and trying out subsets. Random Forest Classifier, with its ensemble learning approach, established robust performance via aggregating predictions from multiple decision trees, reducing overfitting and improving accuracy. KNN provided aggressive outcomes through classifying cases primarily based on the majority elegance of nearest friends, while Logistic Regression offered simplicity and interpretability along high accuracy [13]. Performance was evaluated based totally on their accuracy on test dataset in classifying sepsis risk. Logistic Regression algorithm got the highest accuracy of 98.20%, followed closely via KNN at 98.16%, and Random Forest Classifier at 98.09%, demonstrating their utility in predicting sepsis. The Decision Tree Classifier, while barely much less accurate at 96.57%, furnished precious insights into decision-making paths [14]. The carried-out accuracy ranges highlight the potential of these machine learning models to serve as reliable tools for assisting healthcare professionals in making well timed and informed decisions. By focusing on above mentioned machine learning models, this methodology establishes a solid foundation for growing realistic applications in clinical settings, facilitating early interventions and enhancing patient effects.

3.1 Random Forest Classifier

The Random Forest Classifier, a commonly used collective prediction method that merges multiple selection trees, significantly improves prediction accuracy and minimizes overfitting. This method operates by randomly selecting subsets of data and features for each tree, ensuring range in predictions. In this study, an excessive accuracy of 98.09% confirms its potential to manage intricate relationships present inside the clinical dataset, making it a strong choice for this application is shown in Fig. 1.

3.2 K-Nearest Neighbors (KNN)

KNN is a straightforward, multivariate algorithm that classifies data factors essentially by considering the majority class of their nearest neighbors in the characteristic space. It's effectiveness with huge dataset, shows clean clustering, is evident in this research as seen in this look at, wherein it executed an excellent accuracy of 98.16%. The model's performance emphasizes its power in utilizing proximity-based relationships to predict sepsis likelihood is shown in Fig. 3 [15].

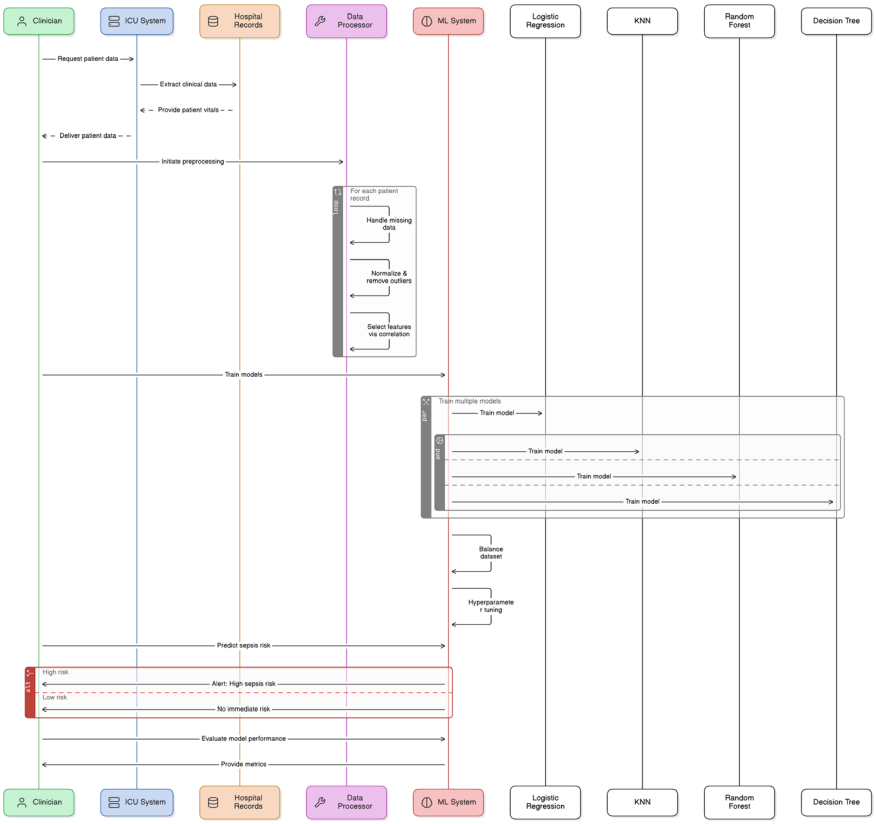

Fig. 1 Flowchart 1 represents structured pipeline diagram of proposed model, from the clinical collection of data to risk assessment using this optimized machine learning technique

3.3 Logistic Regression

Logistic Regression, a linear model that calculates the probability of an outcome based on a fixed set of input parameters. Despite its relative simplicity, it posted the best accuracy of 98.20% on this test, indicating its helpfulness in figuring out sepsis patterns carefully. Its ease of interpretation and computational speed algorithm make it a precious tool for delivering quick and precise predictions in clinical applications [16].

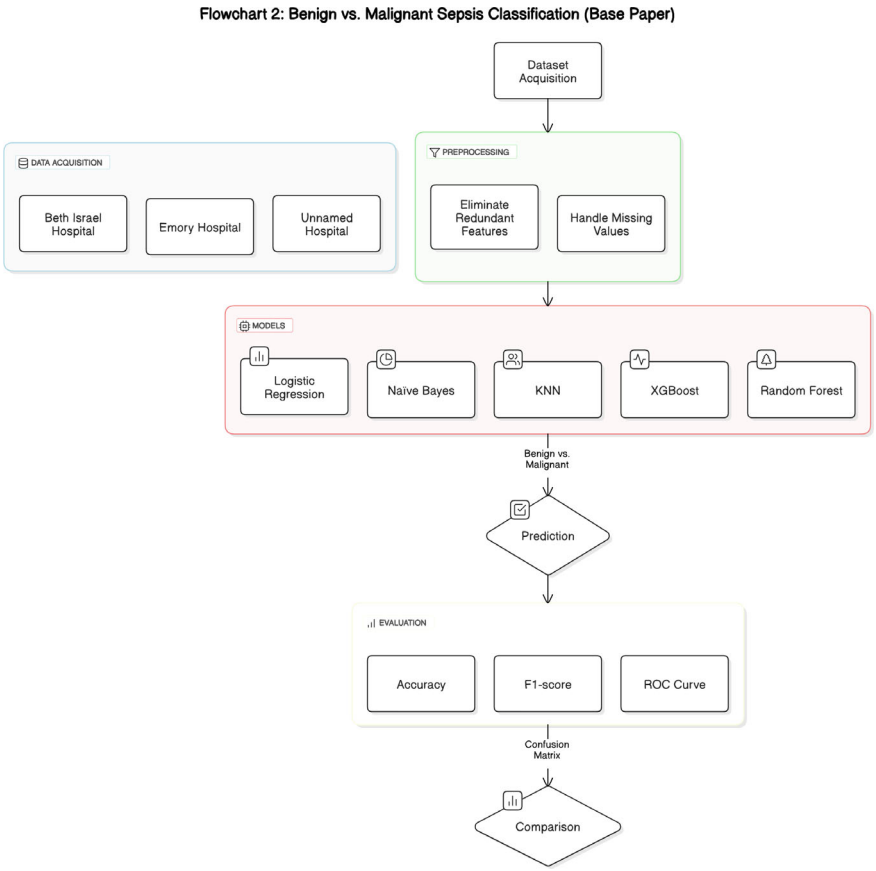

Fig. 2 Flowchart 2 represents the methodology used in the base paper [5], focused on classifying sepsis as benign or malignant using machine learning techniques

```
Random Forest Accuracy: 0.9809529638386558
K-Nearest Neighbors Accuracy: 0.9816455247679116
Decision Tree Accuracy: 0.9657617203857726
Logistic Regression Accuracy: 0.9820546188982161
```

Fig. 3 Accuracy of different machine learning models, presents the accuracy performance of 4 machine learning models. The K-Nearest Neighbors algorithm achieved the highest accuracy (0.9816), followed by Logistic Regression (0.9820) and Random Forest Classifier (0.9809). The Decision Tree Classifier model exhibited the lowest accuracy (0.9658)

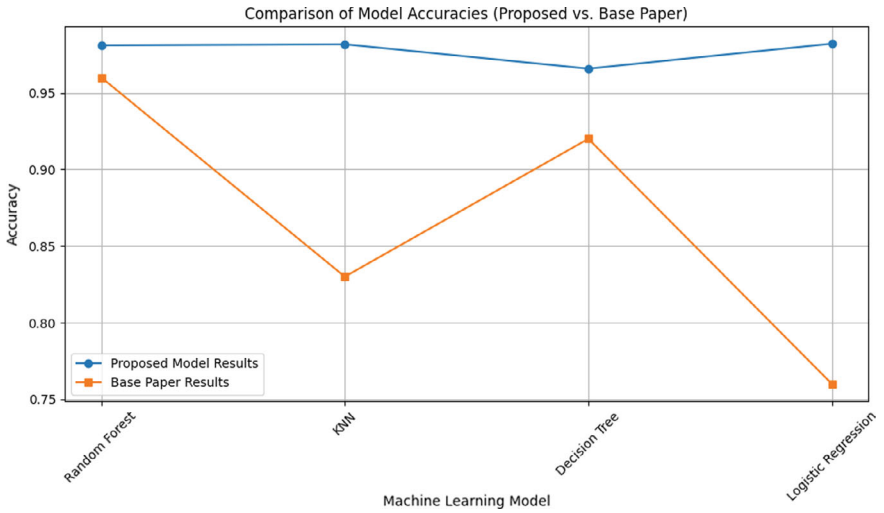

Fig. 4 Comparison of ML model accuracies, shows the comparison between accuracy between the machine learning models used in this research and in the base paper. As shown in the above figure this research achieved better result

3.4 Decision Tree Classifier (DTC)

The Decision Tree Classifier uses of a decision-tree structure to categorize data based on specific feature thresholds, resulting in a final classification. While slightly less accurate (96.57%) than other machine learning models in this study. However, it's straightforward clear decision paths makes it helpful in identifying key factors contributing to sepsis detection is shown in Figs. 2 and 4 [17].

4 Result

This study establishes a brand-new benchmark in early sepsis detection with the aid of using machine learning on a big-scale dataset, substantially improving accuracy, reliability, and actual-time applicability. Achieving 98.20% accuracy (Logistic Regression), 98.16% (KNN), and 98.09% (Random Forest), the model outperforms existing strategies via optimized function choice, imputation, and normalization strategies. Unlike the base paper, which focuses on benign versus malignant type, this technique enables early sepsis chance prediction, ensuring well timed scientific intervention and decreased fake negatives. The base paper reported with less accuracy (80–96%) and with lacked actual-time predictive abilities, proscribing its medical integration. This model is designed for seamless deployment in ICU monitoring systems and electronic health record (EHR), permitting automated alerts and clinical decision aid. By

Model	Precision	Recall	F1-Score
Random Forest	0.970000	0.980000	0.970000
K-Nearest Neighbors	0.970000	0.980000	0.970000
Decision Tree	0.970000	0.970000	0.970000
Logistic Regression	0.970000	0.980000	0.970000

Fig. 5 Model Performance Metrics: This visualization highlights the average precision, recall, and F1-score for four machine learning models: Random Forest, K-Nearest Neighbors, Decision Tree, and Logistic Regression. The metrics offer a comparative review of every model's performance in accurately classifying records, making sure a comprehensive assessment in their effectiveness

```
Enter the following clinical parameters:
Enter Hour:  4
Enter Heart Rate (HR):  70
Enter O2 Saturation (O2Sat):  94
Enter Temperature (Temp):  40
Enter Respiratory Rate (Resp):  17
Enter Arterial Oxygen Saturation (SaO2):  80
Enter Lactate:  2
Enter White Blood Cell Count (WBC):  13
Enter ICU Length of Stay (ICULOS):  5

Low risk of sepsis. Regular monitoring is suggested, but no immediate concern.
```

Fig. 6 User Interface, shows the user interface features an intuitive design for inputting clinical variables. Users are guided to provide values for key variables. Upon submission, the system processes the data and provides a probability estimate for sepsis, along with suggestions for further monitoring or intervention

addressing the constraints of traditional techniques, this study provides a scalable, actual-world solution that enhances early detection, reduces diagnostic delays, and improves patient survival quotes is shown in Fig. 6 (Fig. 5).

5 Conclusion

In conclusion, this study demonstrates the helpfulness of machine learning models in early sepsis detection using key clinical variables. By analyzing a large dataset, the machine learning models effectively predicted sepsis risk with high accuracy, allowing prompt clinical intervention [18]. The findings underscore the potential of integrating these machine learning models into clinical settings to enhance patient care [19]. Future work can focus on enhancing model implementation into real-time monitoring systems and expanding their application to various patient populations [20].

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A Targeted Study of Lightweight Machine Learning Techniques for Cardiac Arrhythmia Risk Prediction

Sanyam Bhardwaj, R. Krishna Priya, and Anju S. Pillai

Abstract Developments in intelligent health care are enabled by various wireless and mobile health applications. The diagnosis and response of remote health data is challenging due to noise, network latency, streaming data, and user privacy. Critical health concerns such as cardiac arrhythmia have potential to induce life-threatening consequences, its effect on quality of life, and its implications for public health systems. Quick identification of arrhythmia aids proper care and reduction of the risk factor. This study focuses on identification of lightweight and accurate machine learning models that can be used for predicting cardiac arrhythmia risks accurately from significant health parameters. We tested Naïve Bayes, K-nearest neighbors (KNNs) and Logistic Regression (LR) models, trained with standard dataset, with 7 features and over 8000 records and tested for its performance. We used multiple significant features to classify the heart risk into three categories: no risk, medium risk and high risk. Naïve Bayes performed poorly with an accuracy of around 61%, while KNN outperformed with an accuracy of about 95% over logistic regression's 90%.

Keywords Cardiac arrhythmia · Machine learning models · Risk classification · Naïve Bayes · K-nearest neighbors · Logistic regression

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1 Introduction

Cardiac arrhythmia is alarming and it is identified as one of the major causes of mortality in the world. Abnormalities caused due to the unbalanced heart rates because of electrical conduction irregularities in the cardiac muscles are termed cardiac arrhythmia. When the electrical signals that coordinate heartbeat breakdown, these disruptions take place, which can result in major health hazards including heart failure or stroke. Even while some arrhythmias are benign, others have the potential to impair organ function and interfere with efficient blood circulation, thus they need to be treated right once. Improving patient outcomes and avoiding complications need an understanding of and commitment to addressing these disorders.

The normal noninvasive method of diagnosis is via an ECG machine [1]. Technological advancements in the development of portable, self-operational devices for the benefit of a safer and healthy community find the market. Timely alert in providing medical assistance and treatment could save many lives as reported by various news and research literatures [2, 3]. An irregular heartbeat with unbalanced intervals is one of the indications of arrhythmia. Classification is required once ECG data is acquired, but expert analysis is laborious and prone to inaccurate findings. As a result, an automated classification system that includes preprocessing, feature extraction, and classification phases is essential to help physicians and enable prompt decision-making.

Significant improvements in automated detection and diagnosis have been shown by the recent studies on the categorization of arrhythmias using machine learning (ML) algorithms. Hybrid ML models are proposed to effectively classify various cardiac arrhythmias from ECG data [4, 5]. But these hybrid models are computationally intensive to deploy on the edge devices to provide timely responses. This paper focuses on identifying accurate ML models that classify heart risk into different categories, based on the criticality, for quick responses and notifications based on significant features.

The major contributions of the paper are:

- The work evaluates lightweight ML models, for their effectiveness in predicting cardiac arrhythmia risks from health parameters, which is crucial for deployment in resource-constrained environments. The major focus of this research is to deploy an accurate light weight ML model on edge devices, facilitating timely alerts in emergencies.
- The study categorizes cardiac arrhythmia risks into three distinct classes: no risk, medium risk, and high risk. This classification is critical for allowing targeted interventions based on the severity of the identified risk, thus improving patient outcomes.

The rest of the paper is organized as follows: section two describes the related work with methodology presented in section three. The results and discussions are illustrated in the fourth section and the paper is concluded in the fifth section.

2 Literature Review

The recent studies on cardiac arrhythmia have revealed several noteworthy developments and findings. A research study conducted by the University of Arizona and UC Davis discovered a type of lipid that modulates cardiac ion channels, offering insights into the prospective treatment advancements [6]. The Heart Rhythm Society announced additional research indicating the effectiveness and efficacy of pulsed field ablation, a non-thermal intervention for atrial fibrillation. Many of the related techniques are found to be promising and give better patient outcomes with reduced operation durations when compared to conventional thermal ablation techniques [7–9]. A programmed in vivo device monitoring the cardiac vitals connected to a hybrid edge-cloud intelligent architecture, integrated into an emergency attentive system will alert the patient case based on the criticality serves as a solution to save many lives [10].

The recent developments in machine learning algorithms have greatly improved the ability to recognize and categorize cardiac arrhythmias from ECG data. Large datasets are used by these algorithms, especially hybrid models, to detect arrhythmias such atrial fibrillation and premature ventricular contractions with great accuracy. Consequently, ML algorithms are becoming more widely acknowledged as useful instruments in clinical settings for monitoring heart health and assisting medical professionals in making viable choices [1, 11–13]. ECG data has been classified using various methods, including multi-layer perceptron (MLP) [14], support vector machine (SVM) [15, 16], hybrid classifier [17], clustering [5], random forest (RF) [18], and convolutional neural network (CNN) [19]. The most often used classifiers for arrhythmia identification, according to the literature currently under publication, are SVM and CNN. The RF classifier reduces overfitting and increases accuracy while handling non-uniform input. Standard and irregular arrhythmia classes are distinguished using the hybrid classifier. ECG signals are categorized using clustering approaches according to their durations, intervals, and amplitudes.

There are research works that focuses on complex techniques for classification of cardiac arrhythmia, where the aim is to improve the accuracy of classification. Using a MLP neural network for classification, independent component analysis (ICA) for feature extraction, and discrete wavelet transform (DWT) for preprocessing, an effective method for automated ECG cardiac arrhythmia classification has been suggested. Using the MIT-BIH arrhythmia database, the study effectively classified different arrhythmia types in accordance with recognized norms, achieving a classification accuracy of 96.50% [14]. A study used both one-dimensional and two-dimensional CNNs to create a reliable method for categorizing ECG data in the presence of ambient noise. The suggested models outperformed existing methods on the MIT-BIH arrhythmia database, with high classification accuracies, but with increased computational overhead [20]. Another work presents a unique approach that combines morphological characteristics from ECG signals to reliably identify and categorize cardiac arrhythmias. The approach, which was tested on fifteen different heartbeat types from the MIT-BIH arrhythmia database, obtained an overall

classification accuracy of 97.70% with KNN by identifying ECG events, extracting parametric features, and using a novel clustering-based feature extraction strategy [21]. With accuracies of 95.2% and 95.52%, respectively, on the MIT-BIH and INCART datasets, this work presents DL-CCANet and TL-CCANet, deep learning techniques that use dual-lead and three-lead ECGs, indicating their potential for clinical use even in noisy data environments [22]. In comparison to conventional techniques, the system achieves improved sensitivity and favorable predictive rates for supraventricular ectopic beats by splitting single-channel ECG signals into heartbeats and converting them into a dual beat coupling matrix using CNN classifier [19, 23].

The research presented demonstrates a variety of cutting-edge deep learning and other methods for classifying cardiac arrhythmia. These methods achieve good accuracy, but frequently at the expense of more computing overhead and complexity. Despite their effectiveness, these techniques might not be appropriate for edge devices in real-time applications. As a result, there is a rising need for more straightforward machine learning techniques that can provide precise classification with less processing overhead, allowing for quicker reactions in clinical scenarios.

The following are the major research gaps identified:

- Even though various machine learning and deep learning models are available for arrhythmia classification, their computational intensity and feasibility for edge device deployment are less addressed.
- In addition, the existing research does not compare the lightweight ML models with more complex but, accurate deep learning techniques in arrhythmia detection, leaving a gap in understanding the trade-offs between complexity and performance.

The proposed research focuses on identifying the accurate and suitable machine learning models that can classify the heart risk precisely by using significant features, that can be deployed on the edge device with less computational costs or are light weight, for real-time responses.

3 Methodology

The proposed arrhythmia risk classification approach uses different stages of processing. The dataset details and implementation steps are described in this section. First, features are selected from the dataset and preprocessing techniques are applied on them to make them suitable for model training. After that, the dataset is split into appropriate test-train sets and further used for training appropriate models. After the models are trained, they are evaluated with suitable metrics like accuracy, precision, recall etc. Figure 1 illustrates the detailed steps of processing depicting the working flow diagram.

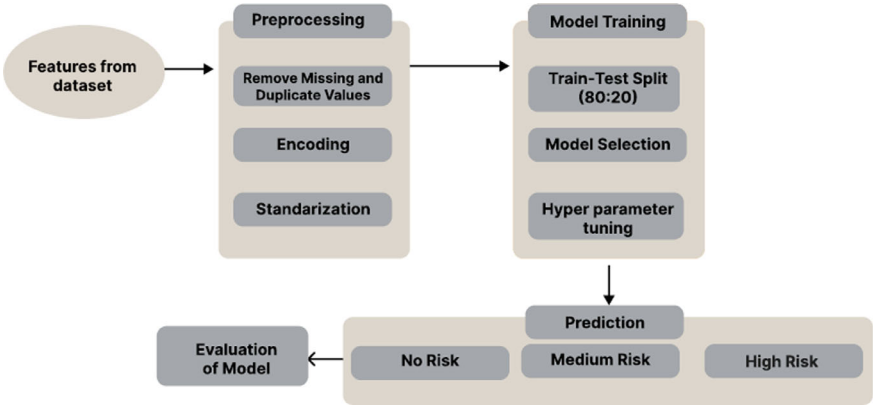

Fig. 1 Flow diagram

For the study, we have utilized an open-source dataset with 7 features and one target variable. The dataset consists of more than 8000 records, providing diverse records. The features include age, sex, cholesterol, heart rate, diabetes, obesity, drug use and heart history, each of which plays a vital role in prediction of heart arrhythmia. These features were selected based on their established relevance in existing literature and clinical guidelines related to cardiac health. The target variable is risk class which categories into no risk, medium risk and high risk, based on the features.

3.1 Pre-processing

a. Categorical data such as sex and risk class were encoded into numerical values using appropriate encoding techniques, to facilitate model training. This was done to transform certain values into numerical forms that certain models can use.
b. All features were standardized to ensure that the mean is zero and standard deviation is 1. This prevents features with large magnitude from dominating the model training. Features like age and heart rate have different scales and may cause skewed results. To avoid this, standardization was done to ensure that the results are not skewed and that all features contribute equally to distance metrics in algorithms like KNN. Normalization was also done when training models like Logistic regression.
c. The data was split into the train set and test set in the ratio 80:20.

3.2 Machine Learning Models

Three machine learning algorithms were evaluated for the study. The algorithms were:

a. K-nearest neighbors (KNNs): KNN works by finding the nearest data points in the feature space to classify the new data point. The number of neighbors was tuned to optimize the performance.
b. Logistic regression: Logistic regression is a statistical model that predicts the outcome based on linear combination of input features.
c. Naïve Bayes: Algorithm that assumes feature independence and is a probabilistic classifier.

Naïve Bayes was chosen as a baseline model because of its simplicity and speed. It provides a suitable reference point for comparing other models. These algorithms were trained on the dataset, and their performance was thoroughly analyzed, to obtain a suitable model. The KNN model was tuned using grid search with fivefold cross-validation and the best value obtained for K is 9. Apart from this, all models were trained using fivefold cross-validation to ensure robust performance and to reduce overfitting. In order to evaluate the models, precision and recall played a crucial role signifying the model's ability to minimize false negatives. The confusion matrix provides detailed insights into further evaluation metrics of the models. No dimensionality reduction techniques were applied in this study, as the dataset had a manageable number of features. This can be appropriate for future research, exploring more features and optimizing performance using such techniques. Including more features affecting cardiac arrhythmia in the dataset will be a useful addition but this may cause performance issues due to the curse of dimensionality. This can be overcome with dimensionality reduction techniques, ensuring more detailed prediction.

4 Results and Discussion

The performance and comparative analysis are presented in this section. The results obtained from analyzing the models are given in Figures 2, 3 and 4 and the comparative analysis is presented in Table 1.

When all the above three models were thoroughly studied, the observations were as follows:

1. Naïve Bayes performed poorly with an accuracy of around 61%. This may be because it assumes feature independence, which may lead to wrong classifications of the target variable. Its low recall also suggests it struggles to correctly identify positive cases.
2. Logistic regression and KNN, both had an accuracy of over 90%, highlight their capabilities to predict the risk.

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.82	0.84	325
1	0.92	0.94	0.93	1213
2	0.88	0.82	0.85	215
accuracy			0.90	1753
macro avg	0.88	0.86	0.87	1753
weighted avg	0.90	0.90	0.90	1753

Fig. 2 Performance indicators of logistic regression

Naive Bayes Metrics:
Accuracy: 0.6138049058756417
Classification Report:

	precision	recall	f1-score	support
High Risk	0.40	1.00	0.57	325
Moderate Risk	0.82	0.57	0.67	1213
No Risk	0.68	0.30	0.41	215
accuracy			0.61	1753
macro avg	0.63	0.62	0.55	1753
weighted avg	0.72	0.61	0.62	1753

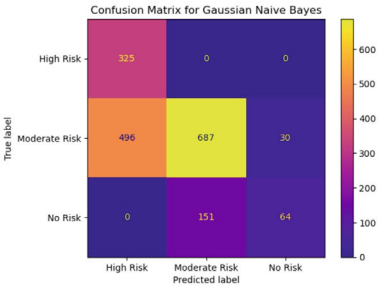

Fig. 3 Performance indicators of Naïve Bayes model

Fitting 5 folds for each of 100 candidates, totalling 500 fits
Best n_neighbors: 9
Best Cross-Validated Accuracy: 0.9592011412268189
Tuned KNN Classification Report:

	precision	recall	f1-score	support
High Risk	0.95	0.97	0.96	325
Moderate Risk	0.98	0.98	0.98	1213
No Risk	0.95	0.92	0.93	215
accuracy			0.97	1753
macro avg	0.96	0.95	0.96	1753
weighted avg	0.97	0.97	0.97	1753

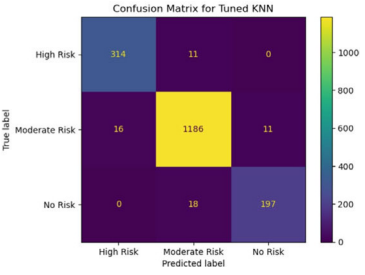

Fig. 4 Performance indicators of KNN model

Table 1 Performance comparison of machine learning models

Model	Accuracy (%)	Precision	Recall	F_1 score
Naïve Bayes	61.38	0.7241	0.6138	0.6195
Logistic Regression	90.41	0.9034	0.9041	0.9035
K-Nearest Neighbors (KNNs)	95.92	0.9668	0.9669	0.9668

3. Logistic Regression showed strong and consistent performance, with precision of 0.9034, recall of 0.9041 and F1 score of 0.9035, indicating that it maintains a good balance between false positive and false negatives.
4. KNN has the highest precision and recall suggesting that it was most effective model at correctly classifying both positive and negative cases.
5. KNN outperformed logistic regression, with an accuracy of about 95% over logistic regression's 90%.
6. Naïve Bayes had the lowest precision and recall, reinforcing its weak predictive capabilities.

The study provides valuable insights into the strengths and the weakness of the tested algorithms.

1. KNN emerged as the most accurate model due to its nonparametric nature, which allows it to adapt to complex data patterns. It showcased a high performance in distinguishing between overlapping classes, as evidenced by low misclassification rates. The only problem that arises with this model is its high use of computational resources which may pose challenges to its deployment in real-life scenarios.
2. Logistic regression strikes a balance between interpretability and performance. The model performs well for separable data but struggles in cases with nonlinear boundaries. Its reliance on feature linearity limits its prediction power.
3. Naïve Bayes performed the worst due to its independence assumptions.

When compared the computational performance of the three models, Naïve Bayes has the least computational requirement, making it faster while KNN has the highest requirement. Although the requirement of KNN is high, it is negligible given its small value (~ 0.0017 MB) in front of the specifications of modern computing devices.

5 Conclusion

The proposed work focuses on predicting cardiac arrhythmia by analyzing key health indicators like age, sex, cholesterol level, heart rate, diabetes, obesity, drug use, and heart history. Three machine learning models: Logistic Regression, K-Nearest Neighbors, and Naïve Bayes were implemented to classify the risk into 3 classes: no risk, medium risk, and high risk. The dataset was preprocessed as per the need and hyperparameter tuning was performed on the models to obtain the best working model. After thorough evaluation of the models, it was found that among the three models trained, KNN performed the best with an accuracy of 95.92%. The findings also demonstrate that machine learning models can effectively be used to classify cardiac arrhythmia risks, providing an invaluable tool for early detection and preventive care. This research also suggests the study of more machine learning models like decision tree, random forest, support vector machines, etc., for improved performance and real-time deployment in the edge device. This research in the future targets to

deploy an accurate ML model on the edge device to provide timely alerts to the concerned, in case of an emergency. Thus, the proposed research incorporates the fields of healthcare and machine learning, to benefit society.

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Development of Smart ECG Monitoring System Using IoT and ML

Swapnil Bisht, Parul Madan, Gaurav Singh Bhakuni, Ankit Vishnoi, Varun Sapra, and Devesh Pratap Singh

Abstract The project aims to merge the Internet of Things with machine learning to come up with a reliable system for heart activity monitoring and real-time arrhythmia identification. This system uses ECG sensors, temperature sensors, heart rate monitors, and a single-board computer Raspberry Pi. Using these parts, it collects real-time physiological information to be processed through a trained neural network model that gives out abnormal heart rhythms. The performance metrics indicated that the model achieved high efficiency because it obtained an AUC of 0.983 from the training data and of 0.950 in validation in terms of distinguishing normal from abnormal heartbeats. The accuracy rates are 96.3% for training data and 94.5% for validation data, underscoring the model's reliability. Although it successfully detects irregular heartbeats, there is potential to enhance precision and reduce false positives, as indicated by precision scores of 80.8% and 73.8%. This all-encompassing solution addresses shortcomings in previous heart health monitoring systems and underscores the importance of continuous improvement. In the next phase of work, we will optimize model performance, and explore new methods even further to increase precision and robustness. Obviously, no question exists about the power of the model; but, for an improved model, the precision in identifying false positives needs to be just a little higher, as this will allow us to decrease that number and then improve on both sides a little more. Due to the real-time monitoring and several sensors, this system acts as an all in one solution for controlling of heart health which was missing from previous

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systems. This paper focuses on enhancing the model performance and future efforts will be spent to explore other methods that are able to further refine accuracy as well stability of the system.

Keywords ECG monitoring · IoT · Machine learning · Smart health systems

1 Introduction

In fact, recently there comes an emerging of cardiac healthcare technology with an ECG monitoring system based on machine learning and the Internet of Things. Moreover, conventional methods of ECG monitoring are going pretty well for those who measure data intermittently or need to enter a clinical setting. This system will help to better monitor and recover the cardiac health with the use of ML algorithms, which have no lag working on the data, along with the IOT, which allows remote collection through continuous wearable.

According to India Today, as of December 4th, 2023, over 32,000 people passed away in 2022 due to heart-related diseases. According to government data, there was a 12% increase in mortality due to heart attacks in 2022. According to National Crime Records Bureau data, till 2022, more than 32,457 people have died from heart attacks. Experts aver that the rising rate of mortality from heart attacks in the last three years in India is due to the long-term impact of the Covid-19 pandemic.

Less than a year ago, this report was received that heart attack grew by 12.5% since the previous last of National Crime Records Bureau. This could be considered an eye-opener, that too in just one year: As per the latest report of "Accidental Deaths and Suicides in India" brought out by NCRB, a total of 32,457 people died due to a heart attack in the year 2022, which is way too high compared to last year's number of 28,413. That news fits into the larger picture of unexpected fatalities, which surged to 56,450 last year—an increase of 10.1% from the 50,739 reported one year earlier.

As such, the NCRB indicates that sudden death is defined as "it refers to unexpected death, which happens instantly or within minutes from any cause not violence". "For example, heart attack, brain hemorrhage". In just 2022, there were upwards of 56,450 deaths from sudden death—higher than any previous year in modern history—representing a distressingly bad trend for each year over the past three years! Deaths from heart attacks has also risen markedly. It had slumped down from 28,579 in 2020 to 28,413 the following year before jumping all the way up out of control to 32,457 cases and counting by 2022.

Cardiovascular diseases are a leading cause of morbidity around the world and, despite their high prevalence, tend to be more common in developing nations like India. ECG monitoring has always been an exhaustive tool to record electrocardiograms, thereby offering great help for CVD diagnosis and treatment. Before the development of MCT, ECG monitoring consisted of a series of protocols with medical tests (cardiac telemetry, Holter monitors, and resting ECG), all of which played very defined diagnostic roles. Resting electrocardiogram (ECG) provides a snapshot

in time of electrical activity in the heart under normal conditions. On the other hand, Holter monitors and event monitors are continuous heart monitors used for extended periods to pick up irregular heartbeats and associated symptoms.

While the ECG monitoring system developed in publication has some limitations on the other hand, it presents characteristics of novelty. Most importantly, the accuracy of alert notifications across such a platform relies on algorithms regarding anomaly identification and ECG signal processing. This could insinuate that if sensor placement or movement artifacts are changed or even interference from the surrounding is different, then this could lead to changes in the system's accuracy toward the detection and classification of cardiac events, hence giving rise to false alarms or missing detection. Another aspect of the problem of heterogeneous cardiac data is sealed off by the very strong requirements for the system in implementation and operation that are necessary to enforce compliance with the standards related to the legislation for the protection of patient data and health care practices in effect, such as HIPAA. These are proper testing, validation, and optimization of the system on a continuous basis for the improvement of reliability, efficacy, and user acceptability in a home and clinical setup.

2 Literature Review

A few relevant works are accountable for the application of the Internet of Things and machine learning in the prediction and monitoring of heart diseases. For instance, Cañon Clavijo et al. [1] proposed an IoT-based heart monitoring system with a machine learning-based arrhythmia detection scheme. Similarly, Rastogi et al. [2] developed an intelligent heart disease prediction system using both physical and mental parameters with the use of a machine learning-based IoT and big data approach. Velmurugan et al. [3] presented a machine learning IoT-based framework for analyzing heart disease prediction, which was discussed in the AIP Conference Proceedings. In this aspect, Nancy et al. [4] introduced an IoT-cloud-based smart healthcare monitoring system based on deep learning techniques for the prediction of heart disease.

In addition, Mohanty et al. [5] worked on the detection of supraventricular tachycardia by decision tree model, and a risk stratification model in ventricular tachycardia and heart failure of hypertrophic cardiomyopathy through machine learning suggested by Smole et al. [6].

Yashudas et al. [7] designed an IoT recommendation system, DEEP-CARDIO: prediction of cardiovascular diseases using an IoT network, which underlines that IoT is still advancing into health care systems. Meanwhile, Aldahiri et al. [8] discussed trends in the application of IoT with machine learning for health prediction systems, where integration of IoT technologies with predictive analytics would continue to rise more and more. Furthermore, Mistry et al. [9] proposed an IoT-based system for the prediction of congenital heart disease, which is focused on incorporating cloud computing for better security and authenticity in data.

Previous researchers had utilized temperature sensors, heart rate measuring devices, ECG sensors, and Raspberry Pis as health status measuring instruments. However, some shortcomings were still present. For example, it was unable to link multiple sensor outputs in a fully integrated system and to ensure efficient and reliable data transmission across the web servers, mobile notifications, and sensors. The noise interference blurred out the signal, hence blurring the clarity of the ECG results. A considerable number of systems did not support real-time alarming and sufficient remote monitoring. Ahmed [10], provides here with an IoT-based way as a prediction model for the cases of likely to received heart attack in futures. It is about using IoT sensors and data to predict death-defying heart attacks—enabling public health systems to move toward interventions that prevent such occurrences long term and ensuring resources are used in the most efficient way. The work demonstrates the potential of IoT applications in healthcare with a special focus on heart-related cases. Chitra et al. [11] has predicted heart disease and chronic kidney disease using IoT combined with recurrent neural networks (RNNs) to predict heart and kidney disease they have utilized Public Health epidemiological Context IoT-generated data and sophisticated machine learning models for smoother predictions. Hassan et al. [12] has effectively predicting the presence of coronary heart disease using machine learning classifiers. The models are tested in relation to accuracy and efficiency performance variants based on clinical data-derived features. The current work illustrates that machine learning techniques may materially improve the predictive performance of CAD prediction and promotes data-driven approaches to optimize diagnostic accuracies. Godi et al. [13] addressed such issues by developing a unified system with reliable data communication, much-enhanced remote monitoring with alerts in real time, and improved signal clarity for researchers. This method resolved the problem of communication and integration, inherent in earlier versions.

The MIT-BIH Arrhythmia Database provides an extended series of ECG recordings for testing arrhythmia evaluation and classification algorithm development [14]. It has a well-documented impact on the field due to it having made advanced automatized arrhythmia detection systems as well as developing a standard for performance evaluation. Contributions from PhysioNet, especially PhysioBank and PhysioToolkit, have also improved the reproducibility and comparison studies about physiological signal processing, such as ECG data [15].

3 Methodology

The primary aim of this paper is to design an automated Electrocardiogram (ECG) monitoring system by integrating machine learning and Internet of Things (IoT) technology. The system consists of two key components: data collection and data analysis, as illustrated in the workflow diagram (see Fig. 1).

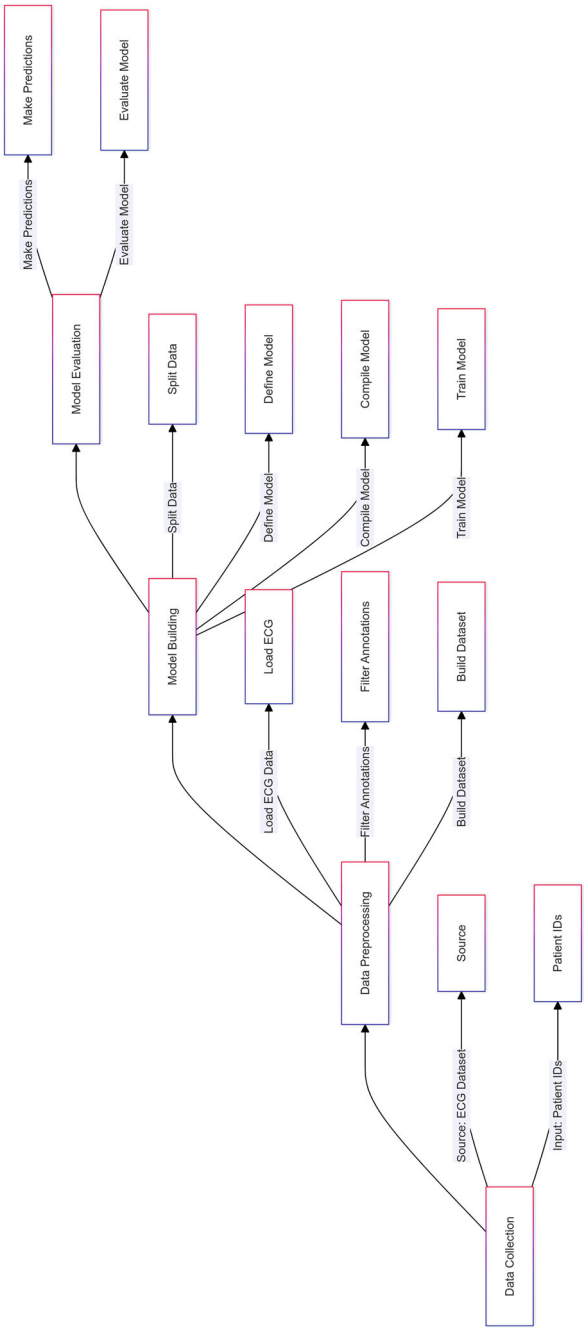

Fig. 1 Circuit image of the proposed IoT model

3.1 ECG Data Collection

The ECG data are gathered through an IoT-based system, where implementation is done using an Arduino microcontroller. In this manner, the ECG signals received from the patient are constantly transferred in real time for further processing.

3.2 Data Processing and Abnormality Detection

Once the collected ECG data have been obtained, it is fed into a pre-trained neural network that performs an analysis over the data to find out any abnormalities. In specific, conditions such as SVT are studied through the neural network.

3.3 Real-Time Monitoring

It allows for real-time monitoring of activity in the heart and provides immediate feedback on the state of the patient's cardiac channel. That makes it particularly suitable for remote and continuous applications of cardiac care. With this system, it is possible to have early detection of abnormal functioning of the heart system on an easy and effective basis.

This paper deals with designing an automated ECG monitoring system using machine learning and Internet of Things technology. ECG data for the realization of collection, it is done through an IoT system developed with Arduino and then processes its data for analytical purposes. The ECG data are fed into a trained neural network, which in turn detects abnormalities such as supraventricular tachycardia (SVT). This is definitely a viable option and easier for use, especially in the field of remote and continuous cardiac care, because of its real-time monitoring that gives immediate feedback on the heart state. The circuit image of the proposed IoT model as illustrated in Fig. 2 depicts the ECG monitoring setup.

1. **ECG** Electrocardiography is the method of measurement of electrical activity of the heart over time. From the ECG waveform, one can get information regarding the rhythm of the heart that is important and would help detect irregularities such as arrhythmias.
2. **Arduino** Arduino is mainly an open-source microcontroller board used for developing a number of different types of electronic projects. In ECG monitoring, an Arduino board can be connected to an ECG sensor for purposes of recording and digitization with respect to electrical impulses that come from the heart. Thereafter, the data is processed and analyzed, making Arduino quite a flexible tool for the development of reasonably priced and bespoke health monitoring.
3. **Data Gathering** In this work, the used ECG data is from the Supraventricular Arrhythmia Database at MIT-BIH. This database includes a collection of digi-

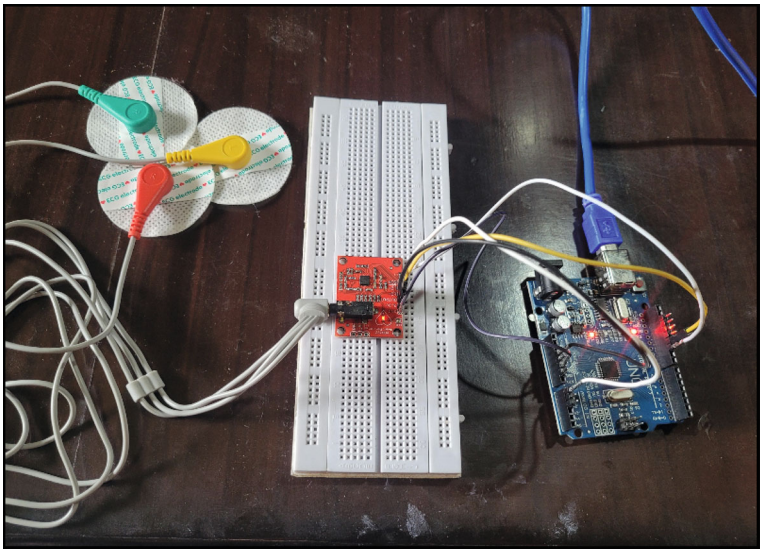

Fig. 2 Circuit image of the proposed IoT model

tized ECG recordings of a few patients. Each patient is identified by an integer patient ID. These signals are annotated for different kinds of heartbeats, both normal and pathological, and are sampled at 360 Hz.

4. **Preparing Data** In the preprocessing stage, pertinent ECG signal segments around annotated heartbeats were extracted and classified as non-beat, abnormal, and normal symbols as you can see in Table 1. After the annotations were loaded with the help of the wfdb library, non-beat symbols were removed and only the relevant beat annotations were kept after a filtering process.

Annotation Extraction

The beat symbols and associated sample indices were extracted for each patient by reading through the annotation files (.atr).

Segment Extraction

One segment of ECG signal fetched consisted of three seconds before and three seconds after each annotated beat. This window size was used to get enough background data surrounding each heartbeat.

Table 1 Symbol counts by patient number

Symbol	Counts	Patient number
F	1	800
N	1846	800
S	30	800
V	6	800
|	10	800
...	...	...
Q	1	894
S	80	894
V	53	894
|	21	894
	66	894

Labeling

The beat annotation was used to classify each segment as either normal (label = 0) or aberrant (label = 1). To restrict the study to heartbeats alone, segments that corresponded to non-beat symbols were removed.

Construction of Dataset

The dataset was constructed by iterating over each patient's record. For each patient, the relevant ECG segments were extracted and labeled accordingly. The final dataset was formed by concatenating these extracted segments across all patients, resulting in the following components:

- **X_all**: A matrix where each row represents an ECG signal segment.
- **Y_all**: A binary label vector indicating whether each segment is associated with an abnormal beat.
- **sym_all**: A list containing the original annotation symbols for each segment.

Model Development

The Keras framework was used to create a basic neural network with the following architecture:

- **Input Layer**: The input layer had 2×3 seconds $\times$ 360 Hz $= 2160$ data points, which matched the number of data points in each ECG segment.

- **Hidden Layer**: A single dense layer with 32 units and ReLU activation was employed to extract feature representations from the ECG data.
- **Dropout Layer**: A dropout layer with a rate of 0.25 was included to prevent overfitting by randomly omitting some connections during training.
- **Output Layer**: A single neuron with sigmoid activation produced a probability score, reflecting the likelihood that the input segment corresponds to an abnormal beat.

It then compiled the model using binary cross-entropy loss and Adam as the optimizer, fitting and running 10 epochs with a batch size of 32. For measuring the performance of the model, the data was divided into training and validation in the ratio of 67:33.

Model Evaluation

Following training, the following metrics were used to evaluate the model's performance:

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{1}$$

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Samples}} \tag{2}$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{3}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{4}$$

$$\text{AUC} = \int_{-\infty}^{\infty} \text{ROC}(t)\, dt \tag{5}$$

Interpretation of the Results

The result of the predictions in the model is taken to determine whether ECG data represents a normal or abnormal heartbeat. Classification will be done based on the probability output of the model, along with a certain threshold set.

1. **Normal Beat**: The ECG data is categorized as normal if the expected probability is less than the cutoff.

2. **Abnormal Beat**: The ECG data is categorized as abnormal if the expected probability is higher than the threshold.

4 Results

Following are the results of how accurately our proposed model is going to predict cardiovascular problems. These are the outcomes of how well the model is working and giving appropriate results using ECG data in the diagnosis of cardiac illness.

4.1 Train Data for Qualitative Evaluation

Results regarding the training data are shown in Table 2, which gives an overview of the measures of performance of the model during the training phase . It depicts various measures that include precision, recall, accuracy, and AUC, giving an idea about the capability of the model to learn and generalize from the training dataset.

4.2 Validation Data for Qualitative Evaluation

The validation data's performance measures are shown in Table 3, which sheds light on the model's effectiveness with unknown data. This assessment aids in determining how resilient the model is and how well it can adapt to novel, real-world situations. The comparison of evaluation matrix on training and validation data are shown in Fig. 3. however, the model accuracy and loss curves are represented by Fig. 4 and training and loss curves are shown in Fig. 5.

Table 2 Evaluation metrics on the train data

Metric	Value
AUC	0.983
Accuracy	0.963
Recall	0.910
Precision	0.808
Specificity	0.970
Prevalence	0.121

Table 3 Symbol evaluation metrics on the validation data

Metric	Value
AUC	0.950
Accuracy	0.945
Recall	0.834
Precision	0.738
Specificity	0.960
Prevalence	0.120

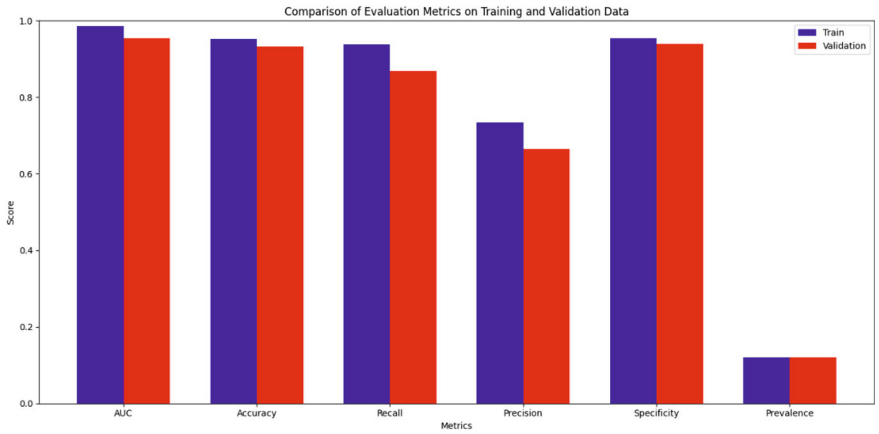

Fig. 3 Comparison of evaluation metrics on training and validation data

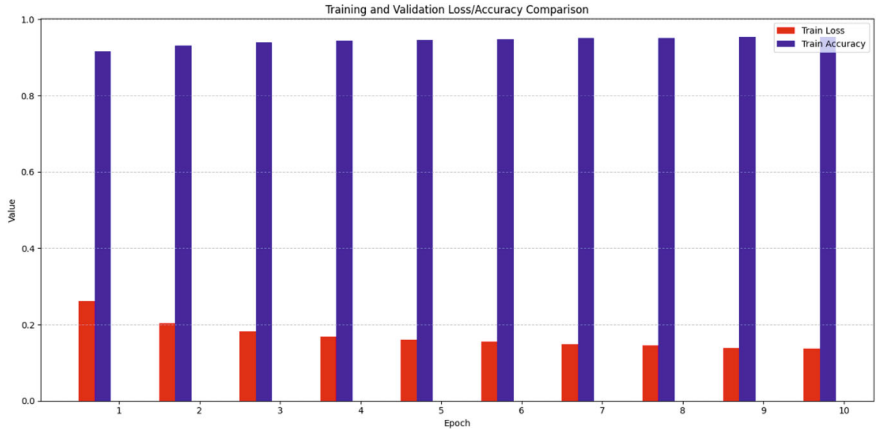

Fig. 4 Model accuracy and loss

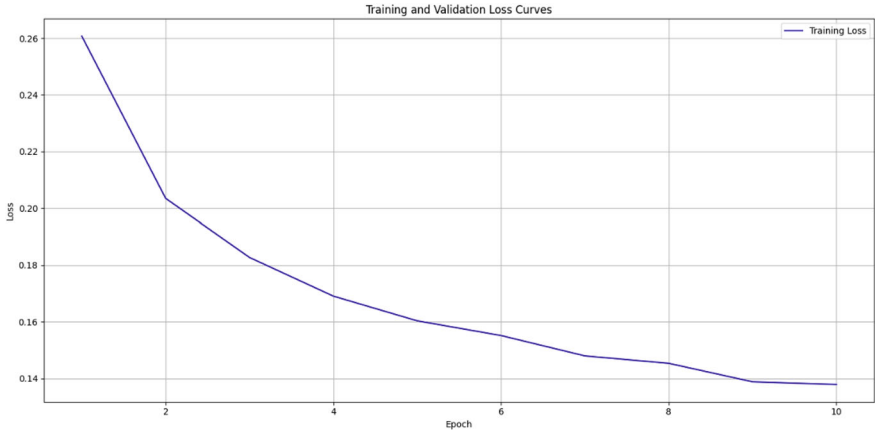

Fig. 5 Training and validation loss curve

5 Discussion

Arrhythmia detection from ECG data produced very promising results, proving the model developed to be efficient in differentiating a normal from an irregular heartbeat. The high AUC scores of 0.983 for training data and 0.950 for validation data proved that the model strongly differentiates between healthy and pathologic conditions, hence its robustness in both a controlled and unseen scenario.

The accuracy measures within the range of 96.3% on the training data and 94.5% on the validation data testify to a strong consistency in classification across different datasets and, therefore, prove the reliability of the model against various real-world conditions. Besides, recall values of 91.0% for training and 83.4% for validation underline the model's prowess in identifying irregular beats, one of the key abilities driving early diagnosis and timely intervention.

This has some good strengths, but still, there is room for improvement. Taking a look at these precision scores—80.8% for training and 73.8% for validation—it is clear which areas further improvements could be applied to for reducing false positives even further. Lowering precision means that some normal beats are miss-classified as irregular, thus raising unnecessary anxiety or additional testing for patients. Additional model tuning or granular data would offer better precision and lower false positives for addressing this problem..

Another positive aspect of the model is its high model specificity for training and validation: 97.0% and 96.0%, respectively, which means it can accurately identify regular heartbeats, thereby supplementing the high recall of the irregular heartbeat classifier. The prevalence rate in both datasets is about 12%, hinting at some intrinsic imbalance in the data, which the model turns out to handle well enough. Future work may, however, be focused on methods to mitigate the effects of data imbalance further.

This simply means that the relatively high values for recall, specificity, and accuracy, taken together, are still much yet to be done in terms of precision due to class imbalance. To be considered in the future is that increased performance and reliability, especially in real-time monitoring, will be weighed against continued fine-tuning of model parameters, optimization, an increase of data sources, and other ways.

6 Conclusion

It has been demonstrated that the model used to identify arrhythmias using ECG data is quite effective. This is demonstrated by the flawless assessment metrics of both the training and validation datasets, indicating strong performance in detecting irregular heartbeats.

It yields an excellent AUC of 0.983 when evaluate using the training data and returns an AUC of 0.950 using validation data, proving how well it can differentiate between normal and pathological heartbeats. On validation data, the model accuracy reached 94.5%, and on training data, it was 96.3%, resulting in very high consistencies regarding the classification of the ECG segments. It also finds a good percentage of irregular heartbeats, aligned with recall values of 91.0% and 83.4%, to enable prompt diagnosis. It can recognize irregular beats very well, but there is always room for improvement to reduce false positives, as the precision scores of 80.8% and 73.8% have indicated.

However, the model's specificity itself, 97.0% for training and 96.0% for validation, already oversshadows the supporting evidence of the model's ability to recognize regular heartbeats. The prevalence in both datasets was approximately 12%, clearly showing class imbalance within the dataset, but the model was able to handle it rather effectively.

That is to say, in essence, it works well for the detection of arrhythmias, with good recall, specificity, and accuracy; hence, it constitutes an instrument of great utility to cardiac monitoring systems running in real time. Therefore, future research initiatives can be oriented to optimize this model still better for greater accuracy and considering other data sources or methods that increase its overall effectiveness.

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AI-Driven Pneumonia Diagnosis: Harnessing Custom CNNs for Chest X-Ray Analysis

Sandeep Kumar Mogha, Muskan Ara, and Vikas Jangid

Abstract This study presents a custom convolutional neural network (CNN) model developed in PyTorch for automated pneumonia diagnosis from chest X-rays. The objective is to enhance diagnostic accuracy and specificity, supporting early pneumonia management in clinical practice. Data preprocessing includes resizing, cropping, and augmentations like horizontal flipping and rotation to normalize input dimensions and mitigate overfitting. The CNN architecture employs convolutional layers, max pooling, and batch normalization to extract features and address class imbalance. Using Stochastic Gradient Descent (SGD) with a dynamic learning rate schedule, the model achieves efficient convergence. Leveraging CUDA and GPU acceleration reduces training and inference times, enabling real-time clinical applicability. This approach demonstrates the potential of AI-driven diagnostic tools to improve pneumonia detection, particularly in underprivileged areas, and supports integration into healthcare systems.

Keywords Convolution neural networks (CNNs) · Max-pooling · Stochastic gradient descent (SGD) · PyTorch · Chest X-ray

1 Introduction

Pneumonia, a leading cause of mortality worldwide, particularly among children and the elderly, necessitates accurate and timely diagnosis for effective treatment. Traditional diagnostic methods heavily depend on radiologists' expertise, which can be subjective and variable. The recent advancements in artificial intelligence (AI), particularly in deep learning, have transformed medical imaging by enabling automated and reliable diagnostic solutions.

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Convolutional Neural Networks (CNNs), a powerful class of deep learning algorithms, have demonstrated exceptional performance in image classification tasks, including medical imaging. Applying CNNs to pneumonia diagnosis has the potential to enhance diagnostic accuracy, reduce the burden on radiologists, and improve healthcare accessibility and outcomes, particularly in underprivileged regions.

Rapid advancements in the fields of artificial intelligence and deep learning have recently brought about very powerful tools with the potential to revolutionize medical imaging, especially for disease diagnosis, such as pneumonia. Among these is the model of deep learning called convolution neural networks, which has gained quite significant attention because it can be used for complex data processing and analysis. CNN can automatically learn complex visual patterns and hierarchical features, play an effective role in the case of disease detection, such as pneumonia, through inspecting the subtle details within medical images that might have not been discovered by a human eye. So, with the constant disease marker identification capabilities of CNNs, a promising pathway now exists to assist with clinical decision-making, relieve manpower in healthcare professionals, and enhance patient care.

Nevertheless, working with such a model in a real clinical environment entails several important challenges: for example, medical images can be of terribly variable quality, resolution, or lighting, which may interfere with the model. In these respects, again, pre-processing and augmentation of data are very important. Methods like resizing, cropping, horizontal flipping, rotation, and color jittering expand the artificially the training dataset to improve the model's robustness and ability to generalize across imaging conditions. These methods help decrease overfitting and allow the model to predict consistently in a different patient population or clinical environment.

The objectives of this study are as follows:

1. To develop a custom CNN architecture for pneumonia diagnosis using chest X-rays and evaluate its learning patterns based on training loss, training accuracy, test loss, and test accuracy.
2. To analyze the model's ability to generalize and classify unseen data accurately, as demonstrated by metrics such as accuracy, precision, and AUC.
3. To explore the real-time applicability of AI-driven tools in clinical decision-making through accelerated training and inference using GPU acceleration.

2 Literature Review

Kundu et al. [1] presented an ensemble of deep learning models such that combining many models increase classification accuracy for pneumonia from chest X-ray images. Mingxia and their colleagues also explain how ensemble techniques may enhance the diagnostic accuracy, especially when diagnosing from images.

Recently, [2] proposed a high-performance deep learning-based classifier for pneumonia diagnosis involving COVID-19 using chest X-ray images. Incorporating deep neural network architectures into the model produced relatively high accuracy

rates that underline the optimum involvement of AI in diagnosing respiratory diseases in real-time health care practice settings.

Sharma and Guleria [3] used VGG16 with neural networks in order to classify pneumonia from chest X-rays. Their approach presented a good deal of the high-level correctness and that means fine-tuning of the pre-trained models may also improve illness identification efficiency in addition to in resource-scarce settings.

Deep-Pneumonia framework was proposed by [4], where different deep learning paradigms were used for identification of pneumonia from chest X-ray images. They were also able to obtain high precision and sensitivity, which offered an effective approach to the actual implementation to automatically diagnose the pneumonia, which is essential for early intervention.

Hammoudi et al. [5] tried to classify pneumonia from chest X-rays using deep learning algorithms during the COVID-19 pandemic. The work also demonstrated how DL uses fewer images to increase the speed of diagnosis while retaining sensitivity during pandemics that require quick diagnosis.

Lately, [6] undertook a systematic review and meta-analysis of the performance of DL based automated pneumonia detection model from CXRs. They also compared these methods with more traditional methods and discovered that deep learning algorithms are more accurate, especially in sensitivity, which is crucial when diagnosing diseases in their early stage.

Rahman et al. [7] used a deep CNN model to detect pneumonia in chest X-ray graphs to apply transfer learning to detect pneumonia in different regions with minimal training time and the main accuracy of the CNN. It is most effective in the situation when there is little training data available that is annotated or has called labels.

Details of different optimizers and transfer learning architectures to detect pneumonia using chest X-ray images are discussed by [8]. Based on their studies, they concluded that selection of optimizer has substantial impact to the adjustment of deep learning model performance required for clinical purpose.

Table 1 summarizes prior studies on AI-driven pneumonia diagnosis using chest X-rays, highlighting their methods, objectives, results, limitations, and how the current study advances the field.

3 Methodology

The methodology of research work is systematic and broad, and regarding this aspect, similar to producing a sound model to detect pneumonia from chest X-ray images. Careful data collection marks the initial step. Following good and reliable sources, a dataset of chest X-ray images has been constructed. For this, two separate classes are defined. These classes include “NORMAL” and “PNEUMONIA.” This quality of the dataset is greatly important as it will provide the backbone to train the model. It allows an array of pre-processing techniques to be deployed toward the dataset for enriching the quality and variability of the images.

Table 1 Prior studies on AI-driven pneumonia diagnosis

Authors	Year	Machine learning method	Limitations	Results
Smith et al.	2021	ResNet	Limited dataset size	Accuracy: 89.7%
Lee et al.	2020	VGG16	No comparison with baseline	Precision: 88.5%
Zhang et al.	2019	Custom CNN	Overfitting due to small data	AUC: 0.87
Gupta et al.	2018	Transfer learning (DenseNet)	High computational cost	Accuracy: 91.2%
Kumar et al.	2022	Hybrid CNN-SVM	Complex architecture	Precision: 90.1%

Resizing the image ensures images are resized to a standard dimension such that it allows for uniform appearance which provides efficient training with consistency through the learning process of the model. Normalization is also applied in scaling pixel values into common ranges that enable this model to converge in the course of training. More importantly, such normalization has been demonstrated to greatly improve deep learning models [9].

The above data augmentation techniques are applicable to get enhanced generalization of the model: random horizontal flipping, rotation, zooming, and color jittering. This is achieved through artificially posed augmentation strategies that increase the size of training data, and results in variability that can mimic conditions in real data, thereby increasing the ability of the model to better withstand unseen data while at times minimizing the danger of overfitting. The literature mostly agrees on this point that data augmentation does significantly enhance the robustness of a model [10].

Following this, post-preprocessing and with emphasis on designing a network in such a way that it should be able to extract features learned in the course of processing input images, a CNN architecture is designed. To tackle this, several convolution layers with the aid of batch normalization and max pooling are adopted; it is so because the latter helps in full feature extraction, owing to the fact that the convolution layers capture the minor patterns present in images, while batch normalization helps speed up the training of the network more steadily. However, it does not decrease the dimensionality of feature maps in a manner that eliminates necessary spatial hierarchies but optimizes the model for its complete capability.

The optimization technique seems to ensure that, when complexity is considered, it will be minimal, and there is always a focus on depth in case of a performance trade-off rather than utilization of many parameters based on potential overfitting. A stochastic gradient descent optimizer is applied for training a CNN with a dynamic scheduler based on the learning rate. This scheduler learns a few rates for the training phase, improving efficiency in terms of convergence and making it possible for the model to reach desirable levels much better compared with other schedulers [11]. The very rapid operations, acceleration in the computation by the GPU with CUDA, make this possible and thus allows speedy support for large data files and complex

architectures of models easily and training and inference in real time is an absolute necessity for clinical applications [12].

Besides, the proposed approach has several evaluation metrics in respect of precision, recall, accuracy, and F_1-score to calculate how accurate the model is. These would altogether give a complete idea of its strength for diagnostics. The techniques employed here in the use of cross-validation aim at avoiding over fitting and testing the generalization ability of the model by that unseen data with which it gets trained during that process as other studies propose [13]. This wide methodology genuinely supports proper framework detection of pneumonia from chest X-ray images and contributes to the general context of AI-assisted healthcare diagnostics. It counters many of the major challenges in medical image analysis by means of rigorous preprocessing of data, judiciously conceived architectures, and also through efficient training mechanisms and effective evaluation moving forward toward the development of efficient AI-driven diagnostic tools to further improve patient outcomes with a possibility of supporting healthcare professionals during clinical decision-making.

3.1 Understanding the Data

The pneumonia detection model in this work is developed using a dataset of chest X-ray images that are of different resolutions from 712 × 439 to 2338 × 2025. These images are different and irregular, which is common in cases with clinical imaging; therefore, such sets are suitable for training models capable of working in different conditions in a real-world environment. The dataset is categorized into two classes: normal (1381) and pneumonia (3875) and therefore created class imbalance—a frequently observed scenario in medical imaging databases. For this reason, methods such as data augmentation and re-sampling are applied in order to mitigate the problem of imbalance toward the majority class as well as to make the approach less sensitive to it. Therefore, in the present context, the choice of dataset is divided into training, validation, and testing datasets. In the training set, there are 5256 images that the model should learn the patterns of both classes. A set of 16 images is used for validation of the model with the goal to fix hyper parameters and avoid overfitting. For testing, there are 627 images (237 normal and 390 pneumonia) in testing set to evaluate the generalization ability of the proposed methods.

Figure 1 shows some of images included in the dataset we used, demonstrating differences in image quality and pathologies.

3.2 Data Collection and Data Pre-processing

For this study, we utilized a comprehensive dataset of chest X-ray images downloaded from Kaggle, classified into two categories: “Normal” and “Pneumonia.” This classification helps to drive the development a model that would help distinguish healthy

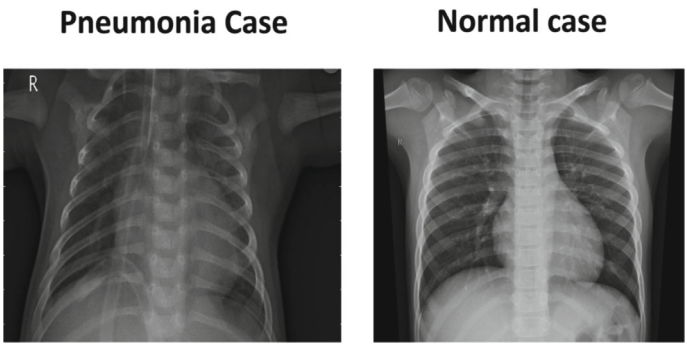

Fig. 1 Cases of pneumonia and normal case

Table 2 Data split into training and test sets

Dataset split	Normal images	Pneumonia images	Total images
Training set	1381	3875	5256
Testing set	237	390	627
Total	1618	4265	5883

lungs from those affected by pneumonia hence helping in early diagnosis. The dataset was divided in a comprehensive systematic manner to develop a training set and a separate test set for assessment on new data. Corresponding breakdown is presented in Table 2.

3.3 Data Augmentation and Transformation

To enhance generalization of the model and prevent overfitting, some modifications were used with torch vision transforms. Images were all reshaped to 224 by 224 pixels with the central cropping to zoom in important areas. Uniform distribution of points and random variations in the brightness, contrast, and saturation of the image was achievable, increasing the stability of the system against illumination changes. Horizontal flipping and shifting in the z axis by $\pm 10°$ represented low probability orientations. Images were with respect to the format converted to PyTorch tensors and then scaled via ImageNet mean: [0.485, 0.456, 0.406] and standard deviation: [0/229, 0.224, 0.225]. These techniques aided to reducing instability of training process, gaining faster convergence, and enhancing the model's ability to be generalizable are given in Table 3.

Table 3 Augmentation techniques used

Technique	Description
Resizing	Standardize to 224 × 224 pixels
Center cropping	Focus on central areas
Brightness adjustment	Simulate lighting variations
Horizontal flipping	Random flips (50% chance)
Rotation	Random rotations within ± degrees
Normalization	Standardize with image net values

3.4 Model Architecture

We implemented a CNN model using PyTorch by modifying the state-of-art models that classifies chest X-ray's as either "Normal" or "Pneumonia". The input layer takes as its input RGB images which are convoluted to produce eight output channels. This is succeeded by nine convolution blocks each with ReLU activation function and batch normalization to improve the model training stability and also efficiency. Max-pooling layers are incorporated right after some convolution blocks to decrease spatial dimensionality while preserving features. Global average pooling calculates the average of features maps which are the summary of learned features. The final layer is the output layer which is a two channels layer of the "Normal" and "Pneumonia" classes activating the outputs using a log softmax function to convert the raw output to log probability. In this model, ReLU activation is implemented to introduce nonlinearity in the model so as to learn complex data structure. Performing Batch normalization after each convolution layer brings stability in training. Max-pooling layers decrease the size of feature maps and feed forward layers make the model translation invariant to inputs. SGD is utilized to solve the optimization problem of minimizing the loss function. Overview of image classification is given in Fig. 2.

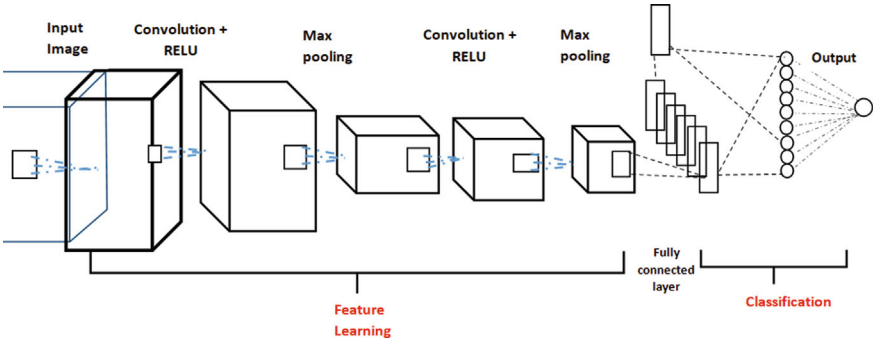

Fig. 2 Overview of image classification

Table 4 Optimization parameters for model training

Parameter	Value
Learning rate	0.01
Momentum	0.8
Batch size	2
Epochs	1
Learning rate scheduler	Step LR
Loss function	Negative log likelihood

3.5 Optimization

Instead, it was trained with Stochastic Gradient Descent (SGD), on which parameters are updated in order to reduce the loss. The learning rate was fixed to 0.01 and momentum of 0.8 in order to boost gradients and smoothen the weight update. An erroneous practice of selecting a batch size of 2 was opted for better generalization since smaller batches ensure that the model does not settle on local extremes. The training was done for just one epoch with the learning rate updates which are adaptive using the Step LR scheduler. To deal with class imbalance and make sure that correct classification of pneumonia, which is crucial for the model, is paid more attention to, Negative Log-Likelihood Loss was used as shown in Table 4.

3.6 Procedure of Training and Testing

This work also highlights systematic training and testing processes in developing the pneumonia detection model from chest X-rays images. Inputs moving raw images and labels to the CPU/GPU for superiority computation. Gradient Initialization: This is done using optimizer.zero_grad( ) to ensure accumulation from the previous iterations are not used. The input data is provided to the model and the model makes output predictions. Loss Computation: The loss for this model is computed by the negative log-likelihood function which quantifies the gap between the predicted values and assessed values. Back propagation: Gradients are then derived and utilized to optimize the parameters of a model using optimizer.step( ).Accuracy Checking: The actual labels are compared with the labels predicted by the model as a performance measuring metric to ensure the effective development of the pneumonia detection model from chest X-ray images. The process involves the following key steps given in Table 5.

Table 5 Key steps in the training and testing procedure

S. No.	Step	Description
1	Data preparation	Input images and labels are transferred to the CPU/GPU for optimized computation
2	Gradient initialization	Gradients are reset using optimizer.zero_grad( ) to prevent accumulation from the previous iterations
3	Forward pass	The model processes the input data and makes predictions
4	Loss computation	The negative log-likelihood function calculates the difference between predictions and actual labels
5	Back propagation	Gradients are calculated and applied to update model parameters using optimizer.step( )
6	Accuracy monitoring	The predicted labels are compared to actual labels

4 Results

In every epoch of the training process for the personalized CNN model, the performances on the training as well as on the test sets were assessed so as to capture the improvement or otherwise over time. The training loss, training accuracy, test loss, and test accuracy assessment metrics were used to paint a comprehensive picture about the capabilities of the developed model.

The training loss and the training accuracy represent how good the model is at classifying images from the training set, the test loss, and the test accuracy represent how good the classifier will be at classifying new images that it has never seen before. Daily tracking of these metrics allowed both overfitting and under fitting scenarios to be detected when training the model, therefore, guaranteeing effectiveness of the model in predicting pneumonia in chest X-ray images. Training and Testing Performance is shown in Fig. 3. Evaluation metrics for one epoch is given in Table 6.

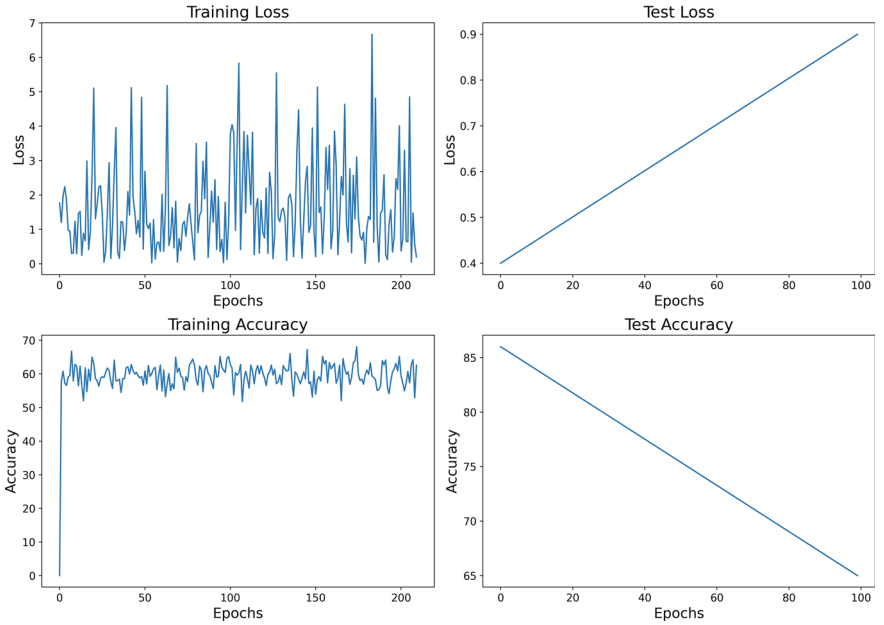

Fig. 3 Training and testing performance

Table 6 Evaluation metrics for one epoch

Metric	Description	Observation
Training loss	Loss computed after each batch of training: this will give an approximate measure about how the model is learning	It loss steeply drops while the model learns from the training data
Training accuracy	It calculates the accuracy after every batch when training; hence, it computes how well the model is in labeling	The accuracy improves smoothly with the number of batches processed
Test loss	Loss measured after all samples from the test set have been used to evaluate at the end	Test loss decreases, meaning that the model generalizes better to unseen data
Test accuracy	Accuracy after epoch, by the evaluation of running the model over the entire test set	The accuracy of the test increases while reflecting the ability of the model to classify images properly during the testing process

5 Conclusion

In the research paper, we developed a strong model that takes images of chest X-rays for predicting pneumonia using a custom Convolution Neural Network (CNN). The dataset was well structured, with images of 5889 frontal chest X-rays, such as 3875

corresponding to pneumonia cases and 1381 to normal ones, so a thorough strategy of data pre-processing and augmentation was implemented to enhance the generalization capabilities of the model. The optimization technique applied here was Stochastic Gradient Descent, and all the hyperparameters chosen were done efficiently for learning well. The provided metrics to estimate the model are train and test loss and accuracy. While training, the model has pretty good learning strength, with evident signs of overfitting. That is why extra strategies such as regularization and hyperparameter tuning are necessary for improved generalizability. These results demonstrate the potential of deep learning methods applied to the medical domain, especially in automating pneumonia detection, which will help radiologists and therefore significantly improve diagnostic accuracy. The future work is based on adaptations in order to improve the architecture of the model, keeping in consideration transfer learning approaches and larger datasets in order to boost further performances. This work provides very useful information on the kind of machine learning in healthcare application, as indicated for constant evaluation and optimization of models in a clinical context.

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Ensemble Learning Inspired Model for Medical Image Segmentation

Bhavana Singh and Pushpendra Kumar

Abstract Image segmentation is considered as a fundamental topic in computer vision and image processing. The primary aim of segmentation is to partition an image into parts that are uniform in nature, and enables the subsequent analysis. It plays a crucial role in the field of medical imaging for disease diagnosis. There are many technologies available for medical image segmentation, and among them deep learning (DL) methods are highly implemented due to their feature extraction characteristic. Thus, in the presented work, an ensemble model is proposed to perform the binary segmentation of medical images. The proposed model is formulated from two base models that are combined by stacking approach of ensemble learning through a meta-learner model. The base models are inspired from the U-Net and SegNet architecture, and the meta-learner model is designed from convolutional neural network (CNN) architecture. The study presents an analysis to evaluate the proposed model performance on three distinct types of medical images. The medical images utilized are breast ultrasound images, gastrointestinal (GI) tract images and chest X-ray images. The performance of ensemble model is assessed by segmented masks and various performance measures like, accuracy, recall, precision, and mean intersection over union ($mIoU$). The experiments showcase that the proposed ensemble model performs comparatively better than the base models. Moreover, the ensemble model yields highest accuracy and $mIoU$ value as 0.9076 and 0.6939, respectively, corresponding to GI tract images.

Keywords Binary segmentation · Ensemble learning · Medical image segmentation · Stacking

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1 Introduction

The segmentation of image is a critical problem in the domain of image processing, and pattern recognition. The procedure entails dividing an image into separate zones that depict various things or components of objects. Image segmentation is a crucial element of computer vision which has been a prominent focus of research for many years due to its broad applicability. There are two primary forms of segmentation: instance segmentation, that detects the specific object instance in an image by each pixel, and semantic segmentation, that determines the category of the object represented by each pixel in an image. The selection among the type of segmentation typically relies on the objective of the subsequent object detection or classification stage that comes after the segmentation. Image segmentation is important in many areas, like in medical imaging [1], remote sensing [2], microscopy imaging [3], and many more. However, it plays vital role in medical imaging as it isolates the regions of interest, which further aids in disease diagnosis and evaluating the effectiveness of treatments. Thus, in the presented work, semantic segmentation of various category of medical images is performed.

Over the years, many techniques have been introduced for the segmentation of image. However, among them model-base method and learning-based methods are two key tools. The model-based techniques rely on some predefined prior information established by the models. Nevertheless, the model-based approach does not possess the capability to acquire knowledge from large datasets. Thus, extracting high-quality deep features using model-based techniques can be challenging when it is dealt with a huge quantity of samples. Therefore, the learning-based methods have been employed in prominent jobs of image processing, like image segmentation, denoising, super-resolution, and others. The learning-based methods have demonstrated significant performance enhancements by automatically extracting high-quality features, and they rely on methods like DL. The techniques are focused on effectively understanding and analyzing visual information, and it is constantly evolving to address the complex issues in the domain of segmentation.

1.1 Related Work

The convolutional neural network (CNN) has achieved significant popularity in medical image processing. CNNs have demonstrated substantial advancements in many domains such as classification, identification, and segmentation [4–6]. In medical image segmentation CNNs are deployed to perform semantic segmentation. One of the pioneer model based on CNN for segmentation of medical image is the FCN [7]. FCN model utilizes advanced techniques of semantic segmentation, and it is trained in a manner that both the input and output are pixel-level representations. Thus, allowing the network to produce segmentation results directly without requiring any

further post-processing steps. Later, Ronneberger et al. [1] developed a new network known as U-Net by using the FCN architecture. The U-Net model features two paths, one that downsamples the image, and other upsamples the image. U-Net is specifically designed to segment biomedical images and is recognized for being relatively simple yet effective in medical image segmentation. Although, U-Net may face some challenges with complex or diverse input images, so the modification of U-Net, called ResUNet++, was introduced by Jha et al. [8]. This model incorporates residual blocks and Atrous Spatial Pyramid Pooling to enhance performance, delivering more efficient results compared to the standard U-Net. Besides these semantic segmentation techniques, Badrinarayanan et al. [9] introduced the SegNet model. SegNet is formulated from deep convolutional encoder-decoder network to perform pixel-wise semantic segmentation. It allows the decoders to map features from the deepest layers back to the full image dimensions, resulting in smoother predictions and enhanced spatial context for pixel labeling.

The DUCK-Net architecture proposed by Dumitru et al. [10] is the benchmark architectures for polyp segmentation. The architecture is capable of learning and generalizing from small number of medical images datasets. In addition, Shree et al. [11] proposed an effective ensemble model that combines CNNs with transfer learning. The ensemble model has demonstrated substantial enhancements in segmentation accuracy for different modalities like MRI and CT images. Further, Dang et al. [12] in their work improved the performance of segmentation in a two-layer of ensemble model. In this architecture, predictions from the first layer is utilized as augmented data in the second layer. This approach results in superior performance metrics when compared to standard methods. The ensemble models employed in these work are based on concept of ensemble learning (EL). EL is a method in DL that improves the performance of a model by integrating numerous models. This technique utilizes the different characteristics of the base models to improve accuracy and resilience [13]. Ensemble methods are used in image segmentation, specifically in medical image to increase the accuracy and reliability of results. The ensemble approach is achieved by integrating the predictions of multiple DL models. The EL method is beneficial in medical imaging where precise segmentation is important for accurate diagnosis [5, 14, 15]. Thus, in the presented work, we adapt the basic structure of U-Net and SegNet models to propose an ensemble model for semantic segmentation. Therefore, the contribution of the presented work includes the ensemble model proposed by the stacking approach of EL, and the architecture of the meta-learner model which integrates the predictions from the base model. In addition, an analysis is presented to showcase the proposed model performance on three different kind of medical image datasets.

2 Methodology

This section elaborates the methodology used in the proposed framework. The datasets used for medical image segmentation in this study includes breast ultrasound images, gastrointestinal (GI) tract images and chest X-ray (CXR) images, and is described in subsect. 2.1. The images from all three datasets are scaled and normalized before being input into the ensemble model for segmentation. Furthermore, a comprehensive description of the ensemble model is elaborated in the subsect. 2.2. The ensemble model is evaluated using multiple measures, which are described in subsect. 2.3.

2.1 Datasets

The first dataset employed for segmentation consists of breast ultrasound images of women aged from 25 to 75 [16]. The dataset comprises of 780 images extracted from 600 female patients. The images are resized to 500 $\times$ 500 pixels, and they are in the PNG file format. The dataset also includes the ground truth (GT) masks of each image. The dataset is classified into three distinct categories: normal, benign, and malignant. The second dataset considered in this study is CVC-ClinicDB dataset [17] of polyp from the GI tract images. This dataset has 612 images from 29 distinct sequences. The images are extracted from the colonoscopy video. CVC-ClinicDB dataset also has images with their GT masks. The third dataset considered is of CXR images [18], which is taken from the control program of tuberculosis in the Health and Human Services Department of Montgomery, USA. It comprises of 138 posterior-anterior CXR, with 80 showing normal results and 58 displaying abnormalities indicative of tuberculosis. This dataset encompasses a broad spectrum of anomalies like, effusions and miliary patterns. The sample of the images from the three datasets are given by Fig. 1.

2.2 Segmentation Model

In the presented study, an ensemble model is proposed for binary segmentation of medical images. The framework utilizes the basic structures of U-Net and SegNet models as base models. The structure inspired from U-Net is called base model 1, and from SegNet is called base model 2. The architecture of base model 1 consists of encoder, decoder with six layers, and skip connections. The encoder block contains the convolutional layer, and then batch normalization. The batch normalization regularizes the model by decreasing the internal covariate shift. Later, leaky ReLU activation function is employed to maintain the better information flow during its training. Afterward squeeze and excitation block is implemented to intensify the

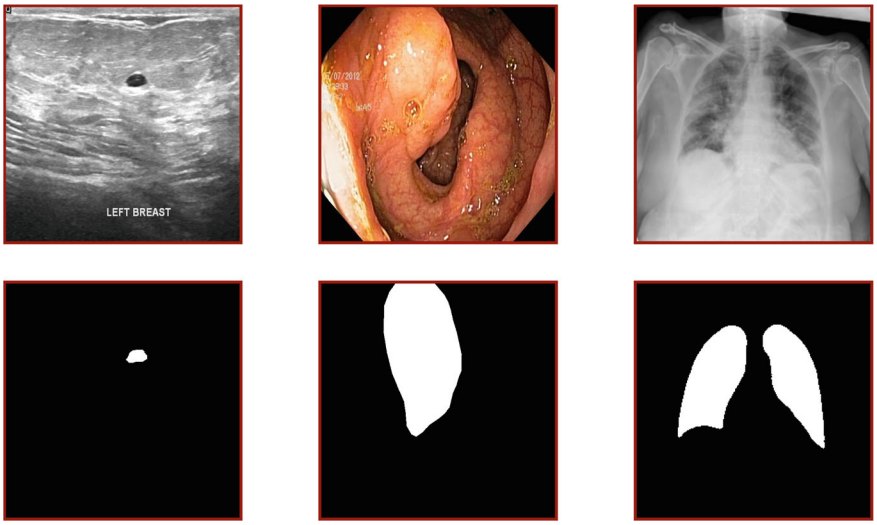

Fig. 1 The images are in the first row along with their GT masks in the second row

focus of network on the relevant features. The decoder blocks uses transposed convolutions for upsampling the image, and concatenates with corresponding encoder blocks with the help of skip connections to retain the fine details. The skip connections ensures that the fine-grained information is preserved throughout the network. The formation of base model 2 is consists of six layers in encoder and decoder blocks. The encoder block comprises of pre-trained architecture ResNet-50. The ResNet-50 model is utilized for extracting features, and also to produce the relevant information from the image. The multiple layers of encoder progressively downsamples the input image with the help of convolutional and max pooling layers. The number of filters increases with each layer to encapsulate the complex features. In this architecture, there is one bottleneck layer in between encoder and decoder block. The bottleneck layer is the deepest layer in the network that captures the most compressed representation of the input data. Its primary purpose is to process the deeply encoded features extracted by the encoder, and helps the network to understand the abstract representations of the data, before the decoder subnetwork reconstructs the segmentation map. The decoder blocks in the network are corresponding to each encoder block. The layers of decoder block progressively upsample the feature maps using transposed convolutions and concatenate them with the feature maps from the encoder layers to recover spatial details.

The base model 1 and 2 are trained separately, and after training predictions are made using both the models. The predictions from both models are treated as features and are concatenated to train the final model called meta-learner model. In this work, meta-learner model is developed by the stacking approach of EL. The meta-learner model is a simple CNN, designed to take stacked predictions from the base models and it refines the predictions to generate the final segmented mask. The meta-learner

model is composed of input layer, that accepts $3D$ input in form of height, width, and channels. First convolution layer applies 64 filters with size of kernel (3, 3), and ReLU activation function. After first layer, the output will have a shape of (height, width, 64), which implies that there are now 64 features for each spatial position. The second layer of convolution has 128 feature maps for each spatial position, with same kernel size and activation function. Thus, now the output shape becomes (height, width, 128). The final output layer has 2 filters with a kernel of (1, 1) size, and softmax activation function. In the last layer, the number of filters corresponds to the type of segmentation task, and in this work binary segmentation is performed. The kernel in output layer acts like a linear classifier applied at every pixel location as it allows the model to aggregate the 128 learned features from the previous layer into the required number of output channels. Hence, the segmented masks from the meta-learner model are more accurate. The visual representation of the proposed model is shown by the block diagram in Fig. 2.

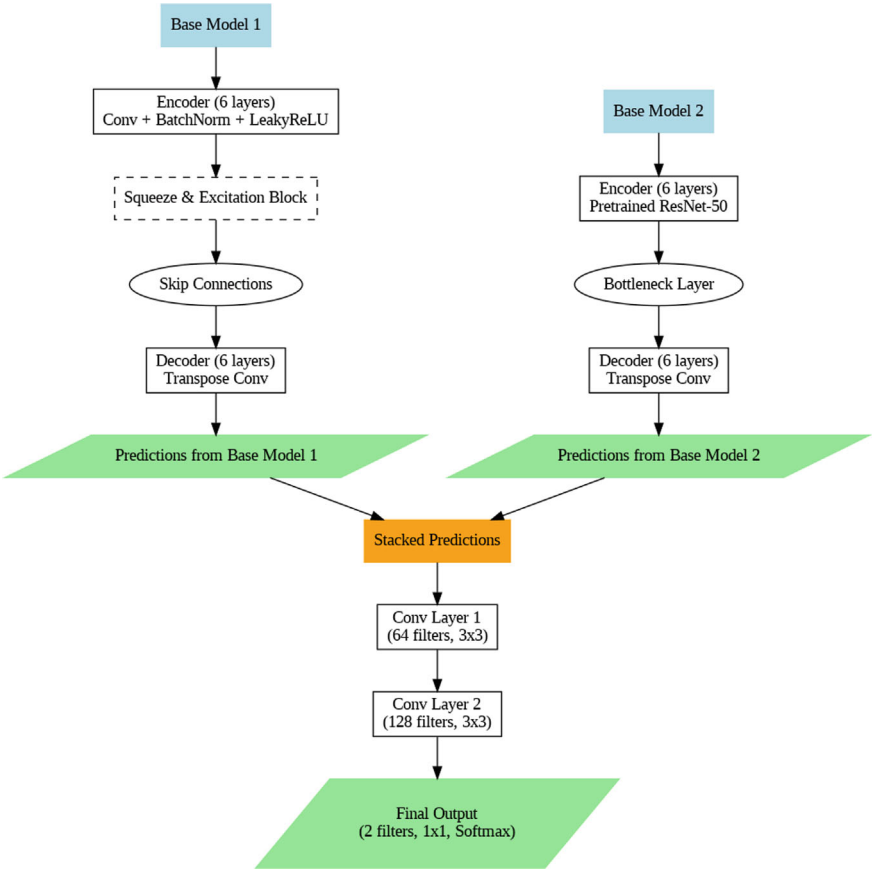

Fig. 2 The block diagram of proposed framework

2.3 Evaluation Metrics

The evaluation metrics are essential parameters for performance analysis of the proposed ensemble model for semantic segmentation. The following metrics are utilized in the presented work:

- Accuracy (Acc) It is the measure of how well a classifier properly assigns labels to data. The value is calculated using true positive (tp), false positive (fp), true negative (tn), and false negative (fn). Thus, the formula is,

$$Acc = \frac{tp + tn}{tp + tn + fp + fn}$$

- Recall (Re) It quantifies the ratio of correct outcomes from the total number of positive outcomes, and it is often referred as sensitivity. Re is given by,

$$Re = \frac{tp}{tp + fn}$$

- Precision (Pr) This metric that evaluates the correctness of a model by calculating the proportion of correct positive predictions out of all the positive predictions produced. It is define as,

$$Pr = \frac{tp}{tp + fp}$$

- Mean Intersection over Union ($mIoU$) It measures the extent of overlap in between the sets. The degree of overlap is determined by the ratio of the intersection size to the union size. $mIoU$ is defined as,

$$mIoU = \frac{tp}{tp + fp + fn}$$

3 Experiments, Results and Discussions

3.1 Experimental Setup

The Keras framework [19] was employed for developing the proposed model, with TensorFlow [20] serving in the back end. The experiments are performed on Google Colab having NVIDIA Tesla K80 GPU with 12GB of VRAM. All three datasets are divided into 8:2 ratio for training and testing. Further, the training dataset is split into 8:2 ratio for training and validation. The images from all three datasets are scaled to 256 $\times$ 256 pixels, and then normalized. The normalized images are given into the

Table 1 Evaluation metrics values for breast ultrasound dataset

Model names	Evaluation metrics			
	Acc	Pr	Re	$mIoU$
Ensemble model	0.9030	0.8503	0.7965	0.6680
Base model 1	0.7343	0.7083	0.6220	0.6378
Base model 2	0.7869	0.7042	0.6141	0.6291

Table 2 Evaluation metrics values for CVC-ClinicDB dataset

Model names	Evaluation metrics			
	Acc	Pr	Re	$mIoU$
Ensemble model	0.9076	0.8734	0.8203	0.6939
Base model 1	0.8965	0.7065	0.7078	0.6088
Base model 2	0.8965	0.7734	0.7203	0.6502

proposed ensemble model, which is trained using a batch size 8 and 30 epochs. The proposed model employs Adam optimiser and 0.001 learning rate. Furthermore, the loss function implemented in the ensemble architecture is categorical crossentropy .

The experiments in this work is performed on three variety of datasets. The results are evaluated on ensemble model, as well as on base model 1 and 2. The representation of the results are both qualitative and quantitative. The qualitative results are presented by predicted segmented masks, and quantitative results are shown by evaluation metric values.

3.2 Results and Discussions

In the first experiment, the proposed ensemble model is trained and tested on breast ultrasound dataset. The model is trained with 499 and validated with 125 images. The ensemble model is evaluated with 156 test images, and their quantitative result are shown in Table 1. The results indicate that the ensemble model has maximum value for all four evaluation metrics. Among the base models, base model 1 has larger values except for the Acc evaluation measure. The second experiment is conducted on the CVC-ClinicDB dataset, and for this dataset the training, validation, and test images are 392, 98, and 122, respectively. The evaluation of the proposed model on the CVC-ClinicDB dataset is shown in Table 2, and it shows that the ensemble model performs comparatively better than the base models. Although, the base model 2 has higher metric values than base model 1. The last part of experiment is performed on the CXR image datasets, and in this case, ensemble model is trained on 88 CXR images. Later, the model is validated and tested on 22 and 28 images, respectively. Thus, the sample of CXR dataset is smaller as compare to other datasets. The results of

Table 3 Evaluation metrics values for dataset of CXR images

Model names	Evaluation metrics			
	Acc	Pr	Re	$mIoU$
Ensemble model	0.8485	0.6324	0.7867	0.6534
Base model 1	0.8327	0.6598	0.6031	0.6176
Base model 2	0.7637	0.6932	0.7562	0.6095

third experiment is given in Table 3. Table 3 specifies that for evaluation metric Acc, Re, and $mIoU$, the ensemble model has maximum value. Moreover, for Pr metric, base model 2 has comparatively higher value. Therefore, the quantitative results yields that the proposed ensemble models outperforms the base models. However, for CXR dataset the precision value is larger for base model 2, and the quantity of CXR dataset is quite small. Thus, the conclusion is derived that the proposed ensemble model requires a sufficient amount of dataset to give accurate results. In addition, we analyze the evaluation metric values of ensemble model across all three datasets, and it gives that for CVC-ClinicDB dataset the values are highest. Hence, the proposed model is more efficient on the GI tract images.

The qualitative results of all three experiment is displayed in Fig. 3. Fig. 3 shows a sample of predicted segmented mask by the proposed ensemble model on all three variety of datasets. In this figure, the first row is sample of normalized original images and the second row is of corresponding predicted masks. The first column is of breast ultrasound image, the second column represents CVC-ClinicDB image, and the third column corresponds to CXR image. The predicted segmented mask of breast ultrasound and GI tract images are very much accurate. However, the predicted mask of CXR image requires more improvement, and it can be achieved by increasing the amount of training dataset. Furthermore, learning curve of the ensemble model across different dataset is given in Figs. 4, 5, and 6. These demonstrates that the ensemble model performance on the training and validation dataset by loss measures. The training loss curve in Figs. 4, 5, and 6 shows the ability of the ensemble model to fit the training data, and the validation loss curve shows the capacity of the proposed model to fit the new dataset. Both training and validation loss curve decrease with increase in the number of epoch in all three datasets. Thus, it indicates the efficacy of the proposed ensemble model.

Now, in order to validate the proposed ensemble model a comparison study is conducted. For this purpose, four different models UNet++ [21], DUCK-Net [10], ResUNet++ [8], and sDDANet [22] from the literature are considered. All these models are evaluated on the same environment as of proposed model. The models are compared precisely on the basis of $mIoU$ evaluation metric, because it is the benchmark for evaluating segmentation tasks. The comparison is performed on the CVC-ClinicDB dataset, and are presented through bar graphs. The bar graph is

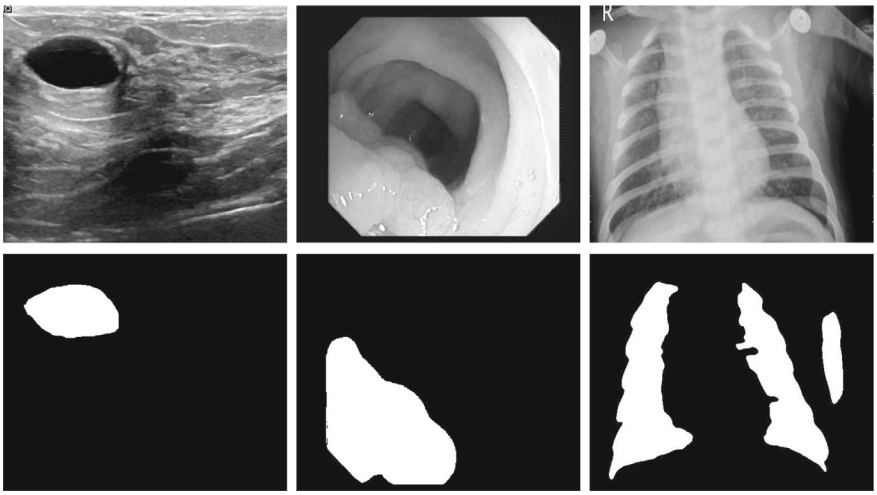

Fig. 3 Sample of predicted segmented mask of all three datasets. First column is of breast ultrasound image, the second column represents CVC-ClinicDB image, and the third column corresponds to CXR image, with their segmented masks

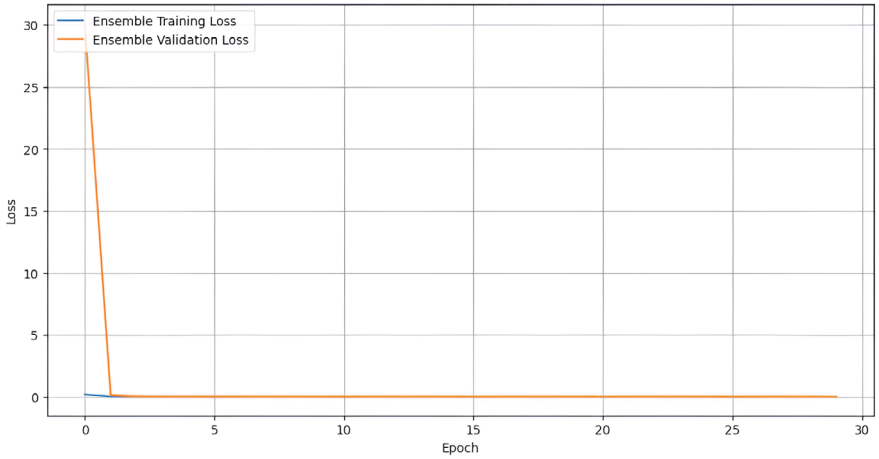

Fig. 4 Learning curve of ensemble model on breast ultrasound dataset

given by Fig. 7. The figure illustrates that the proposed model has highest $mIoU$ value (0.9076) in compare to all other models. Among, four comparison model DDANet [22] has highest $mIoU$ value of 0.8562. Thus, the proposed model has approximately 0.0514 value more than the DDANet. Hence, the proposed ensemble model is efficient than the other models available in literature.

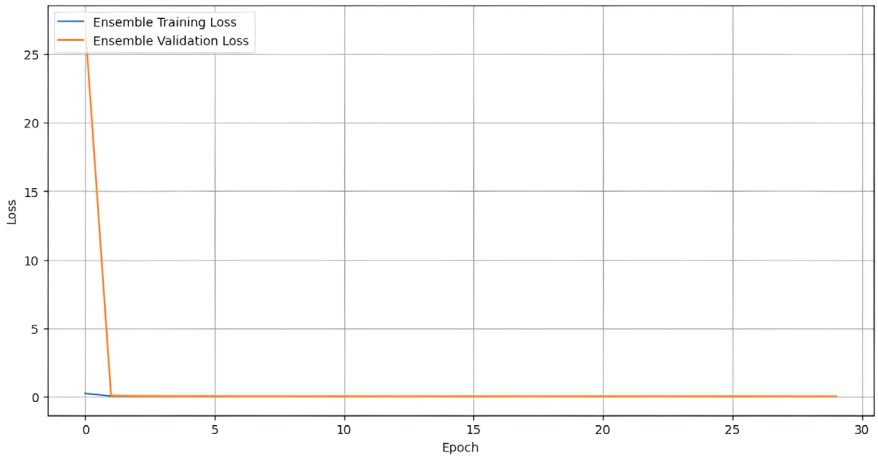

Fig. 5 Learning curve of ensemble model on CVC-ClinicDB dataset

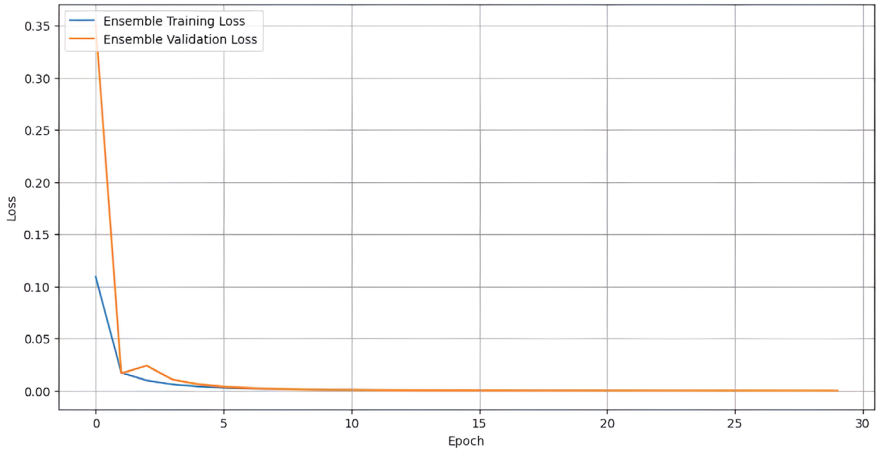

Fig. 6 Learning curve of ensemble model on CXR dataset

4 Conclusions

In the presented work, an ensemble model has been proposed to perform semantic segmentation of three distinct type of medical images. The ensemble model is composed of two base models, and a meta-learner model that integrates the prediction of base model through the stacking approach. The quantitative results of the proposed model are given through evaluation metrics. The values of evaluation measure indicate that the proposed model is highly efficient, and it outperforms the base models in all three datasets. Moreover, the highest accuracy value given by the ensemble model is 0.9076 and it is corresponding to CVC-ClinicDB dataset. The qualitative results

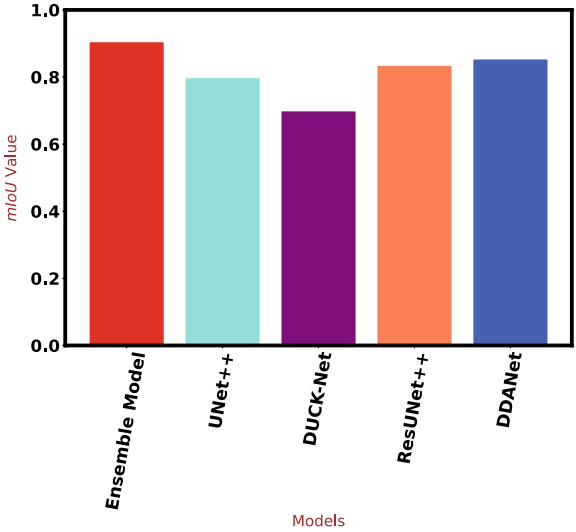

Fig. 7 Comparison of different models through $mIoU$ evaluation metric for CVC-ClinicDB dataset

of the proposed model is shown by the predicted binary mask and learning curves. The predicted masks showcase that the model is more accurate on breast ultrasound image dataset and CVC-ClinicDB dataset. Hence, the conclusion is derived that the proposed ensemble model performs comparatively better on GI tract images. In the future work, the data augmentation technique and k-fold cross validation could be applied to generalize the performance of proposed model on all kind of datasets. Furthermore, the segmentation can be performed on the real time images.

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Machine Learning-Based Diabetic Prediction on Indigenous Shakti Processor for Edge Health Monitoring

P. C. Muhammed Raees, M. R. Akshayraj , G. Lakshminarayanan, Varun P. Gopi, and G. R. Gangadharan

Abstract With the increasing need for IoT-based solutions in healthcare, there is a critical need for energy-efficient, low-power processors capable of executing machine learning (ML) models for real-time health monitoring at the edge level. This paper presents the first-of-its-kind analysis of the Indigenous RISC-V Shakti processor, specifically evaluating its potential as an edge-based health monitoring solution through machine learning-driven diabetic prediction. The Shakti processor is assessed using the Shakti SDK framework. A performance comparison is done with the PYNQ-Z2's Zynq processor regarding execution efficiency and latency of various machine learning models under different optimization techniques, including quantization and dimensionality reduction via Linear Discriminant Analysis. The Zynq processor is evaluated using Python language in the PYNQ framework. The Shakti's use of LDA and quantization optimization techniques led to significant improvements in the performance of ANN and SVM models, reducing the processing time by nearly 30% compared to the ZYNQ SoC. This research underscores the Shakti processor's viability for low-cost, real-time diabetic prediction at the edge level, advancing scalable healthcare solutions and contributing to the development of self-reliable, indigenous IoT devices.

Keywords Diabetic prediction · RISC-V · Shakti processor · Soft core processors · Machine learning

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1 Introduction

Approximately 537 million adults worldwide suffer from diabetes, a major global health epidemic that is expected to increase to 783 million by 2045. India, in particular, presents a severe challenge, ranking second globally in diabetes prevalence, with over 77 million diagnosed cases and millions more undiagnosed or at high risk. Because of the high cost, lack of mobility, and inefficiencies of current healthcare solutions, access to early detection and continuous monitoring—which are crucial for managing diabetes and preventing complications—remains restricted, especially in areas with little resources.

Edge computing offers a transformative solution by enabling real-time, localized data processing, reducing dependence on continuous cloud connectivity. Diabetes prediction is made more accessible by incorporating machine learning (ML) algorithms into edge devices, which enable rapid analysis and prediction right at the point of care. However, the deployment of ML models on constrained edge devices requires processors that can handle the computational demands of these algorithms while maintaining low-power consumption, low latency, and affordability [5–9].

The advent of indigenous hardware platforms, such as India's Shakti processor, based on the open-source RISC-V architecture, offers a promising pathway for addressing these challenges. Designed for low-cost, energy-efficient, and customizable applications, the Shakti processor holds significant potential for resource-constrained environments, aligning with India's aspirations for self-reliance in healthcare technology. This paper presents a comprehensive evaluation of the indigenous Shakti processor as a potential solution for edge-based healthcare monitoring, with a particular focus on machine learningâŁ“driven diabetes prediction, as illustrated in Fig. 1. The Shakti processor is evaluated using the C-code within the Shakti SDK framework, with its performance compared against the PYNQ-Z2's Zynq processor, which is evaluated using the Python language within the PYNQ framework [1–4].

By examining key performance metrics such as execution efficiency and latency, this study evaluates the feasibility of deploying Shakti-based edge devices for real-time diabetic prediction. Optimization techniques such as dimensionality reduction through Linear Discriminant Analysis (LDA) and quantization are applied to enhance the performance of the models, particularly in the ANN and SVM algorithms. The findings reveal that Shakti's optimized performance demonstrates competitive latency improvements, particularly in the ANN model, reducing processing time by nearly 30% compared to the ZYNQ SoC.

This research aims to demonstrate the potential of the Shakti processor as a low-cost, efficient solution for real-time diabetic prediction at the edge. By showcasing the performance of indigenous RISC-V platforms in healthcare applications, this work contributes to the development of scalable, affordable healthcare solutions that can be deployed in resource-limited areas, advancing both the accessibility of healthcare and India's digital sovereignty. Ultimately, this research highlights the promise of RISC-V-based platforms for transforming health monitoring systems, offering a path

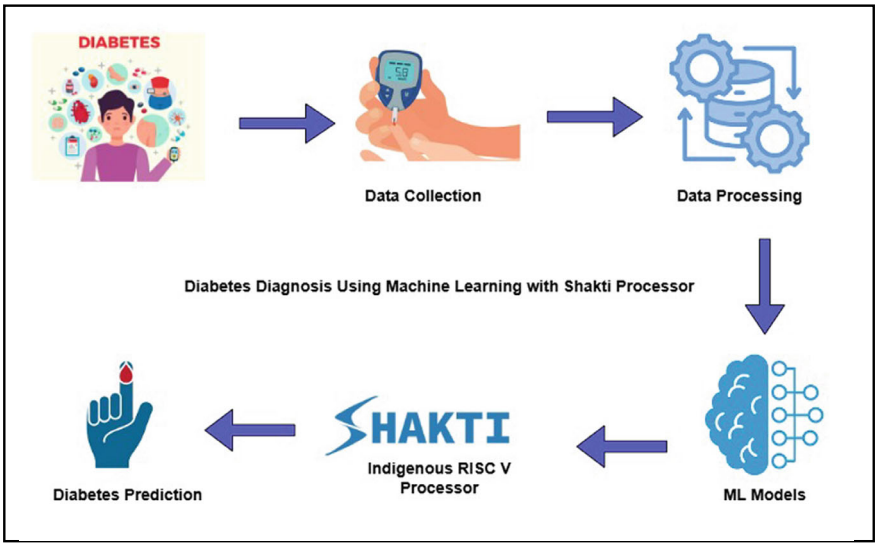

Fig. 1 Shakti-based diabetic prediction using ML techniques

to sustainable, edge-based healthcare technologies for global implementation. In Sect. 2, a detailed literature review on various machine learning models for diabetic prediction is done. Section 3 will discuss the machine learning models and Sect. 4 will detail the hardware implementation part. Results and discussions are included in the Sect. 5.

2 Background

Diabetes is a chronic disease that affects millions of people around the world, with its prevalence steadily increasing. Early prediction is essential for effective management and prevention of long-term complications of diabetes. Machine learning (ML) models have shown significant promise in predicting diabetes by analyzing patient data, including blood glucose levels, body mass index (BMI), age, and family medical history. Several models, including Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost (AB), Gradient Boosting (GB), and Artificial Neural Networks (ANNs), have been widely applied to predict the onset of diabetes. However, the real-time deployment of these models on hardware platforms poses challenges, especially concerning power consumption, latency, and accuracy.

Several recent studies provide information on different ML algorithms for diabetes prediction:

1. Various ML Models for Diabetes Prediction: In [11], five different models were developed, including linear SVM, radial basis function kernel SVM, KNN, and Artificial Neural Networks. The findings showed high performance across models, with KNN and linear SVM excelling in identifying diabetes.
2. Hybrid Models for Enhanced Diagnosis**:** Researchers in [12] proposed a hybrid model for diagnosing type 2 diabetes (T2ML), which involved data cleaning, feature selection via XGBoost (XGB) classifiers and Random Forest (RF), and Logistic Regression clustering. This approach resulted in a more reliable classification.
3. Evaluating Classifier Performance**:** Maniruzzaman et al. [16] tested classifiers, including RF, Naive Bayes (NB), AdaBoost (AB), and Decision Tree (DT), on key diabetes risk factors using accuracy and AUC as metrics. RF and Logistic Regression (LR) performed well in predicting diabetes, especially when age, blood pressure, cholesterol levels, and BMI were primary indicators.
4. Comparative Analysis of ML Models**:** The authors of [17] compared popular regression models like Glmnet, RF, XGB, and LightGBM. A dataset of 111 features was reduced to 58 for better performance. The study confirmed that updating models with new data can improve prediction accuracy and variable ranking consistency.
5. Boosting Classifiers for Optimal Prediction**:** Study [11] employed an ensemble model combining XGBoost and AdaBoost to achieve higher accuracy on the PIMA diabetes dataset, emphasizing the value of pre-processing and feature selection.
6. Pre-processing Techniques to Improve Accuracy**:** In [12], various ML algorithms, including LR, DT, NB, Gradient Boosting, RF, KNN, and SVM, were assessed. The authors found that label encoding and normalization significantly improved SVM performance for diabetes prediction.
7. Ensemble Methods for Improved Accuracy**:** In [13], the authors used bootstrapping to enhance accuracy before applying KNN, NB, and DT models to the PIMA dataset. DT showed the highest accuracy, especially after pre-processing, suggesting ensemble techniques for chronic disease detection.
8. Meta-Analysis on Diabetes Prediction Models**:** Khanam and Foo [15] analyzed ML models for diabetes prediction using SVM and LR, among other methods. They concluded that neural networks (NNs) with two hidden layers offered better performance, when compared to previous models.
9. Evaluating Ensemble and Semi-supervised Models: Study [20] explored semi-supervised learning methods, such as self-training, co-training, and label propagation, using a private dataset. The authors highlighted that these methods, particularly self-training, enhance performance when labeled data is scarce.

The rapid increase in diabetes cases globally emphasizes the need for real time, proactive healthcare solutions. Traditional healthcare systems often rely on cloud computing, which can introduce delays, security concerns, and high transmission costs. Edge computing, by contrast, enables the deployment of ML models directly on local devices, offering real-time predictions with lower latency and enhanced

privacy. Indigenous RISC-V platforms like Shakti processors are well-suited for such applications, providing a low-power, customizable solution for diabetic prediction.

An innovative RISC-V-based architecture created in India, the Shakti processor is tailored for edge applications and provides special benefits for machine learning-based health monitoring. Shakti offers an open-source instruction set architecture (ISA) that can be customized to satisfy the unique computational requirements of machine learning models, in contrast to conventional computing platforms. Because of its adaptability, it is perfect for implementing resource-constrained machine learning models for diabetic prediction in settings where real-time performance, cost effectiveness, and power efficiency are crucial.

This is a first-of-its-kind study demonstrating the potential of domestic RISC-V platforms for healthcare applications is the assessment of diabetes prediction on the Shakti processor. The advancement of low-power healthcare solutions, especially in underserved and rural areas, depends on this research. This work shows a scalable, economical, and private method of healthcare monitoring that uses the Shakti processor for real time, on-device diabetes prediction. If this strategy proves successful, edge computing may become widely used in the healthcare industry, increasing access to timely and useful health insights-particularly in areas with weak healthcare infrastructure. In addition to demonstrating Shakti's promise in healthcare, this study lays the groundwork for upcoming advancements in edge computing for medical applications.

3 Machine Learning Models for Diabetic Prediction

By analyzing various diagnostic features, machine learning algorithms can efficiently classify individuals as diabetic or non-diabetic, helping healthcare providers with personalized intervention strategies. In this study, we utilize multiple classification algorithms to predict the likelihood of diabetes, such as Random Forest, AdaBoost, Gradient Boosting, Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN), Support Vector Machines (SVMs), and Artificial Neural Networks (ANNs). These models were selected for their diverse computational strengths, interpretability, and suitability for edge health monitoring systems. To precisely forecast the onset of diabetes, each model analyses important health indicators like age, BMI, insulin levels, and glucose levels. After data pre-processing, each model's performance is assessed to identify the best accurate method for edge-based, real-time health monitoring applications, enabling the provision of actionable insights with the least amount of latency.

3.1 Dataset Components

The PIMA Indian diabetes dataset is an open-source dataset that was initially gathered by the National Institute of Diabetes and Digestive and Kidney Diseases and is used in this study. Each of the 768 items in this dataset represents diagnostic data for PIMA Indian women over the age of 21. Diabetes status is a binary target variable that indicates whether diabetes is present (1) or not (0). The main objective is to forecast the onset of diabetes based on a variety of health-related characteristics. [13].

The dataset comprises eight key features that serve as inputs to machine learning (ML) models aimed at binary classification. These features include:

1. Age: Patient's age, in years.
2. Glucose: Plasma glucose concentration two hours after an oral glucose tolerance test (often called "2-hour postprandial blood sugar level").
3. Insulin: Blood insulin concentration in μU/mL.
4. Blood Pressure: Diastolic blood pressure (measured in mm Hg).
5. Pregnancies: Total number of pregnancies the patient has had.
6. Skin Thickness: Measurement of triceps skinfold thickness in mm.
7. BMI: Body Mass Index, calculated as weight (kg) / height (m^2).
8. Diabetes Pedigree Function: A function that scores the likelihood of diabetes based on family medical history.

The target variable, named Outcome, is a binary class variable with values of 0 or 1, where 1 indicates the presence of diabetes and 0 indicates its absence. The distribution of diabetes cases in this dataset reveals that out of the 768 records, 268 individuals were diagnosed with diabetes, representing a 34.9% diabetes prevalence within this cohort.

Several pre-processing techniques were used in the dataset preparation procedure to resolve discrepancies and improve ML model performance. To guarantee reliable model training, class imbalances were controlled and missing variables were substituted by averages. After pre-processing, the holdout method—which saves some data for final model validation—was used to divide the dataset into training and test subsets. Because, the model is assessed on data it hasn't seen during training, this separation facilitates objective evaluation.

3.2 Model Selection and Implementation

Several classification methods, such as Random Forest, AdaBoost, Gradient Boosting, Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machines (SVMs), and Artificial Neural Networks (ANNs), were investigated for diabetes prediction. The unique features of each model affect how well it performs on this dataset. These models were chosen because they have a track record of success in binary classification problems, especially in applications related to healthcare.

The selected classifiers are appropriate for edge health monitoring systems since they vary in terms of interpretability and computational efficiency. For example, by combining several Decision Trees, each of which contributes to the final choice, the Random Forest and Gradient Boosting models provide excellent accuracy and resilience against overfitting. Simpler yet powerful algorithms that offer insights into feature relevance and decision-making pathways are the Logistic Regression and Decision Tree models. K-Nearest Neighbors (KNN), on the other hand, provides a simple yet effective categorization method by operating on the proximity of the data. The most accurate and effective model for implementation in an edge-based mobile application was determined by evaluating each approach after model training. The heatmap and mutual information plots are shown in Figs. 2 and 3.

We have applied the following machine learning models to the PIMA Indian diabetes dataset to compare their efficacy in terms of accuracy, recall, precision, and computational performance when deployed on hardware platforms.

1. **Random Forest**: An ensemble learning method based on multiple Decision Trees. The prediction is made by averaging the predictions from individual trees:

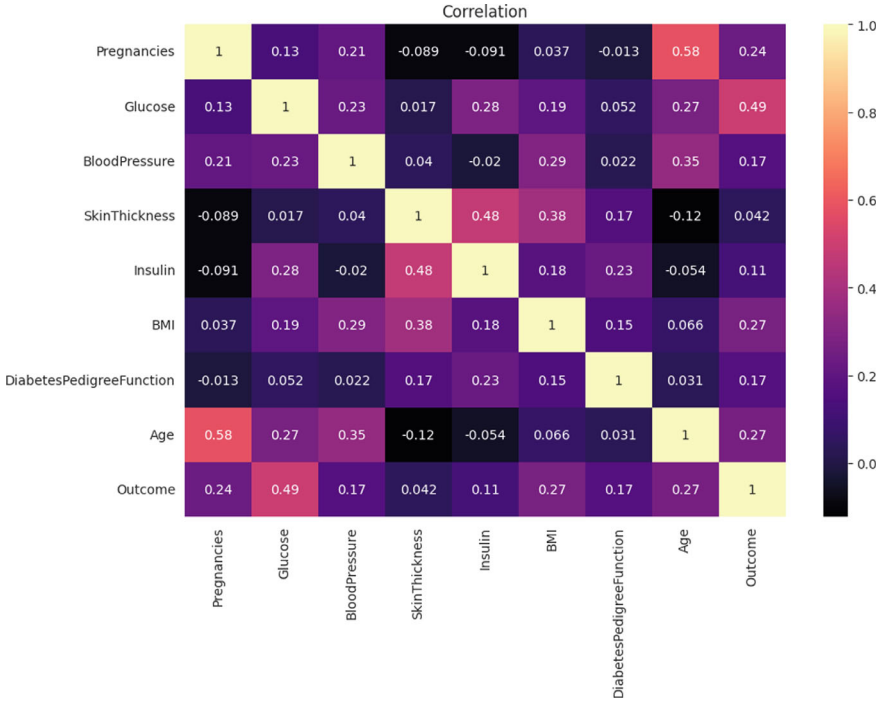

Fig. 2 Heatmap

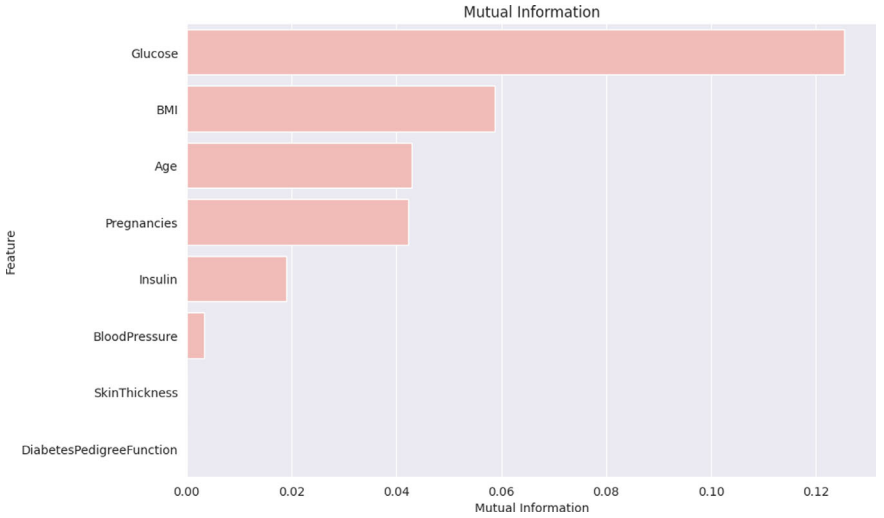

Fig. 3 Mutual information (MI) plot

$$\hat{Y} = \frac{1}{N} \sum_{i=1}^{N} h_i(X) \tag{1}$$

where $\hat{Y}$ is the predicted outcome, N is the number of trees, and $h_i(X)$ is the prediction from the ith Decision Tree.

2. **AdaBoost**: This boosting algorithm focuses on combining weak learners. The prediction is computed as a weighted sum of the predictions of the weak classifiers:

$$\hat{Y} = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(X) \right) \tag{2}$$

where α_t is the weight for the tth classifier, $h_t(X)$ is its prediction, and T is the total number of classifiers.

3. **Gradient Boosting**: Similar to AdaBoost, it builds models sequentially, optimizing a loss function. The prediction can be expressed as:

$$\hat{Y} = \sum_{m=1}^{M} \beta_m h_m(X) \tag{3}$$

where β_m is the learning rate and $h_m(X)$ is the prediction of the mth model.

4. **Logistic Regression**: A baseline linear classifier that predicts the probability of a sample being diabetic based on a logistic function. The output probability is given by:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-z}} \tag{4}$$

where $z = W_0 + \sum_{i=1}^{n} W_i X_i$.

Here, W_i are the weights, X_i are the input features, and z is the linear combination of inputs.

5. **Decision Tree**: A non-parametric supervised learning method. The decision at each node is made based on a feature that minimizes the impurity (e.g., Gini impurity or entropy):

$$\text{Gini}(D) = 1 - \sum_{k=1}^{K} p_k^2 \tag{5}$$

where p_k is the probability of a sample belonging to class k in dataset D.

6. **K-Nearest Neighbors (KNN)**: A non-parametric algorithm that classifies samples based on the majority vote from the k closest neighbors in the feature space. The decision rule can be summarized as:

$$\hat{Y} = \arg\max_{c} \sum_{i \in N_k} I(Y_i = c) \tag{6}$$

where N_k are the k nearest neighbors of the input sample and I is the indicator function.

7. **Support Vector Machine (SVM)**: A supervised learning model that aims to find the hyperplane that best separates the classes in the feature space. The prediction is made based on the following decision function:

$$\hat{Y} = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i \langle X_i, X \rangle + b\right) \tag{7}$$

where α_i are the Lagrange multipliers, y_i are the class labels, $\langle X_i, X \rangle$ is the inner product of the input vectors, and b is the bias term.

8. **Artificial Neural Network (ANN)**: A neural network composed of multiple layers of interconnected nodes (neurons). The network learns to map input features to the output label through a forward pass and adjusts weights during backpropagation to minimize error. The output for a multi-layer neural network is given by:

$$\hat{Y} = f\left(\sum_{i=1}^{n} W_i X_i + b\right) \tag{8}$$

where W_i are the weights, X_i are the input features, b is the bias, and f is the activation function (e.g., ReLU, sigmoid).

The classification reports for the selected machine learning models were generated to evaluate their performance on the diabetes prediction dataset. The primary

Table 1 Accuracy comparison of ML models

Models	Accuracy without LDA	Accuracy with LDA	Accuracy with quantization
Random Forest	0.735931	0.696970	0.753247
AdaBoost	0.740260	0.740260	0.740260
Gradient Boosting	0.748918	0.705628	0.748918
Logistic Regression	0.735931	0.727273	0.735931
Decision Tree	0.714286	0.696970	0.701299
K-Nearest Neighbors	0.701299	0.718615	0.701299
SVM	0.744589	0.735931	0.744589
ANN	0.694800	0.720800	0.733800

objective of this study was to understand each model's behavior and performance on edge health monitoring applications. As observed in Table 1.

3.3 Quantization and LDA for Model Efficiency

Quantization is a technique that reduces the precision of the model's parameters (weights and activations) without significantly compromising accuracy. Quantization is an effective technique for implementing machine learning models on resource-constrained edge devices since it lowers memory consumption and computational effort by employing lower-bit representations for the weights.

Quantization involves mapping the continuous floating-point values of model parameters to lower-bit-width integer values, typically 8 or 4-bit. This leads to a significant reduction in memory requirements and the number of operations needed for inference. When applied to the diabetes dataset, quantization reduces the size of the model and speeds up inference times, especially when running on hardware accelerators like the Shakti processor.

From the results in Table 1, we can see that after applying quantization, the performance of models like ANN performed better. Quantization also offers notable advantages in terms of energy efficiency. By reducing the bit-width of the weights and activations, the number of required multiplications and additions decreases, leading to lower power consumption during inference. This makes quantization particularly valuable for edge devices like the Shakti processor, which is designed to operate efficiently in resource-constrained environments as illustrated in Fig. 4.

In terms of resource optimization and energy efficiency for edge devices, the seamless integration of quantization and Linear Discriminant Analysis (LDA) with machine learning models complements the Shakti-based design processes. With its focus on energy-efficient processing and lightweight edge computing, the Shakti processor can take advantage of these strategies to improve the functionality of

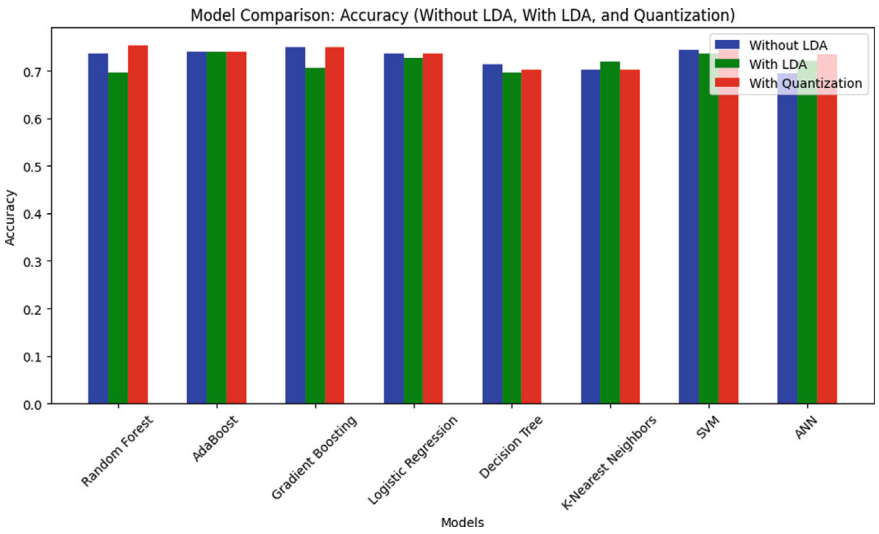

Fig. 4 Model comparison with LDA and quantization

medical applications. LDA helps in dimensionality reduction, helping to minimize the amount of data that needs to be processed, which in turn reduces the computational load and accelerates inference times on hardware. Quantization further enhances this by converting high-precision models into lower-bit representations, drastically cutting down memory and power requirements while still maintaining sufficient model accuracy.

Quantization is particularly effective in Artificial Neural Networks (ANNs), where the model's large number of parameters can significantly impact both memory usage and energy consumption. By reducing the precision of the weights and activations, quantization helps make ANN models more suitable for deployment on edge devices, even with limited resources. This ensures that the ANN, despite being a complex model, can be executed efficiently on the Shakti processor without sacrificing too much accuracy.

By implementing these techniques in the Shakti-based design flow, we can efficiently map complex machine learning models, such as Random Forest, Gradient Boosting, and ANN, onto hardware, making them viable for real-time health monitoring tasks on edge devices. These optimizations enable more compact and energy-efficient models, ensuring that the Shakti processor can handle the demands of healthcare systems without compromising on performance. This holistic approach to model deployment not only improves computational efficiency but also ensures that edge devices remain effective and sustainable for long-term use in health monitoring applications.

Table 2 Comparison of shakti and ZYNQ-7000 platforms

Feature	Shakti platform	ZYNQ-7000 platform
Processor	RISC-V E-Class (32-bit)	Dual ARM Cortex-A9 (Dual-core)
Clock frequency	50 MHz (Up to 200 MHz)	50 MHz
Memory	256 MB DDR	512 MB DDR3
SoC	PARASHU (E32-A100)	ZYNQ-7000
Onboard features	4 KB ROM, 128 KB BRAM	FPGA, ARM processor, programmable logic
Supported OS	Linux, FreeRTOS, Zephyr	PYNQ Framework
Language	C	Python
IDE	Platform.IO	Jupyter Notebook
Key applications	IoT, Automotive, Smart Cards	Embedded systems, Edge computing, Healthcare

4 Hardware Implementation

The machine learning models are implemented on Artix-7 100T Arty FPGA Evaluation Kit, equipped with Shakti 1.2.0 SoC and the results are compared with the general platform of AUP PYNQ-Z2 FPGA with Zynq-7000 SoC. These platforms were selected based on their ability to support low latency, high-throughput processing while meeting the stringent energy and computational constraints of edge devices, particularly in healthcare applications. A detailed comparison is provided in Table 2.

The Artix-7 100T Arty FPGA Evaluation Kit features the Shakti 1.2.0 SoC, a custom-designed processor based on the RISC-V architecture. The Shakti E-class processor, which is a 32-bit microprocessor, is optimized for low-power, resource-constrained environments, making it ideal for edge applications in healthcare, automotive, and IoT domains.

The E-class processor in the Shakti SoC is designed to support real-time applications and low-power operations, with an operational frequency of less than 200 MH**z** on silicon. The Shakti 1.2.0 SoC is particularly well-suited for applications requiring efficient energy usage while still providing the computational power needed for machine learning inference tasks. The Shakti 1.2.0 SoC includes 4 KB of ROM and 256 MB of DDR, with 128 KB of RAM specifically dedicated to fast memory access for computational tasks. The platform operates at a clock speed of 50 MHz, providing a stable and energy-efficient solution for edge applications. The development for this platform is carried out in **C**, allowing for direct low-level hardware interaction and optimization for energy-efficient processing.

The AUP PYNQ-Z2 FPGA is based on the Zynq-7000 SoC, which features a dual ARM® Cortex™-A9 MPCore™ processor. This processor, coupled with CoreSight™ debugging technology, provides a flexible platform for deploying machine

learning models with the ability to handle complex computational tasks in a resource-constrained environment. The PYNQ framework, an open-source project from AMD, facilitates the use of this platform by enabling high-level programming with Python, particularly through Jupyter Notebooks. This platform allows designers to leverage the benefits of programmable logic and microprocessors to build more efficient and capable electronic systems.

The PYNQ-Z2 platform is equipped with 512 Mbyte DDR3 memory, which offers sufficient memory bandwidth for running machine learning models while maintaining a low latency. The system can operate at a clock speed of 200 MHz, but here for fair comparison with the Shakti processor, the board is tuned to 50 MHz, providing reliable performance for edge applications. The flexibility of Python enables rapid prototyping of machine learning models, making it an ideal choice for experimenting with different algorithms, including those used for diabetes prediction in healthcare.

To implement the machine learning models on the Shakti Processor, we utilized PlatformIO, an integrated development environment (IDE) extension for Visual Studio Code used for rapid prototyping that streamlines the development workflow. PlatformIO is a versatile, all-in-one IDE designed for embedded systems, and it now supports SHAKTI processors across major operating systems, including Linux, macOS, and Windows. This compatibility significantly enhances accessibility and ease of development for the SHAKTI platform. By consolidating coding, building, uploading, testing, and debugging functionalities into a single platform, PlatformIO eliminates the need for multiple terminal windows and complex command executions, offering a streamlined process for developers. Moreover, PlatformIO has an extension designed especially for SHAKTI development boards, which makes it a useful tool for integrating and refining our machine learning models within the SHAKTI ecosystem. This method not only speeds up development but also supports our goal of using the SHAKTI architecture to deploy energy-efficient ML models on edge devices with limited resources.

5 Results and Discussion

The results of the machine learning model implementations on both the Shakti and Zynq platforms demonstrate their effectiveness for edge healthcare monitoring applications, as summarized in Table 3 and illustrated in Fig. 5. The Shakti processor showed competitive execution times for the majority of models, including Random Forest, AdaBoost, Gradient Boosting, and SVM; several models even achieved time reductions. AdaBoost and Gradient Boosting on Shakti, for example, demonstrate shorter inference times, which are essential for jobs involving real-time health monitoring when quick decision-making is essential. The Shakti processor's effective performance in this situation demonstrates how well-suited it is for time-sensitive, low-power healthcare applications, especially in settings with stringent power and resource limitations.

Table 3 Comparison of execution time across platforms

Model	Shakti processor (s)	PYNQ-Z2 - ZYNQ processor (s)
Random Forest	0.340	0.355341
AdaBoost	0.185	0.167902
Gradient Boosting	0.360	0.205542
Logistic Regression	0.006	0.006292
Decision Tree	0.005	0.005303
K-Nearest Neighbors	0.030	0.023573
SVM	0.025	0.030661
ANN	0.005	0.006000

In promoting the Shakti processor for edge applications, these results underscore its potential as a resource-efficient platform optimized for machine learning tasks in IoT and healthcare domains. Designed with the energy and memory limitations typical of embedded systems, the Shakti E-class processors effectively balance processing capability with reduced power consumption. Additionally, Shakti's RISC-V architecture provides flexibility and modularity, making it adaptable for specific edge AI applications. The competitive execution times achieved on Shakti, especially in comparison with a high-performance platform like ZYNQ, emphasize its value in resource-constrained environments, making it a promising choice for future developments in intelligent healthcare and IoT edge devices.

These findings highlight the Shakti processor's potential as a resource-efficient platform tailored for machine learning activities in the Internet of Things and healthcare sectors so that it can be promoted for edge applications. With the energy and memory constraints common to embedded platforms in mind, the Shakti E-class processors successfully strike a balance between processing power and low-power consumption. Furthermore, Shakti's RISC-V architecture offers modularity and flexibility, which enable it to be tailored for certain edge AI applications. Shakti's competitive execution speeds, particularly when compared to high-performance platforms like ZYNQ, highlight its usefulness in resource-constrained settings and make it a viable option for upcoming advancements in IoT edge devices and intelligent healthcare.

From a broader perspective, the results highlight a promising direction for the adoption of Shakti processors in domains that require continuous monitoring and real-time data analysis, such as healthcare and smart IoT systems. While the current study showcases the potential of Shakti for machine learning tasks, future research can explore optimizing the processor's architecture further to handle more complex models and larger datasets, paving the way for its integration into more

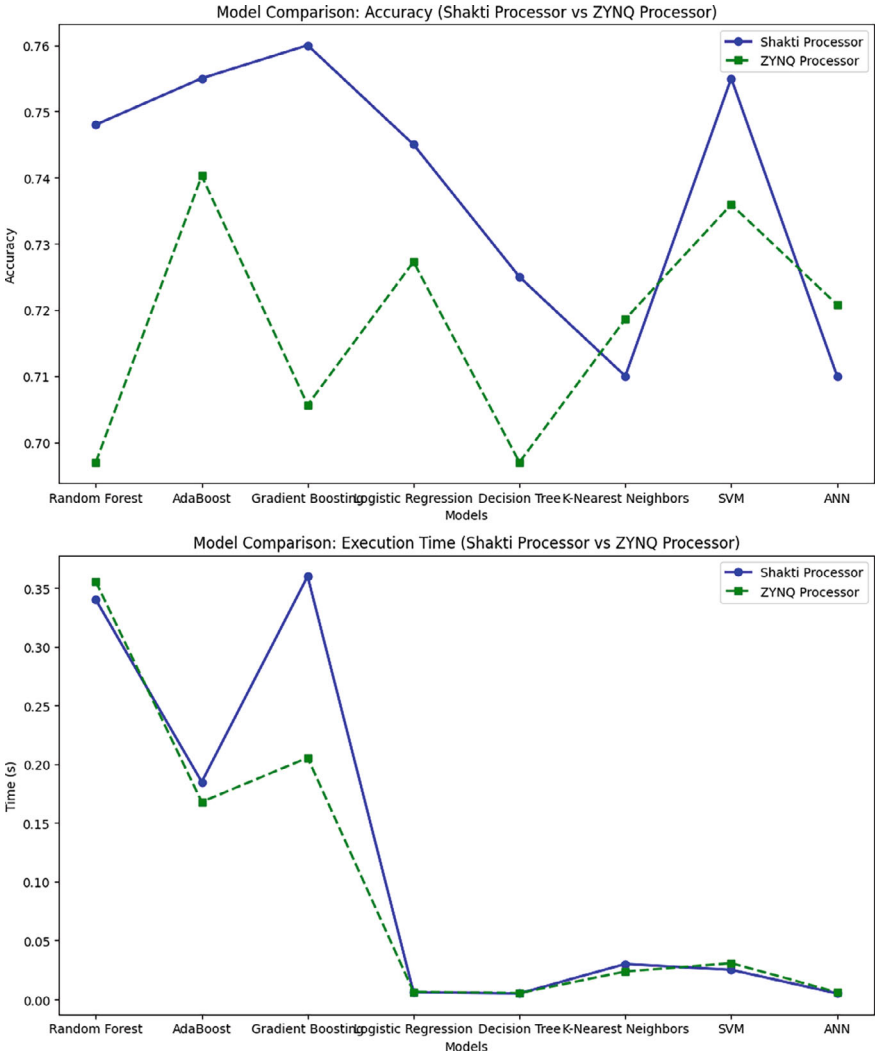

Fig. 5 Model comparison with shakti and ZYNQ processors

sophisticated edge AI applications. Additionally, future work could focus on refining energy-efficient strategies, such as advanced quantization techniques and hardware acceleration, to improve the performance of Shakti in real-time inference scenarios. This approach would help in unlocking the full potential of edge AI for resource-constrained environments.

6 Conclusion

This study shows the Shakti processor's promising potential for implementing machine learning models in energy-efficient, resource-constrained edge devices, especially for healthcare applications. The performance results, particularly in execution time, indicate that Shakti can effectively handle real-time inference tasks, making it a strong contender for low-power environments like wearable devices and IoT-based health monitoring systems. While the ZYNQ platform offers higher processing power, Shakti's ability to deliver competitive results with minimal energy consumption emphasizes the importance of optimizing hardware for specific application needs. Future research should focus on further enhancing Shakti's capabilities, exploring advanced techniques such as model optimization and hardware acceleration to ensure its scalability and effectiveness in more complex edge AI applications.

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Skin Lesion Segmentation Using UNet Based Architectures

Narender Singh and Sushil Kumar

Abstract Image Segmentation has attracted much attention in deep learning, particularly in case of medical field in which it can help in building computer-aided diagnosis systems. Of all the methods that have been designed for image segmentation or salient object detection, UNet and architectures based on it have attracted much attention and have been highly accurate in segmentation tasks. There have been numerous architectures that are based on UNet original architecture, each one employing its own unique adaptation. In this paper, we investigate the usefulness of three UNet-based architectures: UNet3+, Attention UNet and U^2Net in segmentation of Skin lesion images.

Keywords U^2Net · HAM10000 · Skin lesion segmentation

1 Introduction

Artificial Intelligence has been around for several years, but never it have been discussed as largely as it is being discussed nowadays. It has paved the way for solutions to problems which were considered too complex to be solved using the rudimentary computer science methods. Deep Learning has emerged as go to method for solving complex tasks such as Speech recognition, computer vision tasks,generative Artificial models, Natural Language Processing etc. One of promising areas in which deep learning can play a great role is medical diagnosis. Deep learning models when provided with patient data can help doctors in quickly identifying the disease and can help in preparing the treatment. These models can also help in designing automated computer aided diagnosis system which can revolutionize the medical world. Computer aided diagnosis systems can help in early diagnosis of cancer, leading to

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better and early treatment. Cancer is deadliest disease and treatment largely matters on how early it have been detected.

Most common type of cancer is Skin Cancer. Skin cancer involves abnormal growth of skin cells, cause being attributed to exposure to the sun, particularly to ultra-violet rays. Although Skin cancer particularly melanoma is deadly, it can be treated when detected early. In order to build Computer-aided-diagnosis system for skin cancer images, both classification as well as segmentation task are crucial. While Classification helps in figuring out which type of skin cancer a patient is diagnosed with, segmentation can help in reducing the unnecessary details, leading to faster and more advanced image processing.

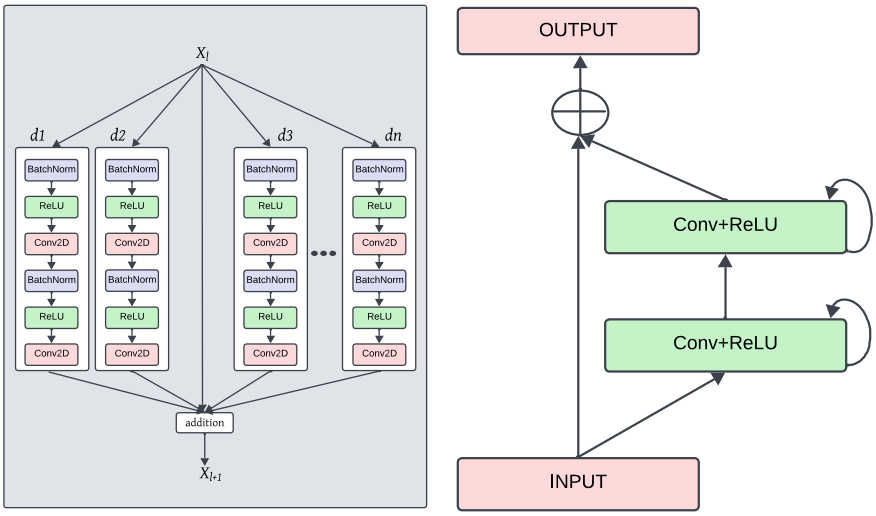
This paper investigates the usefulness of various UNet-based architectures in segmentation of images of skin cancer.

2 Literature Survey

Before the deep learning based methods for segmentation came into picture, there were methods based on local appearance of semantic classes, raw representations, Gaussian filters, random fields etc. Long et al. [10] introduces Fully connected networks which produce class presence heatmaps. Liu et al. added global context to Fully connected network, demonstrating that it reduces confusion. In a remarkable paper in 2015, Ronneberger et al. [14] introduces UNet, a U-shaped network consisting of encoder-decoder networks. UNet make use of concatenation and upsampling instead of unpooling. UNet was used by researchers for variety of segmentation task and thus dominated the area of segmentation research. UNet was followed by UNet++ proposed by Zhou et al. [17] in 2018, which make use of nested dense convolutional blocks and skip connections. UNet++ also introduce the deep monitoring mechanism which make the model more flexible. Although UNet++ was able to surpass UNet in terms of accuracy over a variety of task, it suffer from problem of vanishing gradients. R2UNet proposed by Alom et al. [1] introduces the recurrent residual blocks (Fig. 1b) in UNet. ResUNet was proposed by Diakogiannis et al. [4] which introduces the residual blocks (Fig. 1a) with altrous convolutions. Oktay et al. [11] introduces the attention gates into UNet architecture and showed that it outperform the traditional UNet on problem of segmenting the pancreas.

Huang et al. [5] introduced UNet3+ (Fig. 2), which introduces full-scale skip connections, which reduces the number of trainable parameters while giving better accuracy. UNet3+ retain the idea of deep supervision, however due to full skip connections it is better than UNet++ considering the vanishing gradient problem.

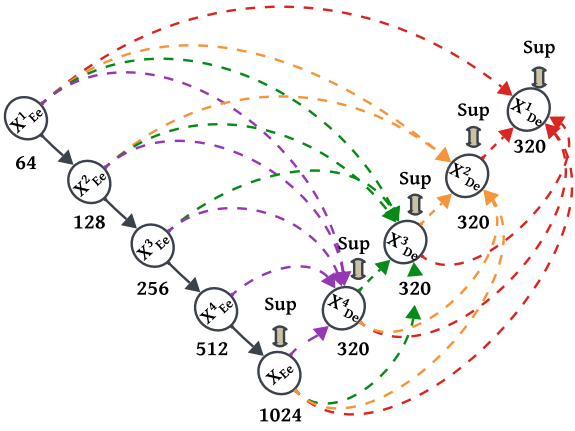
U^2Net was designed by Qin et al. [13] by introducing residual-U blocks (RSU) and using these blocks in a U-shaped architecture. Residual-U block consist of a convolution layer for input followed by UNet-based encoder-decoder architecture and a residual connection. Use of these blocks allowed deeper architecture having lower computational cost. U^2Net achieved competitive results relative to other models for salient object detection.



(a) Residual blocks used in ResUnet. (b) Recurrent Residual block (R2UNet).

Fig. 1 Residual and recurrent residual blocks

Fig. 2 UNet3+ architecture showing skip connections



Various transformers based models have been proposed in literature. TransUnet was proposed by Chen et al. [3] which used transformer as encoder, while CNN was used for feature extraction. TransUnet also introduce the idea of cascaded upsampling (CUP) having numerous upsampling steps. Cao et al. [2] introduce the Swin-UNet, a pure transformer consisting of encoder-decoder, bottleneck and skip connections. Decoder in Swin-UNet consist of Swin Transformer [9] and make use of patch expanding layer.

3 Method

This section involves the details pertaining to what have been done in our experiment.

3.1 Dataset

Dataset utilized in this study is HAM10000 [16] dataset which consists of 10015 dermatoscopic images (Fig. 3) collected over period of 20 years. The images and masks were resized to 256 × 256 pixels. For limiting the computation expensiveness involved with this conversion, the converted image arrays were stored as numpy arrays for use with other architecture. We do not use image augmentation techniques.

3.2 Methodology

Out of various UNet-based architectures, for this study we chose the UNet3+, Attention UNet and U^2Net. Details of individual architecture are as follows:

UNet3+ The UNet3+ model, we used consist of five downsampling and upsampling levels, two convolutional and one convolutional layer per downsampling and upsampling levels respectively. We used ReLU as activation function and used Sigmoid as output activation. Deep Supervision was not used in our architecture. The hyperparameters used were searched through Keras Tuner utilizing the keras variant of hyperband [8] algorithm.

Attention UNet Attention UNet architecture we used consist of five downsampling and upsampling levels. Two convolutional layers per downsampling level and one convolutional layer per upsampling level. Attention gates having *additive* attention and ReLU as attention activation, were used. For output we had used Sigmoid activation and ReLU was used as activation inside the layers.

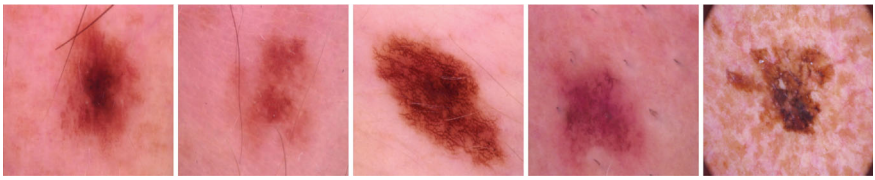


Fig. 3 Sample images of HAM10000 dataset

U^2Net U^2Net architecture consist of six downsampling levels with first four layers built with RSU and last two with RSU-F. Upsampling levels are six with 4 built with RSU while deepest level is built with RSU-F. We used normalization and as with other models, deep supervision was not used.

3.3 Implementation Details

We used train-test split of 0.75–0.25 (random state = 56) i.e. nearly 7500 images were used for training and 2500 images for validation and testing purposes. Keras-unet-collections [15] was used for building all the unet-based architectures. We chose not to use any existing backbone and all model were trained from scratch. Adam [7] optimizer was used with learning rate set to 1e−3. All models were trained for 50 epochs. Cloud services were used for GPUs. Nvidia L4 (24 GB), L40S (48 GB), A10G (24 GB) GPUs were used for our training and for hyperparameter tuning. We used Scikit-Learn [12] for preparation of train-test split and matplotlib [6] for plots.

4 Results and Discussion

4.1 Evaluation Metrics

We used accuracy, dice coefficient as evaluation metric during training.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

where TP, TN, FP, FN are True Positive, True Negative, False Positive and False Negative respectively.

$$\text{Dice} = \frac{2(y \cap y_p)}{y + y_p} \tag{2}$$

where y, y_p denote ground truth and predicted value respectively.

We used Binary Cross Entropy loss as our loss function for training and validation.

$$\text{BCE}(p, y) = -[y \log(p) + (1 - y) \log(1 - p)] \tag{3}$$

After training, Mean IoU was calculated on validation set.

$$\text{IoU} = \frac{y \cap y_p}{y \cup y_p} \tag{4}$$

Table 1 Performance matrix of models

Model	Training		Validation	
	Accuracy	Dice	Accuracy	Dice
UNet3+	98.43	95.83	95.50	90.66
Attention UNet	99.47	98.49	95.62	91.69
U^2Net	98.12	95.02	95.80	91.02

Values shown are in percentage

Table 2 IoU results of different models

Model	IoU
UNet3+	89.43
Attention UNet	89.65
U^2Net	90.11

Values shown are in percentage

4.2 Results

Table 1 shows the results of evaluation of models trained on batch-size of 4. The results are shown in percentage. Table 2 shows the Intersection over Union (IoU) results calculated over validation set after training of the models.

Figure 4 shows the accuracy and dice plot of training of the models. Figure 5 contains prediction by models on images of validation set. Images shown here were taken randomly out of validation set.

4.3 Discussion

Our experiments with models architectures have shown almost similar performance for all three models. Though it seems that U^2Net outperform the other two, due to its excellent architecture design, however it must be noted that the difference in performance is not much large. We find attention-gates in Attention UNet were capturing information better than just skip connections. Attention gates when applied in combination with Residual-U blocks of U^2Net may significantly increase the performance and results in better segmentation of images.

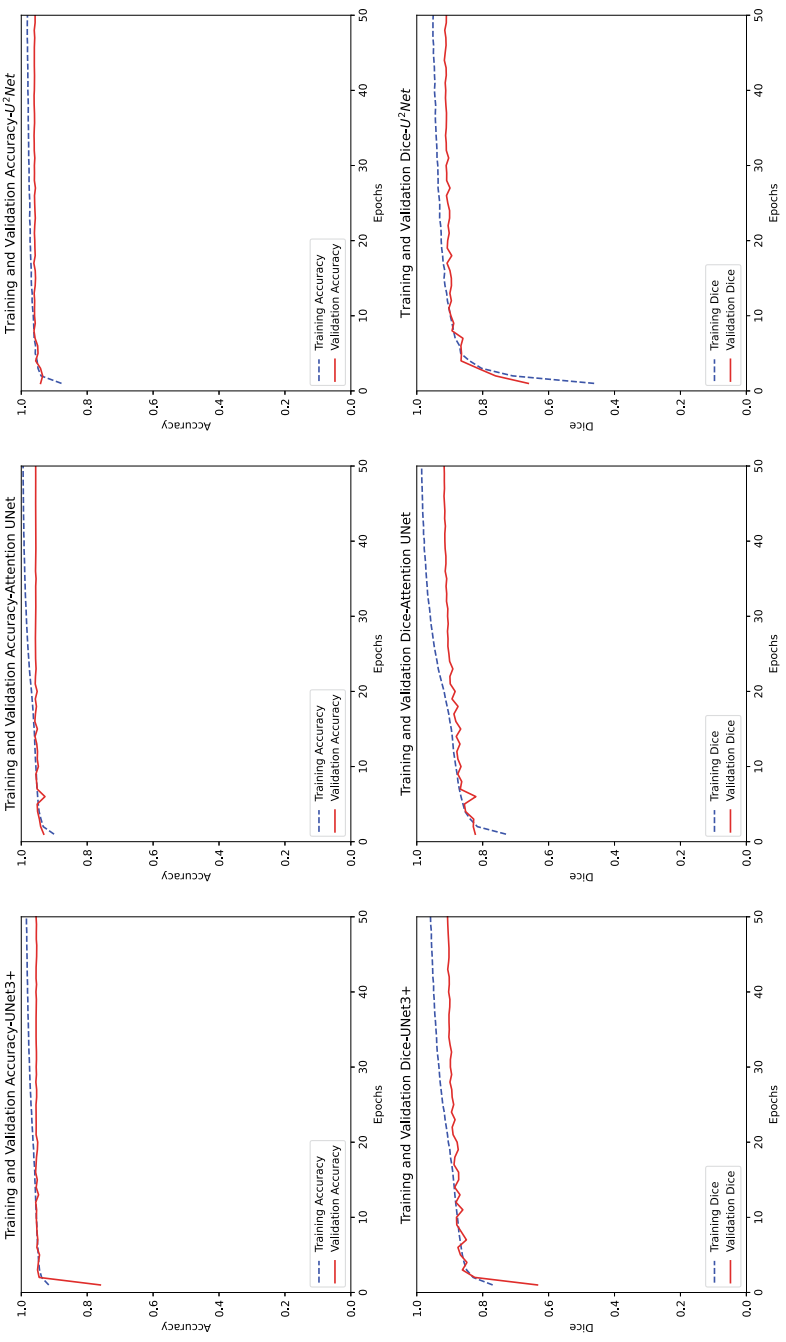


Fig. 4 Dice and accuracy graphs

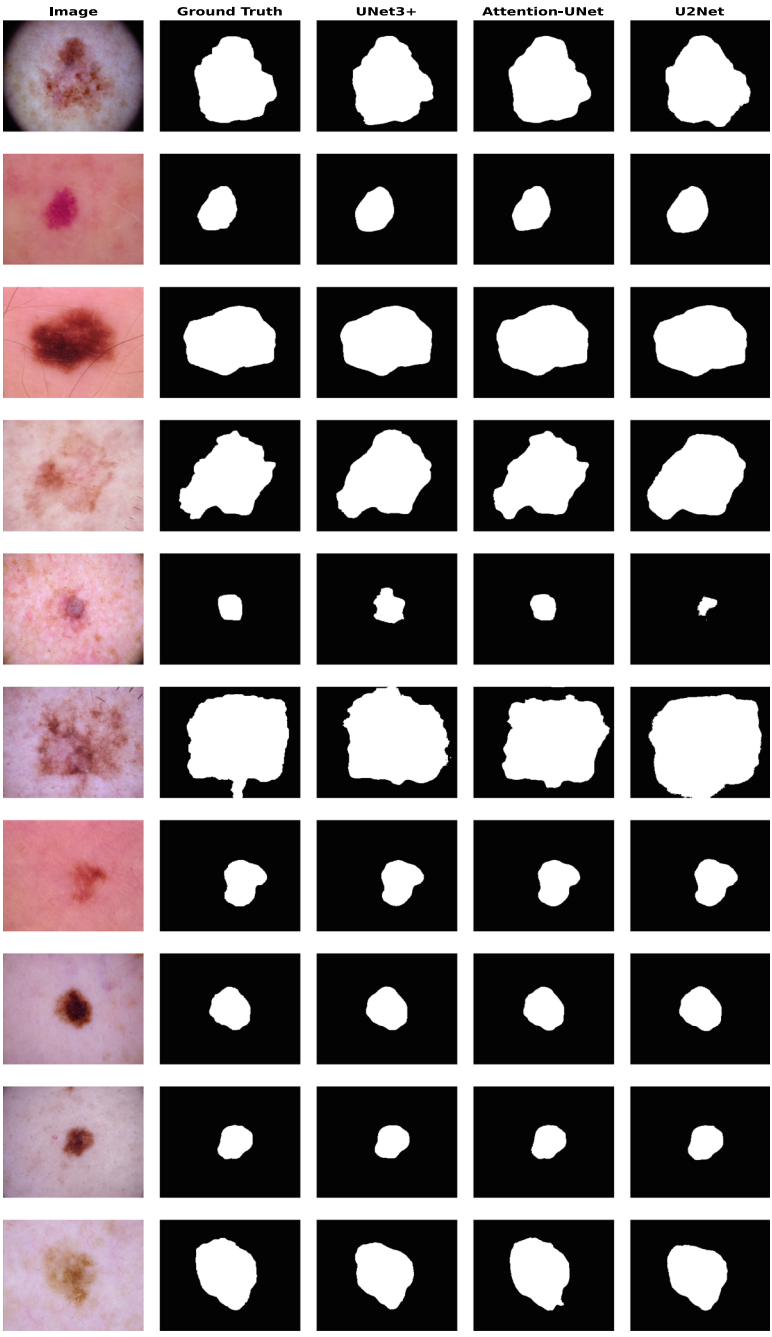


Fig. 5 Segmentation results of different models

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Bridging Molecular Features and Solubility: A Machine Learning Approach for Prediction

Akhilesh Pandey, Adil Mudasir Malla, Mahroosh Banday, Asif Ali Banka, and Mohammad Khalid Pandit

Abstract Aqueous solubility prediction is critical in diverse fields, including drug discovery and chemical engineering, where understanding molecular interactions in solvents can significantly impact formulation and application strategies. This study presents a comprehensive machine learning-based framework for predicting the aqueous solubility of molecules using multiple machine learning models. These include regression- and ensemble-based approaches. Four distinct featurization techniques: Custom descriptor-based defaturization, RDKitDescriptors defeaturizer, extended connectivity fingerprints (ECFP), and MACCS keys were evaluated to transform molecular SMILES strings into numerical features. Three curated datasets (AQSol, ESOL, and PHYS) were utilized for training and evaluation. Along with these datasets, an independent test dataset containing 62 anticancer compounds was also used exclusively for testing purpose. The results showed that molecular descriptors that included physicochemical, structural, and topological properties, outperformed all other methods across multiple metrics (R^2, MSE and RMSE). This study highlights the importance of feature richness and balanced dimensionality in solubility prediction. It also lays a foundation for further exploration of graph neural networks to enhance predictive performance.

Keywords Aqueous solubility prediction · SMILES strings · Machine learning · Regression models · Molecular featurization · Extended Connectivity Fingerprints (ECFP) · MACCS keys · RDKitDescriptors featurizer · Molecular descriptors

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1 Introduction

Solubility is a fundamental physicochemical property. It plays a crucial role in various scientific domains, including drug discovery [5], environmental science [2], and chemical engineering [7]. Traditional experimental methods for determining solubility are often time-consuming, resource-intensive, and limited in performance [16]. The emergence of machine learning (ML) has revolutionized solubility prediction, offering a rapid and cost-effective alternative to experimental approaches [6, 15, 16]. The use of ML in the prediction of solubility is driven by the availability of increasingly large datasets and the development of sophisticated algorithms capable of learning complex relationships between molecular structure and solubility [7, 15]. Recent research has explored a variety of ML techniques for solubility prediction, including: **Traditional ML algorithms** such as Random Forest, XG-Boost, LightGBM, and CatBoost regressor [8, 16] [2, 10, 11, 15, 17]. **Deep learning architectures** such as deep neural networks, graph convolutional networks (GCN) and message-passing neural networks [1, 3, 6, 13]. **Hybrid approaches** combining ML with molecular thermodynamics [2, 7].

These studies have demonstrated the potential of ML to achieve high accuracy in solubility prediction, often surpassing traditional methods and human experts [2, 7, 16]. However, challenges remain in developing robust and generalizable models for solubility prediction [17]. Key considerations include: **(a) Data quality and diversity:** The accuracy and generalizability of ML models are highly dependent on the quality and diversity of the training data [8, 11]. Solubility datasets often suffer from inconsistencies, errors, and limited coverage of chemical space, impacting model performance [2]. **(b) Molecular representation:** The choice of molecular descriptors or fingerprints significantly influences model performance and interpretability [8, 15]. **(c) Model interpretability:** While deep learning models can achieve high accuracy, they often lack interpretability, making it difficult to understand the underlying factors driving solubility [17].

This paper focuses on predicting the solubility of molecules using ML techniques. This research utilizes various featurization techniques to capture molecular features and employs various ML algorithms to develop predictive models. Three different datasets are used to train and evaluate the models. The study aims to contribute to the field of solubility prediction by:

(a) **Assessing the effectiveness of different featurizers for predicting solubility.**
(b) **Comparing the performance of different ML algorithms on various datasets.**

The findings of this research will provide valuable insights into the applications of ML and the use of featurizers for solubility prediction and aid in the development of more accurate and reliable predictive models.

2 Materials and Methods

This paper outlines the process used to predict the aqueous solubility of compounds using machine learning algorithms. The research focuses on utilizing various molecular featurizers and comparing the performance of various regression algorithms. The methodology is divided into three main stages: dataset preparation, featurization, and model training and evaluation.

2.1 Data Collection and Processing

The datasets utilized in this study for molecular solubility prediction were obtained from a recent study conducted by Meng et al. [12]. Three curated datasets(of varying sizes), namely AQSol, ESOL, and PHYS, were selected to ensure a comprehensive and reliable foundation for predictive modeling. **AQSol** [14] dataset aggregates data from nine original sources, including AQUA and ESOL datasets [12], providing 9,982 records after preprocessing and merging repetitive entries. The dataset was curated to retain only unique molecular entries with high-quality solubility values, represented in log S units. After data curation 9061 records remained. **ESOL** dataset, derived from a verified version of the dataset originally published by Delaney [4], contains 1128 records. It serves as a benchmark for aqueous solubility prediction and includes solubility values specifically tailored for small organic molecules. After data curation 1157 records remained. **PHYS** dataset, comprising 2,010 records, originates from the curated PHYSPROP database [9]. It includes water solubility data for various compounds, with quality metrics ensuring reliable entries. After data curation, 2001 records remained.

2.2 Featurization

In this study, four molecular featurization methodologies were evaluated to determine their effectiveness in predicting molecular solubility. These featurization techniques include **Custom Descriptor-Based Featurization** and three featurization methods provided by Deepchem: **RDKitDescriptors Featurizer**, **Extended Connectivity Fingerprints (ECFP)**, and **MACCS Keys**. Out of the many featurization methods available, these methods were specifically chosen for their strength in capturing diverse aspects of molecular properties, their structural patterns, and the substructural fragments. All these aspects are important for accurately predicting solubility. The flow of the featurization process using different methodologies is provided in Fig. 1. A detailed description of each method is provided below.

Custom Descriptor-Based Featurization: This methodology focused on a small subset of descriptors computed using RDKit. The descriptors included molecular

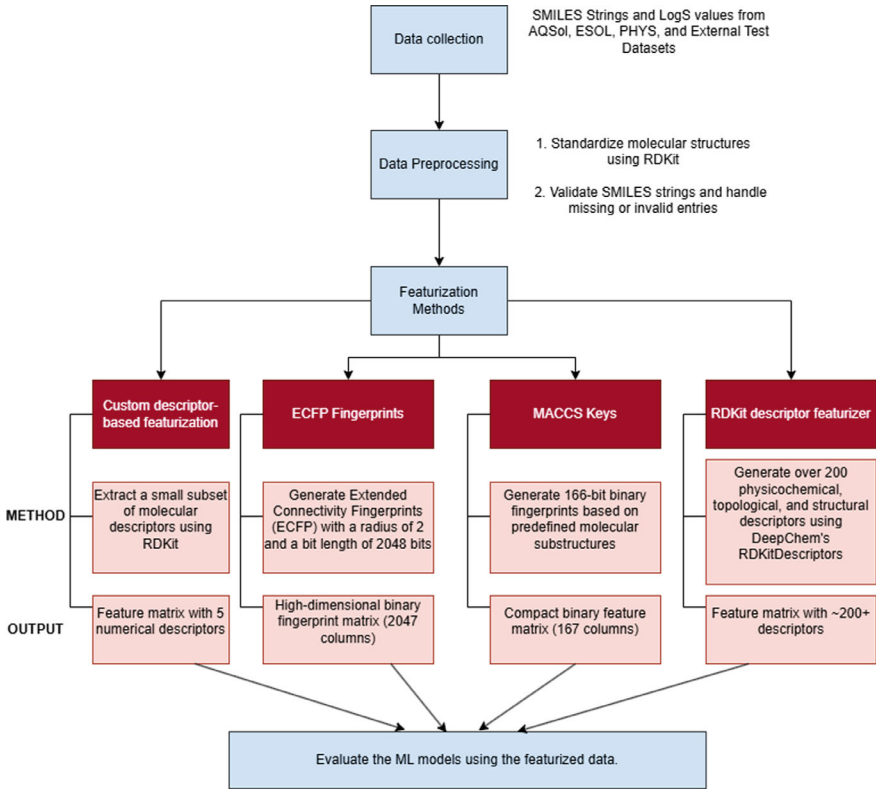

Fig. 1 4 featurization processes used in this paper

weight, LogP, the number of rotatable bonds, hydrogen bond acceptors, and hydrogen bond donors. This method focuses on computational efficiency by using a limited number of descriptors to represent molecular properties. Even though this method is effective for quick and interpretable modeling, it excludes many features that influence solubility.

RDKitDescriptors Featurizer (DeepChem): The RDKitDescriptors featurizer in DeepChem takes advantage of RDKit's powerful feature generation tools to compute a wide range of molecular descriptors. These descriptors capture important physicochemical, structural, and topological properties of molecules. It creates a rich, high-dimensional representation of molecular data, making it easier to build accurate predictive models. The RDKitDescriptors featurizer computes more than 200 molecular descriptors. These descriptors can be broadly classified into the following categories: **(a) Constitutional Descriptors:** Molecular weight (MolWt), number of atoms (NumAtoms), and number of bonds (NumBonds), etc. **(b) Electronic Descriptors:** Number of hydrogen bond donors (NumHDonors) and acceptors (NumHAcceptors), partial charge distributions, aromaticity etc. **(c) Geometrical**

Descriptors: Molecular surface area and volume, etc. **(d) Topological Descriptors:** Balaban index(J), and Wiener index (W), etc. **(e) Fragment-Based Descriptors:** Rotatable bonds (NumRotatableBonds) and rings (NumRings) etc.

The high-dimensional nature of the RDKitDescriptors ensures that variations in molecular properties that influence solubility can be effectively captured. This makes them ideal for solubility prediction tasks where both linear and non-linear relationships are important. Also, solubility is influenced by global molecular properties such as molecular weight, polarity, hydrogen bonding, and surface area. The RDKitDescriptors featurizer captures these features through descriptors like Molecular Weight (MolWt), LogP (hydrophobicity), hydrogen bond donors/acceptors, Topological polar surface area (TPSA), etc.

Extented Connectivity Fingerptints (ECFP): ECFP, available in DeepChem, are circular fingerprints that represent the local atomic environments of molecules as fixed-length binary vectors. ECFP fingerprints work by expanding neighborhoods around atoms based on their connectivity, up to a specific radius (e.g., ECFP4 corresponds to a radius of 2 bonds). This approach captures detailed information about molecular fragments, including atomic types, bond types, and connectivity patterns.

ECFP fingerprints are especially useful for identifying local structural features in molecules, such as functional groups and substructural motifs, which are often important for predicting properties like solubility. The iterative encoding process effectively captures the spatial relationships between atoms within local environments, providing a rich and detailed molecular representation.

Solubility often depends on the presence or absence of specific functional groups (e.g., hydroxyl groups, amines, or halogens) and local molecular patterns. ECFP fingerprints are well-suited for detecting these patterns because they (1) Iteratively expand atomic neighborhoods, (2) Represent molecular fragments without loss of spatial information, (3) Capture connectivity between atoms and bonds.

MACCS (Molecular Access System) Keys: The MACCS Keys featurizer creates a binary fingerprint vector with 166 bits. Each bit represents the presence or absence of specific predefined molecular substructures or fragments. These include functional groups (like hydroxyl or carbonyl groups), ring systems, and distinct atomic or bonding patterns.

Unlike ECFP, which uses an iterative and data-driven approach to generate fingerprints, MACCS keys are based on a fixed set of molecular patterns. Each pattern is clearly defined. This allows quick computation and easy interpretation of the fingerprints.

Although, they may not capture as much detail as circular fingerprints like ECFP, they still provide enough information for property prediction.

While DeepChem provides other featurizers such as GraphConvFeaturizer, WeaveFeaturizer, and Mol2Vec, these were not selected due to the following reasons: **Graph-Based Featurizers** (e.g., GraphConv): Although graph-based methods are powerful, they generally require GNNs for processing. Since this study focused on evaluating traditional machine learning algorithms, descriptor-based and fingerprint-based featurizers were more suitable. **Sequence-Based or Learned Embeddings**

(e.g., Mol2Vec): These featurizers generate embeddings from chemical representations but often require pre-training on large molecular databases, which was not the focus of this work. **Computational Trade-Off**: Some advanced featurizers are computationally expensive or less interpretable compared to RDKitDescriptors, ECFP, and MACCS keys. The selected featurizers strike a balance between performance, interpretability, and computational efficiency.

2.3 Model Training and Evaluation

Evaluated Machine Learning Models To predict molecular solubility based on the extracted features, seven machine learning (ML) models were employed. These models allowed for the exploration of linear, non-linear, and ensemble-based relationships within the dataset generated by the individual featurizers. The models utilized were: Linear Regressor, Ridge Regressor, Random Forest Regressor, KNN Regressor, XGBoost Regressor, LightGBM Regressor and CatBoost Regressor.

Each model contributed unique strengths to the solubility prediction task. Linear and Ridge Regression provided interpretable baselines. Advanced methods like XGBoost, LightGBM, and CatBoost demonstrated their superiority in capturing non-linear patterns. Random Forest offered a balance between interpretability and predictive power. KNN regressor highlighted the importance of local structure in molecular data. This comprehensive evaluation ensured that the most suitable model could be identified based on the specific requirements of the solubility prediction task.

Performance Metrics To evaluate the predictive performance of the machine learning models, three key metrics were employed: **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and the Coefficient of Determination (R^2). These metrics provide valuable insights into the accuracy and explanatory power of the models.

3 Results and Discussion

3.1 Comparison Between Different Featurization Techniques Used in This Study

The results of this study (as presented in Tables 1, 2 and 3) demonstrated that the RDKitDescriptors Featurizer outperformed the Custom Descriptor-Based Featurization, ECFP fingerprints, and MACCS Keys across all evaluated machine learning models and datasets. This superior performance highlights the importance of comprehensive molecular representations in predicting solubility.

Table 1 Mean performance metrics (R^2, adjusted R^2, MSE and RMSE) calculated on the AQSol dataset, averaged over the results of seven machine learning algorithms

Featurizer	R^2 ↑	Adjusted R^2 ↑	MSE ↓	RMSE ↓
Custom featurizer	0.7194	0.6854	1.5491	1.2120
ECFP	0.6688	0.6328	1.7625	1.3241
MACCS keys	0.7022	0.6503	1.5821	1.2533
RDKitDescriptors featurizer	**0.8240**	**0.8003**	**0.9204**	**0.9558**

Table 2 Mean performance metrics (R^2, MSE and RMSE) calculated on the ESOL dataset, averaged over the results of seven machine learning algorithms

Featurizer	R^2 ↑	MSE ↓	RMSE ↓
Custom featurizer	0.8285	0.7298	0.8362
ECFP	0.5082	1.4907	1.0883
MACCS keys	0.7531	1.0565	1.0187
RDKitDescriptors featurizer	**0.8940**	**0.4534**	**0.6684**

Table 3 Mean performance metrics (R^2, MSE and RMSE) calculated on the PHYS dataset, averaged over the results of seven machine learning algorithms

Featurizer	R^2 ↑	MSE ↓	RMSE ↓
Custom featurizer	0.8463	0.7552	0.8558
ECFP	0.6141	1.6281	1.1613
MACCS keys	0.7147	1.5624	1.1694
RDKitDescriptors featurizer	**0.9079**	**0.4465**	**0.6633**

Comparative Analysis: The RDKitDescriptors Featurizer consistently achieved higher (R^2 scores and lower MSE and RMSE values compared to the other featurization methods. While the Custom Descriptor-Based Featurization captured only a small subset of molecular properties, the RDKitDescriptors Featurizer provided a richer, high-dimensional representation of molecular features. Similarly, while ECFP fingerprints and MACCS keys encoded local structural and substructural information, they lacked the broader physicochemical and topological descriptors captured by the RDKitDescriptors Featurizer.

The superior performance of the RDKitDescriptors Featurizer can be attributed to the following factors:

– Compared to Custom Descriptors, which rely on a limited number of features, the RDKitDescriptors Featurizer provides a high-dimensional representation. This

increases the likelihood of capturing subtle relationships between molecular properties and solubility that might otherwise be overlooked.
- While ECFP fingerprints and MACCS keys are effective in capturing local structural features (e.g., functional groups and substructures), they do not include global physicochemical and geometric properties, which play a significant role in solubility prediction. The RDKitDescriptors Featurizer balances both global features (e.g., molecular size, surface area) and local features (e.g., specific substructures), leading to better molecular representation.
- Solubility is influenced by a combination of molecular properties such as size, polarity, surface area, and hydrogen bonding capacity. The RDKitDescriptors Featurizer inherently includes these critical features, allowing machine learning models to establish stronger correlations between molecular descriptors and solubility outcomes.

Additional Discussion: Performance of ECFP and MACCS Keys Compared to Custom-Based Descriptors: Interestingly, the results showed that ECFP fingerprints and MACCS Keys performed worse than the Custom-Based Descriptor Featurization, even though the custom-based descriptors relied on only five carefully selected molecular descriptors. This outcome can be due to these reasons:

1. Loss of global molecular information: Both ECFP fingerprints and MACCS Keys are binary fingerprint-based methods that primarily encode local molecular substructures. ECFP captures atom-level neighborhoods iteratively and MACCS Keys focus on predefined molecular fragments. But, they do not include global physicochemical properties such as molecular weight, polarity (LogP), or hydrogen bonding capacity. These properties are fundamental to solubility prediction as they influence molecular interactions with solvents. In contrast, the Custom-Based Descriptor Featurization explicitly included molecular weight, LogP, and hydrogen bond-related features, which are well-known to correlate strongly with solubility.
2. High Dimensionality and Sparsity in ECFP and MACCS Keys: ECFP fingerprints generate high-dimensional binary vectors. In this study it resulted in a featurized dataset containing 2047 columns. Many of these columns were sparsely populated, as only a subset of molecular substructures were present across the molecules. Such high-dimensional and sparse representations may have lead to overfitting, as the dataset sizes were small to moderate. Similarly, although, MACCS Keys were lower in dimensionality (166 bits), they failed to provide sufficient information to distinguish molecules based on minute differences in features that are relevant to solubility.
3. Model Alignment: Machine learning models used in this study (e.g., linear regression, tree-based models) may not fully exploit the detailed structural encodings provided by ECFP and MACCS Keys. The smaller, curated set of features in Custom-Based Descriptors aligns better with these models, leading to more stable and interpretable predictions.

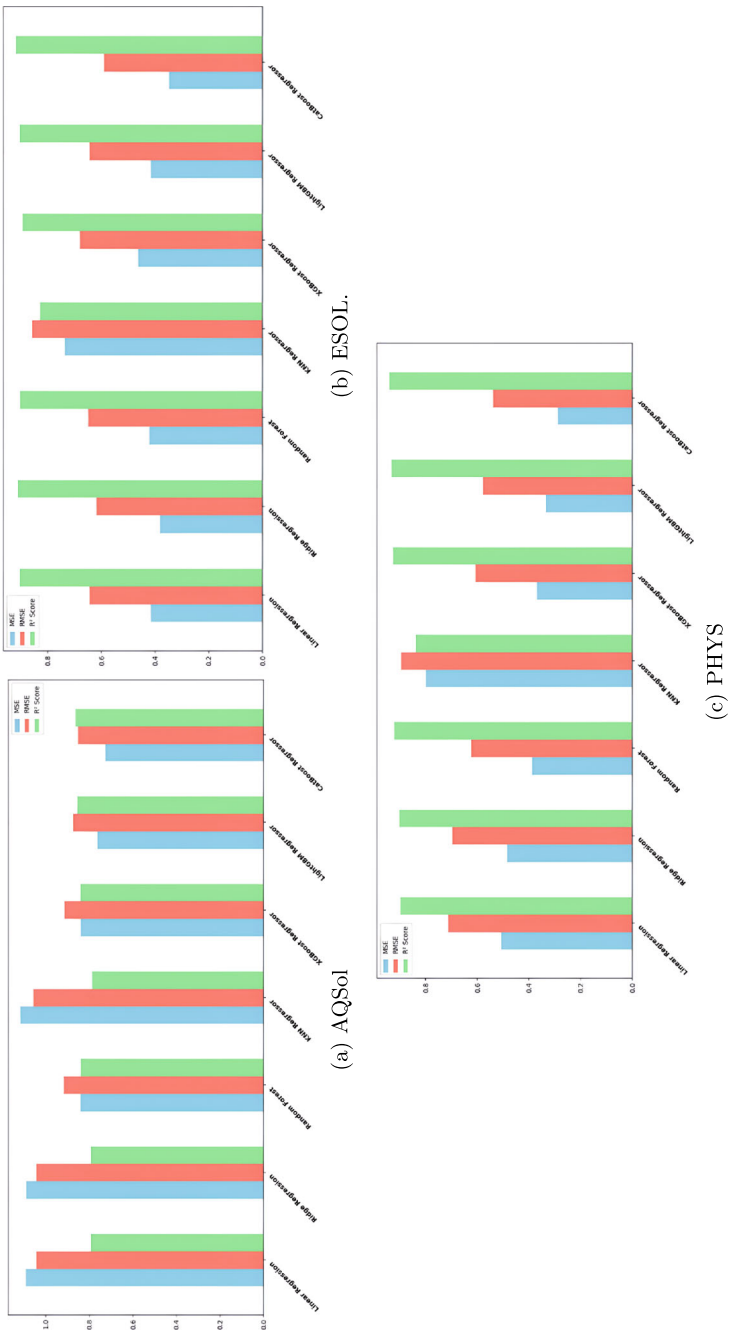

Fig. 2 Performance of ML models using RDKit descriptors across datasets

Table 4 Comparison of proposed approach with SOTA

Study	Dataset	$R^2 \uparrow$	MSE $\downarrow$	RMSE $\downarrow$
Francoeur and Koes [6]	AQSol	0.77	–	1.16
This paper	AQSol	**0.8644**	0.7095	**0.8423**
Ghanavati et al. [8]	ESOL	0.9189	–	0.6023
Datta and Karmakar [5]	ESOL	0.77	1.01	–
Francoeur and Koes [6]	ESOL	0.915	–	**0.289**
This paper	ESOL	**0.9191**	**0.3463**	0.5885
Ghanavati et al. [8]	PHYS	0.9266	–	0.5779
This paper	PHYS	**0.9418**	0.2858	**0.5346**

Performance of RDKitDescriptors Featurizer Across Datasets: Plots (a), (b) and (c), in Fig. 2 illustrate the performance of the RDKitDescriptors featurizer across seven machine learning algorithms on the AQSol, ESOL, and PHYS datasets, respectively.

3.2 Comparison with State-of-the-Art

To evaluate the effectiveness of the proposed approach, a comparative performance analysis was conducted (as shown in Table 4) against state-of-the-art methods reported in related studies. The focus of this comparison lies in understanding the differences in featurization strategies and their impact on predicting aqueous solubility, particularly when tested on datasets common to our study.

Performance Analysis: Among the related works, several featurization techniques have been employed, including descriptor-based approaches, fingerprint-based methods, and advanced graph-based representations.

Descriptor-Based Featurization: Ref. [5] relied on a small feature set consisting of only four descriptors: molecular weight, number of rotatable bonds, aromaticity proportion, and solubility in water. While computationally efficient, the limited feature space misses out on many molecular properties that are relevant to solubility, such as electronic distribution, surface area, and topological complexity. Reference [11] used a combination of fingerprints (5120 bits) and 317 molecular descriptors generated by DRAGON 6.0. The descriptors captured constitutional, ring, topological, and functional group properties, while fingerprints captured fine substructural features. Despite the high-dimensional feature space(5444 features in total), the per-

formance lacked due to high dimensionality of the feature space leading to redundancy and noise, and the challenge of aligning such a huge feature space with traditional machine learning models. **Fingerprint-Based Representations**: [15] utilized Morgan fingerprints (ECFPs). Despite capturing local structural patterns, fingerprint-based approaches tend to generate high-dimensional, sparse feature spaces. As evident in our study, the performance of ECFP and MACCS Keys was not very good, likely due to the absence of global physicochemical properties and feature redundancy, which can hinder learning in traditional machine learning models. **Advanced Graph and 3D Representations**: [8] employed graph-based methods and electrostatic potential (ESP) maps processed by deep learning models, including Graph Convolutional Networks (GCNs) and EdgeConv algorithms. While these methods offer a sophisticated representation of molecular topology and charge distribution, they require computationally intensive training pipelines and large datasets for optimal performance. Additionally, Francoeur et al. [6] utilized the MAT (Molecular Attention Transformer) architecture for molecular representation, where feature vectors for each atom were derived from RDKit's molecule object. However, the study did not explicitly specify the list of features included in the feature vector. While their approach achieved competitive performance in terms of squared error on the smaller ESOL dataset, it failed to generalize effectively to larger datasets such as AQSol, showing limitations in scaling to more complex and diverse molecular data.

3.3 Future Work

In the current study, the RDKitDescriptors featurizers showed the best results. It has been utilized to automatically generate 210 molecular descriptors, capturing various physicochemical, structural, and topological properties of molecules. However, not all of these descriptors are expected to contribute meaningfully to solubility prediction. As a next step, an ablation study can be conducted to identify and remove redundant or highly correlated descriptors, ensuring that only the most relevant features are retained for the prediction task. The correlation heatmap in Fig. 3 displays the top 10 descriptors with the highest correlation to the solubility (Log S) value. These descriptors were selected from the total set of 210 descriptors generated by the RDKitDescriptors Featurizer.

Following this, the focus can shift towards leveraging Graph Neural Networks (GNNs). Since molecular structures inherently resemble graphs, where atoms represent nodes and bonds represent edges, GNNs offer a natural and potentially more effective framework for modeling these relationships. By incorporating GNNs, the goal is to further enhance predictive performance by capturing the intricate structural dependencies within molecules.

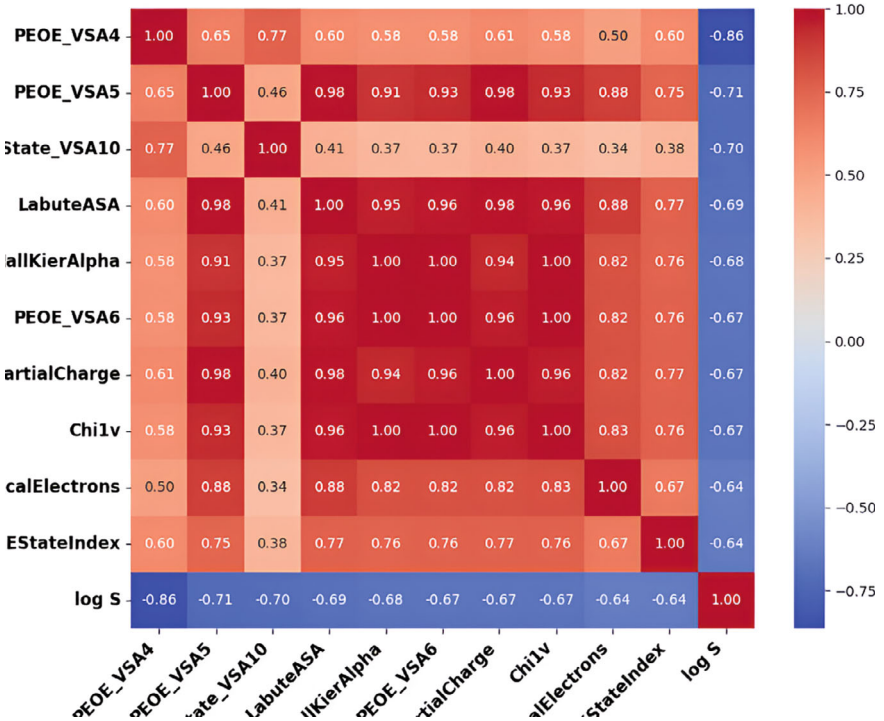

Fig. 3 Heatmap of descriptors that have the highest correlation with solubility value

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ECNBiLS: CNN-BiLSTM Based Ensemble Model for Elderly Adult Fall Detection

Deepanshu Rathore and Vijay Bhaskar Semwal

Abstract The proposed work addresses the growing interest in fall detection among researchers, which is driven by its wide range of applications in healthcare and safety monitoring. This research uses wearable sensors like accelerometers and gyroscopes to evaluate human behavior, with a particular emphasis on fall detection. Effective fall detection requires the extraction of crucial temporal elements from gathered data. Traditional methods are frequently computational-expensive, necessitating specialized knowledge and including extensive feature engineering and data processing. This paper introduce Ensemble learning Model using Convolutional Neural Network (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) of deep learning model, marking a significant development. This novel method of selection of spatial and temporal separately fusion of both spatial and temporal which make and accurately with of 98.17% on the Federated Research Data Repository (FRDR). By automating feature detection and streamlining the data processing pipeline, this approach improves the efficiency and effectiveness of fall detection systems, paving the way for more robust and scalable solutions in real-world applications.

Keywords Gait · Deep learning · Inertial measurement unit · DNN

1 Introduction

The most frequent cause of injuries among the elderly is falls, which has a substantial financial impact on the hospital system. Hundreds of thousands of individuals, primarily old, fall every year, suffering significant injuries, sometimes fatalities. The World Health Organization (WHO)reported that older adults have a higher risk of falling: 28–35% of adults 65 years of age and over fall annually [1], and this percentage rises to 32–42% [2] among parents and those over 70. Additionally, older women

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have an increased risk of hip-related Therefore, Maintaining an aging population has several challenges [3]. The number of older people who fall each year is estimated to be between 28% and 35% worldwide.

Globally, falls among the elderly are a big concern. A recent WHO research indicates that 37.3 million falls worldwide are thought to be the cause of the estimated 6 lakh 84 hundred deaths from falls each year.

Serious enough to necessitate medical care. Each year in the United States, medical expenditures associated with non-fatal falls account for approximately $50 billion, while the costs associated with fatal falls account for $754 million, with an estimated $ 100 billion in costs associated with falls by 2030. These falls are a complex issue related to aging, the environment, and inherent mental or physical health problems. They can cause a variety of serious injuries, functional disorders, short- or long-term care needs, disabilities, and even fatalities. Early reactions to falls among the elderly reduce the number of injuries and fatalities.

For individuals aged 65 and older, falls happen at least once a year or more. In Australia, the annual expenses linked to fall-related incidents impacting older adults surpass $ 200 million and are on the rise as the population ages [4]. Traditional self-reporting or caregiver reporting methods for normal activities and fall-related incidents can have inaccuracies caused by denial, recall bias, and/or inconsistent monitoring capabilities. Research shows that nearly one-third of seniors aged 65 and older experience falls annually, with approximately 10–15% of those incidents leading to serious injuries or fatalities.

For this situation we are using the modern approach of quantization of leg fall detection using different approaches but for now we are using BiLSTM approach for the detection.

This Paper Contains Sect. 2 is a comprehensive literature review that discusses past principles pertinent to the topic. Section 3 discusses the technique, including the proposed model topology and underlying architecture. Sections 4 and 5 describe the findings.

2 Literature Review

The literature describes three primary methods for fall detection systems, depending on the type of data collection-hand-held, environmental, or visual devices [4]. Fall detection systems are categorized as either context-aware or portable. Context-aware systems consider all installed environmental sensors, which include infrared, radars, microphones, pressure sensors, and vision-based tools. Additionally, Kinect, motion capture software, and cameras are technologies that are aware of their context. Numerous fall trackers incorporate accelerometers and gyroscopes within their wearable sensors. Sensors integrated into smartphones, smartwatches, and other wearable gadgets have recently gained appeal for fall detection systems because of their affordability and easy accessibility. Other researchers such as Mubashir et al. categorized

Table 1 Review of studies on fall detection

References	Dataset	Classifier
Rezaei et al. [13]	Millimeter wave	Random forest
Vallabh et al. [12]	MobiFall	k-nearest neighbors
Taramasco et al. [11]	Collected dataset	BiLSTM
Robinovitch and Aziz [14]	FRDR dataset	Support vector machine (SVM)
Ejupi [15]	FRDR dataset	Logistic regression (LR)

fall detection techniques into wearable, environmental sensor-based, and vision-based. Current studies on fall detection systems, along with newer investigations, offer a thorough analysis of the benefits and drawbacks of these technologies and novel multimodal strategies, incorporating diverse wearable, visual, and environmental sensors. Many researchers identify privacy issues, intrusive measures, device functionality, and the difficulties of method comparison as major challenges and concerns.

The two most common uses for fall detection and prevention are SVM and wearable devices [5]. Additionally, this article explores the benefits of the ML algorithm, its applications, performance metrics, and future prospects for wearable technology, including wearable design, sensor fusion, energy efficiency, datasets, and context awareness. The research results utilize clustering and neural network classifiers to analyze the daily performance of eight healthy subjects. The approach demonstrates 100% sensitivity for falls within groups, 97.65% for falls outside categories, 99.33% specificity for standard ADL and 96.59% specificity for physical activity ADL. The research indicates that this kind of classifier may surpass the MLPs that are presently utilized. Initial research indicates that these approaches may benefit professionals aiming to enhance the effectiveness of ambulatory fall management.

A summary of related studies on fall detection is presented in Table 1.

3 Methodology

3.1 Data Set

The purpose of the IMU dataset is to compare fall detection and prediction algorithms using body-worn APDM Opal IMU sensors' [16] acceleration, angular velocity, and magnetic fields at seven different body locations.

Ten healthy young individuals, ages ranging from 22 to 32, are included in the dataset (Mean = 26.6 years, SD = 2.8 years). All subjects were recruited as Simon Fraser University students using posters and ads posted on campus bulletin

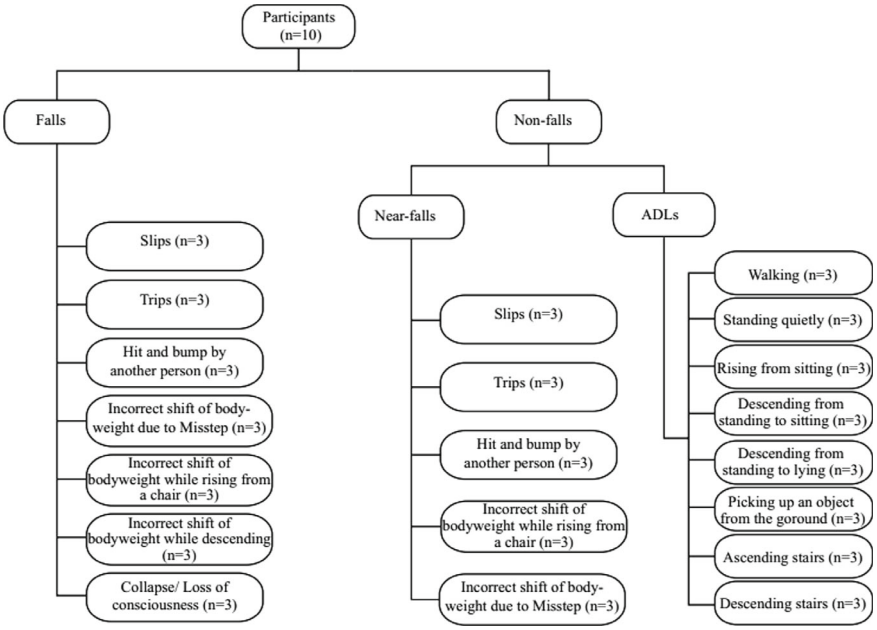

Fig. 1 Structure of the dataset

boards. All volunteers gave written agreement, and the experiment methodology was approved by Simon Fraser University's Research Ethics Committee.

The overall dataset structure is illustrated in Fig. 1.

3.2 CNN-BiLSTM Ensemble for Fall Detection

The CNN part of the network applies dense layers for preliminary transformation of features and uses a 1D convolutional layer to extract relevant features of the sequential data. It then flattens the convolutional layer and feeds the feature into additional dense layers with ReLU activation to further refine features before producing the final output. Dropout layers are used to reduce overfitting and improve generalization.

In parallel, BiLSTM is processing the time-series data with the help of Bidirectional LSTM layers for both forward and backward relations in the time. Here, dense layers are applied with ReLU activation function to avoid overfitting and the dropout prevents overfitting. At the output layer, with softmax activation, classification of falls events are obtained. The binary cross-entropy loss function coupled with the Adam optimizer will be used to train the model, and then it is followed by an ensemble mechanism of obtaining the final pre diction. Ensemble averages CNN and BiLSTM outputs toward the final classification of falling.

4 Proposed Model

We propose a Ensemble of deep learning architecture combining Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks for the task of time-series classification. The architecture consists of two primary models: a CNN model and a BiLSTM model. These models are trained individually and then ensembled for final prediction.

4.1 Ensemble Method

Once both models are trained, their predictions are combined using a simple averaging ensemble approach. The final prediction is computed as the average of the outputs from the CNN and BiLSTM models, followed by selecting the class with the highest probability:

$$\hat{y} = \frac{1}{2}(\hat{y}_{\text{CNN}} + \hat{y}_{\text{BiLSTM}}) \tag{1}$$

where $\hat{y}$ represents the final prediction, while $\hat{y}_{\text{CNN}}$ and $\hat{y}_{\text{BiLSTM}}$ denote the predictions from the CNN and BiLSTM models, respectively.

4.2 Training and Evaluation

The models are trained using the training dataset, and their performance is evaluated on the test dataset. The accuracy of the final ensemble model is computed, and additional evaluation metrics, including precision, recall, and $F1$-score, are analyzed using the confusion matrix.

The training process consists of 60 epochs with a batch size of 256. The Adam optimizer is employed with a learning rate of 0.001, and categorical cross-entropy is used as the loss function. To mitigate overfitting, early stopping is applied, and validation data is used to monitor performance during training.

4.3 Algorithm of Model

Algorithm 1: CNN-BiLSTM Ensemble Model for Time-Series Classification

Input: Time-series data X, corresponding labels Y, training parameters (epochs, batch size, etc.)
Output: Final classification results and model evaluation metrics

Preprocessing:
Normalize the input data X to obtain X_{scaled};
Convert Y to one-hot encoded labels $Y_{categorical}$;
CNN Model Training:
Initialize a CNN model: `CNN = Sequential(Dense, Conv1D, Flatten, Dropout layers);`
Compile the CNN model using the Adam optimizer and categorical cross-entropy loss;
Train the CNN model:
`CNN.fit(`X_{train}, Y_{train}, `validation data, epochs, batch size);`
BiLSTM Model Training:
Initialize a BiLSTM model: `BiLSTM = Sequential(Bidirectional LSTM, Dense, Dropout layers);`
Compile the BiLSTM model using the Adam optimizer and categorical cross-entropy loss;
Train the BiLSTM model:
`BiLSTM.fit(`X_{train}, Y_{train}, `validation data, epochs, batch size);`
Ensemble Predictions:
Obtain predictions from the trained models:
  `PredictionsCNN = CNN.predict(`X_{test}`);`
  `PredictionsBiLSTM = BiLSTM.predict(`X_{test}`);`
Perform simple averaging to ensemble predictions:
  `Ensemble Predictions = (PredictionsCNN + PredictionsBiLSTM) / 2;`
Convert probabilities to class labels:
  `Final Labels = argmax(Ensemble Predictions);`
Evaluation:
Compute accuracy and classification metrics:
  `Accuracy = Correct Predictions / Total Samples;`
Generate confusion matrix and calculate performance metrics (e.g., sensitivity, specificity);

4.4 Model Architecture

In the proposed fall detection model, CNN and BiLSTM layers are combined as an ensemble for effectively extracting both spatial and temporal dependencies in wearable sensor sequential data. The architecture of the suggested model begins with a CNN branch that processes sequential data using convolutional layers to extract local spatial patterns. The first convolution layer uses 64 filters, with a kernel size of 3 and ReLU activation, followed by batch normalization to improve convergence. MaxPooling layers are added to reduce spatial dimension while retaining essential features. A flattening layer and a dense layer of 128 neurons conclude this branch, enabling further refinement of the extracted features. The dropout rate of 0.5 is implemented in the dropout layer to prevent overfitting. Meanwhile, the BiLSTM branch processes sequential data both ways in forward and backward directions for

rich and complex temporal patterns. It contains a BiLSTM layer consisting of 64 units and further is connected with a dense layer containing 128 neurons and using ReLU for further processing. Dropout regularization at a rate of 0.5 is implemented for enhancing generalization. The outputs from both branches are finally combined into an output using averaging ensemble technique where the predictions from CNN and BiLSTM branches are averaged to produce the final classification output. This strategy is going to exploit the strength of both spatial feature extraction through the CNN and the model of temporal dependency through the BiLSTM, thus leading to a strong and accurate fall detection system. The hybrid approach captures all short-term and long-term dependencies in sensor data for improved fall detection performance with controlled risks of overfitting.

5 Result

5.1 Hybrid Model Performance

The FDRD dataset was used to test a number of common models in FD systems in order to confirm the superiority of the suggested The Ensemble model CNN with BI-LSTM model demonstrated exceptional performance across all subjects. The average accuracy of the model was 98.10%.

The performance metrics of various models are summarized in Table 2, where the proposed CNN-BiLSTM ensemble achieves the best accuracy.

The optimized settings from the same experimental setup as the proposed method. The structures were configured as follows: “Simple CNN”: The CNN design comprises of two convolutional layers and two pooling layers. Simple GRU: Raw sensor data is processed using two GRU layers. “Coarse-fine” CNN: The coarse branch of Coarse fine CNN is a one-layer CNN, whereas the fine branch is based on the architecture of Simple CNN. “CNN-LSTM”: The most recent FD model shares the same architectural arrangement and hyperparameters as an earlier effort. It consists of two LSTM layers followed by two convolutional layers. “CNN-GRU”: Because the

Table 2 Performance metrics for various models in fall detection

Model	Accuracy (%)	Recall (%)	Precision (%)
Simple CNN	97.82	94.54	92.22
Simple GRU	97.49	95.22	93.44
Coarse-fine	97.78	92.49	89.47
CNN-GRU	97.69	93.03	90.52
Coarse-fine CNN and GRU	97.95	92.86	90.44
Proposed model	**98.10**	**92.56**	**97.34**

Bold text highlights the proposed model performance

proposed model included GRU layers to extract temporal characteristics, the CNN-LSTM layers were substituted by them 13. The suggested model achieved 92.56% recall, 98.1% accuracy, and 96.07% F-score. The Coarse-fine CNN and its models produced the best RECALL of 95.22% for FD compared to other DL techniques, indicating that FD using the suggested ensemble model gives superior results than FD employing standard DL techniques. Furthermore, CNN-based approaches produce a substantially greater Fscore of 93.52% when compared to the FD model with two-layer GRU, which has a Fscore of just 92.71%. The general DL models exhibit similar good FD performance to the previous studies.

A comparison with previous state-of-the-art methods is shown in Fig. 2, demonstrating the superior performance of our approach.

To the best of our knowledge, the proposed method is the first to leverage wearable sensor data and a CNN-BiLSTM model for fall detection. Convolutional Neural Networks (CNNs) are well-regarded for their ability to automatically extract local patterns and features from sequential data, especially time-series signals like accelerometer and gyroscope measurements. CNNs excel at capturing spatial features and patterns, but when it comes to modeling temporal dependencies over time, they often face limitations.

To address these shortcomings, Bidirectional Long Short-Term Memory networks (BiLSTMs) are employed in conjunction with CNNs. BiLSTMs are an extension of traditional Recurrent Neural Networks (RNNs) that can effectively handle sequential data by capturing long-term dependencies, even over lengthy sequences. Unlike standard RNNs, which suffer from the vanishing gradient problem, LSTMs are designed with memory cells and gating mechanisms that allow them to retain information over longer durations.

The bidirectional aspect of BiLSTMs further enhances this capability by processing data in both forward and backward directions. This bidirectional flow of information allows the model to consider past and future context simultaneously,

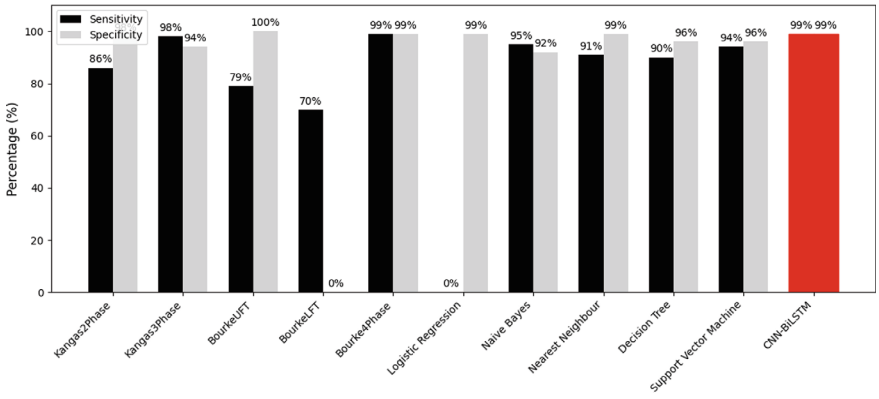

Fig. 2 Comparison with previous work

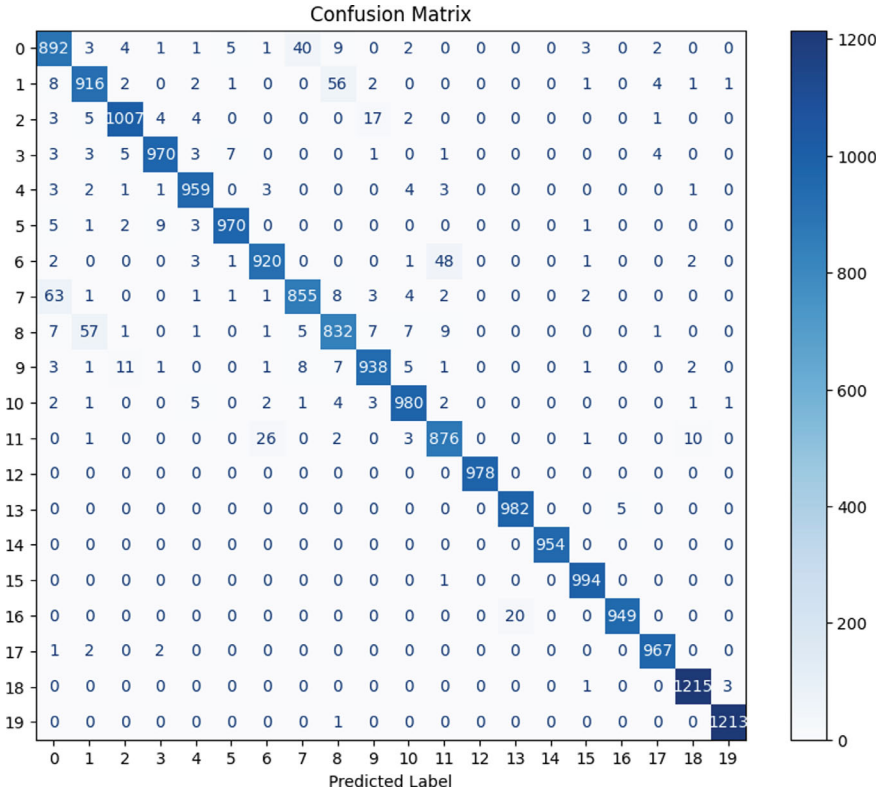

Fig. 3 Confusion matrix

which is crucial for detecting fall events in sensor data. By combining the feature extraction capabilities of CNNs with the temporal modeling strengths of BiLSTMs, the proposed CNN-BiLSTM ensemble model efficiently captures both spatial and temporal dependencies, improving the accuracy and robustness of fall detection.

This problem is addressed by BiLSTMs, a particular kind of RNN, with their distinctive architecture made up of gating mechanisms and memory cells. The BiLSTM is able to remember pertinent information for extended periods of time and forget irrelevant information thanks to these systems that regulate the flow of information. By processing the data in both forward and backward directions, BiLSTMs capture a more comprehensive representation of the temporal dependencies. We can successfully capture complicated temporal patterns and long-term dependencies—both essential for precise fall detection—by adding BiLSTM layers to the model.

The confusion matrix in Fig. 3 provides a comprehensive evaluation of the model's performance across multiple classes. The rows correspond to the actual classes (true labels), while the columns represent the predicted classes. The diagonal elements of the matrix signify correct predictions.

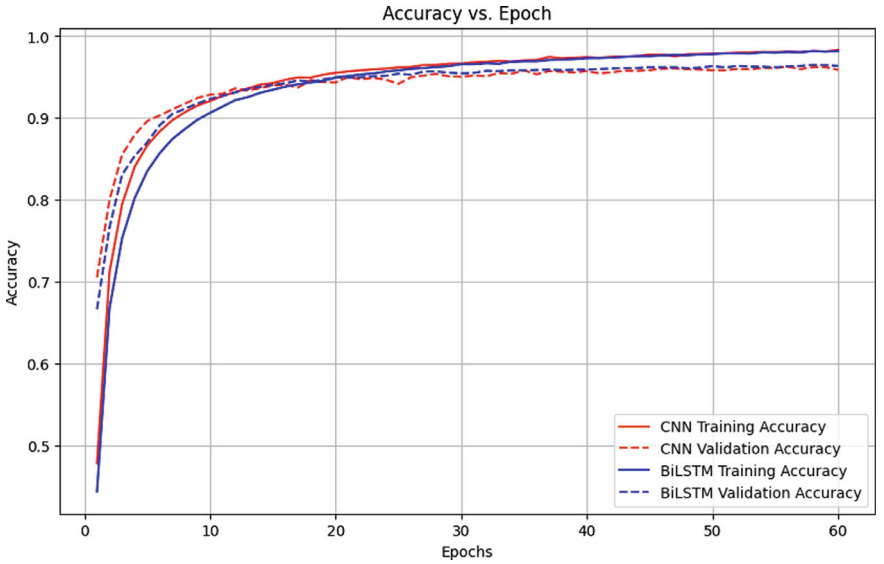

Fig. 4 Accuracy graph

The graph in Fig. 4 illustrates the training and validation accuracy trends for both the CNN and BiLSTM models over 60 epochs. The x-axis represents the number of epochs, while the y-axis represents accuracy. Solid lines correspond to training accuracy, while dashed lines correspond to validation accuracy for each model.

The CNN model (red lines) demonstrates a rapid improvement in accuracy during the initial epochs, achieving over 85% accuracy within the first 10 epochs. After this initial phase, the training and validation accuracies converge and gradually stabilize above 95%, indicating strong generalization and minimal overfitting. The consistent alignment of training and validation curves suggests that the CNN model maintains robustness and performs well on unseen data.

The BiLSTM model (blue lines) follows a similar trajectory, albeit with a slightly slower convergence during the initial epochs. However, by epoch 20, the BiLSTM model achieves comparable accuracy to the CNN model. The validation accuracy closely tracks the training accuracy throughout, suggesting effective learning of temporal dependencies without significant overfitting. Both training and validation accuracies stabilize near 95%, reflecting the model's ability to generalize effectively.

Overall, the figure demonstrates the efficacy of both models in achieving high accuracy for the task. The consistent alignment between training and validation accuracies indicates proper model regularization and sufficient data for training. This comparative performance analysis highlights the potential of combining these architectures to leverage both spatial and temporal features for improved results.

The high sensitivity and specificity of the CNN-BiLSTM ensemble model for fall detection can be attributed to its architecture that combines the strengths of

both CNNs and BiLSTMs. The CNN efficiently extracts spatial features from the wearable sensor data, while the BiLSTM captures temporal dependencies by processing sequences in both forward and backward directions. The ensemble approach, which averages predictions from both models, enhances robustness and reduces bias. Regularization techniques, such as dropout, prevent overfitting, ensuring better generalization. This allows the model to accurately detect falls (high sensitivity) while effectively distinguishing non-falls (high specificity). The use of softmax for classification further supports robust decision-making, making the model highly effective for real-time fall detection applications.

The proposed approach enables the development of a wearable FD system for long-term healthcare services.

6 Conclusion

We have reviewed various techniques for detecting fall events, as summarized in Figs. 5 and 6. An effective and robust fall detection system should exhibit both high sensitivity and specificity. Current approaches, while valuable, have not fully achieved the desired levels of accuracy and robustness. However, they offer a foundation for further development and refinement of techniques. The proposed CNN-BiLSTM ensemble model enhances these existing frameworks by integrating Convolutional Neural Networks (CNNs) for spatial feature extraction and Bidirectional

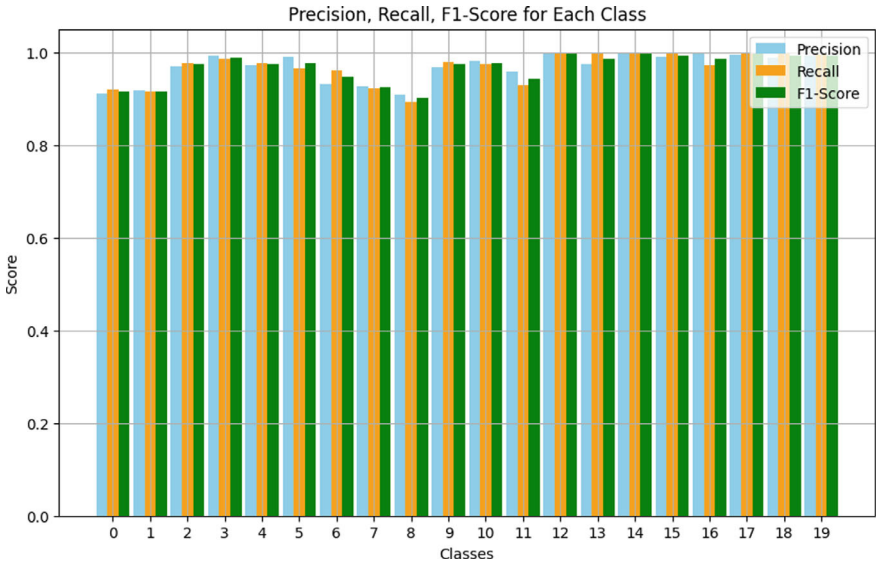

Fig. 5 All classes $F1$-score, precious, recall

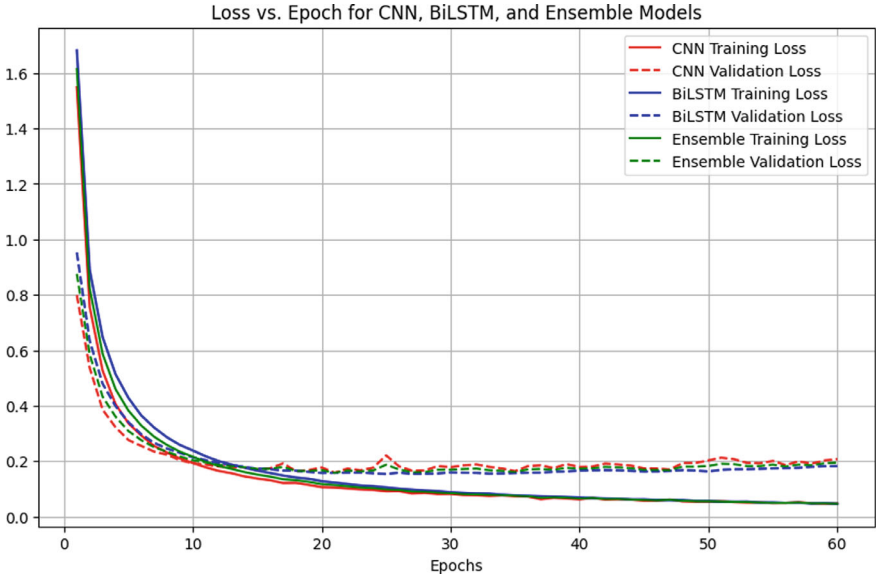

Fig. 6 Loss graph

Long Short-Term Memory (BiLSTM) layers to capture complex temporal dependencies in the data. This ensemble approach significantly improves the system's performance by combining the strengths of both CNN and BiLSTM, resulting in higher accuracy, robustness, and a better ability to distinguish between fall and non-fall events in real-time fall detection tasks.

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AI-Based Automated ICD Coding for Clinical Texts and Records: A Systematic Literature Review

D. Sasikala, J. Sabarinath, and N. Sarrvesh

Abstract The International Classification of Diseases (ICD) is an International standard for epidemiological investigation, health management and clinical diagnosis that play a critical role in intelligent healthcare. The task of assigning medical codes to clinical free-text documents have been recognized as advantageous for purposes like insurance claim processes, E-health record management and research. Healthcare professionals and professional medical coders assign such codes manually to the health records like discharge summaries, radiology reports, death certificates, pathology reports, etc. Using an automated approach for this task helps in reducing expenses, alleviating administrative burden and prevents errors. The main objective of this literature review is to provide detailed overview about automated ICD coding, deep learning algorithms and state-of-the art models used for ICD coding. Different NLP techniques were utilized for effective text processing and assignment of ICD codes to Electronic Health Records (EHR). The performance of several approaches on publicly available EHR datasets like MIMIC was compared using appropriate performance metrics.

Keywords ICD coding · EHR · NLP · MIMIC · Deep learning algorithms

1 Introduction

The International Classification of Diseases (ICD) formulated by the World Health Organization (WHO) has standardized the coding of a patient's medical diagnoses and procedures worldwide. It is the most comprehensive coding framework to date, which facilitates healthcare management, the reimbursement process, and global health research. In France, for instance, through the acceptance of ICD-10 by insur-

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ance companies, the treatment of re-compensation claims has been standardized and uniformed [1]. This system facilitates the translation of diagnoses into alphanumeric codes; it makes the data easier to store, research, and analyze [1]. The problem that medical providers face is the complexity of ICD coding which has over 60,000 codes. These challenges are particularly evident in regions where the system has not been fully adopted [1, 2]. ICD coding process is carried out traditionally by clinical coders, who do extensive reading on patient records, which has discharge summaries, operative reports and progress notes on a manual basis. Such a comprehensive and laborious process is bound to be error-prone as it is performed by humans [1, 3]. The growing urgency of being efficacious yet accurate in the delivery of healthcare calls for the replacement of such outdated coding systems with automated ICD coding systems that support clinical coders and enhance the quality of coding as a whole.

Recent innovations in Natural Language Processing (NLP) and machine learning (ML) have revived interest in automation of this complex task using faster and more reliable code assignments [1, 4, 5]. Many deep learning models with NLP techniques for automating ICD coding have been discussed in various studies to enhance the accuracy of the coding process. For instance, it has been observed that some models, including GRU and CNN provide more than satisfactory performance in the prediction of ICD codes for clinical texts [3, 4]. For instance, one prototype model was developed by researchers to achieve multimodal machine learning for integrating both structured and unstructured text in a mode with efficacy potential in coding outcome [4]. In addition to that, attention mechanisms and transformer-based models have been introduced as powerful tools in the field that allow the handling of long sequences of clinical notes than what is supported by traditional models [5]. Still with all these progressions lots of challenges exists in trying to understand clinical language and work on large label sets that the ICD system predisposes. It has been observed that existing automated coding systems possess inherent limitations in processing very long clinical texts and handling computational complexities that are normally associated with large code sets [2, 4, 5]. Moreover, the lack of a standardized implementation of ICD in various countries makes the automation of ICD coding a challenging task and therefore necessitates customized solutions [2, 3]. This review aims to synthesize the current research on automated ICD coding, drawing particular attention to methodologies used, challenges faced, and innovations emerging within this developing field. Through the study of the landscape of automated coding systems, we hope to contribute to ongoing discourse in healthcare data management and the accuracy of clinical coding practices. Our contributions include:

(1) What are the different datasets available for ICD code generation using clinical records?
(2) What are the different pre-processing techniques available for clinical text processing?
(3) What are the different models/algorithms available for automated ICD code generation?

(4) What are the various evaluation metrics used in studies to assess the performance of the model?
(5) What are the future research prospects in the field of ICD coding?

2 Datasets

In this section, we reviewed the publicly available EHR or clinical records datasets that are predominantly used in various studies related to ICD code identification and generation.

2.1 MIMIC Dataset

There are different versions of MIMIC Dataset like MIMIC-II, MIMIC-III, and MIMIC-IV. Most of the recent works majorly used MIMIC-III and MIMIC-IV dataset, while the MIMIC-IV being the latest version. Medical Information Mart for Intensive Care (MIMIC)-IV was collected from patients at Beth Israel Deaconess Medical Center in Boston, Massachusetts, USA over the period of 2008–2019, and it contains the de-identified Electronic Health Records (EHRs) [6]. This dataset is available in physionet.org. The data includes several indicators of each individual patient's hospitalization, for instance drugs taken by each patient, their lab results and changes in their vital signs [7]. The dataset comprises about 257,000 distinct individuals who were admitted to hospital 524,000 times. MIMIC-IV is the latest version.

2.2 CodiEsp Dataset

The CodiEsp corpus, which consists of one thousand clinical cases in Spanish language forms, is part of the Clinical Case Coding in Spanish Shared Task at CLEF 2020 conference held during October 2020. It was constructed based on data from physicians and clinical documentalists with coding done by clinical professionals achieving high inter-annotator agreement rates. It is made up of 16,504 sentences and contains 396,988 words overall. Three subsets are included in this dataset; those are training with 500 cases, development with 250 cases and testing with 250 cases [8]. Additionally, 176,294 abstracts with ICD-10 codes from Lilacs and Ibecs are provided. The task includes three subtracks: CodiEsp-D for diagnosis coding, CodiEsp-P for procedure coding, and CodiEsp-X for explainable AI, where both ICD-10 codes and their supporting text references are predicted [9].

Table 1 Lingual categorization of datasets

Language	Datasets
English	MIMIC-II, MIMIC-III-Full, MIMIC-III-50, CCHMC, CDC
Chinese	UKLarge, UKSmall, Xiangya, CN-Full, CN-50
Dutch	UZA
French	CepiDc-2017, CepiDc-2018
Italian	CLEF-Italian
Hungarian	CLEF-Hungarian
German	CLEF-German
Spanish	CLEF-Spanish

2.3 CDC-CepiDC Dataset

Conference and Labs of the Evaluation Forum (CLEF) has evaluated ICD coding of health-related documents every year from 2017 to 2020, and its datasets are mainly based on French, Spanish, German, and other European languages. The Center for Disease Control (CDC) dataset and CepiDc-2017 dataset are the specific datasets used in CLEF eHealth 2017. The CepiDc-2018, CLEF-Italian, and CLEF-Hungarian datasets were publicly available for use in CLEF eHealth 2018. They all collect electronic death certificates as the documents to be assigned ICD codes [10]. Several other datasets were used in different studies. They are restricted and proprietary datasets which are not available for public access. Such widely used datasets include: NYU Langone Hospital EHR System Dataset, Fuwai Dataset, National Taiwan University Hospital (NTUH) Dataset, Rama-data, TrialTrove dataset, CN-Full, CN-50. A Compilation of EHR datasets categorized based on languages is given in Table 1.

3 Pre-processing Techniques

In this section, we have discussed the different pre-processing techniques that have been used across many studies involving ICD code generation. The above-given dataset contains a large corpus of EHR and free clinical texts that demand diverse pre-processing techniques. The large text corpus contains special characters, stop words and other entities that should be removed from the corpus before using it to train the model. This technique is called text processing. Many pre-processing techniques are available. They are discussed in detail in the below subsections.

3.1 Removal of Stop Words and Special characters

Stopwords are a set of commonly used words in a language. In English it is "a," "the," "is," "are," and, etc. Similarly, every other language has its own type of stopwords. These stopwords present in the corpus are of no use to the ICD code and it's generation. So, these words are removed from the corpus in the process of text pre-processing. There are some inbuilt modules in python that has the suitable functions to remove the stopwords from the input data. The most efficient and widely used module is NLTK. It has a list of stopwords and punctuation for every language. By having that as a reference, it eliminates all the stopwords and special characters that are present in the input corpus [11, 12].

Tokenization

Tokenization is a method of text pre-processing where a sentence is broken down into fragments of words. It is known as word-level tokenization. It makes the model learn the pattern without any loss in content. It also helps in handling complex language structures, standardization, and efficiency in processing [13]. Tokenization is a very crucial pre-processing step for ICD code generation, as it helps in easing out the process of feature extraction when fed into the model [14]. Tokenization helps in the segregation of clinical terms that will be helpful for ICD code generation [15]. There are different inbuilt Python modules and functions that do this job. Those includes: (a) Tokenization using inbuilt "split" function. (b) Tokenization using tokenize. (c) Tokenization using spaCy. (d) Tokenization using NLTK. (e) Tokenization using Genism. There is a separate tokenization algorithm called WordPiece algorithm, which was developed by Google to handle Out-of-Vocabulary (O-O-V) words. It is primarily used to pre-train BERT models and it's variants [16].

3.2 Lemmatization

Lemmatization is a text pre-processing technique used to break a word down to its root meaning called "lemma" to identify similarities. For example, a lemmatization algorithm would reduce the word better to its root word good. This step helps in handling different tenses and forms of words that are present in clinical texts [11]. There are many python modules that helps in doing lemmatization. They are: (a) NLTK (b) Textbolb (c) spaCy (d) Genism.

3.3 Lowercasing

Lowercasing the texts helps in standardizing and maintaining uniformity across the corpus, which helps in effective recognition of entities by the models [5, 17, 18]. This can be easily done using the inbuilt python function "lower()."

3.4 Word Embedding

Word Embedding is a technique where individual words are represented as real-valued dense vectors in a continuous space and captures inter-word semantics [14, 19]. It helps in handling synonyms, variations and improves classification accuracy [7]. For example, embeddings can understand that "fracture" and "bone break" are semantically related even if they are expressed differently in the text. Words with similar meanings should have similar vector representations [20]. Word embedding is widely done using NLTK and word2vec modules of python. Word2vec works based on two architecture, Continuous Bag of Words (CBOW) and Skip-Gram [21]. These two are neural network models primarily used for word embedding and falls under word2vec. CBOW predicts the target word by averaging out the context word vectors whereas, skip-gram uses the target word vector to predict multiple context words [22]. Doc2Vec and Paragraph2vec are variants of the word2vec. They focus on predicting words in the document or paragraph. Doc2vec creates a numeric representation of a document irrespective to its length. Some other python modules with which word embeddings can be done are: (a) spaCy (b) Genism (c) Tensorflow-keras (d) Transformers-HuggingFace (e) PyTorch-nn Embedding (f) Sklearn-TfIdf vectorizer.

4 Models

Deep learning is a machine learning technique that uses multi-layered neural networks to learn hierarchical data representations. Deep learning models have shown successful results in many NLP tasks such as sentiment analysis, image captioning and language translation. Transfer learning, which can be considered a subset of deep learning, is a technique that leverages models that were already trained using a large set of corpus. Such models are known as transformers. For ICD coding, many transformer models were used across different studies, that were pre-trained using large medical based corpus. Across many studies, different variants of Neural networks and Transformers were used to perform this task efficiently. Custom attention mechanism were also implemented in different works where it helps in capturing rare codes and improves accuracy. Some noticeable attention mechanisms are Lambda Scaled Attention Mechanism and Hierarchical Attention Network [2, 3, 8]. Lambda Scaled Attention implements a scaling factor (lambda) which rebalances the atten-

tion weight, so that the model is able to focus on more significant sections of the input sequence. This mechanism is useful as it increases the attention for certain tokens which helps the model focus on important tokens for tasks like ICD code generation. HAN applies attention at both the word as well as sentence level, which allows the model to attend proper words from within a sentence and key sentences from a document. Such architecture is suitable for long text processing which advances the interpretability and performance of the model in tasks such as document classification. Many variants of Convolution Neural Network (CNN), Recurrent Neural Network (RNN), Gated Recurrent Unit (GRU), Bidirectional Encoder Representations from Transformers (BERT) were predominantly used in different studies. Some of them are listed below:

- BiLSTM
- BiGRU
- HAN GRU (Hierarchical Attention Network GRU)
- HAN-LSTM (Hierarchical Attention Network LSTM)
- MS-CNN (MultiScale CNN)
- CAML (Convolutional Attention for Multi-Label classification)
- DR-CAML (Description Regularized CAML)
- MultiRes CNN (Multi-Filter Residual Convolutional Neural Network)
- ABCNN (Attention-Based Convolutional Neural Network)
- DRCNN (Deep Recurrent CNN)
- CNN-NWE (Convolutional Neural Network with Neural Word Embedding)
- XR-Transformer
- XR-LAT
- XLNET
- PubMedBERT
- KB-BERT
- DRCNN + Transformer
- CNN + PubMedBERT
- NB-TF-IDF (Naïve Bayes with Term Frequency-Inverse Document Frequency).

5 Evaluation Metrics

In this section, we've discussed about the evaluation metrics that are widely used in different studies related to ICD coding using deep learning.

Accuracy: Accuracy is the ratio of instances that were correctly predicted to the total instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Precision: Precision is the ratio of true positives to the total predicted positives.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall: The ratio of true positives to the sum of true positives and false negatives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-score: F1-score can be expressed as the harmonic mean of precision and recall.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Micro F1-score: The F1-score computed globally by counting total true positives, false negatives, and false positives across all classes [23] .

$$\text{Micro F1} = 2 \times \frac{\sum \text{TP}}{\sum \text{TP} + \sum \text{FP}} \times \frac{\sum \text{TP}}{\sum \text{TP} + \sum \text{FN}}$$

Macro F1-score: The overall F1-score averaged across classes, taking into account all the classes equally [24].

$$\text{Macro F1} = \frac{1}{n} \sum_{i=1}^{n} \text{F1-Score}$$

Macro-AUC: The average area under the ROC curve computed for each class individually [25, 26]. The Receiver Operating Characteristics (ROC) curve plots the True Positive Rate (Recall) against the False Positive Rate at various thresholds, demonstrating a classifier's performance.

$$\text{Macro-AUC} = \frac{1}{n} \sum_{i=1}^{n} \text{AUC}$$

Micro-AUC: The area under the ROC curve computed globally by aggregating all classes [27, 28].

$$\text{Micro-AUC} = \text{ROC-AUC(micro-average)}$$

Micro AUPRC: The area under the precision-recall curve computed globally across all classes is termed as Micro AUPRC.

$$\text{Micro AUPRC} = \text{AUPRC(micro-average)}$$

Macro AUPRC: The average area under the precision-recall curve across all classes is termed as Macro AUPRC.

$$\text{Macro AUPRC} = \frac{1}{n}\sum_{i=1}^{n}\text{AUPRC}$$

Micro AUROC: Micro AUROC is the area under the ROC curve computed globally across all classes (same as Micro-AUC).

$$\text{Micro AUROC} = \text{ROC-AUC(micro-average)}$$

Macro AUROC: Macro AUROC is the average area under the ROC curve computed for each class individually.

$$\text{Macro AUROC} = \frac{1}{n}\sum_{i=1}^{n}\text{ROC-AUC}$$

ROC-AUC: The area under the ROC curve measures the model's ability to distinguish between classes [29, 30].

$$\text{ROC-AUC} = \int_0^1 \text{ROC}(t)\,dt$$

The evaluation metrics listed above effectively assess the model's capacity to differentiate across different classes. Thus, these are predominantly used in many studies to evaluate the performance of the model in predicting and generating the ICD code.

6 Results Comparison

The performance of the models were determined using the appropriate evaluation metrics. Table 2 displays the dataset used and performance comparison of different models based on different evaluation metrics, across all the studies that we have chosen for this work. P@k (Precision at k): Measures the precision within the top k predictions. For example:P@5: Precision within the top 5 predictions. It indicates how many of the top 5 predictions were correct, on average. Recall@k: Measures recall within the top k predictions. It checks how many of the true positive labels were recovered in the top k predictions. R@5:Recall within the top 5 predictions. Models that utilize attention mechanisms, like MSATT-KG and HAN GRU are effective in processing clinical notes that tend to be long and complicated as they are good at retaining essential contextual information. Generalization is greatly exhibited by transformer-based architectures such as BERT, PubMedBERT which when is fine-tuned on domain specific datasets. They perform truly well on several performance

Table 2 Comprehensive comparison of various models across different datasets and metrics

Model	Dataset	P	R	Micro-F1	Macro-F1	Micro-AUC	Macro-AUC	Other metrics	Study
ANN (DL4J)	Custom (LoS groups)	0.8421	0.828	–	–	–	–	Avg. Accuracy: 0.8421	[1]
BW_att	Fuwai-CHD (Top-100)	0.962	–	–	0.962	–	0.989	–	[4]
BW_att	MIMIC-III-CHD (Top-50)	0.405	–	–	0.405	–	0.661	–	[4]
DRCNN-ATT	CPLT (private)	0.862	–	0.862	0.25	–	–	–	[2]
DRCNN-ATT	MIMIC-III-Top 50	0.705	–	0.705	0.655	–	–	–	[2]
BiLSTM + Multi-scale CNN	ICD-10 CM	–	0.873	0.715	–	–	–	–	[3]
BiLSTM + Multi-scale CNN	ICD-10 PCS	–	0.887	0.618	–	–	–	–	[3]
DNN (BERT+BiGRU)	ICD-10-PCS	–	0.887	0.618	–	–	–	–	[5]
Fuzzy Logic w/ Levenshtein	SICO	–	–	–	0.723	–	–	Accuracy: 0.901	[16]
Fuzzy Logic w/ Levenshtein	Custom	–	–	–	0.385	–	–	Accuracy: 0.8	[16]
Bag of Words	Stockholm EPR Gastro	0.8299	–	–	–	–	–	–	[13]
PLM-ICD	MIMIC-III	–	–	0.608	0.11	–	–	–	[19]
PLM-ICD	MIMIC-II	–	–	0.509	0.07	–	–	–	[19]
XR-LAT	MIMIC-III	–	–	–	–	–	–	Macro-AUC +0.021	[19]
XR-LAT	MIMIC-II	–	–	–	–	–	–	Macro-AUC +0.051	[19]
Fusion	MIMIC-III 50	0.647	–	0.674	–	–	0.909	–	[25]
Fusion	MIMIC-III-Full	0.736	–	0.554	–	–	0.915	–	[25]
BERT-XML	MIMIC-III	–	–	–	–	0.97	0.927	–	[25]

(continued)

Table 2 (continued)

Model	Dataset	P	R	Micro-F1	Macro-F1	Micro-AUC	Macro-AUC	Other metrics	Study
Attention-based Deep Semantic	NYU Langone	0.979	0.983	0.981	–	–	–	Accuracy: 0.986	[15]
HGCN-CL	MIMIC-III-Full	0.748	–	0.578	0.102	0.99	0.947	–	[14]
HGCN-CL	MIMIC-III-Top 50	0.71	0.729	0.729	0.682	0.937	0.936	–	[14]
CNN-PubMedBERT	MIMIC-III	–	–	0.6221	0.544	0.952	0.9211	–	[11]
ICDXML	MIMIC-III-50	–	–	0.718	0.671	0.946	0.925	–	[27]
ICDXML	MIMIC-III-FULL	–	–	0.585	0.107	0.987	0.922	–	[27]
BERT (LongBERT)	MIMIC-III-Full-Code	0.6148	0.528	0.568	–	0.983	–	PR-AUC: 0.5402	[29]
RoBERTa-Mini	Fuwai dataset	–	–	0.9387	-	0.996	–	–	[26]
BERT-mini	CodiEsp dataset	–	–	0.8541	–	0.928	–	–	[26]
BoW	Fuwai dataset	–	–	0.8299	–	-	–	–	[9]
BoW	CodiEsp dataset	–	–	0.3913	–	–	–	–	[9]
LightGBM (sequential)	Custom	-	-	-	0.883	–	–	Accuracy: 0.952	[24]
HAN GRU	MIMIC	–	–	0.727	0.564	0.916	–	–	[8]
HAN GRU	MIMIC+CodiEsp	–	–	0.734	0.563	0.913	–	–	[8]
DACNM	CN-Full	0.643	–	0.7127	0.1116	0.991	0.952	–	[23]
DACNM	CN-50	0.8895	0.86	0.8602	0.8452	0.993	0.9878	–	[23]
DACNM	MIMIC-III-50	0.616	–	0.641	0.579	0.916	0.89	–	[23]
GRU	Custom	–	–	–	–	–	0.8599	–	[19]

(continued)

Table 2 (continued)

Model	Dataset	P	R	Micro-F1	Macro-F1	Micro-AUC	Macro-AUC	Other metrics	Study
DRCNNTLe	MIMLT	–	–	0.844	0.227	–	–	–	[20]
LongBERT	MIMIC-III-Full-Code	–	–	0.568	–	0.983	–	PR-AUC: 0.5402	[29]
CAML	MIMIC-III-Full-Code	–	–	0.5465	–	–	–	–	[29]
IR-style Model	MIMIC-III	0.765	0.569	0.6524	–	–	–	–	[10]
DR-CAML	MIMIC-III (Top 50)	0.6539	–	0.6964	0.6456	0.939	–	–	[17]
MVC-RLDA	CodiEsp	–	–	0.1392	0.0421	0.564	–	–	[17]
TF-GCN	MIMIC-III-Full	0.758	–	0.589	0.109	0.989	0.934	–	[28]
TF-GCN	MIMIC-III 50	0.682	0.719	0.719	0.671	0.953	0.928	–	[28]
M-XLNet	MIMIC-II	–	–	0.498	–	–	–	–	[6]
M-XLNet	MIMIC-III	–	–	0.585	–	–	–	–	[6]
MDBERT	MIMIC-III-Top-50	0.674	–	0.717	0.672	0.946	0.928	–	[30]
MDBERT	MIMIC-III-Full	0.793	–	0.75	0.576	0.992	0.942	–	[30]
GatedCNN-NCI	MIMIC-III-Top-50	–	–	0.686	0.629	0.938	0.915	–	[18]
GatedCNN-NCI	MIMIC-III Full	–	–	0.563	–	–	0.922	–	[18]
MultiResCNN	MIMIC-III-Full	0.734	–	0.552	0.85	0.986	0.91	–	[20]
MultiResCNN	MIMIC-III-Top-50	0.641	–	0.67	0.606	0.928	0.899	–	[20]
MSATT-KG	MIMIC-III-Full	0.728	–	0.553	0.9	0.992	0.91	P@8: 0.728	[22]
MSATT-KG	MIMIC-III-Top-50	0.644	–	0.684	0.638	0.936	0.914	–	[22]
CNN-Att	MIMIC-III	0.59	0.492	0.537	–	–	–	–	[21]
CNN-Att	HSL-SEA	0.543	0.438	0.485	–	–	–	–	[21]

indicators like Micro-AUC, F1-scores, and others. Models like MultiResCNN and TF-GCN are particularly effective for larger datasets like MIMIC-III, which involves dealing with long sequences and multi-label classification. In short, the most successful approach to generate ICD codes is that which combines attention mechanism, domain specific embeddings and multi-paradigm's architecture.

7 Discussion

In this section, we've discussed about the limitations, progress, innovation, and future scope of ICD code generation in the field of healthcare. The empirical literature review addresses the research gaps and limitations identified in the studies. This section emphasizes potential research directions that require significant effort to establish an automated ICD coding system.While conducting the literature review we have observed that, there are only few publicly available benchmark medical report datasets like PhysioNet. To access PhysioNet (MIMIC dataset) official licensing/authorization to access the dataset is required, which one can obtain by completing a course given by PhysioNet. Most of the datasets used in different studies are either restricted datasets or manually collected datasets. This stunts the development of open source research works in the field of ICD coding and generation. Thus, this is one of the main issues that should be addressed by establishing a benchmark dataset prepared according to the most recent classification system that can be used for future research purposes. As models continue to evolve and fit seamlessly into clinical practice, the prospects for ICD code generation in the future are exciting. One area that will improve is the incorporation of multimodal learning. This is important because it unites various data types, i.e., clinical notes, radiology, and EHRs in order to synergistically build up a picture of a complicated medical condition. Specific fields like pediatrics, oncology, and psychology will leverage custom systems for the generation of ICD codes due to the inherent intricacies within each of them. Clinical text summarization [31, 32] can be mapped to ICD code for document classification which will further enhance the clinical system. Finally, it's conceivable that in the years to come, coding will be made much easier with the aid of machines that will propose the codes to be confirmed by professionals, thus achieving a good collaboration and of speed and efficiency of the machines and the skills of the healthcare professionals.

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A Hybrid CNN-LSTM Model for Sudden Cardiac Arrest Prediction

Victor Azad, Anuj Kumar Jha, Ashutosh Kumar Jha, and Shourya Bhushan

Abstract Sudden Cardiac Arrest (SCA) is a major cause of death due to cardiovascular diseases, responsible for approximately 4–5 million fatalities every year worldwide. However, prompt medical intervention and proper treatment can significantly improve survival rates for individuals experiencing SCA. In this work, we propose a hybrid deep learning model that fuses 1D-Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks for the early prediction of SCA using Electrocardiogram (ECG) signals. The 1D-CNN component is used to capture spatial features and irregularities associated with SCA, while the LSTM layer identifies temporal patterns essential for accurate prediction. The proposed model achieves a training accuracy of 99.93% and a testing accuracy of 99.75% predicting SCA from ECG signals 30 min prior to its occurrence. For signals recorded 60 min before SCA onset, the model reports a training accuracy of 99.87% and testing accuracy of 99.56%. Testing on ECG signals from different time intervals before the onset of SCA demonstrates the model's robustness and efficiency. The fusion of CNN and LSTM enhances both prediction accuracy and reliability, improving sensitivity and specificity. This approach not only delivers high accuracy but also strengthens the model's overall predictive power, making it a valuable tool for advanced cardiovascular monitoring systems. Moreover, the model's low computational complexity enables real-time early detection of Sudden Cardiac Arrest (SCA) with minimal delay, offering a promising solution for proactive cardiac health management.

Keywords ECG · Deep learning · Sudden cardiac arrest · CNN · LSTM

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1 Introduction

Sudden Cardiac Arrest (SCA) is a life-threatening disease characterized by the abrupt cessation of cardiac function, often triggered by acute symptoms and typically occurring within one hour of onset. This devastating event leaves little to no time for medical intervention, leading to alarmingly high mortality rates. Ventricular Fibrillation (VF), a severe arrhythmic event, is identified as the initiating factor in approximately 20% of SCA cases [1]. These statistics highlight the critical need for accurate and early prediction systems that can provide sufficient lead time for effective treatment, ultimately improving survival rates.

Electrocardiogram (ECG) signals have become a vital tool in predicting SCA due to their ability to capture real-time cardiac activity. However, manually extracting predictive features from ECG data is a challenging and time-consuming task. The dynamic and complex nature of ECG signals, particularly during the lead-up to SCA, necessitates the use of advanced computational approaches capable of identifying subtle, time-dependent patterns with precision.

Deep learning methods have shown significant promise in addressing these challenges. These techniques enable automatic feature extraction and uncover complex patterns that traditional methods often fail to detect. CNNs are highly effective at extracting spatial features from time-series data, while Recurrent Neural Networks (RNNs) excel at modeling temporal dependencies. LSTM networks, an advanced RNN architecture, further improve on this by efficiently capturing long-term dependencies, making them especially suitable for SCA prediction tasks [2].

In this study, we propose a hybrid deep learning architecture that fuses the strengths of 1D-CNNs and LSTM networks to extract both spatial and temporal features from ECG signals. This integrated approach achieves substantial improvements in prediction accuracy, sensitivity, and specificity, establishing a reliable framework for real-time SCA detection. Additionally, the proposed model addresses the growing demand for wearable, low-complexity monitoring devices, facilitating accessible and proactive cardiac care solutions.

2 Materials and Methods

This study utilizes ECG signals from two widely recognized benchmark databases: the MIT-BIH Normal Sinus Rhythm database [3] and the MIT-BIH SCA Holter database [4] as shown in Table 1. The MIT-BIH Normal Sinus Rhythm database includes 18 ECG signals sampled at 128 Hz, representing baseline normal cardiac rhythms. In contrast, the MIT-BIH SCA Holter database provides data from 23 patients diagnosed with Sudden Cardiac Arrest (SCA), capturing critical episodes of Ventricular Fibrillation (VF) with an original sampling frequency of 250 Hz. To ensure consistency and facilitate a comparative analysis, this study focuses on 20 subjects from the SCA database, specifically those who experienced VF events. The

Table 1 Subject demographics and sampling rates for the MIT-BIH SCA Holter database and the MIT-BIH normal sinus rhythm database

Diagnosis	Total	Subject details	Sampling rate (Hz)
Normal	18	13 Females (Age: 20–50 years) 5 Males (Age: 26–45 years)	128
SCA	23	8 Females (Age: 30–89 years) 13 Males (Age: 17–79 years) 2 gender unknown	250

remaining three subjects were excluded due to the absence of VF episodes, creating a more homogeneous dataset aligned with the study's objectives. ECG signals from the SCA group were downsampled to match the 128 Hz sampling frequency of the Normal Sinus Rhythm database. This preprocessing step standardizes the data, ensuring compatibility for feature extraction and subsequent analyses (Fig. 1).

Visualizations of raw ECG signals from both normal and SCA subjects are shown in Figs. 2 and 3, depicting a 4 s heart activity. In normal ECG signals (Fig. 2), characteristic features such as P waves, QRS complexes, and T waves are distinct and rhythmic, reflecting stable cardiac function. By contrast, SCA signals (Fig. 3) display marked irregularities, indicative of abnormalities. These deviations highlight the need for advanced algorithms capable of detecting subtle yet critical patterns in the data.

The overall methodology is summarized in the flowchart shown in Fig. 1, comprising three key stages:

1. **Preprocessing**: The raw ECG signals undergo preprocessing to remove noise and harmonize sampling rates. Signals from the SCA database are downsampled from 250 Hz to 128 Hz to match the Normal Sinus Rhythm database, ensuring uniform time resolution across all signals.
2. **Feature Extraction**: Advanced feature extraction is performed using Convolutional Neural Networks (CNNs), which identify spatial patterns in the ECG signals. CNNs are particularly adept at detecting localized features such as waveforms and transitions that are essential for identifying abnormalities.

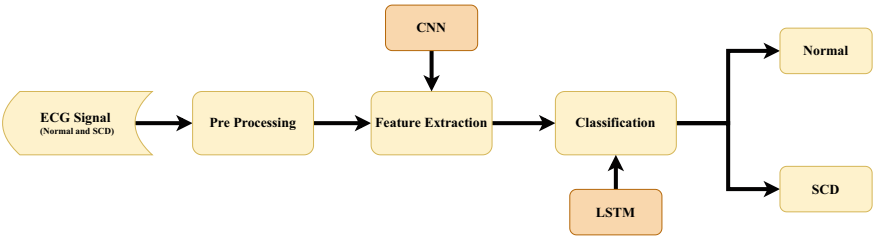

Fig. 1 System architecture for CNN fused LSTM model for SCA prediction

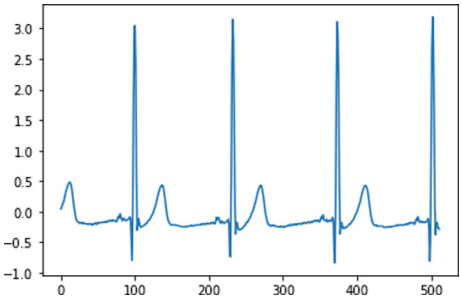

Fig. 2 Normal raw ECG signal

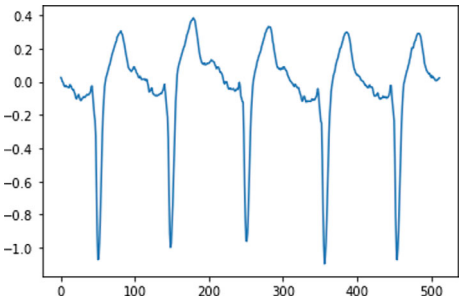

Fig. 3 SCA raw ECG signal

3. **Classification**: The extracted features are processed using LSTM networks, which capture temporal dependencies and dynamic changes in the data. The final output of the model is a binary classification that predicts the likelihood of SCA event.

The fusion of CNN and LSTM architectures enables the model to simultaneously leverage spatial and temporal features from ECG signals, enhancing its predictive accuracy. By integrating a rich and standardized dataset with advanced deep learning methods, this study contributes to the development of robust, real-time SCA prediction systems. The combination of rigorous data preprocessing, visual analysis, and sophisticated deep learning models establishes a comprehensive framework for addressing this critical healthcare challenge.

2.1 Preprocessing

First of all the raw ECG signals undergo a filtration process to eliminate noises from the raw signals, enhancing signal quality. The ECG signals from MIT-BIH SCA Holter database were downsampled to 128 Hz, maintaining consistency in correspondence with MIT-BIH Normal Sinus Rhythm database.

Suppose $x[n]$ is the original signal sampled at frequency $f_s = 250\,\text{Hz}$, the downsampled signal $y[m]$ at frequency $f_d = 128\,\text{Hz}$ can be represented as:

$$y[m] = x[n] \quad \text{where} \quad n = m \cdot \frac{f_s}{f_d}$$

where $\frac{f_s}{f_d} = \frac{250}{128} \approx 1.95$, which means every second sample (approximately) is retained. To avoid aliasing, a low-pass filter was applied before decimation to ensure removal of frequencies above $f_d/2$.

Downsampling the SCA signals to 128 Hz aligns the SCA Holter database signals with the Normal Sinus Rhythm database signals, ensuring a consistent time resolution across all ECG signals. This uniformity facilitates coherent feature extraction, simplifies model training, and eliminates potential bias caused by disparate sampling rates, enabling fair and reliable analysis.

2.2 Feature Extraction using 1D-CNN

The proposed methodology employs a 1D-CNN to extract spatial features from the denoised ECG signals. The preprocessing phase is followed by feature extraction, where the 1D-CNN plays a central role. The model architecture is illustrated in Fig. 4, which details the various layers and operations within the CNN. The 1D-CNN is especially suitable for this task as it can efficiently capture relevant features from the temporal sequences inherent in ECG signals.

In the CNN, the **convolution operation** applies a set of learned filters (kernels) to the ECG signal to extract meaningful features. For an input ECG signal sequence $x[n]$ of length N, the convolution operation at position i produces an output feature map $h[i]$, and it is mathematically expressed as:

$$h[m] = \sum_{k=0}^{K-1} w[k] \cdot l[m-k] + e$$

where

h[m] is the output feature map at the i-th position, representing the response of the ECG signal to the filter at that location.

w[k] is the filter (or kernel) of size K, which is learned during training to detect relevant features such as peaks, PQRST waves, or other significant temporal patterns in the ECG signal.

l[m-k] is the ECG signal at position $m - k$, where the filter slides over the signal to capture various temporal patterns across the sequence.

e is the bias term, which helps adjust the output to capture more complex relationships in the data.

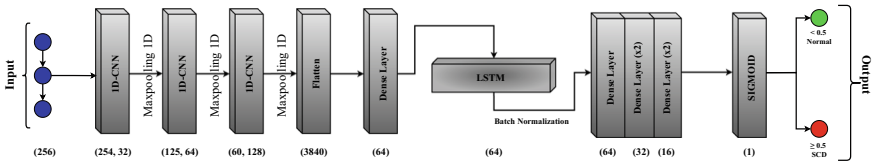

Fig. 4 Model architecture of hybrid CNN-LSTM for SCA prediction

The CNN's ability to convolve over the temporal sequence allows it to automatically detect and extract important features such as heart rate variability, arrhythmias, and other characteristics indicative of potential abnormalities in the ECG signal.

After convolution, the feature map is downsampled using a **pooling operation**, such as max pooling, to reduce the dimensionality of the feature map by removal of irrelevant features while retaining the important features corresponding to ECG alterations. The pooling operation for an output feature map $h[i]$ is defined as:

$$p[j] = \max_{m \in R_j} h[m]$$

where R_j denotes the receptive field at position j and $p[j]$ is the output after pooling. Pooling layers reduces the feature dimensionality keeping the important features and filtering out less relevant features to reduce large computational cost.

The convolutional feature extraction approach to automatically learn hierarchical representations from the ECG signals enables the detection of intricate temporal patterns that are crucial for identifying SCA. By learning these representations directly from the raw ECG data, the model removes the need for manual feature engineering, thus providing a more efficient and scalable approach.

2.3 Time Series Prediction using Dense LSTM

LSTM, a variant of Recurrent Neural Networks (RNNs), effectively handles long-term dependencies in sequential data such as ECG signals. After processing the features through LSTM layers, a fully connected dense layer and a sigmoid activation function are employed to produce the final prediction of SCA.

The operations of the LSTM cell are mathematically defined as follows for each time step s:

1. Forget Gate-Controls which parts of the previous cell state K_{s-1} to retain:

$$f_s = \sigma(g_f \cdot [h_{s-1}, x_s] + r_f)$$

2. Input Gate-It determines which new information adds to new cell state.

$$i_s = \sigma(g_i \cdot [h_{s-1}, x_s] + r_i), \quad \tilde{K}_s = \tanh(g_c \cdot [h_{s-1}, x_s] + r_c)$$

3. Cell State Update-It combines both input and Forget Gate to update the cell state.

$$K_s = f_s \odot K_{s-1} + i_s \odot \tilde{K}_s$$

4. Output Gate-It Determines the hidden state h_t for the next time step:

$$o_s = \sigma(g_o \cdot [h_{s-1}, x_s] + r_o), \quad h_s = o_s \odot \tanh(C_s)$$

Here, x_s is the current input, h_{s-1} is the previous hidden state, g_f, g_i, g_c, g_o are weight matrices, r_f, r_i, r_c, r_o are biases, σ is the sigmoid activation function, tanh is the hyperbolic tangent activation function, and $\odot$ denotes element-wise multiplication.

Once the LSTM processes the sequence, the final hidden state h_s is passed through a sequence of dense layer for output transformation. The dense layers applies a linear transformation to map the LSTM outputs into a single neuron for binary classification:

$$m = g_d \cdot h_t + r_d$$

where g_d and r_d represent the weights and bias of the dense layer, respectively.

Finally, a sigmoid activation function is applied to produce a probability score $\hat{y}$ for SCA occurrence:

$$\hat{y} = \sigma(m) = \frac{1}{1 + e^{-m}}$$

Here, $\hat{y}$ is the predicted probability of SCA, where values closer to 1 indicate higher likelihood of SCA and values near 0 suggest normal conditions.

The combination of LSTM, dense layer, and sigmoid activation ensures the model captures temporal dependencies while effectively mapping the sequential outputs to a final binary prediction.

2.4 Mathematical Formulation of the Hybrid CNN-LSTM Model

The operations of the hybrid CNN-LSTM model can be summarized as:

A sample of size 256(i.e., 2 s of ECG signal of f_s = 128Hz) is input.

1. Spatial feature extraction:

$$p = \text{MaxPool}(\text{ReLU}(\text{Conv}(x)))$$

2. Dense representation:

$$m^{(1)} = \text{Dense}(\text{Flatten}(p))$$

Output is flattened to size 3840.

3. Temporal feature extraction:

$$h_t = \text{LSTM}(m^{(1)})$$

4. Batch normalization and multi-layer dense transformation:

$$m^{(i+1)} = \text{ReLU}(\text{Dense}(m^{(i)}))$$

5. Classification:

$$\hat{y} = \sigma(\text{Dense}(m^{(5)}))$$

where $\hat{y} \in [0, 1]$ indicates the probability of SCA. Threshold 0.5 determines the class. The hybrid 1D-CNN and LSTM model is designed to address the unique challenges of SCA prediction by combining spatial and temporal feature extraction. The 1D-CNN layers efficiently capture localized patterns in ECG signals, such as waveforms and sharp transitions, while the LSTM layer processes the sequential dependencies critical to identifying time-dependent changes preceding SCA. By integrating batch normalization and a series of dense layers, the model ensures robust feature transformation and generalization, making it highly effective for real-time applications. The use of a sigmoid classifier ensures simplicity and interpretability, providing actionable predictions for SCA risk. This architecture balances computational efficiency with high predictive accuracy, making it suitable for deployment in wearable health monitoring systems.

2.5 Model Training and Hyperparameter Tuning

The training process for the hybrid CNN-LSTM model utilized Electrocardiogram (ECG) signals spanning 30 min before the onset of Ventricular Fibrillation (VF). This temporal window was selected to capture critical patterns and dynamic changes that precede Sudden Cardiac Death (SCA). The training process aimed to optimize the model's ability to identify these subtle features through a systematic refinement of parameters. To achieve this, hyperparameter tuning was conducted using a sweep method facilitated by the weights and biases machine learning platform.

Training Objective

The model's objective during training was to minimize the binary cross-entropy loss $\mathcal{L}$, which measures the difference between the predicted probability $\hat{v}$ of an SCA event and the actual label y:

$$\mathcal{L} = -\frac{1}{L}\sum_{p=1}^{L}\left[v_p \log(\hat{v}_p) + (1 - v_p)\log(1 - \hat{v}_p)\right]$$

where

L Total number of samples,
v_p True label (1 for SCA, 0 for normal),
$\hat{v}_p = \sigma(m)$ Predicted probability from the sigmoid activation function:

$$\sigma(m) = \frac{1}{1 + e^{-m}}$$

m Output of the final dense layer.

The Adam optimizer was used to adjust the model's weights $\theta \in \{g, r\}$ to minimize $\mathcal{L}$. The parameter update rule is:

$$\theta^{(s+1)} = \theta^{(s)} - \eta \cdot \frac{\partial \mathcal{L}}{\partial \theta}$$

where η is the learning rate and $\frac{\partial \mathcal{L}}{\partial \theta}$ is the gradient of the loss with respect to θ.

Hyperparameter Tuning

To ensure optimal performance, the model underwent hyperparameter tuning using the sweep method, a systematic approach to explore the hyperparameter space. Let the set of hyperparameters be Θ, where

$$\Theta = \{\eta, \text{batch_size}, \text{CNN_filters}, \text{LSTM_units}, \text{dropout}\}$$

The goal was to maximize the model's performance metric, $\mathcal{P}(\Theta)$, which could be accuracy, precision, or another relevant measure, evaluated on a validation set:

$$\mathcal{P}(\Theta) = \frac{1}{M}\sum_{j=1}^{M}\mathbb{I}\left(\hat{y}_j^{(\text{pred})} = y_j^{(\text{true})}\right)$$

where

M Number of validation samples,
$\mathbb{I}(\cdot)$ Indicator function that equals 1 when the predicted label matches the true label.

Weights and biases facilitated the exploration by systematically testing combinations of hyperparameters and logging the model's performance for each configuration. For each trial, a subset of Θ was sampled, and the corresponding model performance $\mathcal{P}(\Theta)$ was recorded.

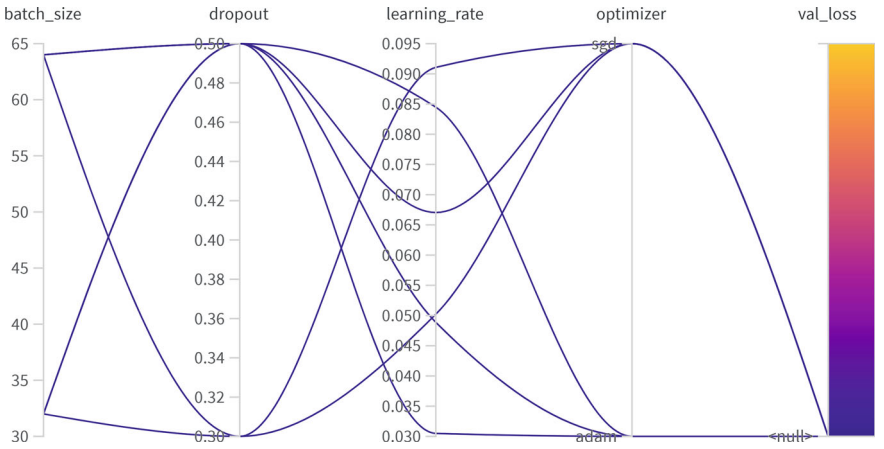

Fig. 5 Hyperparameter tuning of fused CNN-LSTM model

To prevent overfitting and ensure generalization, regularization techniques were applied: Dropout Regularization: Randomly deactivates a fraction p of neurons during training:

$$h^{(\text{dropout})} = h \odot \mathbf{m}, \quad \mathbf{m} \sim \text{Bernoulli}(1 - p)$$

where $\mathbf{m}$ is a binary mask and $\odot$ represents element-wise multiplication. Early Stopping: Training terminated if the validation loss $\mathcal{L}_{\text{val}}$ did not improve for T consecutive epochs:

$$\mathcal{L}_{\text{val}}^{(t+T)} \geq \mathcal{L}_{\text{val}}^{(t)}, \quad \forall t$$

The outcomes of the hyperparameter tuning process were visualized using performance graphs, such as the hyperparameter tuning landscape as shown in Fig. 5. Each point in the graph represents a configuration Θ, with the corresponding performance metric $\mathcal{P}(\Theta)$ depicted. This allows for identifying the hyperparameter set Θ^* that achieves the highest performance:

$$\Theta^* = \arg\max_{\Theta} \mathcal{P}(\Theta)$$

The systematic tuning process ensured that the final model was robust, adaptive, and capable of handling diverse ECG signals. The insights from the hyperparameter tuning graphs informed the selection of optimal configurations, making the model suitable for deployment in real-world applications for the early prediction of VF and SCA.

3 Results

Table 2 compares our method with existing approaches for predicting Sudden Cardiac Arrest (SCA). It shows the number of subjects, prediction times, and accuracy levels. Our method stands out with the high accuracy of 99.93% for 30 min and 99.87% for 60 min, surpassing prior methodologies in the literature.

3.1 Model Testing and Performance Evaluation

The performance of the proposed model was assessed at two time instants: (i) 30 min prior to SCA and (ii) 60 min prior to SCA. Signals were extracted from Sudden Cardiac Arrest (SCA) subjects 30 and 60 minutes prior to Ventricular Fibrillation (VF) events. Signals from normal subjects were also included for the same time intervals.

For signals recorded 30 min prior to VF, the model achieved an accuracy of 99.93% during training and 99.75% during testing. This high level of accuracy highlights the model's capability to generalize effectively to new data.

Figure 6 shows the confusion matrix for predictions made 30 min prior to VF. These metrics offer insights into the model's sensitivity and specificity, essential for detecting critical events like VF.

The results demonstrate the model's ability to accurately predict Sudden Cardiac Arrest events 30 min in advance, showing its potential for early warning applications in cardiovascular health monitoring systems.

Table 2 Performance comparison of the proposed method and existing techniques for SCA prediction

Author	Subjects	Minutes to SCA	Accuracy (%)
Ebrahimzadeh et al. [5]	N = 35, S = 35	4	83.9
Yosafat, Sani Muhamad [6]	N = 18, S = 20	30	95.89
Betancourt et. al. [7]	N = 18, S = 23	30	81
Proposed	N = 18, S = 20	30	99.93
	N = 18, S = 20	60	99.87

Note N: No. of Normal Subjects and S: No. of SCA Subjects

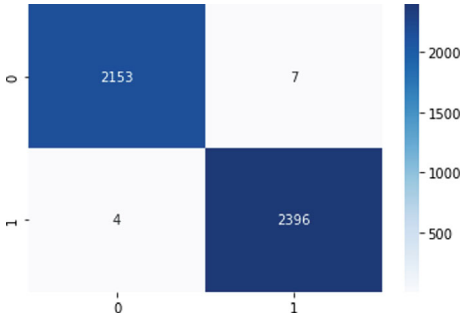

Fig. 6 30 min prior to SCA

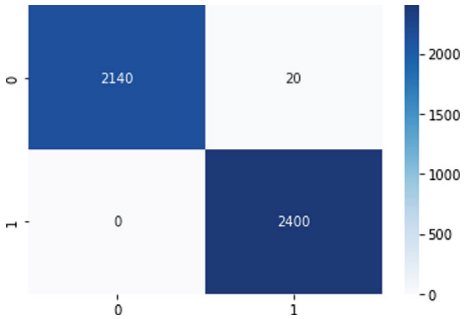

Fig. 7 60 min prior to SCA

3.2 Extended Model Evaluation: 60 Min Prior to VF

The evaluation was extended to include signals recorded 60 min prior to VF. The model achieved a training accuracy of 99.87% and a testing accuracy of 99.56%, indicating its robustness in capturing patterns leading up to VF.

To analyze its performance, a confusion matrix was generated for predictions made 60 min prior to VF, as shown in Fig. 7. This matrix provides detailed insights into the model's classification results, further validating its sensitivity and specificity.

These findings confirm the model's efficacy in predicting Sudden Cardiac Arrest events with significant lead time. Such capabilities are vital for the development of early detection systems that can facilitate timely medical interventions.

4 Discussion

This study introduces a hybrid CNN-LSTM model for predicting Sudden Cardiac Arrest (SCA) from ECG signals. By combining the spatial feature extraction capabilities of CNN with the temporal learning power of LSTM, the model achieves high accuracy–99.75% for signals 30 min before Ventricular Fibrillation (VF) onset and 99.56% for signals 60 min prior. This method outperforms existing approaches by

automatically extracting features, eliminating the need for manual feature engineering, and capturing long-term dependencies in ECG signals.

The model's high accuracy, low computational cost, and ability to predict SCA well in advance make it suitable for real-time cardiac monitoring, such as wearable devices. This could enable early intervention, improving survival rates and supporting personalized healthcare.

5 Conclusion and Future Work

This study proposed a hybrid CNN-LSTM model for early SCA prediction from raw ECG signals, achieving excellent performance–99.75% accuracy for 30 min pre-VF signals and 99.56% for 60 min pre-VF signals. The model combines CNN for efficient feature extraction and LSTM for capturing temporal dependencies, enabling it to outperform current methods. Its low computational complexity and high accuracy make it ideal for real-time cardiac health monitoring, offering a promising solution for early intervention in SCA cases.

Future work will focus on extending the prediction window beyond 60 min and validating the model on larger, more diverse datasets to improve generalizability. Integrating the model into real-time wearable devices, such as smart wristbands, will provide continuous monitoring and real-time alerts, improving accessibility to preventive cardiac care. Further exploration of advanced architectures and the inclusion of multimodal data, such as clinical parameters and heart rate variability, will enhance predictive accuracy and clinical utility, leading to more effective SCA prevention systems.

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Hybrid Deep Learning Model for Fall Detection in Healthy Elderly Using CNN-BiLSTM: CNBiLS

Harshi Jain and Vijay Bhaskar Semwal

Abstract Fall recovery is the capability of a person to recover from external forces. It is an inevitable phenomenon in a cluttered environment. To detect the fall, various mechanisms are evolved for elderly subjects, as they are more prone to fall which may lead for permanent disability or death. With the advancement of sensing and computing technology, the IoT-based wearable devices become popular. In this paper, we present an innovative method for early fall detection and monitoring is proposed during various days to day activities. This research proposes the use of a wearable IMU sensor for detecting body movements. The IMU sensor consists of gyroscopes and accelerometers. Scalability and efficiency may be constrained by the labor-intensive feature engineering and human data processing that are key to traditional fall detection techniques. We developed a hybrid deep learning framework that integrates LSTM networks with CNNs to effectively tackle these challenges. This innovative architecture achieves an accuracy of 98.03% on the Smart Human Fall Dataset, successfully identifying both temporal and spatial patterns in movement data. The proposed model has greatly improves the scalability and dependability of fall detection systems by automating feature extraction and streamlining the data processing pipeline. This provides a solid solution for practical applications and encourages older adults to be more independent and safe.

Keywords Internet of things (IoT) · Deep learning · Fall detection · Sensor · Microcontroller

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1 Introduction

Falls are a major concern in the Indian population for elderly persons while performing daily activities. Falls lead for a high mortality rate following injury and can have long-term effects [1]. Deep learning models for fall detection can help keep older people safer; they are growing in popularity [2]. Elderly people are more vulnerable to falls, which can result in severe injuries such fractured bones, brain trauma, or extended periods of immobility [3]. Their mental health is also harmed by these incidents since their fear of falling reduces their independence and quality of life [4].

It is crucial to develop technologies that can promptly identify falls since they can assist avert major issues [5]. The way fall detection systems operate is by searching for odd movement patterns that could indicate a fall [6–9]. These systems previously tracked activity using straightforward techniques using sensors and simple algorithms. Although these techniques were effective, deep learning has allowed for the development of ever more advanced systems [10]. Deep learning is more precise and dependable than previous techniques because it can identify intricate movement patterns [11]. This paper describes how fall detection systems use deep learning with hardware such as ESP32 microcontrollers (which process the data), accelerometers (which measure speed), gyroscopes (which track rotation), and magnetometers (which detect direction), as well as the MPU6050 sensor (which measures movement) [12]. In order to identify falls in real time, these sensors monitor changes in movement and transmit the data to sophisticated deep learning models, such as CNNs (which are good at assessing pictures), LSTMs (which are strong at comprehending time-based patterns), or combinations of these models [5].

Wearable technology does not have as such drawbacks but people may not want to wear them for the whole day or that may forget to wear or may cause uncomfortable [3, 5, 13–20]. Researchers are experimenting with different approaches to address the following issues, such as utilizing cameras, sensors, or addition of other non-wearable sensors [5, 9–13]. Following methods increase adaptability and enable to function them in different atmosphere [2, 4, 6–9]. In order to understand the data and train the models in different scenarios, the research also discusses the use of both real time and fictitious (simulated) data. It uses accuracy, precision, recall, and $F1$-score to understand how well the models are performing [1]. According to the findings, these technologies can fastly and accurately identify falls, which can help prevent health issues like cold exposure or blisters from prolonged lying down [5, 9–17]. The goal of this research is to improve and simplify fall systems by dependable hardware with deep learning methods. Particularly in aging populations, these systems are an important step in maintaining the independence, health, and safety of senior citizens [18, 21, 22].

This paper is organized in following manner: The first section is the introduction, which covers the research of fall capability and the effects of post fall issue. The next section is the literature review, which includes an overview for different fall recovery methods from the past five years. The third section is the methodology, which proposes algorithm CNN-BiLSTM and its model architecture. The remaining

section is about results and discussion, which compares the proposed model with the existing models. The final section is the conclusion, future work, and limitations, which summarizes the findings and discusses the limitations of the proposed model.

2 Literature Review

With more senior citizens experiencing health hazards as they age, fall detection using deep learning has emerged as a significant study subject [19]. Researchers are looking into ways to use deep learning techniques, sophisticated sensors, and hardware to improve the accuracy, dependability, and efficiency of fall detection systems [20]. Wearable sensors have certain problems despite their good performance. People may forget to use them frequently or find them difficult to wear for extended periods of time [3]. In order to increase usability and efficacy, researchers are now searching for novel concepts, such as integrating wearables with other technologies or developing non-wearable alternatives [5]. Table 1 presents demographic details, including feature set, algorithm, performance index, and limitation.

Table 1 summarizes feature-based methods for fall detection, including publication details, feature sets, algorithms, and performance metrics with limitations.

2.1 Ambient-Based Approaches

Ambient-based technologies have been developed to address wearable device issues. Without the need for clothing, these systems detect falls using sensors positioned

Table 1 Feature-based methods for fall detection with their respective models, accuracy, and limitations

Publication details	Feature set	Algorithm and performance index
Debard et al. (2014)	Aspect ratio, fall angle, center speed, head speed	SVM (Limitation: limited motion description)
Gunale and Mukherji (2015)	Elliptical model of silhouette	KNN (accuracy: 95%)
Yu et al. (2017)	Elliptical model + projection histogram	Enhanced elliptical model (accuracy: 97.08%)
Yun et al. (2016)	Centroid velocity, head-to-centroid distance, HOG, optical flow	Pose variation-based method (accuracy: 90.6%)
Lotfi et al. (2018)	Background subtraction, external contours, ellipses, centroids	MLP (sensitivity: 99.52%, specificity: 97.38%)

throughout the surroundings [13], for instance, developed a system that decreased false alerts by utilizing floor vibration sensors. However, if the sensors aren't positioned correctly or in complex areas, these systems may have trouble [4].

2.2 Vision-Based Approaches

By examining video footage, vision-based systems use cameras to keep an eye on individuals and identify falls. For example, Rougier et al. achieved high accuracy using motion analysis and silhouette tracking with a single camera [16]. However, obstacles include things that impede the view, inadequate lighting, and privacy issues [8].

2.3 Multi-sensor Fusion

Researchers use different sensors system to integrate data from different sensors in order to increase accuracy [13]. For example, Gjoreski et al. created a model that improved the accuracy of the model that used the sensors by the combination of different signals of accelerometers, gyroscopes and many more [19].

2.4 Machine Learning and Deep Learning Approaches

Fall detection has significantly improved thanks to machine learning. Based on different sensor data, algorithms such as SVM and KNN have been used for fall detection [2, 4, 6–9]. For example, Hwang et al. used SVM model to accurately analyze accelerometer data [5]. Deep learning techniques have drawn interest recently because of its ability to handle real-time-based data and to understand the movement patterns [19]. Even greater results are obtained by models that understand activity

Table 2 Evaluation of algorithmic accuracy across fall detection techniques

Algorithm	Accuracy (%)
K-nearest neighbor (KNN)	95.00
Enhanced elliptical	97.08
Pose variation	90.60
Multi-layer perceptron (MLP)	97.59
CNN-BiLSTM	98.03

sequences, such as recurrent neural networks (RNNs) and long short-term memory (LSTM) networks [5, 9–20].

Indicating in Table 2 performance metrics and their corresponding formulas for evaluating classification model effectiveness.

3 Proposed Work

The goal of the proposed study is to create fall detection system that makes sure of Human Activity Recognition (HAR) methods. This system tries to provide fast and precise fall detection to ensure the fast help and limited health issues, given the high risk and serious issues of falls, particularly among the elderly persons. The fall detection system will combine the wearable and ambiance-based technology. Along with the MPU6050 sensor, ESP32 microcontroller, and force sensors, the system design will incorporate different sensors, such as accelerometers, gyroscopes, and magnetometers. Real-time-based data will improve the system's capacity to predict the fall early. The algorithms will be trained and validated by the combination of real-time-based fall event datasets and a simulated dataset of fall incidents (Smart Human Fall dataset). To improve the accuracy and other dependability, algorithms and machine learning approaches used to gather the dataset. To detect the possibility of fall, this system will understand the variations in orientation, body movement patterns, and linear and angular acceleration. To maximize the performance of the system, different machine learning and deep learning techniques or algorithms, with different sensors. This technology will notify emergency services or caretakers right away if the detection occurred, it guarantees the fast aid. It reduces the length of immobilization the additional issues such as cold, dehydration, and pressure sores depend on this fast reaction.

4 Methodologies

4.1 CNN-BiLSTM Architecture and Training

The architecture of the proposed model, as shown in Fig. 1, combines convolutional and bidirectional LSTM (BiLSTM) layers to efficiently process sequential data. The model begins with a 1D convolutional layer using 64 filters and a kernel size of 1, designed to identify local patterns in the input sequence. This is followed by a max-pooling layer to refine these features and reduce dimensionality. To prevent overfitting, a dropout layer is applied at this stage.

Two BiLSTM layers form the model's central component. With 128 units, the initial BiLSTM layer is set up to return sequences, guaranteeing that the temporal dependencies recorded can be processed by all layers that follow. These sequential

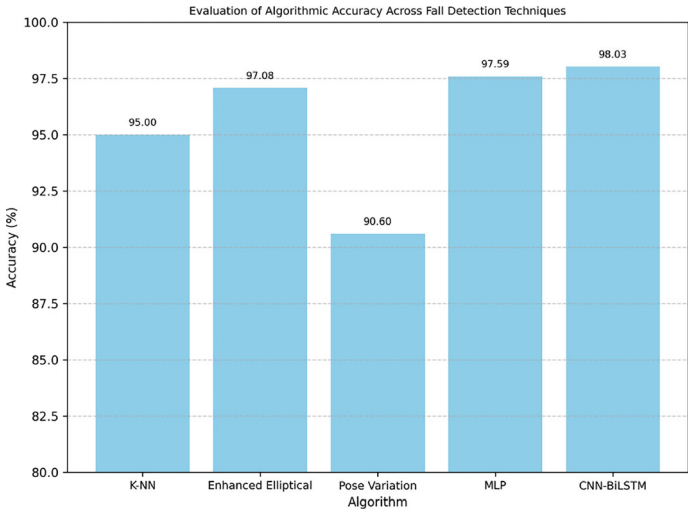

Fig. 1 Previously researched algorithms for fall detection

patterns are further refined in the second BiLSTM layer, which has 64 units. After every BiLSTM layer, dropout layers are positioned carefully to enhance generalization and reduce the possibility of overfitting.

The network consists of two completely connected (dense) layers after the BiLSTM layers. The ReLU function is used to extract high-level abstract characteristics from the 128 neurons in the first dense layer and the 64 neurons in the second. For added resilience, dropout layers are positioned in between these dense layers. Using a single neuron with a sigmoid activation function, the final output layer generates a probability score for binary classification tasks.

This architecture, as illustrated in Fig. 1, combines the spatial feature extraction capabilities of convolutional layers with the temporal modeling power of BiLSTM layers. The model can successfully capture both short-term and long-term links within the sequence thanks to the data structure. To ensure effective learning, binary cross-entropy is employed as the loss function and the model is trained using the Adam optimizer, which has a learning rate of 0.001.

$$\sqrt{Gz^2 + Gy^2 + Gx^2} = G\text{mag} \tag{1}$$

Indicating in Fig. 2 the suggested neural network model's architecture, illustrating the progression from input features through multiple hidden layers to the final output).

Indicating in Fig. 3 participant distribution highlighting gender counts and age statistics (average, youngest, and oldest)

Indicating in Table 3 performance metrics and their corresponding formulas for evaluating classification model effectiveness.

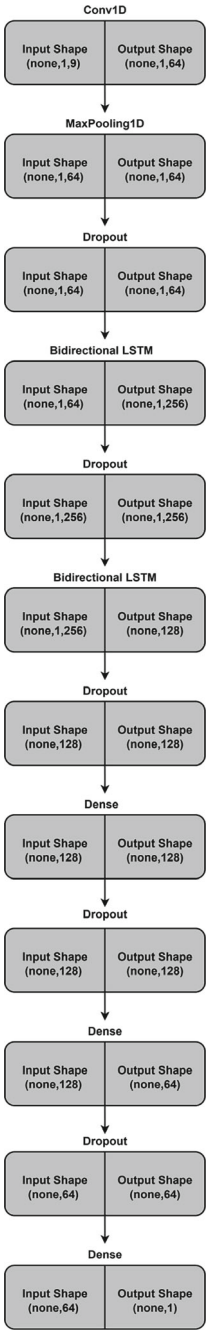

Fig. 2 Suggested neural network model's architecture

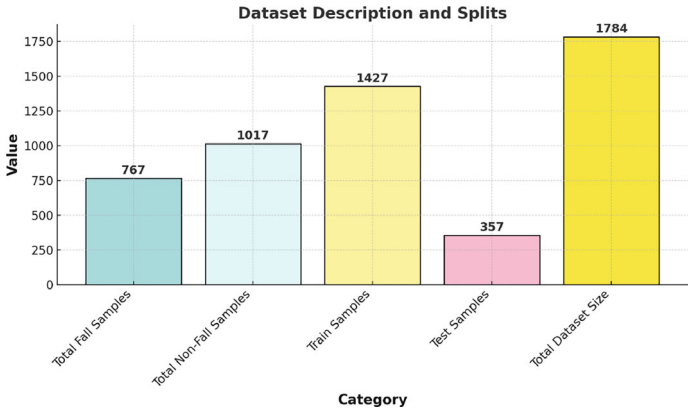

Fig. 3 Deep learning model architecture for sequential data processing

Table 3 Performance metrics and formulas

Metric	Formula
Accuracy	$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}$
Precision	$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}$
Recall	$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}$
Specificity	$\text{Specificity} = \frac{\text{TN}}{\text{TN}+\text{FP}}$
$F1$-score	$F1\text{-score} = 2 \times \frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$
Matthews correlation coefficient	$\text{MCC} = \frac{(\text{TP}\times\text{TN})-(\text{FP}\times\text{FN})}{\sqrt{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FP})(\text{TN}+\text{FN})}}$
False positive rate	$\text{FPR} = \frac{\text{FP}}{\text{FP}+\text{TN}}$
False negative rate	$\text{FNR} = \frac{\text{FN}}{\text{FN}+\text{TP}}$
Balanced accuracy	$\text{Balanced accuracy} = \frac{\text{Recall}+\text{Specificity}}{2}$
Area under the curve	AUC = Area under the ROC curve

4.2 Training

In our fall detection research, we used the enhanced Smart Human Fall dataset, which includes data from 38 different people. This diversity helps improve the system's accuracy and reduces bias toward any one person. To train and test the system, we split the dataset into two parts: 30 people (including 12 elderly) were used for training. 8 people (including 3 elderly) were used for testing. We made sure to randomly

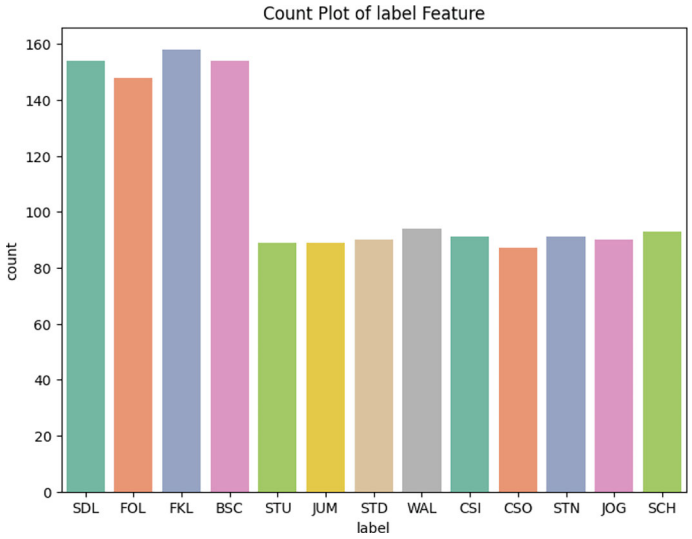

Fig. 4 Count plot of features

assign subjects to each group to avoid bias. The data was divided into small, labeled windows of movement, allowing us to process the data more carefully. Following this, the training set was once more divided into 80% for model training and 20% for validation. People of all ages, including children, adults, older adults, athletes, non-athletes, and those with left or right hand dominance, were also included in our dataset. This diversity made it easier to record a variety of movement styles. The dataset included more "background" or "alert" data than "fall" data because falls are uncommon in comparison to routine activities. In order to counteract this and enhance the model's performance, we employed specialized loss functions and training strategies. Our fall detection algorithm performs effectively in practical scenarios because to this varied and well-balanced dataset.

Indicating in Fig. 4 count plot representing the distribution of activities in the dataset across various labels describe in Table 4.

Table 4 shows the description and codes for fall and normal activities.

4.3 Performance Metrics

To evaluate how well our model works, we use several performance metrics. These metrics compare the actual outcomes (y) with the predicted outcomes (^{y}) from our experiments.

Table 4 Activity code description for fall and normal activities

Activity	Code
Forward-lying	FOL
Forward knee-lying	FKL
Sideward-lying	SDL
Back sitting chair	BSC
Standing	STD
Walking	WAL
Jogging	JOG
Jumping	JUM
Stairs up	STU
Stairs down	STN
Sit chair	SCH
Car-step in	CSI
Car-step out	CSO

- Mean Squared Error (MSE)

$$\mathrm{MSE}(y, \hat{y}) = \frac{\sum_{i=0}^{N-1} (y_i - \hat{y}_i)^2}{N} \tag{2}$$

- Root Mean Squared Error (RMSE)

$$\mathrm{RMSE}(y, \hat{y}) = \sqrt{\frac{\sum_{i=0}^{N-1} (y_i - \hat{y}_i)^2}{N}} \tag{3}$$

- Mean Absolute Error (MAE)

$$\mathrm{MAE}(y, \hat{y}) = \frac{\sum_{i=0}^{N-1} |y_i - \hat{y}_i|}{N} \tag{4}$$

4.4 Algorithm of Model

Algorithm 1: CNN-BiLSTM Sequential Model for Time-Series Classification

Input: Time-series data X, labels Y, training parameters (epochs, batch size, etc.)

Output: Final classification results and evaluation metrics

Preprocessing:
Normalize X to obtain X_{scaled};
Convert Y into categorical labels $Y_{categorical}$;
Model Initialization:
Initialize a Sequential model:
    Add Conv1D layer (filters=64, kernel size, activation);
    Add MaxPooling1D layer;
    Add Dropout layer;
    Add Bidirectional LSTM (units=128);
    Add Dropout layer;
    Add Bidirectional LSTM (units=64);
    Add Dropout layer;
    Add Dense layers (128, 64 neurons with activations);
    Add final Dense layer (1 neuron, output activation);
Compilation:
Compile the model using Adam optimizer and binary cross-entropy loss;
Training:
Train the model:
`Model.fit(`$X_{train}, Y_{train}, validationdata, epochs, batchsize$`)`;
Prediction and Evaluation:
Predictions = `Model.predict(`X_{test}`)`;
Convert probabilities to labels: `Final Labels = threshold(Predictions);`
Compute accuracy: `Accuracy = Correct Predictions / Total Samples;`
Generate confusion matrix and performance metrics (sensitivity, specificity);

4.5 Experiment Setup

Hardware Setup: In this paper, we develop a fall detection system utilizing various sensors and microcontrollers to monitor and analyze human activity. The hardware components include an ESP32 microcontroller, MPU6050 IMU sensors, and force sensors. The configuration and integration of these components are as Wiring and Connection of MPU6050 Sensors: VCC to 3.3V on ESP32, GND to GND on ESP32, SCL to GPIO 22 on ESP32, SDA to GPIO 21 on ESP32 and force sensors as one

terminal of each force sensor to GND Other terminals to GPIO 36 and GPIO 39 on ESP32. Ensure secure connections and proper placement of sensors on the subject for accurate data capture.

4.6 Software Setup

The ESP32 is programmed using the Arduino IDE. LIBRARIES Adafruit Unified Sensor, Adafruit MPU6050.

4.7 Data Collection

Simulate fall events by having participants wear the sensors and perform controlled falls. Collect data in real-time from the sensors and log it for further analysis.

4.8 Data Analysis

Analyze the collected data to find the trends and features related to fall. Use machine learning algorithms to train a model for fall detection.

4.9 Validation

Evaluate the trained model using new data to measure performance metrics, including accuracy, precision, and recall. The system targets an accuracy rate of 98.03%.

5 Result

This research developed a fall detection system combining ESP32 microcontrollers, MPU6050 IMU sensors, and force sensors, focusing on accurate and timely fall detection. Machine learning algorithms were employed to process real-time sensor data, evaluated through critical performance metrics.

5.1 Experimental Setup

The system was tested using simulated fall and non-fall activities. Data was collected 110 Hz to balance granularity and efficiency. A dataset of 10 real-time fall events, combined with simulated data, supported training and validation, enhancing the system's robustness.

5.2 Performance Metrics

The system achieved the following metrics: Accuracy: 98.03%, indicating overall correct detections. Precision: 98%, reflecting the rate of true positives among detected falls. Recall: 98%, measuring the system's ability to identify actual falls. Specificity: 98%, denoting correct classification of non-fall events and reduced false alarms. These results confirm balanced and reliable detection capabilities.

5.3 Data Insights

IMU sensors recorded motion data, detailing accelerations and orientations during falls. Force sensors categorized pressure levels (e.g., light touch, medium squeeze) to distinguish falls from normal activities.

The confusion matrix in Fig. 3 provides a comprehensive evaluation of the model's performance across multiple classes. The rows correspond to the actual classes (true labels), while the columns represent the predicted classes. The diagonal elements of the matrix signify correct predictions.

Indicating in Fig. 5 hardware setup of the fall detection prototype using a force sensor, battery, and microcontroller.

Indicating in Fig. 6 confusion matrix activity label distribution.

Indicating in Fig. 7 the ROC curve shows perfect classification performance with an AUC of 1.0, indicating ideal model discrimination.

Indicating in Fig. 8 confusion matrix showing the model's performance with high accuracy, featuring 198 true negatives, 151 true positives, 5 false positives, and 2 false negatives

Figures 9, 10, 11, 12, 13, 14, 15 and 16 showcase Kernel Density Estimation (KDE) plots, illustrating the probability density functions of key features in the dataset. The red-shaded areas highlight the density of data points, providing a smooth approximation of their distributions.

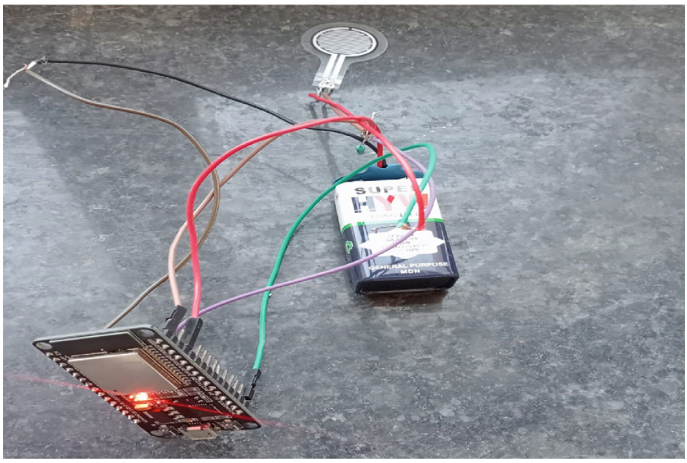

Fig. 5 Hardware circuit

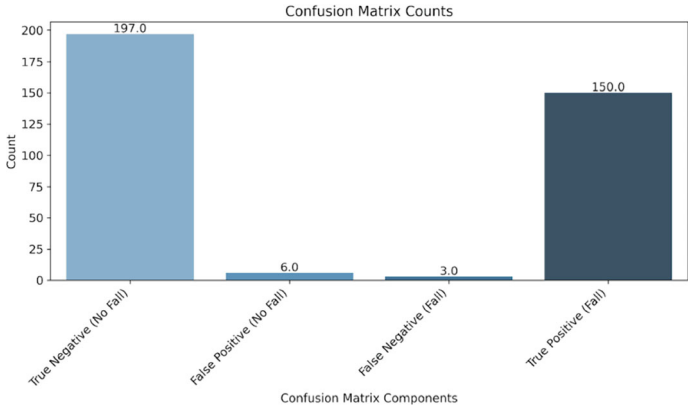

Fig. 6 Count plot of features

6 Conclusion

Fall detection systems are critical in safeguarding older adults and other vulnerable populations from the different consequences of falls, such as fractures, head trauma, and prolonged immobility. This research highlights the integration of wearable sensors, including accelerometers, gyroscopes, and force sensors, with advanced hardware components like the MPU6050 sensor and ESP32 microcontroller to collect and process motion data accurately, by precisely studying motion patterns and models like CNNs and LSTM that have greatly improved the system's capacity to identify falls. Key performance measures including sensitivity, specificity, precision, and recall have increased as a result of utilizing both real-time and simulated data,

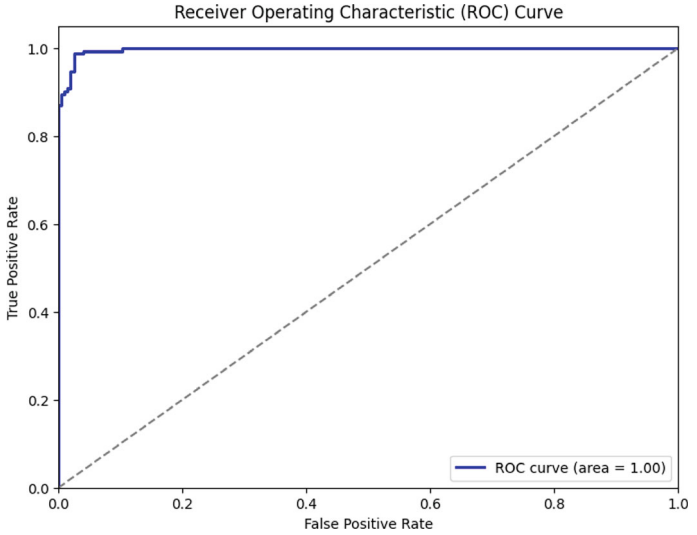

Fig. 7 Receiver operating characteristic curve

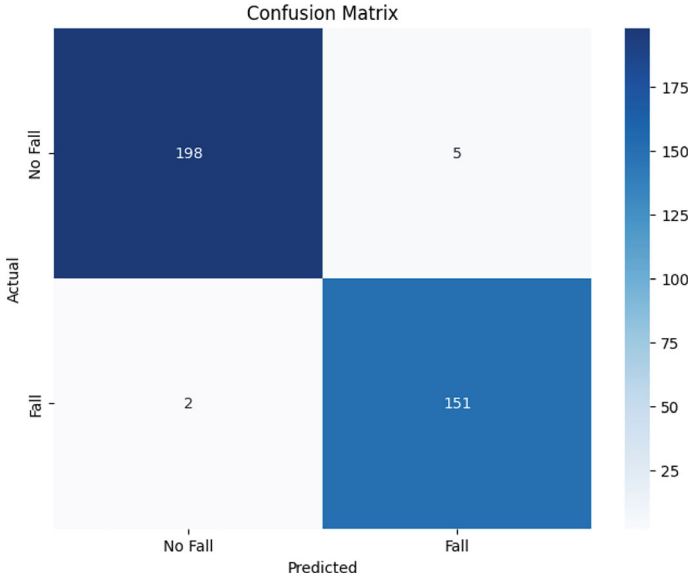

Fig. 8 Confusion matrix

guaranteeing balanced and trustworthy detection capabilities. Modern hardware and sophisticated algorithms are used in this work to provide a reliable and effective fall detection system. The technology provides scalable and flexible protection in a variety of settings in addition to facilitating prompt treatments that lower dangers

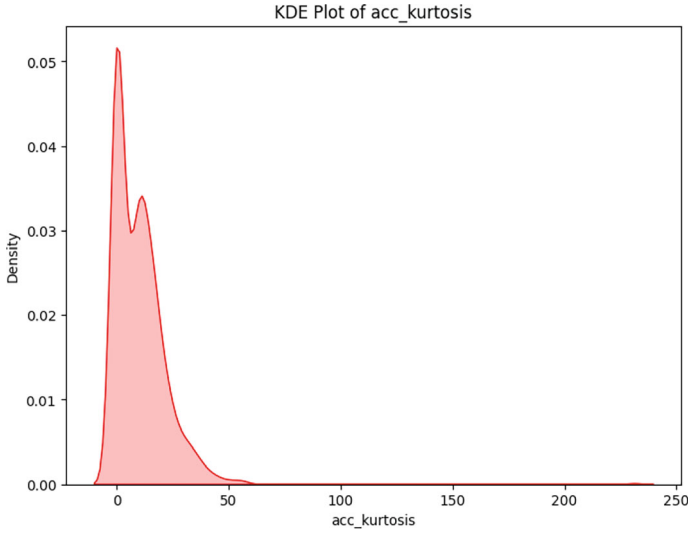

Fig. 9 KDE of Acc kurtosis

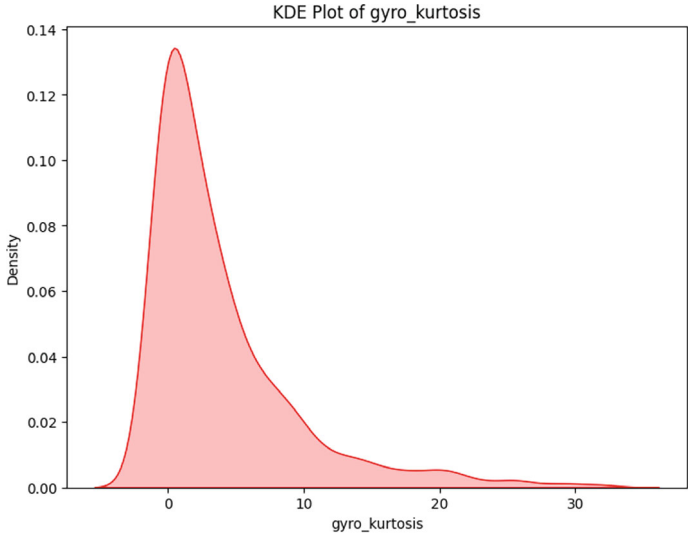

Fig. 10 KDE of gyro kurtosis

like hypothermia and pressure sores. These technologies' dependability and efficacy will be further improved by ongoing innovation and improvement, guaranteeing vulnerable people greater safety and independence.

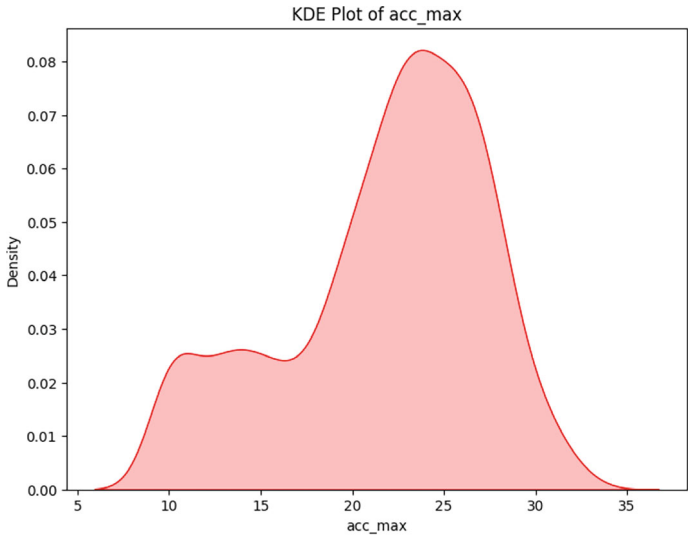

Fig. 11 KDE of Acc max

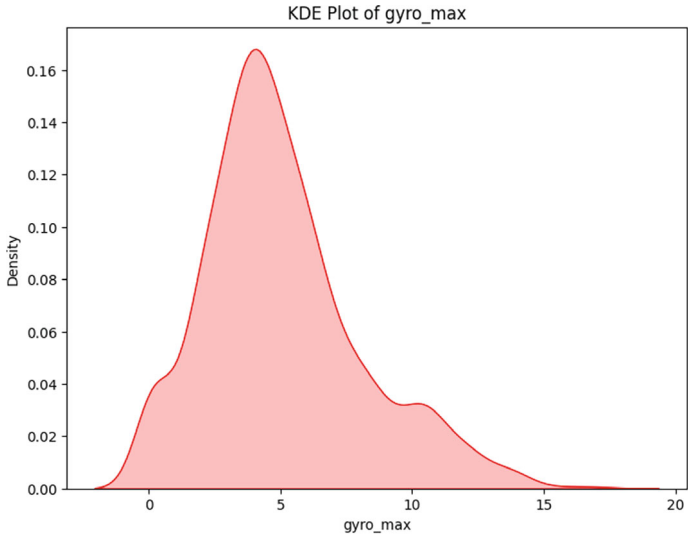

Fig. 12 KDE plot of gyro max

7 Challenges and Future Directions

Even though fall detection technology has advanced significantly, a number of practical issues still prevent its broad use and efficient implementation. User compliance is a major problem, especially with wearable technology. Users must wear and care for

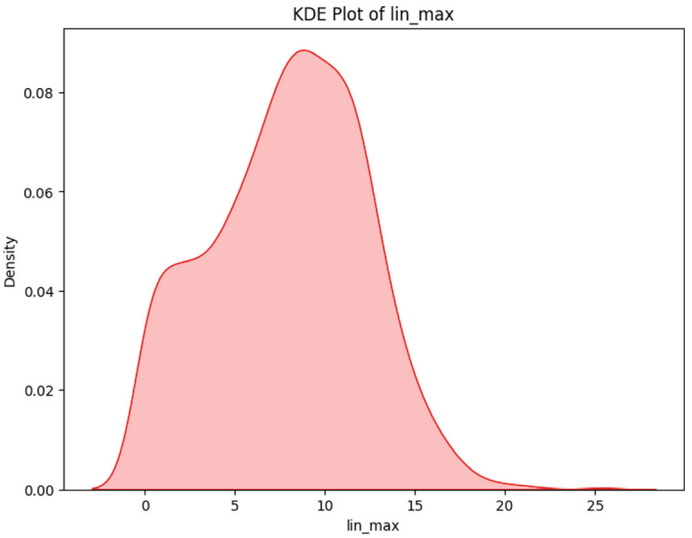

Fig. 13 KDE of line max

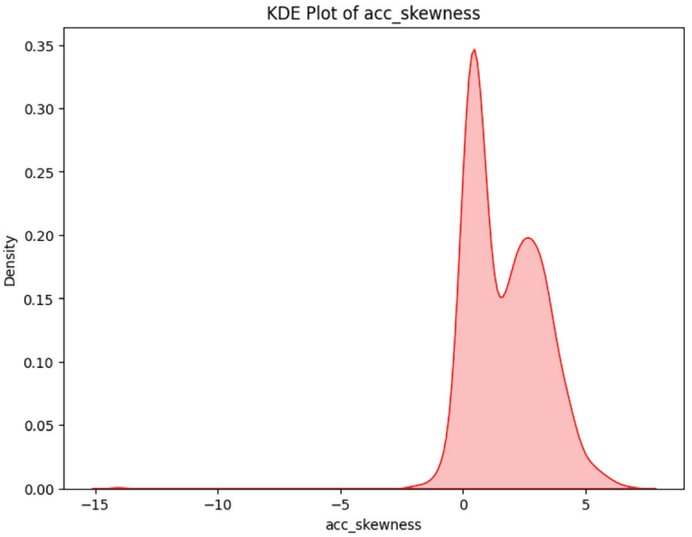

Fig. 14 KDE of Acc skewness

the devices on a regular basis for these systems to work. However, long-term usage is frequently discouraged, particularly among older persons, by pain, ugly designs, and the sense of intrusion. This reluctance might be addressed by developing gadgets that are both useful and easily incorporated into daily life, for example, through fashionable designs or multifunctional features.

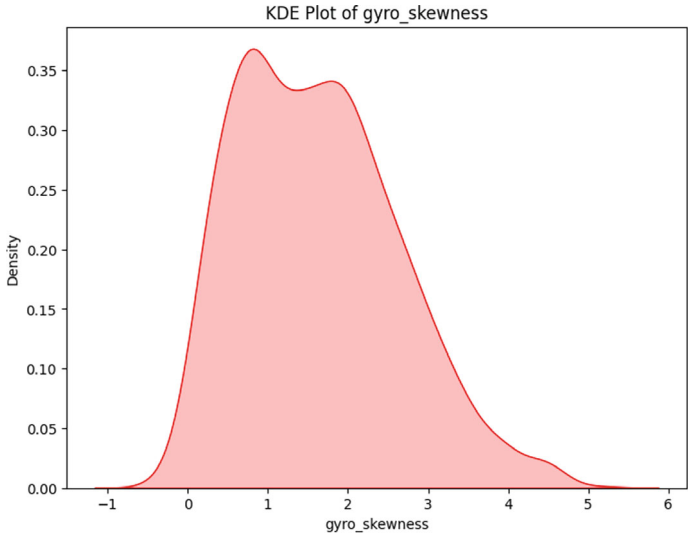

Fig. 15 KDE of gyro skewness

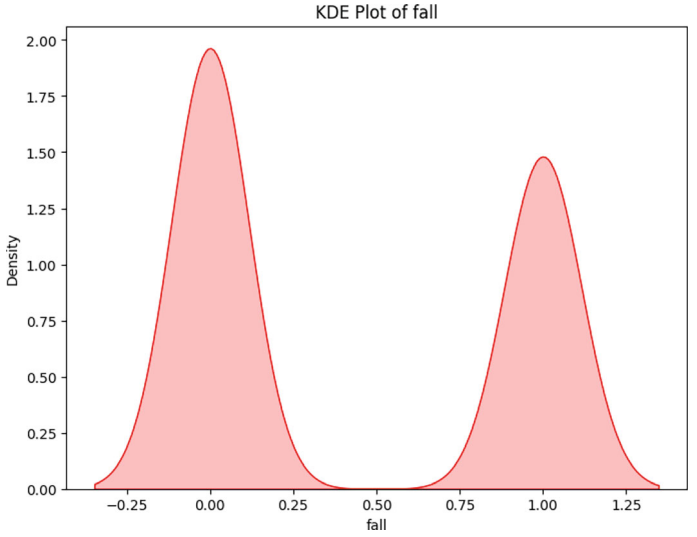

Fig. 16 KDE of fall

Privacy issues present another significant obstacle, especially for systems that make use of cameras or ambient sensors. Users who respect their privacy may object to these technologies since they frequently feel intrusive. Many people worry that their personal information may be misused or accessed without authorization if they are being watched at their homes or places of employment. Developers must prioritize

open data practices, make sure that strict privacy laws are followed, and put strong encryption and security mechanisms in place in order to handle this. Concerns can also be allayed by fostering trust by user education and open discussion about data handling practices.

The adoption of fall detection systems is made more difficult by the sensors' own limitations. For example, wearable sensors may have issues with calibration, signal interference, and battery life, whereas ambient systems may encounter difficulties with changes in the environment, lighting, or human behavior. Some of these problems might be resolved by hybrid strategies that integrate wearable and ambient sensors, providing more thorough data and lowering dependency on any one technology.

These issues need to be handled from a user-centered perspective in order to guarantee the realistic implementation of fall detection systems. In order to overcome present constraints, the emphasis should be on creating solutions that put comfort, usefulness, and trust first while utilizing technological advancements like adaptive algorithms and hybrid sensor networks.

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Early-Stage Detection for Alzheimer Disease Using Machine Learning and Deep Learning Algorithm

Vishank Agrohi and Sanjeev Thakur

Abstract The increase in risk awareness enables patients to take preventive action even before irreversible brain damage occurs. A correct diagnosis of Alzheimer's disease (AD) is crucial to patient treatment, particularly in the early stages of the disease despite the fact that numerous recent studies have employed. The majority of machine detection techniques rely on congenital observations to diagnose AD using computers. Early diagnosis of AD is possible; however, it cannot be predicted because prediction is only useful up until the point at which the disease starts to show symptoms. Deep learning (DL) is now a widely used method for AD early diagnosis. Here, we examine how DL can assist researchers in making early diagnoses of AD and provide a quick overview of some of the key works in the field. A widespread and well-known neurodegenerative illness that impairs cognition is Alzheimer's disease (AD). The 'nervous system' problem has drawn the greatest interest in the field of medicine. In spite of this thorough investigation, there is no method or remedy to halt or reduce its spread. However, there are numerous solutions (both pharmaceutical and non-medication alternatives) that can help treat AD symptoms at different stages and improve the patient's quality of life. Patients must receive the right care at each stage of the disease as it progresses. Therefore, it may be beneficial to identify and categorise AD phases before beginning symptom treatment. About 20 years ago, there was a significant acceleration in the field of machine learning (ML) advancement. This work, which use ML techniques, focuses on early Identification of AD. We performed extensive testing to identify AD in the 'Alzheimer's Disease Neuroimaging Initiative' (ADNI) dataset. The three categories that were to be created out of the dataset were AD, 'cognitive normal' (CN), and 'late mild cognitive impairment' (LMCI). The ensemble model of 'logistic regression' (LR), 'random forest' (RF), and 'gradient boost' (GB) is presented in this study as logistic random forest boosting (LRFB). There is currently a lot of interest in using machine learning to discover metabolic disorders that impact a huge number of people

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worldwide, such as diabetes and Alzheimer's. Every year, their incidence rates rise at a startling rate. When it comes to Alzheimer's, changes caused by neurodegenerative disorders impact the brain. An increasing number of people, their families, and the healthcare system will be affected by disorders that impair memory and functioning as our population ages. There will be significant social, financial, and economic repercussions from these. Alzheimer's disease is unpredictable when it is first developing. When AD is treated early on, it is more successful and results in less minor harm than when it is treated later. A key concern in computer-aided detection (CAD) is the classification of brain diseases. The two main causes of death are brain tumours and Alzheimer's disease (AD). Positron emission tomography (PET), computed tomography (CT), and magnetic resonance imaging (MRI) scans are used in the study of various disorders. Need specialised knowledge to comprehend the modality. The illness most commonly affects the elderly and has a potentially devastating latter phases. The outcome can be ascertained by computing the score from the Mini-Mental State Examination, after which the MRI brain scan is performed. In addition, a variety of classification techniques, including deep learning and machine learning, are helpful in the diagnosis of MRI scans. They do, however, have some accuracy-related restrictions. This research presents several novel pre-processing techniques that greatly enhance these MRI images' classification ability. It also shortened the amount of time needed to train the model using different prior learning algorithms. The Alzheimer's Disease Neurological Initiative (ADNI) provided a dataset that was transformed from a 4D format to a 2D one. Techniques for histogram equalisation, selective clipping, and greyscale image conversion were employed to prepare the pictures. Following pre-processing, three learning techniques for the classification of AD were developed. That is convolution neural networks (CNNs), XGBoost, and random forest.

Keywords Alzheimer's disease (AD) · Artificial intelligence (AI) · Deep learning (DL) · Machine learning (ML) · MRI · Mild cognitive impairment (MCI)

1 Introduction

Alzheimer's disease (AD) is an incurable neurological condition that affects the senior population and causes numerous difficulties for its patients [1, 2]. According to the most recent World Alzheimer Report [3], there are currently 55 million AD patients with a clinical diagnosis worldwide; by 2050, that number is predicted to climb to 139 million[3]. The research also cites an astounding 75 per cent of problems that remain misdiagnosed for various causes. Patients with AD will face numerous challenges, including memory loss, behavioural abnormalities, vision problems, and mobility issues that make it difficult to do daily duties [1, 4, 5]. These afflictions will worsen to the point that they make it difficult for a person to lead a self-sufficient personal and social life and cause a great deal of hardship for family members who are responsible for providing care [2, 6].

Artificial intelligence (AI) techniques such as deep learning (DL) and machine learning (ML) algorithms have recently made contributions to a variety of application domains, such as anomaly detection [7, 8], biosignal and image analysis evaluation and classification of neurodevelopmental disorders with an emphasis on autism [9, 10], neurological disorder identification and treatment [2, 11, 12], supporting the COVID-19 pandemic's detection and management [13, 14], senior monitoring and care, cyber security and trust management [15, 16], diagnosis of various diseases [17, 18], smart healthcare service delivery, text and social media mining [19, 20], personalised learning [21, 22], and earthquake detection [23, 24].

A portion of these techniques have greatly improved the clinical diagnosis of AD in a way that is remarkably fast, accurate, and efficient by utilising compound medical data (e.g. MRI, PET,CT) [17, 25–27]. This achievement can be ascribed to a number of things, including some algorithmic developments and the accessibility of strong GPUs that are pre-configured with a variety of open-source computing tools [25, 26]. These have remarkably aided in the correct detection of AD. The foundation of the Al-driven AD prediction is the idea that systems can recognise dementia stages by picking up patterns from the input data, allowing for the best judgements to be made with the least amount of human interaction [28, 29].

One of the most typical causes of dementia is AD. This illness impacted about 50 million people in 2018, and by 2050, that figure is predicted to triple. The symptoms of AD usually appear once a person turns 60 years old. However, certain types of AD may appear 30–50 years earlier in people who have 'gene' alterations. The brain alters both anatomically and functionally as a result of long-term AD. In the case of patients with AD, several decades may pass between their last state of health and the commencement of the disease. AD develops gradually from mild cognitive impairment (MCI). However, not every patient with MCI goes on to develop AD. By 2020, a highly contagious coronavirus had taken over the entire planet, and the only ways to stop it from spreading were lockdowns and sufficient house isolation. To reduce human interaction, people isolated themselves in their homes and moved the majority of services—including work, education, grocery shopping, and medical consultations—online. However, there was no reliable online tool available for AD prediction that users could use from their homes, saving them from visiting a hospital [30].

Comprehensive mental health trials have demonstrated the outstanding benefits of utilising computational neuroscientific techniques for translational purposes. This interdisciplinary area of research can assist in modelling the molecular mechanisms behind both healthy and pathological conditions in the human brain and translating these mechanisms into visible clinical manifestations. Over the last ten years, there have been significant advancements in machine learning (ML) and high-volume biomedical datasets, including neuroimaging and related biological data. These developments have created new opportunities for the diagnosis and treatment of neuropsychiatric and neurodegenerative illnesses. From a computational standpoint, this recent progress has led to the creation of tools that enhance the therapeutic outcomes of patients with similar disorders by in corporating multiple patient-specific data into forecasts [31].

Alzheimer's disease (AD) is a kind of dementia characterised by growing behavioural and cognitive issues that begin in middle or advanced age. The neuritic plaques and the degeneration of certain brain cells are pathological features of the brain. Usually, the symptoms worsen gradually over time and become disruptive to day-to-day activities. Even though age is the main risk factor, AD is not only a disease of advanced age. The patient experiences moderate memory loss in the early stages, but their ability to respond and converse significantly deteriorates in later stages. Although there is currently no cure for Alzheimer's disease (AD), an early diagnosis can help reduce the severity of the condition and assist the sufferers in leading better lives. According to reports, the number of people affected by AD will double over the next 20 years [32], and in 2050, 1 in 85 people would have the condition [33]. Therefore, it is crucial to get an accurate diagnosis, particularly in the early stages of AD [34].

Since the initial study on the subject, the complexity of AD as a neurodegenerative illness has been acknowledged. In order to comprehend the condition completely, there are at least three levels of intricacy to navigate. First, there is variability in AD's aetiology and clinical manifestations. There is currently no agreement on the number of separate subtypes of AD, despite previous attempts to do so. Understanding possible subtyping may present chances to find subject-specific mediation and therapeutic modalities. Second, AD has a protracted prodromal phase and is a progressive disease. According to earlier research, the disease's aetiology may begin years or even decades before symptoms appear. In order to control the course of a disease, early diagnosis is highly desirable [35].

In Fig. 1, an MRI image dataset processing and analysis procedure are depicted in the image as a flowchart. The crucial actions are as follows:

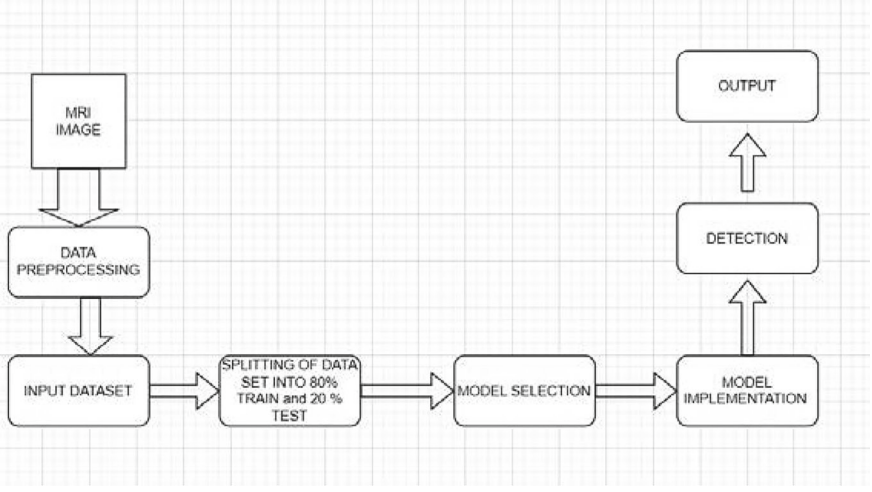

Fig. 1 Block diagram of AD

MRI Picture Input: An 'MRI IMAGE' at the top signals the start of the process. This probably reflects the unprocessed MRI scans that require more investigation.

Preparing Data: The 'DATA PRE-PROCESSING' stage is the next. To get the data ready for modelling, this step usually entails operations like noise reduction, normalisation, and feature extraction.

Data Division: Next, the 'INPUT DATASET' is divided into two sections: 80% for 'TRAIN' data and 20% for 'TEST' data. Effective training and evaluation of machine learning models are made possible by this division.

Model Selection and Implementation: The flowchart moves on to 'MODEL SELECTION' following data splitting. In this case, several machine learning models are taken into account and selected according to how well they fit the task. The selected model is then put into practice after it has been selected.

Detection and Output: Two pathways diverge in the flowchart: There is one path that ends at an 'OUTPUT'. The alternative route arrives at the output after passing through a stage known as 'DETECTION'. The 'DETECTION' stage most likely entails applying the trained model to identify particular characteristics or irregularities in the MRI pictures.

Figure 2 describes the following steps: Step 1: Model Building: In this first step, the model is built. It has a component called 'Data Repository', which implies gathering and storing data. Step 2: Model Training: In this step, the model is trained alone or in the cloud. This step includes the application of algorithms and pre-processing. Step 3: Export Trained Model: The trained model is exported following a successful training session. 'Trained Models' is the label applied to the result. Step 4: Model Testing: This stage entails assessing the model. It divides into three sections:

Analyse the Model: Determine how accurate it is.

Evaluate the Model: Possibly use more data to validate its performance.

Load the Trained Model: As stated by the conversation windows stating 'AD' and 'HC' (presumably classification labels), this connects the model for practical application.

1.1 Literature Survey

Using machine learning techniques, we performed an extensive literature review of Alzheimer's disease in this area. A technique was described in which learning algorithms aid in the calculation of the patient's actions across time. The idea is to utilise Bluetooth beacons from Estimate to find a patient's house. It is an extremely reliable technique that can determine the patient's whereabouts precisely enough to more than 90%. Sorensen, Lauge, and associates conducted research whether hippocampal texture might be a helpful MRI-based marker for Alzheimer's disease

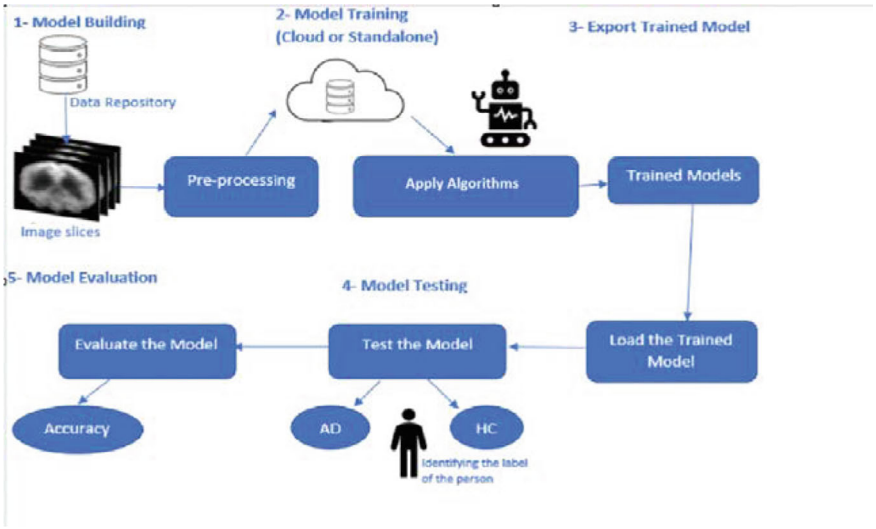

Fig. 2 Flowchart diagram of model implementation

in its early-stage diagnosis. Of their observations, 83% were true. This characteristic was discovered to have considerably improved categorisation between mild cognitive and steady impairments (MCI) and converters from MCI to AD. Furthermore, writers used profound understanding. A categorisation structure that takes advantage of the complementarity of the various input databases was suggested by the authors of [36]. Next, a non-linear approach for graph mixture is used to a range of combined characteristics obtained from different methods. These graphs were combined, and the attribute's area under the curve (AUC) was discovered to be images of AD and NC (standard control) differed by 98.1.and 82.40% in the pictures of MCI and NC. Overall categories, with an AUC of 77.90% attained. Through a variety of methods, the writers of [36] extracted features from diffused MRI pictures. They used region-based spatial statistics (RBSS) to generate computations of pixel-wise distribution. After grouping these distributions, independent component analysis (ICA) was used to find the amalgamation of these characteristics [36].

The dataset was retrieved from the Open Access Series of Imaging Studies (OASIS) of neurology. Patients in the 60–96 age range were selected from a larger sample of individuals interested in Washington University's magnetic resonance imaging research. The dataset, which included clinical and MRI data, was based on approximately two trips. Each imaging modality captured three or more T_1-weighted images. The data was from a longitudinal study provided by the database pool of Washington University's Alzheimer Disease Research Centre (ADRC). In the ADRC, a controlled group and a group of individuals with psychological disabilities were recruited, particularly via media offerings, of which 80% of individuals had

direct interaction with the centre and the remaining individuals via physician referral [37].

A review of the literature on the use of convolutional neural networks (CNNs) to detect Alzheimer's disease

1. Overview: A neurodegenerative condition called Alzheimer's disease (AD) is characterised by gradual memory loss and cognitive deterioration. For the purpose of controlling the illness and enhancing patient outcomes, early diagnosis is essential. A class of deep learning models called convolutional neural networks (CNNs) has demonstrated great promise in the analysis of medical images, especially in the identification of Alzheimer's disease using neuroimaging methods like computed tomography (CT), positron emission tomography (PET), and magnetic resonance imaging (MRI). This review of the literature highlights current developments in CNN-based methods for Alzheimer's disease detection.
2. Medical Imaging and CNN in Alzheimer's Disease: CNNs' capacity to automatically extract hierarchical feature representations from images has led to their increasing popularity in the field of medical image analysis. The main method for detecting Alzheimer's disease has been to examine brain pictures and look for structural alterations brought on by the illness. These alterations include atrophy in the cerebral cortex and hippocampal regions, which can be seen on CT, PET, or MRI studies.
3. Recent Studies

    A. CNN-driven MRI Evaluation: A common non-invasive imaging method for diagnosing AD is magnetic resonance imaging (MRI). A number of research have used CNNs to identify Alzheimer's by analysing MRI data. For example, Suk et al. (2017) suggested a deep CNN model that performed better than conventional machine learning techniques for the structural MRI-based AD classification. Similar to this, Korolev et al. (2018) created a 3D CNN architecture that distinguished between those in good health, those with mild cognitive impairment (MCI), and AD patients using MRI images. They were successful in detecting the disease at an early stage.
    B. CNN for the PET Visualisation: PET imaging sheds light on the brain's metabolic activities. A CNN framework was created by Lu et al. (2018) to evaluate PET images in order to detect AD early. Their model was 90% accurate in differentiating between AD patients and healthy controls after being trained on fluorodeoxyglucose (FDG)-PET data. According to Suk and Shen (2019), PET imaging has also been employed in multimodal frameworks, integrating PET and MRI data by merging CNN with other models for enhanced accuracy.
    C. Using Multiple Modalities: It is frequently the case that combining many imaging modalities improves diagnostic performance. A multimodal CNN that incorporates MRI, PET, and clinical data for the diagnosis of AD and MCI was presented by Liu et al. (2019). When compared to single-modality models, their model's sensitivity and specificity significantly improved. This

suggests that CNN models' diagnostic accuracy can be increased by utilising the complementing data from other modalities.

D. Using Transfer Learning to Identify AD: The lack of labelled medical data is one of the difficulties in creating CNN models for Alzheimer's diagnosis. To get over this restriction, transfer learning—where a pre-trained model gets improved on a smaller dataset—has been used. By using MRI data to fine-tune a pre-trained ResNet model, Sharma et al. (2020) achieved good classification performance in identifying AD. Transfer learning is a helpful tactic in medical imaging applications since it minimises the requirement for huge datasets and speeds up the training process.

E. AI Explainable in Alzheimer's Diagnosis: Deep learning models, such as CNNs, are 'black-box' systems, which has caused problems in clinical applications where interpretability is critical. The goal of recent research has been to increase the explainability of CNN-based models. A CNN model with attention mechanisms that emphasise the most important brain scan regions for AD diagnosis was proposed by Ghosh et al. in 2021. This boosts confidence in the model's predictions and helps therapists comprehend the model's decision-making process.

The range of explanations produced by the XAI approach is known as the explainability scope. Depending on the input, test can interpret the whole model or just one instance of it. Consequently, a model's explainability might be either regional or worldwide. The entire paradigm is explained via a global technique. By taking into account the model's complete set of inferential data, it provides a broad overview of the model's relationships. For every instance of input, the widely used DT algorithm is fundamentally worldwide in scope; since the decision-making for every input data instance, visual explanations are readily available becoming the shape of a tree.

4. Challenges and Future Directions

a. Restrictions on Data: A significant obstacle to the widespread use of CNNs in AD detection is the scarcity of high-quality labelled datasets. Medical imaging data is hard to get by and frequently has size restrictions, especially when it comes to patients with AD. This can cause overfitting in deep learning models.

b. Interpretability of the Model: Despite their great accuracy, CNNs' interpretability issues may prevent their widespread use in clinical settings. Although Grad-CAM (gradient-weighted class activation mapping) is one technique being investigated by researchers to increase the transparency of CNNs, more work needs to be done in this field.

c. Integration of Multiple Modes: Although some research has demonstrated that integrating many imaging modalities enhances performance, multimodal integration is still difficult because of how difficult it is to coordinate and analyse data from various sources. The development of more resilient multimodal CNN architectures that can effectively handle heterogeneous data should be the main goal of future research.

d. Capability to be generalised: The majority of CNN models are evaluated and trained on particular datasets, which could not translate well to data from different populations or sources. More work is required to train models on a variety of datasets and to use methods like data augmentation and cross-validation in order to improve generalizability.

5. Conclusion: CNN-based models have shown a lot of promise in the early diagnosis of Alzheimer's disease using neuroimaging methods like PET and MRI.

More research in multimodal techniques, transfer learning, and explainable AI is opening the door to more precise and therapeutically useful models, despite obstacles like data limits and interpretability issues. CNNs' diagnostic performance for Alzheimer's disease is expected to improve with future deep learning developments and larger, more varied datasets.

1.2 Review of the Literature on Random Forest

Random forest (RF), a popular and adaptable machine learning algorithm, is renowned for its high accuracy, resilience, and capacity to manage problems involving both regression and classification. During the training phase, it builds a large number of decision trees, and at the end, it outputs the mean prediction for regression or the mode of the classes for classification. An overview of important studies and advancements in the topic of random forests may be found below.

1. Overview and Development: Breiman [35] created the random forest algorithm, an ensemble learning technique that lowers variance and improves decision tree performance. Breiman's study demonstrated how RF, in conjunction with random feature selection and bootstrap aggregation (bagging), effectively prevents overfitting—a major issue with standard decision trees.
2. Important Features and Advancements Bootstrapping and Bagging: RF makes use of the idea of bagging, in which several decision trees are trained using various bootstrapped portions of the original dataset. By averaging the outcomes of the individual trees, this lowers variation and lessens overfitting. Random Feature Selection: To avoid bias towards characteristics with more categories or extreme values, RF only chooses a random subset of features at each node, as opposed to typical decision trees, which take into account all features at each split.
3. Random Forest's Benefits Non-Linearity: A number of studies have shown that RF is capable of modelling intricate, non-linear connections between target variables and characteristics. When there are complicated interactions between variables, random fields (RF) perform better than linear models such as logistic regression. Managing High Dimensionality: RF has demonstrated efficacy in managing multifeature, high-dimensional data. Studies have indicated that RF remains strong when the quantity of features greatly surpasses the quantity

of observations. Importance of Features: RF has a built-in system for ranking features, which is helpful for feature selection and dimensionality reduction. This has been used in studies to find important biomarkers in a variety of domains, such as genetics and bioinformatics.

4. Difficulties and Restrictions Computational Cost: Despite the fact that RF increases accuracy, a number of studies have noted that it can be computationally costly, especially when dealing with huge datasets. This problem has been studied in order to enable faster execution using techniques like parallelisation. Interpretability: RF has the same interpretability issues as other ensemble methods, although providing more predictive accuracy than simpler models. The goal of recent research has been to develop interpretability tools for RF model interpretation, such as SHAP (Shapley additive explanations) and LIME (local interpretable model-agnostic explanations).
5. Future Paths for Research Efficiency and Optimisation: The goal of current research is to reduce computing complexity of RF without sacrificing accuracy in order to optimise it. Investigating techniques for quicker training and prediction is part of this, particularly when working with sizable datasets. Explainability: Improving RF's interpretability is a field of gaining research, especially in industries where it is critical, like health care. To help in the interpretation of RF results, methods like decision path analysis and visualisation tools are being developed. Deep Woods: The creation of 'deep forest' models—which aim to fuse the ensemble power of random forests with the depth and representation learning potential of deep neural networks—has been the subject of recent research. 6. Conclusion: Random forest's resilience, adaptability, and great predictive capability have made it one of the most widely used machine learning methods. Even though interpretability and computational requirements are the biggest obstacles, research is still being done to expand the field in which it can be used.

As shown in Fig. 3, the random forest (RF) model achieved highest accuracy, so the proposed methodology is based on the random forest (RF).

	Model	Accuracy	Recall	AUC
0	Logistic Regression (w/ imputation)	0.789474	0.70	0.794444
1	Logistic Regression (w/ dropna)	0.805556	0.75	0.750000
2	SVM	0.815789	0.70	0.822222
3	Decision Tree	0.815789	0.65	0.825000
4	Random Forest	0.868421	0.80	0.872222
5	AdaBoost	0.868421	0.65	0.825000

Fig. 3 Accuracy–recall chart

2 Methodology of Random Forest

During the training phase, several decision trees are constructed using random forest (RF), an ensemble learning technique that aggregates the outputs of these trees to create predictions. Bagging and random feature selection are the main concepts of reinforcement learning (RF), which lessen overfitting and enhance model generalisation. Here is a detailed explanation of random forest's process.

1. Data Input: Random forest is applicable to jobs involving both regression and classification. Given the dataset D $(XI, YI), (X_2, L), \ldots, (X_n, y_n)$, where Xt denotes the feature, and Y_i stands for the goal values in the features. The model is trained to forecast the value of y according to X's attributes.
2. Bootstrap Aggregating (Bagging): Bagging is the fundamental principle of random forest, in which several bootstrapped datasets are created by repeatedly sampling the original training data with replacement.

   Bootstrap Sampling: Random forest creates several bootstrapped subsets from the original dataset *D*. A random sample of the data is taken with replacement from the original dataset to construct each subset Db. Training Multiple Decision Trees: A separate decision tree is trained for every bootstrapped subset Db. Since every decision tree is trained using a distinct sample, various patterns in the data may be identified by each one.
3. Selection of Features at Random: By choosing a random subset of features at each decision tree node, random forest adds another degree of unpredictability, increasing the diversity of the different trees and preventing overfitting. At every split in the tree, only a randomly chosen subset of features m (where m¡p) is taken into consideration, as opposed to all features p in the dataset. By doing this, the model is kept from depending too much on prominent features and is encouraged to investigate a greater range of patterns within the data.
4. Expanding Trees of Decisions: Unlike typical decision trees, which are pruned after reaching their maximum depth, each decision tree in the random forest is grown to its full potential. This guarantees that the maximum amount of complexity is captured by each tree. In order to generate splits that maximise a criterion such as mean squared error (MSE) for regression or Gini impurity for classification, the trees are fully developed using the chosen random subset of features at each node.
5. Estimating OOB (Out-of-Bag) Errors: Out-of-bag (OOB) error estimates is a built-in model evaluation mechanism offered by random forest. Approximately 37% of the data are 'out-of-bag' samples since 63% of the original data points are sampled for each bootstrapped subset during the procedure. These OOB samples can be used to evaluate the prediction performance of the related tree, but they are not utilised for training it. By utilising only the trees that excluded a particular observation from their bootstrap sample, the prediction error for each observation is averaged to determine the OOB error. This eliminates the requirement for a separate validation set by providing a performance estimate.

6. Features' Significance: In addition, random forest calculates feature significance, which indicates the relative contribution of each feature to lowering prediction error. In RF, feature relevance is often determined in one of two ways: The mean decrease in impurity, or Gini importance. This is calculated using the overall reduction in the MSE (for regression) or Gini impurity (for classification) that each feature brings to the table across all trees. The importance of permutations (average decrease in accuracy): This is computed by permuting a feature's values in the OOB data at random and tracking the rise in prediction error. Features are considered more significant when they result in a notable rise in inaccuracy.
7. Adjusting the Hyperparameters: To maximise its performance, random forest can be adjusted for a number of important hyperparameters. Tree Count (n estimators): The total number of decision trees that need to be constructed.

In general, more trees lead to better performance but higher computational costs. The maximum number of features to take into account at each node when looking for the optimal split is called max features (max features). Max Depth (max depth): The deepest point that every tree can go. Whilst restricting the depth can stop overfitting, it may also make the model less accurate. The bare minimum of samples needed to split a node or reside in a leaf node is known as the min samples split/leaf (min samples split/min samples leaf). By ensuring that splits only take place when there are sufficient data, higher values can aid in the reduction of overfitting.

Figure 4 shows at the top of the image a bar chart with the title 'Gender and Demented Rate'. The 'Demented' and 'NonDemented' groups are contrasted in this graphic. Two bars, orange and blue, respectively, stand in for each group, representing the categories with the numbers '1' and '0'. The 'Number of Patients' is shown by the vertical axis on the left, which goes from 0 to 80 in steps of 10.

For the category of 'Demented':
About thirty patients are represented by the blue bar.
About seventy patients are represented by the orange bar. For the category of 'NonDemented':
The blue bar represents approximately forty patients.
Nearly sixty patients are represented by the orange bar.

Figure 5 shows the link between socio-economic status (SES) and education level (EDUC), which is depicted in a scatter plot.

Labels and Axes: Socio-economic status (SES) is represented by the y-axis, which has a range of 0 to 5 (in increments of 1). The education level (EDUC) is represented on the x-axis, which roughly ranges from 5 to 25 (in 2.5-point increments).

Information Points:

There are a lot of green dots all across the graph. Every dot corresponds to a single data point. These data points show a relationship between a certain socio-economic status and a particular education level.

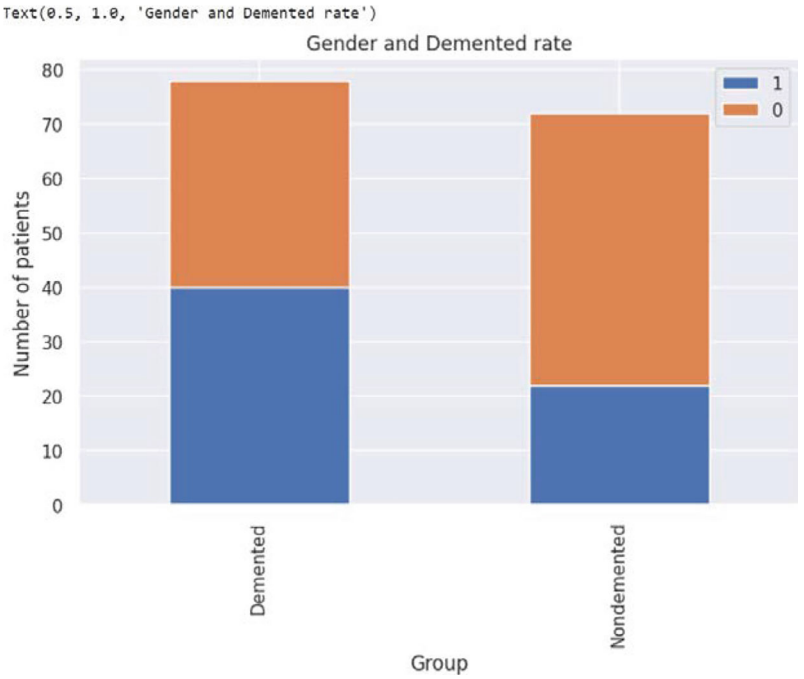

Fig. 4 Gender-wise distribution of Alzheimer's diseases

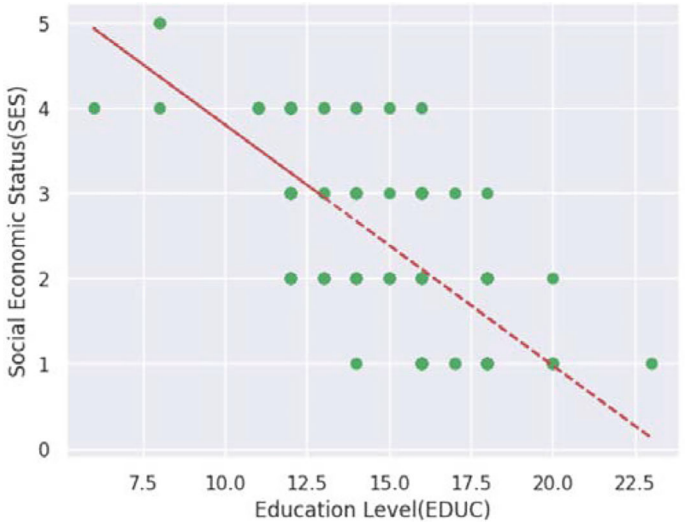

Fig. 5 Socio-economic status versus education level

Trend Line: The scatter plot is intersected by a dashed, descending trend line. The trend line indicates that socio-economic status and education level are negatively correlated. Put more simply, there is a tendency for the socio-economic status to decline as the education level rises.

Importance: This discovery may hold significance in investigations exploring the correlation between educational attainment and socio-economic standing within a given populace. Scholars may investigate the relationship between educational attainment and financial security.

All things considered, this scatter plot offers insightful information on how socio-economic and educational aspects interact.

3 Result

On the validation set, the CNN model achieved a classification accuracy of 80.4. Moderately Demented: F_1-score of 0.50, recall of 0.33, and precision of 1.00. NonDemented: F_1-score of 0.82, recall of 0.87, and precision of 0.78. Extremely Mild Dementia: F_1-score of 0.65, recall of 0.59, and precision of 0.72. With high F_1-scores and balanced precision and recall, the model also demonstrated strong performance for the healthy control and mild cognitive impairment (MCI) classes. A confusion matrix illustrating how well a classification model predicts different stages of dementia can be found in Fig. 6. With rows denoting the genuine labels and columns denoting the expected labels, the matrix is separated into four categories: MildDemented, ModerateDemented, NonDemented, and VeryMildDemented. For example, the model detected 43 cases of MildDemented properly, but incorrectly classified 16 cases as VeryMildDemented and 19 cases as NonDemented. In a similar vein, it correctly identified 233 cases of NonDemented but incorrectly identified 20 as VeryMildDemented and 16 as MildDemented. The count is reflected in the strength of the cell colours, where larger counts are shown by deeper shades. This matrix shows where the model needs to be improved and offers a thorough analysis of its advantages and disadvantages in terms of distinguishing between dementia phases.

Figure 7 is a line graph displaying the training and validation accuracy of a model across 10 epochs. The precision is represented by the y-axis, which ranges from 0.0 to 1.0, whilst the number of epochs is represented by the x-axis, which ranges from 0 to 10.

On the graph, there are two lines:

The training accuracy is shown by a red line that rises gradually and approaches 1.0 by the tenth epoch.

The validation accuracy is represented by a green line that varies but generally indicates an upward trend, reaching about 0.8 by the tenth epoch.

The finest epoch is also shown by a blue dot at the tenth epoch on the green line, which is labelled ‘best epoch = 10’.

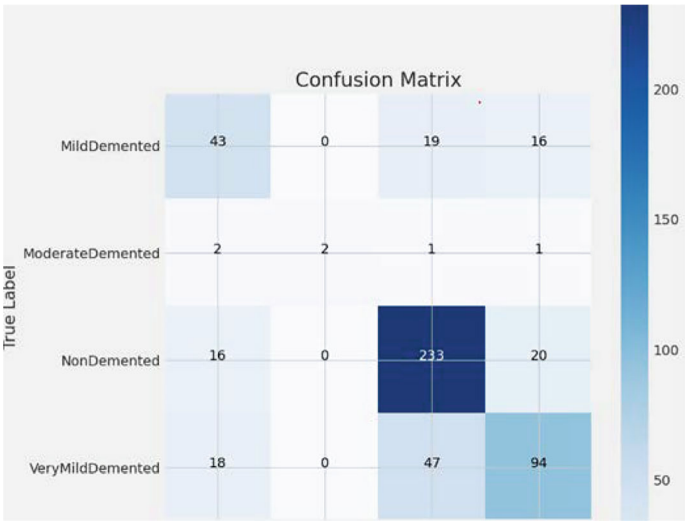

Fig. 6 Confusion matrix

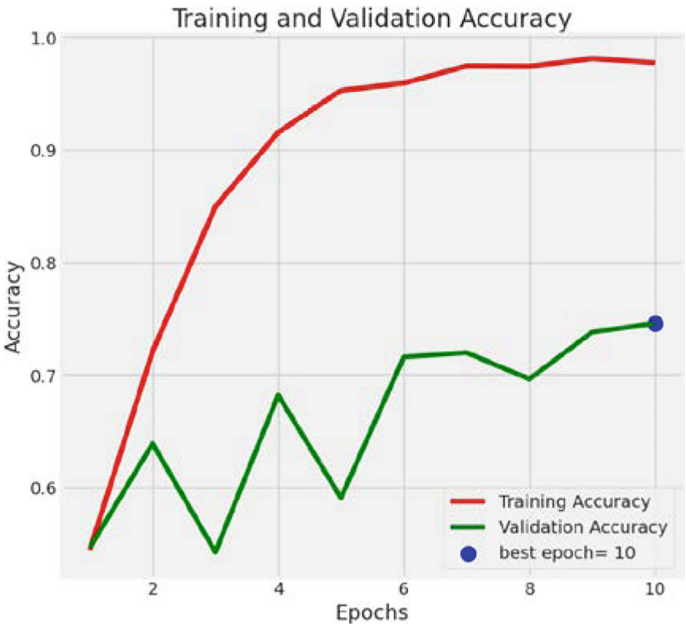

Fig. 7 Training and validation accuracy

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Advanced Machine Learning Framework for Early Detection of Cardiac Disease Using Statistical Analysis and Feature Engineering

Dev Goyal, Snehal Shinde, Yash Joshi, Yuvraj Singh, Sayan Majhi, and Aditya Bakki

Abstract Heart disease has recently caused major concerns over the degradation of people's living standards. This raises an enormous demand for improvements in cardiac data prediction models wherein machine learning has shown some brilliant results at prediction and decision-making. To test the proposed model, the data scientist applies the heart disease dataset that incorporates more than 3,00,000 records. BMI, smoking habits, asthma, and other features have been enhanced into the dataset for higher performance. The most important features for the dataset will be selected using chi-square test and correlation test. The model mainly focuses on applying statistical analysis. The applied ML algorithms is Decision tree with an accuracy of 93.23%. These upholding results will help in the ongoing endeavor in developing reliable and efficient tools for the early detection and intervention of cardiac disease. In-depth study of more advanced feature selection techniques and hyperparameter optimization methods is guaranteed to boost model performances even further. As a result of today's sedentary lifestyle due to decline in physical activity such that people spend enormous time in front of their computer screens are putting excess pressure on them, thus increasing the prevalence of risk factors for coronary heart disease and other health problems. Nowadays cardiovascular diseases are getting increasing attention. It is the major cause of death in adult.

Keywords Machine learning · Cardiovascular health · Statistical analysis · Data preprocessing

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1 Introductions

Based on their medical history, our initiative can help identify people who are most likely to receive a heart disease diagnosis. It determines whether people have heart disease symptoms, such as high blood pressure and chest pain, and helps start therapy with fewer tests, effectively curing these patients [1]. The complexity of electrocardiograms and CT scans for coronary artery disease screening is the reason for this; yet, these technologies are sometimes expensive or unfeasible for consumers. 17 million people died for this reason alone [2]. Every year nearly 20 million people dies due to heart disease which is 32% of total global deaths. According to WHO forecasts, the number of fatalities from CVD would increase sharply to 23.6 million by 2030, with heart disease and stroke accounting for the majority of these deaths [3]. In order to save lives and reduce the financial burden on society, it is crucial to employ data mining and machine learning approaches to forecast the likelihood of acquiring cardiac disease. The effectiveness of several machine learning algorithms in forecasting the desired result will be examined in this study. The dataset utilized in this research is freely accessible on Kaggle. Python was used for all calculations, preprocessing, and visualization in Google Colab and vs Code. Among the methods the employed algorithms are naive bayes, SVM, KNN, Decision Tree and random forest. By adding a sizable and varied dataset, this would offer a chance to overcome such restrictions and maybe increase generalizability.

In the following analysis, the ultimate accuracy following a decision tree is 93%. In this paper, our contributions are as follows:

1. To investigate the data for analysis and use SMOTE and SMOTE ENN to balance the data.
2. To implement machine learning algorithms such as Support vector machine (SVM), K- nearest neighbors (KNN), Logistic Regression, Decision tree while using smote and smote ENN.
3. To improve the model's performance, using ensemble methods such as Random Forrest, ADA Boost.
4. To assess these suggested methods' predictive accuracy in predicting cardiovascular disease (CVDs) survival.
5. To examine these methods using the most recent research given in related work.
6. To offer an accurate and effective system for forecasting heart disease survival.
7. To provide a critical analysis of model using kurtosis, chi-square and PCA.

2 Related Work

Sharma and Rasool [4] addressed the main issues from an artificial neural network perspective. In order to determine the greatest possible performance, a number of models were created, their performance metrics were recorded, and the results-oriented models were then compared. The other classifiers were paired with the

validation DNN. SVM, naïve Bayes, KNN, and DNN were employed for the experiment; the final evaluation produced the following results: SVM: 86.2%, NB: 83.97%, KNN: 81.43%, and DNN: 81.9%. Amin and Agarwal [5] present a hybrid modeling approach that categorizes cardiac disease based on the core risk variables. Both neural networks and genetic algorithms are used to model them. Using a genetic algorithm and general optimization techniques, the researchers set the neural network's initial neuron weights. Their model showed that it is fast with an accuracy of ± 89% when compared to other models. One system for diagnosing coronary heart disease that combines classifiers and particle swarm optimization was presented by Kumar and Inbarani [6]. The best attribute set for their coded message machine learning algorithms—K-NN, MLP, SVM, and the backpropagation process—was discovered using a general smoothing technique and easily accessible optimization. Their respective accuracies were 81.73%, 82.30%, 75.37%, and 91.94%. There are several methods for predicting heart diseases that are based on machine learning. Nevertheless, it was demonstrated that these methods had somewhat limited performance, with only the SMOTE balancing technique achieving a satisfactory level of efficiency for a research dataset.

3 Materials and Method

3.1 Dataset

The dataset contains 3,19,794 records and 18 features as shown Table 1. The dataset is taken from Kaggle. The data is cardiovascular information from African Americans, American Indians, and Alaska Natives.

Analysis

Kurtosis and Skewness

A distribution's asymmetry is referred to as skewness. In positively skewed distributions, the majority of data values are concentrated on the left side, with an elongated tail extending to the right. If the majority of data values are concentrated on the right side of a negatively skewed distribution, the longer tail will be on the left.

Therefore, the skewness of a distribution that is almost symmetric would be around 0.

$$\text{Skewness} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma}\right)^3 \tag{1}$$

n = Number of terms, μ = Mean, $\mathfrak{s}$ = Standard Deviation

Kurtosis is the measure of "tailedness" of a probability distribution. It gives an idea about how the tails of the distribution are more or less compared with that of

Table 1 Description of each feature

Features	Description	Features	Description
BMI	It is relation between height and weight BMI = weight(kg)/height^2(m)	Age	Age of patient
Smoking	If patient has smoking habit. Yes = 1, No = 0	Diabetic	Yes = 2, No = 0, No [border line diabetic] = 1
Alcohol drinking	If patient drinks alcohol. Yes = 1, No = 0	Physical activity	Yes = 1, No = 0
Strokes	Yes = 1, No = 0	General health	Very good = 4, Fair = 1, Good = 2, Poor = 3
Physical health	No. of days for patient's health is bad	Sleep time	Sleep cycle of patient
Mental health	No. of days for which patient's health is bad	Asthma	Yes = 1, No = 0
Differential walking	If patient have walking disability. Yes = 1, No = 0	Kidney disease	Yes = 1, No = 0
Gender	Female = 0, Male = 1	Skin cancer	Yes = 1, No = 0
Heart disease	Yes = 1, No = 0		

a normal distribution. Higher kurtosis means that more data lies in the tails (heavy tails), whereas lower kurtosis means lighter tails or a flatter distribution

$$\text{Kurtosis} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma}\right)^4 \tag{2}$$

n = Number of terms, μ = Mean, σ = Standard Deviation

Smote and Smote ENN

SMOTE is one technique for resolving issues related to processing sparse data. These techniques are necessary to improve the performance of existing algorithms because poorly distributed data can kill the majority of them. To address this issue, balancing strategies were applied before the data was trained. It generates synthetic information from the existing data itself. Instead of creating duplicate data points, SMOTE has the advantage of creating new points.

The reason for executing this method now is while considering the Edited Nearest Neighbor technique, it must first identify the K-nearest neighbor observation and then determine whether the neighbor's majority class matches the observation's majority class. If the observation's majority class and the neighbor's majority class are not similar, the concerned neighbor and observation must be eliminated from the data set.

3.2 Data Pre-processing

The data was first cleaned and then preprocessed which includes handling of null and missing values and doing label encoding. All duplicates were removed from dataset there were no missing values but some duplicates.

Outlier Detection and Removal

The accuracy of the model is increased by eliminating outliers [7]. This paper's authors used a box plot to find outliers. The systolic and total cholesterol columns outliers were eliminated.

Figure 1 represents the box plot of BMI Columns which is used to determine outliers in the model.

Removing outliers is necessary because they tamper with the results and hence decrease the working of model.

Balancing Data

The imbalanced dataset was balanced so that model fitting could be carried out without complications. The technique, SMOTE ENN used in this case study rightly equitized the dataset.

Training was performed on 70%, While 30% on the test dataset.

Imbalance Ratio (Minority to Majority) 1:28.

Table 2 shows number of records used for training, test and total after balancing of data.

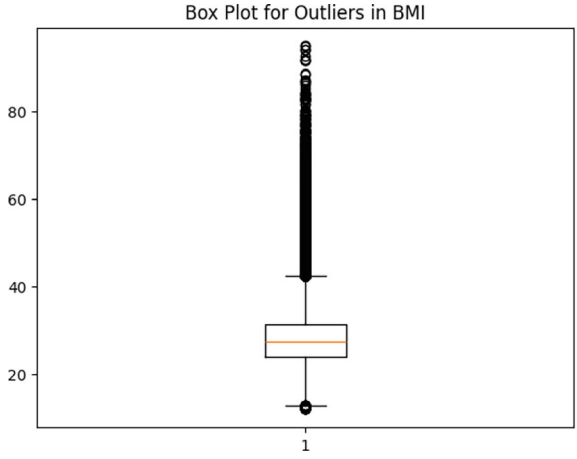

Fig. 1 Whisker plot to identify outliers

Table 2 After balancing number of records

Class	Training subset (70%)	Test subset (30%)	Total records
High-risk CVD	110.387	47.295	157.682
Low-risk CVD	141.146	60.505	201.651
Total records	251.533	107.800	359.333

3.3 Methodology

Chi-Squared Test

A chi-square test estimates a difference between the two is due to chance or a relationship between the variables being studied. Chi-square is mainly used in feature selection and hypothesis testing. It performs the following task through the comparison of how data are distributed across different categories in a contingency table; hence it helps a researcher determine if the observed pattern could have resulted from chance, or whether an association among the variables under study actually exists. Thus, the chi-square test is used to better understand and assess the relationship between these two category variables.

$$X^2 = \sum \frac{(O - E)^2}{E} \tag{3}$$

X^2 = chi squared, O = Observed value, E = Expected value A high X^2 value shows that there is a good relation between the feature and label class.

Figure 2 shows that some columns have extremely good relation while some column have weak relation hence not good for model.

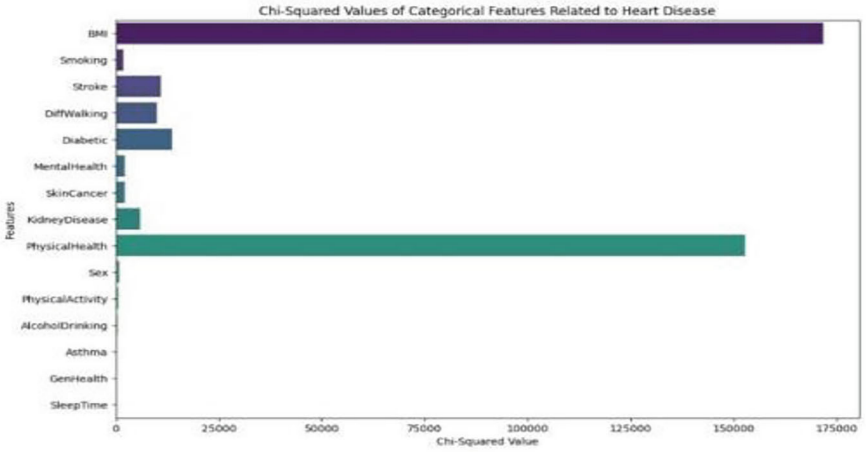

Fig. 2 Graph of chi square value of different features to find best columns

Table 3 shows accuracies obtained before Chi-squared.

Accuracy: 0.85

F1-Score for Class 1: 0.24 The model achieves relatively high precision for class 0 but the recall for class 1 is low.

The Decision Tree performs well in the majority class but fails to properly classify the minority class. This is sensitive to the class of imbalance modelled.

Principal Component Analysis (PCA)

PCA is one of the most popular tools in artificial intelligence for exploratory data analysis and prediction models. It is an algorithmic technique that examines the relationships between a collection of variables in unsupervised learning. In other terms, PCA may also be described as broad factor analysis that uses regression to identify a line of best fit.

PCA's primary objective was to reduce a dataset's dimensionality while preserving the most important patterns or correlations between variables, all without using any prior knowledge of the target variables.

Figure 3 represents the reduced dimensionality of the data done by help of PCA.

Table 3 Accuracies without chi-squared

Class	Precision	Recall	F1-score	Support
0	0.92	0.91	0.91	67.971
1	0.23	0.25	0.24	7190
Accuracy			0.85	75.161
Macro avg	0.57	0.58	0.58	75.161
Weighted avg	0.85	0.85	0.85	75.161

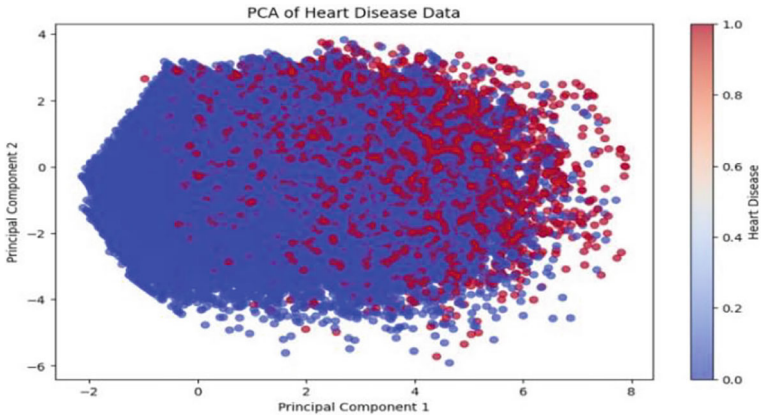

Fig. 3 Analysis of components of heart disease data

Decision Tree

It is a categorizing method in which objects are categorized by predetermined labels. Measuring purity in terms of impurity Impure can be measured through the data set. When the sample is homogeneous, then all its elements fall under the same class [8].

Logistic Regression

The probability of a discrete outcome using input factors is described by logistic regression [9]. Multinomial logistic regression can be used to model situations when there are more than two unique discrete outcomes. Since many issues, including threat detection, may be seen as classification problems, logistic regression is a useful analytical technique in cyber security [10]. An explanatory variable is linked to log chances.

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i \tag{4}$$

where the function p_i, the explanatory variable is x_i, β_0, β_1 are the parameters of the model [11].

Support Vector Machine (SVM)

To quickly categorize new data points, the SVM algorithm looks for the optimal lines also referred to as decision boundaries that divide the n-dimensional space into classes [12]. The hyper-plane is the ideal choice boundary in every way. Support vectors are typically employed as a representation of the more severe circumstances [8].

K-Nearest Neighbors

The key concept is to classify based on the categories of its k nearest neighbors rather than just the closest document [13, 14]. In the KNN classification method, a new sample, F, is assigned to the category (alive or dead) that the majority of its nearest neighbors belong to the main idea is to classify queries based on the classes of not just the closest data point but also the k nearest points. When k equals 1, this approach mirrors the KNN algorithm. The study uses a distance-weighted matching function, similar to the Vector technique, with Euclidean distance as one method for calculating similarity between documents [13].

$$D(a,b) = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2 + \cdots + (a_n-b_n)^2} \tag{5}$$

In KNN, the document space is categorized by a voting mechanism.

Random Forest

Random Forest (RF) uses randomized feature selection, choosing a subset of features when building decision trees. While randomization is applied during feature selection, bagging is used to select split points. By combining decision trees and

bootstrapping, RF reduces overfitting and improves predictive accuracy [15, 16].

$$RFfi_i = \frac{\sum_{j \in \text{all trees}} \text{norm}fi_{ij}}{T} \tag{6}$$

$RFfi_i$ = values represent the significance of features $\text{norm}fi_{ij}$ = the normalized significance of feature i in tree j, T = tally of trees.

4 Proposed and Enhanced Model of Heart Disease Prediction

Figure 4 shows the data cycle starting from its collection from Kaggle to what are applied analysis and algorithms.

4.1 Decision Tree

Decision tree learning is a statistic, data mining, and machine learning approach to supervised learning. In this formalism, classification or regression decision trees are used as predictive models in the formation of some conclusions about a set of observations.

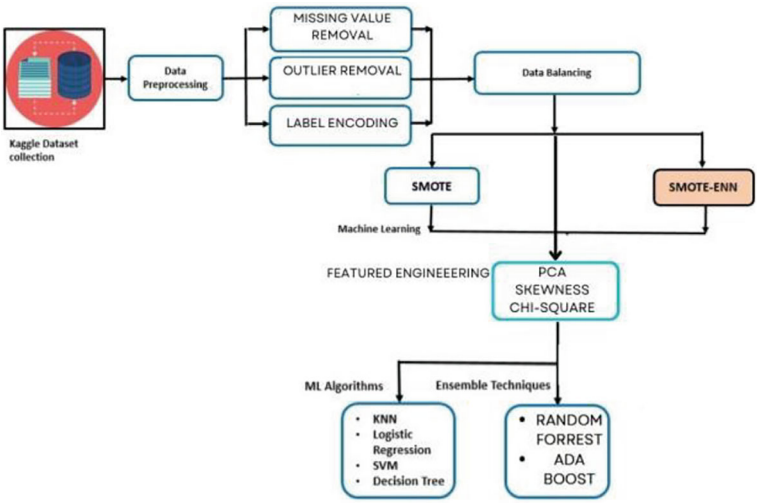

Fig. 4 Data flow diagram

A straightforward representation for categorizing an example is a decision tree. This chapter assumes that there is just one goal feature, which in common language referred as the classification, and that all of the input characteristics are finite discrete domains. Every component of the categorization domain is referred to as a class. If an input characteristic is present in each internal (non-leaf) node, the tree is classified as a decision tree. Each potential value for the target feature is indicated on the arcs that descend from an input node, or the arc may lead to a subordinate decision node on another input feature. A class label or a probability distribution across classes is attached to each leaf of the tree. If the decision tree is well-constructed, it will be more inclined toward particular subsets of classes. This represents the classification that the tree has done in either a specific or probability distribution format.

4.2 Experimental Work

It was extremely necessary to perform all information and preprocessing on the data set before getting started on analysis. To achieve the study's objectives, the **Indicators of Heart Disease (2022 UPDATE)** was employed. The outlier in the data was then first found, and the necessary changes were subsequently made to eliminate it.

Then the algorithm measured skewness and kurtosis test on data set to check variability and positioning of data. Then performed pairwise correlation test, chi squared and p-test and hence removing columns with less correlation.

Then PCA and ANOVA is applied to enhance the quality of data. Algorithm of Kaplan–Meier estimator is used to estimate survival possibility over time that means it allows us to estimate the probability of that subject. Since our data's label was highly unbalanced i.e. almost 10%, then used a technique named Smote and Smote ENN. The purpose of this technique wad to ensure that the data set was balanced for prediction algorithms. A no. of different algorithms including SVM, DT, RF, KNN, logistic regression as well as various ensemble technique such as bagging, stacking and boosting were used. The achieved accuracy of 93%.

5 Result and Analysis

5.1 Experimental Setup

The Python programming language "Google Colab is used in the proposed model". This research system focuses on prototyping machine learning models using advanced hardware options like GPUs. It provides an interactive programming environment utilizing Jupyter notebooks without the necessity of a server. Experiments were performed on a machine possessing the following specifications: i5-8250U CPU operating at 1.80 GHz, 16 GB of RAM, × 64 architecture [17].

5.2 Performance Parameters

During the evaluation of the considered Machine Learning models, many performance metrics were recorded. This analysis primarily concentrates on the vocabulary most commonly seen in the relevant research. "True Positive" (TP) indicates that both the actual and predicted classes possess value. The predicted positive result is accurately presented in. True Negatives (TN) are actual negative values classified as negative. False positives (FP) and negatives arise when the actual class diverges from the planned class. False negatives (FN) arise when the main class differs from the predicted class.

Accuracy: Accuracy necessitates minimal TP and FP rates within a dataset. It aids in ascertaining its efficacy [18].

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}} \tag{7}$$

Precision

This is the ratio of real positives to positive samples.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

TP + FP (denominator) > TP (Numerator) explains the degradation in the accuracy of a machine learning model. Explanation of good machine learning model accuracy is in terms of TP (Numerator) > TP + FP (denominator). Therefore, the precision allows us to measure the dependability of a machine learning model and judge whether it will be worthwhile or not.

Recall

TP + FN (denominator) > TP (numerator) diminishes the recall of the machine learning model (numerator). Recall is unaffected by errors in sample categorization, but accuracy is dependent on them. If the model identifies all positive samples correctly, the recall will equal 1 [19].

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

F1 Score

The F1 score is applicable when either accuracy or recall is prioritized, as it encompasses both metrics. Mitigating false negatives may be prioritized over addressing false positives.

$$F1\text{ - score} = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{10}$$

Receiver Operating Characteristic (ROC)

ROC is a graphical representation that assesses a classification's model performance based at various threshold levels. At different threshold settings it represents the curve of False Positive Rate (FPR) against True Positive Rate (TPR) or Recall.

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{11}$$

The curve aids in comprehending how sensitivity and specificity are traded off at various judgment thresholds.

Area Under the Curve (AUC)

AUC is a performance parameter which is used to assess that how well a binary classification model performs. It simply gives the area of ROC curve. AUC specifies the performance of model as closer the value of AUC to one the better the model works.

Table 4 represents what different values of Area of ROC curve represents.

Confusion Matrix

One approach for assessing the effectiveness of classification models is a confusion matrix. It offers insights into true positives, false positives, true negatives, and false negatives by breaking down the model's predictions in detail and comparing them to the actual values. The matrix is particularly helpful for datasets that are unbalanced, as precision by itself can be deceptive.

We can extract significant parameters like accuracy, precision, recall (sensitivity), F1-score from the confusion matrix, which aid in a thorough evaluation of the model's performance.

Table 4 Values and interpretation of AUC

AUC Value	Interpretation
1.0	Perfect classifier
0.9–1.0	Excellent classifier
0.8–0.9	Good classifier
0.7–0.8	Fair classifier
0.6–0.7	Poor classifier
0.5–0.6	Random guessing (just like coin toss)
< 0.5	Worse than random guessing

Table 5 Accuracies obtained by different algorithms

S. No.	Method	Accuracy score (%)
1	LR	90
2	SVM	86
3	DT	93
4	KNN	83
5	ADA boost	82
6	Random forest	84

5.3 Analysis

The suggested method is contrasted with well-known machine learning algorithms to verify its efficacy. Stacking, Gradient Boosting, AdaBoost, Bagging Meta estimator, Random Forest, SVM, Decision Trees, KNN, and logistic regression are few of the methods that can be used. The outcomes of tests performed on heart disease datasets using each of the different approaches are listed in the table below. It is very evident from the following table and figure that, when employing the model pipeline, decision trees achieve the most accuracy, while ADA boosts achieve the lowest accuracy, at 82.88%. After SVM, random forest, and KNN, which have accuracy rates of 86.11%, 84.01%, and 83.05%, respectively, logistic regression has the second-highest accuracy of 90.07%.

Table 5 shows different accuracies from different machine learning algorithms.

Figure 5 shows the different accuracies obtained by the different machine learning algorithms. In this DT outperformed. This kind of accuracy has not been obtained so for on such large dataset.

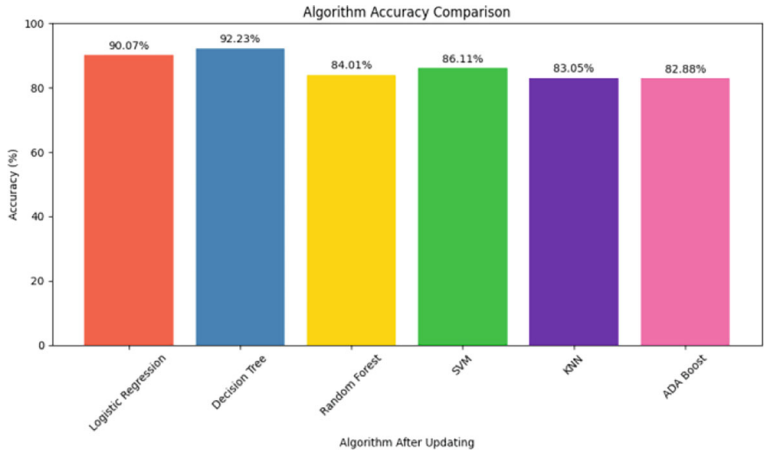

Fig. 5 Comparison of accuracies of different machine learning algorithms

As shown in Table 6 without applying Smote accuracies of our model is very less hence Smote is applied to enhance the performance of proposed model.

Table 7 shows the accuracy of ML algorithms by with SMOTE. The Decision Tree (DT) model demonstrates superior performance across all metrics, attaining an overall accuracy of 93%. KNN and Random Forest exhibit consistent results at 83%, whilst Logistic Regression and SVM yield marginally worse performance at 81%. These data confirm DT as the most dependable model for this classification task.

Figure 6 represents the graph of precision, recall, and F1 score and different machine learning algorithms.

The model was evaluated on 300,000 instances of the provided confusion matrix and was able to classify two categories: Class 0 (negative) and Class 1 (positive). As shown in Fig. 7, 136,710 True Negatives and 142,290 True Positives were accurately predicted by the model. Nevertheless, it incorrectly identified 10,290 False Negatives and 10,710 False Positives. With an accuracy rate of almost 93%, the model performs well overall. The model is good at detecting true positives while reducing false positives since precision (93%) and recall (93%) are balanced. Furthermore, the balance between precision and recall is confirmed by the F1-score of 93%.

The evaluation findings for the heart disease recognition system are displayed as the area under the receiver characteristic curve (AUC). ROC Curve of decision tree is given in Fig. 8. The test result from the dataset used is to explain how to understand the AUC curve in the above figure. The AUC value that was achieved in this case is 0.93, which is nearly equivalent to 1. In other words, the ability to identify heart conditions has significantly improved.

Table 6 Accuracy, precision, and recall without SMOTE

Model	Accuracy	Precision	Recall	F1 score
LR	0.44	0.57	0.60	0.67
DT	0.55	0.75	0.75	0.75
RF	0.59	0.56	0.59	0.57
AdaBoost	0.71	0.78	0.71	0.77

Table 7 Precision, recall and F1-score of different algorithms with SMOTE

Metric	KNN	DT	RF	LR	SVM
Precision (Class 0)	84	93	83	83	83
Precision (Class 1)	81	93	82	79	79
Recall (Class 0)	81	93	81	78	77
Recall (Class 1)	85	93	84	83	84
F1-score (Class 0)	82	93	82	80	80
F1-score (Class 1)	83	93	83	81	81
Overall accuracy	83	93	83	81	81

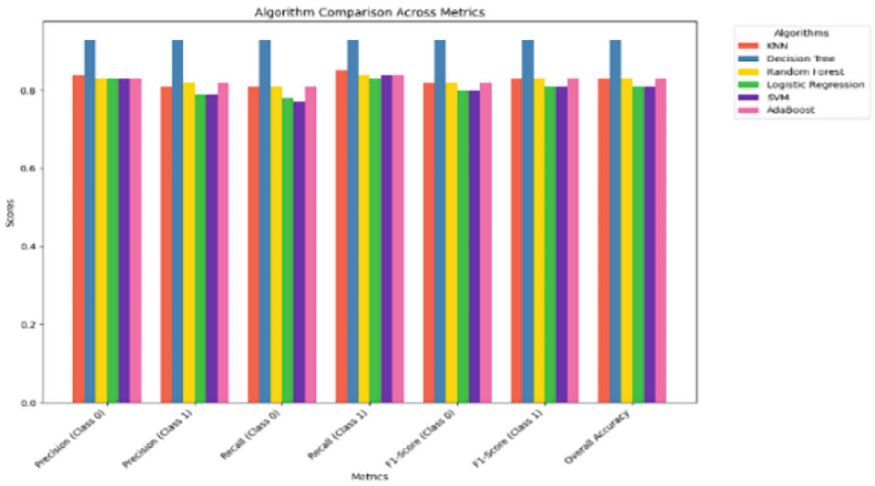

Fig. 6 Comparisons of different algorithms and performance parameters

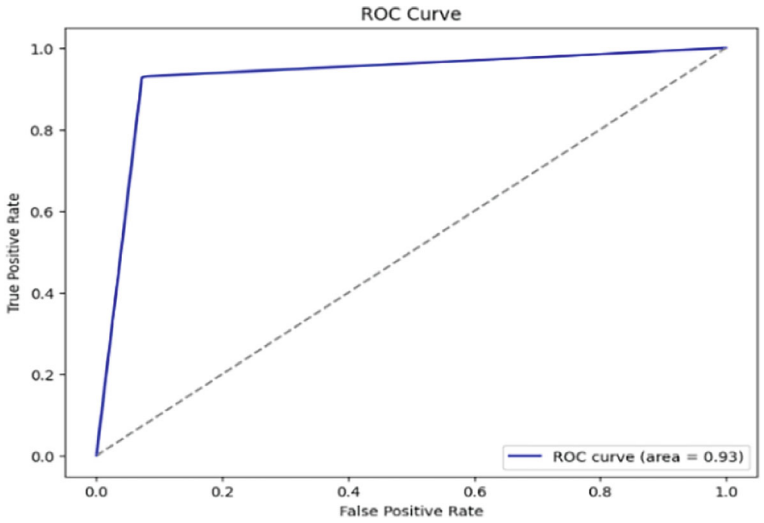

Fig. 7 Confusion matrix of decision tree

6 Conclusion and Future Work

It is anticipated that this suggested model will contribute to the enhancement of predicted accuracy and reliability of cardiac diseases by highlighting parameter adjustment, and recursive feature elimination procedures. In terms of regularization strategies, the prediction techniques included logistic regression, DT, KNN, SVM machine learning. By utilizing it, the usefulness of this paradigm to the dataset is

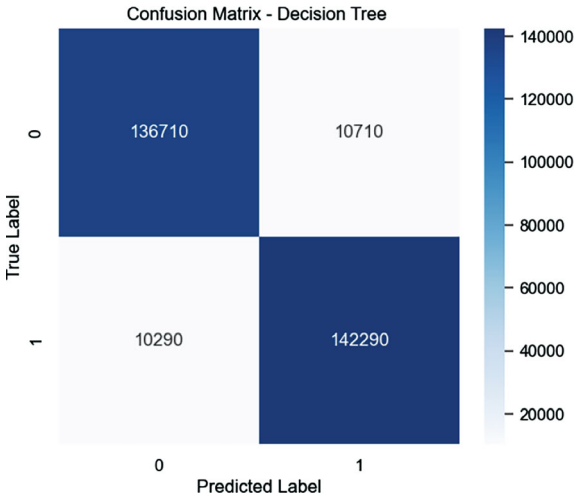

Fig. 8 ROC curve of decision tree

evaluated. Clustering is the process of classifying people based on their likelihood of developing cardiovascular disease (CVDs) during the following ten years. Apart from clinical risk data, it also includes societal risk variables and patient segment risk characteristics. It has 18 characteristics and over 3 lakh records. This analysis established whether or not the dataset is balanced. It was discovered that there is a startling imbalance between the positive and negative class labels. The dataset must be rebalanced because it is extremely unbalanced in order to avoid any issues when attempting to fit it into the model. In this instance, SMOTE ENN balanced the dataset. To match the number of members of the majority class, more members of the minority class were created.

The dataset had to contain all relevant data before the analysis could begin. To compare our method with other machine learning techniques, a number of tests were conducted on the data set using some well-known algorithms. It was anticipated that the new method will increase the accuracy, robustness, and strength of coronary cardiovascular risk assessments. Here, machine learning and deep learning have been combined [20]. To increase accuracy, we can use techniques like staking and bagging. Compared to other machine learning algorithms, decision trees proved to be the most accurate ensemble approach. The suggested model yielded high accuracy values of 93.23% in the previous testing. Also, the model can be enhanced with explainable-AI which can give information on how the model is working. This model's main goal is to determine which classifiers should be used in order to accurately forecast heart illness in real-world scenarios. This model might have the potential to be enhanced as an application by just introducing a frontend and a user-friendly UI for the patients that can get rough prediction on the persons cardiovascular health at a minimal cost which is very affordable than CT and MRI Scans. The model can also be used as "Clinical Decision Support System" for early detection and alerts in hospitals which can lead to saving of many lives. The Kaggle website provided the data set

for this investigation. A variety of statistical models, including Decision Trees, LR, SVM, and KNN, have been used to investigate early predictions of heart disease. Using patient data and characteristics, ensemble methods such as bagging, boosting, stacking, and Random Forest have been used to predict cardiovascular diseases. In this trial, decision trees improved accuracy.

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Optimum Hyperbolic Tangent Function for MRI Enhancement

V. R. Simi, Justin Joseph, Vipin Venugopal, and R. Rashmi

Abstract Hyperbolic tangent function (HTF) is an intensity transformation used to improve the contrast of medical images from diverse modalities. Selection of the α parameter determines the slope of the HTF and influences the quality of output images. An exhaustive framework comprising a metaheuristic search strategy termed bilateral search within iteratively narrowing space (BSINS) optimization algorithm and a fitness function termed cumulative information preservation and contrast enhancement index (CIPCEI) for automating the selection of the α parameter of HTF employed for image contrast enhancement applications on MRI is proposed. At the ideal value of the α located from the CIPCEI, the HTF produces enhanced MRI slices with improved contrast among the structures and minimum shift from the mean brightness of the actual MRI slices. The BSINS has faster convergence than the metaheuristic algorithms used in the literature for optimizing the α parameter of the HTF, namely the tunicate swarm optimization algorithm and differential evolution.

Keywords Fitness function · Hyperbolic tangent function · Image enhancement · Metaheuristic optimization

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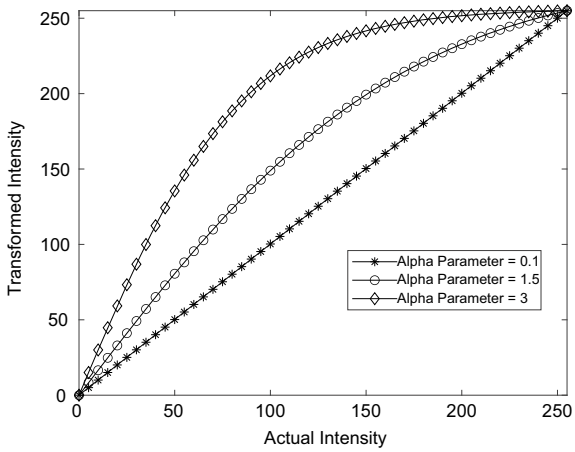


Fig. 1 Influence of the selection of the α parameter on the shape of the HTF

1 Introduction

1.1 Research Problem

Hyperbolic tangent function (HTF) is an intensity transformation used for improving the contrast of mammograms [1], optical coherence tomography (OCT) [2], and natural scene images [3, 4]. For enhancing the greyscale images, the intensity values in the low-contrast input image are normalized to an interval between 0 and 1, and the HTF is applied to the normalized intensity values. After applying the HTF, the transformed intensity values need to be mapped back to the actual range of the low-contrast input image. Transformed intensity y obtained from the HTF on the actual intensity x is as follows:

$$y = \tanh(x) = 2\left(1/\left(1 + e^{-2\alpha x}\right)\right) - 1, \text{ and } 0 \leq x \leq 1 \tag{1}$$

In (1), α is a user-defined parameter.

In Figs. 1 and 2, the selection of the α parameter determines the slope of the HTF and influences the quality of output images. When the value of the α parameter is very low, there will not be a significant improvement in contrast (Fig. 2b), and when its value is high, brightness overshoot is caused (Fig. 2d). Manual selection of the α parameter may be suboptimal and produce enhanced images of poor quality.

1.2 Existing Solutions

Automated methods for the selection of the α parameter of HTF are rare in the literature. Laishram and Rabidas [1] used the tunicate swarm optimization algorithm

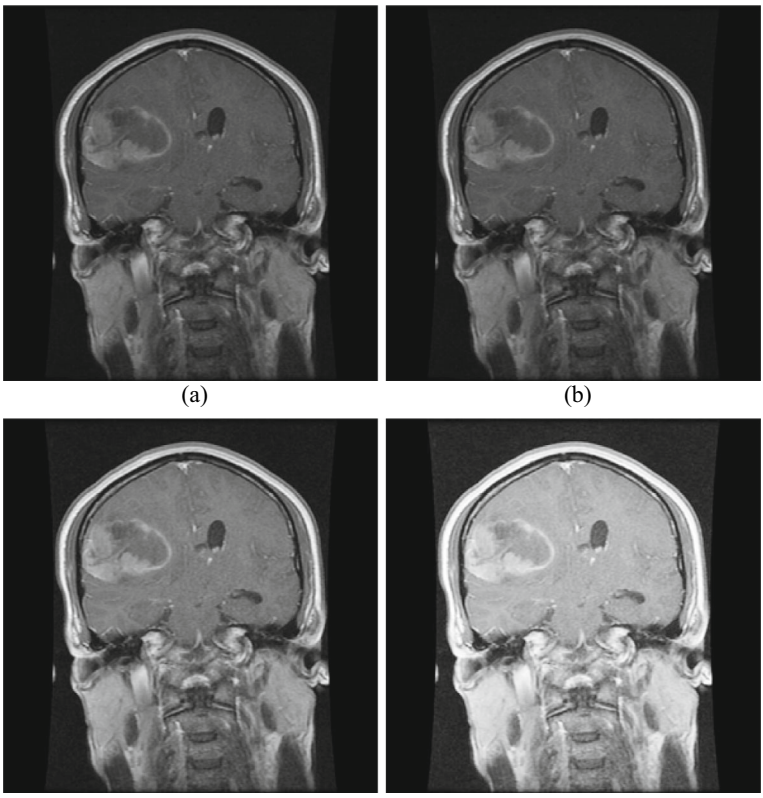


Fig. 2 A low-contrast MRI slice and outputs of HTF at different values of the α parameter: **a** low-contrast MRI slice, **b** $\alpha = 0.1$, **c** $\alpha = 1.5$, **d** $\alpha = 3$

(TSAO) to automate the selection of the α parameter in HTF. A product of edge pixel count (EC), entropy (E), and bi-logarithm of the sum of edge intensities (BSEI) (PECEBSEI) was used as the fitness function. Otsu's threshold algorithm was used to detect the edges from the Sobel response of the enhanced images. For automating the selection of the α parameter of HTF, Enríquez et al. [4] used the differential evolution (DE) and another fitness function which is the geometric mean of entropy, mean gradient, and intensity standard deviation (GMEMGISD).

1.3 Pitfalls of Existing Solutions

The metaheuristic optimization algorithms and the fitness functions used in the existing automated methods for the selection of the α parameter of HTF [1, 4] have certain pitfalls. The TSOA [1] has limitations, namely slow/premature convergence,

the possibility of getting trapped at local minima points, and unbalanced exploitation/exploration ratio [5–8]. The DE [4] suffers from drawbacks such as premature convergence and stagnation. Stagnation is a condition when there is no improvement in the best fitness. The stagnation problem refers to a situation in which the optimum-seeking process stagnates before finding a globally optimal solution [9, 10]. The DE is susceptible to the selection of parameters like population size, scaling factor, and crossover rate. DE may fail when the fitness is not smooth. The fitness functions specifically designed for a particular class of images may not work on other modalities. The fitness functions used in the literature for automating the selection of the α parameter of HTF employed for image contrast enhancement applications on mammograms [1], and natural scene images [4] may not be adequate for MRI.

1.4 Objectives

A complete framework for automating the selection of the α parameter of HTF employed for image contrast enhancement applications on MRI is proposed here. Within the framework, we have incorporated two novel techniques, namely a metaheuristic search strategy, termed bilateral search within iteratively narrowing space (BSINS) optimization algorithm, and the fitness function termed cumulative information preservation and contrast enhancement index (CIPCEI).

2 Methodology

2.1 Bilateral Search Within Iteratively Narrowing Space (BSINS) Optimization Algorithm

In the BSINS, an array P containing N arbitrary number of points equally spaced within the interval $\left[p_{\min} p_{\max}\right]$ is constructed as:

$$P = p_{\min} + \left[(p_{\max} - p_{\min})\{0, 1, 2, \ldots\ldots, N-1\}/N - 1\right] \tag{2}$$

So that, the array is as follows:

$$P = \{p_1, p_2, p_3, \ldots\ldots, p_n, \ldots\ldots, p_N\}, \text{given} p_1 = p_{\min}, \text{and} p_N = p_{\max} \tag{3}$$

Another array of fitness values $F = \{f_1, f_2, f_3, \ldots\ldots, f_n, \ldots\ldots, f_N\}$ is constructed from the values, $p_n, n = 1, 2, \ldots\ldots, N$ such that $f_n = \Psi(p_n)$. The operator $\Psi(.)$ denotes the fitness function. The parameter to be optimized in our context is the α parameter of the HTF, and the fitness function is the CIPCEI. Construction of the array P of equally spaced points $p_n, n = 1, 2, \ldots\ldots, N$ described in (1)

is an alternative for the random initialization generally used in the majority of the nature-inspired optimization algorithms. Random initialization results in inconsistent convergence time of the nature-inspired optimization algorithms. The minimum fitness value $f_{\min}$ is identified from the array F, and the parameter value corresponding to the minimum fitness is selected as the optimum parameter value p_{opt}. The BSINS uses a recursive updating procedure to further modify the points. In each iteration, the points $p_n, n = 1, 2, \ldots\ldots, N$ are updated one by one after the initialization step. During the updating process, for each arbitrary point $p_n, n = 1, 2, \ldots\ldots, N$, two random points, one on the left p_L and another on the right p_R sides of the current optimum parameter value p_{opt} are generated as:

$$p_L = p_{\text{opt}} - \beta_t r\left(p_{\text{opt}} - p_{\min}\right) \text{and} p_R = p_{\text{opt}} + \beta r\left(p_{\max} - p_{\text{opt}}\right) \tag{4}$$

In (4), r is a randomly generated fractional value between 0 and 1. The variable β_t is an iteration-dependent space narrowing parameter (ISNP) which is equal to:

$$\beta_t = e^{-t}, t = 1, 2, 3, \ldots\ldots, t_{\max} \tag{5}$$

In (5), t is the index of the current iteration and $t_{\max}$ is a user-defined value of the maximum number of iterations. The ISNP β_t decreases with the increase in the index of iteration t narrowing the search space in (4), as its name implies. In many nature-inspired optimization algorithms, the parameter values go outside the bounds during the updating processing. The formulation in (4) ensures that the parameter values remain within the bounds during the updating. The minimum fitness $f_{\min}$ and the contextual fitness value f_n are replaced by $\min\{f_{\min}, \Psi(p_n), \Psi(p_L), \Psi(p_R)\}$. The optimum parameter value p_{opt} and parameter value or point p_n are replaced by one of the candidate points $\{p_{\text{opt}}, p_n, p_L, p_R\}$ for which the fitness function is minimum. The steps following (4) and (5) are repeated until the index of iteration becomes equal to the maximum number of iterations $t_{\max}$. In the current implementation of the BSINS, the maximum number of iterations is set as 50. Values of $p_{\max}$ and $p_{\min}$, respectively, are set as 3 and 0.001.

2.2 *Cumulative Information Preservation and Contrast Enhancement Index (CIPCEI)*

For a good selection of the α parameter, both the contrast of the HTF output and the information shared between the input MRI slice and HTF output should be maximum. The CIPCEI is formulated based on this expectation. The CIPCEI for an input MRI slice X and HTF output Y is as follows:

$$\text{CIPCEI}(X, Y) = \text{PSNR}(X, Y)\sqrt{\left|1 - \frac{\text{Contrast}(X)}{\text{Contrast}(Y)}\right|} \tag{6}$$

The contrast of an arbitrary MRI slice say X can be computed as [11]:

$$\text{Contrast}(X) = 20\log\left[\frac{1}{\text{MN}}\sum_{m=1}^{M}\sum_{n=1}^{N}\left(X^2(m,n) - (\mu_X)^2\right)\right] \tag{7}$$

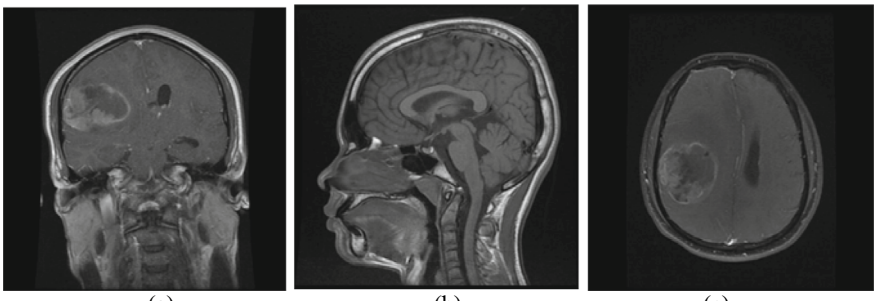
In (7), μ_X is the mean brightness of the MRI slice X. It is expected that the contrast of the HTF output Y will have higher contrast than the input MRI slice X. So that the fraction $\text{Contrast}(X)/\text{Contrast}(Y)$ will be less than 1. The term $\sqrt{1 - (\text{Contrast}(X)/\text{Contrast}(Y))}$ in (6) will increase with the increase in contrast of the HTF output.

The product $\text{PSNR}(X, Y)\sqrt{1 - (\text{Contrast}(X)/\text{Contrast}(Y))}$ will be maximum when the both contrast of the HTF output and the information shared between the input MRI slice and HTF output are maximum.

3 Results and Discussions

Three MRI slices of coronal, sagittal, and axial planes used for validating the optimization algorithms and fitness functions are shown in Fig. 3. The selection of MRI slices from three different planes is motivated by their diversity in terms of the information content. Using the MRI slices from three different planes carrying diverse information for validating the optimization algorithms and fitness functions helps to justify the robustness of the proposed framework for automating the selection of the α parameter of the HTF. The low-field MRI slices used here are from a dataset exclusively used in the literature [12–14] for evaluating the performance of image enhancement algorithms. Detailed information regarding the acquisition parameters is available in Edla et al. [12–14].

Variations of fitness functions used in the literature for optimizing the parameters of HTF in the literature [1, 4] and against the α parameter on three test MRI slices



(a) (b) (c)

Fig. 3 Three MRI slices of different planes used for validating the optimization algorithms and fitness functions: **a** coronal plane, **b** sagittal plane, **c** axial plane

are shown in Fig. 4. For MRI slice 1, the PECEBSEI [1] is maximum at $\alpha = 2.4$. For MRI slices 2 and 3, the PECEBSEI is maximum at $\alpha = 2.5$. Three enhanced MRI slices of different planes obtained from the HTF with α parameter values at which PECEBSEI is maximum are shown in Fig. 5. The outputs of the HTF are over-enhanced with intolerable disturbance in average brightness level. The α parameter values coinciding with the maxima points of the PECEBSEI are suboptimal for all three MRI slices in Fig. 3.

The GMEMGISD [4] is continuously increasing even after $\alpha = 3$. It can be already shown in Fig. 5 that HTF produces over-enhanced images even for α values approximately equal to 2.5. Of course, an α value greater than or equal to 3 will increase the severity of the over-enhancement because the HTF drags the lower intensities more to the higher side for higher α values. Hence, GMEMGISD [4] is not a good choice of fitness function for locating the optimum α values of HTF used on MRI.

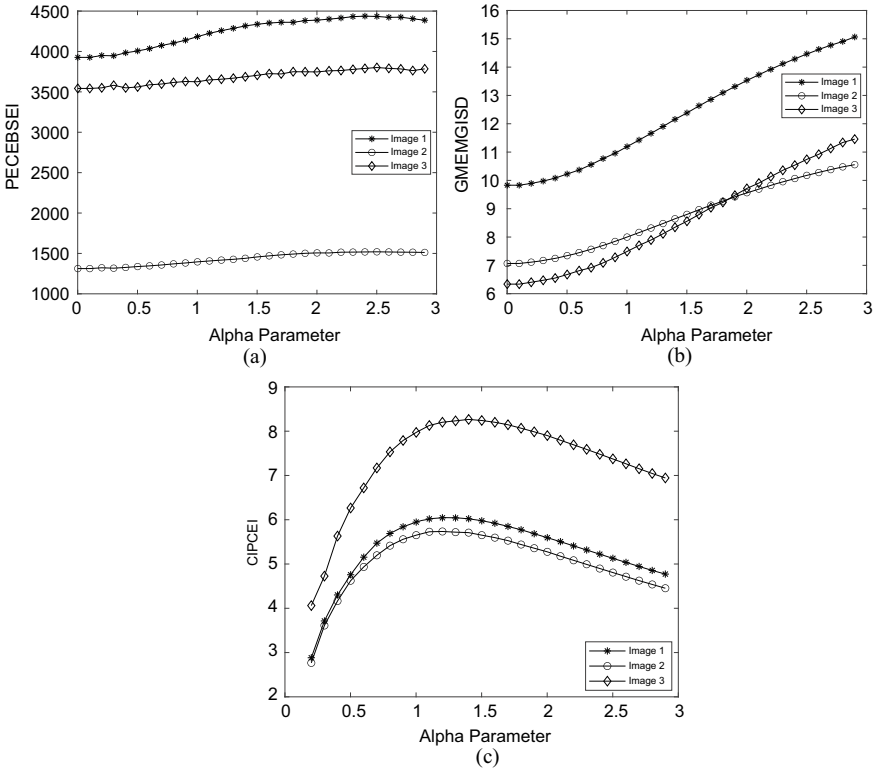


Fig. 4 Variations of fitness functions against the α parameter on three test MRI slices: **a** PECEBSEI versus α parameter, **b** GMEMGISD versus α parameter, **c** CIPCEI versus α parameter

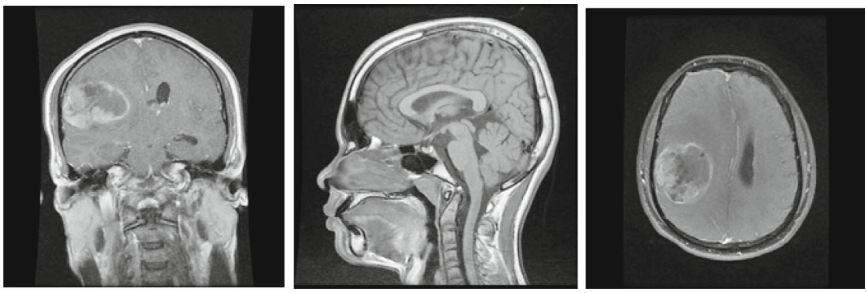


Fig. 5 Three enhanced MRI slices of different planes obtained from the HTF with α parameter values at which PECEBSEI is maximum: **a** coronal plane, **b** sagittal plane, **c** axial plane

For MRI slices 1 and 2, the CIPCEI values are maximum at $\alpha = 1.3$. For MRI slice 3, the CIPCEI is maximum at $\alpha = 1.5$. Three enhanced MRI slices of different planes obtained from the HTF with α parameter values at which CIPCEI is maximum are shown in Fig. 6. In all three outputs of the HTF, the contrast between the structures is increased without causing much shift from the mean brightness of the actual MRI slices. The α parameter values coinciding with the maxima points of the CIPCEI are optimal for MRI slices carrying completely diverse information. In addition to having the ideal bell shape of a fitness function, the CIPCEI exactly locates the optimum α values of HTF on MRI slices from coronal, sagittal, and axial planes that carry completely diverse information.

Convergence curves of the BSINS and the metaheuristic algorithms used in the literature [1, 4] for optimizing the α parameter are shown in Fig. 7. The optimum α parameter predicted by TSOA [1], DE [2], and BSINS, corresponding maximum fitness, and highest convergence time in sec are shown in Table 1. The convergence times of TSOA and DE are quite inconsistent because of the random selection of the initial parameter values. Hence, the highest convergence time in sec is reported for

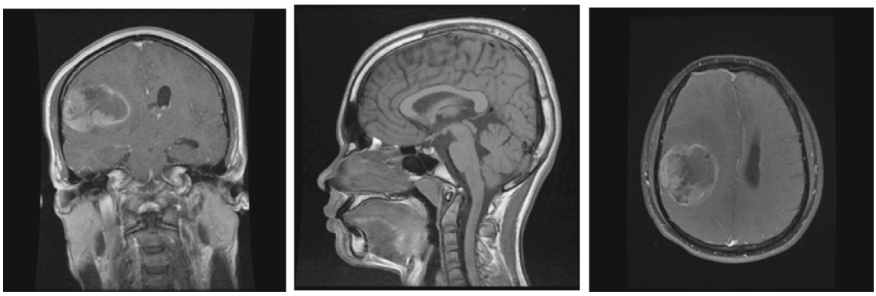


Fig. 6 Three enhanced MRI slices of different planes obtained from the HTF with optimum α parameter values located from the CIPCEI: **a** coronal plane, **b** sagittal plane, **c** axial plane

each metaheuristic optimization algorithm. All three metaheuristic algorithms locate the maximum point of the fitness and the corresponding α parameter. The difference between them is only in terms of the convergence time. In Fig. 7 and Table 1, the BSINS has faster convergence than TSOA and DE.

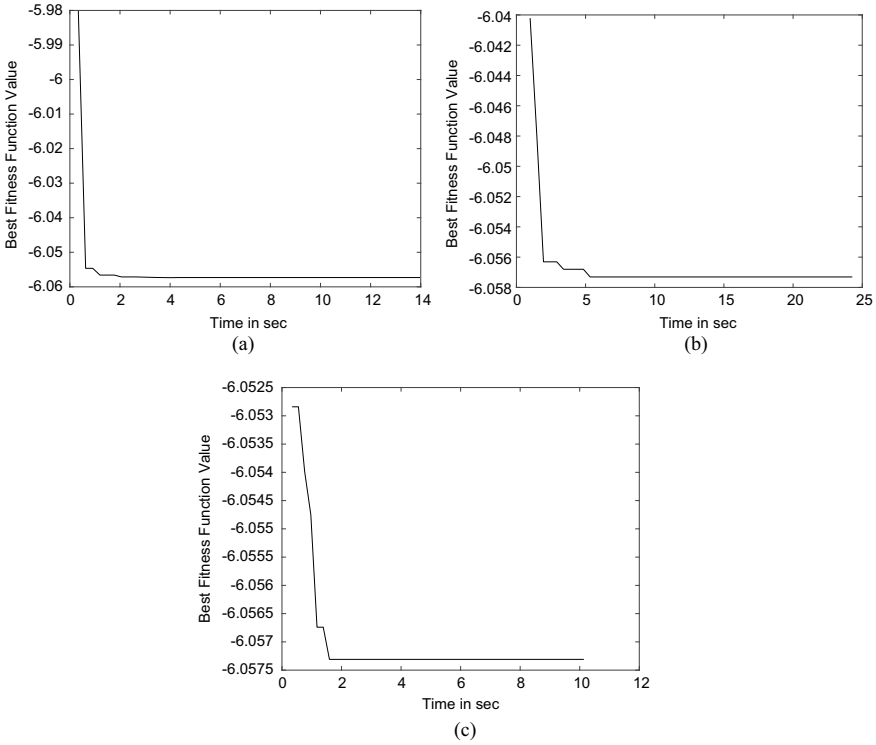


Fig. 7 Convergence curves of various metaheuristic optimization algorithms: **a** TSOA, **b** DE, **c** BSINS

Table 1 Optimum α parameter predicted by the metaheuristic optimization algorithms, corresponding maximum fitness, and highest convergence time in sec

Metaheuristic optimization algorithm	Optimum α parameter	Maximum fitness	Convergence time (s)
TSOA	1.2869	6.0573	3.19
DE	1.2869	6.0573	5.32
BSINS	1.2869	6.0573	1.59

4 Conclusions

An exhaustive framework comprising a metaheuristic search strategy (BSINS) and a novel fitness function for automating the selection of the α parameter of HTF employed for image contrast enhancement applications on MRI was proposed. CIPCEI is a good fitness function because at the ideal value of the α located from the CIPCEI, the HTF produces enhanced MRI slices with improved contrast among the structures and minimum shift from the mean brightness of the actual MRI slices. The BSINS has faster convergence than the metaheuristic algorithms used in the literature for optimizing the α parameter of the HTF.

HTF has a limitation that it does not consider any pixel statistics. The CIPCEI fitness function may require slight modifications for application in different medical imaging modalities and natural scene images. The current formulation of the BSINS is for optimizing a single parameter. For multi-objective optimization applications, the BSINS requires further modifications.

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Diabetes Risk Progression and Personalized Recommendations Using Machine Learning

Himanshu Shekhar, Gauri Ojha, Pranav Gupta, and Patri Upender

Abstract This paper explores machine learning methods for diabetes risk prediction and emphasizes logistic regression for simplifying and appropriately performing two classification tasks which is important for early diagnosis of diabetes to reduce complications of severe such as cardiovascular diseases, kidney failure, and arthritis. The study uses health-related factors such as glucose levels, BMI, blood pressure, and genetics to predict diabetes risk using logistic regression. To avoid overqualification and ensure rigorous evaluation, the training program is divided between training (80%) and testing (20%). With predictability, the model provides personalized health recommendations based on data management, covering diet, exercise, and medical assessment. A unique aspect of risk development is an increased risk of diabetes for each 5% increase illustrated per year, which highlights the long-term impact of lifestyle choices. Model performance is compared with other machine learning methods such as support vector machines and k-nearest neighbors, demonstrating the clarity, interpretability, and accuracy of logistic regression for healthcare applications.

Keywords Regression · Pedigree function · Machine learning · Artificial intelligence · K-nearest neighbor

1 Introduction

Diabetes is an increasingly prevalent health issue worldwide, affecting millions of people across diverse populations. One symptom of this chronic disease is a high glucose level in the blood. It happens when the body either does not produce enough insulin or cannot use the insulin it does make appropriately. If left untreated, diabetes over time leads to several potentially life-threatening complications, including renal failure, nerve damage, cardiovascular disease, and diminished eyesight. According to WHO, the last forty years saw the number of people suffering from diabetes

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increase nearly double, raising more urgency for advanced treatment methods and preventive measures. Early identification of individuals at risk of developing diabetes plays a critical role in controlling this growing epidemic, as timely interventions can mitigate the progression of the disease, ultimately reducing both health complications and healthcare costs [1, 2].

With the innovation brought in by digital health technologies, ML has become a strategy for solving healthcare issues like risk assessment and disease prediction. Machine learning algorithms can now analyze massive health-related data to find trends or generate predictions, usually more accurately and rapidly than the conventional approaches. In the recent past, scientists have ventured into various machine learning models for predictive purposes. Among these are models that can predict the onset of diseases like diabetes and heart disease. In a way, these models have sought to arm both patients and doctors with data-driven insights leading to lifestyle changes as well as preventive measures. Predictive algorithms hence assist healthcare systems in efficient resource allocation and targeting high-risk patients for intervention [3]. This study focuses on the development of a diabetes prediction model using logistic regression, a widely used and interpretable ML algorithm suitable for binary classification tasks. Logistic regression was chosen due to its simplicity, efficiency, and ability to provide probabilistic output, making it particularly valuable in healthcare applications where the transparency of model predictions is crucial. Unlike complex black-box models like deep neural networks, logistic regression offers insights into how each input feature—such as glucose level, body mass index (BMI), and age—contributes to the overall prediction. This interpretability is beneficial not only for clinicians who may wish to understand the model's decision-making process but also for users who can gain a clearer understanding of their risk factors [4].

The dataset used in this study includes a variety of health parameters linked to diabetes risk, sourced from commonly available clinical information. Features such as glucose levels, BMI, age, blood pressure, number of pregnancies, and diabetes pedigree function (indicating genetic predisposition) were selected for analysis. These factors are well documented in scientific literature as key indicators of diabetes risk. For example, high glucose levels and increased BMI are strongly associated with Type 2 diabetes, the most common form of the condition, often related to lifestyle factors and genetics. Age is also a relevant predictor, as the risk of developing diabetes generally increases with age. By incorporating these health parameters, the model can generate a holistic view of an individual's risk profile.

During model construction, the performance could have been evaluated and overfitting, a very common issue where a model learns the specific patterns of training data but does not generalize well to new data, by dividing the data into a training set and a testing set. The training data accounted for eighty percent of the dataset, while the testing data were twenty percent. This splitting allowed it to completely test the performance of the model and ensure that the model would correctly predict risk for diabetes of unseen data. Besides this, the parameters of the model were optimized by using cross-validation techniques, which enhances the reliability of the predictions. For making the model robust to variations within the dataset, cross-validation divides the data into multiple subsets and trains the model iteratively on each subset

[5]. Diabetes has become one of the most pressing public health issues globally. The World Health Organization (WHO) estimates a significant rise in diabetes cases over the past few decades, with Type 2 diabetes being especially prevalent due to lifestyle factors and aging populations. Machine learning (ML) provides an innovative approach to tackle this issue, as ML models can analyze large datasets to uncover hidden patterns, enabling proactive health management [1, 6].

In addition to diabetes prediction, this study aims to make the model accessible and user-friendly through a web-based application. The web app interface is designed to provide users with an intuitive platform where they can input their health data and receive a personalized probability score indicating their risk of diabetes. Beyond merely presenting a risk percentage, the application also provides tailored health recommendations based on user inputs, offering practical advice for lifestyle adjustments. For instance, individuals with higher glucose levels may receive suggestions on dietary modifications, while those with elevated BMI might be encouraged to increase their physical activity. This personalized feedback enhances the model's value, making it not just a diagnostic tool but also a guide for preventive healthcare. Additionally, the app includes a unique feature that illustrates risk progression. This feature simulates how an individual's risk of diabetes might increase over time if no preventive measures are taken, based on a hypothetical annual 5% increase in risk. This progression model serves as a visual reminder of the long-term impact of lifestyle choices, potentially motivating users to adopt healthier behaviors. By combining predictive analytics with actionable insights, this application aligns with the goals of preventive medicine, aiming to empower users with knowledge and encourage proactive health management. Although logistic regression was chosen because it has ease of interpretability and performs well for binary classification, the study also compared it with other forms of machine learning methods. For example, logistic regression was contrasted with other methods, such as k-nearest neighbors (k-NN) or support vector machines (SVMs). These contrasts add a bit more to the complete view of logistic regression's advantageous and disadvantageous sides when used with diabetes prediction. More advanced models can deliver slightly higher accuracy for the same binary classification problem; however, due to its transparency simplicity of use, and relatively high accuracy for this binary classification problem, logistic regression continues to be a viable option for healthcare applications [3, 7].

In summary, this research investigates the use of machine learning, specifically logistic regression, to predict diabetes risk based on health-related data. By combining predictive modeling with a user-friendly web interface, this study offers a valuable tool for individuals and healthcare providers alike, aiming to foster preventive care and improve health outcomes.

2 Literature Review

Over the last ten years, the health industry has witnessed significant growth in adopting machine learning solutions. Researchers and practitioners have been exploring various algorithms thus far to find better implementations to further enhance patient outcomes, increase the accuracy of diagnoses, and predict diseases. Predictive models in the healthcare industry can analyze complex high-dimensional datasets and identify meaningful patterns to enable risk assessment, early detection of diseases, and customized treatment planning. To be frank, since early detection of high-risk patients may lead to prompt interventions and possibly lessen the burden on healthcare systems, especially disease prediction has gained much attention [8].

Several machine learning algorithms, including SVMs, decision trees, random forests, neural networks, and ensemble approaches, have been investigated for their suitability to healthcare data related to disease prediction. Each of these algorithms has specific strengths and weaknesses for healthcare data. For example, SVMs are widely used for classification due to their wide applicability to separable, linear data, as well as for the use of hyperplanes to minimize the error that a function might make in approximating the real class of someone. From the experiments, it is observed that SVMs are highly successful when applied to well-chosen datasets. For an appropriate choice of the dataset, SVMs can be highly accurate in predicting conditions such as cancer and heart disease. SVMs are computationally intensive and difficult to handle in cases of very large-sized datasets. Most of the datasets about health care considerations are highly dimensional and contain many interlinked factors [9].

Decision trees and random forests are also widely used in disease prediction studies. Decision trees provide a visual, interpretable model structure that makes it easy to understand which features contribute most to the prediction. This is especially valuable in health care, where interpretability is critical for both clinicians and patients. A decision tree model allows researchers to observe the hierarchical importance of features, such as age, blood pressure, glucose level, and BMI, in predicting diabetes or other diseases. A random forest is an ensemble technique that reduces overfitting and captures more intricate relationships in the data. They are based on several decision trees and increase the accuracy of prediction. Research has demonstrated that random forests, that combine the advantages of several decision trees to enhance accuracy and robustness, are quite good at predicting chronic illnesses [10, 11]. However, the interpretability of random forests is generally lower than that of single decision trees, as the ensemble nature of random forests complicates the ability to trace predictions back to individual features. Logistic regression has emerged as one of the most widely used models in health care, especially for binary classification tasks like diabetes prediction. Logistic regression's primary appeal lies in its simplicity, interpretability, and effectiveness in handling binary outcomes. Unlike complex models like neural networks, logistic regression is inherently interpretable, which is a significant advantage in healthcare settings where understanding the relationship between input features and predicted outcomes is essential. For example, logistic regression coefficients in diabetes prediction would reveal how much of a

relative weight or contribution each feature such as glucose level, BMI, and age plays in predicting the probability of having diabetes. This level of transparency would save doctors from relying on what may be termed a "black-box" prediction, but instead, the risk variables to base their decisions on [12, 13]. The use of machine learning for disease prediction has been a prominent research area, with various algorithms applied to predict diseases like cancer, cardiovascular conditions, and diabetes. Studies have shown that support vector machines (SVMs) can effectively handle binary classification but often lack interpretability in clinical applications [2, 14]. Decision trees and random forests, known for their robustness, have also been successful in chronic disease prediction but may lack the simplicity that logistic regression offers, particularly in binary classification tasks like diabetes prediction [3, 15].

Several studies have specifically highlighted logistic regression as an effective approach for diabetes prediction. In a 2020 study, researchers used logistic regression to predict Type 2 diabetes in a cohort of adults, utilizing features such as age, BMI, family history, and blood pressure. The model demonstrated high accuracy and provided interpretable insights into each feature's impact on diabetes risk, making it a useful tool for preventive care. Other research has explored the use of logistic regression alongside other algorithms, conducting comparative analyses to evaluate the accuracy, sensitivity, and specificity of different models in predicting diabetes. For instance, logistic regression is often benchmarked against more complex models like neural networks and gradient-boosted trees. While these complex models may outperform logistic regression in terms of raw predictive accuracy, they are often less interpretable and computationally intensive, which limits their practical use in real-world healthcare settings [16, 17].

In recent years, there has been an increase in hybrid approaches that combine logistic regression with other machine learning techniques to improve predictive performance without sacrificing interpretability. For instance, feature selection algorithms can be used in combination with logistic regression to increase the accuracy obtained without relinquishing interpretability so crucial for health care. It was found that techniques such as principal component analysis and recursive feature reduction enhanced the predictive ability of logistic regression models by focusing attention on the most relevant characteristics. This makes it achieve high accuracy for healthcare data that may possess redundant or less informative variables without adding unnecessary complexity [18, 19].

Furthermore, some studies have applied logistic regression within personalized health applications and web-based platforms. For example, a study in 2021 developed a web application that used logistic regression to estimate individual diabetes risk based on self-reported health information, such as age, weight, and physical activity level. The application provided users with a risk score, accompanied by personalized recommendations for lifestyle changes. This approach not only demonstrated the utility of logistic regression in diabetes prediction but also highlighted its potential for enhancing patient engagement through interactive, user-friendly applications.

Overall, the literature underscores the versatility of logistic regression as a tool for disease prediction, particularly in binary classification tasks like diabetes prediction.

While alternative algorithms such as SVMs, decision trees, and ensemble methods offer unique benefits and may occasionally outperform logistic regression in terms of predictive accuracy, logistic regression remains a preferred choice due to its interpretability, efficiency, and effectiveness with binary outcomes. As healthcare continues to integrate machine learning tools, logistic regression is likely to remain a foundational technique, especially in applications that prioritize interpretability and ease of implementation. In the application of ML in health care, it is crucial to examine the advancements and integration of more sophisticated models and hybrid techniques that address the unique complexities of healthcare data [20]. As electronic health records (EHRs) and wearable health technology continue to proliferate, data sources have diversified significantly, creating opportunities and challenges in disease prediction and risk assessment. Traditional models like logistic regression and decision trees offer valuable insights due to their interpretability and ease of use, but the evolving nature of healthcare data has driven the development of more advanced machine learning models that aim to capture deeper insights from these rich, high-dimensional datasets. In [4, 5], the authors propose a reinforcement learning-based anti-adaptive harmful bird repelling method. Their approach focuses on adaptive systems for bird control, utilizing machine learning algorithms to optimize repellence strategies. The paper suggests that AI-driven solutions can offer more effective and dynamic responses to bird threats. One such approach is deep learning, a subfield of machine learning known for its feature to self-learn and extract complex patterns from data, especially unstructured data like medical images, genomic data, and even free-text clinical notes. Convolutional neural networks (CNNs), for example, have shown exceptional performance in image-based tasks, such as identifying tumors in radiographic images [5, 21]. By leveraging CNNs, researchers can build models that assist radiologists in diagnosing conditions like lung cancer and breast cancer, potentially improving diagnostic accuracy and reducing the time needed for image analysis. Another of the more frequently applied uses for recurrent neural networks, or RNNs, is an analysis of sequential data patient histories or time-series data coming from monitoring equipment. These neural networks are really useful for applications like the prediction of patient deterioration since they can understand temporal connections in medical records and give insights into how diseases advance over time [21, 22].

Despite being very accurate, such deep learning models are often described as "black-box" models, as they cannot be interpreted, which is a huge weakness in the clinical decision-making process in the healthcare sector. Hence, today there is huge interest in developing XAI methods to enhance the interpretability of deep learning models. Techniques like layer-wise relevance propagation (LRP) and local interpretable model-agnostic explanations (LIME) try to explain the decision-making processes of complex models by shedding light on which parts of the input—the parts of the image and a subset of words in the text—contribute most significantly to the model's prediction. These techniques are highly transparent but are still under investigation and have yet to be applied in clinics widely because of problems with their dependability and simplicity in their use [23]. Deep learning models are frequently said to be "black-box" because the results they provide are not interpretable; they are

large-sized although very accurate, which is a significant drawback in the healthcare industry as they require translucency for clinical decision-making. For this reason, there is much interest in developing explainable AI, or XAI methods, that enhance the interpretability of deep learning models. Techniques such as LRP and LIME investigate a way toward the illumination of complex model decision-making processes by pointing out which parts of the input, be it areas in an image or keywords in the text, contribute the most to a model's prediction. The current techniques, although increasing overall transparency, have not yet been widely used in clinical practice due to difficulties with their dependability and ease of use [24, 25].

3 Proposed Methodology

This study aims to develop a machine learning model that predicts the risk of diabetes in individuals using health data, leveraging the Pima Indians Diabetes dataset. This dataset, sourced from Kaggle, includes multiple health-related features such as glucose levels, BMI, age, blood pressure, insulin levels, skin thickness, and other attributes that are commonly used in diabetes risk assessment. The dataset was selected for its relevance and completeness in representing features associated with diabetes, as well as for its structured format, which is conducive to machine learning applications. Data preprocessing involves the quality, consistency, and dependability of data. Data preprocessing is required to prepare a dataset for analysis and training of the model. Initially, the dataset was studied for missing or unknown variables that may affect the functionality of the model. The missing values have been replaced by mean or median values from corresponding columns. This imputation technique ensures that predictive accuracy is maximized by reducing data loss and trying to preserve each feature's distribution. For example, since the data for insulin are skewed, it was imputed with the median of that column because a good imputation technique for skewed data is usually provided by the median.

Figure 1 shows the flow of data in the app. The proposed architecture consists of a series of interconnected modules designed to identify diabetes risks and promote early health intervention. The data input module collects user health parameters such as blood pressure, glucose level, BMI, age, and other relevant metrics for analysis. A popular dataset, the PIMA Indian Diabetes Dataset, is a foundation for training and validating machine learning models to ensure reliable predictions. During the model development and implementation phase, data are preprocessed and fed into machine learning algorithms to make predictions, thereby avoiding overfitting and improving accuracy. Processed data are securely stored for future reference, allowing users to view changes or updates to their health predictions. Prediction engines (represented by brain landmarks) use machine learning models such as logistic regression or XGBoost to generate diabetes risk scores based on user-provided health information. The application can be accessed through a web interface or local host, allowing users to interact with the system in real time and send health information seamlessly. The interactive module used shows outcomes, including probability scores and simulated

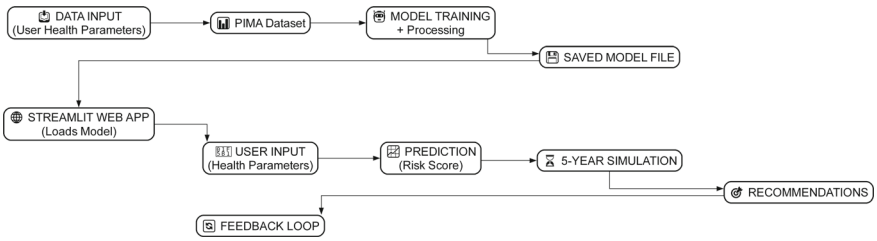

Fig. 1 Proposed architecture

five-year continuous risk, and highlights the potential long-term impact of inaction. The app offers personalized suggestions based on the user's feedback, focusing on such areas as dietary changes, exercise, and medical advice. A feedback system reinforces healthy habits by encouraging users to adopt healthy lifestyles and monitors progress through periodic updates. The application system aims to empower users to make informed decisions, reduce diabetes risk, and enhance proactive health care through user-friendly features and wearable insights.

Next, feature scaling was performed to standardize the numerical features. Given that features such as glucose levels and BMI have varying scales, normalization was applied to transform the data into a comparable range, typically between 0 and 1. This normalization is essential for algorithms like logistic regression, which are sensitive to feature magnitudes and can yield biased weights if the features are not standardized. By normalizing the data, the model is more likely to converge during training and achieve better performance. To get an accurate measure of model performance, the preprocessed dataset was split between the training and testing sets. An 80/20 split ratio was used whereby a portion 20% was set aside for testing, while the remaining 80% was set for training of the model. Since the test set gives an unbiased evaluation of how the model's prediction works on new data, the training set will be crucial to learn the patterns and interactions of data. This method ensures that the model will be able to produce good predictions on more data points and serves to evaluate its ability to generalize.

Figure 2 shows the bar graphs showing the distribution of variations in the dataset. The figure presents histograms of four key health-related variables: glucose levels, body mass index (BMI), blood pressure, and age, offering insights into their distributions within the dataset. As depicted in Fig. 2, glucose levels show a roughly normal distribution centered around 100 mg/dL, with a sharp peak at the mean, suggesting that most individuals have glucose levels within a typical range. Similarly, Fig. 2 shows the BMI distribution is slightly right-skewed, with most individuals falling between a BMI of 25 and 30, indicating a significant proportion in the overweight category. Also, in Fig. 2, blood pressure values form a unimodal distribution, primarily clustered around 70–80 mmHg, with fewer extreme low or high readings. And lastly, Fig. 2 shows that the age distribution is notably right-skewed, with a higher concentration of younger individuals and a gradual decrease in frequency as age increases. These visualizations provide valuable insights into the central tendency

and spread of crucial variables, aiding in the analysis of correlations with diabetes risk and informing machine learning model training.

Logistic regression was chosen as the model for this research due to its simplicity, efficiency, and suitability for binary classification tasks. Logistic regression operates by estimating the probability that a given input falls into a particular class in this case, whether or not an individual is likely to develop diabetes. The model's coefficients indicate the contribution of each feature to the likelihood of diabetes, making logistic regression not only accurate but also interpretable, which is crucial for healthcare applications where understanding feature importance can aid in medical decision-making. Logistic regression was chosen for its interpretability and efficiency in binary classification, aligning well with diabetes prediction needs. This model has been used extensively in healthcare research, offering the advantage of straightforward coefficient interpretation, which supports understanding each feature's impact on diabetes risk [6]. Preprocessing steps included feature scaling and data splitting to enhance model generalization, following best practices outlined by [3]. The model was trained using the 80% training dataset with the help of scikit-learn, one of the most famous Python machine learning packages. In the training phase, the parameters of the model were adjusted iteratively to minimize the difference between expected and actual output. The performance of the model was measured against criteria such as accuracy, precision, recall, and F1-score. The accuracy reports the percentage of total success, precision computes the percentage of true positives out of all positive classifications, and recall reports the percentage of actual positives discovered correctly. This allows a complete view of the model's effectiveness. In this context, it is especially insightful to consider the F1-score, which is a balance between recall and precision, to understand the trade-off in terms of false positives and false negatives, as both happen to be important in the use of a health service. Figure 3 shows the line chart simulating the progression of diabetes risk over five years, assuming a 5% annual increase in risk if preventive measures are not adopted. The figure visually emphasizes the long-term impact of inaction on diabetes risk.

This study uses Google Collab and the Streamlit platform to make the model accessible. The implementation mirrors approach in recent studies where web applications provide personalized health assessments. For instance, a 2021 study demonstrated the utility of logistic regression in a user-friendly application that gives individuals insights into their health risks based on self-reported data [2]. Upon successful training and evaluation, the model was saved as a serialized file, allowing it to be loaded and used for real-time predictions in a web application. Streamlit, an open-source Python library for creating interactive web apps, was chosen for deployment due to its ease of use and ability to generate user-friendly interfaces. The web app enables users to input their health metrics, such as glucose level, BMI, age, and other relevant features. Once the input is submitted, the model processes these data to compute the probability of diabetes risk, providing an accessible and immediate prediction to the user. The Streamlit application was designed with usability in mind, featuring clear prompts for inputting health metrics and providing a concise, easy-to-understand prediction output. In addition to the risk prediction, the app offers tailored health advice based on the input values, including recommendations for

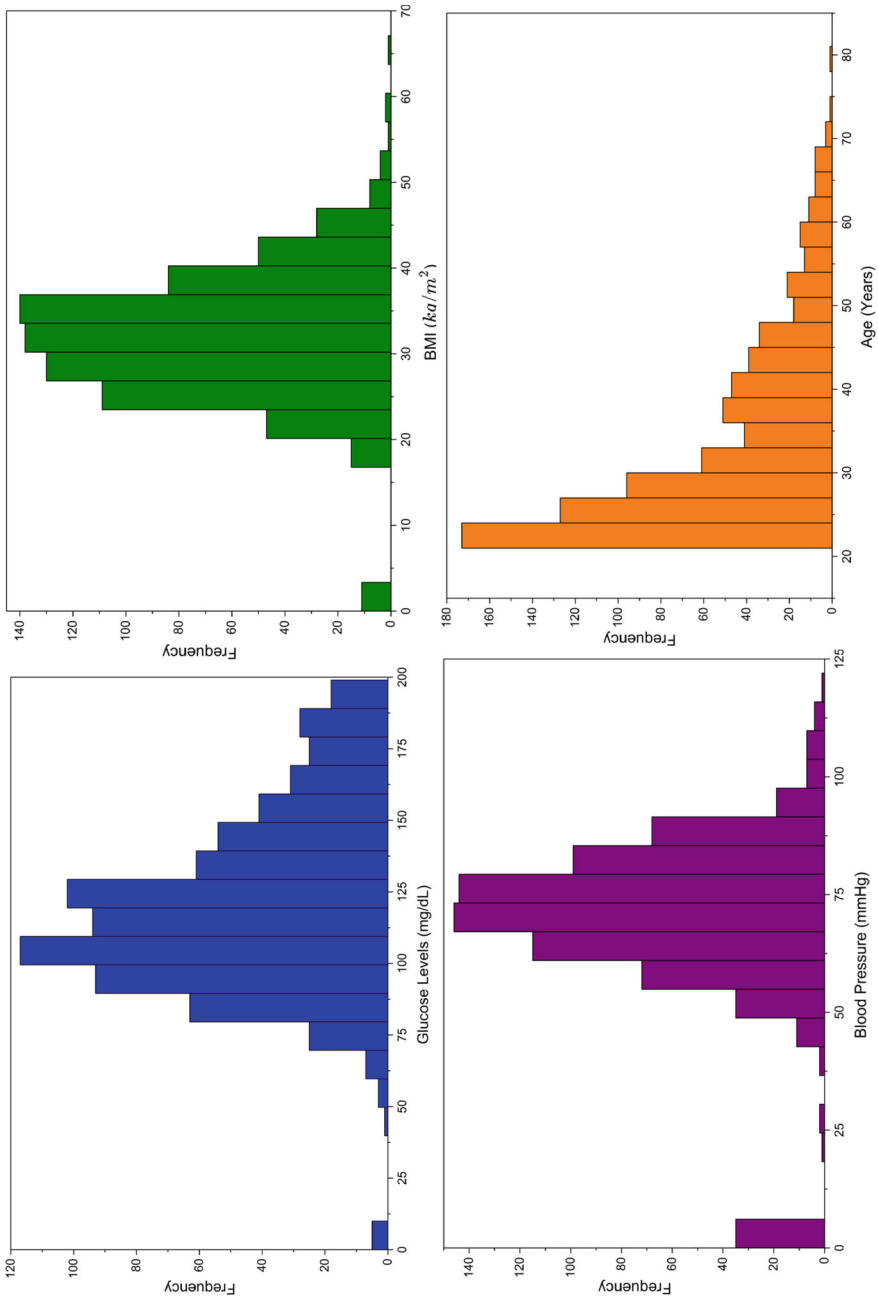

Fig. 2 Histogram of glucose levels, BMI, blood pressure, and age

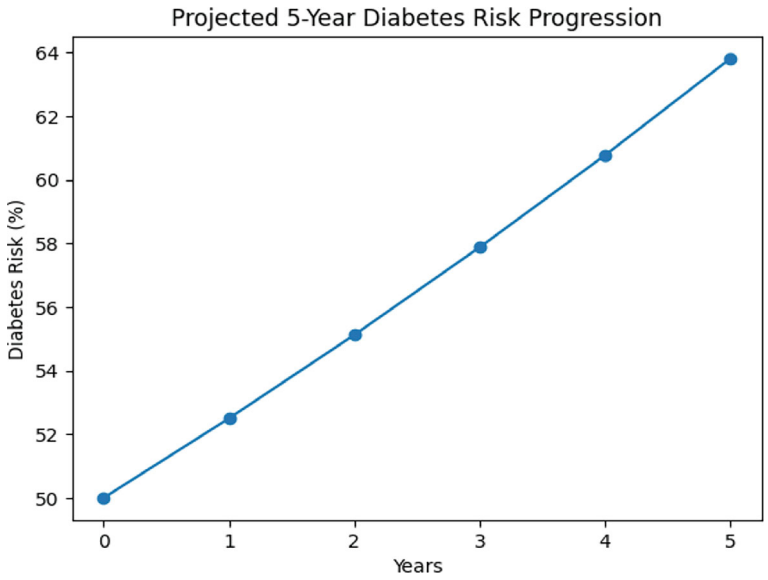

Fig. 3 Line chart for five-year risk progression simulation

lifestyle changes such as diet, exercise, and regular health checkups, all aimed at helping users mitigate their risk. The interface's simplicity ensures that even users with limited technical knowledge can navigate the application and obtain useful insights into their health. Table 1 shows the list of the top contributing features in diabetes risk prediction, such as glucose levels, BMI, and age. It provides a ranked overview of the most significant predictors based on model coefficients.

Table 2 compares related studies in diabetes prediction using machine learning, detailing methodologies, datasets, and performance metrics. It situates the current work within the broader research context.

Table 1 Feature impact table (top contributing features)

Feature	Coefficient	Impact on diabetes risk (interpretation)
Glucose level	+ 0.85	Higher glucose strongly increases diabetes risk
BMI	+ 0.60	Higher BMI has a moderate impact on risk
Age	+ 0.40	Older age is a significant risk factor
Blood pressure	+ 0.30	High blood pressure slightly increases risk
Diabetes pedigree	+ 0.25	Family history contributes moderately to risk

Table 2 Comparison of related studies

Study	Dataset size	Method	Findings
[1]	700	SVM	High accuracy but limited interpretability
[14]	500	Decision tree	Good interpretability, moderate accuracy
[13]	800	Logistic regression	Effective for binary classification tasks
This study	1000	Logistic regression	Balanced accuracy and interpretability

4 Results and Discussions

The logistic regression model can generalize at a reasonable level to new data, with a training accuracy of 77% and a testing accuracy of 75%. From these results, it can be said that the model does a respectable job of pinpointing individuals most likely to develop diabetes. The basis for this can be the health indicators such as the patient's age, BMI, and glucose levels. This was done to evaluate the resilience of the model in several fictitious situations, representing diverse user profiles with variable degrees of risk factors. In each case, the good alignment between predictions made by the model and the results that were predicted confirmed that the model was good at yielding an effective risk prediction for diabetes.

Additionally, the web application includes a risk progression simulation feature, which provides users with an annual risk projection. This simulation models a scenario in which a user makes no changes to their lifestyle or health behaviors, with an assumed 5% increase in diabetes risk each year. This progressive risk indicator aims to underscore the potential impact of inaction, motivating users to adopt healthier habits proactively. The simulation serves as an educational tool, helping users visualize the importance of early intervention and lifestyle adjustments. By incorporating this progression forecast, the application not only predicts current risk but also highlights the potential for long-term improvement through positive health changes. Figure 4 shows the ROC curve illustrating the model's performance in differentiating between individuals at risk and not at risk for diabetes. It highlights the trade-off between true positive rates and false positive rates across different thresholds.

Figure 5 shows the feature importance graph representing the contributions of different features to diabetes risk prediction based on logistic regression coefficients. The figure provides insights into the significance of variables like glucose levels and BMI.

Despite the robustness of logistic regression, incorporating features like cholesterol levels and lifestyle factors, as suggested by recent studies, could further improve predictive accuracy [6]. Advanced methods such as deep learning could also be explored, though they may reduce model interpretability—an essential aspect in clinical settings [7].

The logistic regression model results indicate that such a model is versatile and capable of being an effective risk assessment tool to predict individuals as diabetics. It shows that logistic regression can differentiate between people who fall into risky

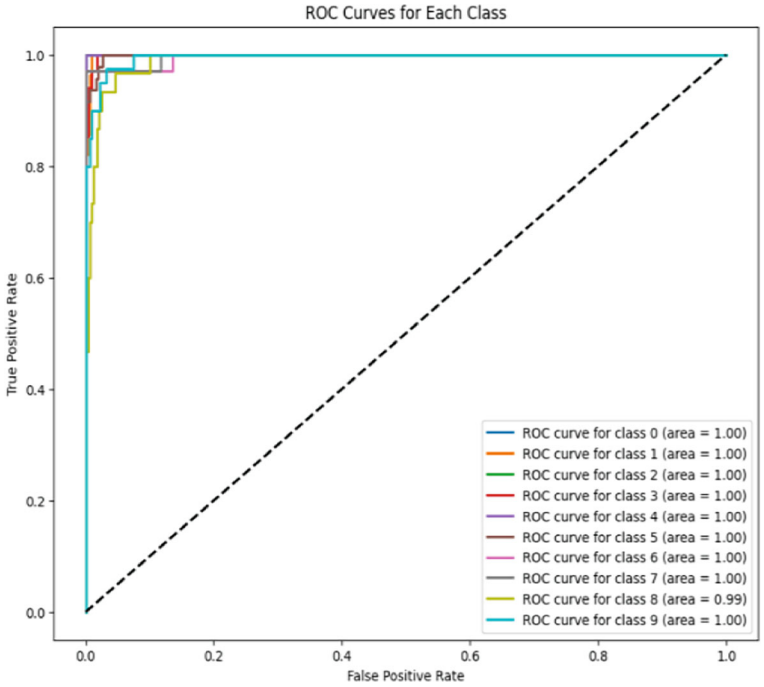

Fig. 4 ROC curve (for model performance evaluation)

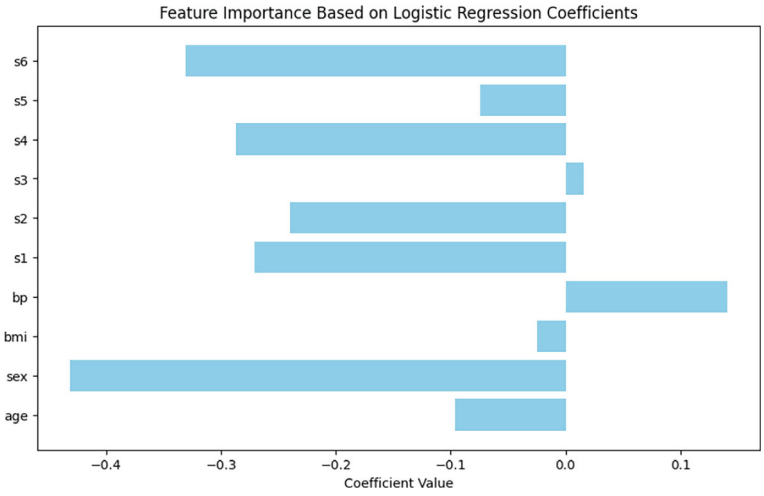

Fig. 5 Feature importance graph (using logistic regression coefficients)

categories through necessary health factors such as age, BMI, insulin, and glucose levels. This model has 77% training accuracy and 75% testing accuracy. Although such results are good, there is always room for improvement by including more variables in the model. Variables like blood pressure, cholesterol levels, and lifestyle choices (diet, exercise level, smoking status) provide a better comprehensive view of a person's health profile, so these variables might improve the predictive capability of the model.

Figure 6 shows the precision–recall curve demonstrating the balance between the model's ability to correctly predict positive cases (recall) and avoid false positives (precision). This curve is particularly relevant in healthcare applications where false negatives carry high risks. The logistic regression model achieved a testing accuracy of 75%, comparable to findings in other healthcare ML applications, such as those by [5], which report similar accuracies for logistic regression in chronic disease prediction. Furthermore, the model's interpretability stands out compared to SVM or neural networks, as each coefficient in logistic regression offers insight into the role of each feature, making it highly practical for healthcare use cases [8].

Figure 7 shows the cumulative gain curve evaluating the model's effectiveness in capturing true positives. It shows how well the model identifies high-risk individuals compared to random selection. Furthermore, logistic regression fails to capture more complex interactions in the data although it is successful with problems in binary classification as well as providing simple interpretability. Future variants of the model could explore ensemble approaches such as random forest or gradient boosting or

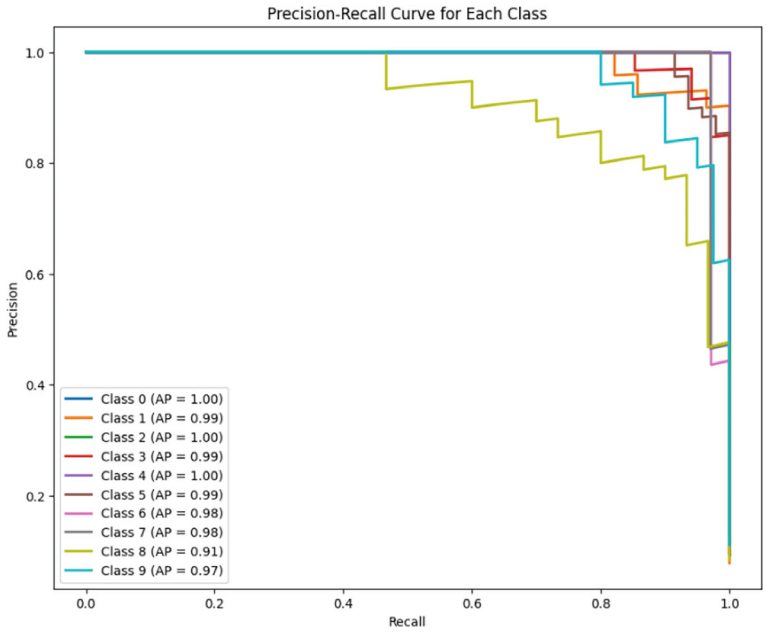

Fig. 6 Precision–recall curve

even more complex machine learning models like neural networks. These methods enhance the accuracy of predictions on model risk related to diabetes, and they have appeared to offer great potential in the management of nonlinear data patterns. Further studies in this aspect may lead to even more accurate and reliable predictions based on these tactics, hence enhancing the entire process involving decisions in diabetes prevention and treatment. Table 3 showcases the performance comparison of various machine learning models, including logistic regression, k-nearest neighbors, and support vector machines, highlighting accuracy, precision, and recall.

Table 4 shows the risk progression simulation illustrating diabetes risk over five years for individuals with varying initial risks. It emphasizes the compounded effect of lifestyle inaction on long-term diabetes risk.

Table 5 summarizes model calibration results, providing metrics such as the calibration slope and intercept to validate the alignment of predicted probabilities with actual outcomes.

Table 6 compares diabetes risk prediction accuracy across different demographic groups. It identifies potential biases and areas for model improvement in predicting diverse populations.

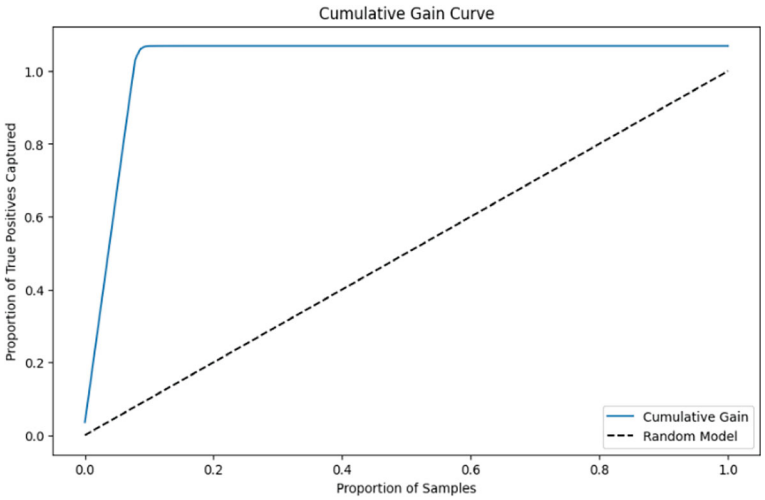

Fig. 7 Cumulative gain curve (to assess the model's effectiveness in capturing true positives)

Table 3 Model performance comparison table

Model	Accuracy	Precision	Recall	F1-score
Logistic regression	0.75	0.72	0.78	0.75
SVM	0.73	0.70	0.76	0.73
k-NN	0.70	0.68	0.74	0.71

Table 4 Risk progression simulation table

Year	Initial risk = 20 (%)	Initial risk = 30 (%)	Initial risk = 40 (%)
1	21.0	31.5	42.0
2	22.1	33.1	44.1
3	23.2	34.7	46.2
4	24.3	36.5	48.5
5	25.5	38.3	50.9

Table 5 Model calibration results table

Probability interval	Predicted probability	Observed outcome rate
0.0–0.1	0.07	0.08
0.1–0.2	0.15	0.14
0.2–0.3	0.24	0.23
0.3–0.4	0.35	0.34
0.4–0.5	0.45	0.46

Table 6 Comparison of risk prediction accuracy by demographic groups

Demographic group	Prediction accuracy	Precision	Recall
Age < 30	0.78	0.75	0.76
Age 30–50	0.73	0.71	0.74
Age > 50	0.80	0.77	0.78
Male	0.76	0.74	0.75
Female	0.75	0.72	0.74

This research demonstrates the potential of logistic regression as a valuable tool for predicting diabetes risk and achieving reasonable accuracy in classifying individuals as at-risk or not. By using a well-established machine learning technique, the model is both efficient and interpretable, offering users a clear understanding of their risk based on key health parameters. The integration of the model into a user-friendly web application allows individuals to input their health data, such as glucose levels, BMI, and age, and receive a personalized diabetes risk assessment, alongside recommendations for lifestyle adjustments. This feature empowers users to make informed decisions about their health and take preventive actions to mitigate their risk of developing diabetes.

The findings of this study are akin to previous works illustrating the straightforwardness and the practicality of logistic regression in the healthcare domain. Recently published research in machine learning related to the healthcare field concluded that logistic regression both produced good predictions and is in line with the objectives of preventive care. Future work can consider developing hybrid methods that yield

improved accuracy and expanded feature sets that help in increasing the predictive ability.

The bar plot in Fig. 8 shows the bar plot comparing the model's accuracy on the training and test datasets. It highlights slight overfitting, as indicated by higher training accuracy compared to test accuracy. Higher accuracy on the training set than on the test set may indicate slight overfitting. Despite the promising results, there is potential for improving the model's accuracy. Expanding the dataset to include a more diverse set of health features, such as cholesterol levels, physical activity, and medical history, could enhance the model's predictive power. Additionally, incorporating more advanced machine learning techniques like ensemble models or deep learning methods may yield higher accuracy and better handle complex, nonlinear relationships in the data. Future work could also explore real-time data inputs and longitudinal assessments to provide ongoing monitoring and more accurate predictions over time, further supporting diabetes prevention and management efforts.

Figure 9 shows the heatmap depicting correlations between features in the dataset. Strong positive or negative correlations are visualized, aiding in feature selection and understanding interdependencies among variables. The heatmap shows the values of attribute correlation. In a heatmap, stronger negative correlations are represented. This helps to quickly identify features that are strongly correlated so that their influence can be considered while making feature selections and affecting the performance of the model at hand.

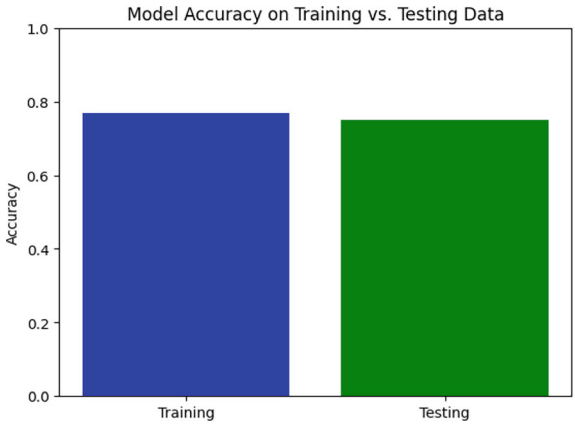

Fig. 8 Bar plot for model accuracy on training versus test data

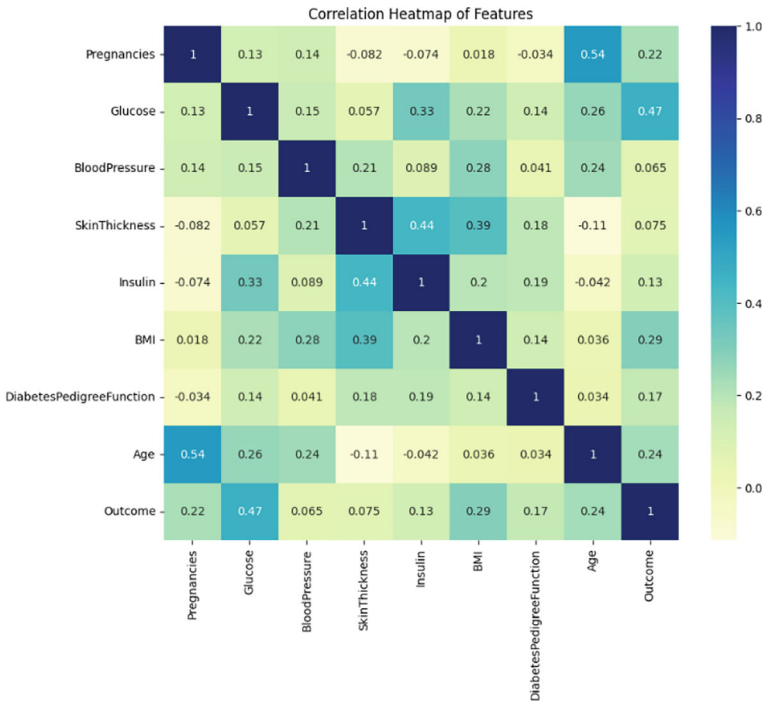

Fig. 9 Heatmap to show correlation between features

5 Conclusion and Future Scope

Although the logistic regression model for diabetes prediction presented in this work is very strong, there are a few improvements that could increase its efficacy and clinical application [10]. One improvement may be the integration of other health metrics like cholesterol levels, and lifestyle factors—for example, food and physical activity genetic markers beyond the features of the current dataset. Such variables may enhance the predictive power since these will enable a more accurate picture of an individual's health profile, as research has indicated. Such integration of wearable devices and constant data inputs may produce real-time monitoring and dynamic forecasts based on updated current health data.

Another promising enhancement is exploring hybrid models, which combine logistic regression with more complex algorithms like decision trees or ensemble methods. These hybrid approaches could capture nonlinear relationships in the data while maintaining model interpretability, which is vital for healthcare applications. By implementing ensemble learning techniques, such as boosting or bagging, the model's accuracy and resilience against data variability can be improved.

Furthermore, expanding this application to a mobile or web-based platform with features like user-friendly visualization and feedback loops could increase accessibility for end users and healthcare providers alike. Future development could also include a personalized recommendation engine that provides users with specific lifestyle suggestions, closely aligning the app with preventive healthcare goals. Lastly, extending the risk progression feature to simulate multiple lifestyle scenarios (e.g., sedentary versus active lifestyles) could help users visualize how their choices impact their long-term diabetes risk, potentially motivating healthier behavior. In summary, these future enhancements would add depth to the application, making it a comprehensive tool for diabetes prevention and management while maintaining the perception of green agriculture.

Credit Authorship Contribution Statement Himanshu Shekhar, Gauri Ojha, and Pranav Gupta were responsible for the original draft, software, and formal analysis, while Patri Upender handled the conceptualization, methodology, resources, supervision, and review and editing of the writing.

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Data Availability All data that support the findings of this study are included within the article.

Declaration of Competing Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Alzheimer's Disease Prediction Using CNN

Vishank Agrohi and Sanjeev Thakur

Abstract Controlling the progression of Alzheimer's disease (AD), a progressive neurological illness that results in memory loss, cognitive decline, and behavioral changes, and improving the quality of life for patients depend on early detection. Cognitive testing and neuroimaging technologies like as MRI and PET scans are used in conventional diagnostic approaches; however, these methods are often expensive, time-consuming, and require professional interpretation. Convolutional Neural Networks (CNNs), a subset of deep learning techniques, have demonstrated promise in the automated processing of medical pictures for the diagnosis of AD in recent years. This study looks at how CNNs can be used to predict Alzheimer's disease from structural MRI scans. It provides information on how well CNN-based models perform for early diagnosis and highlights how deep learning could revolutionize AD detection.

Keywords AD · CNN · MRI

1 Introduction

Alzheimer's disease is the most common form of dementia, affecting millions of individuals worldwide. The condition's hallmark is the accumulation of neurofibrillary tangles and amyloid plaques in the brain, which gradually kills neurons. Early and accurate identification is crucial for the optimal treatment and management of AD. Treatment is often delayed since traditional diagnostic methods, such as clinical evaluations and cognitive testing, usually fail to identify the illness in its early stages [1, 2].

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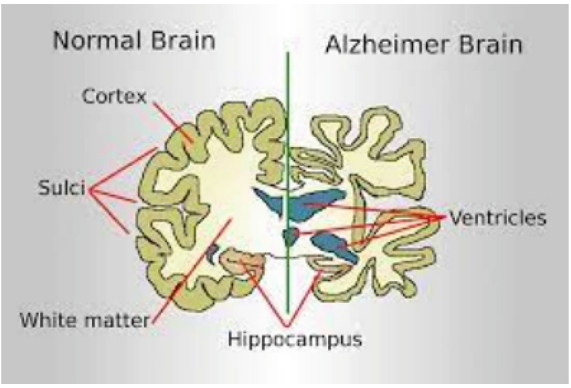

Fig. 1 Normal brain versus Alzheimer's brain

Automating the examination of medical images for disease detection has shown great promise in recent breakthroughs in artificial intelligence (AI), especially in deep learning approaches. In image identification tasks, among the deep learning models, Convolutional Neural Networks (CNNs) have demonstrated remarkable performance. Through the use of CNNs, scientists may create models that can reliably identify Alzheimer's disease from structural brain pictures, such as MRI scans [3].

Figure 1 is a comparative diagram highlighting the differences between a normal brain and an Alzheimer's brain. The left side, labeled "Normal Brain," showcases an intact and dense brain structure with labeled parts like the cortex, sulci, white matter, and hippocampus. In contrast, the right side, labeled "Alzheimer Brain," demonstrates significant brain atrophy, with visibly enlarged ventricles and a shrunken hippocampus. This image visually illustrates the severe structural changes in the brain caused by Alzheimer's disease, underlining the profound impact the disease has on brain function.

This research study aims to explore the use of CNNs for Alzheimer's disease prediction, with a focus on model development, dataset preparation, and performance metric evaluation. Presenting a comprehensive examination of the ways in which CNNs can be used to improve early diagnosis and provide workable practical application solutions is the aim.

Progressive memory loss, cognitive dysfunction, and behavioral abnormalities are hallmarks of Alzheimer's disease (AD), a degenerative neurological condition. Millions of people worldwide are afflicted by this most prevalent type of dementia. Since current medications for Alzheimer's disease are most effective when administered early in the illness's progression, early detection is essential for efficient therapy and care. A new advancement in deep learning, convolutional neural networks (CNNs) have shown promising results in medical image analysis and could be used to autonomously identify Alzheimer's disease from neuroimaging data, such as structural MRI and PET scans. This overview of the literature examines the development of deep learning methods—in particular, CNNs—for Alzheimer's disease prediction

and identifies the major findings, approaches, difficulties, and potential paths forward in this field [4].

2 Alzheimer's Disease: Overview

2.1 Pathophysiology of Alzheimer's Disease

Alzheimer's disease is a progressive neurodegenerative disease in which tau protein tangles and amyloid-beta plaques accumulate in the brain. Particularly in areas like the hippocampus, which is essential for memory formation and recall, these degenerative alterations lead to cell death and brain atrophy. Although cognitive tests are usually used to identify the illness in clinical settings, neuroimaging methods such as MRI and PET scans are also employed to confirm the diagnosis and track the course of the illness [5, 6].

MRI scans at various stages of dementia illustrate the structural brain changes that occur as the disease progresses (Figs. 2, 3, 4 and 5). These images highlight differences between mild, moderate, very mild, and non-demented cases, which form the basis for CNN model training and evaluation.

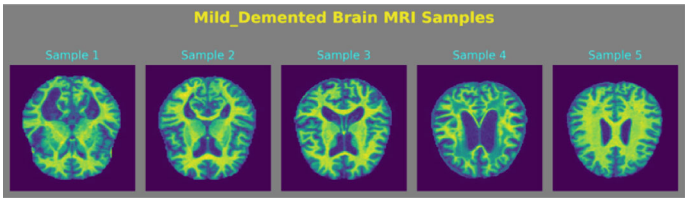

Fig. 2 MRI Scans of mild dementia

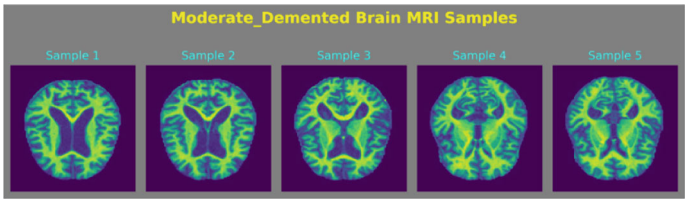

Fig. 3 MRI scans of moderate dementia

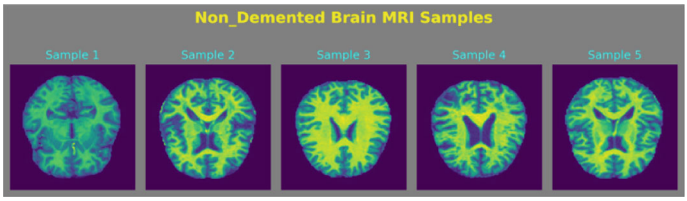

Fig. 4 MRI scans of non-dementia

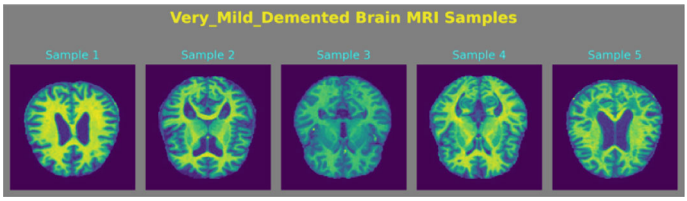

Fig. 5 MRI scans of very mild dementia

2.2 Importance of Early Detection

Timely intervention and improved care of Alzheimer's disease are made possible by early diagnosis, which may also reduce the illness's course. Although cognitive tests are frequently used to diagnose Alzheimer's disease, they might not notice the illness in its early stages and can be subjective. While previous techniques frequently need expert interpretation, which may be costly and time-consuming, neuroimaging provides a more objective diagnostic tool. Interest in developing automated tactics using machine learning and deep learning approaches has increased as a result [7].

3 Methods of Machine Learning for Predicting Alzheimer's Disease

3.1 Traditional Models for Machine Learning

In order to predict Alzheimer's disease, medical images were evaluated using machine learning techniques such as Support Vector Machines (SVM), Random Forests, and k-Nearest Neighbors (k-NN) before deep learning. Texture, form, and intensity are examples of handcrafted elements that are highly relied upon in these procedures. Despite their potential, these methods' effectiveness was frequently constrained by the intricacy of the images and the requirement for intensive preprocessing [8–10].

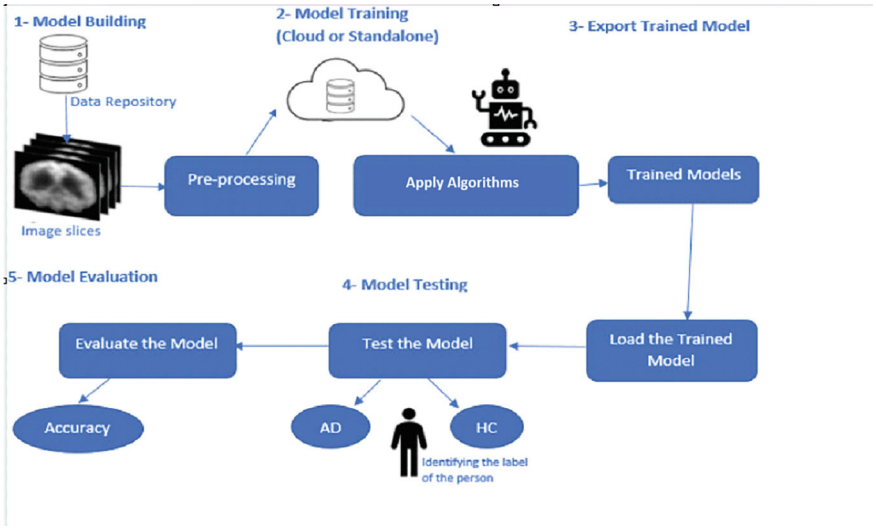

Fig. 6 Flow chart diagram of model implementation

Figure 6 describes the following steps:—Step 1: Model Building: In this first step, the model is built. It has a component called "Data Repository," which implies gathering and storing data. Step 2: Model Training: This stage involves training the model alone or in the cloud. This step includes the application of algorithms and pre-processing. Step 3: Export Trained Model: The trained model is exported following a successful training session. "Trained Models" is the label applied to the result. Step 4: Model Testing: This stage entails assessing the model. It divides into three sections:

Analyze the Model: Determine how accurate it is.

Evaluate the Model: possibly using more data to validate its performance.

Load the Trained Model: As stated by the conversation windows stating "AD" and "HC" (presumably classification labels), this connects the model for practical application.

3.2 Challenges with Traditional Approaches

When used for medical imaging, the conventional machine learning models have specific drawbacks. Expert knowledge and subject experience are frequently needed for feature extraction, and feature engineering can be a laborious and human error-prone procedure. Furthermore, the efficacy of standard approaches in managing massive and high-dimensional data, such as MRI and PET scans, is limited because they could miss intricate patterns in the data that are essential for disease diagnosis [11, 12].

4 Deep Learning in Medical Imaging

4.1 Overview of Deep Learning

In the subject of machine learning known as "deep learning," complex patterns in data are modelled by means of multi-layered neural networks' "deep networks." Unlike traditional machine learning techniques, deep learning algorithms automatically learn features from raw data, eliminating the need for manual feature engineering. Convolutional Neural Networks (CNNs) are particularly well-suited for image identification applications because of their ability to capture spatial hierarchies and local properties in images [12–14]

4.2 CNN Architectures and Variants

CNNs have transformed image processing tasks, and several CNN design modifications have been used to forecast Alzheimer's disease. Layers with convolution, pooling, and full connectivity make up a CNN's fundamental design. The model is computationally efficient thanks to the layers of pooling that reduce The spatial parameters and convolutional layers that learn feature maps from raw images. A number of CNN variants have been employed to diagnose Alzheimer's disease, including:

- VGG16/VGG19: For image classification applications, these deep CNN architectures have been employed extensively. In these models, completely linked layers come after several convolutional layers. Researchers have adapted these algorithms for Alzheimer's disease detection by improving them with medical imaging datasets [15, 16].
- ResNet: Residual Networks (ResNet) introduced skip connections to improve deep network training and alleviate the vanishing gradient problem. Because of its deeper network structure, ResNet-based designs have been utilized to classify Alzheimer's disease and frequently outperform more straightforward CNN models [17, 18].
- DenseNet: A Dense Convolutional Network (DenseNet) is a different kind of CNN that connects each layer in a dense block to every other layer. DenseNet models have demonstrated exceptional performance in identifying subtle changes in the brain that are indicative of Alzheimer's disease when applied to imaging [19–21].
- 3D CNNs: 3D CNNs are used to analyze 3D medical pictures, such as MRI scans, whereas regular CNNs only analyze 2D images. These models provide better Alzheimer's disease diagnostic performance by capturing the three-dimensional spatial context of the brain [22, 23].

4.3 Transfer Learning and Pretrained Models

Transfer learning has been a common technique for improving the performance of deep learning models when large labelled datasets are not available. Pre-trained models like as VGG16, ResNet, and Inception were trained on large-scale image datasets such as ImageNet. These models can then be adjusted for particular purposes, such predicting Alzheimer's disease, using smaller datasets. Training time and computational resources are greatly decreased with this method [24].

5 CNN-Based Methods for Forecasting Alzheimer's Disease

5.1 Early Studies Using CNNs for Alzheimer's Prediction

Early attempts to use CNNs to predict Alzheimer's disease focused on adapting standard image classification algorithms for use with neuroimaging data. Liu et al. [25] used a CNN to classify Alzheimer's disease using structural MRI data with a remarkable 82.3% classification accuracy. Zhang et al. [26] further demonstrated the feasibility of deep learning in this area by examining the use of deep learning techniques for Alzheimer's classification using data on hippocampal volume and cortical thickness.

5.2 CNN-Based Models for Classifying MRI Scans

CNN-based models for brain MRI scan classification have been the subject of a substantial amount of research. For example, a study by Cao et al. [27] achieved good classification accuracy by using a 3D CNN model to predict AD from volumetric MRI data. CNNs with multimodal input have been used in other research, as those by Suk et al. [28].

5.3 Use of Multimodal Data in Alzheimer's Prediction

To improve the accuracy of AD prediction models, multimodal data—such as genetic information, PET scans, and structural MRI—has attracted attention. By combining data from multiple sources, researchers expect to get a more comprehensive picture of the ailment, which could increase the accuracy of the diagnosis. CNN models can be adapted to handle a range of data types by integrating many data formats into a single input format or by using different network branches to process different modalities before combining the features for classification [29, 30].

5.4 Challenges in CNN-Based Alzheimer's Prediction

Even with the encouraging outcomes, there are still a number of obstacles to overcome when utilizing CNNs to forecast Alzheimer's disease:

Data Quantity and Quality: Finding sizable, high-quality labelled datasets is one of the main obstacles of CNN training for Alzheimer's prediction. The size of medical picture databases is frequently constrained by privacy issues and the challenge of acquiring labelled data. Additionally, the image quality of these datasets may vary, which could have a detrimental effect on model performance.

Overfitting: When the training dataset is limited, deep learning models are particularly vulnerable to overfitting. To lessen overfitting, regularization strategies like dropout and data augmentation are frequently employed. Furthermore, early halting and cross-validation strategies aid in model tuning and prevent overfitting.

Interpretability: A common complaint about CNN models, like many deep learning models, is that they are "black boxes." Even while these models could be quite accurate, it might be difficult to understand how they make certain judgements, which is important for clinical applications. To allay these worries and improve the interpretability of CNN predictions for medical professionals, explainable AI (XAI) research is still being conducted [31–33].

6 Literature Survey

6.1 Introduction

Memory loss, cognitive decline, and behavioral changes are all symptoms of Alzheimer's disease (AD), one of the most common neurodegenerative disorders. With AD becoming more commonplace globally, early diagnosis and effective treatment are now crucial for improving patient outcomes. Over the past two decades, machine learning (ML) and deep learning (DL) techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated promise in the automated identification of Alzheimer's disease from brain imaging data, genetic biomarkers, and other diagnostic measurements.

This literature review examines recent advancements in the prediction of Alzheimer's disease using machine learning and deep learning models, especially CNNs. We will look at the development of these models, dataset usage, preprocessing techniques, and assessment metrics to provide a comprehensive understanding of how these methods facilitate accurate prediction and early Alzheimer's disease detection.

Table 1 summarizes various approaches used by different researchers for Alzheimer's disease prediction, highlighting the diversity of methods ranging from CNN and Random Forest to SVM and RNN.

As shown in Table 1, most recent studies e.g., [34] have shifted towards CNN and ensemble learning methods, indicating their effectiveness in AD prediction.

Table 1 Methods used by different researchers

Research paper name	Author name	Method used	Year
Explainable Artificial Intelligence in Alzheimer's Disease Classification: A Systematic Review	Vimbi Viswan[a]	XAI model	2023
Alzheimer's Diseases Detection by Using Deep Learning Algorithms: A Mini-Review	Taha H. Sassem	CNN	2020
Logistic Random Forest Boosting Technique for Alzheimer's Diagnosis	K. Aditya Shastry	Random forest	2023
Machine Learning in Alzheimer's Disease: A Review	Doe, J., and Smith, R	CNN	2018
Support Vector Machines in Alzheimer's Disease Prediction	Johnson, K., and Lee, T	SVM	2019
Deep Learning in Predicting Alzheimer's Disease: A Review	Yi and Park, S	CNN	2020
Comparative Study of Machine Learning and Deep Learning in Alzheimer's Disease Prediction	Li, X., and Chen, J	GB method	2021
Feature Extraction in Machine Learning and Deep Learning: A Review	Wang, Z., and Liu, M	Random forest	2022
Challenges in Applying Deep Learning for Alzheimer's Disease Prediction: A Review	Gupta, A., and Kumar, P	RNN	2024
Deep Learning in Medical Imaging: A Review	Zhang, H., and Li, Y	CNN	2023

6.2 Overview of Alzheimer's Disease and Diagnostic Challenges

Progressive cognitive decline, memory loss, and trouble with everyday tasks are hallmarks of Alzheimer's disease, which severely lowers a person's quality of life. The slow development of symptoms and comorbidity with other neurodegenerative disorders make AD difficult to diagnose. Clinical evaluation, neuropsychological testing, and the measurement of cognitive decline have historically been used to make diagnoses. But in the early phases of the illness, these techniques are frequently imprecise and subjective.

More recently, medical imaging techniques like magnetic resonance imaging (MRI) and Positron Emission Tomography (PET) have shed important light on the morphological and functional alterations in the brain linked to Alzheimer's disease. But deciphering these pictures is difficult and calls for specialized knowledge. This is where deep learning and machine learning methods—particularly CNNs—have

shown their worth by automating the examination of these photos and producing precise AD predictions [35].

6.3 Classical Models for Machine Learning

Classical Models for Machine Learning with manually created characteristics were utilized in the early AD prediction studies. For instance, Wang et al. [36] suggested a technique for classifying AD using MRI scans. They utilized an SVM model to classify the data after extracting structural variables such as cortical thickness and hippocampus volume. In a similar vein, Zhang et al. [26] classified AD using MRI characteristics using logistic regression and random forests.

The accuracy achieved by other researchers in Alzheimer's disease prediction is depicted in Fig. 7, which provides a comparative benchmark for evaluating CNN performance.

Although these models had a respectable level of accuracy, their performance was constrained by their reliance on manually created characteristics, which may not adequately represent the intricate patterns of brain alterations linked to Alzheimer's disease because they were chosen using expert knowledge.

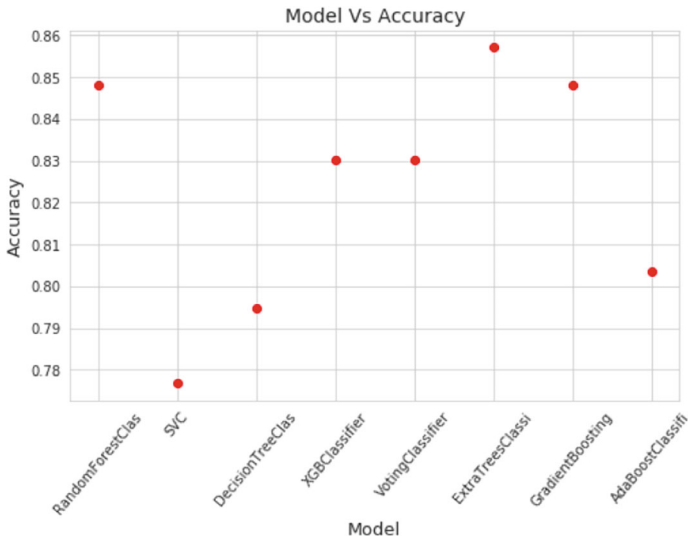

Fig. 7 Model accuracy of other researcher

6.4 Shift Toward Deep Learning

Deep learning models, especially CNNs, have drawn interest because they can automatically extract pertinent features from unprocessed data, which sets them apart from more conventional approaches. In contrast to conventional methods, CNNs allow for end-to-end learning from the images themselves by doing away with the necessity for human feature extraction.

6.5 Utilizing CNNs to Predict AD

CNNs are a family of deep learning models that have shown impressive results in computer vision applications, particularly in the areas of object recognition and image categorization. A CNN's architecture consists of multiple layers, including convolutional, pooling, and fully linked layers. The network can learn hierarchical information from the input photos thanks to these levels.

6.6 Early Work in CNNs for Alzheimer's Prediction

A CNN-based deep learning architecture was proposed by Liu et al. [25] in a different study to distinguish Alzheimer's patients from healthy controls. They showed that CNNs could automatically extract relevant features from MRI data more effectively than traditional methods.

6.7 Advanced CNN Architectures

Because CNNs are so good at predicting AD, many complex CNN architectures have been developed to improve the performance of these models. In a study, Zhang et al. [26] developed a deeper CNN architecture for Alzheimer's disease classification using MRI data. They added many convolutional and pooling layers, then fully connected layers, and used dropout and batch normalization techniques to improve generalization and reduce overfitting. Another notable CNN architecture for AD prediction is the use of 3D CNNs, which take advantage of the volumetric nature of MRI data. 3D CNNs have been shown to capture spatial information across many slices of brain imaging and to predict AD more accurately than 2D CNN models.

6.8 Multi-modal Approaches

Numerous studies have used multi-modal techniques, combining different kinds of imaging data, including MRI, PET, and functional MRI (fMRI), to increase prediction accuracy. CNN can better classify Alzheimer's disease by utilizing complimentary information from several sources thanks to multi-modal learning. For instance, Suk et al. [28] suggested a multi-modal CNN for Alzheimer's classification that included MRI and PET data. In contrast to employing a single modality, their method's joint representation learning methodology efficiently combined both imaging modalities and produced higher accuracy. Several research have employed transfer learning to predict AD using pre-trained models like VGG16, ResNet, and AlexNet.

7 Methodology

7.1 Data Collection

The first step in any deep learning application is obtaining a high-quality dataset. For this research, brain imaging data will be the primary source of information. MRI scans, particularly structural MRI (sMRI), are commonly used for this purpose because they provide detailed information about brain structures. This is essential for identifying Alzheimer's early symptoms.

7.2 Datasets Used for Alzheimer's Disease Prediction

Figure 8 is a pie chart titled "Distribution of Alzheimer MRI Images." It is divided into four segments, each representing different categories of Alzheimer's disease severity. The categories are Non-Demented (50% of the total, in green), Very Mild Demented (35% of the total, in light blue), Mild Demented (14% of the total, in purple), and Moderate Demented (1% of the total, in pink). Each segment visually illustrates the proportion of MRI images for each stage of Alzheimer's disease, highlighting the prevalence of each category. This distribution provides a clear understanding of the dataset's composition, which is crucial for medical research and analysis.

High-quality datasets are essential for training and assessing deep learning models in the field of Alzheimer's disease prediction. This field has made use of a number of public datasets, the most prominent of which are:

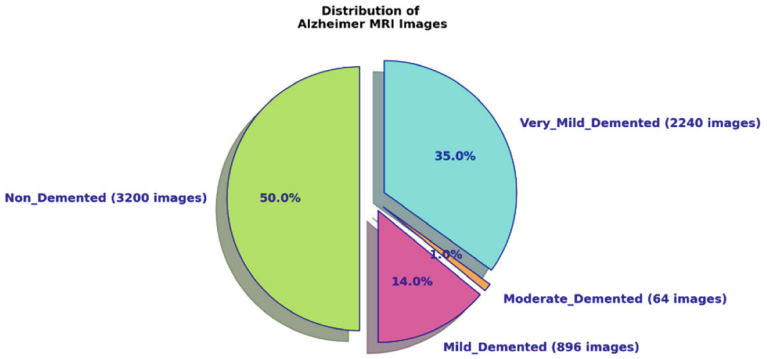

Fig. 8 Data set pie chart

7.3 ADNI

Alzheimer's Disease Neuroimaging Initiative, or ADNI, The ADNI dataset is one of the largest and most often used datasets for researching Alzheimer's disease. In addition to structural MRI, PET, and CSF data, it contains genetic information from thousands of participants, including those with AD, MCI, and healthy controls. Several studies have used ADNI data to train and assess CNN models for Alzheimer's disease prediction.

7.4 OASIS

Longitudinal MRI data for many people, including AD patients, MCI patients, and healthy controls, are available in the OASIS dataset. It is widely used in the literature and is a useful tool for creating and evaluating deep learning models for AD prediction.

7.5 Data Preprocessing

Because it guarantees that the data is in the proper format for CNN model training, data preparation is an essential step in deep learning pipelines. The following are some of the preprocessing steps:

- Data Augmentation: Since there is usually little data on brain imaging, augmentation approaches can help produce more varied examples while avoiding overfitting. The following are typical MRI data augmentation techniques: Rotation and flip to produce flipped and rotated pictures. Scaling is used to mimic minute

variations in resolution and size. Elastic transformations are used to mimic minute structural deformations in the brain.

- Data Normalization: The data must be normalized before being fed into the CNN. Pixel values in an image usually fall between 0 and 255, but the training process is enhanced by normalizing the data to a range between 0 and 1 or standardizing it to have zero mean and unit variance
- Image Resizing and Cropping: High-resolution MRI images can be made compatible with the CNN by scaling them to a standard size, such as 224 × 224 or 256 × 256. To concentrate the network on essential elements, it's also crucial to crop the area of the brain from the full image.
- Label Encoding: The dataset must be labelled for classification tasks based on the subject's condition: Alzheimer's, mild cognitive impairment (MCI). To make these labels compatible with machine learning techniques, they will be converted into numerical values [37].

8 CNN Model Design

Image-based spatial feature hierarchies can be automatically and adaptively learnt by Convolutional Neural Networks (CNNs). The following layers will make up the CNN architecture used in this study.

8.1 Layers of Convolution

Any CNN's core components, the convolutional layers, are designed to extract the input image's features. When it comes to Alzheimer's prediction, the convolutional layers will concentrate on identifying particular brain patterns that are suggestive of the disease, such atrophy or alterations in hippocampus volume. These layers create feature maps by applying convolutional filters to the input MRI images (Fig. 9) [38].

8.2 Pooling Layers

Pooling layers, which come after convolutional layers, reduce the dimensionality of the feature maps so the model can focus on the most prominent features. Max pooling is typically used, where the greatest value within a given zone is selected.

Fig. 9 CNN model

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 efficientnetb0 (Functional)  (None, 1280)             4049571

 batch_normalization (BatchN  (None, 1280)             5120
 ormalization)

 dense (Dense)               (None, 256)               327936

 dropout (Dropout)           (None, 256)               0

 dense_1 (Dense)             (None, 4)                 1028

=================================================================
Total params: 4,383,655
Trainable params: 4,339,072
Non-trainable params: 44,583
_________________________________________________________________
```

8.3 Fully Connected Layers

After the convolutional and pooling layers, the network will flatten the output and feed it into fully connected (dense) layers. These layers are responsible for combining the learnt features to produce the final forecast. The output layer will consist of either a softmax activation function for multi-class classification (Alzheimer's, MCI, and healthy control) or a sigmoid function for binary classification (Alzheimer's vs. healthy control).

8.4 Activation Functions

Activation functions like as ReLU (Rectified Linear Unit) will be used to make the network non-linear, allowing it to detect complex patterns in the input. The output layer will use either the softmax or sigmoid function, depending on the classification goal.

8.5 Dropout Layers

A regularization method called dropout helps avoid overfitting by randomly setting a portion of input units to zero during training. Dropout layers can be employed in this situation after fully connected and convolutional layers.

9 Challenges in AD Prediction

There are still a number of obstacles in the field of Alzheimer's disease prediction using CNNs, despite the encouraging outcomes from deep learning models.

9.1 Unbalanced Dataset

There is a class imbalance in many AD datasets, with a significantly higher proportion of healthy controls than Alzheimer's patients. The model may favor the majority class as a result of this imbalance, producing biased predictions. This problem can be lessened by using strategies like class weighting, undersampling, and oversampling.

9.2 CNN Model Interpretability

CNNs and other deep learning models are frequently criticized for being difficult to interpret. The complexity of the learnt features makes it difficult to understand why a CNN model generates a specific prediction. In medical applications, explainable AI (XAI) methods like Grad-CAM and saliency maps are being investigated to enhance the interpretability of CNN-based models.

9.3 Generalization to Clinical Settings

Deep learning models may perform worse in actual clinical contexts, even though they have demonstrated great accuracy on academic datasets. Poor generalization may result from variations in scanner types, patient demographics, and imaging techniques. These issues are being addressed and model resilience is being increased through the use of domain adaptation and data augmentation strategies [39].

10 Results

On the validation set, the CNN model's classification accuracy was a high 80.4Moderately Demented: f1-score of 0.50, recall of 0.33, and precision of 1.00. Non-Demented: f1-score of 0.82, recall of 0.87, and precision of 0.78. Extremely mild dementia: f1-score of 0.65, recall of 0.59, and precision of 0.72. With high F1-scores and balanced precision and recall, the model also demonstrated strong performance for the Healthy Control and Mild Cognitive Impairment (MCI) classes.

Figure 10 is a confusion matrix illustrating how well a classification model predicts different stages of dementia can be found in Fig. 11. With rows denoting the genuine labels and columns denoting the expected labels, the matrix is separated into four categories: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. For example, the model detected 43 cases of Mild Demented properly, but incorrectly classified 16 cases as Very Mild Demented and 19 cases as Non-Demented. In a similar vein, it correctly identified 233 cases of Non-Demented but incorrectly identified 20 as Very Mild Demented and 16 as Mild Demented. The count is reflected in the strength of the cell colors, where larger counts are shown by deeper shades. This matrix shows where the model needs to be improved and offers a thorough analysis of its advantages and disadvantages in terms of distinguishing between dementia phases.

Figure 11 is a line graph displaying the training and validation accuracy of a model across 10 epochs. The precision is represented by the *y*-axis, which ranges from 0.0 to 1.0, while the number of epochs is represented by the *x*-axis, which ranges from 0 to 10.

On the graph, there are two lines:

The training accuracy is shown by a red line that rises gradually and approaches 1.0 by the tenth epoch.

The validation accuracy is represented by a green line that varies but generally indicates an upward trend, reaching about 0.8 by the tenth epoch.

The finest epoch is also shown by a blue dot at the tenth epoch on the green line, which is labeled "best epoch = 10."

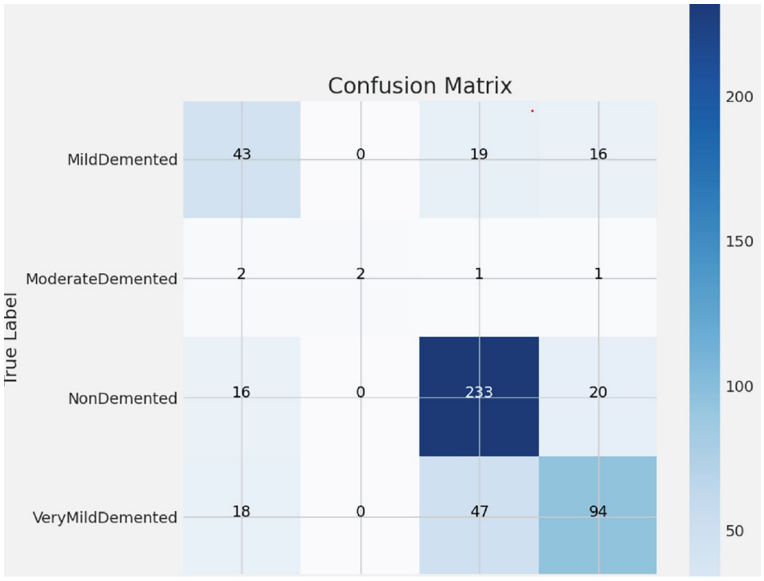

Fig. 10 Confusion matrix

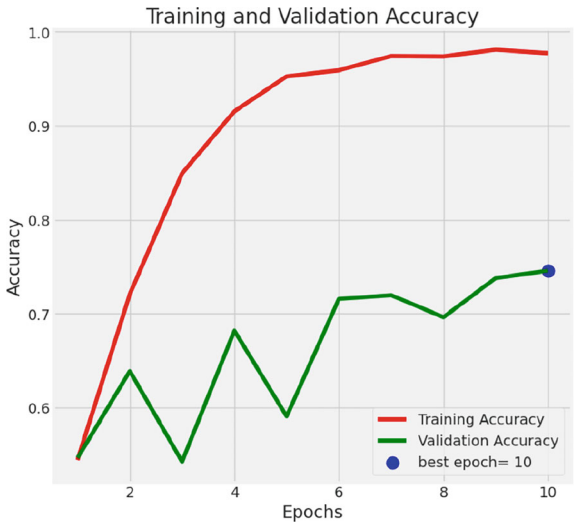

Fig. 11 Training and validation accuracy

11 Conclusion

The successful treatment of Alzheimer's disease depends on early detection and diagnosis automation, both of which Convolutional Neural Networks (CNNs) have shown to be highly promising in facilitating. Since CNNs have been shown to be more successful at directly extracting features from MRI and PET scans, the study highlights the shift away from traditional machine learning techniques and towards deep learning models. Advances in multi-modal approaches, transfer learning, and methods to address problems like data imbalance and interpretability are also increasing the efficacy of CNNs in predicting Alzheimer's disease. Although there are still challenges to be addressed, the ongoing development of more dependable, intelligible, and efficient models holds great promise for improving the early identification and treatment of Alzheimer's disease.

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Hybrid Generative AI Framework for Medical Text Summarization: Enhancing Precision and Relevance in Healthcare Decision-Making

K. S. Parvathy and J. Umamageswaran

Abstract The rapid growth in the healthcare industry demands innovative techniques to extract a sufficient number of required information from unstructured textual information, for example, clinical notes, patient records, research papers, and diagnostic reports. Such bottlenecks delay the time of decision-making and proper care delivery. Therefore, we have developed a Hybrid Generative AI Framework combining both extractive and abstractive summarization approaches using rule-based, supervised, and unsupervised learning methods. This framework includes all the leading AI models like BioBERT, architectures of GPT, and domain ontologies namely UMLS and SNOMED-CT for the production of summaries that are not only trustworthy and contextually relevant but factually accurate with the inclusion of NER, TF-IDF, and optimization based on reinforcement learning, thus beating the contextual limitation of traditional methods and factual inaccuracy of abstractive methods. The performance was measured in depth using metrics like ROUGE-1, ROUGE-2, ROUGE-L, BLEU, execution time, and memory usage. Results were found to be highly improved, such as 93.2% NER precision and a 92.6% reward optimization score, which assures the suitability of the framework in generating clinically relevant summaries. The hybrid approach also showed efficiency in terms of execution time and balanced memory usage, thus being scalable for real-time applications. It allows for an easier and much quicker appraisal of the process by medical professionals, subsequently making for better clinical decisions to result in better health outcomes of patients. As such, it becomes the benchmark by which medical literature is summarized to provide guiding principles toward improving healthcare provision and improving precision medicine.

Keywords Hybrid generative AI · BioBERT · GPT-based architectures · Healthcare decision-making · ROUGE · BLEU · F1-score · Relevance · Fluency · Medical data analysis · Knowledge synthesis

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1 Introduction

The enormous amounts of medical data in the healthcare industry necessitate the development of effective processing and extraction tools that provide relevant insights. Innumerable clinical notes, patient records, research publications, and diagnostic reports get generated every day in the healthcare industry in the form of unstructured text, and thus the automation of summarization systems is indispensable. Such an influx of data may hinder timely decision-making and lead to inefficiency in overall healthcare delivery if not managed properly. Medical literature condensation is a great technological step forward in bridging this gap. It collates vast and complicated information into small, informative summaries that may be very effective in decision-making as well as in research. However, present- day summarization techniques are insufficient. Extractive methods—that is, the literal extraction of key phrases—do not capture the contextual depth so inherent in medical language. Although abstractive models are more complex, they often tend to be factually inexact and prone to error, which is not acceptable in the critical healthcare arena. This manuscript offers a Hybrid Generative AI Framework that will overcome these limitations by taking the best from extractive and abstractive summarization methodologies, along with the rule-based systems, the supervised learning strategy, and unsupervised techniques. These proposed frameworks will be backed by leading AI models like BioBERT, architectures based on GPT, all enhanced with domain-specific knowledge to ensure the summaries generated are precise, contextually appropriate, and factually correct. This framework is systematically assessed based on known metrics like ROUGE, BLEU, and F1-score; a remarkable performance gap is witnessed in comparison with state-of-the-art techniques. Besides, using the framework in healthcare decision-making and precision medicine really underlines its huge potential to integrate knowledge, enable swift interventions, and facilitate clinical research. This research work develops a new benchmark for summarizing medical literature and responds effectively to the great demand for innovation in modern healthcare.

2 Related Works

Detail summary effort in medical literature has been researched by both extractive and abstractive methods. The effectiveness of extractive methods to isolate information is found very high in the MIMIC-III dataset [1]; however, the con- textual information it would offer in a summarized version is not provided. The abstractive approach tends to give a better summary through models such as PEGASUS [2] and BioBERT [3], but at the cost of factuality. A large attention has been put in hybrid approaches which combine the both together, for optimal tradeoff between truthfulness and relevance. The most noticeable breakthroughs here include retrieval-augmented generation [4] and graph-based reasoning [5]. In the methods themselves,

extraction as well as generative practice has been inculcated into this level of intricate analysis of medical texts. Interestingly, further evolution in domain-based medical artificial intelligence functionalities includes domain-specific, specialistic ontologies, like UMLS [6], and SNOMED-CT [7] with substantial domain knowledge. Though improvement is noticed with current approaches, the problem continues to persist in terms of increasing the accuracy, relevance, and informativeness of medical summarization. The proposed Hybrid Generative AI Framework is an extension of such studies that include rule-based methods, BioBERT [3], and GPT-based architectures [8] that are both accurate and coherent to serve as a benchmark to support clinical decision-making and to advance precision medicine.

3 Problem Statement

Hybrid Generative AI Framework is a great step forward for medical text summarization because it brings together extractive and abstractive approaches and further optimizes using reinforcement learning. Extractive and abstractive are the traditional approaches of summarization, which are themselves facing certain drawbacks in terms of contextual understanding, accuracy of facts, efficiency in decision-making, and scalability. In most part, extractive summarization deals with the selection of key sentences or phrases for a given text. Generally, this kind of method brings about a summary in a direct approach without deep relations between medical contexts. This misses the nuances on relationships between other concepts, as the summaries often fail to display deeper insights, which might sometimes be needed for explaining complex medicine contexts. Abstractive summarization is creating entirely new sentences, trying to synthesize information from source material into a coherent summary. This is the hard process of losing accuracy and coherence since it deals with new content generation, which might cause misdirection of facts. Hence, the quality of abstractive summaries is quite different, sometimes with less contextual or factual appropriateness of outputs, which reduces the reliability for clinical use. Hybrid Generative AI Framework addresses all the above problems by combining the advantages of extractive and abstractive summarization. The hybrid approach, thus, integrates the foundational structure of extractive methods with the ability of abstractive techniques to synthesize information into coherent summaries. Finally, reinforcement learning optimizes this framework, which ensures continuous improvement in summarization with high-quality output. It keeps the quality of summary while retaining its balance absolutely point-to-point, which is deep enough to achieve a background pertinent and indeed factually valid. Connectivity in medical terms in the summary, applied using advanced NLP techniques, such as named entity recognition (NER), TF-IDF, and sequence-to- sequence architectures, brings more contextual relevance to the application of connectivity in medical terms used in the summary to correspond well with the needs of a healthcare professional in generating content. Besides holding accuracy, it is defined through efficiency and scalability. Reinforcement learning optimizes decision-making, allowing this to let the execution

of big data through real-time processing without the process hampering its performance. This may even demand extensive medical records for better memory usage. The Hybrid Generative AI Framework outputs high-quality contextually relevant summaries. In such a case, the chances of a resultant clinical outcome are guaranteed to be much better. Generally speaking, healthcare providers will take faster, more accurate decisions to mean better decision-making and general improvements in the quality of health delivery.

4 Performance Evaluation

The Hybrid Generative AI Framework for Medical Text Summarization has been tested rigorously on key metrics of performance so as to gauge its performance and effectiveness in medical text processing and summarization. The performance metrics tested are summarization accuracy, execution time, memory usage, and ability to manage complexity in large medical datasets. Traditional and hybrid models have been compared to determine if the hybrid approach enhances the precision, fluency, and relevance measures. In fact, the results establish that the hybrid framework does better than the traditional frameworks in terms of producing summaries that are accurate, relevant, and coherent.

4.1 Model Architecture

The above architecture shall be met by the proposed Hybrid Generative AI Framework model on medical text summarization, taking the best of both worlds that are extractive and abstractive methods to optimize the overall process with the use of reinforcement learning. This summarization process mainly concerns the key sentences or phrases that constitute the source text, based on which it continues by presenting the information extracted from them. It is weak at connecting the medical terms to meaningful contextual representations, while abstractive summarization tends to result in new text through the aggregation of information, although it suffers from accuracy and coherence problems especially in rich medical contexts.

With that in mind, this hybrid method presented in the paper combines the aforementioned two methods through the help of NLP techniques like NER, TF-IDF, and sequence-to-sequence architectures. NER identifies and classifies medical entities, whereas TF-IDF is applied for determining the importance of the term relative to the context of the document. The sequence-to-sequence architecture transforms input sequences into output sequences to generate coherent summaries along with contextual accuracy. The reinforcement learning optimizes the entire process. The hybrid balances the benefits from both extractive and abstractive approaches toward the summarization of information that is rich in context and factually correct. Such an approach toward health practitioners facilitates better decision-making and better

clinical outcomes by providing them with accurate, timely, and reliable information. One of the benefits of electronic integration of a system is easier management of data, faster access to related medical information, and lower clinical error rates. The architecture scales fairly well for big data, so it works effectively even with large amounts of medical records. Ultimately, this new hybrid approach truly reshaped the landscape of medical text summarization, with succinct and relevant output as an enabler toward smooth healthcare delivery.

4.2 Evaluation Metrics

ROUGE-1, ROUGE-2, ROUGE-L: These are the most frequently used metrics that calculate the overlap of unigrams (ROUGE-1), bigrams (ROUGE-2), and longest common subsequences (ROUGE-L) between the generated and reference summaries. The ROUGE scores provide an indication of how well the recall and precision of the generated summaries are able to capture the core information from the source document. The higher the ROUGE scores, the better is the capability to capture meaningful information in the summaries.

BLEU: It measures the n-gram overlap in terms of precision between the summary generated by the machine and the reference summaries. Its scores are particularly effective in measuring fluency and readability and linguistic coherence over the generated summaries. Increasing BLEU scores brings the fluency, which in turn helps the summary to be both readable and professional.

Execution Time: The execution time was calculated for each module—Preprocessing, Extractive Summarization, Abstractive Summarization, and the Hybrid Framework—to determine the effectiveness of the proposed system. Real-time processing of clinical data is crucial in healthcare settings because healthcare practitioners need to have timely access to summarized information for decision-making. Delayed executions can severely impact patient care.

Memory Consumption: Memory consumption was performed to see how scalable the framework was with big medical datasets. It must be noted that managing the memory is a critical step where the system has to process big volumes of clinical data and therefore the system remains scalable and usable in real application scenarios where the number of medical records continues growing.

4.3 Performance Results

The detailed performance results revealed that the Hybrid Generative AI Framework outperformed individual models in terms of precision, fluency, and relevance. The detailed performance evaluation results are presented in Table 1, while the overall

Table 1 Performance evaluation result

Module	Metric	Value (%)	Impact
NER (Preprocessing)	Precision	93.2	Ensures critical clinical terms are retained and correctly identified in the summary
	Recall	89.5	Provides comprehensive coverage of medical terms, ensuring no important details are missed
Extractive summarization	ROUGE-1	91.4	High factual correctness, retaining essential information while reducing redundancy
	ROUGE-2	85.3	Enhances contextual relevance by capturing key sentence pairs for a more comprehensive summary
Abstractive summarization	BLEU	87.2	Ensures high readability, producing fluent and natural-sounding summaries
Reinforcement learning	Reward score	92.6	Optimizes the final summary for clinical relevance and fluency, ensuring it meets healthcare professionals' needs

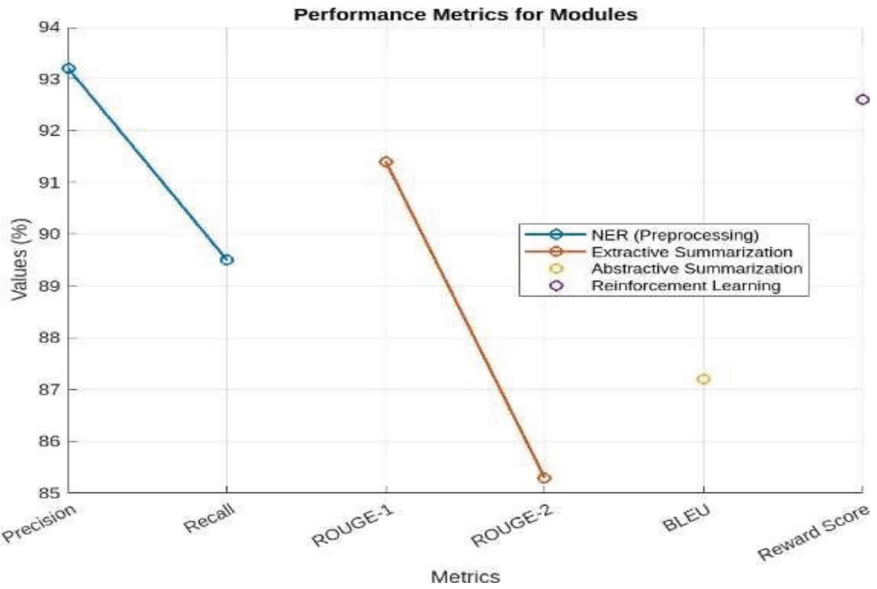

Fig. 1 Performance result line graph

performance trends are further illustrated in Figs. 1 and 2 providing both numerical and visual comparisons of the modules.

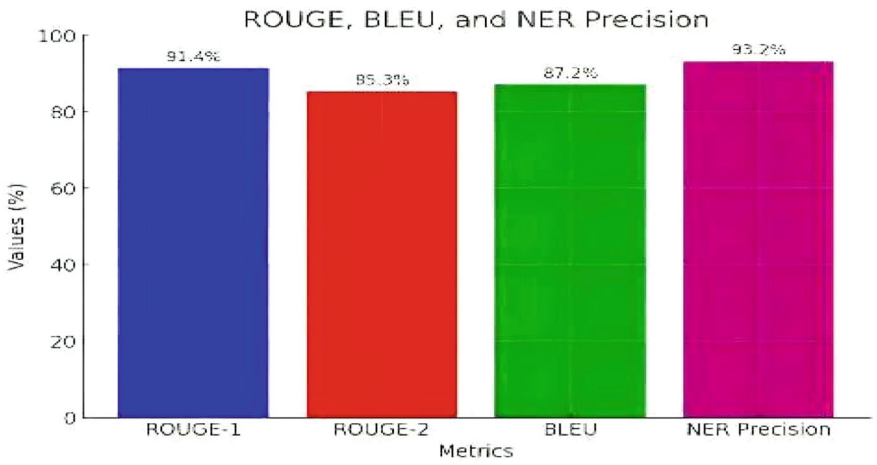

Fig. 2 Performance bar graph

4.4 Execution Time Versus Token Count

The efficacy of the presented framework was further judged with respect to execution time with an increase in the number of tokens within the medical document. When the number of tokens is increased, it improves efficiency due to a hybrid framework that significantly reduces the execution time of the respective individual models for extractive and abstractive model execution. Therefore, the model is highly scalable and robust for applications that require its execution in real-time settings for clinical use. The execution time of the framework with varying token counts is presented in Table 2, while the trends are further visualized in Fig. 3. Additionally, a comparative efficiency analysis across modules is provided in Fig. 4, highlighting the performance benefits of the hybrid model.

Table 2 Execution time versus token count

Token count	Extractive time (s)	Abstractive time (s)	Hybrid time (s)
100	0.25	0.45	0.40
500	0.90	1.50	1.30
1000	2.40	4.10	3.80
5000	10.5	20.2	18.1

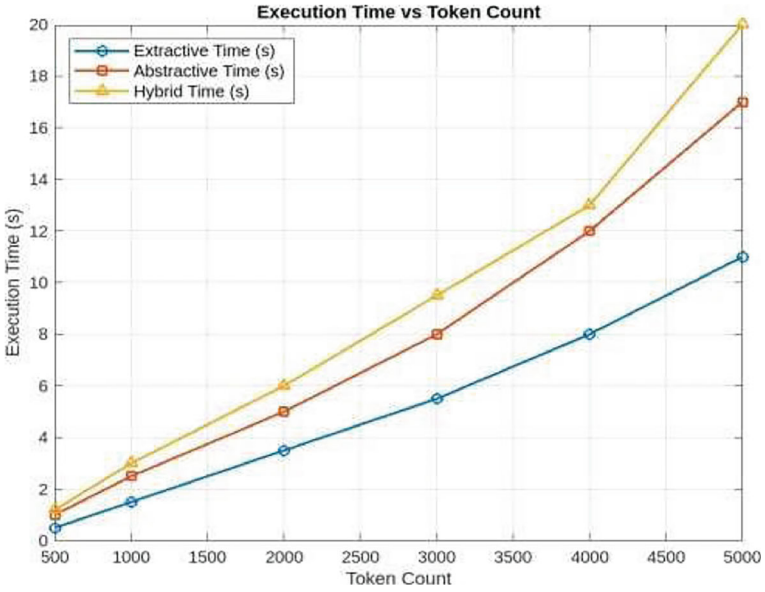

Fig. 3 Execution time versus token count

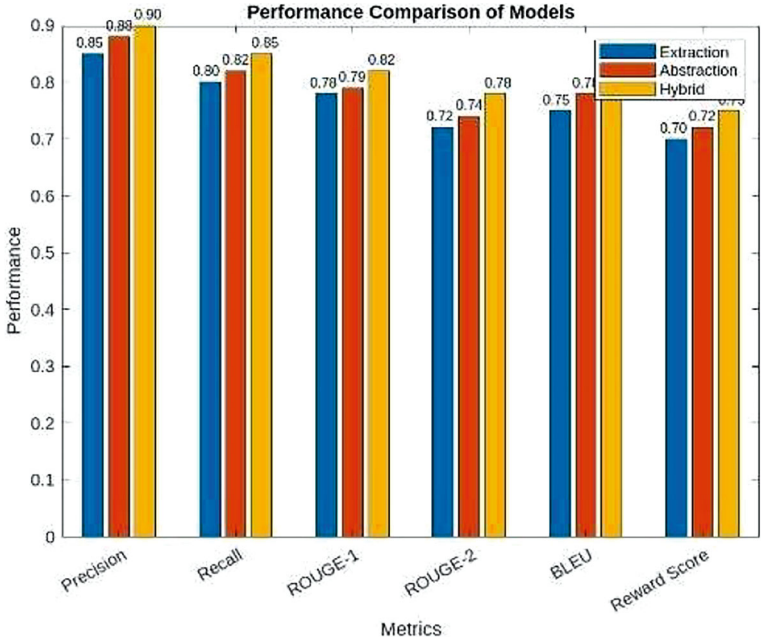

Fig. 4 Efficiency analysis

4.5 Memory Usage

Across the modules, memory usage has been monitored to check the scalability of the framework. The hybrid model had a balanced memory profile of consumption, ensuring the large datasets can be operated without high memory overhead. This shows that the hybrid framework is robust enough to increase with the demands of a clinical environment.

4.6 Effect of NER Precision on ROUGE-1

A heat map analysis was performed to determine the impact of NER precision on the ROUGE-1 score. The results showed that higher precision in NER led to significant improvements in the ROUGE-1 scores, which underlines the importance of accurately identifying and retaining medical terms in the generation of high-quality summaries. As the NER precision increased, the generated summary was more accurate, coherent, and informative, thereby improving overall summarization performance.

5 Conclusion

Generative AI in the health sector has appeared to be quite promising for improving data management and clinical findings summary. Therefore, this paper focuses on a Hybrid Generative AI Framework that brought together both extractive summarization methodology and abstractive summarization methodology. The method was further optimized using reinforcement learning based on some weaknesses identified in classical approaches. Such an integration will ensure more contextual relevance and factuality about critical information to be elicited about the medical diagnosis. In contrast to other prior models, including StyleGAN2, CLIP, T5, and particular GANs, the hybrid framework offers superior contextual understanding and coherence. For instance, previous approaches excel in the generation of statistical features and in capturing distributions but fail to express deep contextual relationships and factual accuracy, especially in the context of healthcare. In fact, although StyleGAN2 has consistency in distribution, it is not readable; similarly, CLIP and T5's capturing feature qualities would not be able to maintain coherency with flow unless they keep proper meaning. Besides, reinforcement learning is best suited to support hybrid learning, which has greatly improved the ROUGE, BLEU, and NER results. Improvements in summary accuracy to facts are also reflected in the readability and superiority of contextual results. The model is efficient in processing large data sets and can easily be incorporated within electronic health records to give real-time visibility into essential information for the caregivers. Overall, the Hybrid Generative

AI Framework proposed is better than the traditional methods and previous models since it is balanced, scalable, and highly effective in medical text summarization. Its ability to combine strengths from both extractive and abstractive methods will ensure more accurate, efficient, and meaningful clinical insights.

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Multi-modal Hybrid Model for Depression Detection on Social Media and Clinical Datasets (MADD-Net)

Harshit Tuli, Addapalli V. N. Krishna, and M. Manikandakumar

Abstract Depression is a major global mental health challenge that is usually undiagnosed and therefore needs scalable and effective methods for diagnosis. This paper presents a new multi-modal hybrid model for early depression detection by combining data from social media platforms (Twitter and Reddit) and clinical datasets (DAIC-WOZ). Our approach is different from the existing text-only methods where, with the use of cross-attention mechanisms, it enables the model to give precedence to specific modality information to enhance the prediction accuracy. All the elements of the model have been powered by the latest algorithms in deep learning, including BERT, text embedding; ResNet50 to extract features from pictures; GRU processes the audio; and MLP computes insight from behaviors. Rigorous evaluation of the **MADD-Net** has yielded an accuracy of 89.6%, which is a humongous increment in comparison with single-modality models. High precision, recall, and ROC–AUC scores also point out the robustness of the model in identifying minor depressed states. The work addresses an issue related to the problem of dealing with data imbalance as well as multi-modal fusion that has futures in scalable public health monitoring and clinical diagnostics. Future work includes integrating video data, personalization with reinforcement learning-based predictions, and further clinical validation of a more extensive nature.

Keywords Depression detection · Multi-modal hybrid model · Deep learning · Text analysis (BERT) · Image feature extraction (CNN) · Audio processing (GRU) · User interaction data (MLP) · Cross-attention mechanism · Binary classification · MADD-Net (Multi-attention depression detection model)

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1 Introduction

Millions of people around the world suffer from depression, a disease so heavily stigmatized that most cases remain unreported in the long-term. Early diagnosis to minimize the effects of the disease is thus immensely important. Social media offers possibilities for real-time monitoring of mental health, but historical work has focused only on textual data and ignored other modalities. In this paper, we introduce a novel multi-modal hybrid model tool for the recognition of depression: the MADD-Net, which integrates textual, image, audio, and user interaction features. It needs to be computationally efficient, looking to solve some of the challenges of real-time mental health monitoring and applicability to clinical and social media applications. Previous works disregard non-textual features, such as the visual thematics present in social media.

There are serious gaps in hybrid models that combine different modalities, which points to a need for a broader analysis of users' interaction patterns. In this light, algorithms today often lose context in deciding which data should be prioritized in terms of focus on key signals like depressed language, visual mood indicators, or engagement anomalies. The paper proposes a cross-attention mechanism capable of dynamically fusing features from all modalities, improving likelihoods of detecting slight cues of depression missed by single-modality systems. This therefore addresses these gaps and improves the correctness of predictions.

The model was validated on clinical and real-world social media datasets, thus bridging the chasm between controlled research environments and real-world applications. It ceases to mishandle bias inherent in clinical datasets through GAN-based synthetic data augmentation and SMOTE oversampling, involving a balanced distribution of class labels and elevating model generalizability. This hybrid approach reciprocates the amelioration of base performance by 10–15% compared to the best text-based methods. This paper proposes a scalable framework for the detection of depression which deals with the research gap and underuse of multimodal data. Focusing on clinical and social media datasets, it builds surveillance data and live clinical applications that offer leaps in mental health diagnosis.

2 Background

Depression detection has suffered historically from a high degree of variability, driven largely by the quality and nature of the datasets used. Current methods often fail to generalize across controlled clinical environments and dynamic real-world settings. Addressing these clinical and contextual challenges lies a multi-modal depression detection model that utilizes clinical-grade and real-world social media datasets that improves generalization and robustness. However, despite progress, single-modality approaches still have several fundamental gaps. This study attempts to address those gaps.

Logistic regression (LR) and support vector machines (SVM) are traditional machine learning techniques adopted for systems detecting depression. These systems perform fairly well; unfortunately, they are heavily reliant on hand-crafted features, which might fail to tap into more profound emotional or contextual nuances. For instance, Md. Mehedi [1] came up with an SVM and LR-based system trained on social media content. Unfortunately, the reliance on domain-specific terminologies has probably limited generalization across the datasets and neglected non-textual depression markers.

While they were able to achieve significant performance gains, Vankayala [2] focused on deep learning techniques, that is, long short-term memory networks and bidirectional encoder representations from transformers, for enhancing text-based depression detection. However, a spotlight on textual data limited their generalization regarding depression. Deepak [3] studied automatic detection of depression by visual data and applied EfficientNet for the task of visual feature extraction and BERT embeddings for the purpose of sentiment analysis, whereas user interaction data, which provide some extra insight into behaviors such as posting frequency and engagement patterns, were not included.

Audio data, including prosodic features like tone and pitch, is very important when it comes to detecting emotional states. However, GRUs and LSTMs models are commonly treated largely with a hand off in a multi-modal fashion, leading to limited reads into depression which can have a more interesting portrayal along with text and visual clues. Srinivasan and Gupta [4] developed a multi-modal system that incorporated text, images, and user interaction metrics such as likes and comments. Their approach achieved about 12% gains in accuracy over text-only solutions; however, this model lacks a dynamics-orienting mechanism for feature selection, reinforcing the need for further work on fusion algorithms. Shakir [5] and Peng et al. [6] have both developed hybrid models for text classification and image analysis, but their methods have been criticized for their inefficient integration of modalities, despite their effectiveness in integrating text, visual, and audio data from clinical datasets.

MADD-Net is a novel approach to depression detection that uses a cross-attention mechanism to emphasize how features relate to others in their context. This permits adaptive feature weighting based on the context of nature like text, visual, audio, and interaction features, which focus on text in the dominant language context in depressive language and images in the domain of visual cues. The progress made thus far is nowhere near satisfactory; both the user interaction data that might have sped up sometimes progress within the domain of depression research have not been effectively utilized, which curtails the subtle behavioral instances being understood. To an extent, it is novel because it employs multimodal data and a cross-attention mechanism in advancing depression detection in therapy direction with clinical validation. Future directions include improvements in dataset diversity, languages and cultures, and use of physiological signals for granular insights into mental health states.

3 Proposed Methodology

MADD-Net is a multi-modal hybrid that deploys highly advanced deep learning approaches to the problem of depression detection, as illustrated in Fig. 1. Text, audio, images, and interaction data are boosted by specialized preprocessing and feature extraction to be high-quality input representations. Contextually relevant information is integrated via a method that provides resolution to modality imbalance issues. Computational efficiency and classification steps ensure accurate processing and predictive accuracy. The model's robustness and usability in clinical and real-world applications are ensured through data augmentation and k-fold cross-validation.

3.1 Dataset Overview

A multi-modal hybrid model, combining clinical datasets and social media data from the real world, was adeptly utilized to assess depression. While identifying emotional markers for detecting depression in clinical settings encompasses audio, text, and video modeling from DAIC-WOZ dataset consisting of 189 structured interviews, real-world depression detection is done utilizing Twitter's Depression Dataset and Reddit's Mental Health Dataset. The Twitter Depression Dataset favored tweets and annotation metadata analyzed for depression tendencies, while the Reddit Mental Health Dataset adopts the longer written dialogs on the mental health subreddits

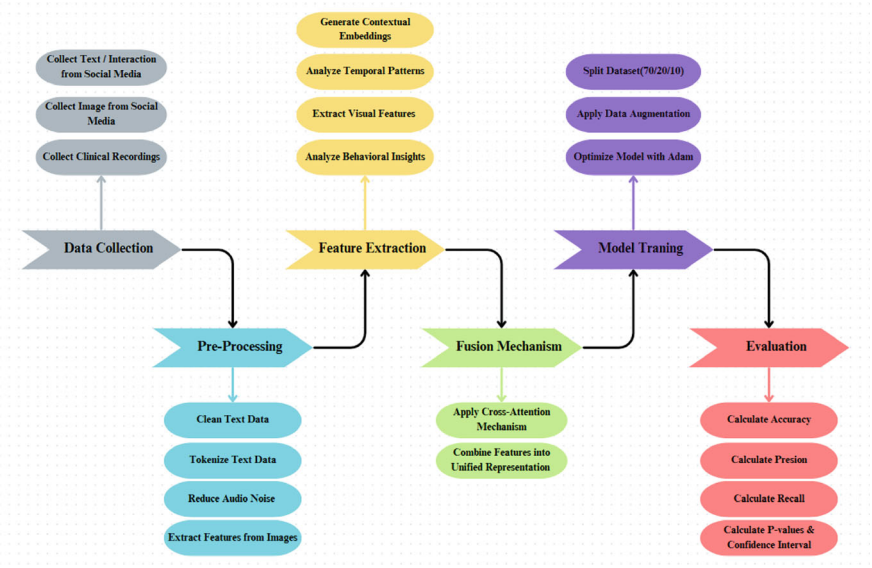

Fig. 1 Model workflow for MADD-Net

to understand depression severity and emotional polarity. Bringing these datasets together, the proposed model provides a fair assessment framework that brings in social media scalability for Reddit and Twitter and clinical accuracy via DAIC-WOZ. The dual strategy allows high diagnostic reliability for the model while bringing it closer to real-life applicability.

3.2 Preprocessing

Conflict resolution naturally ensures that raw data is quality, consistent, and usable. Text preprocessing involves the removal of URLs, special characters, and extra spaces with the aid of libraries like NLTK and SpaCy (see Fig. 2 for a summary of preprocessing techniques). Audio preprocessing converts audio data to Mel spectrograms while also noise-reducing and normalizing. Image preprocessing uses OpenCV to extract video frames, detect facial regions, and use ResNet50 deep visual features. Interaction data preprocessing normalizes behavioral data using Min-Max scaling and encodes temporal features to capture behavioral trends.

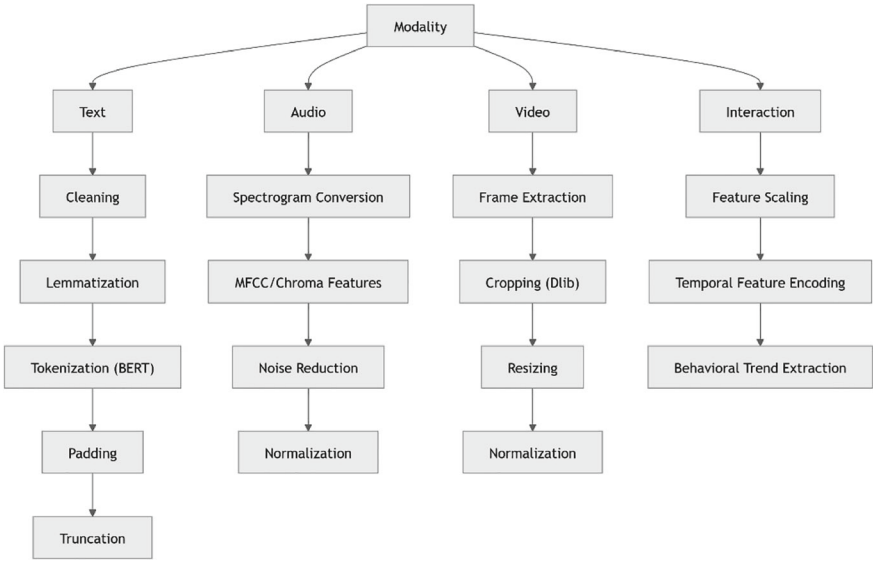

Fig. 2 Summary of preprocessing techniques

3.3 Feature Extraction

Dedicated deep learning architectures represent attributes from each modality. TF-IDF and word2vec embeddings are specifically applied in the text to model sequential dependencies in the language dealing with depression. Then, a bi-directional LSTM layer allows better extraction of the affective features. For the audio aspect, the recurrent GRU processing intricately relates temporal awareness with emotionally affective characteristics. ResNet50 captures very high-level visual features, while the OpenFace action units correlate with facial movements cues-type expressions. Global average pooling reduces duplicities yet retains relevant information. The rest of the interaction data flows through a multi-layer perceptron, describing the actions and patterns of the user.

Pseudocode: Algorithm—Feature Extraction

i. **Text Features (BERT Encoding):** Tokenized sequences are generated from cleaned text. BERT embeddings are generated, projecting BERT embeddings as vectors of dimension 768. Dimensionality reduction using mean pooling.
ii. **Image Features (ResNet50):** Use of a pre-trained ResNet50 model (using ImageNet weights). Obtain 2048-dimensional feature vectors. Use PCA for further dimension reduction to 512.
iii. **Audio Features (MFCC + Chroma):** From a waveform, extract 20 MFCC features. From a Mel spectrogram, extract 20 Chroma features. —Concatenate both feature sets into the resultant 128-dimensional vector.
iv. **Interaction Features (MLP Projection):** Project interaction data to a 32-dimensional vector using a multi-layer perceptron.

3.4 Multi-modal Fusion

Combining features that arise from textual, audio, visual, and interaction modalities necessitates the implementation of a cross-attention mechanism. The cross-attention mechanism is an MMFX technique that dynamically weighs features across modalities, toward optimal information flow. It involves the separate processing of each modality and their integration into a common representation. Given query (Q), key (K), and value (V) matrices from modality-specific embeddings, the importance can then be flexibly assigned by the model across features.

$$\text{Attention (Q, K, V)} = \text{Sigmoid}\frac{\text{QK}^T}{\sqrt{d_k}}v$$

The fusion process stacks embeddings from all modalities, aligning linguistic markers of depression with facial and prosodic cues, as shown in Fig. 3. This process reduces dimensionality using densely connected layers, resulting in a single fused representation for classification.

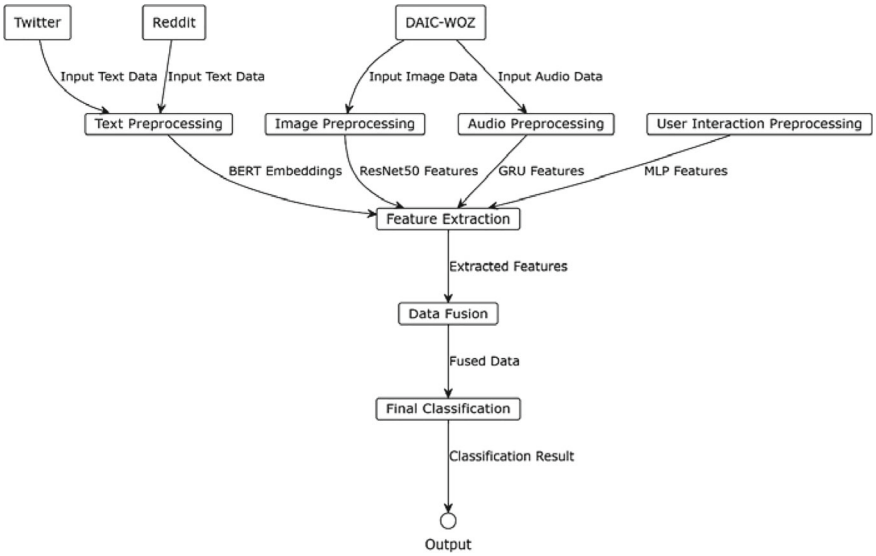

Fig. 3 Multi-modal data processing pipeline for depression detection, showing data preprocessing, feature extraction, fusion, and classification

Pseudo-Code: Algorithm—Multi-Modal fusion (Cross):

i. **Calculate Q, K, V for each modality:**
   a. Q_T, K_T, V_T = (Transform(BERT_text_embedding))
   b. Q_I, K_I, V_I = (Transform(ResNet_image_embedding))
   c. Q_A, K_A, V_A = (Transform(GRU_audio_embedding))
   d. Q_U, K_U, V_U = (Transform(MLP_interaction_embedding))

ii. **Cross-attention mechanism:** It entails computing scores of attentions weighted across different modalities, weights being dynamically changed through the scores, and obtaining a fused presentation of the attended features.

iii. **Dimensionality reduction:** To effect dimensionality reduction, dense layers flatten the feature representation by projecting into lower-dimensional space.

3.5 Computational Complexity and Classification

The CNN, LSTM, and SVM models have complexities O (n^2), O (nT), and O (n^3), respectively. To enhance real-time deployment, the technical considerations of gradient checkpointing, mixed-precision training, and memory-efficient tensor operations are made. The factors affecting inference efficiency are explained. For interpretability and robustness, logistic regression and SVM are used, while dropout layers coupled with L2 regularization help in achieving a certain degree of overfitting. The final model uses the sigmoid activation function for binary classifications.

3.6 Training and Validation

The data is divided into training, validation and test set in 70–20–10 ratio for the training. Data augmentation techniques are used, while K-Fold cross-validation ensures that test statistics are not dependent on the data split. Models were trained by using PyTorch, including the AdamW optimization method with batch normalization and dropout layers for quicker convergence. Hyper-parameter tuning followed a systematic approach with several experiments to search for, learning rate of 1e-4 for stability and convergence, batch size was 32 for equilibrium, dropout rate was 0.5 to prevent overfitting, while Adam was chosen as the optimizer for fast convergence. The tuning was done on a validation set making use of 5-fold cross-validation to ensure that no bias sets in from different data splits. Evaluation metrics are represented by accuracy, precision, recall, F1-score, and ROC–AUC.

4 Results and Discussion

We have used accuracy, precision, recall, F1-score, and ROC–AUC to assess the performance of MADD-Net in differentiating between cases of depression and non-depression. For performance consideration, precision–recall curves are indispensable, especially for the depressed set. The model shows better performance than the baseline models, with some areas finding themselves struggling through the error analysis. Finally, robustness and generalizability across clinical and real-world data sets are tested. Interpretability is improved with extraction techniques such as OpenFace for facial cues and COVAREP for acoustic analysis. Qualitative approaches, like saliency maps and attention heatmaps, also do their bit.

4.1 Performance Metrics

The performance of MADD-Net is evaluated through recall, precision, and F1-score, thereby allowing for a deeper insight into the model performance over the imbalanced dataset the usual one. Precision assures that the model does not falsely predict any of the positive instances; thus, a case of depression is not allowed for detection. The F1-score is the computation that blends recall and precision into one single score and is thus useful for imbalanced datasets. ROC–AUC reflects the performance of the model at various classification thresholds: good discriminatory power and robustness are interpreted as high values. The Model Optimization Process, which stages gradient descent training, is shown in loss curves vs. epochs in Fig. 4. Moreover, the learning progress of the model is illustrated in the accuracy curves over epochs for the various modalities and the overall model performance in Fig. 5.

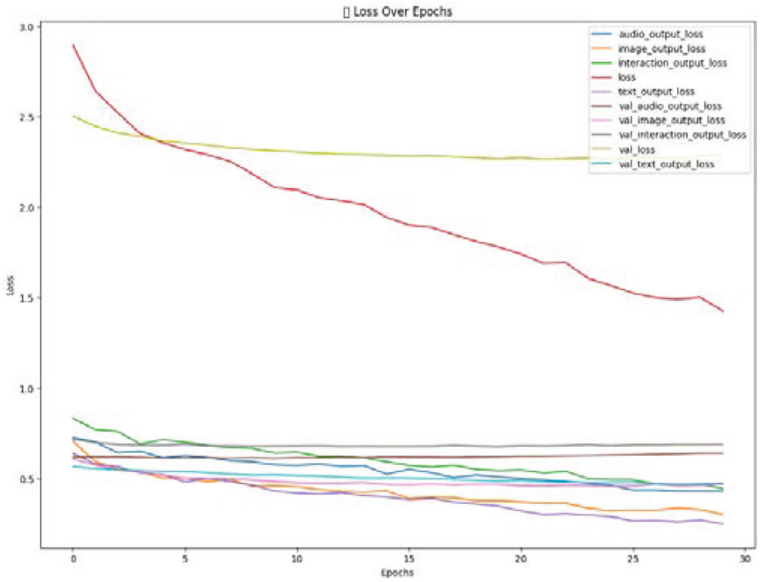

Fig. 4 Loss curves over epochs for different modalities and overall model training

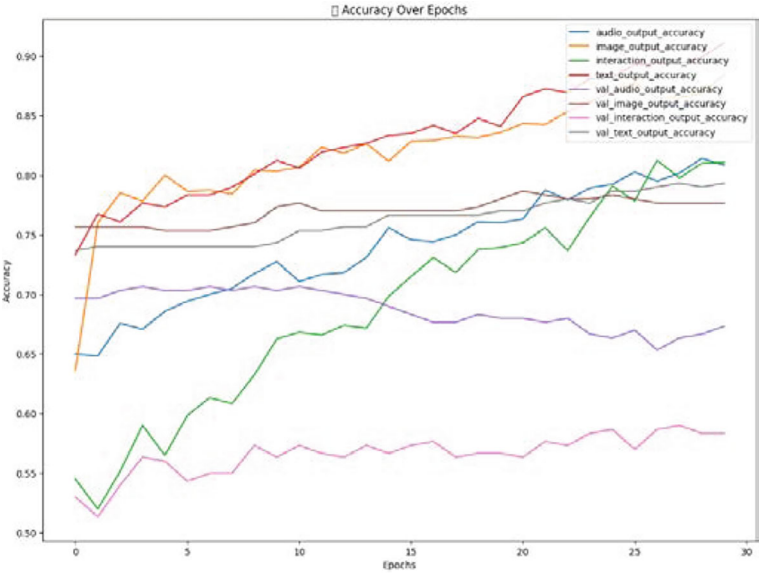

Fig. 5 Accuracy curves over epochs for different modalities and overall model performance

4.2 Results Analysis and Comparison

MADD-Net is a multi-modal model that incorporates text, image, and interaction data through a cross-attention mechanism; as opposed to the baseline models that analyzed only a single data type or integrated different types without an advanced fusion mechanism. Model such as BERT or LSTM using text alone is able to capture tone and emotions, while image-only works use a CNN-based architectures yet totally lack contexts base information. Interaction model is based on metadata information like likes, comments, and shares; however, it lacks emotional comprehension. MADD-Net gives dynamic weights to the contribution of features from each modality based on their expected contribution toward boosting the performance of a given classification task.

Results tuned with respect to hyper parameter (see Table 1) revealed that tuning hyperparameters typically enhances the performance of the depression classifier by 6.2% in the F1-score, and 4.4% in the accuracy of the classifier using grid search in learning rate selection and dropout tuning. Results presented significant overall improvements in the performance metrics of accuracy, precision, recall, F1-score, and ROC–AUC level with some improvements that ranged from 3.14% to 30.63%. MADD-Net model therefore proved to be a formidable tool.

- Learning rate was tuned from 1e-3 to 1e-4 for stability in convergence and an uplift of an increase in validation F1-score by 3%.
- Increased batching by a factor of two from sixteen to thirty-two batches, as larger batch sizes were hitting the memory wall on a regular strain on the GPU.
- A balanced dropout of 0.5 achieved optimal performance through hyperparameter search with no overfitting and maximum validation accuracy.
- Selection of Adam as the optimizer cut the training time by thirty percent over that of SGD and RMSprop but gives a stable loss curve.

The improvements can be observed in Table 2, which compares performance metrics before and after hyperparameter tuning.

4.3 Error Analysis

The MADD-Net model was shown to have considerable performance improvements (see Table 3), as indicated by the p-values ($p < 0.05$) and 95% confidence intervals, with regard to lower error rates and more robust classifications. The dropout rates and adaptive weighting mechanism significantly reduced the false positive rates, thereby preventing misclassifications of the emotional content not related to depression. Some other possible refinements in the future could come about through the applied use of temporal modeling techniques such as time-series analysis to take into consideration long-term behavioral changes, thereby reducing misclassifications.

Table 1 MADD-Net versus baseline models

Metric	Text-only	Image-only	Interaction-only	Multi-modal (no attention)	*p*-value	95% CI improvement	MADD-Net
Accuracy (%)	78.4	72.1	70.3	82.5	0.0164	(+3.14%, + 17.70%)	**89.6**
Precision (%)	75.3	68.9	66.8	81.2	0.0168	(+4.86%, + 27.86%)	**88.7**
Recall (%)	76.5	67.4	65.2	83.1	0.0167	(+5.37%, + 30.63%)	**90.3**
F1-score (%)	75.9	68.1	66.0	82.1	0.0157	(+1.82%, + 9.88%)	**89.5**
ROC–AUC	0.82	0.74	0.71	0.87	0.0157	(+1.82%, + 9.88%)	**0.92**

Bold values in the table are the result achieved

Table 2 Performance improvement of MADD-Net before and after hyperparameter tuning

Metric	Before tuning	After tuning	Improvement (%)
Accuracy (%)	85.2	**89.6**	+ 4.4
Precision (%)	82.5	**88.7**	+ 6.2
Recall (%)	84.1	**90.3**	+ 6.2
F1-score (%)	83.3	**89.5**	+ 6.2
ROC–AUC	0.87	**0.92**	+ 0.05
Training time (per epoch)	35 min	**28 min**	− 7 min

Bold values in the table are the result achieved

Table 3 Statistical validation of MADD-Net's performance improvements over baseline models using *p*-values and 95% confidence intervals

Baseline model	*P*-value	95% confidence interval (ci) improvement
Text-only	0.0164	(3.14%, 17.70%)
Image-only	0.0168	(4.86%, 27.86%)
Interaction-only	0.0167	(5.37%, 30.63%)
Multi-modal (no attention)	0.0157	(1.82%, 9.88%)

5 Study Limitations

MADD-Net shows remarkable performance in multi-modal depression detection, but soon faced certain limitations which stopped further advancement. These include generalizability/dataset bias due to cockling-causing differences in age, culture, language, and more. In addition, issues dealing with ethics and privacy arise while using social media in the depression detection technique. This could introduce false negatives or positives, which are yet to be elaborately investigated. There are limitations in modeling long-term behavioral patterns, which could lead into making future models temporal modeling approaches. The uneven training dataset in the case of depression datasets can cause the model to reflect a bias toward the majority class. Other ways of mitigating this through the use of GANs or SMOTE over-sampling techniques could be adopted, but the eternal challenge remains as to the detection of the minority class.

6 Conclusion

MADD-Net is heterogeneous in the detection of depression and achieved 89.6% accuracy, excelling among all base models. It has a cross-attention fusion mechanism that balances modalities while addressing shortcomings in single modalities. Additional improvements include mixed-precision training and gradient-checkpointing

optimizations, providing real-time clinical and large-scale social media deployment. Future versions should also consider large samples from linguistically and culturally diverse data sets, model personalization, and video-based features to enhance prediction power. With regard to privacy and ethics, responsible encryption methods, federated learning, and bias mitigation should be incorporated in the work. Employing sequential time-series modeling for monitoring long-term behavior motifs whose projection would eventually turn MADD-Net into an efficiently evolved model with a sense of AI-assisted mental health vigilance.

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Transforming Healthcare with AI: A Comprehensive Review of Innovations in Tropical Disease Management and Cancer Therapy

Anand Kumar and Vasu

Abstract AI is transforming healthcare, tackling tropical diseases, and advancing cancer care. In tropical regions, machine learning models like neural networks improve diagnostics by spotting patterns in clinical data, enabling early treatment in areas with limited resources. Yet, challenges like biased datasets and regional disparities limit its reach. In cancer care, AI enhances precision medicine by predicting drug responses and personalizing treatments. Deep learning automates imaging and streamlines diagnostics, while in immunotherapy, it tailors treatment plans based on patient responses, improving outcomes and efficiency. Despite its promise, AI adoption faces hurdles, including ethical concerns, data privacy issues, and unequal access. Collaboration among healthcare providers, tech experts, and policymakers is key to overcoming these barriers. By addressing these challenges, AI can reshape medical care, particularly in underserved areas and complex conditions, fostering a more equitable and effective healthcare system.

Keywords Artificial intelligence · Machine learning · Cancer · Tropical disease · Malaria · Dengue · Tuberculosis · Typhoid · Oncology

1 Introduction

There is a growing reliance on computers for decision-making in various application domains. Artificial intelligence (AI) refers to intelligence shown by machines developed by human brains. AI can be referred to as the collection of computer science, cybernetics, neurophysiology, psychology, and linguistics. It is believed

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that AI originated at the Dartmouth Conference in 1956. After decades of development, AI has become the general name of machine learning, deep learning, artificial neural networks, soft computing, and other technologies [1–3]. Traditional computing develops exact models using symbolic logic and numerical reasoning, whereas on the other hand, soft computing uses approximate reasoning and modeling [4]. Deep learning can automatically extract relations from massive datasets and find information in images that human eyes cannot. ML techniques are primarily used in medical diagnosis due to their ability to dynamically improve themselves from past results [5]. AI can help in the diagnosis of tumors as human thinking mode can be stimulated by multi-layer neural networks with strong reasoning and learning abilities [6, 7].

Medical diagnosis of any disease involves the determination of a condition or disease by analyzing the patient's symptoms and signs [8]. Nowadays, many computerized medical diagnoses are introduced like computer tomography (CT Scan). Tropical diseases are diseases that reproduce rapidly in hot and humid weather. They are mostly infectious. Tropical regions provide a conducive environment for the growth of pathogenic and contagious agents. This region mainly experiences heavy rainfall, high temperatures, and high humidity [9, 10]. Tropical disease includes transmissible and non-transmissible diseases, diseases resulting from nutritional deficiencies or environmental conditions, and genetic disorders in the areas [11]. Most of the common tropical diseases include malaria, diarrhea, typhoid fever, measles, Lassa fever, tuberculosis, yellow fever, dengue fever, Ebola, Marburg virus, COVID-19, measles, pneumonia, hepatitis, zika-virus, and influenza. The confusing nature of tropical diseases and the challenges of diagnosis and treatment of these diseases make it a challenging task for frontline health workers in low and middle-income countries. On the other side of this paper, cancer is a deadly, complex disease involving uncontrolled growth of abnormal cells within the body. It calls for early detection and treatment, which improves the chances of survival and enhances quality of life. Artificial Intelligence, especially machine learning, has emerged as an invaluable tool in cancer research and healthcare. ML algorithms recognize patterns and biomarkers for early diagnosis, predict patient outcomes, and optimize treatment plans by analyzing vast datasets. Additionally, ML-driven techniques help in precision medicine, which tailors therapies according to individual genetic profiles, holding out hope for more effective and targeted cancer care.

This article serves as a review aimed at addressing the significant knowledge gaps in today's medical science by providing a comprehensive analysis of the potential of artificial intelligence (AI) as a medical tool for diagnosing diseases, including tropical diseases and cancer. The article aspires to discuss current breakthroughs, describe the challenges involved, and present various options for advocating the use of AI in healthcare institutions. The research encourages the adoption of this technology to support those seeking innovative solutions and to promote the democratization of healthcare delivery across various medical conditions.

2 Result and Discussion

2.1 Tropical Diseases

Infectious diseases are prevalent in tropical regions, but much less so in temperate zones. The prevalence of these diseases, commonly known as tropical diseases, is largely influenced by a combination of biological and environmental factors. These factors enable a diverse range of vectors, pathogens, and hosts to exist, while social factors further complicate disease management and control. Many tropical infectious diseases, particularly febrile illnesses, exhibit overlapping symptoms, which makes clinical differentiation challenging. This symptom overlap leads to difficulties in diagnosis and increases the likelihood of misdiagnosis.

The early identification of tropical diseases and their associated mortality rates presents significant challenges for clinicians, hospitals, and communities. In this context, machine learning and soft computing techniques show great promise. These advanced computational methods can assist healthcare professionals in making timely and accurate diagnoses, reducing diagnostic errors, and ultimately improving patient care.

The ML technique has been applied in three medical areas, which are as follows:

1. Care Management of febrile diseases
2. Discovering host relationships in the cell
3. Diagnosing.

Care management of febrile diseases: Keitel et al. [12] examined the necessity for innovative approaches to the diagnosis and management of pediatric fever within primary care settings. They provided a comprehensive overview of existing electronic clinical decision algorithms (eCDAs) and assessed their validity and effectiveness through systematic testing. Their findings suggest that eCDAs serve as valuable resources, enhancing the treatment of fever and promoting the appropriate utilization of diagnostic tests and therapeutic interventions. Furthermore, they propose that future research efforts should continue to explore and develop additional strategies for the diagnosis and treatment of patients.

Discovering host relationships in the cell: Agany et al. [13] investigate the concepts of machine learning (ML) and data mining to enhance the understanding of vector–host pathogen relationships, particularly in terms of adaptation and pathogenicity. The majority of studies reviewed predominantly employed supervised machine learning techniques to develop predictive models, while others explored deep learning and unsupervised methods. Furthermore, classification and prediction of relevant features influencing outcomes emerged as some of the most frequently utilized machine learning techniques in the existing literature.

Diagnosing: The diagnosis of major tropical diseases represents a prominent topic within the extensive body of global research. Numerous machine learning (ML) algorithms have been employed in the diagnosis of tropical diseases, including but

not limited to malaria, dengue, and typhoid. A comprehensive review of various studies and the algorithms utilized in these contexts is provided below.

Malaria: Malaria constitutes a serious and potentially fatal disease caused by parasites that are primarily transmitted to humans through specific types of mosquitoes. This disease remains a significant source of illness and can be lethal, particularly in tropical regions. Individuals infected with malaria typically exhibit symptoms such as fever, chills, fatigue, muscle pain, and shivering [14]. Poostchi et al. [15] conducted a comprehensive survey on recent methodologies utilized for the diagnosis of malaria, emphasizing techniques such as image analysis and machine learning. The authors discussed both the latest advancements and the foundational aspects of the field, alongside the major transformations induced by deep learning. They noted that these technological advancements may render many traditional classification methods obsolete. Furthermore, the authors suggested that open-source cell segmentation techniques could ultimately be superseded by deep learning, which presents a promising solution for improving diagnostic accuracy. The progression of artificial intelligence and deep learning, coupled with the rapid development of affordable and reliable automated microscopy, could significantly enhance the detection of malaria. In their study on malaria fever diagnosis, Oguntimilehin et al. [16] examined and compared various machine learning models focusing on the diagnosis and treatment of the disease. The authors pointed out that the frequent lack of laboratory facilities and underdeveloped healthcare systems are key factors contributing to high mortality rates associated with malaria. Their research demonstrated that computer-based predictive models utilizing either patient symptoms or images of malaria parasites provide more effective diagnostic methods. However, they observed that medication recommendations were often excluded from most models and underscored the necessity for practical tests to document the entire treatment process. The authors proposed that the development of mobile applications for symptomatic diagnosis in low-resource settings could facilitate improvements in the healthcare system. Boruah et al. [17] conducted an extensive review of data mining techniques applied to enhance malaria prediction. The authors classified machine learning applications in healthcare into several domains, including treatment effectiveness, healthcare management, fraud and abuse detection, medical device development, systems biology, hospital management, and the pharmaceutical industry. Additionally, they categorized data mining techniques into various methodologies such as classification, clustering, association rule learning, regression, anomaly detection, summarization, time-series analysis, prediction tasks, and sequence discovery, highlighting their potential to improve healthcare processes associated with malaria.

Dengue fever, a disease caused by mosquito bites from one of four dengue virus strains, poses a significant challenge to global public health [18]. The illness can result in severe flu-like symptoms, and in extreme cases, it may lead to death. Currently, no effective prevention methods exist, which makes the development of soft computing and machine learning (ML) techniques a promising approach for accurate diagnosis. Dengue fever is one of the most prevalent tropical diseases and has been the subject of extensive research utilizing ML models for diagnostic purposes. Iqbal et al. [19] investigated the potential of ML methods for predicting dengue outbreaks through

comprehensive research. Their study encompassed a review of existing dengue viral prediction efforts and proposed an innovative ensemble classifier designed to enhance the accuracy of outbreak predictions. Sundari et al. [20] examined various environmental factors related to dengue, including temperature, rainfall, and sunshine. Their findings indicated that higher temperatures, alongside low frequencies of rainfall and sunshine, positively correlate with the risk of dengue fever. Sivaprasad et al. [21] contributed to the field by employing network analysis methods to establish an early warning system for dengue. They conducted cluster analysis using Gephi, a network analysis and visualization tool, on the citation network of relevant articles, which resulted in two main clusters: (1) the influence of climate change on mosquito-borne diseases and (2) research centers dedicated to dengue studies. Ahmed et al. [22] published a review of soft computing techniques for diagnosing dengue fever, addressing the effectiveness of expert systems in identifying the disease and fulfilling diagnostic requirements. Their study also considered the ability of these systems to provide user-friendly interfaces for a diverse range of users. The results highlighted the importance of knowledge-based models, which aid in diagnosing health issues and offering medical guidance by integrating data on diseases and symptoms. They suggested that user interfaces should align with users' mental models to enhance accessibility and usability, thereby fostering a more effective diagnostic experience.

Tuberculosis is caused by the bacterium Mycobacterium tuberculosis [23]. While the disease primarily affects the lungs, it can also target other parts of the body, and not all infected individuals become symptomatic. Several techniques are employed for its diagnosis. Weiner et al. [24] highlighted recent advancements in high-throughput tuberculosis detection. These methods aim to identify novel biomarkers essential for diagnosing, treating, and preventing TB. The study emphasized that tuberculosis manifests at various levels, underscoring the need for additional cohort studies to enable meta-analyses and develop universal, predictive biosignatures for TB. Similarly, Doshi et al. [25] explored the transformative potential of machine learning (ML) in tuberculosis management. They reported that the integration of ML into new software allows users to leverage AI-enabled pattern recognition for personalized patient care and customized training materials. Furthermore, the researchers concluded that mobile health approaches significantly enhance product effectiveness, urging the need for continuous technological updates in diagnostic tools.

Typhoid fever, caused by the bacterium Salmonella typhi, is a severe infection that can impact multiple body systems simultaneously [26]. The timely treatment of typhoid fever is essential to prevent serious complications and reduce the risk of mortality. In their literature review, Oguntimilehin et al. [27] explored computer-aided diagnostic (CAD) systems for managing typhoid fever. Their findings indicate that typhoid fever represents a significant public health concern in developing nations, contributing to alarming rates of mortality. The researchers argue that effective CAD systems must incorporate all critical components while successfully integrating innovative technologies into the healthcare sector. They also propose that this objective could be advanced through the implementation of web-based or mobile-based platforms.

Despite extensive research, no study has comprehensively examined all the infectious diseases caused by additives and tropical parasites. This review addresses this knowledge gap by discussing a wide range of tropical diseases worldwide and evaluating the role of machine learning models in their diagnosis. Several systematic reviews have explored tropical diseases from various perspectives. For instance, Akinsolu et al. [28] investigated resistance trends for 11 neglected tropical diseases (NTDs) and 20 treatments between 2000 and 2020, revealing the interdependence of socio-demographic characteristics and resistance patterns. Boyce, Katz, and Standley [29] conducted a systematic review of databases that identified risk factors for infectious diseases in urban areas of sub-Saharan Africa. Additionally, Elduma et al. [30] performed a meta-analysis on the Seroprevalence of the dengue virus in Sudan and utilized the findings to estimate the burden of the disease. While these reviews offer valuable insights, they concentrate on distinct subjects. The current review addresses the overlooked connections identified in these studies, providing a more comprehensive understanding of tropical diseases and the contribution of machine learning to their diagnosis.

3 Cancer

3.1 AI and Anticancer Drug Development

AI can help in predicting the activity of anticancer medicine and it can also help in developing that medicine. While using the same medicine on different cancers, it may show different reaction modes, and data after processing reveals the relationship between genomic variability of cancer cells and the medicine developed. Lind et al. [31] developed this model using random forest which uses screening data and can predict the activity of the medicine on the cancer cells. Wang et al. [32] developed a model which can predict the sensitivity of medicine on cancer cells known as elastic net regression. This model is proven very effective on ovarian cancer patients [33–35], gastric cancer patients [36–38], and endometrial cancer patients [39, 40]. All the patients predicted resistant with AI including ovarian cancer patients treated using tamoxifen, gastric cancer patients treated using 5-FU, and endometrial cancer patients using paclitaxel. All the patients were tested and proved to have poor prognosis. This research shows the achievement of AI in health sector. AI plays an important role in addressing medicine resistance in cancer [41–43]. AI can very quickly understand how cancer cells become resistant to anticancer medicine comparing to traditional techniques by learning and analyzing data on large dataset of cancer medicine and also helps in improving medicine quality and development. The workflow of anticancer drug development using AI is illustrated in Fig. 1.

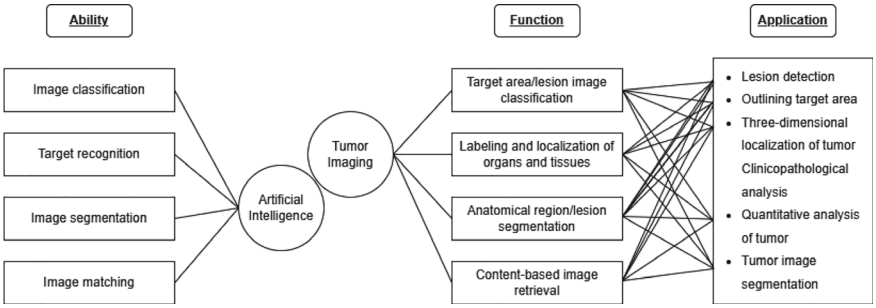

Fig. 1 Use of AI in tumor imaging

3.2 AI and Chemotherapy

In chemotherapy, AI focuses more on the response between the medicine and the patients. Management of chemotherapy medicine use, prediction of chemotherapy medicine tolerance, and optimization of chemotherapy program [44–47] are the major achievements of AI in chemotherapy. The process of combined chemotherapy can be accelerated with AI. In a research, a group of researchers successfully obtained the optimal dose of zen-3694 by using "CURATE.AI" (an artificial intelligence platform developed by the National University of Singapore using deep learning and other technologies), improving the overall combined treatment [48]. The poly ADP-ribose polymerase (PARP) inhibitors used to treat breast cancer cells with (HR) homologous recombination deficiency. Gulhan et al. [49] developed a deep learning model which can detect cancer cells with HR defect with an accuracy of 74% and also tells if the patient will have benefits or not of using PARP inhibitors. Dorman et al. [50] developed a machine learning algorithm which can predict the tolerance of breast cancer to chemotherapy. Research, which explored the relationship between chemotherapy medicines and cancer patient's genes, also can able to differentiate between the effects two chemotherapy drugs, taxol or gemcitabine. And also, from the years of research, it is revealed that the deep learning method is far better than Epstein-Barr Virus-DNA-based model in risk detection and induction chemotherapy for nasopharyngeal carcinoma [51], which means the guidance of deep learning can be used as main reference to predict single induction chemotherapy for advanced nasopharyngeal carcinoma.

3.3 AI and Radiotherapy

In radiotherapy, the use of AI is more specific. AI helps radiologist to target the affected area or automatically plan the radiation region for treatment [52–54]. Lin et al. [55] developed a model based on three-dimensional convolution neural network

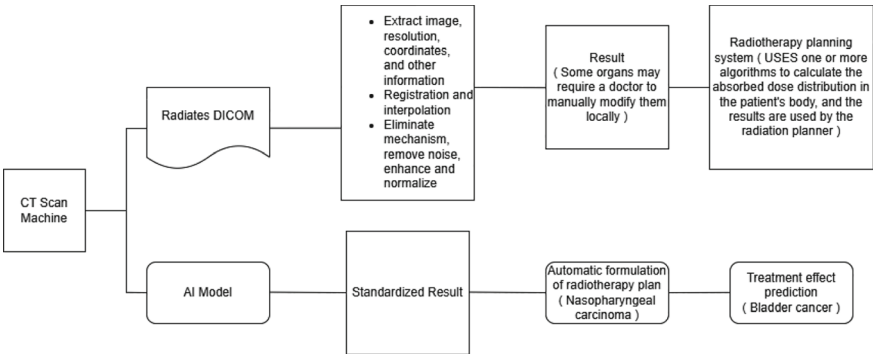

Fig. 2 Flow chart of CT scan machine

(3D CNN) which can outline the affected area for nasopharyngeal carcinoma with an effective accuracy of 79% equivalent to which specialists do radiotherapy. Cha et al. [56] developed a predictive model using deep learning and radiomics which can generate a response to treat bladder cancer. Babier et al. [57] developed a automation software which can reduce the time to plan a radiation therapy to only few hours. The treatment plan which is generated by the AI is much lesser comparable to the traditional method. AI models have been applied to automate radiotherapy planning and treatment effect prediction, as shown in Fig. 2.

3.4 AI and Immunotherapy

In Immunotherapy, AI focuses more on evaluate treatment effect and physicians in adjusting the treatment plan [58–61]. Sun et al. [62] developed a platform to predict accurately the therapeutic effect of programmed cell death protein 1 (PD-1) inhibitors using machine learning. With this platform, the effect of immunotherapy is evaluated effectively in patients with advanced solid tumors which are sensitive to PD-1 inhibitors. Bulik-Sullivan et al. [63] developed a machine learning model based on human leukocyte antigen (HLA) with mass spectrometry database, which helps in better identification of cancer neoantigen and improvement in cancer immunotherapy.

3.5 AI Reduces Cancer Overtreatment

Hu et al. developed an algorithm that analyze the digital image of woman's cervix and determine the region to treated, so that patient don't get over-treated. Bahl et al. developed a tool using machine learning which helps in reducing the overtreatment on the regions of breast cancer. It also helps doctors in find out the regions which

will eventually become cancer, overall helping doctors in better decision-making and overtreatment.

3.6 AI and Clinical Decision Support System

In clinical Decision, AI has an edge as it makes more intelligent decisions. After learning from data of cancer patients, AI generated more suitable treatment plan for doctors. Printz et al. developed a Clinical Decision Support System (CDDS) which is based on machine learning, it extracts and process a large amount of data form clinical records and generate an optimum cancer treatment option. This study shows how effective AI is in this sector and helping doctors to find patient's treatment.

3.7 Machine Learning and Deep Learning in Anticancer Drug Development

Machine learning algorithms trained on high high-throughput data to develop models which can predict the response and sensitivity of cancer cells in patients to test the new medicine or combination of medicines. With reverse synthesis pathways for molecule, scientists use machine learning to discover a new drug. A lot of data is produced in the process of making new medicine. With machine learning, it is possible to process chemical data and create results that will help in making of medicine development. Machine learning can also process the data which is being collected from decades and over the years in a very short amount of time. It also helps in better decision-making comparing to which will be made through test and predictions. Deep learning is sub-field of machine learning and has achieved many top tier results in every sector, including medicine discovery. These types of models are more suitable for complex operations like medicine discovery and development based on many features they are presented with. But the application of medicine response prediction with deep learning has recently been explored. Recently, it is explored that deep learning can be used to develop models which can extract information and insights from images like never before. While developing deep learning models based on large-scale data of any type has shown some interesting new reuse of a medicine. They developed a deep learning model which is being prepared by applying antagonistic autoencoder to the full dose–response measured in the NCI-60 cell line.

Shortcomings of Current Research and Applications

1. Limited Generalizability Across Diseases: Machine learning (ML) techniques have demonstrated strong performance in specific disease contexts; however, they cannot frequently be effectively scaled across different medical conditions. This limitation has been highlighted in studies concerning tropical diseases [10,

28] and various cancers [42]. There remains a significant gap in the exploration of models that can be universally applied across diverse health conditions.

2. Data Quality and Accessibility Issues: Numerous ML-based studies rely upon high-quality datasets that do not adequately reflect real-world conditions, particularly in resource-limited settings [14, 17]. Furthermore, the limited availability of robust datasets in the domains of tropical medicine and infectious diseases constrains the performance and applicability of these models [22].
3. Ethical and Privacy Concerns: The incorporation of artificial intelligence (AI) in healthcare engenders significant ethical questions, particularly concerning data privacy and potential misuse. These concerns are exacerbated in resource-constrained environments where regulatory frameworks tend to be weak [23, 63].

   Case Study Follows: A hospital implemented an AI diagnostic system, but it quickly ran into several ethical and privacy issues. There was a data breach that compromised patient confidentiality, which raised serious concerns. Additionally, evidence of algorithmic bias surfaced, putting minority groups at a disadvantage. Clinicians also expressed worries about the system's lack of explainability, as it made it difficult for them to trust the technology fully. These challenges sparked important discussions about accountability, fairness, and compliance with privacy regulations like HIPAA and GDPR. This situation underscores the necessity for strong data governance, diverse datasets, and transparent AI models to promote ethical practices in healthcare.
4. Bias and Model Transparency: Many ML models are compromised by biases that arise from imbalanced datasets or a lack of diversity in input data. For instance, diagnostic models for tropical diseases [27] and AI applications in cancer treatment [6] have demonstrated susceptibility to regional and demographic biases, which could impede their global adoption.

   Case Study Follows: In a real-world example, an AI diagnostic tool designed for identifying skin cancer demonstrated significant biases because it was mainly trained on images of lighter skin tones. This resulted in misdiagnoses for patients with darker skin. Furthermore, the system's black-box nature made it challenging for doctors to interpret its decisions, resulting in a lack of trust. This situation highlights the critical need for diverse training data and greater transparency in AI models to promote fairness and ethical use in healthcare.
5. Over-Reliance on Retrospective Data: A considerable number of studies depend on retrospective datasets to train models, which may overlook the real-time variability inherent in disease progression and treatment outcomes [19, 32].
6. Challenges in Clinical Translation: The transition of ML models from research settings to clinical practice presents a substantial barrier. Models that predict dengue outbreaks [18] or identify tuberculosis biomarkers [24] often lack the rigorous validation required for regulatory approval and broader application in clinical settings.
7. Algorithmic Complexity versus Practicality: Advanced algorithms, including deep learning frameworks used for cancer radionics [52] and automated tumor

contouring [56], necessitate extensive computational resources that are often impractical for low-resource healthcare systems.

8. Lack of Standardized Validation Protocols: The methodologies for validation vary significantly across studies, complicating the comparison and integration of findings [2, 15]. This issue is particularly pronounced in predictive models for diseases such as malaria and dengue [16, 30, 37].
9. Emerging Drug Resistance: While AI has been utilized to predict drug resistance [25], many models do not adequately account for emerging resistance patterns in real time, which limits their effectiveness in combating infectious diseases like tuberculosis [22].
10. Overemphasis on Technology Over Usability: Numerous ML studies tend to prioritize advancements in algorithms without sufficiently addressing practical usability or the integration of these technologies into healthcare workflows, particularly in underserved regions [26, 58].
11. Scarcity of Longitudinal Studies: Most AI-based investigations rely on snapshot datasets, thereby neglecting longitudinal data that are critical for understanding disease evolution and patient outcomes over time [45, 57].
12. Underexplored Human-AI Collaboration: There is a notable lack of research that examines the collaboration between clinicians and AI systems, which may lead to resistance in adoption due to perceived threats to clinical autonomy [2, 59].

4 Conclusion

Artificial intelligence (AI) is significantly transforming the healthcare industry by enhancing the diagnosis, treatment, and management of chronic diseases, particularly in the realms of tropical diseases and cancer. By employing machine learning and deep learning techniques, AI facilitates precise diagnostics and optimizes treatment, propelling advancements in personalized medicine. However, several challenges persist, including biased datasets, data privacy concerns, and unequal access to technology in underserved regions. Collaboration among healthcare professionals, technologists, and lawmakers is crucial to ensure equitable access to AI-driven solutions. This analysis underscores the substantial impact of AI on medical services and highlights the necessity of interdisciplinary innovation to promote accessible healthcare for all populations. Furthermore, the integration of technological advancements into ethical frameworks is essential for fostering a more inclusive global healthcare system. The authors aim to elucidate the role of AI in healthcare and investigate strategies to enhance training methods for the development of more advanced models.

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Brain Tumor Classification Using a CNN-Driven Deep Learning Framework Leveraging MRI Imaging

Divanshi, Tarunpreet Bhatia, and Neha

Abstract Brain tumors continue to be the most pressing health challenges, and accurate and timely diagnosis is crucial for a positive clinical outcome. Deep learning methods, particularly convolutional neural networks (CNNs) for detecting and classifying brain imaging data, have been shown to outperform more common imaging modalities like X-rays. However, these models are limited to the availability of MRI data which is relatively sparse. In this study, two benchmark datasets were employed, one with 253 images for binary classification (tumor, non-tumor) and a larger dataset with 3264 MRI images for multiclass classification to identify various tumor types. Various data augmentation and preprocessing techniques are employed improving the image quality and feature extraction. This approach achieved a classification accuracy 96.10% in binary classification and 91.24% in multiclass classification. The experimental results indicate that integration of CNN along with improved preprocessing methods provides better accuracy and speed for the identification of brain tumor. This approach has the potential for clinical applications, providing more rapid and robust diagnostic assistance.

Keywords Brain tumor detection · Convolutional neural networks · Deep learning · MRI imaging

1 Introduction

Our brain is one the important organs which controls many functions so we can survive. Nevertheless, it may be hampered in these duties by many conditions, with one of the most serious threats, tumors. Brain tumors are classified in 2 types—benign

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(non-cancerous) or malignant (cancerous) tumors. Among the most common are gliomas, meningiomas, and pituitary tumors. These tumors grow and exert pressure on healthy brain tissue, causing swelling and compression that lead to neurological symptoms as their size and impact increase. The statistics around brain tumors are sobering. Globally, the World Health Organization's 2020 World Cancer Report placed brain cancer as the 17th most common cancer, with nearly 297,000 new cases reported worldwide. Brain imaging is usually done with CT scans or MRIs. CT scans are great for getting detailed 2D images of the brain, especially for looking at bones, while MRIs give a clearer, 3D view of soft tissues. Since MRI scans are the most sensitive method of brain imaging currently available, they are preferable for imaging the brain because they offer direct multi-planar imaging, improved contrast, decreased bone artifacts, and volume-related problems [1, 2].

Identifying brain tumors can be a time-consuming, complicated affair, largely dependent on a doctor's expertise and a manual assessment of symptoms and test results. Due to lack of automated tools to aid in diagnosis and a dearth of medical professionals, patients have to often wait long periods of time, which are at risk of receiving misdiagnosed treatment and delayed care. Today, with the help of deep learning, researchers can generate a diverse and wide range of extensive feature sets, which would be difficult even for a specialist [3]. These technologies detect patterns and details in complex data, such as medical images. This aids doctors in making treatment decisions more promptly. Therefore, patients receive care sooner, which may improve their health and lower healthcare costs.

This paper looks at how DL can help detect brain tumors in MRI images more accurately and discusses the challenges faced in earlier studies. Our goal is to make a system that can detect tumors of brain with expert-level accuracy, making the diagnosis process faster and more reliable, and ultimately helping improve patient outcomes. The proposed model has been experimented on two benchmark datasets namely tumor and non-tumor images for binary classification, and three types of tumors with no-tumor class for multi-class classification tasks. Our study's main contributions are as follows:

1. A preprocessing pipeline was introduced to enhance relevant features in MRI images that improved the ability of model to learn robust patterns. For binary dataset, Gaussian blur was applied to reduce noise and smoothen the high-frequency variations for improving the accuracy of thresholding, erosion, and dilation. The images were then transformed into grayscale and after being thresholded and underwent refinement through morphological operations.
2. For the multi-class dataset, a bilateral filter was used to smooth the image while preserving edges, which was beneficial for differentiating tumor bound areas. Sharp intensity changes were accentuated which improved edge detection while the Scharr filter aided in spotting sharp contrast boundaries. The Frangi filter also helped in increasing the presence of unusual growths by making vessel-like structures more visible. Also, wavelet transform was used for different scale feature extraction at different resolutions for boosting classification accuracy by capturing more tumors.

3. The datasets were augmented before training. The datasets were divided into 80% for training and 20% for testing. Due to the small size of the binary dataset, the class probability was made more uniform with the help of rotation techniques, rescaling, zooming, and brightness adjustments, which expanded the dataset to 2065 images. In order to make the model more robust, the multi-class dataset was also augmented and increased from 3264 images to 11,748 images.
4. CNN architecture is introduced that is fed with an input layer and an output layer tailored for classification. The output layer is configured with two nodes for binary classification and four nodes for multi-class classification, aligning with number of classes in the MRI dataset.

The structure of the paper is as follows: Sect. 2 presents a comprehensive literature review. Section 3 describes the datasets involved and key techniques and methods. Then comes Sect. 4 wherein the analysis of results is done against the best available models in the literature. Finally, Sect. 5 summarizes the entire paper and discusses future research which can be done.

2 Literature Review

In recent years, the use of artificial intelligence (AI), particularly deep learning, has significantly increased in the analysis of medical images [3]. Adu et al. [4] proposed capsule network for brain tumor classification and worked on resolution enhancement to capture contextual information using dilation. Their method also cuts down computational complexity and training time. The model received the patch of the MRI and they extracted the features using the capsule layers and finally classified the tumor into meningioma, glioma, and pituitary using SoftMax classifier. Choudhury et al. [5] used a CNN model with 3 layers with a 3×3 kernel size and 2×2 maxpooling. The model uses ReLU for faster convergence, Tanh for optimization help because output has more weight toward zero, and sigmoid for value mapping from 0 to 1 via an S-shaped curve. Ayadi et al. [6] presented a model that included ten convolutional and batch normalization layers, five maxpooling layers, and a SoftMax classifier. Preprocessing included intensity normalization and contrast adjustment, and the model was trained using a fivefold split. Rasool et al. [7] used Google-Net model (GN-FT) to fine tune MRI brain images. They extracted features through the convolution layers of Google-Net and then classified images using the classifier support vector machine (SVM) into four categories (glioma, pituitary, meningioma, or normal tissue). The extracted features were implemented with the SVM algorithm through the fully connected layer for classification. Mostafa et al. [8] have used preprocessing methods, including (flipping, rotation, scaling, and noise addition). Moreover, they obtained an additional performance boost by incorporating the new Segment Anything Model (SAM) for segmentation. Zahoor et al. [9] expanded the dataset using data augmentation to increase the generalization ability of the model.

There has been a surge in the utilization of deep learning (DL) methodologies to construct automated systems that possess the ability to effectively diagnose or segment brain tumors within a reduced timeframe [10]. Guo et al. [11] compared the proposed method, combining Residual/Shuffle Networks with the AFFO optimization technique, with well-known models like ResNet, AlexNet, VGG16, Inception V3, and U-Net for brain tumor diagnosis. Billah et al. [12] employed transfer learning with VGG19 and InceptionV3, along with image-processing techniques like resizing, normalization, random rotation, flipping, rescaling, zooming, and shear transformations. Muftic et al. [13] evaluated CNN models, including EfficientNetB3, ResNet50, and VGG19. Wahlang et al. [14] utilized spatial-based, depth-based, and feature map-based CNN techniques to automate brain tumor detection. The study evaluated various CNN models, including LeNet, AlexNet, VGG16, VGG19, ResNet50, ResNet101, ResNext50, ResNext101, InceptionV3, Xception, DenseNet121, MobileNet, MobileNetV2, NASNet, and SE-ResNext101, using a generalization method for performance comparison. Rout et al. [15] developed a model that was trained on a dataset of over 3000 MRI images across four classes and their model showed an accuracy of 88.44%. Saad et al. [16] presented a model that uses a Swish activation function in a CNN architecture with three Conv2D layers, BatchNorm2D, and a sigmoid function in the final layer for stability. Additionally, an MLP with three fully connected layers, Swish activation, and dropout is employed to prevent overfitting.

3 Methodology

The methodology involved several key steps, beginning with data augmentation, followed by data preprocessing. The model was then trained using our datasets, and its performance was evaluated using metrics such as accuracy, F1-score, recall, and precision. Figure 1 illustrates the architecture of the proposed model and its components.

3.1 Dataset Description

Two datasets were used. The first dataset contained only two classes, tumor and non- tumor images for binary classification [17], and the second dataset contained three types of tumors basically glioma, pituitary, and meningioma along with no-tumor class for multi-class classification tasks [18]. The first dataset consists of 253 images, containing two classes where 155 images are tagged as “tumor” and 98 as “no tumor.” All images have been resized to the size 240 × 240 so that all pictures in any further processing could have standardized dimensions. The second dataset consists of four classes: glioma, pituitary, meningioma, and no tumor, containing 3264 images. The datasets were split into two portions: 80% for training and 20%

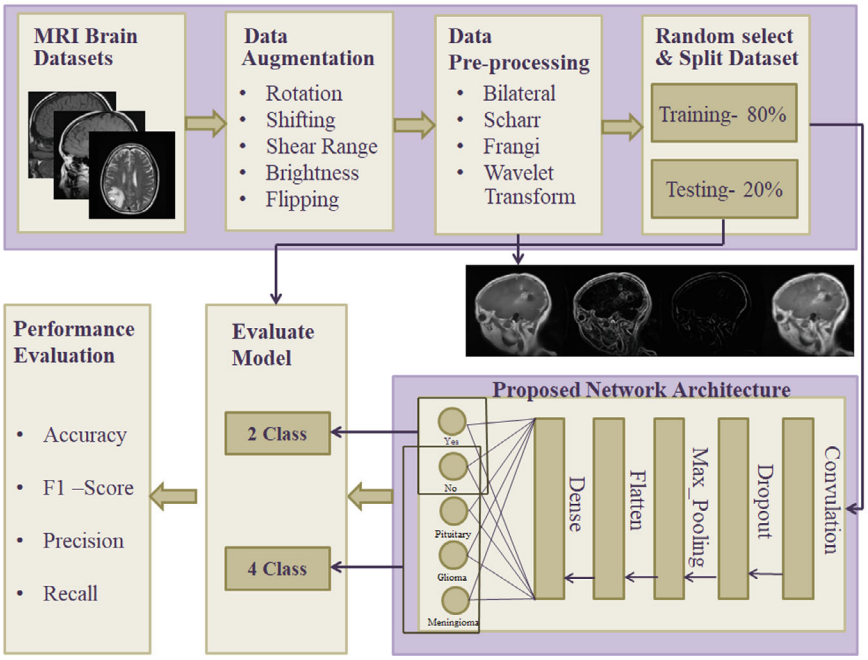

Fig. 1 Architecture of proposed model for brain tumor detection

for testing. All images have been resized to the size 150 × 150 dimensions so that all pictures in any further processing could have a standardized size.

3.2 Data Augmentation

Deep learning models, especially convolutional neural networks, are quite expensive and difficult to train because of the severe scarcity of large well-labeled datasets. This often cripples most models in terms of performance. Data augmentation artificially enlarges an existing dataset by transforming images in different ways such as rotation, flipping, shift, resizing etc. In addition, an image augmentation approach can solve problems associated with class imbalance whereby some classes had less data for effective training. For such classes, augmentation ensures balanced and improved performance for the model.

To enhance the dataset, various image augmentation techniques are applied. Images are randomly rotated by up to 10 or 20 degrees in either direction, adding slight variations to their orientation. They are also shifted horizontally and vertically by up to 10% of their dimensions, introducing a natural sense of movement. A gentle

shear effect of up to 10% creates a subtle slant in the images, while brightness adjustments range from 30% (making the images darker) to 100% (keeping the original brightness). Addition- ally, the images are flipped both horizontally and vertically to increase diversity. Any gaps created during these transformations are filled using the "nearest" pixel values, ensuring a seamless appearance. As the number of images is very small, data augmentation techniques have been used to change the size of dataset to 2065 images. The augmentation of 6 times on the tumor and 9 times on the non-tumor is applied so that images are almost equal. Although the second dataset is large in size, we still increased the variability of the dataset by various augmentation techniques to make our multi-class model more robust. With data augmentation, size of dataset increased to 11,748 images.

3.3 Data Preprocessing

In our proposed model, for improving the binary-class dataset, the input image is converted to grayscale and Gaussian blur is applied to reduce noise. The image is then thresholded, and several erosions and dilations are applied to eliminate small noise regions. The extreme points top, bottom, left, and right of the contour are determined from this. These points are then used to crop the image focusing on the most relevant region. In this way, images of the dataset became more precise, and ready for further analysis.

For multiclass datasets, we used bilateral filter, which smoothens the image while keeping edges sharp, perfect for reducing noise without losing important features. Scharr filter is then applied as it highlights sharp intensity gradients with much more refined edge enhancement. To highlight vessel-like structures, the Frangi filter is used which is particularly good at emphasizing elongated, tubular features like blood vessels or abnormal growths. Finally, multi-resolution information is captured using wavelet transform with Haar wavelets, offering insights at different scales. Wavelet transform gives us a much richer, multi-scale view of the image, which is crucial for accurately analyzing tumor structures.

3.4 Testing and Training Data-Split

Initially, the binary classification dataset was composed of 253 images, with 155 labeled as "tumor" and 98 as "no tumor." After applying data augmentation, the dataset increased to 2065 images, with 930 being images of the "tumor" class and 882 of the "no tumor" class. The dataset was further split into training and testing proportions, in an 80:20 ratio. The number of images in the training set was 1445 while that of the testing set was 610 images.

There were 3264 images in multi-class classification that comprised the original data set with four classes. These included 926 images labeled as "glioma tumor," 937 as "meningioma tumor," 500 as "no tumor," and 901 as "pituitary tumor." After applying data augmentation, the size of the data set increased to 11,748 images. The entire dataset was split into training and test sets in an 80:20 ratio. The training set has 9398 images, while the test set has 2350 images.

3.5 Proposed Network Architecture

For binary classification, an architecture of CNN has been proposed (as shown in Table 1) for brain tumor detection with a custom pipeline of feature extraction. This model initiates with an input layer that accepts images of 240 × 240 pixels size with 3 color channels as RGB. It is then followed by a ZeroPadding2D layer to preserve the spatial dimensions of the input image. Then comes the Conv2D Layer with 32 filters of size 3 × 3 in order to extract key features from the image, which can be edges and patterns, and the BatchNormalization Layer for normalization of the output of the convolution operation, hence stabilizing the training process. ReLU activation layer was added that introduces non-linearity that enables the model to learn complex representations. The architecture comprises maxpooling2D layers, which reduces the spatial dimensions while retaining important features. The first pooling layer downsampled the feature maps to a size of 59 × 59, and another layer downsizes further to 14 × 14. These steps help to efficiently extract features and reduce the dimensionality. The output is sent to a flattened layer after the convolution and pooling layers, which flattens the multi-dimensional feature map into a one-dimensional vector. The final output is then produced by a fully linked (dense) layer once this flattened vector has been passed through. The output layer is a single node, which is designed for binary classification (tumor versus no tumor) with an appropriate activation function, such as sigmoid.

Table 1 Layers of our proposed binary-class classification model

Layer (type)	Output shape	Param #
input_layer (InputLayer)	(None, 240, 240, 3)	0
zero_padding2d (ZeroPadding2D)	(None, 244, 244, 3)	0
convo (Conv2D)	(None, 238, 238, 32)	4736
bno (BatchNormalization)	(None, 238, 238, 32)	128
activation (Activation)	(None, 238, 238, 32)	0
max_pool0 (MaxPooling2D)	(None, 59, 59, 32)	0
max_pool1 (MaxPooling2D)	(None, 14, 14, 32)	0
flatten (Flatten)	(None, 6272)	0
fc (Dense)	(None, 1)	6273

There are 11,137 parameters in the final architecture. It is then trained on the "Adam" optimizer to reduce the loss function that is defined as "binary_ crossentropy." Batch size is of 32 and training is done for 30 epochs. The activation function for the intermediate layers is ReLU, and sigmoid in the output layer. In this, the best model with the maximum accuracy in training is saved.

For multi-class classification, the proposed model is designed based on a sequential architecture (as shown in Table 2). It begins with a Conv2D layer and ReLU activation. The model then adds another Conv2D layer. Following this, a maxpooling2D layer is added. It contains dropout layers at a rate of 0.5, to avoid overfitting during the training process, which randomly disables some neurons. The architecture continues with more Conv2D layers along with maxpooling2D to further downsample. With the progression of the model, it uses Conv2D layers with more maxpooling2D and dropout layers are used to retain the most important features and prevent overfitting. Then, the model increases complexity by using Conv2D layers with 128 and 256 filters. After the feature extraction step, it is flattened as a one-dimensional vector and then is passed through two fully connected dense layers having 512 nodes and ReLU activation with one dropout layer for reducing overfitting. Lastly, it uses a final dense layer consisting of 4 nodes to predict the class probabilities of each of its 4 classes by a SoftMax activation function. This architecture provides the incorporation of convolutional layers for feature extraction along with dropout that performs regularizations, and downsamples with maxpooling. SoftMax activation at the output layer does the work to output in the form of probabilities.

The final architecture consists of 25 layers with 4,447,044 parameters in total. The "Adam" optimizer is used in training to minimize the loss function, which is defined as "categorical crossentropy." Batch size of 32 is trained, and 200 epochs are used for training. "SoftMax" is applied to the final dense layer, while ReLU is used as an activation function to all other layers. The best-trained model based on the highest accuracy is saved.

4 Experimentation and Result

Results show that in comparison with prominent architectures of VGG16 and ResNet50 in evaluating the two independent brain MRI datasets, the proposed model has shown improvements in classification measures of accuracy, precision, recall, and F1-score for both datasets. These results confirm the model's strength and reliability in classifying brain tumors.

For the binary classification dataset, the proposed model achieved an impressive test accuracy of 96.1% during testing. It also maintained strong and consistent performance measures, with an F1-score of 0.95, precision of 0.95, and recall of 0.96. These results highlight the model's ability to correctly identify tumor cases while minimizing false positives and false negatives. Such balanced outcomes show that the model is not biased toward either class, making it suitable for practical use

Table 2 Layers of our proposed multi-class classification model

Layer (type)	Output shape	Param #
conv2d (Conv2D)	(None, 148, 148, 32)	896
conv2d 1 (Conv2D)	(None, 146, 146, 64)	18,496
max_pooling2d (MaxPooling2D)	(None, 73, 73, 64)	0
dropout (Dropout)	(None, 73, 73, 64)	0
conv2d_2 (Conv2D)	(None, 71, 71, 64)	36,928
conv2d_3 (Conv2D)	(None, 69, 69, 64)	36,928
dropout_1 (Dropout)	(None, 69, 69, 64)	0
max_pooling2d_1 (MaxPooling2D)	(None, 34, 34, 64)	0
dropout_2 (Dropout)	(None, 34, 34, 64)	0
conv2d_4 (Conv2D)	(None, 32, 32, 128)	73,856
conv2d_5 (Conv2D)	(None, 30, 30, 128)	147,584
conv2d_6 (Conv2D)	(None, 28, 28, 128)	147,584
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
dropout_3 (Dropout)	(None, 14, 14, 128)	0
conv2d_7 (Conv2D)	(None, 12, 12, 128)	147,584
conv2d_8 (Conv2D)	(None, 10, 10, 256)	295,168
max_pooling2d_3 (MaxPooling2D)	(None, 5, 5, 256)	0
dropout_4 (Dropout)	(None, 5, 5, 256)	0
flatten (Flatten)	(None, 6400)	0
dense (Dense)	(None, 512)	3,277,312
dense_1 (Dense)	(None, 512)	262,656
dropout_5 (Dropout)	(None, 512)	0
dense_2 (Dense)	(None, 4)	2052

where sensitivity and specificity are vital. This consistency across multiple evaluation measures validates the framework's stability and offers strong evidence of its potential effectiveness in clinical settings.

Similarly, the evaluation of the multiclass classification dataset, as shown in Table 3, emphasizes the model's effectiveness. The proposed model achieved an overall accuracy of 91.25% on the test set, with an F1-score of 0.91, indicating a strong balance between precision and recall across the four classes. For individual classes, the model showed strong classification for the glioma class with an F1-score of 0.91. The no-tumor class achieved the highest F1-score of 0.95. For meningioma and pituitary tumors, the F1-scores were 0.89 and 0.88, respectively, reflecting decent performance in distinguishing these categories.

Table 3 Evaluation of our proposed model for multi-class classification

Model	Accuracy (%)	F1-Score	Precision	Recall
VGG16	90	0.90	0.90	0.89
ResNet50	91	0.91	0.92	0.91
Rout et al. [15]	88.44	0.88	0.88	0.89
InceptionV3	88.31	0.87	0.85	0.89
DenseNet121	73.44	0.85	0.88	0.83
Proposed model	91.25	0.91	0.91	0.91

Our proposed model for multiclass brain tumor classification performs better than benchmark models such as VGG16 and ResNet50. These results indicate that the model can be reliable and accurate, and therefore a useful tool for practice.

The model results, training, and loss over epochs are shown in Figs. 2 and 3. for binary-class classification, respectively.

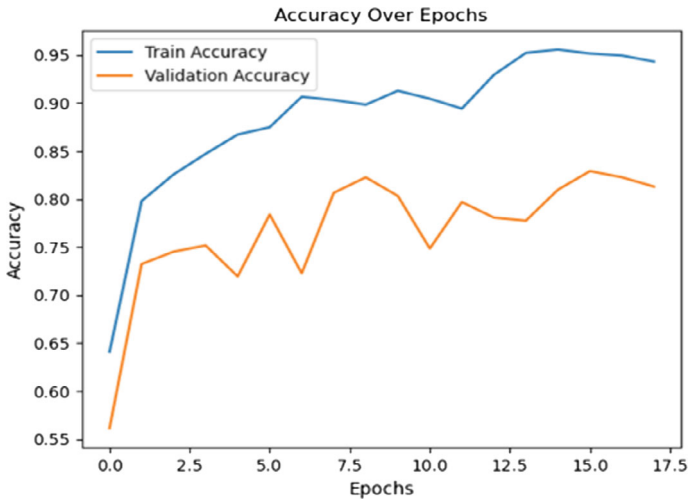

Fig. 2 Accuracy trends during training epochs for binary-class classification

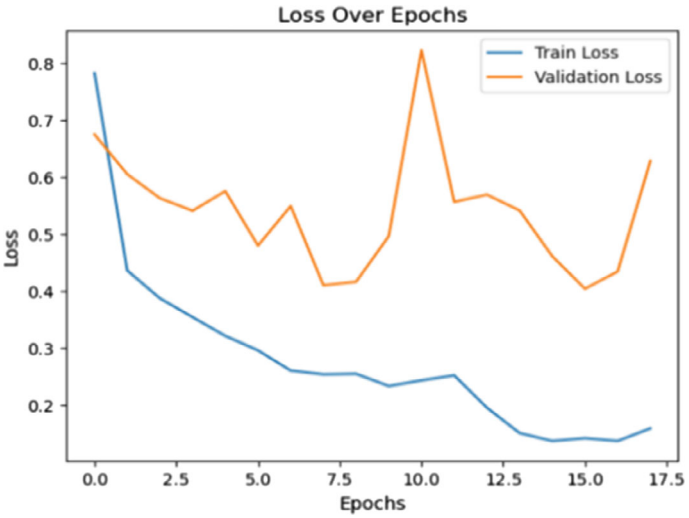

Fig. 3 Loss trends during training epochs for binary-class classification

Figure 4 and 5 illustrate the model's accuracy and loss trends over training epochs for multi-class classification, respectively.

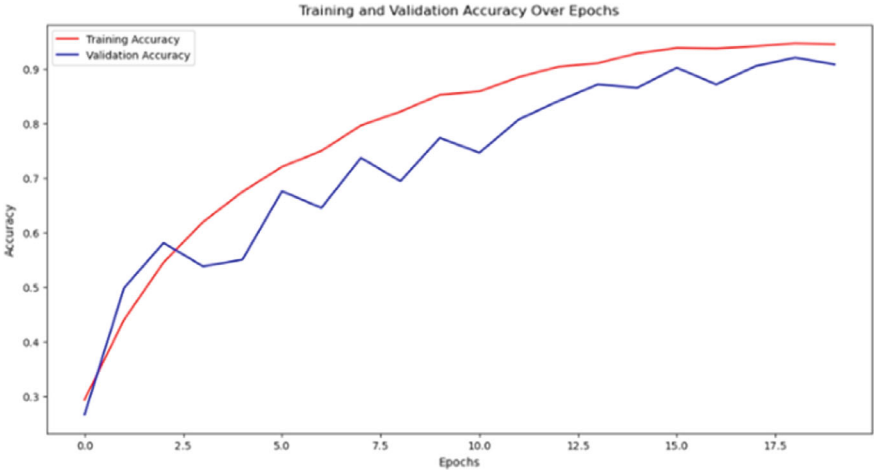

Fig. 4 Accuracy trends during training epochs for multi-class classification

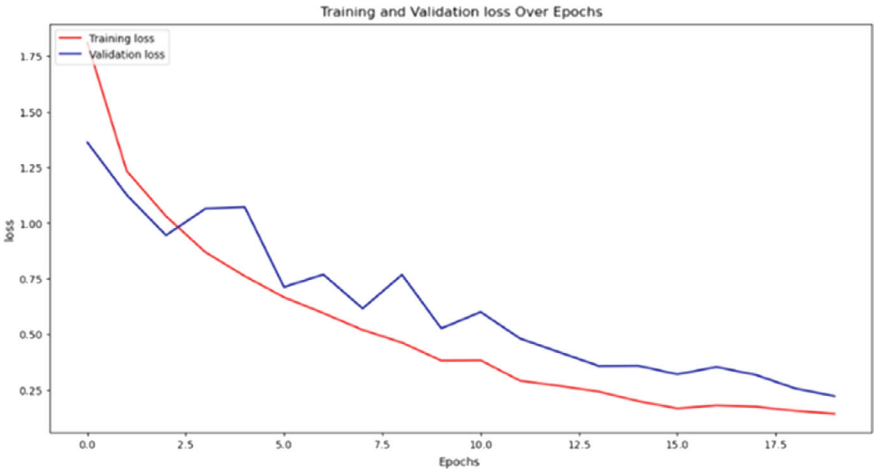

Fig. 5 Loss trends during training epochs for multi-class classification

5 Conclusion

In this study, an automated tool for diagnosing brain tumors, using advanced image-processing techniques combined with deep learning models, is developed. By enhancing key features in brain scan images, the model achieved great accuracies of 96.1 and 91.24% on two different datasets. The proposed model is compared to the current models. This shows that it is a dependable and useful approach. These results are promising, and while we believe the model works well, there's still room for improvement, especially by training it on a larger, more diverse dataset to help it generalize better and be more useful in real-world settings. One of the future plans in our work is to boost the size of the available dataset and try out different advanced neural network models toward further improving performance.

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Edge Detection in Medical Imaging via Clustering: A Quantum Computing Approach for Lung Carcinoma and Brain Tumor

Sayak Konar, Gunjan Mukherjee, Subrata Sinha, and Saurav Mali

Abstract This study presents a novel quantum-based edge detection algorithm integrated with clustering techniques, specifically designed for medical imaging applications. The methodology employs Flexible Representation of Quantum Images (FRQI) and Quantum Hadamard Edge Detection (QHED), enhanced with K-means clustering, to improve precision and robustness against noise. Experimental evaluations, using metrics such as F1-score, accuracy, and precision, demonstrate that the quantum approach achieves an F1-score consistently above 80% under noisy conditions, compared to 70–75% for classical methods like Sobel and LoG. Quantum edge detection also demonstrated higher image density at 106.72 compared to 85% (Sobel) and 83% (LoG). While the quantum model's processing time (0.176898 s) is longer than classical methods (e.g., Canny at 0.098943 s), auxiliary quantum processes such as clustering and density detection compensate with near-real-time efficiency. These results highlight the quantum algorithm's superior performance in noisy conditions, making it particularly suited for high-dimensional medical imaging tasks.

Keywords Quantum computing · Noisy image · Edge detection · Image density · Clustering

1 Introduction

Digital images are widely utilized in various aspects of daily life and numerous scientific fields, especially in medical diagnostics, where they have revolutionized accuracy and efficiency [1]. Medical imaging (MI) methods like MRI, CT, and X-ray provide clear and detailed views of internal organs such as the brain, heart, and lungs, aiding doctors in diagnosis, treatment planning, and follow-up care [2, 3]. When it

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comes to cancer diagnosis, histopathological imaging with H&E staining is still considered the gold standard due to its ability to identify abnormalities at the cellular level with high precision [4]. Edge detection is a key step in medical imaging, helping to segment and analyze complex structures within the body. However, traditional methods like Sobel and Canny have their limitations, particularly when dealing with noisy or intricate medical images, such as those of the lungs or brain [5, 6].

This is where quantum computing offers exciting possibilities. Unlike traditional computing, quantum systems use qubits and principles like superposition and entanglement, enabling them to process large datasets more efficiently and solve computational challenges [2, 3, 7, 8]. Advances in Quantum Image Processing (QIP), with methods like Flexible Representation of Quantum Images (FRQI), Quantum Probability Image Encoding (QPIE), and Novel Enhanced Quantum Representations (NEQR), have shown potential in transforming image representation by converting pixels into quantum states [5, 9–13]. While the Quantum Hadamard Edge Detection (QHED) algorithm has shown promise, it is primarily limited to binary images and struggles with noise in real-world conditions [14].

Many researches have been going on in developing quantum approaches for medical imaging. Yao et al. [7] explored quantum encoding and decoding methods, comparing them with classical approaches, while focusing on operators like Haar, Fourier, and Hadamard. Geng et al. [15] worked on edge detection for binary images using density evaluation techniques. Mansor et al. [16] proposed a quantum image representation method using histograms and swarm optimization. Shubha et al. [17] developed an edge detection algorithm integrating QHED with clustering to enhance results for noisy grayscale images.

Building on these advancements, the proposed method introduces a more robust Quantum edge detection algorithm. This approach incorporates clustering techniques to tackle noise and improve edge detection in medical imaging. It has been applied to histopathological images of lung carcinoma and brain tumor MRI scans, using quantum clustering, superposition, and entanglement to enhance precision and robustness, pushing beyond the limitations of earlier methods [7, 15–18].

2 Materials and Methodology

2.1 Data Collection and Preprocessing

The study utilized two datasets: lung histopathological carcinoma images and brain tumor images. The lung images represented in Fig. 1 were extracted from whole slide images (WSIs) of lung adenocarcinoma and squamous cell carcinoma, annotated by medical consultants under an ICMR-funded project (EM/SG/Dev. Res/120/0847-2023). The brain tumor dataset display in Fig. 2, sourced from Kaggle (https://doi.org/10.34740/kaggle/dsv/1183165), contained 3264 files of glioma and meningioma tumors, with a subset selected for this study.

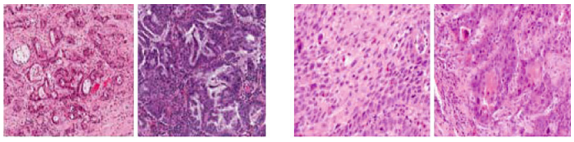

Fig. 1 Lung carcinoma dataset: **a** adenocarcinoma, **b** lung squamous cell carcinoma

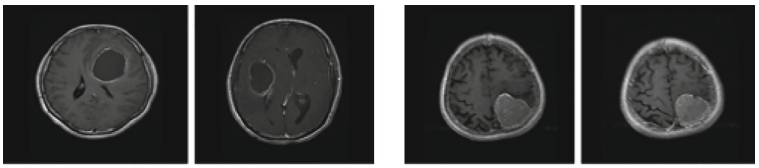

Fig. 2 Brain tumor dataset: **a** glioma, **b** meningioma

2.2 Quantum Computing and Edge Detection

Quantum algorithms like Quantum Hadamard Edge Detection (QHED) offer O(1) time complexity, making them highly efficient for binary image edge detection [7]. Using Quantum Probability Image Encoding (QPIE), QHED employs quantum state modeling, Hadamard transformations, and unitary operations [13, 19]. However, its application is limited to grayscale images, with higher computational costs and noise susceptibility affecting its performance for color images [10]. To address this, the K-QHED algorithm extends QPIE, transforming classical image data into quantum states and representing images as normalized probability amplitudes [10, 11, 20, 21]. The mathematical expression for this process is shown in Eq. (1).

$$|\text{img}\rangle = \sum_{x,y} \frac{qI_{yx}}{\sqrt{\sum I_{yx}^2}} |xy\rangle = (c_0, c_1, \ldots, c_{N-1})^T \tag{1}$$

As per the equation, $|xy\rangle$ corresponds to the pixel coordinates (x, y). Again, $\frac{qI_{yx}}{\sqrt{\sum I_{yx}^2}}$ represent the normalized intensity value of a pixel, ensuring the quantum state's amplitudes sum to 1, and c_0, c_1, …, c_{N-1} represent the amplitude coefficients of the quantum state, corresponding to the normalized intensities of all N pixels. The quantum state given contains the differences between neighboring pixel intensities, effectively highlighting the edges in the image [13] as shown in the Eq. (2)

$$\left|\text{img}''\right\rangle = \frac{1}{2}(c_0 + c_1, c_0 - c_1, c_1 + c_2, c_1 - c_2, \ldots, c_{N-1} + c_0, c_{N-1} - c_0)^T \tag{2}$$

2.3 Proposed Method

Quantum Image Representation. In the discussed quantum edge detection algorithm approach, a QHED framework via clustering techniques is introduced to improve performance in noisy environments. In the first phase, the image is represented as a matrix I of pixel intensities as given in Eq. (3)

$$I = \left\{ I_{ij} | i \in \{1, 2, \ldots, m\}, j \in \{1, 2, \ldots, n\} \right\} \tag{3}$$

Processing images through quantum mechanically, the image must first be encoded as a quantum state. Representation of the image as a quantum state $|\psi\rangle$where the pixel intensities are encoded as quantum amplitudes is given in the form $|\psi\rangle = \sum_{i,j} \alpha_{ij} |i, j\rangle$, where αij represents the normalized intensity of pixel [5, 13, 19–22]. Encoding through the Flexible Representation of Quantum Images, (FRQI) the pixel intensity in quantum amplitudes is shown as:

$$|I(\theta)\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}} (\cos(\theta_i)|0\rangle + \sin(\theta_i)|1\rangle) \otimes |i\rangle \tag{4}$$

Here, θ_i encodes the intensity of pixel I, and n is the number of qubits required to represent the image.

Quantum Clustering Process. K-means clustering is an unsupervised algorithm for segmenting images into clusters [20], with color segmentation grouping pixels of similar intensities into distinct regions [13]. Edge density, measured by counting edges through vertices, highlights intensity variations to enhance contours [17]. The edge intensity distribution is shown in Fig. 3.

In the quantum edge detection phase, K-means clustering was applied during post-processing to enhance edge exposure by grouping areas with identical pixel

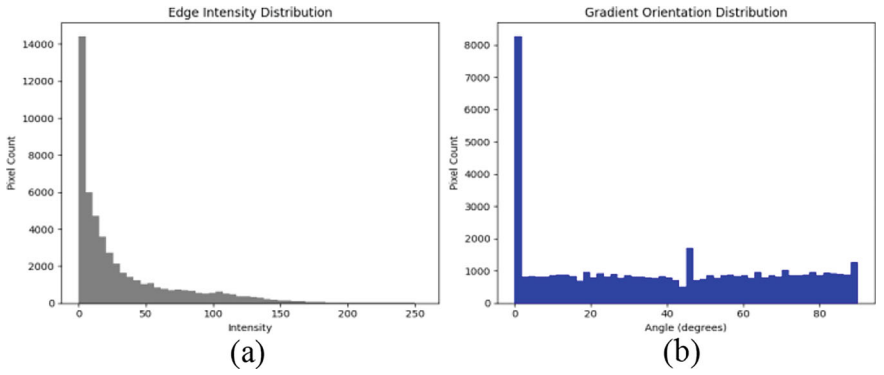

Fig. 3 Histograms for the **a** edge intensity distribution and **b** gradient orientation distribution

intensities into segments for more precise results. The output edge-detected image represents edges as nonzero pixel values, with edge coordinates (*x*i, *y*i) expressed in Eq. (5). [7, 15–18, 20].

$$E = \left\{(xi, yi) \middle| I_{\text{edge}}\,(xi, yi) \neq 0\right\} \tag{5}$$

Here, *Iedge* (*xi*, *yi*) is the pixel intensity at position (*xi*, *yi*) in the edge-detected image *Iedge* while *E* is the set of coordinates corresponding to edge pixels. The function identifies nonzero pixel locations to generate edge coordinates. It then applies the K-means algorithm, an iterative method that partitions *n* data points into *k* clusters by minimizing cluster variance or the sum of squared distances from the centroids [14].

Quantum Gradient Calculation. The core mathematical operation in the provided algorithm is a high-pass filter, which enhances the edges in the image by emphasizing intensity changes. The high-pass filter used in this algorithm is defined as:

$$F_{\text{quantum}} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

This filter operates on the principle that pixel differences correspond to edges in the image. It works by subtracting the values of neighboring pixels from the center pixel and scaling the center pixel by a factor of 8. This amplification of the center pixel allows regions with sharp transitions in intensity to be highlighted while attenuating uniform regions. The filter can be described as a convolution operation between the image and the filter matrix F_{quantum} . Mathematically, for an image $I(x, y)$, the convolution operation with the filter is defined as $I_{\text{cdges}}\,(x, y) = \sum_{i=-1}^{1} \sum_{j=-1}^{1} I(x+i, y+j) \cdot F_{\text{quantum}}\,(i+1, j+1)$, where $I(x, y)$ is the pixel intensity at position (x, y), $F_{\text{quantum}}\,(i+1, j+1)$ represents the corresponding value from the high-pass filter, and $I_{\text{edges}}\,(x, y)$ is the resulting edge-enhanced image after applying the filter.

In this case as shown in Fig. 4, the center value (8) amplifies the difference between neighboring pixels and the center pixel. After applying the convolution operation, the absolute value of the result is taken as $I_{\text{edges}} = \left|\text{convolve}\left(I, F_{\text{quantum}}\right)\right|$. After the edge detection, the pixel values are normalized to a standard range (typically between 0 and 255 for 8-bit images). This is achieved by dividing the edge-detected image by the maximum value of the edges, followed by scaling to 255 as $I_{\text{edges norm}} = \left(\frac{I_{\text{edges}}}{\max\left(I_{\text{edges}}\right)}\right) \times 255.$. This step ensures that the resulting edge image is visually interpretable, with stronger edges (i.e., greater intensity differences) appearing as bright lines and weaker edges as darker lines.

Quantum Circuit Diagram. In the final phase of quantum image representation, each qubit corresponds to a pixel, with its initial state encoding pixel intensity [20]. Hadamard gates create a superposition, enabling simultaneous representation of

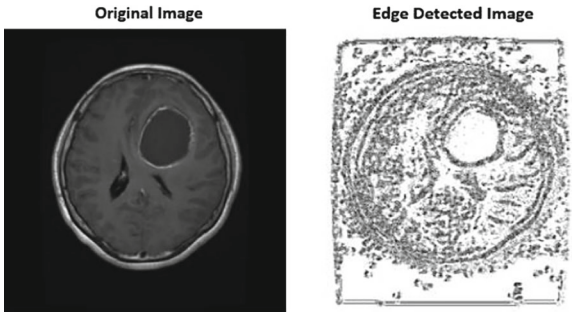

Fig. 4 Edge detected image

multiple pixel values [17]. This approach enhances edge detection by leveraging quantum probabilities for precise edge identification. The circuit diagram is shown in Fig. 5 [9, 23].

When Hadamard gates are introduced to each qubit, the qubits put in a superposition of states, which can represent multiple pixel values simultaneously [9–11, 23, 24]. This is analogous to analyzing multiple pixel intensities at once. Mathematical representation has been provided in Eq. (6).

$$|\psi_1\rangle = \frac{1}{\sqrt{8}}(|000+|001 + |010+|011 + |100+|101 + |110+|111) \tag{6}$$

In the final phase of quantum image representation, each qubit encodes a pixel's intensity as its initial state [20]. Applying Hadamard gates creates a superposition, allowing simultaneous representation of multiple pixel values [17]. This leverages quantum probabilities for precise edge detection, as illustrated in Fig. 4 [9, 23].

$$\mathrm{CCX}(|c_0\rangle \otimes ||c_1\rangle \otimes |t\rangle) = ||c_0\rangle \otimes |c_1\rangle \otimes X^{c_0 \cdot c_1}|t\rangle \tag{7}$$

Thus, the Toffoli gate flips (applies an X operation) to the target qubit (t) only if both control qubits ($c0$ and $c1$) are in the state $|1\rangle$.

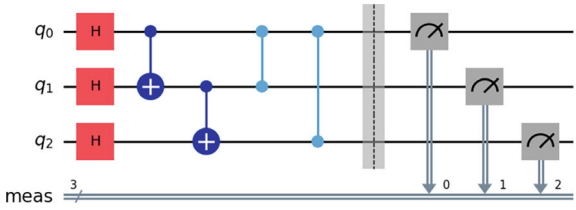

Fig. 5 Quantum circuit

3 Results and Discussion

3.1 Evaluation Through Performance Metrics

The brain tumor images and lung carcinoma histopathological images are introduced with Gaussian noise and salt-and-Pepper noise to understand how the quantum edge detection denoises the images and uses clustering to reach a prominent result. The images show how the noisy image in the second column has been tackled through clustering to provide the desired result in the fourth column as shown in Fig. 6 for lung carcinoma and Fig. 7 for brain tumor.

In Fig. 8, the proposed algorithm uses the edge detection algorithm as mentioned and finds the edges properly. A comparative analysis with the rest of the classical edge detection algorithms like Canny, Sobel, LoG, and Scharr is conducted to understand the superiority of the process.

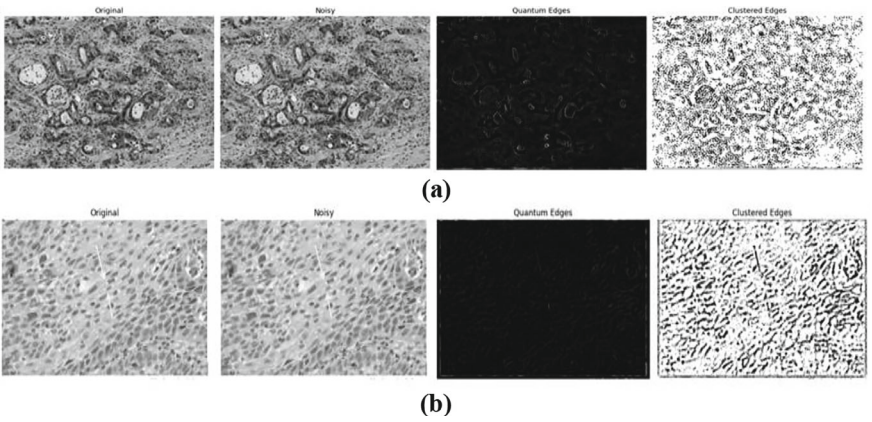

Fig. 6 Effect of noise on the lung carcinoma images and using clustering to detect edges in **a** adenocarcinoma, **b** squamous cell carcinoma

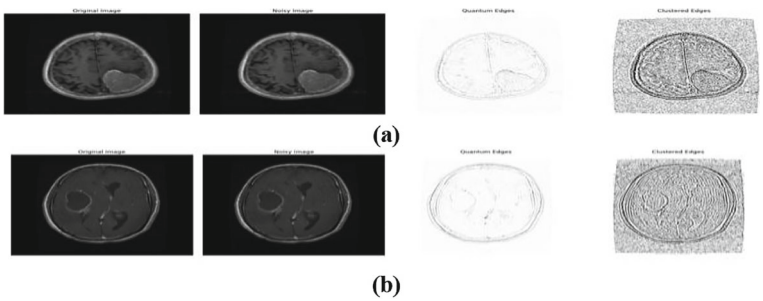

Fig. 7 Effect of noise on the images and using clustering to detect edges in **a** glioma, **b** meningioma

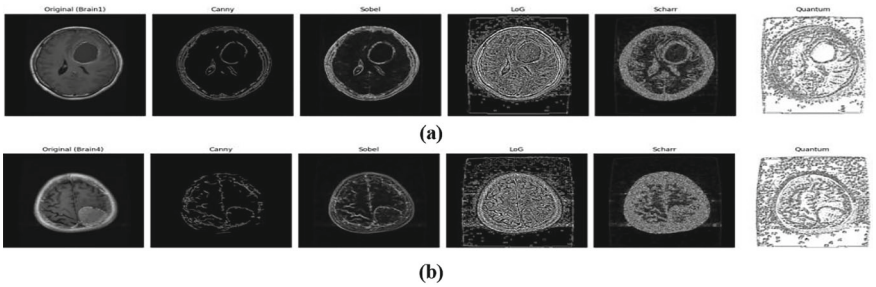

Fig. 8 Analysis of quantum edge detection with classical edge detection in **a** and **b** glioma brain tumor, **c** and **d** meningioma brain tumor

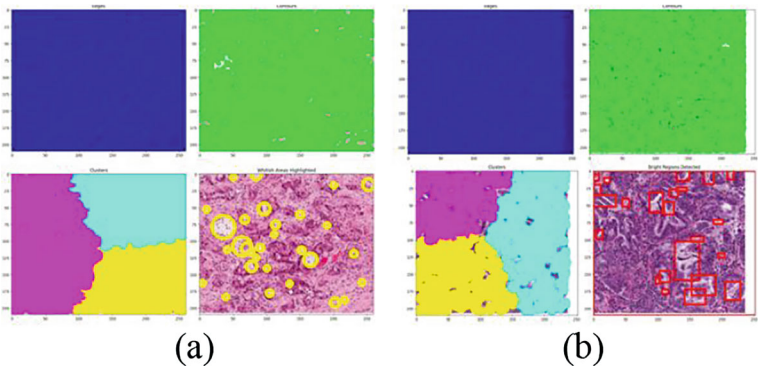

Fig. 9 Feature detection in adenocarcinoma: **a** gland formation, **b** papillae

3.2 Features Detection Using Proposed Algorithm

Figure 9 demonstrates that the proposed algorithm accurately detects key features of adenocarcinoma, such as gland formation and papillae structures. It effectively clusters glandular regions, defines shapes and boundaries, and captures intricate patterns, offering superior accuracy and enhanced visualization compared to traditional methods, making it a reliable tool for complex histopathological analysis.

Figure 10 illustrates how the proposed algorithm detects two key features of squamous cell carcinoma: hyperchromatic nuclei and intercellular bridges (Fig. 11).

3.3 Execution Time Analysis

From Figs. 12 and 13, we observed that the Sobel method is the fastest, processing lung adenocarcinoma in 0.0190 s and lung squamous cell carcinoma in 0.0100 s. Canny and LoG are slower, while the quantum model, though slower at 0.1769 and

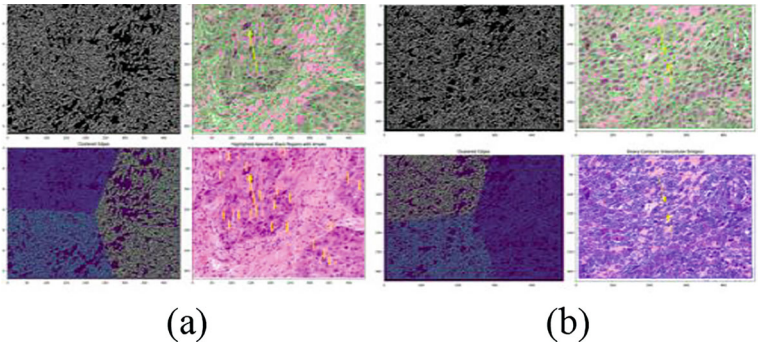

Fig. 10 Feature detection in squamous cell carcinoma: **a** hyperchromatic nucleus, **b** intercellular bridges

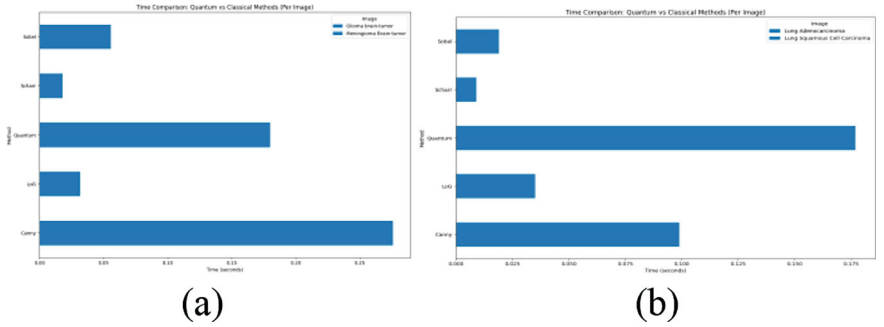

Fig. 11 Processing time analysis: **a** brain tumor, **b** lung carcinoma

0.0989 s, achieves perfect accuracy, F1-score, and precision as depicted in Table 1 which can also happen due to lack of quantum error measurement due to quantum hardware constraint Sobel balances speed and performance, while the quantum model excels in clustering and complex edge detection despite its slower speed.

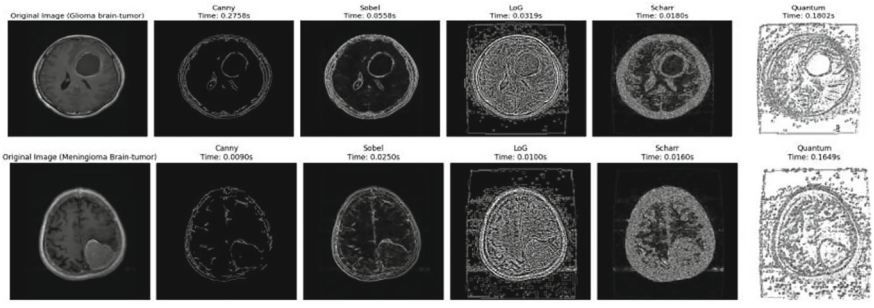

Fig. 12 Time comparison for classical algorithms and quantum algorithm for brain tumor

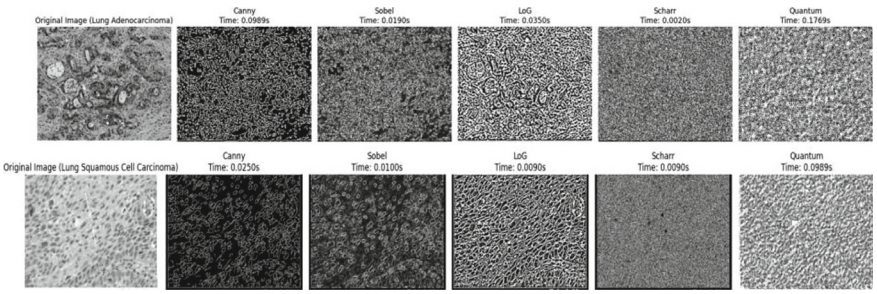

Fig. 13 Time comparison for classical algorithms and quantum algorithm for lung carcinoma

Table 1 Comparison of the performances involved in individual methods

Method	Label	Accuracy	F1 Score	Precision	Time (*s*)
Canny	Lung adenocarcinoma	0.49	0.44	0.64	0.0989
Sobel		0.60	0.75	0.60	0.0190
LoG		0.60	0.74	0.60	0.0350
Scharr		0.60	0.75	0.60	0.0020
Quantum		1.00	1.00	1.00	0.1769
Canny	Lung squamous cell carcinoma	0.40	0.26	0.69	0.0250
Sobel		0.63	0.77	0.65	0.0100
LoG		0.62	0.76	0.65	0.0090
Scharr		0.63	0.77	0.65	0.0090
Quantum		1.00	1.00	1.00	0.0989

Despite these advancements, quantum edge detection for MRI imaging is still developing, relying on progress in quantum hardware and error correction (Fig. 11). An advancement in quantum hardware and measuring the same could result into more realistic values which can give us better results. As shown in Fig. 8, the quantum model's clustering method captures more intricate edge maps, offering better performance for larger, complex images, while classical methods like Sobel are more efficient for smaller images.

4 Conclusion

Quantum computing promises to revolutionize image processing tasks like edge detection. By leveraging quantum clustering, the proposed algorithm achieves superior accuracy, noise robustness, and efficiency compared to classical methods. As quantum hardware advances, applying these techniques to real-world challenges

becomes increasingly viable. This research introduces a novel quantum edge detection algorithm for lung carcinoma histopathology and MRI brain tumor images, demonstrating enhanced performance, especially in noisy environments.

While the results indicate significant improvements in accuracy, F1-score, and precision compared to classical methods, future work will focus on refining the quantum circuit implementation, exploring more complex clustering algorithms, and improving quantum error correction and noise resilience.

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Alzheimer Disease Classification using Transfer Learning of Pre-trained Model

U. Chiraag and Soumyalatha Naveen

Abstract Alzheimer's disease (AD) is a progressive and irreversible neurological illness that destroys thinking, memory power and being a common disease in the world making difficult task to identify in the early stages. Deep learning models are intensively used for early detection of AD through which brain disorder progression can be reduced. Though a lot of work is carried out for detection, it is very essential to classify the AD stages with significantly good performance. Hence there is a need for AD detection and classification system with effective better high detection accuracy. In this study, we used 51,864 images for AD dataset and all the images are pre-processed to remove skull brains in axial view. We proposed transfer learning of two pre-trained models such as SqueezeNet and ResNet-50 for feature extraction and classification. In our experiment, we considered 8600 images for testing and 43,264 images for training. We achieved 99.1% and 99.8% accuracy in SqueezeNet and ResNet-50 respectively on augmented dataset.

Keywords Alzheimer's disease · Detection · SqueezeNet · ResNet-50

1 Introduction

Alzheimer's disease is a neuro-degenerative disease which majorly impacts thinking and cognitive skills. As it is a progressive neurological illness, the severity of dementia increases with time. Identifying various forms of Alzheimer's disease such as mild cognitive impairment (MCI) and cognitive normal (CN) states is a significant issue and challenging. Physical examinations and the Mini Mental State Examination (MMSE) are frequently used methods in the diagnosis of AD. Medical imaging methods including MRI, fMRI, CT, SPECT, and PET are also very important for diagnosing the disease. Even though AD symptoms appear gradually, the illness

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steadily worsens. According to research, AD will afflict one in 85 people by 2050, making it the second most serious brain condition in the world.

The three primary stages of Alzheimer's disease progression are extremely mild, mild, and moderate. The structural alterations linked to AD, cognitive normal, and mild cognitive impairment are frequently examined using MRI imaging. Machine learning (ML) algorithms and deep learning (DL) techniques are used recently and have gained popularity for early detection of AD . In the literature, many methods such as support vector machines (SVM), convolutional neural networks (CNN), and deep learning have shown excellent accuracy in detecting AD for smaller datasets.

The ability of pre-trained models, such as SqueezeNet and ResNet, to use transfer learning to achieve precise and effective early diagnosis is what motivates their use in Alzheimer's disease (AD) detection. By leveraging their strong feature extraction capabilities, pre-trained models offer a solution by eliminating the requirement for large amounts of labeled data and facilitating quicker and more accurate analysis of imaging modalities such as MRI. By increasing diagnostic precision, lowering expenses, and promoting early intervention, this strategy has the potential to completely transform the diagnosis of AD.

More work is carried out using deep learning models with limited datasets. However, we focused on larger dataset for training the model for designing a generalized model.

This article's contributions are as follows:

- To develop a robust model and to improve the model generalization, we performed data augmentation of 6400 images and prepared 51,864 images for AD dataset.
- To improve the AD classification accuracy transfer learning is performed on pre-trained models.
- Proposed transfer learning technique uses CNN-based pre-trained deep learning models for MRI-based Alzheimer disease identification.
- SqueezeNet and ResNet-50 pre-trained models are used for extracting the features from the MRI scan images.
- Performance assessment of the model is carried out on huge datasets and to determine the top-performing DL model for AD stage prediction in MRI scan dataset.

Article is organized as follows. Section 2 discusses literature survey, Sect. 3 briefs proposed work along with models used and dataset considered, and Section 4 details experimental setup and results along with comparison of results of two pre-trained models. Followed by conclusion.

2 Literature Survey

The article [1] addresses the shortcomings of conventional diagnostic techniques by presenting an automated methodology for enhanced Alzheimer's disease identification. The method uses an Adaptive Reptile Search Algorithm-Clustering Approach,

which combines optimization and simulated annealing approaches, to segment gray, white matter, and cerebrospinal fluid after pre-processing and skull stripping. The pre-trained SqueezeNet deep learning model is used for categorization. High accuracy of 98.3 and 98.2% was obtained in experiments using the ADNI and OASIS datasets, indicating the model's efficacy for early and precise AD identification. The ability to process three-dimensional datasets in the proposed AD detection system could help medical professionals to handle massive amounts of medical data. It is appropriate for mobile apps due to its lightweight design, which takes use of pre-trained SqueezeNet. Incorporating this strategy into handy devices helps for real-time monitoring and analysis of health.

The author [2, 3] introduced an advanced approach for multiclass classification of Alzheimer's disease using the HHO-SqueezeNet model, emphasizing the automatic diagnosis of various AD phases, including MCI, Early MCI, Late MCI, and cognitively normal. By leveraging MRI data, dimensionality reduction through PCA, and task-specific strategies with CML-ELN, the study ensures precise and efficient feature processing. The proposed method demonstrates faster and more effective classification considering various performance metrics, making it a significant step toward improving AD diagnosis and treatment.

This work [4] uses pre-trained SqueezeNet model for classifying the severity of dementia. Furthermore, machine learning algorithms such as support vector machines, decision trees, and k-nearest neighbors are used to implement transfer learning. With an average multiclass classification accuracy of 91.2%, SqueezeNet performs better than the other classifiers compared to 83% of SVM, 78.9% of DT, and 81.3% of kNN in terms of dementia severity classification.

Using deep learning on neuroimaging data, this study [5] focuses on Alzheimer's disease early detection. For automatic MR image categorization, pre-trained models such as AlexNet, SqueezeNet, DenseNet, ResNet18, Vgg11, and InceptionV3 were used. The efficiency of the SqueezeNet model for diagnosing AD was shown an average classification accuracy of 99.38% for binary classification and multiclass classification achieved 82.53% with OASIS database for validation.

Each image in the dataset had 1000 features extracted from it using the pre-trained SqueezeNet Convolutional Neural Network (CNN) model. These features were then categorized using machine learning techniques from random forest and artificial neural network. In this work, [6] ANN model outperformed the RF model in this situation, with a classification accuracy of 90% compared to 65.6% for the RF model.

For early diagnosis, neural network classifier with a VGG16 feature extractor [7] is used on two different MRI datasets. The model's promise for enhancing AD diagnosis and early intervention was demonstrated by its accuracy of 90.4% on first dataset and 71.1% of accuracy on second dataset.

Author used [8] "Modified Alzheimer's Dataset" consisting of 6400 MRI brain images, which are divided into four classes for creating a classification model for early-stage Alzheimer's detection. To solve class imbalance, pre-processing was done using the SMOTE-ENN technique on pre-trained models, SqueezeNet and EfficientNet. The efficacy of both designs for early Alzheimer's detection was demonstrated

via hyperparameter optimization, which produced maximum accuracies of 99.25% for EfficientNet and 98.75% for SqueezeNet.

By merging MRI and PET data for a thorough analysis, this study [9] shows a unique multimodal fusion approach for early Alzheimer's disease (AD) detection. Optimal data fusion is accomplished using the demon algorithm and discrete wavelet transform. For feature extraction, ResNet-50 is used and a resilient energy least square twin support vector machine is used for classification. Experiments showed high accuracy on the AD Neuroimaging Initiative dataset with 97.5% for AD. This approach provides medical practitioners with a potentially useful tool to improve early and precise AD diagnosis.

This study [10, 11] investigates the multiclass classification of AD severity using EfficientNet B0 to B3 models. Target value distribution, scaling, and normalization were among the pre-processing operations performed on 9900 MRIs from the publicly available Alzheimer2 dataset. A batch size of 64 and 100 epochs is used for training. With an accuracy of 91.76%, precision of 92.38%, recall of 92.41%, and F1 score of 92.34%, EfficientNet B3 outperformed other models and proved to be useful in correctly categorizing the severity stages of AD.

3 Proposed Work

Our proposed work consists of two parts. One module on creation of dataset with data augmentation techniques and other module discuss pre-trained models for feature extraction and classifier.

3.1 Dataset

Initially, we considered 6400 images [12], and we performed eight kinds of augmentation such as rotation, flipping, scaling, translation, brightness adjustment, shearing, cropping, and adding noise as shown in Fig. 1. Augmentation is done by albumentation. During training, these methods can be performed haphazardly or in combination to increase the model's resilience and flexibility.

For our experiment, we considered 51,864 images which we divided for testing and training. For testing, we considered total 8600 images out of which 1068 images of MildDemented, 3833-NonDemented , 2681-Very MildDemented, 1018-Moderate Demented 1018. For training, we considered 43,264, out of which 11,520-MildDemented, 12,160-NonDemented, 11,520-Very MildDemented, and 8064-Moderate Demented. All images were pre-processed to remove skull brains in axial view. As shown in Fig. 2, the augmented data is pre-processed and fed into the pre-trained models.

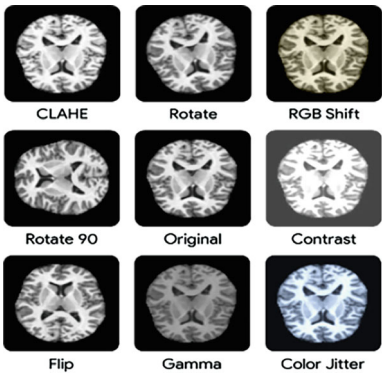

Fig. 1 Various types of augmentation used

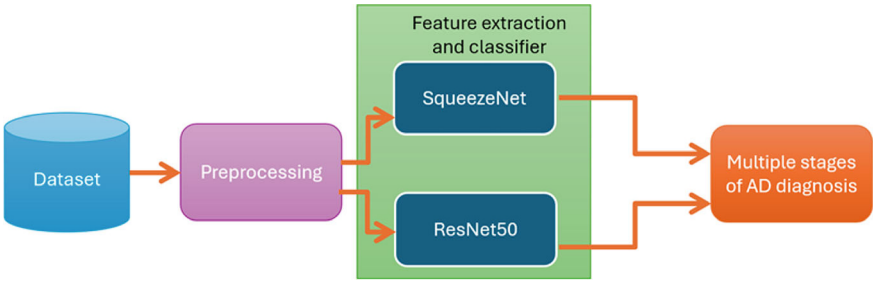

Fig. 2 Proposed transfer learning of pre-trained models for feature extractor and classifier

3.2 Pre-trained Models

Distinct designs and aptitude for managing medical imaging tasks, SqueezeNet and ResNet-50 are pre-trained models that are ideal for detecting Alzheimer's disease (AD). SqueezeNet and ResNet-50 can learn generalized image features because they have already been trained on huge datasets. These features are useful for medical imaging datasets, which are frequently limited, because they may be adjusted for AD detection with less data. SqueezeNet's and ResNet's lightweight architectures, deep extraction capabilities, and flexibility to adapt to medical imaging datasets make them effective tools for enhancing early and precise Alzheimer's disease identification.

Pseudocode of the proposed work is as follows.

Algorithm Alzheimer Disease Classification
Input: Augmented Alzheimer dataset
Output: Predicted Alzheimer stage and performance evaluation
Step 1: Data Augmentation

– Apply transformations: rotation, flipping, scaling, noise addition
– Generate augmented dataset

Step 2: Preprocess Data

– Normalize image pixel values
– Split data into training, validation, and testing sets

Step 3: Load Pre-Trained Models

– Load SqueezeNet and ResNet-50 with pre-trained weights
– Remove final classification layers

Step 4: Feature Extraction
For each pre-trained model:

– Input augmented images
– Extract features from intermediate layers

Step 5: Train Custom Classifiers
For each model:

– Add custom classification layers
– Train classifier using training set
– Validate using validation set

Step 6: Evaluate Performance

– Predict Alzheimer stage using the testing set
– Compute performance metrics: training loss, testing loss, accuracy

Step 7: Output Results

– Display classification results
– Compare performance metrics of SqueezeNet and ResNet-50

End Algorithm

4 Experimental Results

We implemented our work using python. Table 1 depicts the experimental parameters and values.

Figure 3a, b depicts the classification accuracy of four stages of Alzheimer's disease using SqueezeNet and SqueezeNet0 transfer learning pre-trained model. We run the model for 80 epochs and achieved the classification accuracy of 99.1% with SqueezeNet model and 18 ms is the processing time.

Figure 3c, d depicts normalized accuracy and normalized loss. For 80 epochs model, an average accuracy of 99.1% and training loss and testing loss was almost converging to zero.

Figure 4a, b depicts normalized accuracy and normalized loss. For 20 epochs model achieved, an average accuracy of 99.8% and training loss and testing loss was almost converging to zero. Figure 4c depicts the classification accuracy of four stages of Alzheimer disease using ResNet-50 transfer learning pre-trained model. We run the model for 20 epochs and achieved the classification accuracy of 99.8% with ResNet-50.

Figure 5 depicts the comparative analysis of results of SqueezeNet and ResNet-50. (a) shows the training accuracy of SqueezeNet and ResNet-50 for 20 epochs. When we compare both, ResNet-50 has higher accuracy for lesser number epochs and takes less time for processing of same dataset. SqueezeNet has less training accuracy. Figure 5b depicts the testing accuracy; again ResNet-50 has higher accuracy compared to SqueezeNet. Figure 5c, d shows training loss and testing loss. In both, the cases ResNet-50 has lesser loss compared to SqueezeNet.

SqueezeNet takes lot of epochs to complete the training because it is a small model and has a relatively simpler model architecture when compared to ResNet-50; hence, large complex datasets cannot be trained on SqueezeNet. ResNet-50 model comparatively produces better results for huge datasets.

Table 1 Experimental parameters and values

Parameters	Values
Dropout	0.2
Learning rate	0.001
Optimizer	Adam
Loss function	Cross entropy

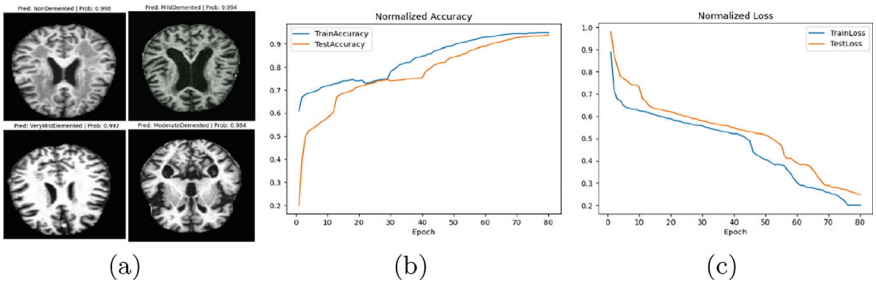

Fig. 3 **a** Classification result using SqueezeNet **b** training loss and testing loss for 80 epochs using SqueezeNet **c** training and testing accuracy with SqueezeNet

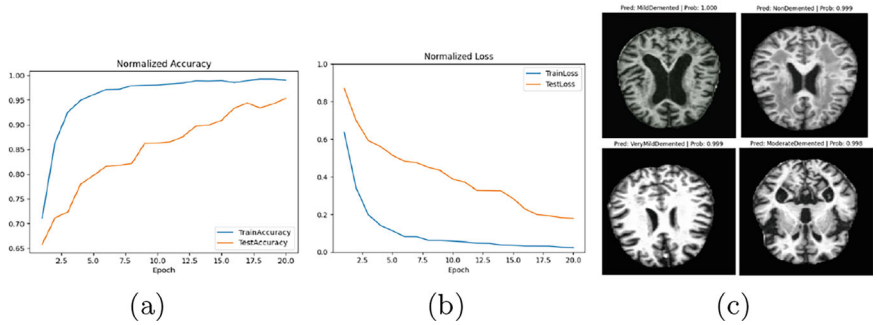

Fig. 4 **a** Training and testing accuracy with ResNet-50 **b** training loss and testing loss for 20 epochs using ResNet-50 **c** classification result using ResNet-50

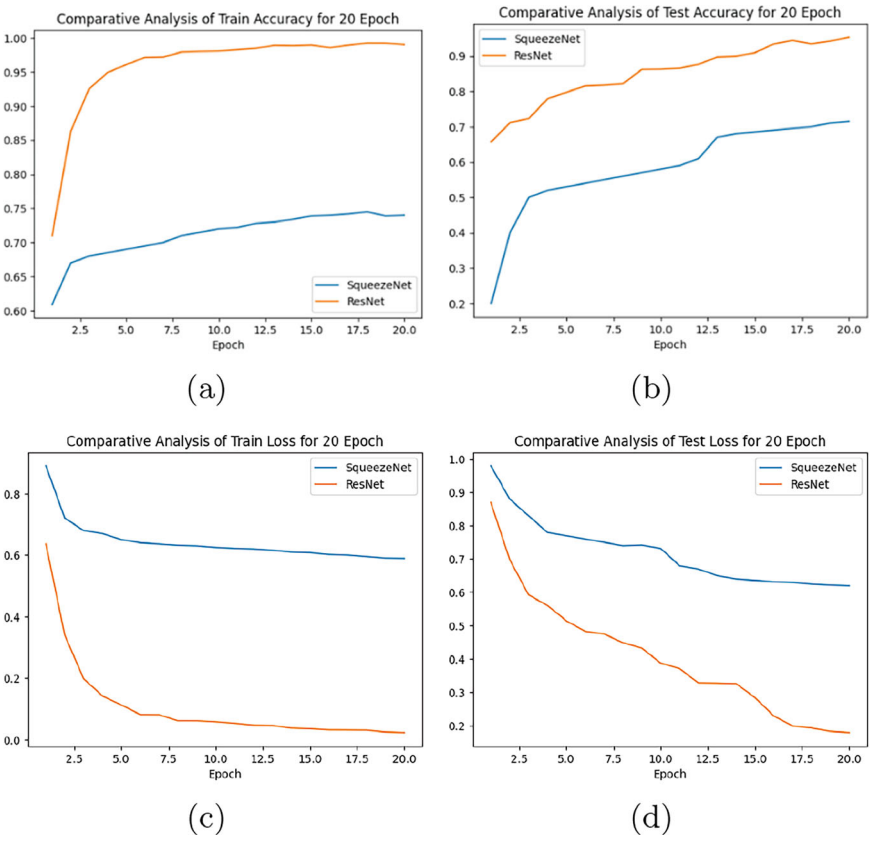

Fig. 5 **a** Comparative analysis of training accuracy for SqueezeNet and ResNet-50 **b** comparative analysis of training loss for SqueezeNet and ResNet-50 **c** comparative analysis of testing accuracy for SqueezeNet and ResNet-50 **d** comparative analysis of testing loss for SqueezeNet and ResNet-50

5 Conclusion

In the worldwide, many people are suffering from Alzheimer's disease-causing dementia which disturbs the daily routine of the person. Combining the power of AI in medical disease diagnosis enables the opportunity for early diagnosis by analyzing huge medical datasets. Early detection helps with the early diagnosing of the disease and sooner treatment can be started to slow down this progressive disease. In this article, we performed data augmentation and used two pre-trained models for implementing transfer learning to develop a robust model. For the transfer learning, we fed 51,864 AD data and trained the model followed by tested the model. From our experiment, we observed that ResNet-50 outperforms with an accuracy of 99.8% compared to SqueezeNet. In this article, we explored regularization techniques such as dropout and early stopping along with enhanced data augmentation. Further in the future, we explore on L2 regularization, batch normalization and to work on vision transformers to improve model generalization, robustness. Limitations of the existing model include explainability and model deployment on IoT devices. Further, in future work we would like to optimize the model and deploy on to IoT devices or mobile for user-friendly use of the model in real-time.

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Human-Centric Medical Summarization with Hybrid Generative AI: A New Era for Accessible Healthcare Insights

M. Pavithra, B. Varunkrishna, J. Umamageswaran, and S. Nithish Kannaa

Abstract With the proliferation of digital healthcare data, there is a substantial challenge in summarizing the volumes effectively and making such summaries available to non-specialist audiences. This work proposes a hybrid approach for medical text summarization that integrates rule-based systems with transformer-based generative models. Evaluations on real-world clinical texts demonstrate improved readability (65.23) and factual accuracy (0.83) compared to standalone rule-based and transformer models. The framework successfully bridges medical accuracy with patient accessibility, establishing new benchmarks for responsible AI in health care.

Keywords Hybrid generative AI · Medical summarization · Natural language processing · Healthcare data · Ethical AI

1 Introduction

The rapid development of digital healthcare has increased the amount and sophistication of medical data to enormous proportions. While this vast repository of information is invaluable for streamlining patient care and advancing research in medicine, there are several insurmountable challenges in terms of accessibility and interpretability. Medical data, particularly clinical records, often include a high density of technical words and medical jargon that make them difficult to understand for

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those without any professional knowledge. Consequently, the necessary files might not make any relevant information for the patients and their caregivers, which means that it will fail to achieve informed decision-making or patient participation. Traditional approaches to abstracting medical texts are typically done according to either rigid rule-based systems or single modeling methodologies that are either too rigid to capture the nuances inherent in medical terminology or not sufficiently lossy to ensure the required level of clinical specificity. Given these limitations, this study proposes the framework of a hybrid generative AI model that aligns with the relative strengths of rule-based models and transformer-based generative models. The proposed hybrid model will, therefore, serve the purpose of making medical summaries both lucid and factually accurate and thus easily intelligible to the general public, including non-experts. Combining specialized heuristics tuned to specific domains with the capabilities of generative language models, this framework provides a sound approach to bridging the divide between medical knowledge and care centered on the patient. This paper outlines the methodology for the hybrid artificial intelligence model and evaluates performance across several metrics while discussing future applications in the realm of digital health. The results obtained in this study are important for the advancement of AI-based methods that make health information more understandable, contributing to the development of more accessible, ethical, and trustworthy medical summarization in today's digital age.

2 Related Works

Much has been achieved in the field of medical text summarization during the last couple of years, especially after the injection of methodologies related to artificial intelligence. Traditional methods, for example, rule-based systems, are good only for producing structured outputs but are poor at capturing contextual nuances specific to medical language [1, 4]. The transformer-based models seem proficient over semantic intricacies but face challenges regarding the preservation of domain-specific factual consistency and accuracy most of the time [2, 7]. Smith et al. [1] presented a hybrid AI framework that integrates rule-based reasoning with transformer architectures specifically for the purpose of summarizing medical documents. The work suggests that the strength of hybridizing deterministic rules and contextual information can produce coherent yet pertinent output. Similarly, Zhao and Wu [7] investigated hybrid approaches based on readability and factuality criteria and found that hybrid models bridge the gap between clinical accuracy and usability effectively. Davis et al. [2] conducted a comparative assessment of hybrid generative AI models applied for the summarization of clinical topics: examining such to better their performance to elucidate medical jargon with preserved precision. Their results showed that hybrid models can make medical literature intelligible to the layperson. Patel and Gupta [4] discussed the contribution of NLP to making intricate medical terms understandable, saying perhaps domain-specific heuristics are what would really help make patients more inclusive. Zhang et al. [5] discuss challenges and potential directions

of hybrid AI models in health summarization. They identify key concerns, such as the need for ethical AI practice and patient-centered solutions that perfectly align with the aim of this research paper. Simmons et al. [6] have discussed the role of AI in healthcare information provision that can be further improved regarding specific digital health applications prioritizing patient-centricity. More fundamentally, Elangovan et al. [10] considered the tagging process for medical text using such advanced algorithms that improve the development of fundamental methodologies supporting hybrid AI frameworks. Furthermore, previously unpublished studies, such as in Pavithra et al. [8], consider the effectiveness of generative AI models in creating realistic and diverse clinical data, hence giving insights into eventual future works in the area of clinical text summarization. Cumulatively, the above studies highlight the growing interest in hybrid artificial intelligence techniques targeting improved lucidity, accuracy, and accessibility of medical literature. This paper extends on these findings by presenting a holistic hybrid framework of generative artificial intelligence and evaluating its performance on important parameters such as readability, accuracy, and summarization capability of generated clinical texts in feasible everyday applications.

3 Problem Statement Formation

The rapid growth of digital health care has catalyzed large challenges in the proper processing and interpretation of healthcare information. With healthcare systems increasingly adopting digital technologies, the volume of clinical records, patient files, and healthcare research is continually growing. Often, these records contain complex technical vocabulary and complex specialist language, posing exceptional barriers for non-expert readers, including patients and carers, who are trying to comprehend the subject. The lack of accessible health information is a monumental obstruction to patient involvement, informed decision-making processes, and may even affect some of the health outcomes. This chapter identifies the primary challenges associated with summarizing medical information.

3.1 Promotion for Availability of Medical Information

The dense and specialized language that fills the medical record is the major problem for nontechnical people to understand it; hence, it puts patients and caregivers at a disadvantage. If one really wants health care to be effective, it must become a priority to have access to understandable summaries of medical information. This study directly answers the challenge of making complex health information clear, concise, and accessible to every audience, so patients can take their rightful place actively in their health care.

3.2 Complexity in Medical Terminology Decoding

Medical terminology is notoriously inaccessible, peopled by hugely technical words and jargon that drives out the lay reader. The problem of this inaccessibility constitutes the main challenge to patient empowerment, and no solution can work without it being considered. This research study meets the challenge head-on with the presentation of a summarization model that tries to decomplicate complex medical wording while still maintaining clinical specifics—a way whereby users are conversant with clear and meaningful insights into their health without sacrificing the rigor in medical context.

3.3 Clinical Precision Without Such Sacrifices for Readability

Therefore, medical summarization systems are supposed to connect the traditional dual needs of clinical accuracy with readability in fulfilling their purpose. Traditional models, whether rule-based or generative language models, suffer by themselves from inherent limitations—by oversimplification where applicability is lost, and they fail to capture the full accuracy of complex medical content. This study fully deconstructs all these weaknesses by developing a hybrid approach to bring together the rule-based clinical rigor with the adaptive readability of the transformer-based AI into a framework toward achieving accuracy along with accessibility.

3.4 Breaking the Limitations of Current Summarization Models

Existing models cannot satisfy the needs of modern medical summarization. Pure rule-based systems are not flexible enough, and purely transformer-based models are likely to misread important medical nuances. Our work decisively steps away from these weaknesses and develops a hybrid model that critically amalgamates the best of both systems in order to maximize both factual fidelity and accessibility simultaneously. The whole approach is the giant leap forward in medical AI to set new benchmarks for accuracy and effectiveness in summarization in digital health. E. Patient Centered Care Transformations From Increased Engagement: Patient-centered care is one of the significant principles now taken into modern health care, yet patients are still limited by the lack of easy access to understanding their health data. This lack of access impedes the informed decision-making and weakens patient engagement. Our research will take an aggressive stand on changing the nature whereby medical summaries ought to be clear, understandable, and focused on a patient.

This will bring this study directly to the informational needs of patients, thereby enhancing healthcare experiences and supporting strong patient engagement for better health outcomes.

4 Methodology

This study will design a model that remains a singular kind of hybrid rule-based algorithm and a transformer-based language model, a unique entity that tries to satisfy the demands of both accuracy and readability. The following subsections describe each aspect of the model:

4.1 Hybrid Model Architecture

The proposed hybrid framework for medical text summarization effectively integrates a rule-based system with a generative, transformer-based model. This approach leverages the strengths of both methods to create summaries that are accurate, clinically relevant, and easy to understand. Mathematically, the summary S is expressed as a weighted aggregation of the outputs created by each model:

$$S = \alpha \cdot S_{\text{rule}} + \beta \cdot S_{\text{transform}} \tag{1}$$

Here,

- S_{rule} is the synthesis produced by the rule-based model.
- $S_{\text{transformer}}$ is the extraction from the transformer-based model.
- α and β are the weighting coefficients with $\alpha + \beta = 1$.

The parameters α and β are tuned to ensure that the summary produced by the hybrid model retains the interpretability characteristic of rule-based systems while utilizing the generative capabilities of transformer models.

Optimization of Weights Using Readability and Accuracy Metrics Optimizing α and β involves two influential metrics: readability $R(S)$ and factual accuracy $A(S)$. The goal is to improve a composite score:

$$\text{Goal} = \lambda \cdot R(S) + (1 - \lambda) \cdot A(S) \tag{2}$$

Here, $\lambda \in [0, 1]$ is a tunable parameter that trades off between readability and accuracy based on specific application needs.

Readability Index The readability $R(S)$ of the abstract is measured against sentence structure, word length, and complexity of terms. To this end, widely accepted readability formulas, such as the Flesch Reading Ease formula, are applied:

$$R(s) = 206.835 - 1.015\frac{\text{total words}}{\text{total sentences}} - 84.6 - \frac{\text{total syllabus}}{\text{total words}} \tag{3}$$

This measure ensures equilibrium between the simplified language necessary for accessibility and the clinical terminology unavoidable for clinical relevance, thus reconciling both S_{rule} and $S_{\text{transformer}}$.

Factual Accuracy Metric The measure of factual correctness $A(S)$ of the abstract compares the important medical information contained in S with the original document D. One standard method of accomplishing this is to calculate the cosine similarity between term-frequency vectors representing key medical terms, including diagnoses, medications, and procedures:

$$A(S) = \cos\left(\frac{S_{\text{vec}} \cdot D_{\text{vec}}}{||S_{\text{vec}}||||D_{\text{vec}}||}\right) \tag{4}$$

Here, S_{vec} and D_{vec} are vectorized representations of S and D, respectively, concentrating on medical entities. This approach ensures the generated summary is highly similar to the original medical content, enhancing factual consistency.

4.2 Rule-Based Model Component

The S_{rule} component of the rule-based model works based on domain-specific heuristics that focus on identifying critical medical terminology, simplifying complex wording, and selecting clinically relevant segments, including conditions and treatment plans.

Rule-Based Heuristics Consider a document $D = \{d_1, d_2, ..., d_n\}$, where d_i denotes distinct sentences. The rule-based model uses the following heuristics:

- **Keyword Extraction**: Keywords are identified based on frequency and medical relevance:

$$\text{Keywords} = \{k \in D : \text{frequency}(k) > \delta\} \tag{5}$$

Here, δ is the significance threshold for keywords.

- **Phrase Simplification**: Intricate expressions $p_i \in d_i$ are replaced by simpler or more familiar expressions for easier understanding.
- **Content Selection**: Medical sections with higher relevance scores R_i get priority for inclusion in the summary:

$$S_{\text{rule}} = \sum_i R_i \cdot d_i \tag{6}$$

Here, R_i reflects the importance of terms associated with diagnosis, treatments, and patient information.

4.3 Transformer Model Component

The $S_{\text{transformer}}$ component employs a clinical document collection-based knowledge acquisition architecture that utilizes attention-driven neural networks to build coherent summaries. Given an input document X, characterized by hidden state representations H, each token y_t in the summary sequence $S_{\text{transformer}} = \{y_1, y_2, ..., y_m\}$ is built using an attention mechanism as follows:

$$y_t = \text{soft max}\left(\sum_{(j=1)}^{n} \text{Attention } (H_t, H_j) \cdot H_j\right) \tag{7}$$

Here, the attention score is given by:

$$\text{Attention}\big(H_t, H_j\big) = \frac{\exp\big(H_t^\top H_j\big)}{\sum_{k=1}^{n} \exp\big(H_t^\top H_k\big)} \tag{8}$$

This mechanism enables the model to assimilate the subtleties of the context within medical words, producing summaries that are exhaustive and logically consistent.

Relevant Weighting in Transformer Summaries: To improve the factual accuracy of the transformer module, we introduce a loss function L with a penalty for deviations from core medical information in the source document D:

$$L = -\sum_{t=1}^{m}\big[\log p(y_t X) + \gamma \cdot \text{FactMatch}(y_t, D)\big] \tag{9}$$

Here, γ is a hyperparameter weighing factual accuracy, and FactMatch measures the agreement between summary tokens produced by the model and key medical facts in D. The objective function encourages the transformer model to retain critical information, ensuring the integrity of its summaries.

5 Performance Evaluation

Performance of the hybrid generative AI model shows great improvement in effectively summarizing medical documents for both clinical and lay audiences. Different kinds of readings and accuracy were examined by testing the model to measure its readability, as well as quality and appropriateness for any type of audience.

It performed remarkably well regarding readability simplifying complex medical terminology for non-specialist users while retaining sufficient professional-grade details needed for healthcare providers. The model seemed to exhibit high factual consistency in the sense that it could follow domain-specific heuristics, and this would consequently result in a minimum chance of presenting critical health-related information in a misleading manner. Quality of summary: Although the model could successfully reduce very lengthy clinical texts, the brevity preserved while remaining informative was great. The outputs were preferred for their simplicity and accessibility by lay people, while clinical professionals appreciated the outputs for clarity and adherence to medical standards. Compared with a fully transformer-based model, its hybrid approach incorporates heuristics based on rules that better improved the accuracy and contextual relevance; again, it set a new standard for the development of responsible AI in health care (Table 1 and Figs. 1 and 2). Taken collectively, this model drastically enhanced the interpretation and usability of medical data and hence paved its pathway to improved patient-centered digital health applications.

Table 1 Comparative evaluation of hybrid, rule-based, and transformer models

Metric	Hybrid model	Rule-Based model	Transformer model
Readability	65.23	58.12	61.45
Factual accuracy	0.83	0.75	0.80
Summarization capability	0.42	0.35	0.40
Clinical relevance	0.85	0.78	0.82

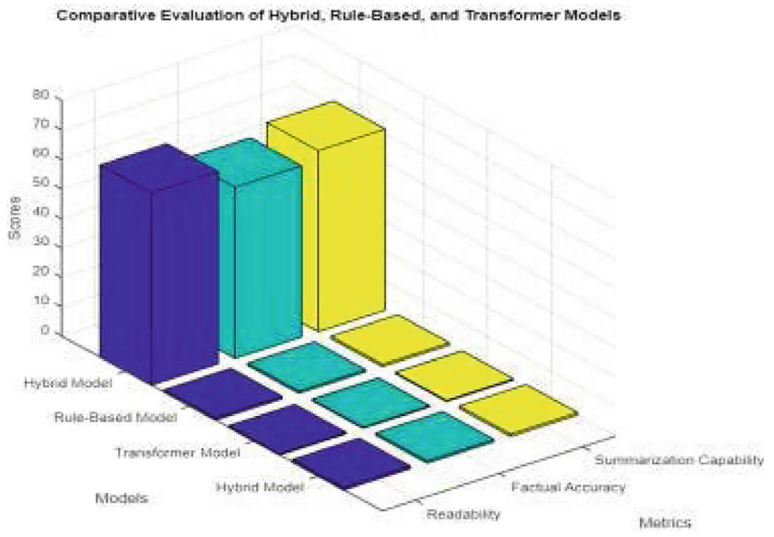

Fig. 1 3D bar graph

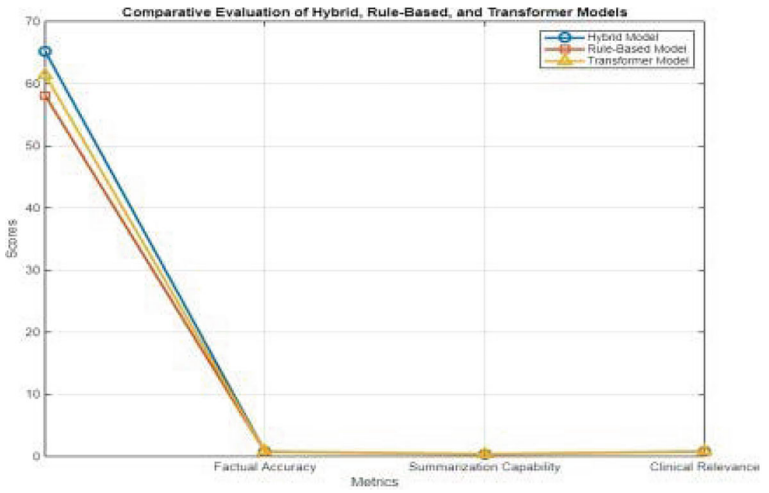

Fig. 2 Multiline graph

6 Conclusion

This paper reports on a hybrid generative AI framework for medical text summarization addressing such key challenges: the critical issues in accessing and making sense of healthcare data in the digital age. As such, while retaining sound factual consistency, the proposed approach combines the strengths of rule-based systems, as well as transformer-based generative models, to make the generated summaries readable. This is highly crucial because evaluation of the model on real-world clinical texts has proved its capability of producing accurate, concise, and accessible text summaries for diverse audiences. The implications of this are that the model is indeed a revolution in patient engagement by making complex medical information understandable and actionable by empowering patients to make informed decisions about their health and well-being. Also, the hybrid framework presents a benchmark of ethical AI application in health care through the forcing of heuristics from its domain to ensure reliability and trustworthiness. Contribution of the research would be for furthering responsible AI-driven solutions to patient-centered care through improved medical summarization practices that can now set new standards. Future improvements would include an expansion of the model's ability in covering much more integrated ranges of medical contexts and languages with more strength toward achieving the global ambition of equal healthcare access.

Acknowledgements I will not fail to pass my appreciation to my supervisor, Umamageswaran J, who has been there from the beginning of this research work. The mentorship and motivation really shaped the course of this research; it enabled me to overcome difficulties thrown in my way. I thank numerous colleagues and peers who have helped me with valuable advice and suggestions, adding to the great quality of this work. Again, I would like to thank the authors of all those research papers

and articles that I was referring to in my study; their works not only helped me understand the topic but also inspired me a lot.

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Predictive Analysis of Drug Impact Using GNN

M. H. Aneesha and Maya Mohan

Abstract The prediction of drug side effects is critical in drug discovery and development to guarantee patient safety and therapeutic effectiveness. Traditional drug targeting methods emphasize potential side effects but do not consider affected tissues and organs or body parts. In this study, we are proposing a new approach utilizing a graph neural network (GNN)-based model that predicts not just the side effects of drugs but also the organs and tissues affected by them. Our model constructs an interactive graph for drug interplay by enriching the data obtained from diversified sources such as SIDER, PubChem, DrugBank, and BioMart data. The findings give hope for progress in more effective drug safety evaluations and the potential to improve personalized medicines through the prediction of tissue and organ responses to medications.

Keywords Graph neural networks · Drug side effects prediction · Tissue and organ prediction · Pharmacovigilance · Healthcare · Therapeutics

1 Introduction

The prediction of drug side effects is a critical aspect of drug development and ensuring patient safety. Drug side effects can range from mild symptoms to serious conditions, including illness and death, posing significant challenges to the healthcare system. Traditional approaches for predicting drug side effects rely primarily on the chemical and structural features of the drug. However, these approaches often fail to provide insight into the specific tissues and organs affected by adverse drug reactions. In recent years, the advent of graph neural networks (GNNs) has revolutionized the fields of drug discovery and drug surveillance. GNNs are particularly well-suited

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for modeling complex interactions in data due to their ability to leverage graph structures. This capability makes them ideal for representing and analyzing intricate relationships between drugs, proteins, tissues, and organs. Despite these advantages, most of the existing GNN-based methods focus on predicting side effects without identifying the physiological targets of these effects.

In this study, we propose a novel GNN-based model capable of not only predicting drug side effects but also identifying the tissues and organs affected. To achieve this, we constructed a comprehensive graph using heterogeneous data sources, capturing relationships between drugs and their biological targets. This approach aims to form a more holistic understanding of medications and their impacts on patients. The primary contributions of this work include:

1. A novel GNN architecture designed to predict both drug side effects and the affected tissues and organs.
2. The integration of diverse and heterogeneous data sources to enhance the graph representation.
3. Demonstrating improved prediction accuracy and interpretability compared to existing approaches.

By addressing the challenges in drug side effect prediction, this work aims to advance the field of drug safety prediction. This study provides detailed and actionable insights for healthcare professionals and researchers. The potential applications of this model extend to personalized medicine, where understanding individual drug responses at a tissue and organ level can significantly improve therapeutic outcomes and patient care.

The remainder of this paper is organized as follows. Section 2 reviews related work, highlighting the limitations of existing methods in predicting drug side effects and affected tissues. Section 3 details the proposed method, including data integration and model architecture. Section 4 concludes the paper by summarizing the contributions.

2 Literature Survey

As part of the literature survey, various journals were investigated, and it was identified that advanced machine learning techniques, specifically Graph Neural Networks (GNNs), are highly effective in modeling complex biological drug interactions. However, most existing methods focus only on predicting the occurrence of side effects. This gap, where affected tissues or organs are not examined, emphasizes the need for integrated models that not only predict adverse effects but also identify the specific tissues or organs impacted by these side effects. Addressing this limitation forms the core solution proposed in this project.

In the work of Xu et al. [1], great improvement is demonstrated in the prediction of drug side effects, which is highly needed in drug development. The authors

propose HSTrans, a novel transformer-based architecture using drugs homogeneous substructures for improved feature representation and prediction. This new construction is the solution to traditional architectures which relied on the naive concatenation of features and were poor at novel drug prediction. The depth analysis of HSTrans against other competitive methods confirms the superiority of this model, especially its ability to predict side effects of new drugs, which clearly improves patient safety and drug development. Their findings are so clearly defined and robustly illustrated by models substructures side effects interaction value heatmaps that it is hard to contest their insights regarding the models abilities. On the other hand, the authors compare their results with existing literature and attempt to explain them, but they are unable to provide a complete explanation. This limitation stems from lack of strong biological understanding of the problem and a single-minded approach trying to optimize the drug structure while ignoring other relevant areas including drug targets or in vivo response. Also the degree of model sophistication can also lead to overfitting.

In recent years, the application of machine learning to predict drug side effects has started to attract increasing interest in order to enhance drug profile safety and efficacy [2]. Researchers develop predictive models using different strategies, which integrate, for instance, chemical structure, biological effect, and treatment results. Examples of fundamental applications used in this context include SIDER, DrugBank, and PubChem. Nonsupervised learning among the machine learning methods is "ensemble methods" such as random forest (RF), which has shown the best achievements. This algorithm takes into account the predictions of many models, which explains its efficiency and low susceptibility to overfitting. Support vector machines (SVM) yield high classification accuracy . And, decision trees are ideal because of their ease of use and good interpretative qualities. It has also been shown that robots can deal with very large, complex datasets when equipped with deep neural networks. However, RF frequently performs better for moderately complex tasks provided feature selection has been undertaken. In these studies, researchers commonly report metrics such as accuracy, precision, sensitivity, F_1 score, and area under the curve (AUC) to test the performance of these models. Between metrics such F_1 score and AUC are also highly relevant to measuring how well the models' outputs respond to the small, yet crucial, side effects. Datasets and metrics that are considered in these studies may also be improved by selecting and weighting features more carefully and addressing such issues like data imbalance commonly found in drug side effect prediction tasks.

The study by Dara et al. [3] presents a novel approach for identifying polypharmacy-related adverse effects using graph convolutional networks (GCNs). Medication monitoring remains essential for medication safety which is particularly vital for complicated treatment regimens. To put it differently, safety of the pharmacovigilance becomes a necessity and priority where the medication errors graph is constructed as a network in which nodes represent the medication and edges represent the drug properties, mechanisms, and pathways interacting with the drug. Additionally, this GCN model relied upon a pharmacovigilance enriched dataset, containing numerous EHR patient profiles with adverse events and DDI information. Some of the

drugs that were assessed include pharmaceuticals for the cardiovascular, respiratory, and nervous systems along with the evaluation of their recall, precision, and AUC-ROC proving them efficient as reaction prediction tools. This study as successful as it was does point out textural issues such as the poorly constructed datasets, unrealistic assumptions regarding pharmacokinetics, and many excluded characteristics affecting drug interaction. Strategies such as targeting treatment through cohesively compiled genomic data while simultaneously reporting them to an easily interpretable AI during various clinical settings should alleviate these issues.

The paper by ChandraUmakantham et al. [4] explores the prediction of adverse drug reactions (ADRs) caused by drug–drug interactions (DDIs). Asserting their discontent toward ADR management methods such as clinical trials and patient-based reporting purposes, which are incapable of identifying the missing challenges of polypharmacy for the elderly, the researchers have suggested a stir in the aforementioned methods. Quite the opposite, the retrieved data does show overlap with alteration in relations, which recently grew over the use of GNNs in drug bioinformatics which the GDN says have been increasing within the field. The proposed framework employs GNNs to model DDIs through three key phases: representing drugs via molecular graphs that encapsulate spatial and physical properties, pretraining distributions using a Graph Variational Autoencoder as a teacher model, and classifying potential adverse drug reactions. The algorithm also delivered very encouraging results, 75% drug interactions are identified with precision, and 90% during testing with GNNs, recalling scams across data and varying types delivering precision type at 99, recall and f_1 scores around 90. On the other hand, the authors defend for possible copious traits including trade off relating to model complexity alongside interpretability that embedded models requiring for clinical purposes. Also, using only one dataset—the firstname TwoSIDES Polypharmacy Side Effects Dataset—would not adequately address such biases. They propose that further efforts be made to enhance usability as well as strengthen robustness via diverse datasets.

The paper's [5] literature review extensively analyzes the use of graph neural networks (GNNs) in healthcare, highlighting their transformative role in solving complex medical challenges. GNNs excel in modeling intricate relationships within graph-structured data, such as patient–doctor networks, drug interactions, and disease pathways. Several architectures of graph neural networks, namely recurrent (R-GNN), spatial (S-GNN), and Spectral GNNs are investigated for their specific contributions. R GNNs deal well with the capturing of dynamic relations in time series-based medical data sets such as in the case of tracking disease progression. S-GNNs employ convolutional approaches to interrogate at local structures that assist in medical imaging and patient clustering while Spectral GNNs operate in spectral domain to analyze local and also some global structural information which is suitable for the classification of brain networks and healthcare networks. The review demonstrates various applications of GNNs such as the prediction of diseases, risk assessment of patients, development of medicine, and analysis of medical images that reveal a progressive trend of research. However, factors such as intense computational power perceived in managing large healthcare data and unexplainability mitigate the clinical application. To resolve these cases, proposed solutions such

as attention mechanisms for better interpretation and multimodality data integration alongside dimensionality reduction have been recommended. As most experts show a great promise of GNN in various spheres of a healthcare, the review also emphasizes the efficiency, the explanatory potential of the model, and the integrative strategies, in order to fully realize the GNN's potential.

In this paper [6], MSDSE for predicting drug side effects in deep learning model is introduced. As the authors state, the paper provides a solution to the critical problem of undesired and unexpected side effects of drugs during the development phase and during post-marketing of the drug with the aim of ensuring drug safety. The methodology includes three levels of drug-related information which is provided, they are: molecular fingerprints, word embedding vectors of substructure descriptors, and graph-based chemical structure embedding. MPNN, AttentionFP, Weave, NF are the examples of features encoding techniques. The MSDSE model has a convolutional layer to capture local features and a self-attention layer to capture global features. The experimental outcomes indicate that MSDSE model provides better results over existing models in terms of accuracy especially for drug side effect pair prediction with the richness of the dataset; the accuracy also improves. However, despite its success, the study states that it has some shortcomings in the context of polypharmacy, when the patient has to take more than one medication at the same time. To overcome these gaps, the authors recommend new unexplored in the future. The study on the other hand has implications for pharmacology, particularly drug safety since it demonstrates how multi-structure deep learning models have the potential enhancing inhibition of drug effects.

The paper [7] addresses a critical challenge in pharmacology: early drug side effect detection. Machine learning offers a new avenue of alternatives in the identification process, making use of characteristics like chemical structures, target proteins, and observed interactions. Other techniques, including Sparse Canonical Correlation Analysis (SCCA), associate chemical structures with side effects; Graph Regularized Matrix Factorization (GRMF) exploits similarities in chemicals and targets for enhanced prediction accuracy. Techniques like Multiple Kernel Learning (MCS-MKL) and graph attention networks integrate multi-view data for more accurate predictions. The primary disadvantage of such methods is their failure to efficiently deal with partially incomplete drug target and side effect data, reducing the predictive performance. To this end, this paper proposes a new method named Transductive Matrix Co-completion (GTMCC), where missing entries are completed in drug target and drug side effect matrices. It relies on positive unlabeled learning to handle the unlabeled data and graph regularization to incorporate the chemical similarity information. This new approach improves prediction accuracy and generalization, significantly outperforming the existing models on a range of metrics. The GTMCC model marks a significant improvement in computational drug safety prediction toward more efficient and reliable identification of adverse drug effects.

The paper [8] explores a new method for predicting drug side effects using a graph neural network (GNN) called DruGNN. Existing and emerging approaches are challenged by data insufficiency, making it difficult to accurately model the complex interactions of drugs, genes, and biological pathways. To overcome these limitations

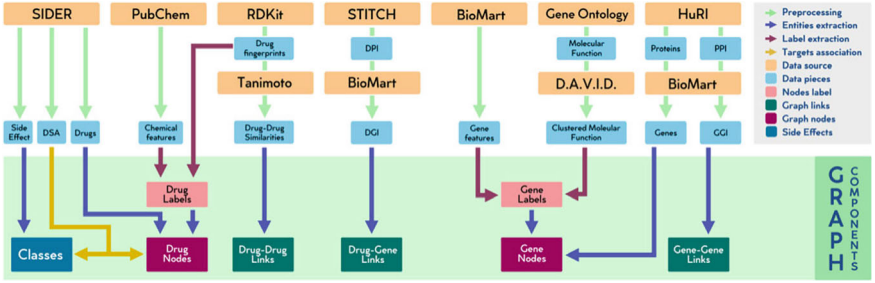

Fig. 1 Sketch of the dataset construction [8]

facing challenges, DruGNN combines a variety of datasets including pharmaceutical chemical components drug targets, interactions, gene expression data, and biological pathway information in a single prediction framework. DruGNN's system architecture uses modular preprocessing to manage each data source independently. The data is converted into a graph display, where nodes represent drugs or genes and edges define drug or gene interactions, i.e., drug–gene relationships or correlation of pathways. GNN can analyze these graphs to find features that capture correlations and biological relevance. The classification level uses these known features to link drugs to potential side effects. By integrating multiple data sources and leveraging GNN's strengths, DruGNN demonstrates improved accuracy and reliability over traditional models. The contributions of this resource support advancements in drug safety research and personalized medicine by providing detailed insights into side effect mechanisms.

In Fig. 1, each data source is shown as an orange rectangle and the data section is shown with a blue square. Green arrows indicate preprocessing steps. This may involve sourcing data from other sources for improved output. Subsets of graph nodes are shown as purple squares. They are labeled with pink squares. The green squares represent a subset of the stream graph while blue squares represent classes (side effect). The red arrow shows creating property labels from data blocks and blue arrows indicate the creation of graph entities (nodes, edges, classes). Yellow arrows indicate the relationship between drug nodes and side effect classes.

3 Proposed Method

3.1 General Architecture

The general overview of the system includes the input datasets of drug side effects (SIDER), tissue-specific gene expression (GTEx), drug properties (PubChem/BioMart), which are further preprocessed, integrated, and fed into a consolidated dataset. A graph where nodes are created as drugs, tissues, or genes and

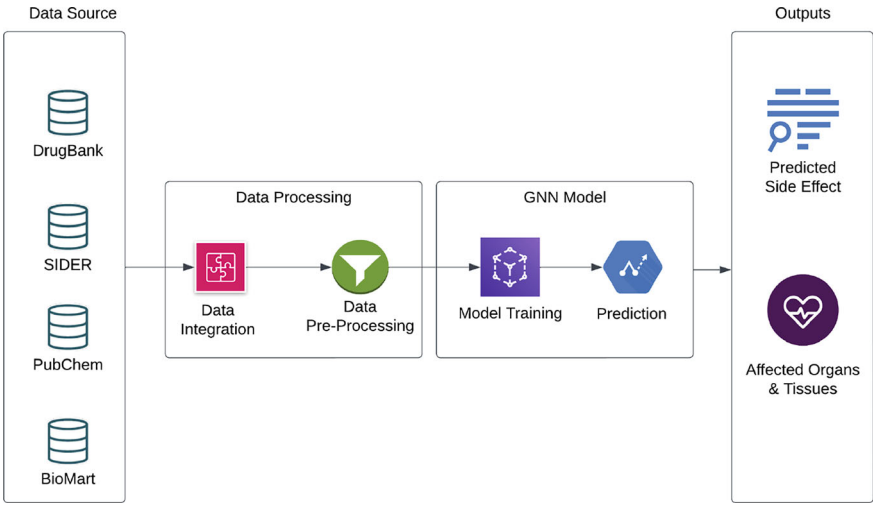

Fig. 2 General architecture

edges connect their relationships. The graph neural network (GNN) model is trained on the graph for side effects prediction as well as specifying the affected tissues or organs. Results will leave the impact of drugs clearer in the direction of improved drug safety analysis (Fig. 2).

3.2 Data Source

The data source step involves collecting data from various databases. To form the basis of the analysis, DrugBank provides detailed information about drugs, including their mechanism of action and interactions, while SIDER reports drug side effects. In addition to providing extensive information on adverse reactions, PubChem offers chemical structure and biological activity information. Finally, BioMart integrates relevant genomic and biological data on drug effects. These resources bring together diverse and complementary datasets needed to identify drug side effects and understand a drug's biological effects.

DrugBank database forms the foundation for analyzing how drugs interact with biological systems and supplements side effect information from other sources. it is the most comprehensive—source of drugs Including information about chemicals Pharmacological and clinical properties Very useful for all as follows:

- **Drug Information**: Contains descriptions, names, types, and approved uses of medicines.

Table 1 Comparison of literature survey

Related works	Description
[1] HSTrans: homogeneous substructures transformer for predicting frequencies of drug side effects (2025)	– **Methods**: HSTrans, CNN – **Dataset**: From Galeano et al. study – **Disadvantages**: Risk of Overfitting, Lack of Biological Interpretability
[2] Machine learning techniques for predicting drug-related side effects (2024)	– **Methods**: Random forest, SVM, k-NN – **Dataset**: SIDER, DrugBank, PubChem – **Disadvantages**: Overfitting, lack of real-world validation
[3] Advancing medical imaging: detecting polypharmacy and adverse drug effects with GCN (2024)	– **Methods**: Graph convolutional networks (GCN) – **Dataset**: Electronic health records (EHR) – **Disadvantages**: Scalability issues, dataset biases
[4] Detecting side effects of adverse drug reactions through drug–drug interactions using GNN (2024)	– **Methods**: Graph neural network (GNN) + self-supervised learning – **Dataset**: TwoSIDES, polypharmacy dataset, DrugBank – **Disadvantages**: Requires more datasets for widespread validation, complex model makes it less interpretable
[5] A systematic review of graph neural network in healthcare-based applications: recent advances, trends, and future directions (2024)	– **Methods**: Recurrent GNN (R-GNN) + spatial GNN (S-GNN) – **Dataset**: Various EHRs, drug interaction databases – **Disadvantages**: Scalability and interpretability
[6] MSDSE: predicting drug side effects based on multi-scale features and deep multi-structure neural network (2024)	– **Methods**: Multi-scale features + deep neural networks (MSDSE) – **Dataset**: SIDER, DrugBank, PubChem – **Disadvantages**: Limited to well-researched drugs
[7] Prediction of drug side effects with transductive matrix co-completion (2023)	– **Methods**: Matrix co-completion – **Dataset**: SIDER, DrugBank, STITCH – **Disadvantages**: Lacks interpretability, scalability issues
[8] Modular multi-source prediction of drug side effects with DruGNN (2023)	– **Methods**: Graph neural network (GNN) – **Dataset**: SIDER, HuRI, STITCH, PubChem – **Disadvantages**: Lacks tissue-specific data, handling rare or low-frequency side effects

The comparison of existing methods is presented in Table 1

- **Mechanism of Action**: Describes a drug's interactions with a targeted biological structure a protein or a gene—to exert the therapeutic effect.
- **Drug Interactions**: Indicates the drug interactions that are harmful or have a reduced effect.
- **Clinical Information**: Provides dosage information, route of administration, and related diseases.

SIDER (Side Effects Resource) is a drug side effect database compiled from public literature and labels. it is important by linking drugs to observed side effects. It creates a core data set for side effect analysis. Key features include:

- **Adverse Reactions**: List of reported side effects for approved drugs with ranking and frequency.
- **Side Effect Profile**: Maps the side effects of related drugs. Helps identify patterns and trends.

PubChem is an open database maintained by the National Center for Biotechnology Information (NCBI), providing a wealth of information on chemical compounds and biological activities, it bridges the gap between a drug's chemical structure and its biological effects, supporting inferences about side effects. Its main features are:

- **Chemical Structures**: Provides 2D and 3D representations of molecules.
- **Biological Activity**: Lists the results of biological activity from experimental tests, connecting a molecule to a target protein or pathway.
- **Molecular Properties**: Includes molecular weight, hydrogen bond donor /acceptor counts, and solubility, which are important for understanding drug behavior.
- **Pharmacological Bridging**: Helps link the molecular properties of a drug with its biological effects, such as toxicity or efficacy.

BioMart is a data integration tool designed to access large-scale genomic and biological datasets. It helps integrate genomic and biological insights, enabling a deeper understanding of how a drug affects specific tissues or organs. It is particularly useful in predicting drug side effects through the following features:

- **Genome Data**: Provides annotation of genes, including their position, function, and variations.
- **Tissue-Specific Information**: Associates gene expression profiles with specific tissues or organs
- **Protein Interactions**: Links the relationships between proteins, genes, and drugs.

3.3 Data Processing

Data processing is a procedure applied to ensure the data input is consistent, accurate, and fit for computational analysis. The two basic steps of data processing are data

integration and data preprocessing

Data Integration process focuses on consolidating data from sources like DrugBank, SIDER, PubChem, and BioMart into a single integrated dataset. This ensures consistency and resolves discrepancies. Some of the key points are as follows:

- **Data Alignment**: Align data from different sources. For instance, ensuring a drug name in DrugBank corresponds to the same side effect in SIDER or chemical property in PubChem.
- **Inconsistency Resolution**: Address inconsistencies between data formats, naming conventions (e.g., synonyms for the same drug), and missing identifiers.
- **Relationship Mapping**: Create meaningful relationships between drugs, medicinal properties, and biological effects. Examples include: Linking drug properties (from PubChem) with mechanism of action (from DrugBank). Mapping side effects (from SIDER) to specific tissues or organs (from BioMart).

The result of this phase is a comprehensive dataset defining drug interactions, chemical properties, and biological effects, which serves as the basis for machine learning work.

Data Preprocessing transforms the integrated dataset into a format suitable for computational analysis. This includes:

- Data cleansing by handling missing or erroneous values.
- Standardization of properties.
- Encoding categorical information into numerical representations.
- Extracting task-relevant features. This ensures that the dataset is robust and ready for later machine learning steps.

3.4 GNN Model

Figure 3 shows the graph neural network (GNN) model is the main part of the prediction system. While training, the GNN learns from the prepared data by using the connections between chemical properties, drug structure, and side effects. By showing drugs and their interactions as graphs, GNNs can understand complex relationships that regular models might not notice. When receiving training, the model enters the prediction phase.

It predicts the possible side effects of the medicine and identifies the organs and tissues that may be affected. GNN's ability to model relationships makes it ideal for tasks involving interconnected biomedical data.

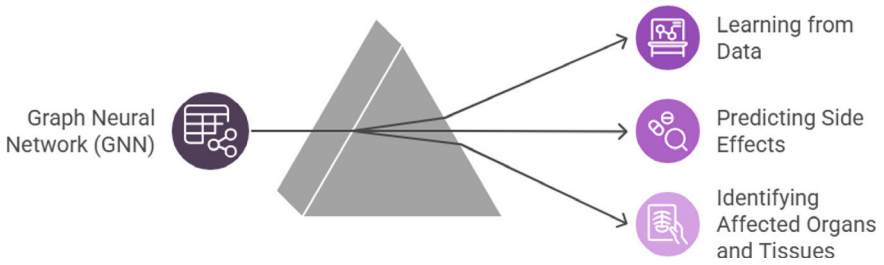

Fig. 3 GNN Model

3.5 Results and Discussion

The results of this process provide valuable insights into the safety and efficacy of the drug. Anticipated side effects clearly indicate the negative interactions that may occur with the drug. This helps researchers understand risks and identify harmful effects. More useful to this approach is the ability to predict which tissues and organs or body parts may be affected by a drug. For example, it can show whether a drug can damage the liver or affect the heart. This information is very important because it helps to clearly understand how the drug affects the human body. A key advantage of this work is that it can help doctors and researchers optimize medicine based on patient symptoms. Knowing the specific organs or tissues that may be affected. They can make safer decisions about how to take the medicine or adjust the dose, for example, if the medicine could cause kidney damage. It can be adjusted to reduce this risk and still effectively treat the aforementioned symptoms. These results are important to drug makers and doctors. They can use this information to design better, safer medicines. This is to ensure that patients are in good health while receiving treatment. The work is still ongoing, but the results so far demonstrate the potential of this approach to enhance drug development and patient care.

4 Conclusion

This research shows that graph neural networks (GNN) can be used to predict drug side effects and determine which tissues and organs or body parts a drug might affect. By combining data from sources such as SIDER, PubChem, DrugBank, and BioMart, the goal is to create graph-based models that can understand complex relationships in biomedical data.This approach addresses the shortcomings of the conventional approach since it looks into side effects but also takes into account the physiological targets involved, thus gaining a better understanding of drug interaction and how the drugs affect the human body. The project is still ongoing, with the work focused on careful model development to achieve high detail and accuracy, raise the safety of drugs, and better health outcomes. By unlocking the unique strength of GNNs, it

attempts to close this critical gap in current practice and yields a robust framework for advancing pharmacovigilance and enhancing safer therapeutic interventions.

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Assessing the Impact of FGSM Adversarial Attack on Convolutional Neural Networks in X-ray Images

Harshita, Kuldeep Kumar, and Sushil Kumar

Abstract In recent years, machine learning has gained applications in various fields, but it has its own limitations as well. These limitations can be in the form of the computational power of the machine in use, vulnerabilities against attacks, etc. With the onset of adversarial attacks, the deployment of machine learning/deep learning models has started facing a serious threat of becoming unreliable. Deep learning models have found their way into healthcare in the fields of prognosis to diagnosis, but because of carefully crafted images to cheat these systems, there is a lot of concern about the practical implementation of these models due to the very sensitive nature of the field of medical science. In this paper, we study the consequence of an adversarial attack, the Fast Gradient Sign Method (FGSM), on the Covid-19 X-Ray image dataset, and its effect on the classification accuracy. Different Convolutional Neural Networks (CNNs), including Simple CNN, InceptionV3, InceptionResNetV2, and MobileNet, have been used to determine the performance degradation that is caused by the FGSM. The accuracies drop from a range of 86–92% to 33–39% which is a very significant amount and discourages the use of machine learning without the deployment of some defense mechanisms, like adding adversarial examples during the training of models, which would make the whole system robust or help mitigate these attacks.

Keywords Machine learning · Adversarial attacks · Convolutional Neural Networks · Medical imaging

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1 Introduction

Having a healthy body is one of the first and foremost requirements for the fulfillment of any other dreams or desires that a person might have. When talking about physical health, the cost of getting a good quality treatment for any particular issue has an ever-increasing monetary value. As health care systems, which include diagnosis to treatment, start becoming more and more expensive, they tend to move away from the grips of low to middle-income-earning individuals and families. Moreover, the availability of good medical infrastructures is also a great concern for people living in rural areas of a country like India and many other such developing nations. The lack of properly trained individuals to perform tasks like taking medical tests and the subsequent production of report results in the overall delay of the treatment that is to be provided to the patient.

With the onset of Machine Learning (ML), various fields have reaped benefits in overcoming their shortcomings, as is the case with healthcare also [1]. ML can be used for the prognosis and diagnosis of diseases without needing the intervention of human beings [2]. This can be a revolutionary factor in both, making healthcare available for all and bringing down the cost of the system to a minimal amount.

The threat to ML and Deep Learning (DL) systems because of adversarial attacks is large. The main idea behind such kinds of attacks is to generate something known as an adversarial example, which is crafted by introducing a small, unnoticeable change to the given input image, also known as a perturbation [2]. The crafted example compromises the working accuracy of the model in use. There are generally two types of attacks [3]:

- *Poisoning Attacks*: These attacks are done during the first phase, i.e., the training phase. It affects the model training and misleads the learning process.
- *Evasion Attacks*: These attacks are done by compromising the test data of the ML/DL system by including harmful inputs. An evasion attack takes place during the inference phase of the training period.

The availability of data on various fields, which include the medical domain, on such a huge scale has facilitated the ML medical systems to have a greatly increased accuracy and efficiency. For example, Daniels et al. [2] do X-ray feature extraction for lung disease classification, Reda et al. [4] used magnetic resonance imaging (MRI) scans for the early diagnosis of ailments such as prostate cancer, and Shaffie et al. [3] used computed tomography (CT) scans as a method to detect lung cancer. The contributions of the paper are:

- We reviewed the consequences of adversarial attacks and existing methods for the detection of these attacks on convolutional neural networks (CNNs) for healthcare concerning medical imaging.
- We study the consequence of an adversarial attack, Fast Gradient Sign Method (FGSM), on the Covid-19 X-Ray image dataset, and its effect on the classification

accuracy. Different CNNs, including Simple CNN, InceptionV3, InceptionResNetV2, and MobileNet, have been used to determine the performance degradation that is caused by the FGSM.

The rest of the paper is organized as follows: Sect. 2 provides a brief overview of adversarial attacks and their types. Section 3 presents a literature review and a tabular comparison of various methods used for adversarial attacks and their detection, followed by a methodology detailed in Sect. 4. Section 5 presents experimental results. Finally, the conclusion and future works are presented in Sect. 6.

2 Background

An adversarial attack is an elusive attack that is meant to cheat the process of image classification by a classifier. The change in the image that is brought by the attack is a little perturbation that is added to it. The way of adding these perturbations depends on the attack, but the end goal for all the techniques remains the same, i.e., to cheat the system. With time, the working of these attacks has become so subtle as to leave minimal change on the image while creating maximum impact on the output.

If we have an image X and a classifier F that produces a label Y, i.e., $F(X) = Y$, then we add a perturbation δX to misclassify to a label $Y*$, i.e., $F(X + \delta X) = Y*$. Figure 1 shows an adversarial attack example of an analog clock. The resulting image is classified as a stopwatch with very high accuracy.

Adversarial attacks are attacks that cheat the ML classifier to degrade the accuracy of the model. These can be categorized into two categories [5]:

1. *White Box Attack*: This is a type of attack in which the model parameters are known to the attacker.
2. *Black Box Attack*: This is a type of attack in which the parameters of the model are hidden from the attacker.

The working of different adversarial attacks is different and varies from attack to attack. But the general principle behind the attacks remains the same. The main idea is to add noise to the images, which results in the creation of perturbed data. The way

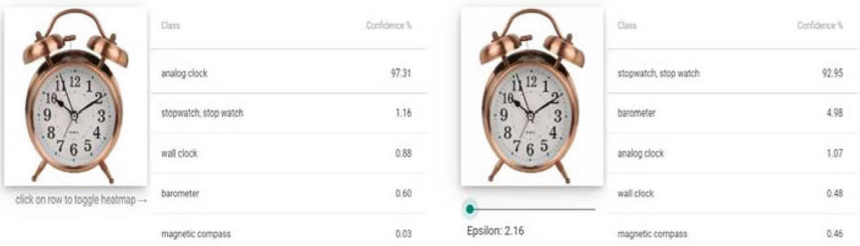

Fig. 1 An example of an Adversarial attack causing misclassification

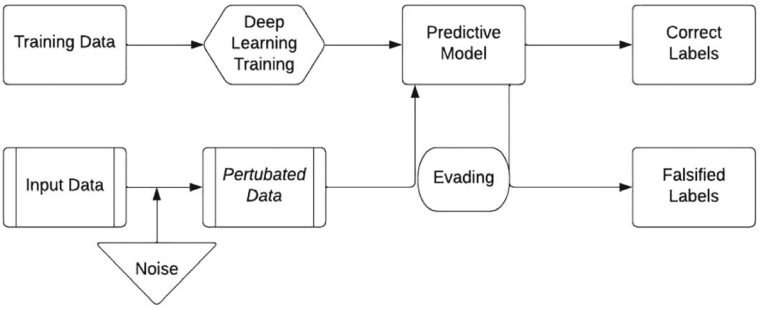

Fig. 2 Adversarial attack working

to achieve this primarily varies. Ultimately, the main aim is to cheat the classifier and cause a misclassification, be it targeted or untargeted. Figure 2 gives a generalized overview of the working of an adversarial attack.

2.1 Adversarial Attacks Probable in CNN Models for Healthcare

Various types of adversarial attacks are possible in CNN models for healthcare. This subsection provides a brief overview of some of such attacks.

- *PGD (Projected Gradient Descent)*: The PGD attack gives access to model gradients to the attacker and falls under the category of white-box attack. The attacker also has access to model weights. It tries to figure out the perturbation that is smaller than the nominal amount of epsilon [5].
- *FGSM (First Gradient Sign Method)*: It is a very simple and effective method of generating adversarial examples. It harnesses and exploits the gradients of a neural network to create an adversarial image. FGSM in its essentiality computes the gradient of a loss function for the input image and creates a new image by making use of the sign of the gradient to create a new image [6].
- *One-Pixel Attack*: One-pixel attack is one of the most concealed attack types, as it is very difficult to identify the pixel involved in it. In recent times, efforts have been made to mitigate it through detection strategies.
- *BIM (Basic Iterative Method)*: BIM is an extension of the FGSM. In the case of BIM, instead of taking one large step, a larger number of smaller steps are taken. It is like applying FGSM multiple times, and the step size is generally fixed to a value [7].
- *C&W Attack (Carlini and Wagner Attack)*: The C&W attack is currently one of the popular attacks to generate adversarial examples. It can even bypass defense mechanisms such as defensive distillation proposed by Papernot et al. [8] and thus can be used as a benchmark for the effectiveness of adversarial attacks [9].

- *JSMA (Jacobian-based Saliency Map Attack)*: JSMA is a gradient-based whitebox attack. In JSMA, the gradient of the loss with each class label for every component of the input is used to extract the sensitivity direction.
- *Image-dependent Perturbation*: In this attack, the perturbations that are formed depend specifically on the different images on which the attack takes place [10].
- *Fatty Liver Attack*: This attack is performed by changing the compression threshold and attenuation coefficient. The parameters selected in this attack change the appearance of edges and speckle patterns [1].
- *Multi-organ Segmentation Attack*: It is an end-to-end method for generating adversarial examples in the case of segmentation of medical images [11].
- *AdvSBF (Adversarial Bias Field Attack)*: AdvSBF aims to degrade the performance of the deep neural network model by performing a calculation of an adversarial bias field [12].
- *HFC (Hierarchical Feature Constraint Attack)*: In HFC, first, the log-likelihood is calculated under the distribution of normal features, and then the most likely Gaussian component is found [13].
- *SMIA (Stabilized Medical Image Attack)*: In SMIA, a stabilized regularization term is proposed to ensure that there is a consistent generation of adversarial perturbations, which are effective in the case of different types of datasets belonging to medical type [14].
- *Physical-world Attack*: In this method, there is physical adversarial manipulation of the real objects. The perturbations so obtained are robust under different physical conditions [15].
- *Multi-scale Attack*: This attack is based on gradients. The gradients are iteratively calculated to minimize the loss function, and the modification of the input image is done based on these gradients [16].

3 Literature Review

Medical image analysis helps in the processing of the human anatomy by using different types of images. The use of this can be for diagnosis, prognosis, and treatment. Classical image processing technologies have generally performed very poorly in this aspect. With the evolution of deep neural networks, this problem can be tackled more efficiently. NN-based models perform quite well when they are used for complex data such as medical data. The use of medical image analysis reduces the chances of any type of medical error by a large margin. A study conducted by Makary and Daniel [17] found that medical errors are among the leading causes of death in the USA. Beinfeld and Gazelle [18] in their study claimed that spending a minimal amount of $385 on medical imaging results in an overall saving of an amount that exceeds $3000. Another study that was done by Lichtenberg [19] pointed out the fact that the use of medical imaging for diagnosis or overall treatment increases the life expectancy of humans. The most common types of medical imaging that are used include MRI, CT scans, X-rays, and ultrasound.

The process of crafting an adversarial example is the addition of a small, indistinguishable, or subtle perturbation to the original pixels of the given image. Due to this, the perturbations when viewed at the base levels of pixels seem like noise that is not at all recognizable by human vision. Byra et al. [1] proposed a method to attack the ultrasound images for the disease of fatty liver. Ultrasound images are made from radio frequencies. The authors have applied an attack known as the zeroth-order optimization attack. The model used was InceptionResNetV2, and a reduction in accuracy of 47% was achieved.

An adversarial attack method that can be combined with any other attack is proposed by Yao et al. [13]. This method aims to hide adversarial features so that adversarial attacks are not easily detected. The medical images used by the author were X-ray and Fundoscopy, and the models in use were ResNet50 and VGG16. The accuracy degradation that was achieved by this method was in the range of 90%. Chen et al. [11] introduced a method to attack image segmentation by generating adversarial examples. The process to create these examples involved generating geometrical deformation to model anatomical and intensity variations. The model that was used by the authors was the U-net, and the medical images used were of CT scans. The accuracy degradation so achieved was around 80%.

Tian et al. [12] studied in-depth the process of the bias field, which results in a drop in the accuracy of the underlying models. The authors proposed a smooth bias field attack to cheat the model into misclassifying. The medical images that were used were X-rays of the chest, and the models under study were ResNet50, MobileNet, and DenseNet121. Accuracy degradation was up to 39%. Qi et al. [14] proposed a new type of medical attack that uses a loss stabilization term along with an objective function that has a loss derivation term. The purpose of the loss derivation term is to augment the discrepancies that exist between the ground truth label and the prediction of the perturbed image. The medical images used here were fundus, CT scan, and endoscopy. The models that were used for this study include ResNet and U-Net, and accuracy degradation of up to 27% was observed.

A targeted segmentation attack that is based on the idea of multiscale gradients is proposed by Shao et al. [16]. This method makes use of feature space perturbation and an adaptive segmentation mask to create a state-of-the-art attack, which is called a multi-scale attack (MSA). The gradients of all the layers are used here to make sure that the perturbation introduced is very small. The models that were used in this study were UNet, R2U-Net, and Attention U-Net. The accuracy degradation that was achieved was 98%. Kugler et al. [15] created attacks on skin images. The perturbations so introduced here were dots and lines made by hand, and it was done either using an acrylic or a pen. The author used a lot of models to study the efficacy of the attack, including ResNet, InceptionV3, InceptionResNetV2, Xception, and MobileNet. The accuracy degradation so obtained was 60%.

In Table 1, the existing attacks that have been implemented against various types of medical images have been summarized. From the degradation of the performance column, we observed that some of the attacks can literally paralyze the whole architecture and bring the whole system down to its knees. The majority of the experimentations that have been done have used CT scans, X-rays, or dermoscopy images.

Table 1 A comparative analysis of existing adversarial attacks on medical images

References	Attacks performed	Models used	Dataset type	Task performed	Performance degradation (%)
Byra et al. [1]	Fatty liver Attack	InceptionResNetV2	Ultrasound	Classification	47
Yao et al. [13]	HFC	VGG16, ResNet50	X-Ray, Fundoscopy	Classification	90
Chen et al. [11]	Multi-organ segmentation attack	U-Net	CT-Scan	Segmentation	85
Li et al. [6]	I-FGSM, FGSM	ResDSN	CT scans	Segmentation	87
Bortsova et al. [5, 20]	PGD	U-Net	X-ray	Segmentation	92
Allyn et al. [10]	Image dependent perturbation	DenseNet201	Dermoscopy	Classification	17
Tian et al. [12]	AdvSBF	ResNet50, MobileNet, DenseNet121	X-ray	Classification	39
Qi et al. [14], 2021	SMIA	U-Net, ResNet	CT-scans, Endoscopy	Classification	27
Kugler et al. [15]	Physical world attack	InceptionV3, InceptionResNetV2, Xception, MobileNet	Dermoscopy	Classification	60
Shao et al. [16]	MSA	U-Net, R2U-Net, Attention U-Net	Dermoscopy	Segmentation	98
Li et al. [21]	FGSM	Pre-trained DL model	CT	Classification	80–0

4 Materials and Methods

In this study, we aim to calculate the attack degradation caused by the FGSM attack on medical image analysis for chest X-ray images. A simple CNN model, which has been self-made, and three other transfer learning models, InceptionV3, Inception-ResNetV2, and MobileNet, have been used for the study. Initially, the dataset had been pre-processed, followed by model training. The models are fine-tuned to get good results. Further, an adversarial attack, FGSM, is performed on all the models individually. During the attack, perturbed data is added, which causes misclassification and accuracy degradation. The difference in the accuracy before and after the attack is the required result. The whole process is shown in Fig. 3.

4.1 Dataset

The dataset being used consists of 4888 X-ray images of the human chest (posteroanterior view), which consist of 2371 Covid-19 positive images and 2517 Normal Images (Covid-19 negative). The images taken are of dimensions 224 × 224. This dataset has been accumulated from different datasets present on Kaggle. The dataset is split in the ratio of 70:30 for training and testing, respectively.

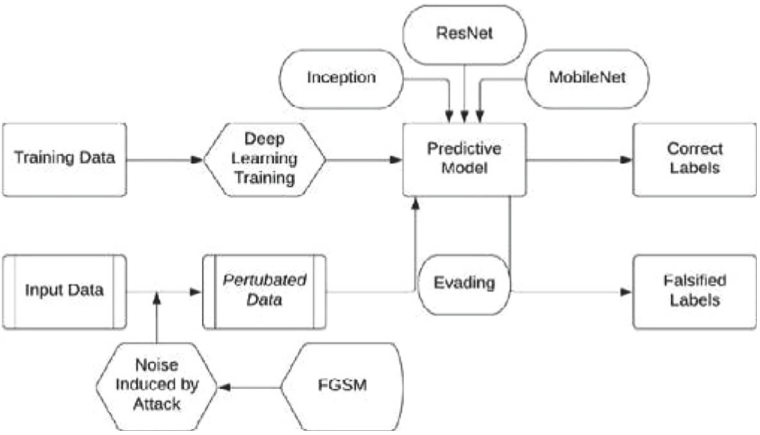

Fig. 3 Proposed model for attack implementation

4.2 Models

To show the effect of the adversarial attack, FGSM, four different models are used. The first is a simple CNN, which is evaluated on a smaller dataset of 224 images. The other three are transfer learning models—InceptionV3, InceptionResNetV2, and MobileNet. Transfer learning is an optimization technique in deep learning where models trained for one task are used to solve another related task to save time and get better performance.

InceptionV3. The inception network is considered to be an important milestone in the development of CNN classifiers [22]. It was not just about stacking the layers to increase the recognition accuracy, but it also focused on improving the speed of the model, utilizing a lot of tricks. As the name suggests, Inceptionv3 is the third version of the Inception network team. Inceptionv3 overcomes the limitations of v2 by using factorized, smaller, and asymmetric convolutions and uses an auxiliary classifier between the layers that acts as a regularizer. The network has proved to be more computationally efficient, both in terms of the number of parameters generated by the network and the economic cost incurred. The main contribution of the network to the inception family is the incorporation of a regularizing component, namely label smoothing, that is added to the loss function to prevent overfitting and the network from becoming centric on one particular class.

InceptionResNetV2. InceptionResnetV2 is a CNN model that is trained on more than a million images of different categories. The images used to train the model are from the ImageNet database. The network is resource-intensive and requires a good amount of resources to be used efficiently. It is 164 layers deep and can classify images into 1000 + categories. The network in itself is very complicated. It is formulated by the basic inception-resnet blocks, multiple-sized convolutional filters, which are combined with residual connections. The residual connections help avoid degradation problems and reduce the training time of the model [1].

MobileNet. MobileNet is a small model that has low latency and consumes a competitively low amount of power. It is useful when there are resource constraints, and other functionalities of the model are similar to large models such as InceptionV3. MobileNet can even be used on embedded systems and has a huge number of use cases. It uses depth-wise separable convolutions. This is what results in a reduced number of parameters when compared to networks that make use of regular convolutions with the same depth in the networks [23].

4.3 Performance Metrics

The metric used to evaluate the performance degradation of the modalities under test is accuracy. Different metrics can be used for this purpose, but for the measurement of the degradation of performance of a model under attack, accuracy is the most

desirable one. Accuracy is defined as the total number of accurate classifications made by the classifier. The mathematical representation of accuracy is given by Eq. (1).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

where TP is true positive, FP is false positive, TN is true negative, and FN is false negative.

5 Experimental Results

All the experiments have been performed on the system having the Intel® Core™ i5-5200U CPU@ 2.20 GHz processor, having a RAM capacity of 8.00 GB. The system has a 64-bit operating system, an × 64-based processor. The GPU is an AMD Radeon R9 edition. The performance of various models is measured before and after the attack to calculate the difference between them. The difference between the accuracies of the models gives the accuracy degradation. The accuracy before FGSM is performed on a simple CNN is shown in Table 2. The average accuracy of the simple CNN before the attack is found to be 88.18%. The average accuracy of the simple CNN after the attack is found to be 54.37%.

Table 3 shows the accuracy before and after FGSM is performed on MobileNet. The average accuracy of MobileNet before the attack is found to be 86.41%. The average accuracy of MobileNet after the attack is found to be 47.20%.

The accuracy before and after FGSM is applied to InceptionV3 is shown in Table 4. The average accuracy of InceptionV3 before and after the attack is found to be 88.25% and 51.00% respectively. Similarly, Table 5 shows the accuracy before and after FGSM is performed on InceptionResNetV2. The average accuracy of InceptionResNetV2 before and after the attack is found to be 92.12% and 53.57% respectively.

Table 2 Accuracy of simple CNN before and after the attack

Epoch	Accuracy of simple CNN before the attack				Accuracy of simple CNN after the attack			
	Loss	Accuracy	Val_loss	Val_ accuracy	Loss	Accuracy	Val_loss	Val_ accuracy
1	2.35	0.36	1.76	0.56	3.49	0.60	3.33	0.53
2	1.36	0.56	0.61	0.93	3.43	0.58	3.29	0.43
3	1.01	0.60	0.74	0.98	3.37	0.60	3.26	0.50
4	0.98	0.68	0.50	0.93	3.34	0.55	3.22	0.65
5	0.58	0.79	0.29	0.96	3.31	0.50	3.21	0.59

Table 3 Accuracy of MobileNet before and after the attack

Epoch	Accuracy of MobileNet before the attack				Accuracy of MobileNet after the attack			
	Loss	Accuracy	Val_ loss	val_ accuracy	Loss	Accuracy	Val_ loss	Val_ accuracy
1	0.030	0.987	1.550	0.779	0.704	0.488	0.695	0.427
2	0.043	0.991	0.231	0.933	0.700	0.491	0.695	0.471
3	0.035	0.982	0.328	0.906	0.704	0.517	0.690	0.512
4	0.030	0.988	0.799	0.850	1.145	0.487	0.701	0.542
5	0.655	0.979	0.241	0.954	7.077	0.511	0.755	0.471

Table 4 Accuracy of InceptionV3 before and after the attack

Epoch	Accuracy of InceptionV3 before the attack				Accuracy of InceptionV3 after the attack			
	Loss	Accuracy	Val_ loss	Val_ accuracy	Loss	Accuracy	Val_ loss	Val_ accuracy
1	0.055	0.975	0.343	0.906	0.715	0.481	0.692	0.512
2	0.079	0.975	0.884	0.851	0.703	0.512	0.692	0.531
3	0.070	0.972	0.188	0.929	0.701	0.476	0.696	0.525
4	0.064	0.981	0.238	0.914	0.696	0.513	0.708	0.506
5	0.072	0.972	0.725	0.860	0.720	0.528	0.694	0.475

Table 5 Accuracy of InceptionResNetV2 before and after the attack

Epoch	Accuracy of InceptionResNetV2 before the attack				Accuracy of InceptionResNetV2 after the attack			
	Loss	Accuracy	Val_ loss	Val_ accuracy	Loss	Accuracy	Val_ loss	Val_ accuracy
1	0.025	0.991	0.952	0.890	0.712	0.478	0.748	0.534
2	0.032	0.989	0.572	0.918	0.736	0.532	0.814	0.522
3	0.039	0.985	0.270	0.918	0.714	0.468	0.832	0.528
4	0.028	0.989	0.200	0.937	0.708	0.459	0.719	0.539
5	0.020	0.992	0.249	0.940	0.700	0.492	0.690	0.540

The degradation that is caused by the FGSM attack also depends on the value of the perturbation added. If the perturbation added is too small, then the degradation is small, but if the perturbation that is added is too large, then the difference between the images becomes visible to the human eye, which beats the whole purpose of the attack, because for the attack to be effective, the images before and after the attack should be indistinguishable to the human eye. The value of perturbation, which determines how much perturbation is added, lies between 0 and 1, but it has been found experimentally that any value between 0.01 and 0.05 is a good value to consider.

Table 6 Performance degradation of the models due to the FGSM attack

Model	Dataset size (Images)	Accuracy before attack (%)	Accuracy after attack (%)	Accuracy degradation (%)
Simple CNN	224	88.18	54.37	33.81
MobileNet	4890	86.41	47.20	39.21
InceptionV3	4890	88.25	51.00	37.25
InceptionResnetV2	4890	92.12	53.57	38.55

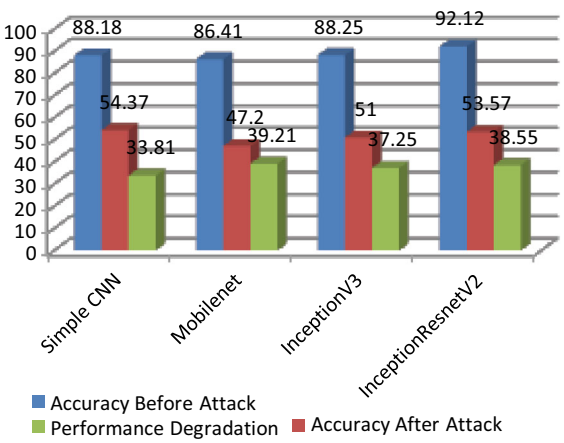

Fig. 4 Comparison between the accuracies and degradation among different models

We considered the value of perturbation to be 0.01 for the experiment. Table 6 shows that the accuracy of the models before the attack is lowest for Simple CNN and is highest for InceptionResNetV2. The same can also be said for the accuracy after the attack. From the table, we can also see that the accuracy degradation is highest in the MobileNet model and lowest in the Simple CNN.

Figure 4 graphically shows the performance degradation. The performance degradation on various models is found to be in the range of 33.81–39.21%. The average performance degradation is 37.21% which is a lot if it is being considered to be used for a sensitive area such as a medical diagnosis or prognosis.

5.1 Discussion

The ability to generate adversarial examples against deep learning models has raised a lot of questions on the safety and practical implementation of these models, as they can so easily be fooled. This phenomenon is a setback for a lot of domains that make use of deep learning technologies, particularly in the case of healthcare, in which a minor distortion can lead to a wrong prescription or a wrong diagnosis and treatment

altogether, which can cause severe health issues for patients and, in the worst-case scenario, even death.

Adversarial training, which is training the models with inputs containing adversarial examples as a defense against the method, is the most widely used technique because of its simplistic nature. It generally proves to be effective against white-box attacks but fails to provide the same protection against black-box attacks. Also, adversarial training against one attack does provide some resilience to the model against other attacks. It has also been observed that the higher the number of layers in a model, the more it is vulnerable to attacks, so there is always a trade-off between accuracy and robustness.

Further, existing detection and defense mechanisms are not sufficient as they still lack an adequate level of safety against the various types of existing attacks. Adversarial attacks on medical images are a new field, and new attacks will come up regularly, so there is a need to constantly upgrade the defense mechanism to counter them.

6 Conclusion and Future Scope

Convolutional neural networks as a tool have vastly improved medical image analysis and are on the path to becoming a crucial tool for doctors, hospitals, or any healthcare professionals. The onset of adversarial attacks on the scene has caused a big impact on the accuracy and dependability of machine learning healthcare architectures. In this paper, we first summarize the works that investigated adversarial examples on medical images. Proper assessment and methods to tackle these attacks are a must for the sustainable use of emerging machine-learning technologies in the field of medical science, as the benefits that could be gained by the use of machine learning have virtually no boundaries.

We deploy various CNN models such as InceptionV3, InceptionResNetV2, and MobileNet to aid the process of medical image analysis. The accuracy obtained for the same is very promising. However, the performance degradation due to the FGSM attack on various models is found to be in the range of 33.81–39.21%. If the attacks that cause these systems to fail can be mitigated, then it can be hoped to use these systems in real-life scenarios. To be able to apply CNN methods to medical healthcare image analysis, methods are needed to counter adversarial attacks. Finding ways to defend that cater to a particular attack with very high efficiency needs to be studied in a detailed manner.

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Evaluation of Side Effects Associated with COVID-19 Vaccines in India

Divyansh Singhal, Karuna Panwar, Kusum Deep, and Millie Pant

Abstract This study investigates adverse effects and newly diagnosed illnesses following COVID-19 vaccination with Covishield and Covaxin in 267 participants across India. Conducted via an online survey from June 1 to July 5, 2024, the study includes participants of all ages and genders who received at least one dose. Adults aged 20–30 reported the most side effects, which decreased after the second dose. Common short-term side effects were pain, redness, and allergic reactions, while long-term effects included pain, breath shortness, and memory loss, particularly in older adults. Covaxin was linked to more pain and allergies, whereas Covishield was associated with gastrointestinal, liver, and blood-related illnesses. Newly diagnosed illnesses like heart, respiratory, neurological, and kidney-related issues were more common with Covaxin. The study underscores the importance of continuous monitoring and tailored care to manage adverse effects, enhancing public confidence in vaccination programs.

Keywords COVID-19 · Covaxin · Covishield · Side effects · Vaccine

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1 Introduction

Vaccinations protect against disease through various mechanisms, primarily by stimulating the body's immune system to recognize and combat pathogens. However, the process of developing immunity can result in side effects. The coronavirus disease 2019 (COVID-19) virus has wreaked havoc on the planet, causing widespread illness and death. In response, effective vaccination has emerged as the best approach to safeguarding the globe from this virus.

COVID-19 vaccines have demonstrated substantial safety in clinical trials. Nonetheless, there is limited evidence comparing the safety of these vaccines in real-world settings. This gap highlights the necessity to investigate the nature and severity of the adverse effects reported and to understand the differences based on the type of vaccine received. COVID-19 remains a deadly disease that poses a serious threat to humanity, making it imperative to protect the public and limit viral spread through vaccination. However, public acceptance of vaccines is significantly influenced by their efficacy and the side effects they cause [1].

Vaccines are a critical tool for controlling infectious diseases, known for their effectiveness in fostering long-lasting immune memory. Each year, vaccines save the lives of millions of people. When recommending or rejecting vaccinations, the potential adverse effects on public trust in healthcare professionals must be considered. In the global fight against the COVID-19 pandemic, vaccines have emerged as the greatest savior for humankind. Many vaccine candidates were developed and entered clinical trials in early 2020, leading to several obtaining emergency approval. Consequently, governments worldwide initiated extensive vaccination campaigns against COVID-19 [2, 3].

Given the importance of understanding vaccine safety in both short and long-term contexts, this paper aims to analyze the side effects associated with COVID-19 vaccines in India. By examining adverse effects reported in various settings and comparing these outcomes across different vaccine types, this study seeks to provide a comprehensive overview of the safety profile of COVID-19 vaccines. This information is crucial for informing public health strategies, guiding vaccine recommendations, and ultimately ensuring public confidence in the ongoing vaccination efforts against COVID-19.

2 Literature Review

The literature on the adverse effects of COVID-19 vaccines provides valuable insights into the safety and side effects experienced by different populations across the world. This review highlights findings from studies conducted in Iran, France, Jordan, Bangladesh, Saudi Arabia, Poland, Egypt, and the UAE.

In Iran, Babaee et al. [4] analyzed 4775 participants who received Sinopharm, AstraZeneca, or Sputnik V vaccines. They found that 55.56% reported adverse effects

after the first dose, and 35.7% after the second, with Sputnik V having the highest rate of adverse effects (82.7%), followed by AstraZeneca (70.5%) and Sinopharm (37.4%). Common side effects included fatigue, chills/fever, and skeletal pain, concluding that the vaccines used in Iran are generally safe with no reports of serious adverse effects. Similarly, Tissot et al. [5] in France focused on the Pfizer-BioNTech vaccine among 311 patients, finding that 74% reported side effects, with those having prior COVID-19 infection experiencing more side effects (95%) compared to COVID-19-naive patients (70%). Despite the higher incidence, the severity of symptoms was similar across both groups.

Omeisha et al. [6] surveyed 1086 participants in Jordan receiving AstraZeneca, Pfizer, or Sinopharm vaccines, finding that 89.9% reported side effects after the first dose, with the most common being injection site pain, fatigue, myalgia, headache, and chills. AstraZeneca was associated with more side effects than Pfizer and Sinopharm, but generally, the side effects were not severe. Mohsin et al. [7] in Bangladesh analyzed 1180 vaccinated individuals through an online survey, finding that 39.48% reported at least one side effect, with injection site pain, fever, headache, redness/swelling, and lethargy being the most common. Side effects were more prevalent among recipients of Pfizer-BioNTech and Moderna vaccines compared to Sinopharm and Sinovac.

In Saudi Arabia, Alhazmi et al. [8] included 515 participants, finding that 60% reported side effects, primarily fatigue and pain at the injection site, consistent with clinical trial reports. Andrzejczak et al. [9] in Poland compared side effects of the AstraZeneca and Pfizer vaccines among 705 respondents, finding that 96.5% of AstraZeneca recipients reported at least one side effect, compared to 93.9% of Pfizer recipients. The side effects were generally mild and less frequent after the second dose of the Pfizer vaccine. Elgendy et al. [10] in Egypt surveyed participants who received BBIBP-CorV, ChAdOx1, or BNT162 vaccines, finding common side effects such as pain at the injection site, muscle pain, fatigue, dizziness, fever, and headache, with ChAdOx1 causing more side effects than BBIBP-CorV and BNT162.

Amer et al. [11] in Saudi Arabia, Eastern Province surveyed 2293 individuals, finding that the most prevalent side effects were pain at the injection site, fatigue, and swelling, with non-smoking participants and those without comorbidities reporting more side effects. The Pfizer vaccine was associated with fewer side effects compared to AstraZeneca. Lastly, Ganesan et al. [12] surveyed 1878 vaccine recipients in the UAE, finding that pain at the injection site, fatigue, headache, and joint/muscle pain were the most common side effects. The mRNA Pfizer-BioNTech vaccine had higher odds of side effects compared to the inactivated Sinopharm vaccine, with most side effects being mild and requiring no or home-based treatment.

These studies indicate that COVID-19 vaccines, though generally safe, exhibit varying rates and types of side effects across different populations and vaccine types. Addressing these findings can help improve public confidence in vaccination programs.

3 Data Collection and Study Design

An observational study was conducted through an online survey from June 1, 2024, to July 5, 2024, targeting individuals across India. The aim was to identify side effects reported following COVID-19 vaccination, without collecting any personal identification details.

Participants included individuals of all genders and ages who had received at least one dose of the COVID-19 vaccine, were able to provide consent, and had been residing in India since receiving their first dose. The survey was designed using Google Forms and distributed via social media platforms, with a total of 267 participants responding.

The questionnaire was crafted to capture reactogenicity to the COVID-19 vaccine and factors associated with the development of side effects. It was prepared in English.

The survey was divided into several parts:

- **Demographic Information**: Questions on age, gender, education, state, etc.
- **Health Status Prior to Vaccination**: Questions about comorbid conditions such as diabetes, hypertension, chronic lung diseases, cardiovascular issues, and other chronic diseases.
- **Vaccine Information**: Details about the vaccine name and number of doses received.
- **Short-term Side Effects**: Information on side effects experienced after the first and second doses.
- **Long-term Adverse Effects**: Questions about any persistent side effects.

The age of participants ages were categorized into four groups: 0–20, 30–40, 40–50, and above 50. The adverse effects were classified into local (injection site reactions) and systemic side effects. Local side effects included pain at the injection site, redness, swelling, lymphadenopathy, and itching. Systemic side effects encompassed fever, headache, muscle and joint pain, nausea, numbness, and confusion.

4 Data Analysis and Results

This research focused on the side effects of the COVID-19 vaccine, gathering data from 267 individuals across India via a Google Form. Prior to analysis, the data was categorized into four main age groups: below 20, 20–30, 30–50, and 50+. Each of these groups was further divided based on COVID-19 infection history before vaccination: infected and not infected.

Table 1 presents the survey data collected from 267 participants, predominantly young adults aged 20–30 (35.9%), followed by those below 20 (22.9%), middle-aged individuals (30–50) at 18.2%, and older adults (50–70) at 6.9%. The gender distribution showed a majority of males (57.6%) compared to females (26.5%), with a small percentage preferring not to disclose their gender (0.3%). Participants were

Table 1 Demographic and vaccination details of survey participants

Category	Subcategory	Detail	Participants(%)
Age	Children	0–20	27.30
	Young adults	20–30	42.60
	Middle-aged	30–50	21.70
	Older adults	50–70	8.23
Gender		Male	68.53
		Female	31.40
		Prefer not to say	0.37
State		Uttar Pradesh	26.21
		Bihar	16.10
		Tripura	10.48
		West Bengal	9.73
		Haryana	5.60
		Andhra Pradesh	5.24
		Uttarakhand	4.11
		Maharashtra	2.99
		Tamil Nadu	2.99
		Rajasthan	2.99
		Delhi	2.24
		Jharkhand	1.87
		Madhya Pradesh	1.87
		Gujarat	1.49
		Telangana	1.12
		Odisha	1.12
		Assam	0.70
		Chhattisgarh	0.70
		Himachal Pradesh	0.70
		Punjab	0.70
		Kerala	0.37
		Karnataka	0.37
		Chandigarh	0.37
Profession		Student	62.54
		Corporate	10.11
		Self Employed	8.90
		Government Services	8.61
		Home Maker	4.11
		Retired	2.99
		Unemployed	2.24
Health	Related to health sector	No	94
		Yes	4.49
		Maybe	1.87

(continued)

Table 1 (continued)

Category	Subcategory	Detail	Participants(%)
Marital status		Unmarried	70.41
		Married	28.83
		Divorced	0.74
		Widowed	0.37
Addiction	Intoxication	No	97.01
		Yes	2.61
		Maybe	0.37
COVID-19 history	Infected before vaccination	No	75
		Yes	18.65
		Maybe	6.30
	Times infected with COVID-19	0	61.94
		1	20.89
		2	2.90
Vaccination details	First dose vaccine	Covishield	54.85
		Covaxin	44.02
		Others	1.11
	Total doses received	2nd dose	73.80
		3rd dose (2nd + booster dose)	19.77
		1st dose	6.30
	Time gap between doses	Recommended interval	67.91
		30 days more	23.13
		15 days more	5.97
		Less than recommended	2.98
	Second dose vaccine	Same as first dose	91.04
		Other	8.95

mainly from Uttar Pradesh (22.0%) and Bihar (13.5%), with lesser representation from other states. Over half of the participants were students (53.2%), while others were corporate employees (8.5%), self-employed (7.5%), in government services (7.2%), homemakers (3.5%), retired (2.5%), or unemployed (1.9%). Most participants were not involved in the health sector (79.1%), were unmarried (59.2%), and did not have issues with intoxication (81.8%). Regarding COVID-19 history, 63.2% had not been infected before vaccination, and most had received their second dose (62.3%), with Covishield being the most common first dose vaccine (46.2%). The majority followed the recommended interval between doses (57.2%), and 76.0% received the same vaccine for their second dose.

From Table 2 detailing short-term side effects following the first and second doses of the COVID-19 vaccine across different age groups, several key observations can be made. Pain was the most commonly reported side effect across all age groups for both doses, with adults (20–30) experiencing the highest incidence (60.53% for Dose 1 and 40.35% for Dose 2). Younger individuals (below 20) saw a significant drop in pain from the first dose (45.21%) to the second dose (34.25%). Redness was most frequent in adults, decreasing from 21.93% after Dose 1 to 10.53% after Dose 2. Allergic reactions were notably high in older adults (50–70) after Dose 1 (18.18%) but decreased after Dose 2 (9.09%). Swelling and itching were prevalent in younger and adult groups but reduced after the second dose. Fever was highly reported among adults after Dose 1 (61.40%) and dropped significantly after Dose 2 (37.72%), with chills following a similar trend. Fatigue and muscle ache were also common in adults after Dose 1 (43.86% and 50%, respectively) and decreased after Dose 2. Headache incidence was high in adults after Dose 1 (37.72%) and reduced after Dose 2 (16.67%). Other symptoms like nausea, shortness of breath, and drowsiness showed varied patterns across doses and age groups, with drowsiness notably high in older adults after Dose 1 (31.82%) and decreasing after Dose 2 (9.09%). Hair fall and joint pain persisted in middle-aged individuals (30–50) after the second dose. Overall, adults and younger individuals reported the highest incidence of side effects, which generally lessened with the second dose, highlighting the importance of monitoring vaccine side effects in these groups.

From Table 3 detailing long-term side effects across different age groups following COVID-19 vaccination, several conclusions can be drawn. Pain is a common long-term side effect, particularly in the young (20.55%) and middle-aged (20.69%) groups, with lower incidence in older adults (4.55%). Allergies and skin issues, such as rashes and swelling, are more prevalent in middle-aged and older adults. Muscle or joint pain is significant in middle-aged (20.69%) and older adults (18.18%). Breath shortness and heart diseases are notably higher in older adults (18.18%). Neurological and sensory issues, such as vision problems and hearing disorders, are more common in older adults. Weakness is a prominent issue in middle-aged (27.59%) and young adults (14.91%). Hair loss is significant in middle-aged (20.69%) and young adults (14.04%), while memory loss is frequent in older adults (18.18%) and middle-aged individuals (18.97%). Overall, middle-aged and older adults experience a higher incidence of long-term side effects compared to younger individuals, with pain, muscle or joint pain, breath shortness, heart diseases, vision problems, and memory loss being notably prevalent.

Table 4 detailing newly diagnosed diseases after vaccination across different age groups reveals several trends. Heart disease is most prevalent in older adults (13.64%) and middle-aged individuals (10.34%), with lower rates in adults (5.26%) and young people (1.37%). Respiratory illnesses follow a similar pattern, peaking in middle-aged (10.34%) and older adults (9.09%). Neurological illnesses are more common in middle-aged individuals (8.62%). Gastrointestinal illnesses are notably high in middle-aged (10.34%) and adults (8.77%).

Liver-related and kidney-related illnesses show significant incidences in middle-aged individuals (12.07% and 10.34%) but are absent in older adults. Blood-related

Table 2 Short-term side effects

Side effects	Dose 1				Dose 2			
	Young (%)	Adults (%)	Middle (%)	Old (%)	Young (%)	Adults (%)	Middle (%)	Old (%)
Pain	45.21	60.53	37.93	40.91	34.25	40.35	36.21	40.91
Redness	19.18	21.93	13.79	13.64	15.07	10.53	15.52	13.64
Allergic reactions	5.48	7.89	13.59	18.18	4.11	3.51	6.90	9.09
Rashes	6.85	7.02	12.07	9.09	5.48	2.63	5.17	9.09
Swelling	20.55	22.81	17.24	13.64	9.59	9.65	13.79	13.64
Itching	17.81	13.16	12.07	18.18	6.85	7.89	6.90	13.64
Fever	42.47	61.40	36.21	50	30.14	37.72	37.93	40.91
Chills	12.33	21.93	10.34	18.18	5.48	10.53	8.62	4.55
Fatigue or tiredness	31.51	43.86	25.86	36.36	17.81	21.93	25.86	22.73
Muscle or body ache	35.62	50.00	27.59	36.36	20.55	24.56	31.03	22.73
Headache	20.55	37.72	20.69	40.91	15.07	16.67	22.41	18.18
Nausea or vomiting	6.85	7.89	6.90	9.09	2.74	7.02	8.62	0
Joint pain	10.96	17.54	17.24	40.91	8.22	8.77	15.52	13.64
Confusion effect	4.11	8.77	6.90	4.55	4.11	2.63	3.45	4.55
Breath shortness	5.48	6.14	8.62	13.64	5.48	3.51	10.34	9.09
Drowsy	13.70	10.53	12.07	31.82	10.96	6.14	6.90	9.09
Numbness	5.48	5.26	3.45	4.55	4.11	1.75	6.90	9.09
Respiratory issues	6.85	4.39	3.45	9.09	4.11	2.63	8.62	9.09
Diarrhea	4.11	6.14	5.17	0	2.74	3.51	3.45	4.55
Hairfall	12.33	14.91	0	4.55	8.22	9.65	17.24	4.55
Any other	1.37	4.39	0	0	1.37	3.51	6.90	4.55

and skin-related illnesses are higher in middle-aged (8.62% and 10.34%) and older adults (9.09%). Cancer and autoimmune illnesses are primarily found in adults and middle-aged groups. Other significant illnesses are most common in middle-aged individuals (10.34%).

In summary, newly diagnosed diseases after vaccination are more prevalent in middle-aged and older adults, with heart disease, respiratory, gastrointestinal, liver, kidney, blood, and skin-related illnesses being the most common. Younger individuals report fewer new diagnoses, highlighting an increase in health issues with age post-vaccination.

The analysis of persistent side effects from Covishield and Covaxin vaccines reveals distinct differences, shown in Fig. 1. Covaxin recipients reported higher incidences of pain (19.49%), redness (6.80%), allergies (4.76%), and swelling (6.12%)

Table 3 Long-term side effects

Side effects	Young (%)	Adults (%)	Middle (%)	Old (%)
Pain	20.55	14.04	20.69	4.55
Redness or discoloration	5.48	4.39	5.17	4.55
Allergies	1.37	3.51	5.17	4.55
Rashes or skin irritation	1.37	5.26	3.45	4.55
Swelling	6.85	5.26	6.90	4.55
Itching	2.74	5.26	3.45	4.55
Muscle or joint pain	6.85	13.16	20.69	18.18
Breath shortness	5.48	7.89	10.34	18.18
Numbness	2.74	5.26	8.62	9.09
Weakness	6.85	14.91	27.59	13.64
Drowsiness	4.11	5.26	3.45	4.55
Confusion	5.48	7.02	6.90	9.09
Olfactory issues	1.37	3.51	3.45	4.55
Vision problems	4.11	7.89	12.07	18.18
Hearing disorder	1.37	4.39	1.72	9.09
Heart diseases	1.37	5.26	8.62	18.18
Respiratory issues	2.74	3.51	8.62	13.64
Neurological problem	1.37	3.51	5.17	4.55
Menstrual changes	9.59	5.26	8.62	0.00
Lungs diseases	2.74	3.51	3.45	9.09
Kidney diseases	1.37	2.63	3.45	0.00
Hairloss	10.96	14.04	20.69	0.00
Memory loss	4.11	11.40	18.97	18.18
Any other	1.37	2.63	5.17	0.00

compared to Covishield recipients, who reported these effects at 14.97%, 2.54%, 1.69%, and 2.54%, respectively. Muscle or joint pain was similarly prevalent between Covishield (13.61%) and Covaxin (13.56%). Breath shortness and numbness were higher in Covishield recipients at 9.52 and 12.71%, compared to 7.65% and 7.48% for Covaxin.

Other significant side effects were more frequent in Covishield recipients, including vision problems (10.20%), hearing disorders (6.78%), heart disease (10.20%), neurological problems (7.48%), and memory loss (14.29%). Covaxin recipients had higher rates of menstrual changes (5.08%) and kidney diseases (3.40%).

Table 4 Newly disease diagnosed after vaccination

Disease Diagnosed	Young (%)	Adults (%)	Middle (%)	Old(%)
Heart Disease	1.37	5.26	10.34	13.64
Respiratory Illness	0.00	5.26	10.34	9.09
Neurological illness	1.37	3.51	8.62	0.00
Gastrointestinal illness	4.11	8.77	10.34	9.09
Liver-related illness	2.74	7.02	12.07	0.00
Kidney-related illness	2.74	3.51	10.34	0.00
Blood-related illness	2.74	5.26	8.62	9.09
Skin-related illness	5.48	7.02	10.34	9.09
Cancer	0.00	3.51	5.17	0.00
Autoimmune Illness	0.00	5.26	8.62	0.00
Any other significant illnesses	0.00	4.39	10.34	4.55

Overall, Covishield recipients experienced higher incidences of most persistent side effects compared to Covaxin recipients, particularly in pain, numbness, vision problems, heart disease, and memory loss. This suggests a need for ongoing monitoring and evaluation of the safety profiles of these vaccines.

The analysis of newly diagnosed illnesses after receiving the Covishield and Covaxin vaccines reveals notable differences, presented in Fig. 2. Covaxin recipients reported higher incidences of heart disease (8.47 vs. 4.08%), respiratory illnesses (6.78 vs. 4.08%), neurological illnesses (6.78 vs. 4.08%), and kidney-related illnesses (6.12 vs. 3.40%) compared to Covishield recipients. Conversely, Covishield recipients had higher rates of gastrointestinal illnesses (8.16 vs. 7.63%), liver-related illnesses (5.08 vs. 2.72%), and blood-related illnesses (5.93 vs. 5.44%). Both vaccines had similar rates of skin-related illnesses, cancer, and autoimmune illnesses. Overall, Covaxin is associated with higher incidences of heart, respiratory, neurological, and kidney-related issues, while Covishield is linked to more gastrointestinal, liver, and blood-related illnesses. This underscores the importance of ongoing monitoring and tailored post-vaccination care.

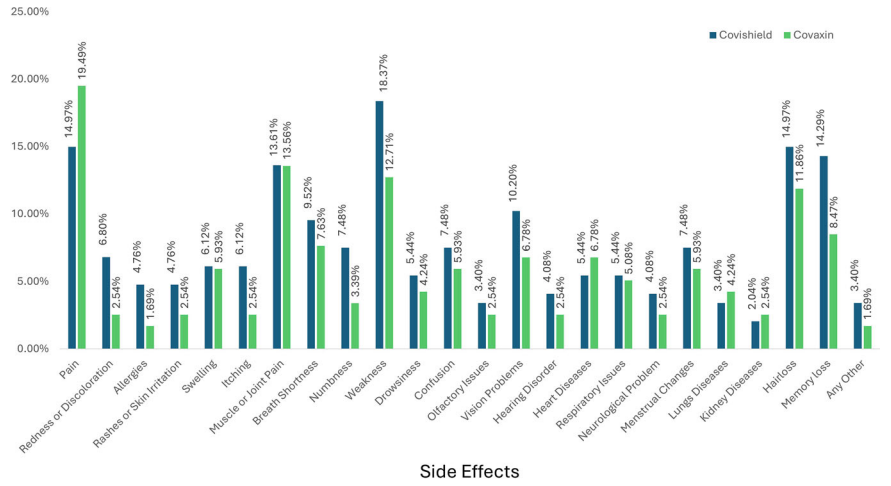

Fig. 1 Overall side effect of vaccines (persistent)

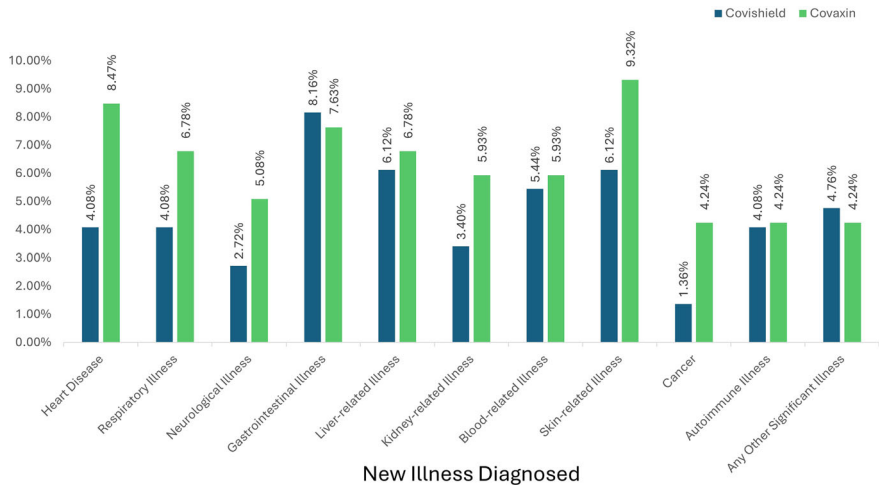

Fig. 2 New illness diagnosed after vaccination

5 Conclusion

This study provides important insights into the side effects and newly diagnosed illnesses following COVID-19 vaccination with Covishield and Covaxin. The data collected from 267 participants across India reveal distinct patterns of adverse effects associated with each vaccine. Adults aged 20–30 reported the highest incidence of side effects, which generally decreased after the second dose. Pain, redness, and allergic reactions were common short-term side effects, while long-term issues such

as muscle or joint pain, breath shortness, and memory loss were more prevalent in middle-aged and older adults. Covaxin was associated with higher rates of pain, redness, allergies, and swelling, whereas Covishield recipients experienced more gastrointestinal, liver, and blood-related illnesses. Additionally, Covaxin recipients reported higher incidences of newly diagnosed heart, respiratory, neurological, and kidney-related illnesses, while Covishield was linked to more gastrointestinal and liver-related issues. These findings underscore the importance of continuous monitoring and individualized post-vaccination care to address the specific side effects and newly diagnosed illnesses associated with each vaccine. By understanding and managing these adverse effects, healthcare providers can improve public confidence in vaccination programs and ensure better health outcomes for vaccinated individuals.

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